~Supplementary Manual~

WuKong: An LLM-Enhanced Framework for Adaptable Proteomics Analysis

Shisheng Wang[1,2]*, XinYuan Wang[3], Wenjuan Zeng[1], Youmei Jin[3], Tao Su[3], Zhongda Chen[5], Yi Zhong[3], Chengpin Shen[5]*, Yansheng Liu[6,7]* and Hao Yang[1,2,4]*

[1] Transplantation Center & Institute of Organ Transplantation, NHC Key Lab of Transplant Engineering and Immunology, Institutes for Systems Genetics, Frontiers Science Center for Disease-Related Molecular Network, West China Hospital, Sichuan University, Chengdu 610041, China

[2] Tianfu Jincheng Laboratory, Chengdu, 610093, China

[3] Proteomics-Metabolomics Platform, Core facilities, West China Hospital, Sichuan University, Chengdu 610041, China

[4] Protein Engineering and Proteomics Center, Sichuan Provincial Engineering Laboratory of Pathology in Clinical Application, West China Hospital, Sichuan University, Chengdu, 610041, China

[5] Shanghai Omicsolution Co., Ltd., Shanghai, 200000, China

[6] Department of Pharmacology, Yale University School of Medicine, New Haven, CT 06520, USA

[7] Department of Biomedical Informatics & Data Science, Yale University School of Medicine, New Haven, CT 06510, USA

*Corresponding authors: Email: shishengwang@wchscu.cn, s@omicsolution.com, yansheng.liu@yale.edu and yanghao@scu.edu.cn.

Catalogue

Supplementary Manual

Abstract

WuKong is a powerful platform designed to elevate the way researchers approach proteomics data analysis. It integrates over 72 well-established analysis modules that cover the entire range of data preprocessing, statistical analysis, functional annotation, visualization, and workflow design. On the one hand, these modules retain the classic analysis frameworks, allowing users to adjust parameters within each module to directly generate results tailored to their specific research needs. This flexibility ensures that both beginners and experienced bioinformaticians can efficiently conduct complex analyses without being constrained by rigid workflows. On the other hand, WuKong incorporates large language models (LLMs) to enhance user experience by facilitating conversational interactions with the platform. Researchers can select from a variety of locally installed LLMs, enabling them to conduct analyses through an intuitive chat interface. This approach not only simplifies the analytical process but also provides real-time guidance and support, making it easier for users to navigate complex datasets and analytical procedures.

The whole source codes: https://github.com/wangshisheng/WuKong. The tool homepage is shown like this:

1. How to run this tool locally?

WuKong is an open source software for non-commercial use and all codes can be obtained on our GitHub: https://github.com/wangshisheng/WuKong. If users want to run *WuKong* on their own computer, they should operate as below:

As this tool was developed with R, you may:

a) Install R. You can download R from here: https://www.r-project.org/.

b) Install RStudio. (Recommendatory but not necessary). You can download RStudio from here: https://www.rstudio.com/.

c) Check packages. After installing R and RStudio, you should check whether you have installed these packages (shiny, shinyBS, devtools, openxlsx, gdata, DT, tidyllm, ollamar, markdown, httr, ggplot2, shinyjs, shinyWidgets, rvest, writexl, shinydashboard, esquisse, ggsci, ggpubr, rstatix, Seurat, data.table, purrr, ceLLama, tidygraph, ggraph, corrr, tidyverse, ConsensusClusterPlus, LSD, pheatmap, grid, ggplotify, akima, ggcorrplot, psych, reshape2, igraph, dplyr, ape, cluster, limma, samr, DNB, ellipse, GGally, patchwork, funnelR, ggseqlogo, ggtree, clusterProfiler, tidyr, GRA, factoextra, alr3, ggpmisc, ggExtra, Mfuzz, impute, Amelia, ropls, FactoMineR, ade4, vegan, ggpie, pwr, ggradar, ggforce, ggdist, gghalves, ggrepel, plotRCS, ggvegan, ggridges, pROC, RRHO2, scales, ggsankey, mdatools, survminer, survival, ggtern, timecourse, Rtsne, tidyestimate, umap, ggupset, ggwordcloud). You may run the codes below to check them:

```
if(!require(pacman)) install.packages("pacman")
pacman::p_load(shiny, shinyBS, devtools, openxlsx, gdata, DT, tidyllm, ollamar, markdown,
httr, ggplot2, shinyjs, shinyWidgets, rvest, writexl, shinydashboard, esquisse, ggsci, ggpubr,
rstatix, Seurat, data.table, purrr, tidygraph, ggraph, corrr, tidyverse, ConsensusClusterPlus,
LSD, pheatmap, grid, ggplotify, akima, ggcorrplot, psych, reshape2, igraph, dplyr, ape,
cluster, limma, samr, ellipse, GGally, patchwork, funnelR, ggseqlogo, ggtree, clusterProfiler,
tidyr, factoextra, alr3, ggpmisc, ggExtra, Mfuzz, impute, Amelia, ropls, FactoMineR, ade4,
vegan, ggpie, pwr, ggforce, ggdist, gghalves, ggrepel, ggridges, pROC, scales, mdatools,
survminer, survival, ggtern, timecourse, Rtsne, tidyestimate, umap, ggupset, ggwordcloud,
coin, survcomp)
```

Please note, if you find some packages cannot be installed directly using the above command, you can find them in the GitHub source and install them by, for example:

```
library(devtools)
devtools::install_github("CelVoxes/ceLLama")
devtools::install_github("anspiess/propagate")
devtools::install_github("gpli/DNB")
devtools::install_github("Nisus-Liu/GRA")
devtools::install_github("ricardo-bion/ggradar")
devtools::install_github("kunhuo/plotRCS")
devtools::install_github("gavinsimpson/ggvegan")
devtools::install_github("RRHO2/RRHO2", build_opts = c("--no-resave-data", "--no-manual"))
```

```
devtools::install_github("davidsjoberg/ggsankey")
devtools::install_github("nicolash2/ggvenn")
```

d) Run this tool locally

```
if(!require(WuKong)) devtools::install_github("wangshisheng/WuKong")
library(WuKong)
WuKong_app()
```

Then WuKong will be started as below, and the detailed operation about WuKong can be found in the subsequent sections.

2. Data Preparation

The uploaded data file formats could be .csv or .txt. Before analysis, users should prepare the proteomics expression data and sample information. The proteomics expression data required here could be readily generated based on results of several popular tools such as Spectronaut, MaxQuant. Users then can upload the data and type in right sample information into WuKong with right formats respectively and start subsequent analysis.

2.1 Expression data

Users can prepare their proteomics expression data as below from any software (e.g. Spectronaut, MaxQuant, etc.) and upload into this tool. In the situation, protein ids/names are sequentially provided in the first two columns of input file. The protein ids/names in the first column could be UniProt ids or protein names. From the second column, proteins expression intensity or signal abundance in every sample should be listed. The data structure is shown as below (rows are proteins and columns are samples):

PG.ProteinGroups	DPC_0h_1	DPC_0h_2	DPC_0h_3	DPC_0h_4	DPC_3h_1	DPC_3h_2	DPC_3h_3	DPC_3h_4	DPC_6h_1	DPC_6h_2
A0A024RB	25189.73	13440.21	19565.14	17490.47	27999.1	11120.79	8287.486	10820.16	21856.66	11681.11
A0A096LP	43690.06	40345.23	37148.71	37204.88	39428.96	39318.33	36062.08	34250.29	34548.79	30284.49
A0A0B4J2/	1060724	836391.3	1103099	1063019	973396.6	821988.6	936122.9	1124465	917335	867221.2
A0A0B4J2I	57807.32	60119.08	63978.09	69136.92	68128.56	31073.24	72903.39	71065.42	45719.14	48018.71
A0AV96	14720.61	16155.04	15082.32	14969.27	14932.68	13164.33	14025.51	12534.12	14209.58	13106.73
A0AVK6	16522.56	18600.74	17865.75	18601.68	18643.31	5125.668	14807.14	16103.46	20163.18	17906.59
A0AVT1	17222.84	24763.66	30250.37	28884.49	16928.29	28179.63	22732.48	16664.99	25190.92	23202.59
A0FGR8	20893.18	17165.7	18176.4	20293.75	18935.83	12087.23	14606.6	11881.54	25152.06	17099.2
A0JLT2	37593.66	28032.06	28467.56	21959.01	34682.7	23316.45	35804.23	32016.78	44664.74	40806.98
A0MZ66	29056.75	32836.46	30119.31	30416.28	[illegible]	[illegible]	33450.51	30839	31779.2	37992.16
A1L0T0	35199.13	35886.66	36962.29	38643.55	29708.71	31485.84	25030.82	31670.05	36484.25	31963.77
A1L390	12021.93	12513.96	11311.04	10823.97	11329.15	9397.313	12337.43	9379.057	12087.48	10786.13
A2RRD8	46519.11	58789.84	54326.02	62714.16	57500.93	64203.45	65464.58	70372.02	79865.82	86481.73
A3KN83	26430.21	25346.23	26637.75	25969.68	26256.35	25992.66	25837	25177.89	24772.02	24262.8
A5YKK6	27706.62	27926.76	26817.8	27684.27	24960.62	26118.65	25467.98	24301.89	31291.05	30275.13
A6NCC3	35944.17	20451.28	27322.03	25434.8	41856.32	7629.363	2581.511	10482.42	26577.31	25864.84
A6NDG6	36172.88	32129.89	29492.28	36041.41	32832.6	28935.81	26838.32	24399.04	38261.67	38333.74
A6NDU8	48340.96	50244.54	40859.61	44103.27	37341.36	42296.37	38651.26	37392.52	47460.55	39691.67
A6NED2	2379.226	9749.278	11220.31	10147.28	8857.214	6130.552	6130.417	3188.316	11724.42	11549.22
A6NFI3	26165.06	32865.58	25063.64	30182.51	29647.58	31624.83	32113.08	31675.63	26228.95	32020.46
A6NHG4	10506.64	6973.583	7115.409	8546.165	3228.381	4923.56	2169.056	6023.871	7572.204	4797.863
A6NHL2	75163.5	66692.17	69335.76	77987.73	69796.09	56109.2	56733.93	61412.86	74370.55	63234.49
A6NHR9	46648.08	49067.9	47149.73	47286.44	49437.65	52611.63	50571.77	49839.58	41032.63	43034.47

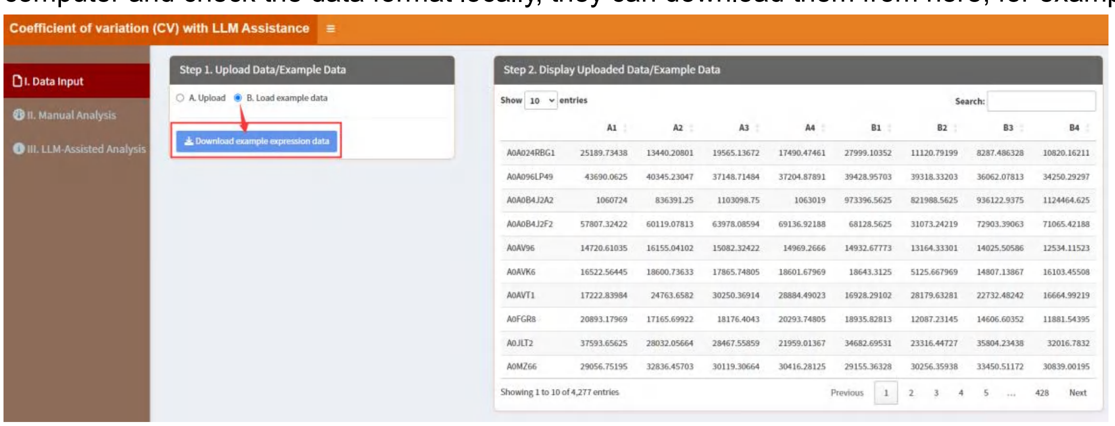
2.2 Samples information data

Sample information here means that users should provide group number, replicate number, and group names, then type them in the corresponding module (see below). This tool will use these information to calculate corresponding results:

2.2.1 Group and replicate number: Type in the group number and replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 2 groups and 4 replicates in each group, so here is “2;4-4”.
2.2.2 Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 2 groups in the example data, you can type in "A;B".

2.3 Download example datasets

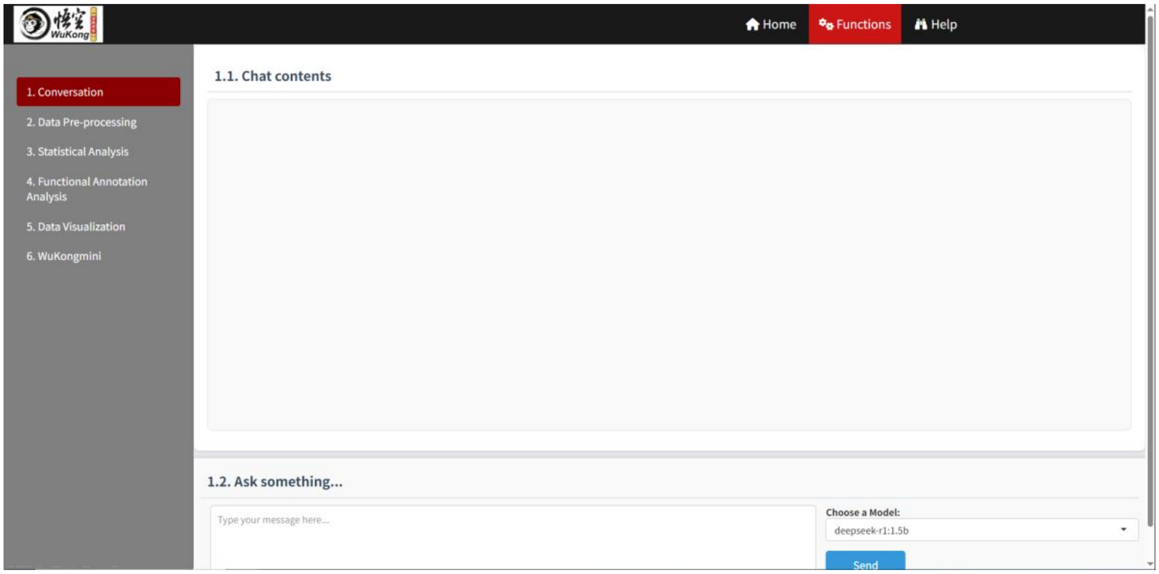
After selecting a module, if users want to download the example datasets to their own computer and check the data format locally, they can download them from here, for example:



	A1	A2	A3	A4	B1	B2	B3	B4
A0A024RBG1	25189.73438	13440.20801	19565.13672	17490.47461	27999.10352	11120.79199	8287.486328	10820.16211
A0A096LP49	43690.0625	40345.23047	37148.71484	37204.87891	39428.95703	39318.33203	36062.07813	34250.29297
A0A0B4J2A2	1060724	836391.25	1103098.75	1063019	973396.5625	821988.5625	936122.9375	1124464.625
A0A0B4J2F2	57807.32422	60119.07813	63978.08594	69136.92188	68128.5625	31073.24219	72903.39063	71065.42188
A0AV96	14720.61035	16155.04102	15082.32422	14969.2666	14932.67773	13164.33301	14025.50586	12534.11523
A0AVK6	16522.56445	18600.73633	17865.74805	18601.67969	18643.3125	5125.667969	14807.13867	16103.45508
A0AVT1	17222.83984	24763.6582	30250.36914	28884.49023	16928.29102	28179.63281	22732.48242	16664.99219
A0FGR8	20893.17969	17165.69922	18176.4043	20293.74805	18935.82813	12087.23145	14606.60352	11881.54395
A0JLT2	37593.65625	28032.05664	28467.55859	21959.01367	34682.69531	23316.44727	35804.23438	32016.7832
A0MZ66	29056.75195	32836.45703	30119.30664	30416.28125	29155.36328	30256.35938	33450.51172	30839.00195

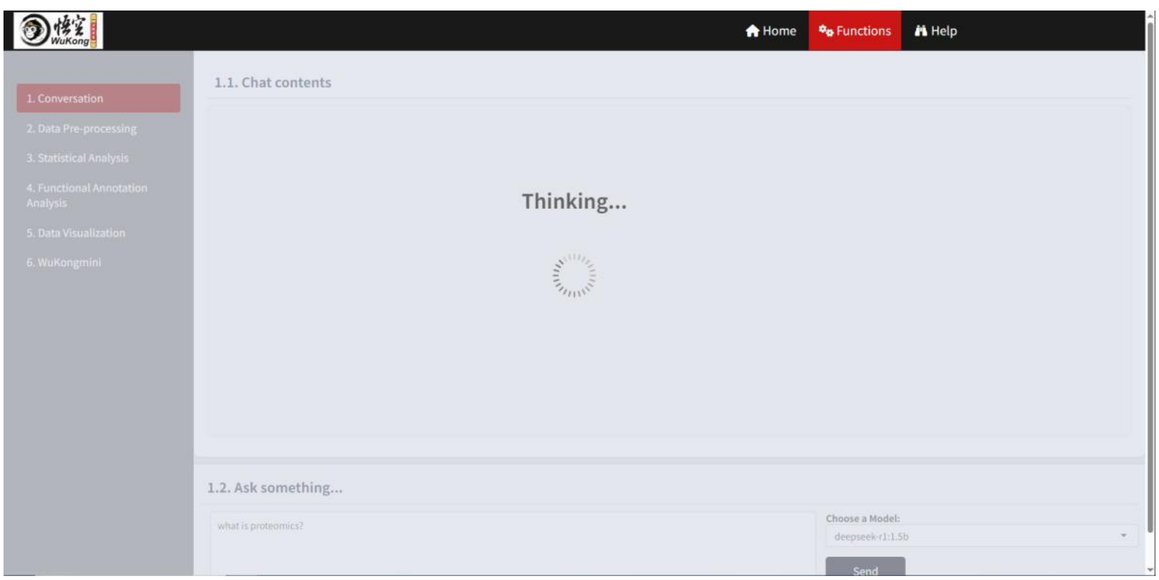
users can download the example data (proteomics expression data) by clicking the “Download example expression data” button. The data are saved as corresponding format and users can open them in other software, such as Excel. Then, users can check the example sample information and understand these parameters better.**3. Conversation**

Here means users can chat with local large language models that are deployed locally. When users click ‘Functions’, like below:

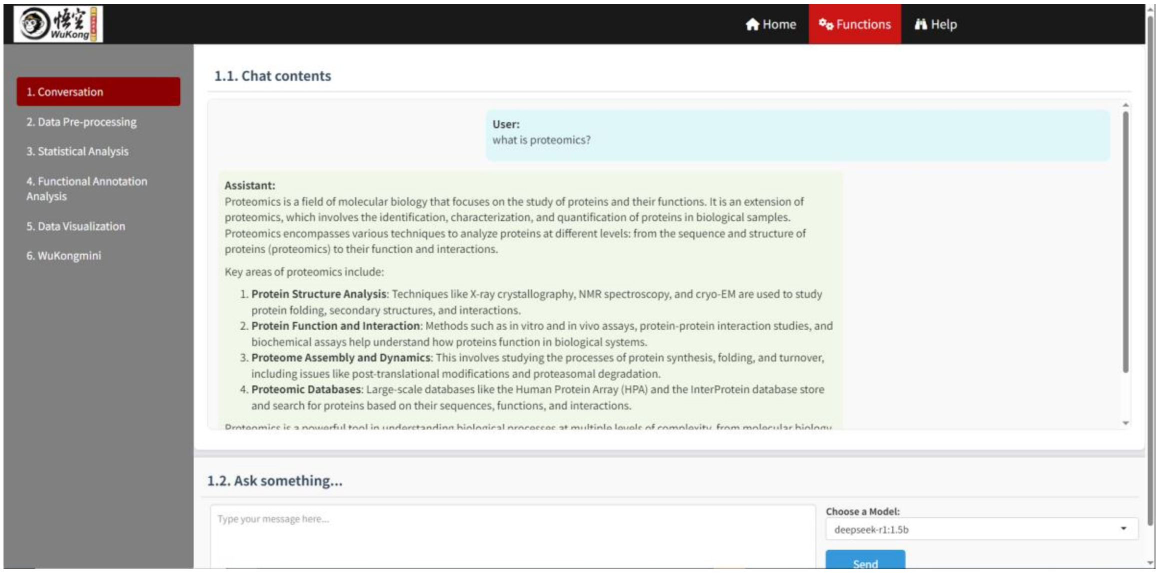


The first function, "Conversation," allows users to engage in interactive discussions. Section "1.1. Chat contents" displays the conversation history, while "1.2. Ask something..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a Model" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part).

For example, when users type one question and click “Send” button, this tool will invoke the corresponding LLM in the background, shown as below:



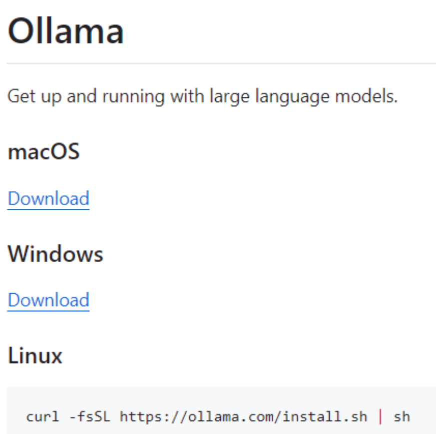
After a while, the conversation will be displayed as below:



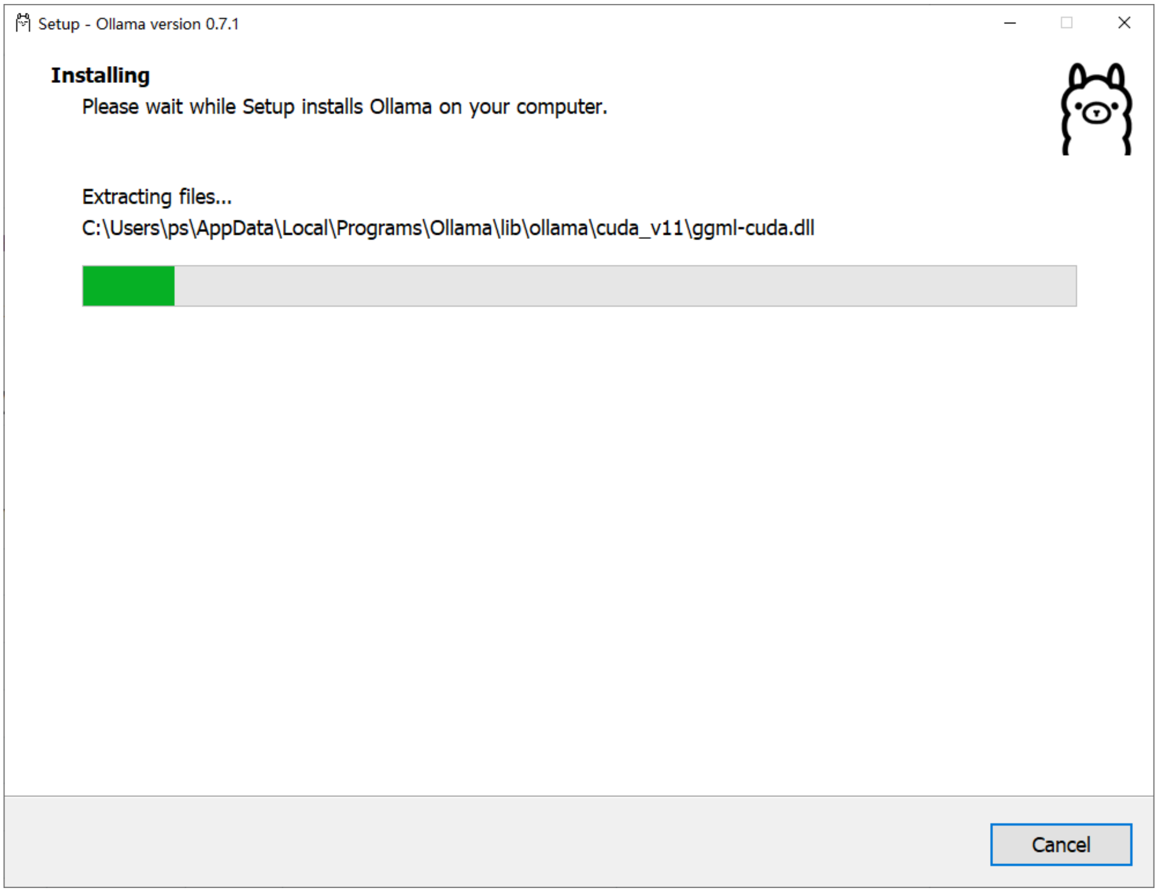
How to install LLMs locally?

Utilizing LLMs on local systems is becoming increasingly popular due to enhanced privacy, control, and reliability. Herein, we primarily introduce how to install LLMs locally using Ollama, a free, open-source tool that enables users to run LLMs on their personal machines.

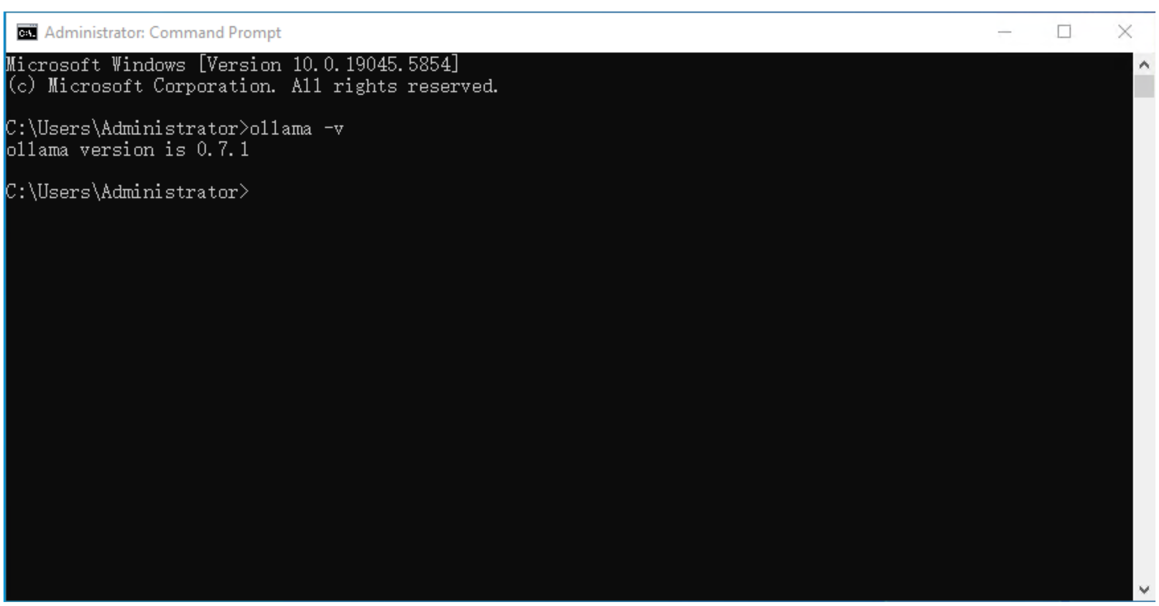
Step 1: Click https://github.com/ollama/ollama, then users can choose which version based on their own system (Windows, Linux or macOS) and download.



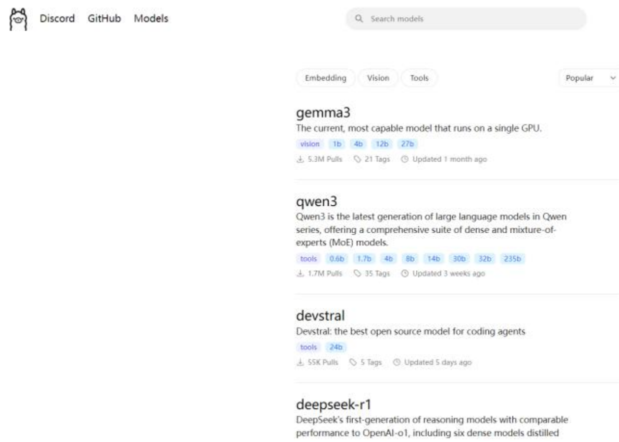
For example, after downloading the ollama for Windows, users can double click the "OllamaSetup.exe" file and the installing process like below:



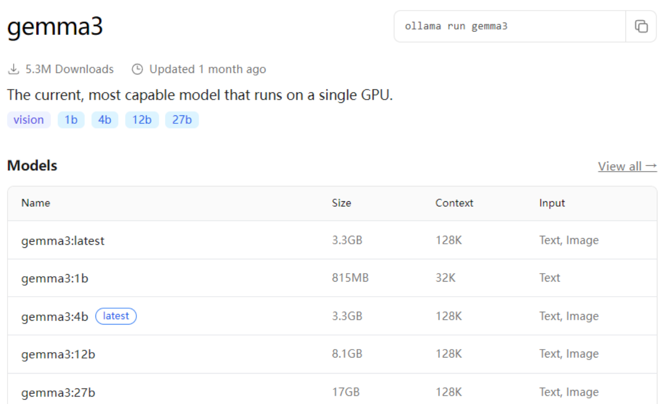
Step 2: After installation, users can open the "Command Prompt", then type "ollama -v" to check the version of ollama.



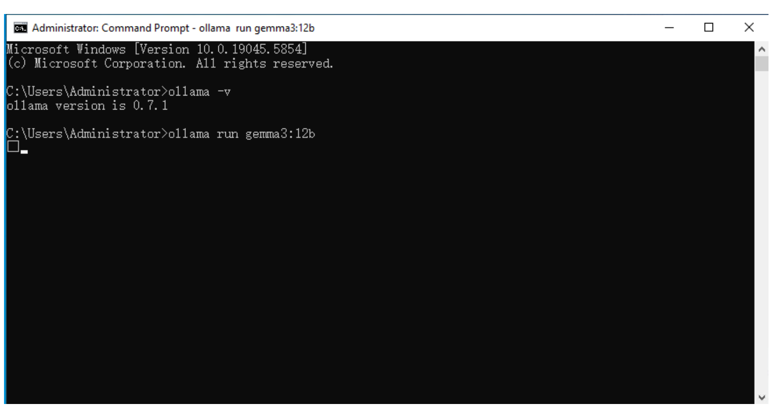
Step 3: Choose one LLM. Users can click https://ollama.com/search, then select any LLM they want.



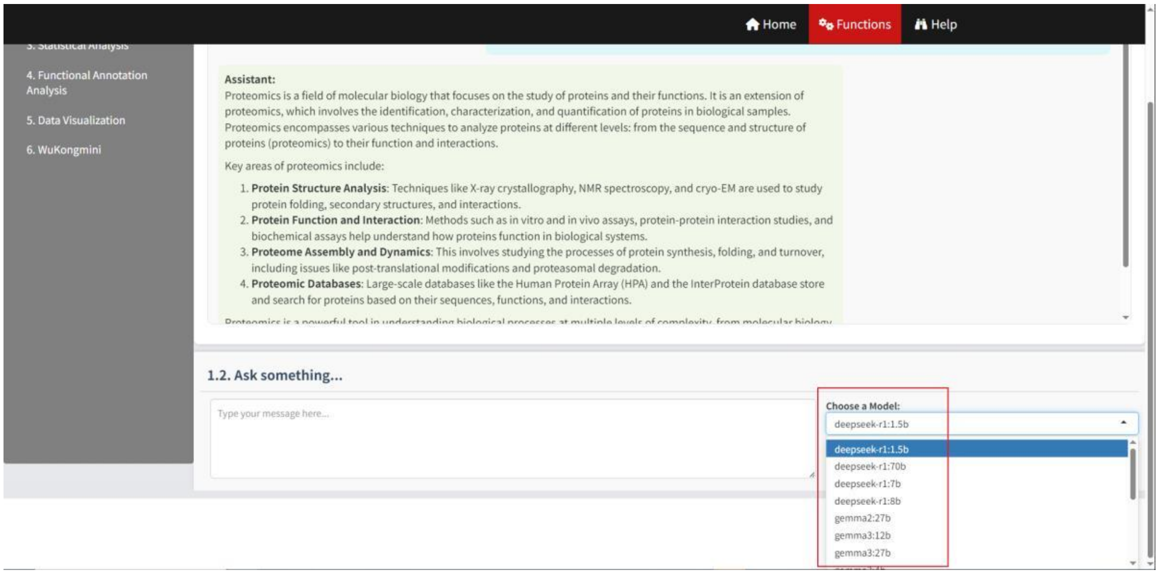
For instance, users select the "gemma3" option and then click on it. As illustrated below, there are numerous versions available with varying parameters. Consequently, users can choose any one that suits their computer configuration.



If users choose “gemma3:12b”, they can copy the command “ollama run gemma3:12b” in the “Command Prompt”. Then the model “gemma3:12b” will be installed successfully in the users' computer.

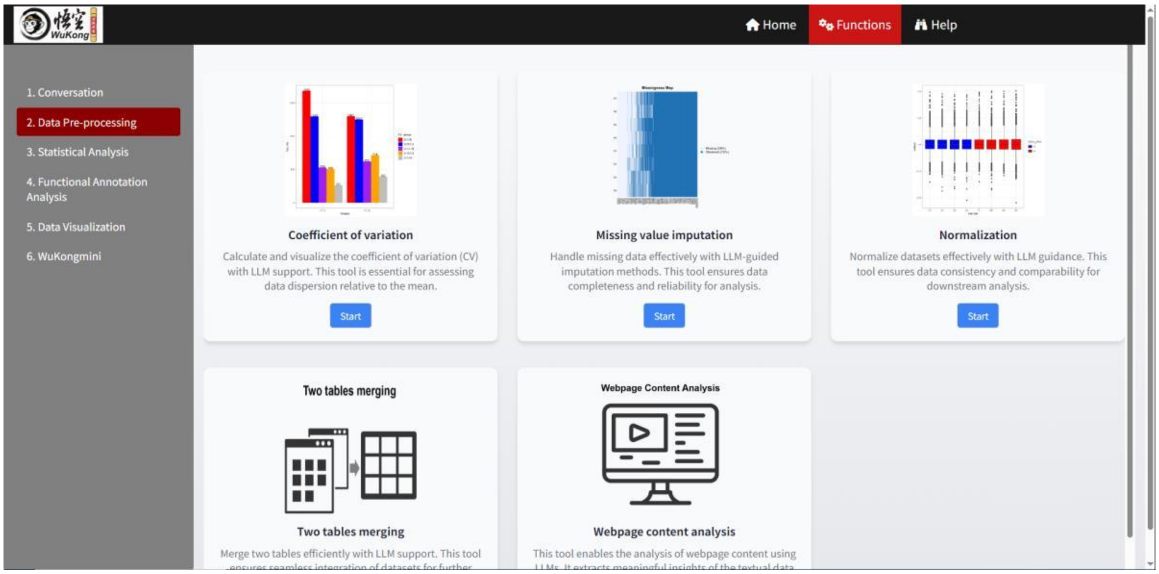


All other LLM installations are similar to the one described above. Subsequently, users will find the installed LLMs displayed in WuKong, where they can select one:



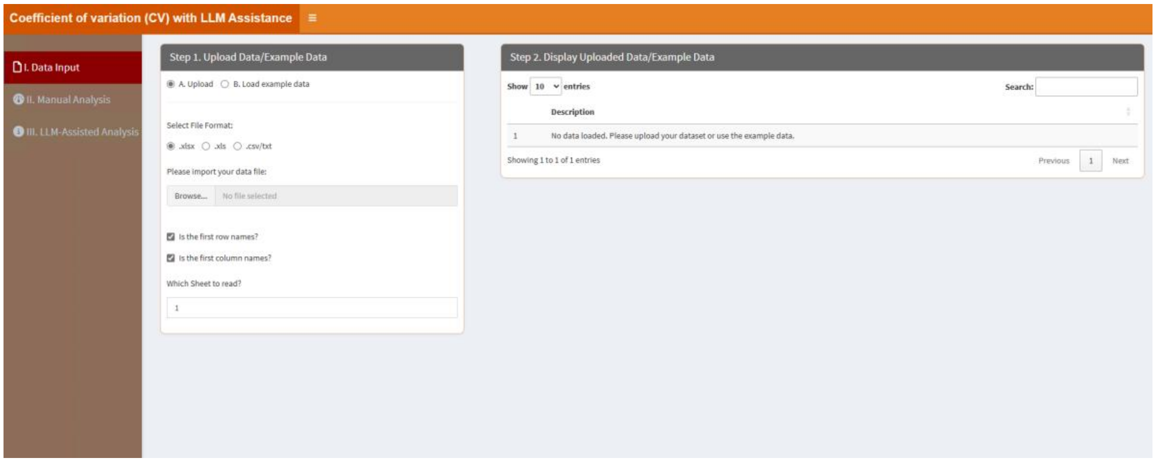
4. Data Pre-processing

Users can pre-process their data in this step, including coefficient of variation, missing value imputation, normalization, two tables merging and webpage content analysis. When users click “2. Data Pre-processing”, it is shown as below. If users want to use one module to analyze their data, just click corresponding “Start” button.

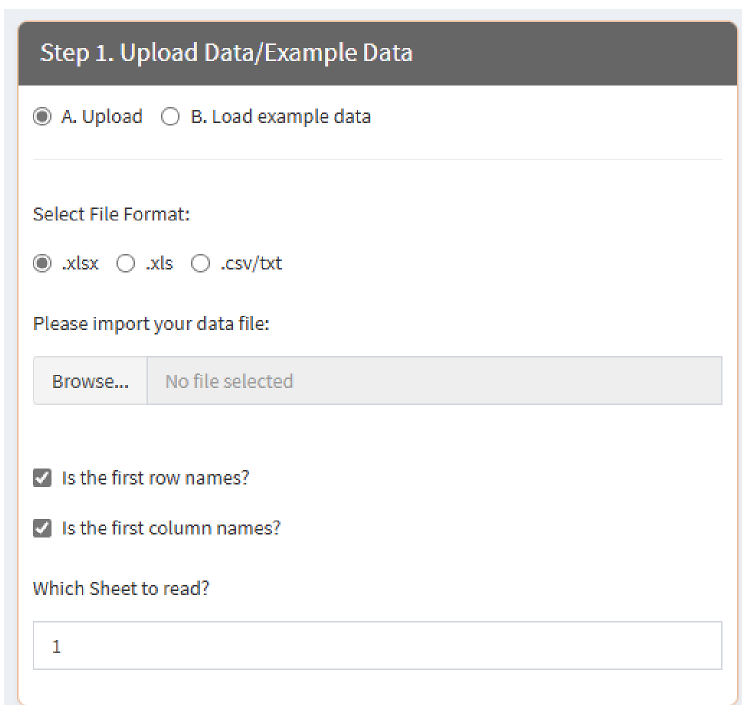


4.1. Coefficient of variation (CV)

CV is a statistical measure that represents the ratio of the standard deviation to the mean. A higher CV indicates greater variability relative to the mean, while a lower CV suggests more consistency. After clicking the "Start" button, it will be started in a new window like below:



"I. Data Input" means users can upload their data or check example data from here. In the "Step 1. Upload Data/Example Data" part, users can upload their data, as below:

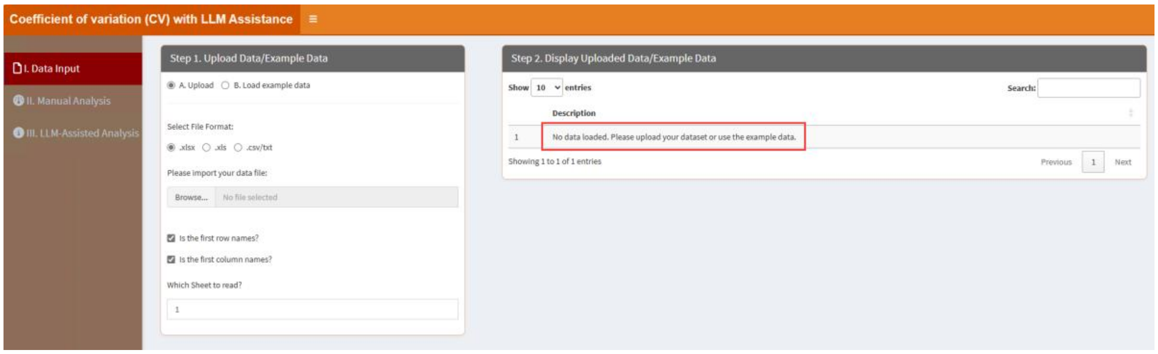


Users can choose "*A. Upload*" to upload their own data here. They should select the right format (*File format*: .xlsx, .xls, .csv, .txt) based on their data and then click "Browse" button to import the data;

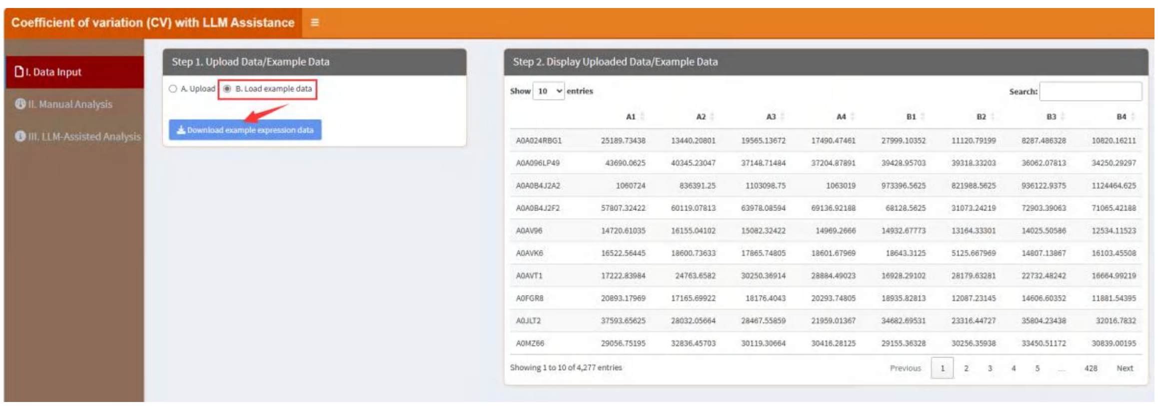
Is the first row names? This means whether the first row is column names. If true, you should choose this parameter.

Is the first column names? This means whether the first column is row names. If true, you should choose this parameter.

If users do not upload any data, the “Step 2. Display Uploaded Data/Example Data” part will show “*No data loaded. Please upload your dataset or use the example data.*” to warn users.



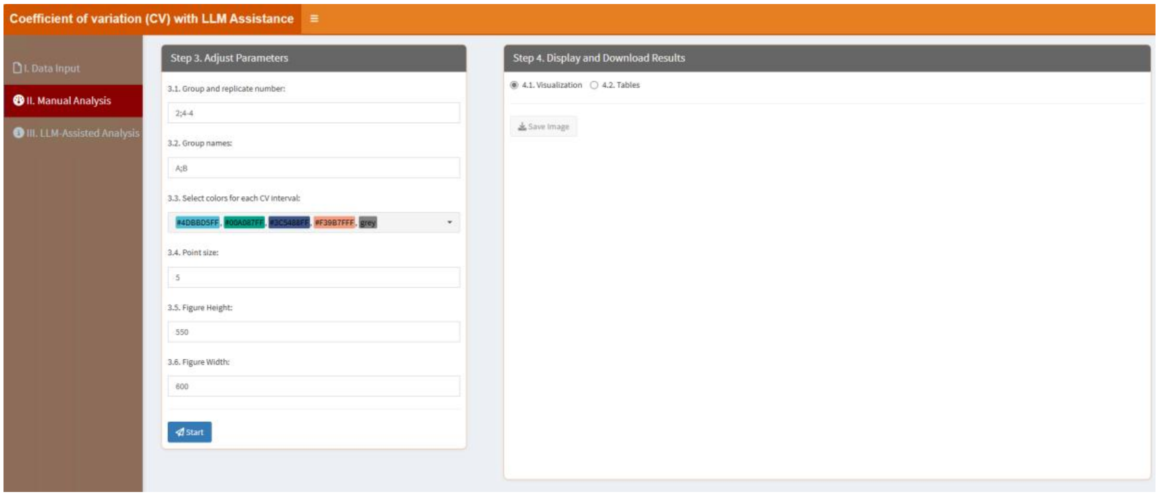
If users want to download the example datasets to their own computer and check the data format locally, they can download them from here:



First, select “B. Load example data” and the example data will be shown on the right panel interactively.

Second, users can download the example data by clicking the corresponding button. The example data is saved as .csv format and users can open them in other software, such as Excel.

"II. Manual Analysis" refers to the process where users can manually analyze their data by adjusting parameters in the "Step 3. Adjust Parameters" section. Based on the adjustments made, the corresponding results will be displayed in the "Step 4. Display and Download Results" section, reflecting the parameters set in Step 3.



The parameters in the "Step 3. Adjust Parameters" part:

3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

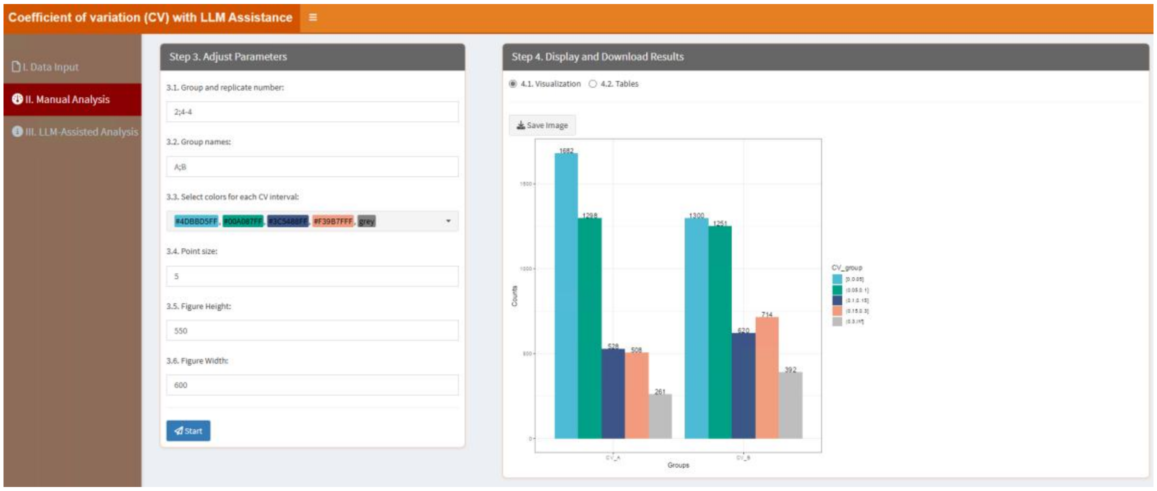
*3.3. Select colors for each CV interva*l: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

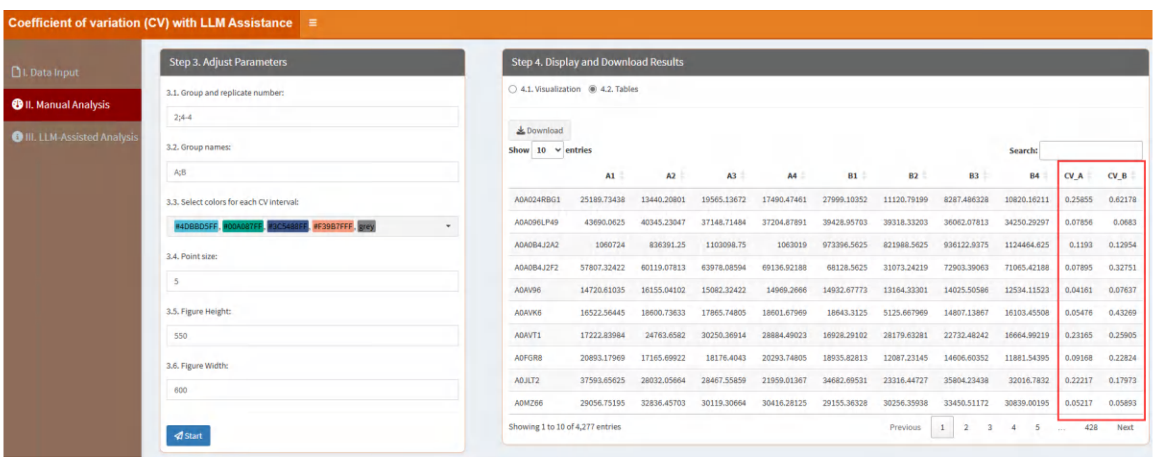
3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

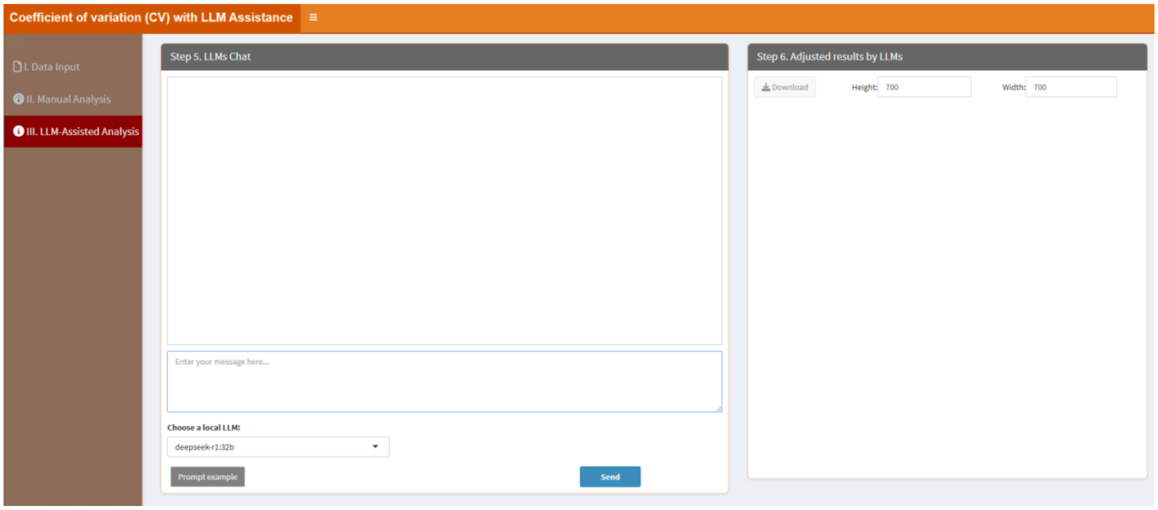


In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the CV of each group:

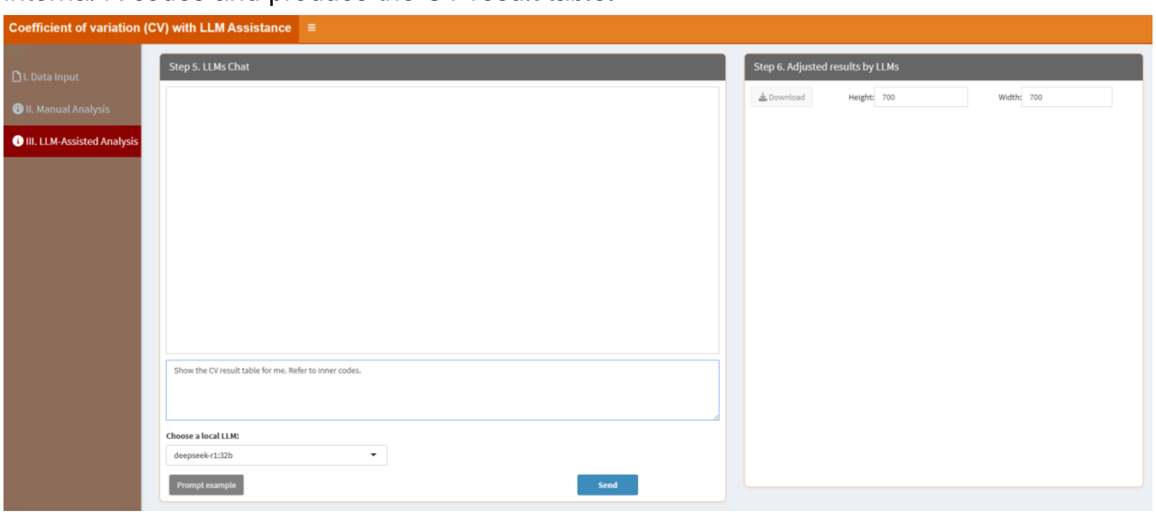


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the

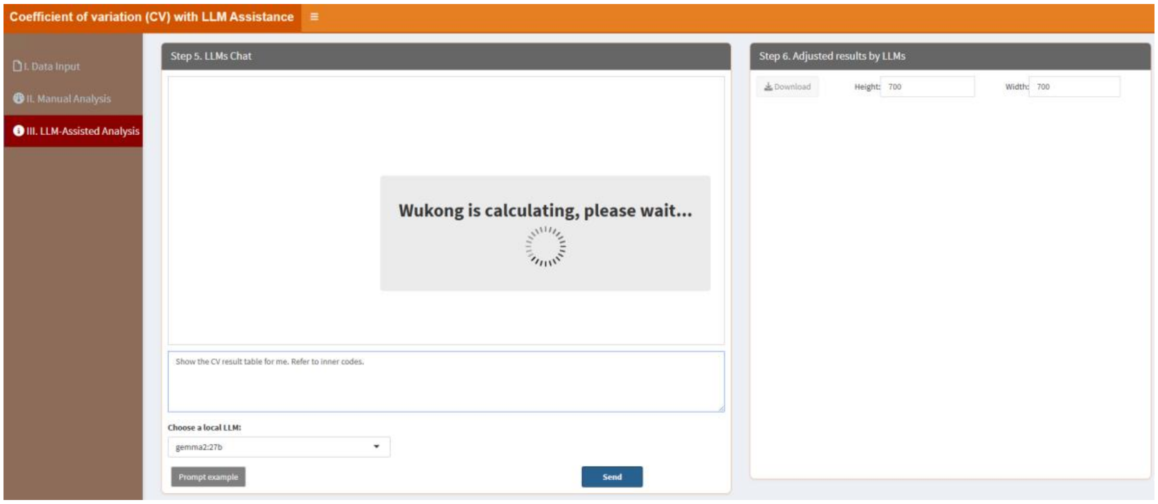
"Step 5. LLMs Chat".



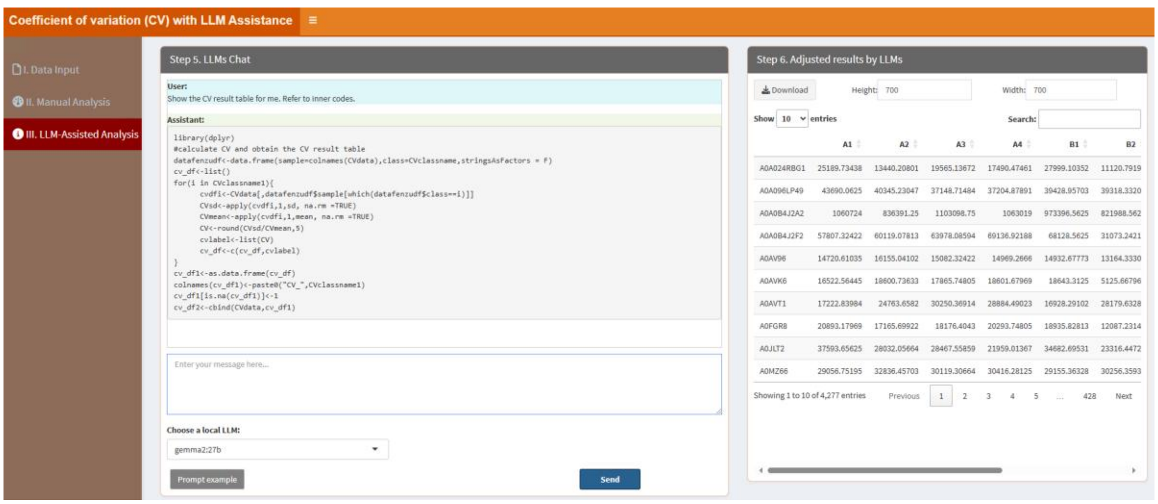
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Show the CV result table for me. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the CV table. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Show the CV result table for me. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the CV result table.



After type the prompt, click “Send” button. It will learn like below:

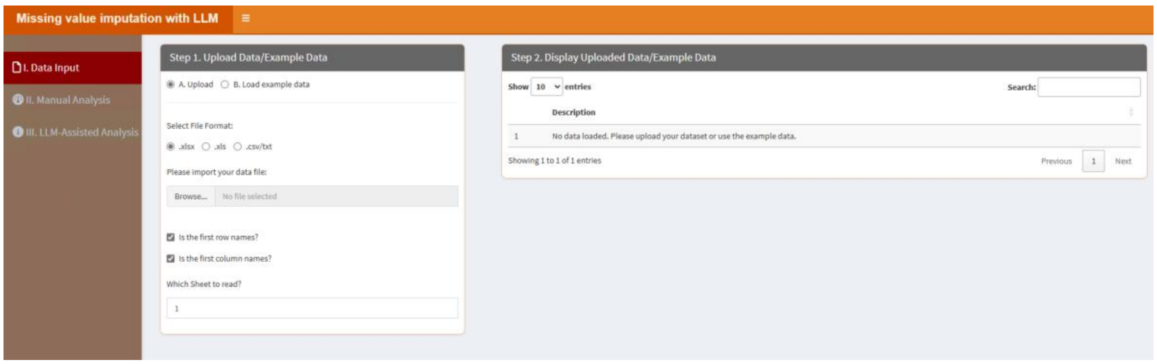


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



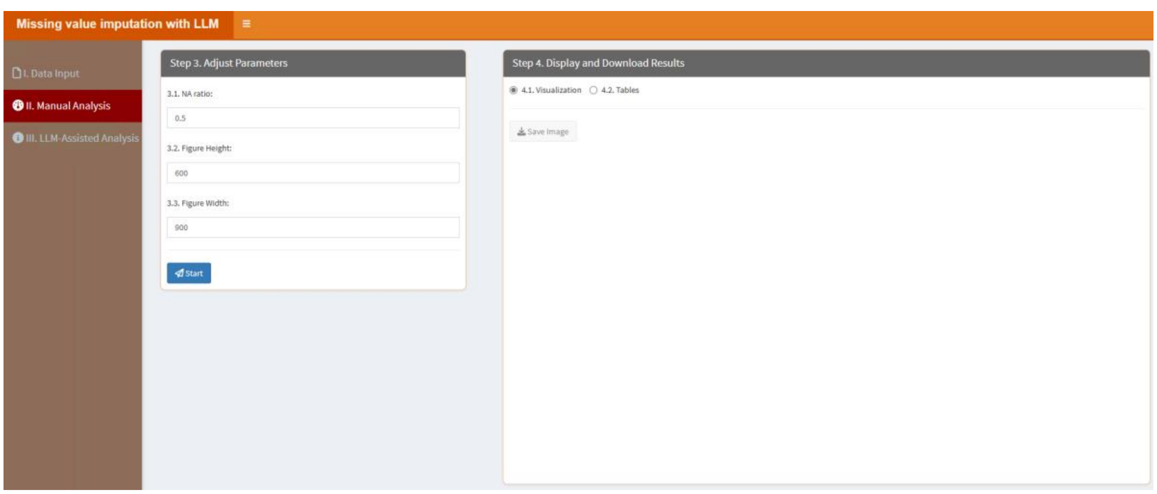
4.2. Missing value imputation

Missing value imputation is the process of replacing missing or incomplete data in a dataset with substituted values to ensure the dataset remains usable for analysis. Common methods for imputation include simple approaches like replacing missing values with the mean, median, or some other advanced methods such as regression imputation, k-nearest neighbors (KNN), and machine learning-based methods. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

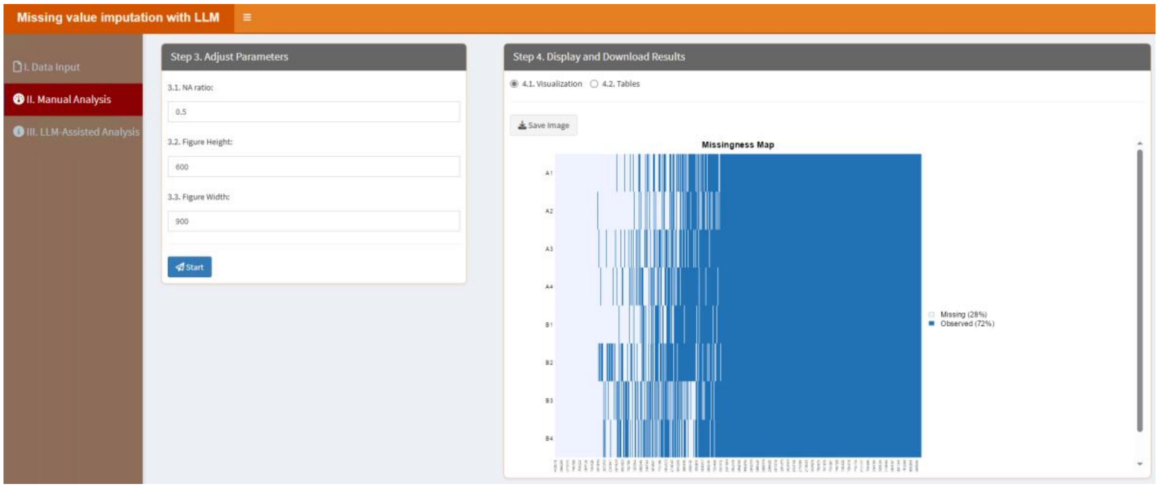


3.1. NA ratio: the threshold of NA ratio. Those peptides/proteins with NA ratio above this threshold will be removed.

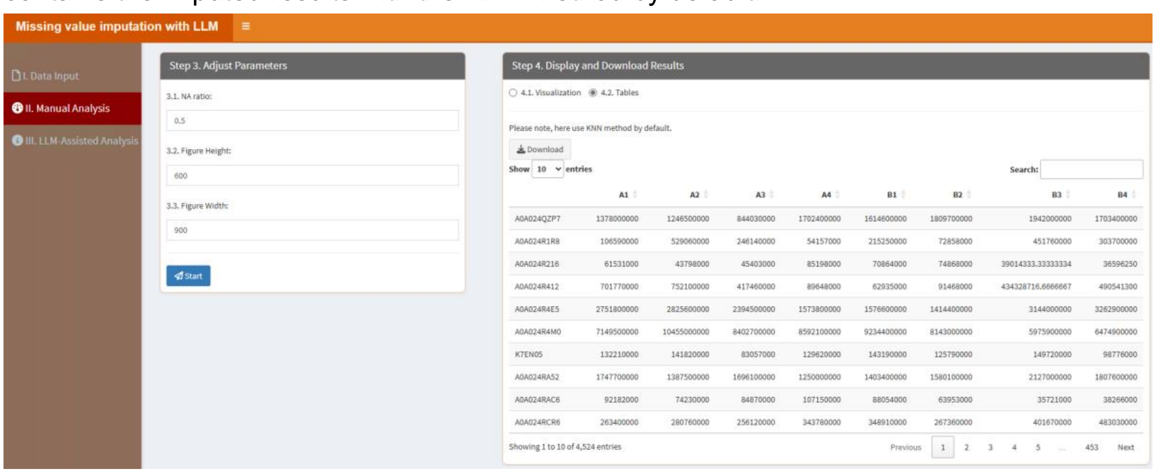
3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the imputed results with the KNN method by default:

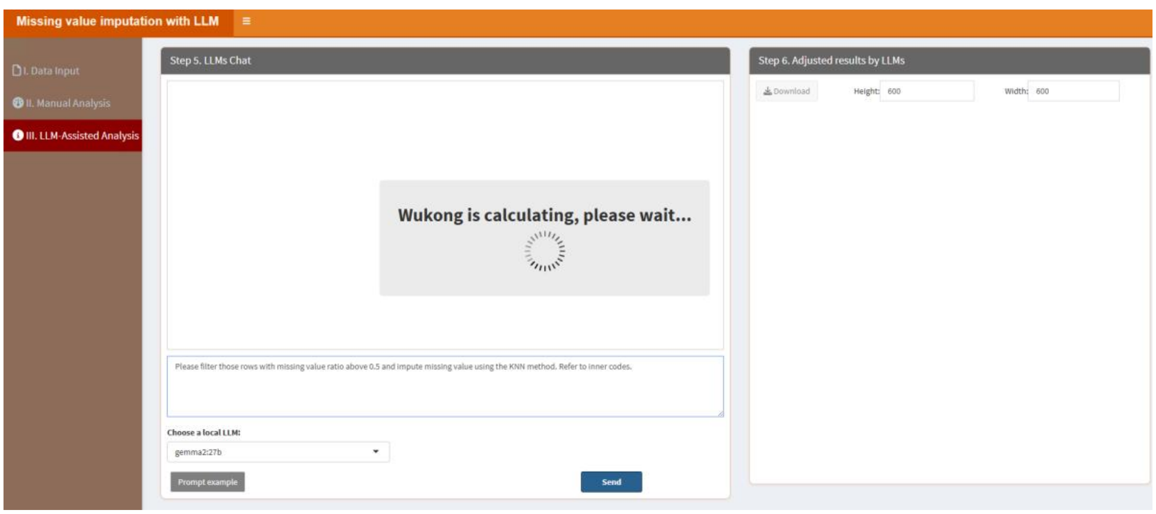


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the

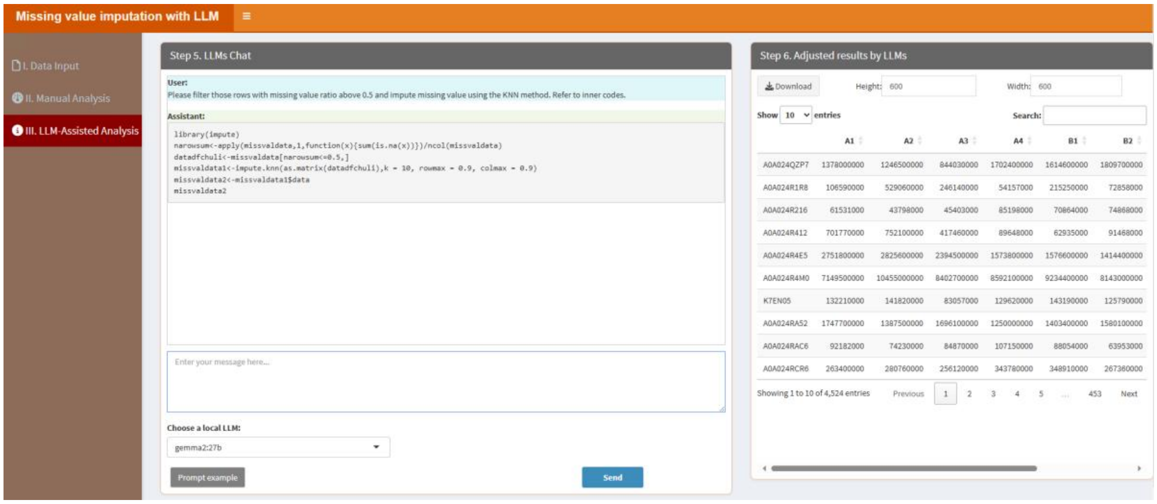
"Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please filter those rows with missing value ratio above 0.5 and impute missing value using the KNN method. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please filter those rows with missing value ratio above 0.5 and impute missing value using the KNN method. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

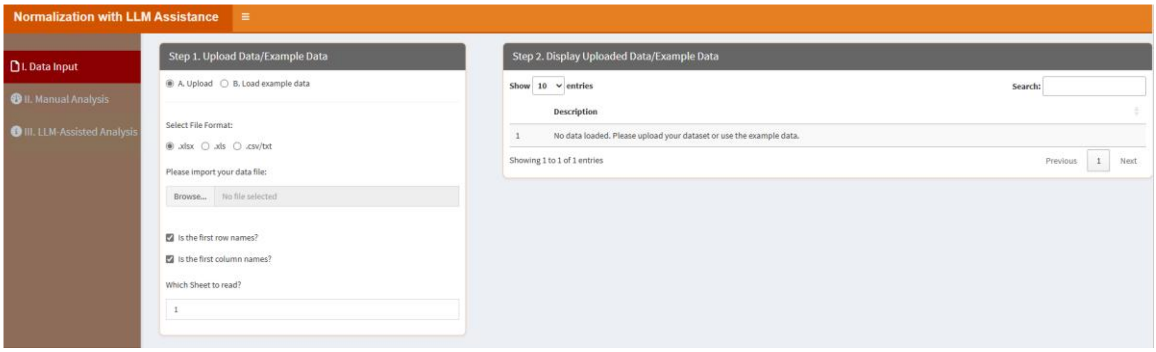


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.


Missing value imputation with LLM
I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis
Step 5. LLMs Chat
User:
Please filter those rows with missing value ratio above 0.5 and impute missing value using the KNN method. Refer to inner codes.
Assistant:
library(impute)
narowsum<-apply(missvaldata,1,function(x){sum(is.na(x))})/ncol(missvaldata)
datadfchuli<-missvaldata[narowsum<=0.5,]
missvaldata1<-impute.knn(as.matrix(datadfchuli),k = 10, rowmax = 0.9, colmax = 0.9)
missvaldata2<-missvaldata1$data
missvaldata2
Enter your message here...
Choose a local LLM:
gemma2:27b
Prompt example
Send
Step 6. Adjusted results by LLMs
Download
Height: 600
Width: 600
Show 10 entries
Search:
A1 A2 A3 A4 B1 B2
A0A024QZP7 1378000000 1246500000 844030000 1702400000 1614600000 1809700000
A0A024R1R8 106590000 529060000 246140000 54157000 215250000 72858000
A0A024R216 61531000 43798000 45403000 85198000 70864000 74868000
A0A024R412 701770000 752100000 417460000 89648000 62935000 91468000
A0A024R4E5 2751800000 2825600000 2394500000 1573800000 1576600000 1414400000
A0A024R4M0 7149500000 10455000000 8402700000 8592100000 9234400000 8143000000
K7EN05 132210000 141820000 83057000 129620000 143190000 125790000
A0A024RA52 1747700000 1387500000 1696100000 1250000000 1403400000 1580100000
A0A024RAC6 92182000 74230000 84870000 107150000 88054000 63953000
A0A024RCR6 263400000 280760000 256120000 343780000 348910000 267360000
Showing 1 to 10 of 4,524 entries
Previous 1 2 3 4 5 … 453 Next

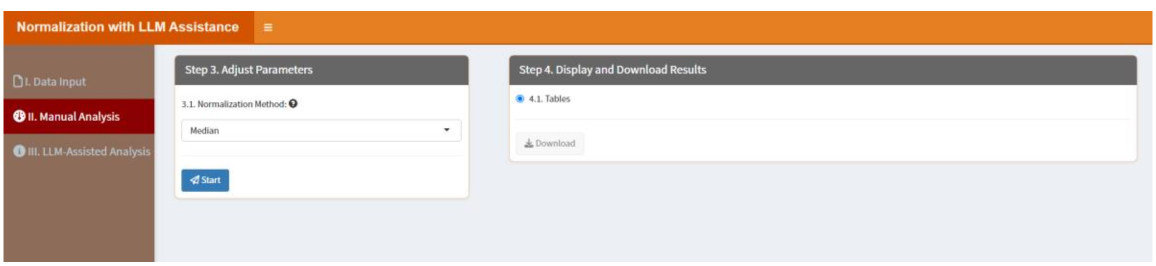
4.3. Normalization

Normalization is a crucial step in proteomics data analysis to ensure the comparability and reliability of results. In proteomics, data often comes from various experimental conditions, instruments, or runs, leading to technical variations that can obscure biological insights. Normalization aims to reduce these unwanted variations while preserving the true biological differences. After clicking the “Start” button, it will be started in a new window like below:



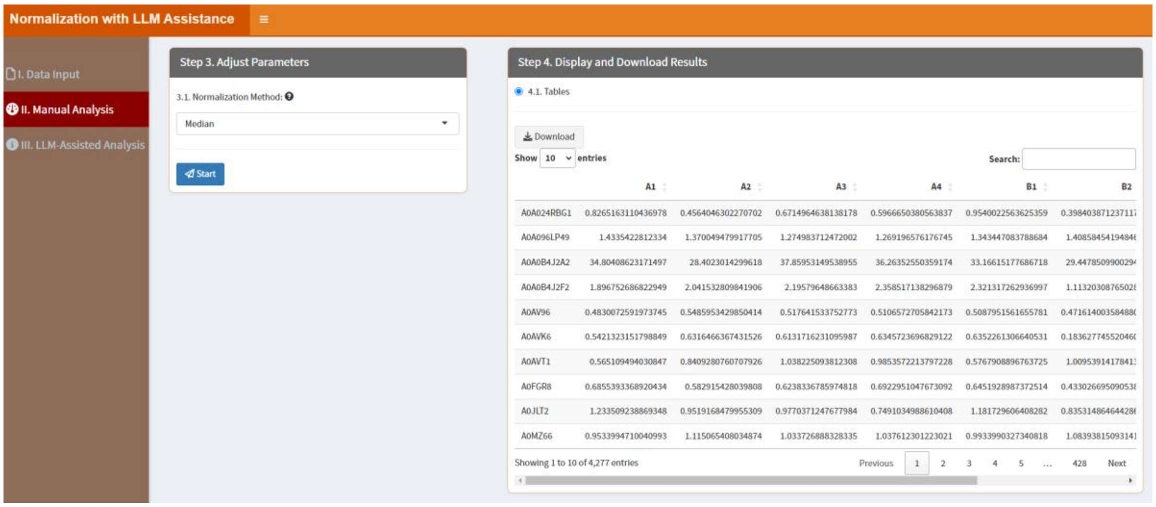
The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Normalization Method: here users can select one normalization method, including: Median, mean, and total normalization involve dividing each value by the median, mean, or total of its column, respectively. Centering adjusts data by subtracting the column mean from each value, resulting in a mean of zero while maintaining the original variance. Range normalization involves subtracting the mean and dividing by the range (the difference between the maximum and minimum values) of the column. Z-score normalization standardizes data by subtracting the mean and dividing by the standard deviation, effectively scaling the data to have a mean of zero and a standard deviation of one. Quantile normalization aligns the distributions of different datasets to match a common distribution. Lastly, scaling to a range involves rescaling data to fit within a specified range, ensuring that all values fall between a predefined minimum and maximum.

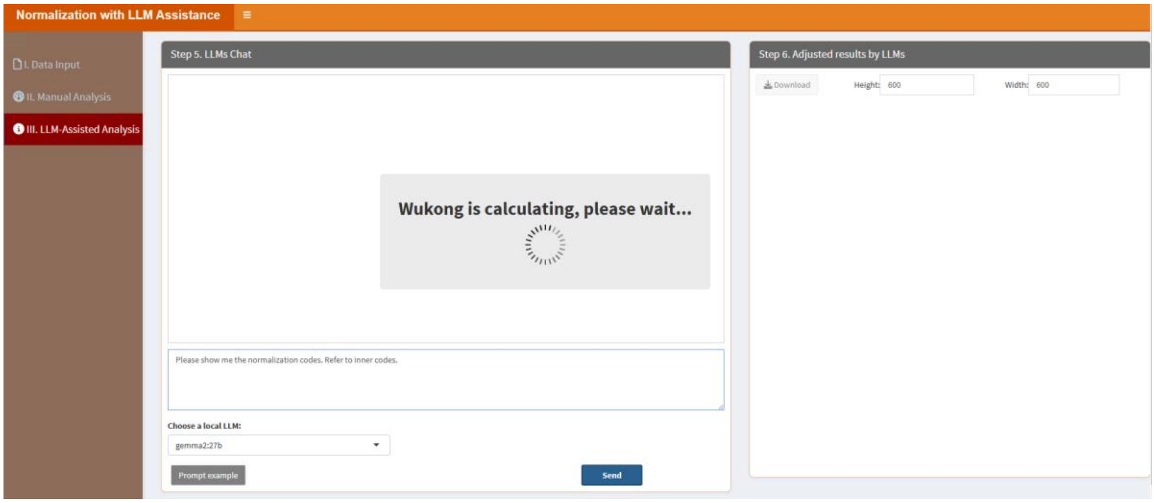
After setting parameters, click “Start” button, the results will be shown as below.
In the “4.1. Tables” part, users can click “Download” button to download the table, which contains the table results with the selected normalization method (e.g. median normalization):



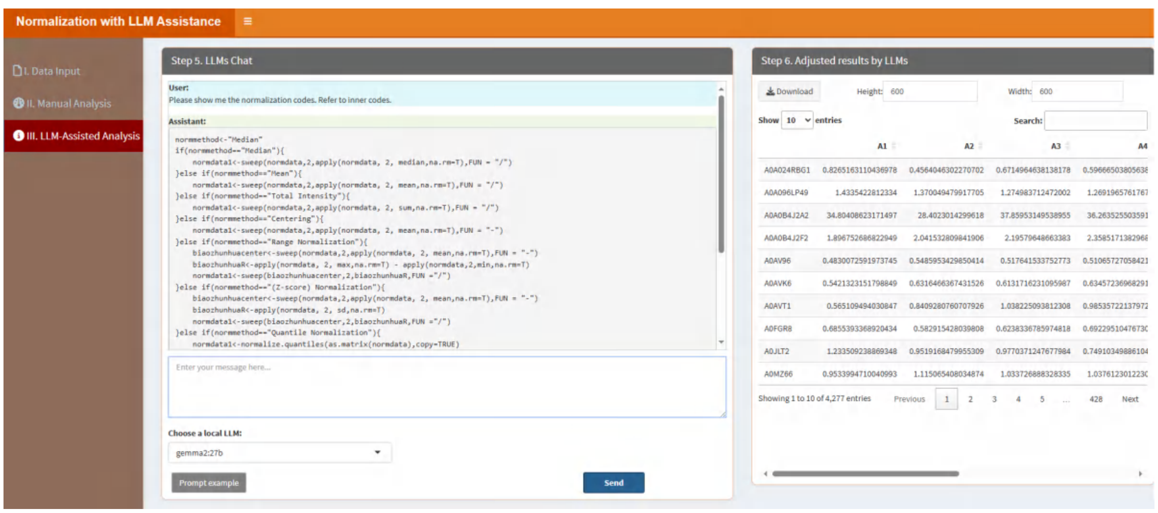
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please show me the normalization codes. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please show me the normalization codes. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

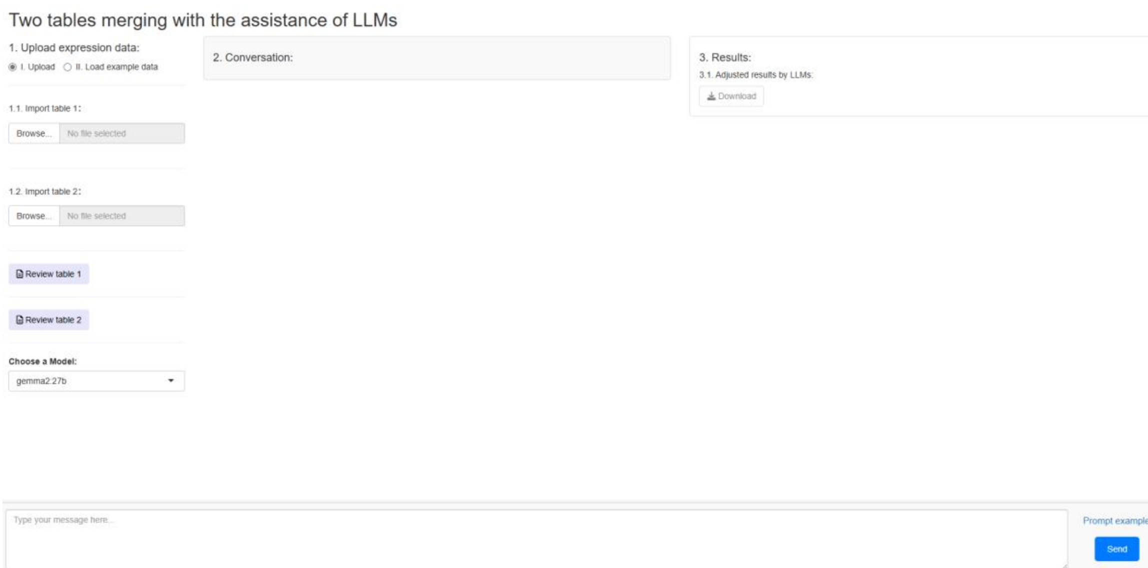


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



4.4. Two tables merging

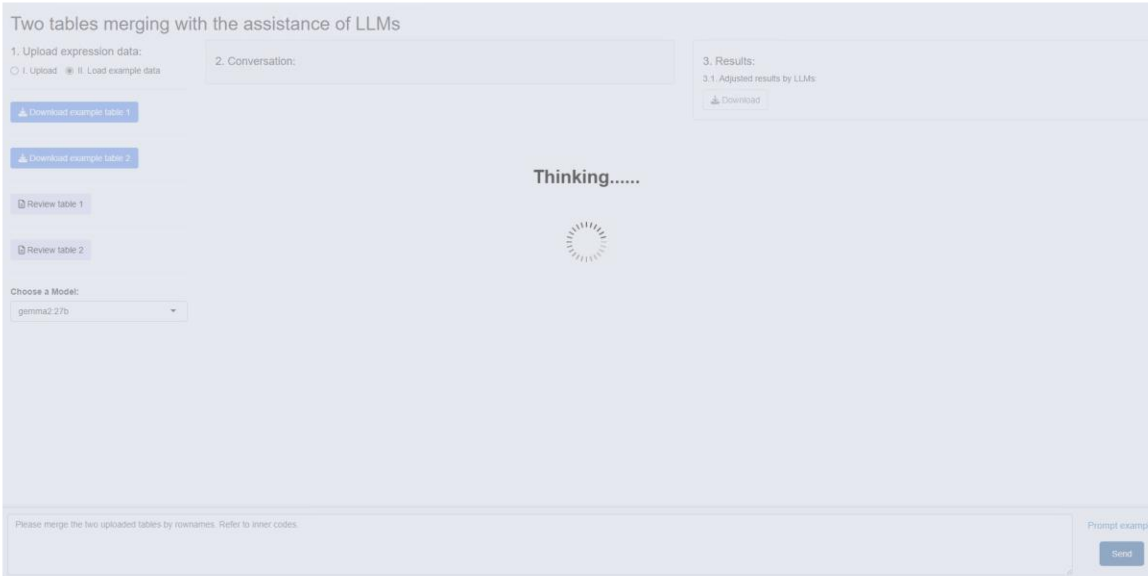
Two tables merging here is the process of combining data from two separate tables into a single, unified dataset based on a common key or shared attributes. After clicking the "Start" button, it will be started in a new window like below:



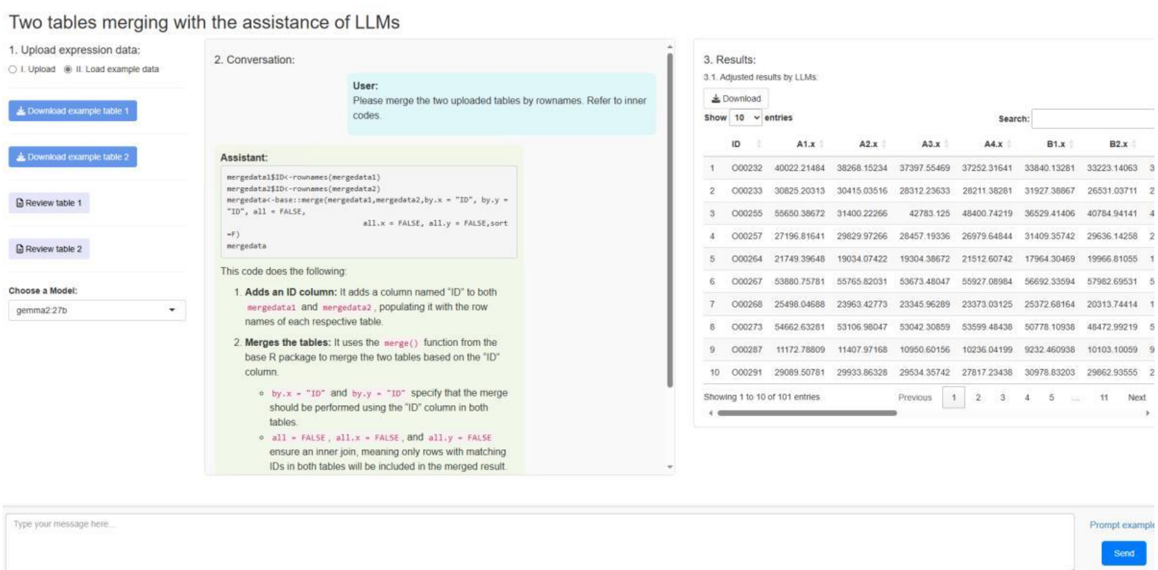
The "1. Upload expression data:" here is similar to the "I. Data Input" part in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here. If users choose "I. Upload", they can upload two data through "1.1. Import table 1" and "1.2. Import table 2" by clicking responding "Browse" button. Then they can click "Review table 1" and "Review table 2" to check the uploaded tables.

Next, users can choose a local LLM model from "Choose a Model" parameter and then type the prompt in the message box on the bottom, for example, the example prompt "Please merge the two uploaded tables by rownames. Refer to inner codes." instructs LLMs to learn the internal R codes and merge the two uploaded tables by the row names. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Please merge the two uploaded tables by rownames. =>", it will similarly prompt LLMs to learn the internal R codes and produce the merged result table.

After type the prompt, click "Send" button. It will learn like below:

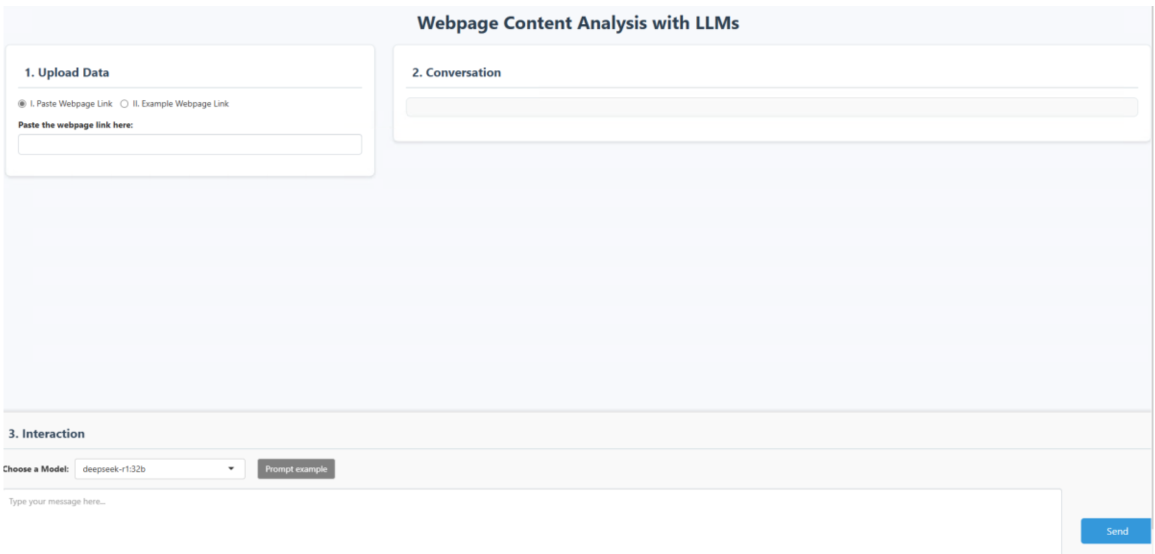


The results are shown as below, then users can download the results by clicking the “Download” button in the “3. Results” section.

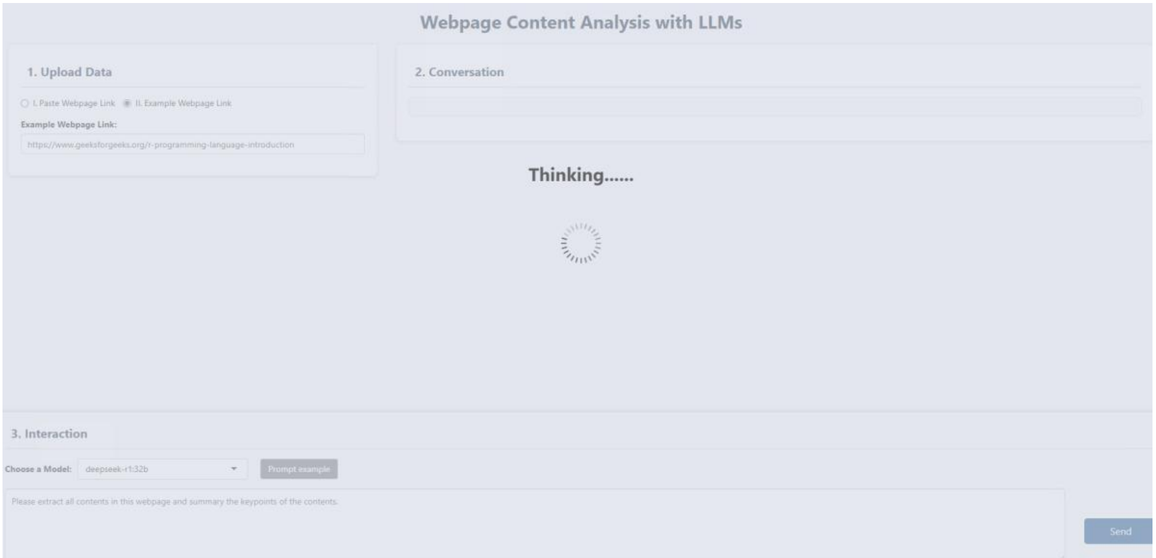


4.5. Webpage content analysis

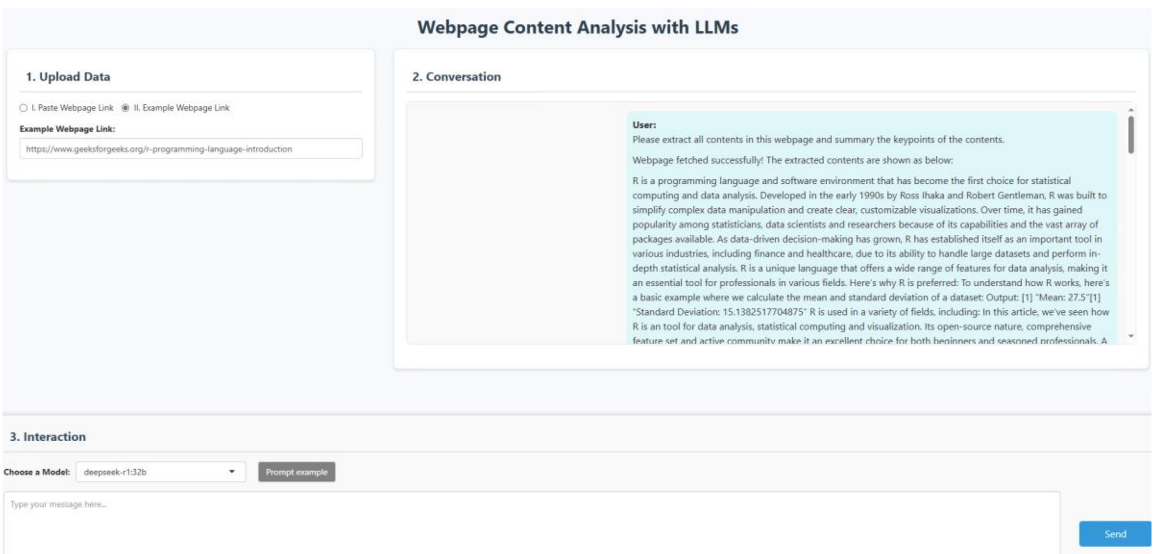
Webpage content analysis is a process that involves extracting the content of a webpage from a user-provided link and analyzing it using local LLMs. This module enables users to retrieve and process textual data, metadata, or other relevant information from webpages, providing insights into the content's structure, themes, or sentiment. By leveraging the capabilities of LLMs, the analysis can include summarization, keyword extraction, sentiment detection, or even deeper contextual understanding. After clicking the “Start” button, it will be started in a new window like below:



“1. Upload data” here means users can type the webpage link or check the example webpage link. After typing the webpage link of interest, users can choose a local LLM from the “Choose a Model” parameter in the “3. Interaction” section and then type the prompt in the message box on the bottom. For example, the example prompt “Please extract all contents in this webpage and summary the keypoints of the contents.” instructs LLMs to learn the contents extracted from the webpage. After type the prompt, click “Send” button. It will learn like below:



The results are shown in the “2. Conversation” section as below:



The conversation history like below:

Webpage Content Analysis with LLMs

2. Conversation

User:
Please extract all contents in this webpage and summary the keypoints of the contents.

Webpage fetched successfully! The extracted contents are shown as below:

R is a programming language and software environment that has become the first choice for statistical computing and data analysis. Developed in the early 1990s by Ross Ihaka and Robert Gentleman, R was built to simplify complex data manipulation and create clear, customizable visualizations. Over time, it has gained popularity among statisticians, data scientists and researchers because of its capabilities and the vast array of packages available. As data-driven decision-making has grown, R has established itself as an important tool in various industries, including finance and healthcare, due to its ability to handle large datasets and perform in-depth statistical analysis. R is a unique language that offers a wide range of features for data analysis, making it an essential tool for professionals in various fields. Here's why R is preferred: To understand how R works, here's a basic example where we calculate the mean and standard deviation of a dataset: Output: [1] "Mean: 27.5"[1] "Standard Deviation: 15.1382517704875" R is used in a variety of fields, including: In this article, we've seen how R is an tool for data analysis, statistical computing and visualization. Its open-source nature, comprehensive feature set and active community make it an excellent choice for both beginners and seasoned professionals. A

Assistant: Okay, so the user has provided a query where they want to extract all contents from a webpage and summarize the key points. The response given is a structured summary with sections like introduction, why R is preferred, example, applications, conclusion, and references.
First, I need to understand what exactly the user is asking for. They seem to be looking for a way to process web content by extracting text and then condensing it into main ideas. This could be useful for someone who wants to quickly grasp the content without reading through the entire webpage.

Looking at the response, it's clear that the assistant has broken down the information into digestible parts. The introduction explains what R is and its creators. Then, it lists reasons why R is preferred, which includes open-source nature, extensive packages, visualization tools, community support, flexibility, and integration capabilities. An example code snippet is provided to show a practical use case.

The applications section highlights various industries where R is used, such as finance, healthcare, marketing, education, and government. The conclusion reinforces why R is valuable for both newbies and experts, mentioning its features and community support.

Now, thinking about the user's possible scenario: they might be someone who needs to analyze web content quickly, perhaps a student, researcher, or professional looking to gather information without manually reading through each webpage. They could be working on a project that requires summarizing data from multiple sources efficiently.

The user didn't specify any particular format for the summary, but the response provided is well-organized with clear headings and bullet points, making it easy to read. This suggests that the user values clarity and structure in their summaries.

I should consider if there are any deeper needs here. Maybe the user wants this process automated for multiple webpages or perhaps they need more detailed analysis beyond just summarizing key points. They might also be interested in how R compares to other tools, which isn't covered in the provided response but could be a valuable addition.

Another angle is whether the user has any specific technical requirements, like handling different types of content or extracting data from images or tables within webpages. The current response focuses on text extraction and summarization, so if the user needs more advanced features, that might require additional functionality.

In terms of potential improvements, adding examples for each application area could make the summary even more informative. Also, including a brief explanation of how R's open-source nature benefits users compared to proprietary software might add value.

Overall, the response seems comprehensive for the given query, but understanding the user's specific needs beyond the initial request would help tailor the solution better. For instance, if they're looking to integrate this into an application or use it for academic research, the approach might vary slightly.

Summary of Key Points:

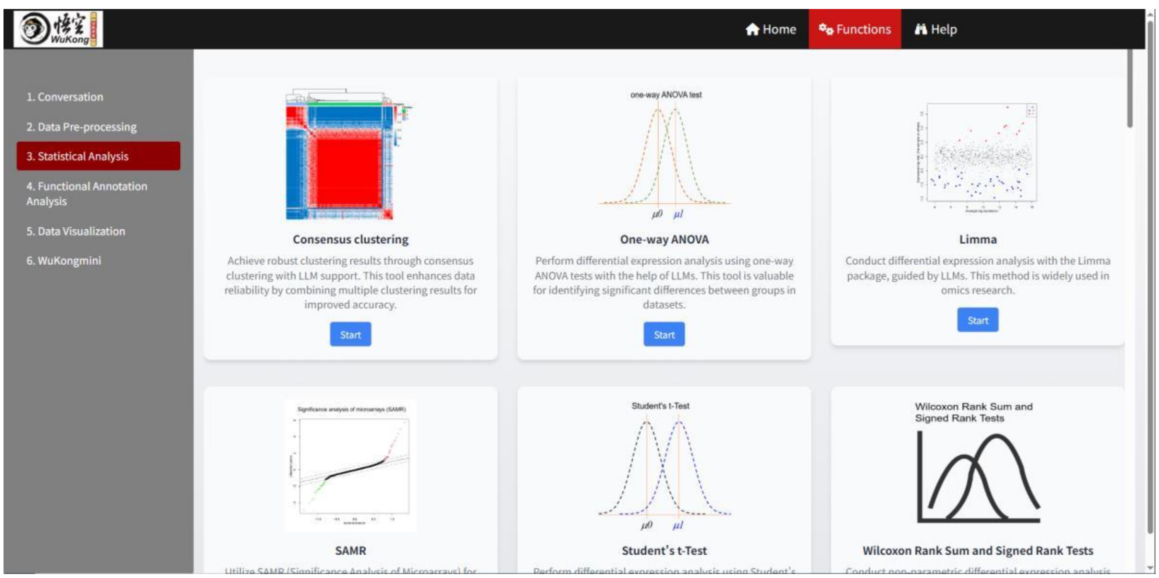
1. **Introduction to R**:
   - R is a programming language and software environment primarily used for statistical computing and data analysis.
   - Developed in the early 1990s by Ross Ihaka and Robert Gentleman.
   - Designed to simplify complex data manipulation and create customizable visualizations.
2. **Popularity of R**:
   - Gained popularity among statisticians, data scientists, and researchers due to its capabilities and extensive array of packages.
   - Widely used in industries like finance and healthcare for handling large datasets and performing in-depth statistical analysis.
3. **Why R is Preferred**:
   - Open-source nature allows for flexibility and customization.
   - Extensive library of packages for various data analysis tasks.
   - Strong visualization tools for creating clear and customizable plots.
   - Active community support and continuous development.
   - Flexibility in handling both small and large datasets.
   - Integration with other tools and platforms.
4. **Example Code**:
   - Example provided to calculate the mean and standard deviation of a dataset.
   - Output demonstrates basic functionality for statistical calculations.
5. **Applications of R**:
   - Used in various fields, including finance, healthcare, marketing, education, and government.
   - Ideal for data-driven decision-making across industries.
6. **Conclusion**:
   - R is a powerful tool for data analysis, statistical computing, and visualization.
   - Its open-source nature, comprehensive feature set, and active community make it suitable for both beginners and experienced professionals.

References:

- The webpage provides an overview of R's features, applications, and why it is preferred in various industries.

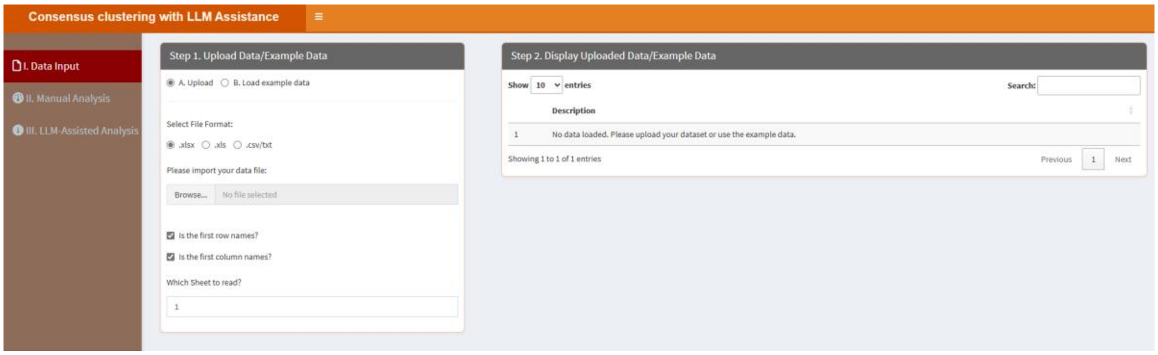
5. Statistical Analysis

Statistical analysis in proteomics plays a pivotal role in extracting meaningful insights from complex biological data. This section encompasses a wide array of models, including Consensus Clustering, One-way ANOVA, Limma, SAMR (Significance Analysis of Microarrays), Student's t-Test, Wilcoxon Rank Sum and Signed Rank Tests, Dynamic Network Biomarkers (DNBs), Factor Analysis, Grey Relational Analysis (GRA), Hierarchical Cluster Analysis (HCA), K-means Clustering, Lack of Fit F-test, Linear Regression, Logistic Regression, Mfuzz, Network Degree Matrix (NDM), OPLS-DA (Orthogonal Partial Least Squares Discriminant Analysis), Principal Component Analysis (PCA), Principal Coordinates Analysis (PCoA), PLS-DA (Partial Least Squares Discriminant Analysis), Power Analysis, Restricted Cubic Spline (RCS) Analysis, Redundancy Analysis (RDA), Rank Rank Hypergeometric Overlap (RRHO) Analysis, SIMCA (Soft Independent Modelling by Class Analogy), Time Course Data Analysis, t-SNE (t-Distributed Stochastic Neighbor Embedding), Tumor Purity Estimation, and UMAP (Uniform Manifold Approximation and Projection). Users can process their data by adjusting parameters in each module through the classic framework, while also leveraging local LLMs to gain deeper insights into each model and enhance their data analysis capabilities.



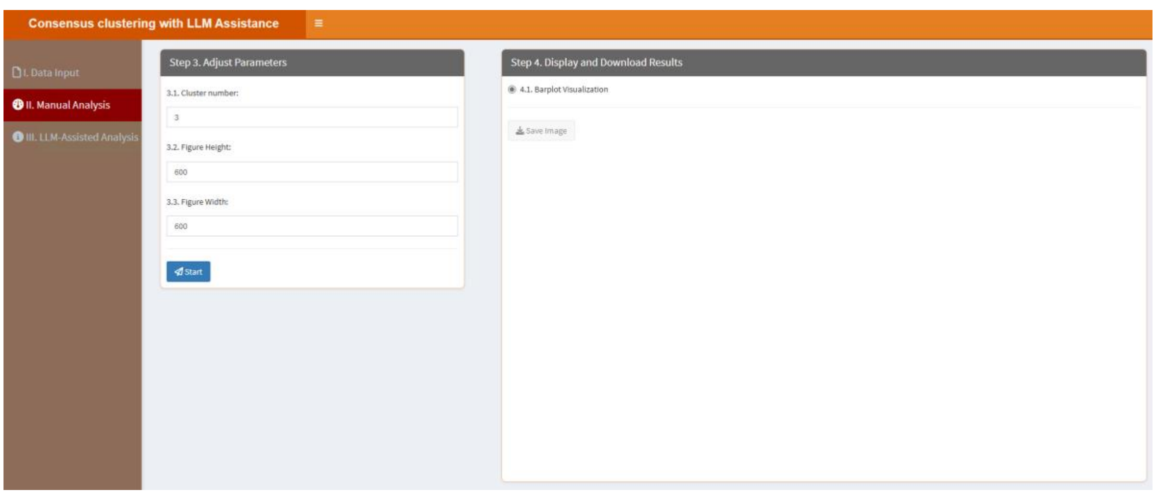
5.1. Consensus Clustering

Consensus clustering allows users to identify robust clusters within their data by aggregating results across multiple iterations. Users can adjust parameters like the number of clusters through the classic framework, while leveraging local LLMs to refine cluster stability and interpret the results. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



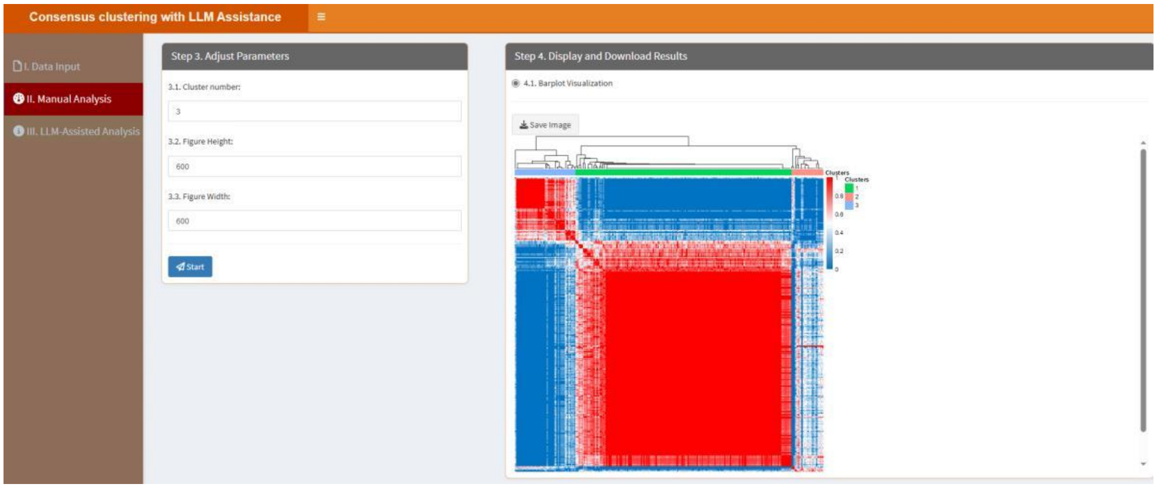
3.1. Cluster number: maximum cluster number to evaluate.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization,

ensuring that it accommodates the data and fits appropriately in the given space.

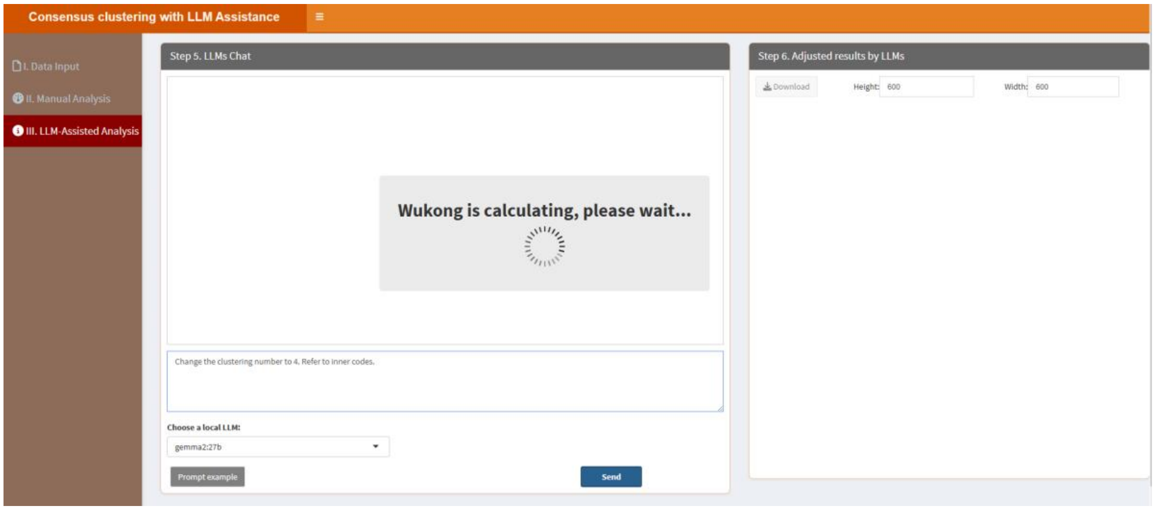
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



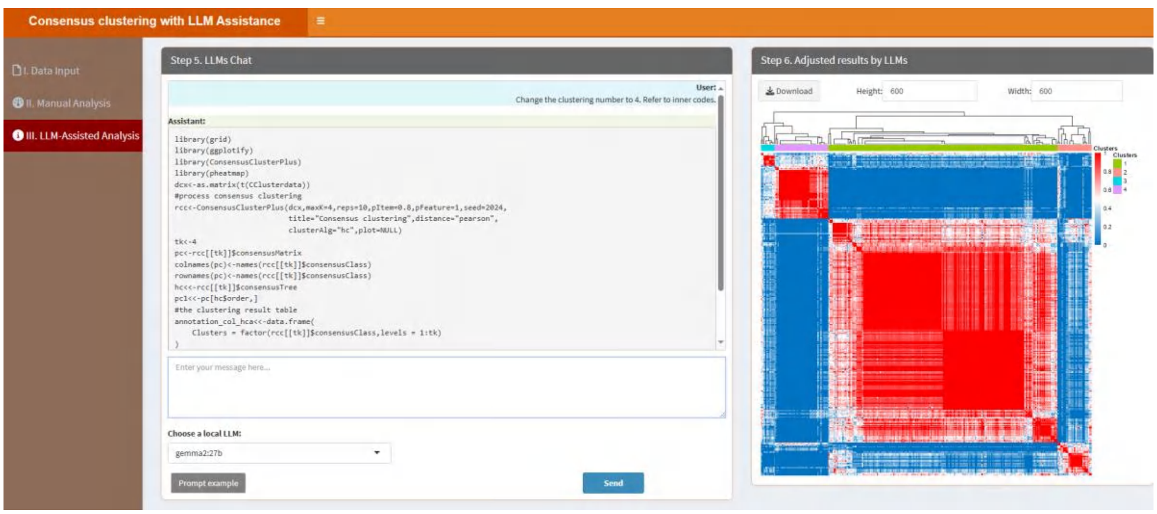
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Change the clustering number to 4. Refer to inner codes.” instructs LLMs to learn the internal R codes and change the clustering number to 4. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Change the clustering number to 4. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

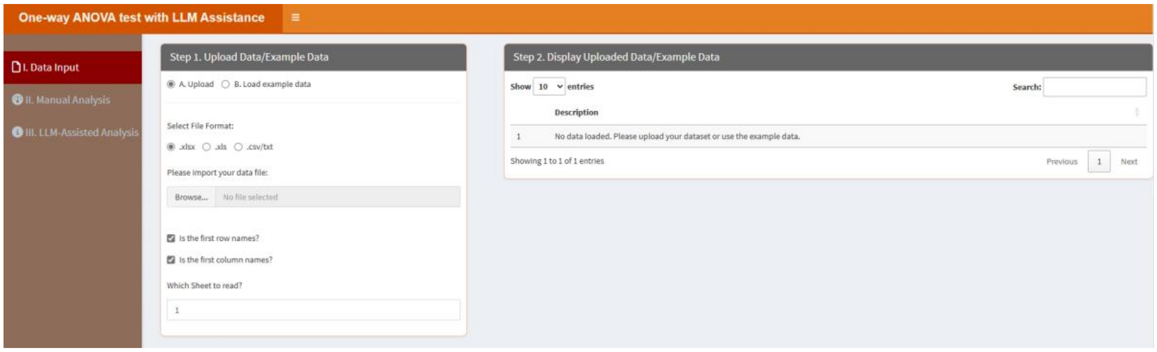


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



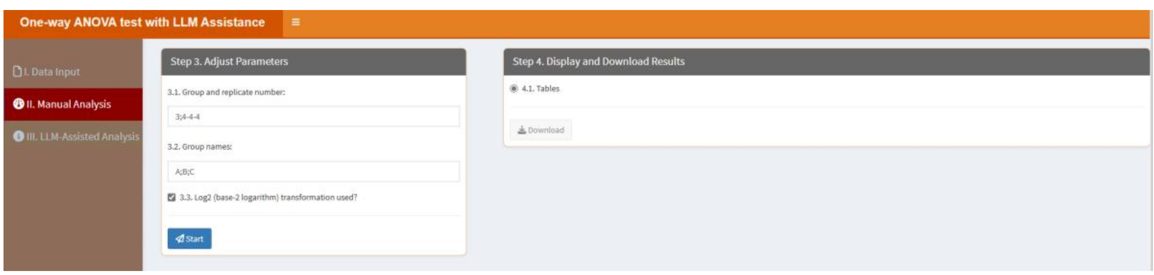
5.2. One-way ANOVA

One-way ANOVA is a statistical method for comparing means across multiple groups. Users can modify parameters such as group definitions using the classic framework and utilize local LLMs to enhance the interpretation of significant differences. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

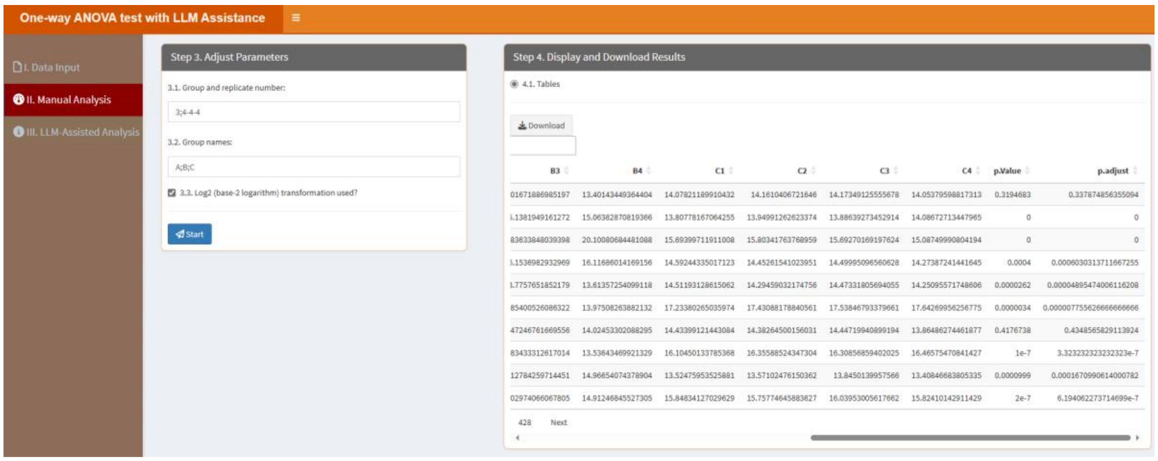


3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Log2 (base-2 logarithm) transformation used? if true, the data will be transformed to the logarithmic scale with base 2.

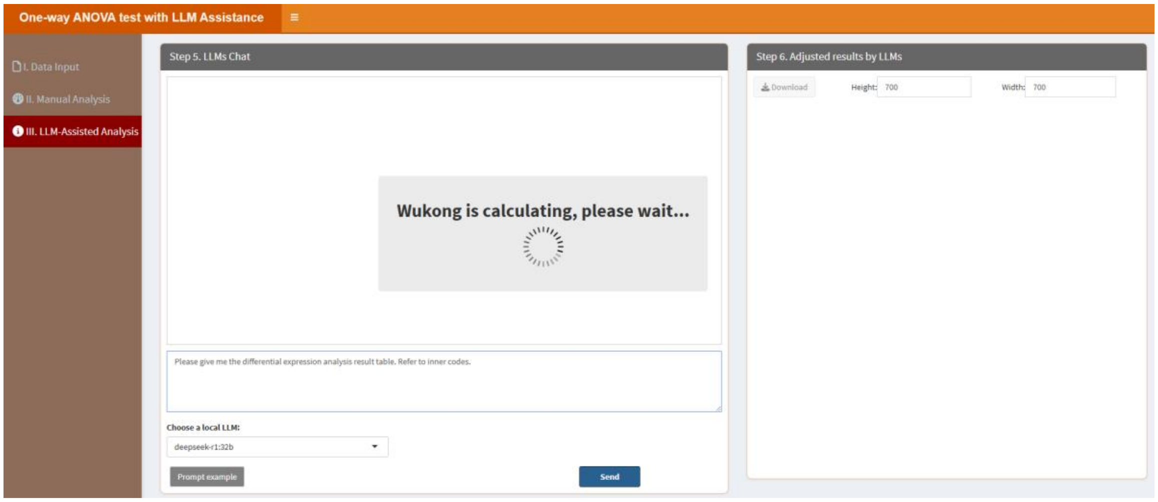
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Tables” part, users can click “Download” button to download the table, which contains the results:



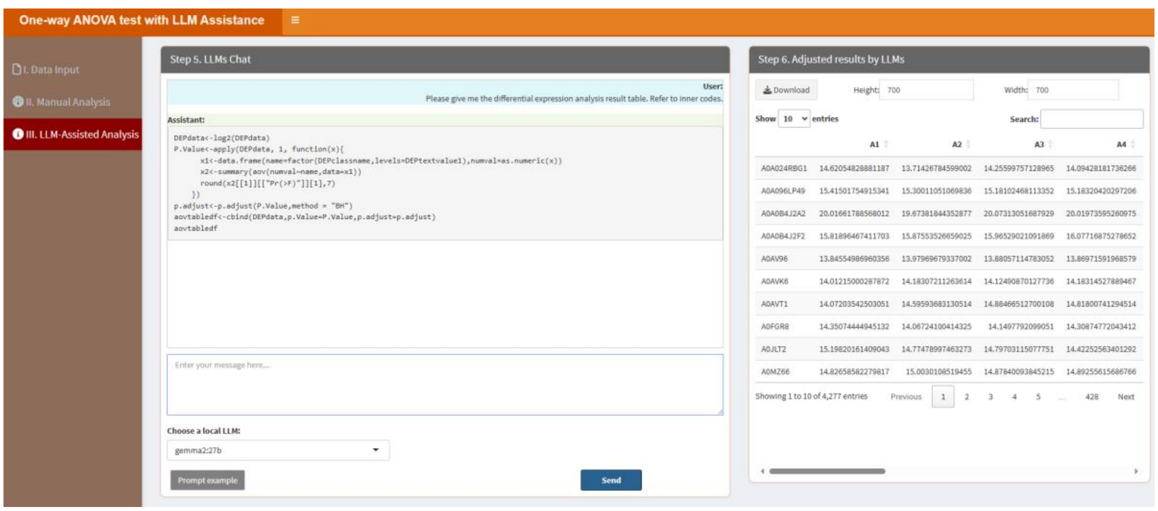
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please give me the differential expression analysis result table. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please give me the differential expression analysis result table. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

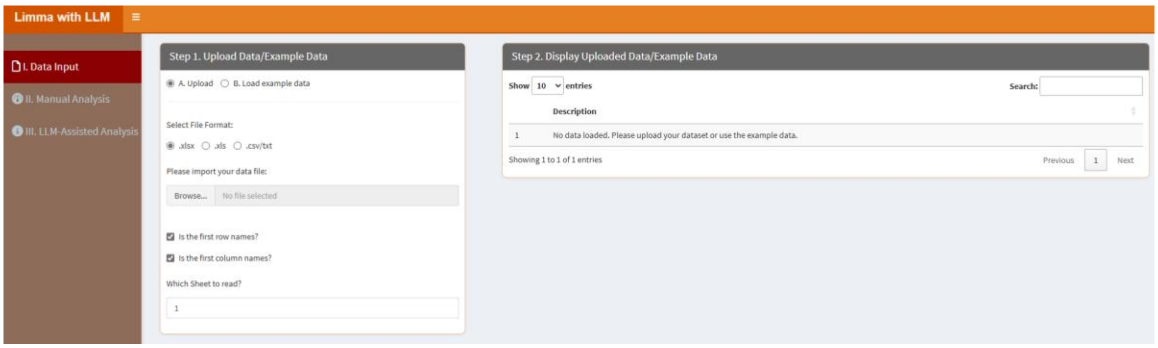


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



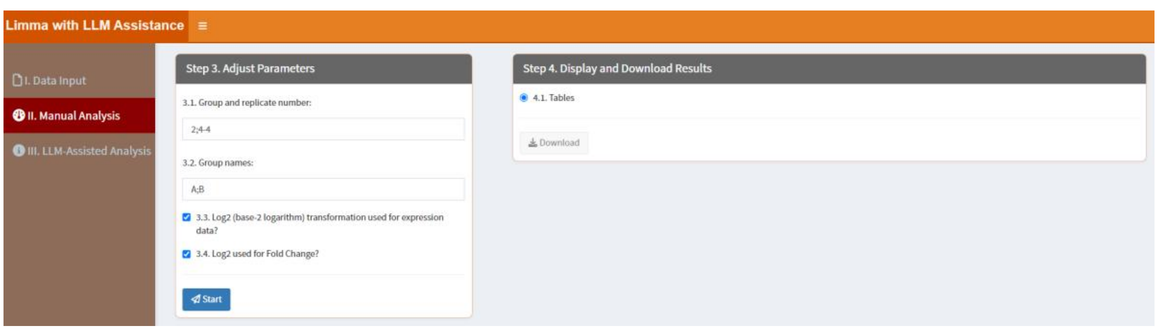
5.3. Limma

Limma is a powerful tool for analyzing differential expression in omics datasets. Users can set parameters like contrasts and Log2 (base-2 logarithm) transformation through the classic framework, while local LLMs can help refine the analysis and provide deeper insights into the results. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

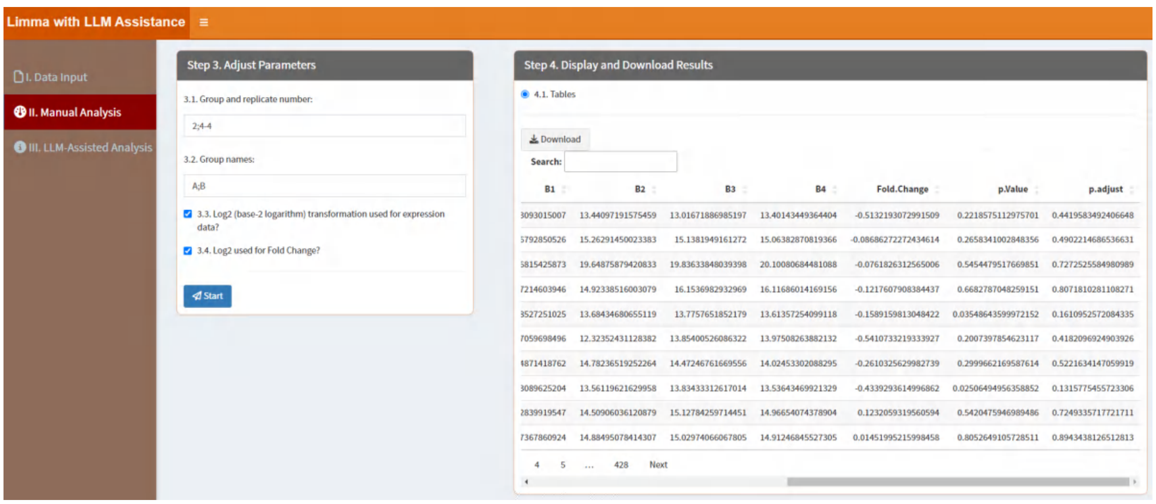
3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Log2 (base-2 logarithm) transformation used for expression data? If true, the data will be transformed to the logarithmic scale with base 2.

3.4. Log2 used for Fold Change? If true, the Fold Change is calculated with log2, for example, Fold Change = log2($Experiment_{mean}$/$Control_{mean}$), otherwise, Fold Change = $Experiment_{mean}$/$Control_{mean}$. $Experiment_{mean}$ is the mean value of the Experiment group, $Control_{mean}$ is the mean value of the Control group.

After setting parameters, click “Start” button, the results will be shown as below:

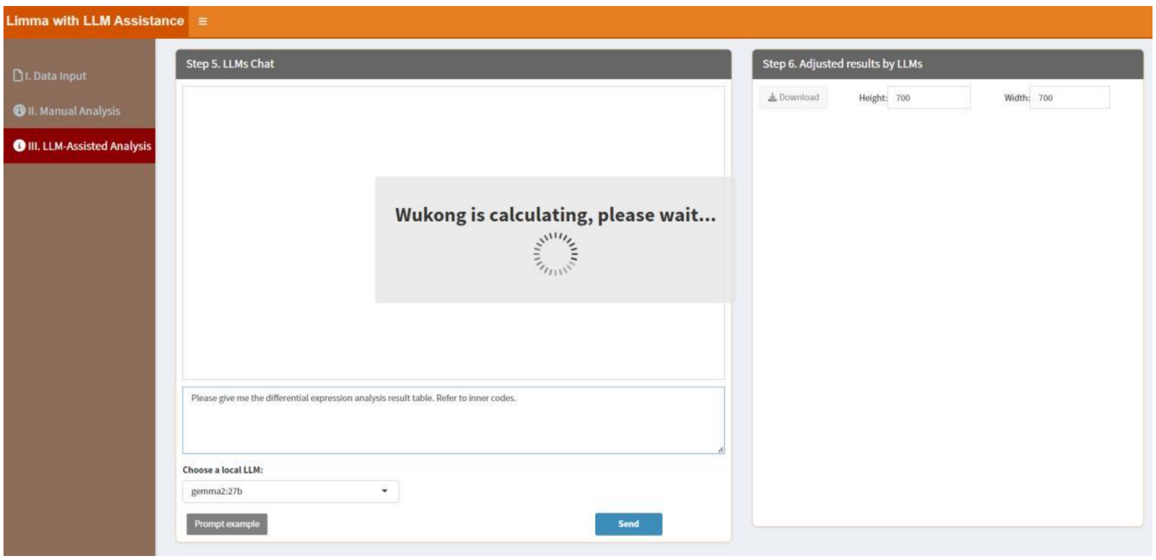
In the “4.1. Tables” part, users can click “Download” button to download the table, which contains the results:



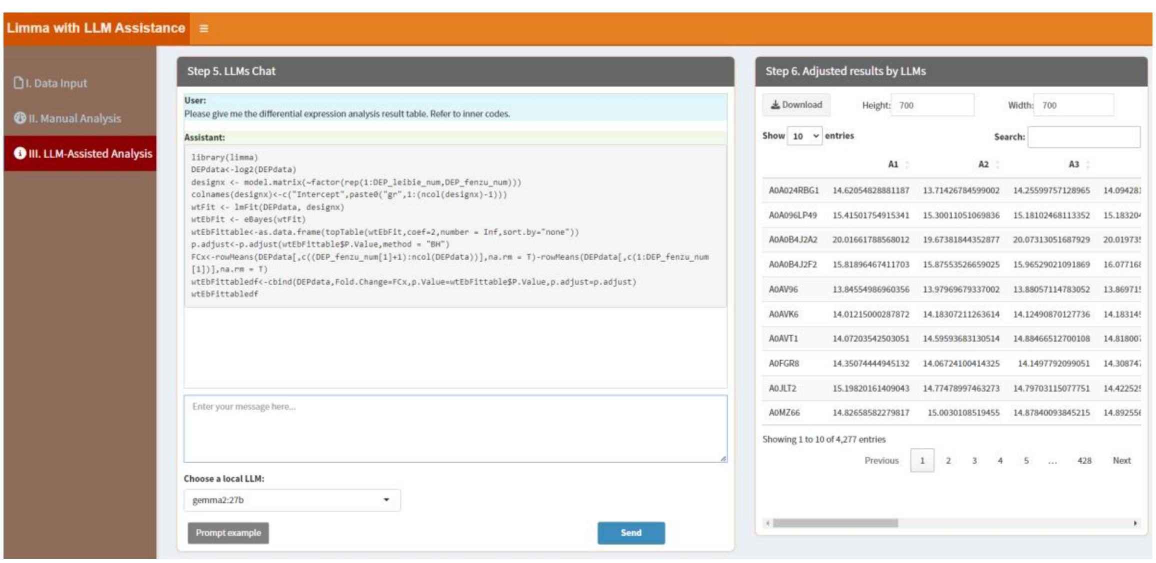
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please give me the differential expression analysis result table. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please give me the differential expression analysis result table. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

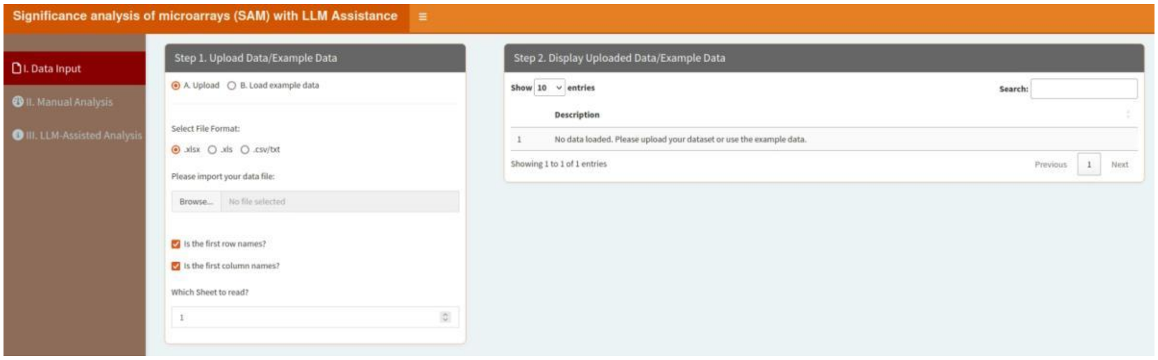


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



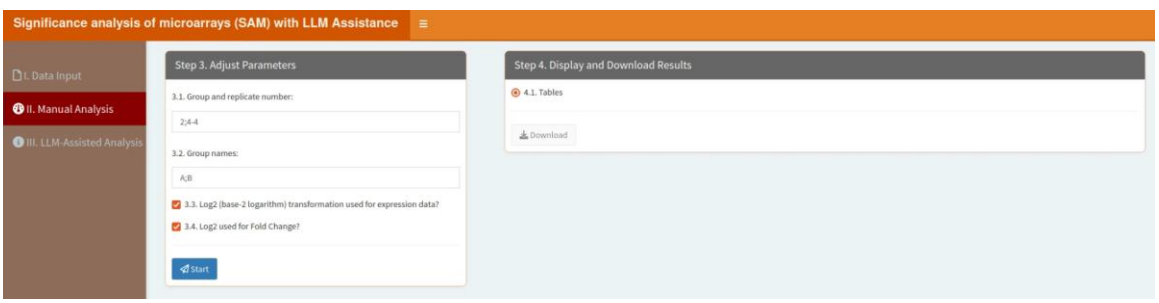
5.4. SAMR (Significance Analysis of Microarrays)

SAMR is a statistical technique to identify significantly expressed features in omics data. Users can adjust sample information and Log2 (base-2 logarithm) transformation using the classic framework, and local LLMs can assist in refining the analysis and interpreting the findings. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

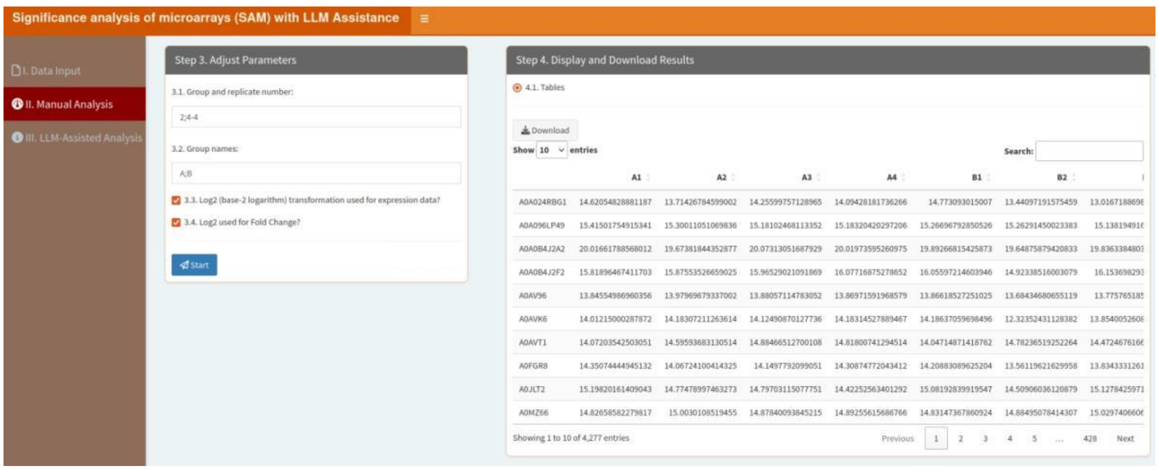
3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Log2 (base-2 logarithm) transformation used for expression data? If true, the data will be transformed to the logarithmic scale with base 2.

3.4. Log2 used for Fold Change? If true, the Fold Change is calculated with log2, for example, Fold Change = log2(Experiment$_{mean}$/Control$_{mean}$), otherwise, Fold Change = Experiment$_{mean}$/Control$_{mean}$. Experiment$_{mean}$ is the mean value of the Experiment group, Control$_{mean}$ is the mean value of the Control group.

After setting parameters, click “Start” button, the results will be shown as below:

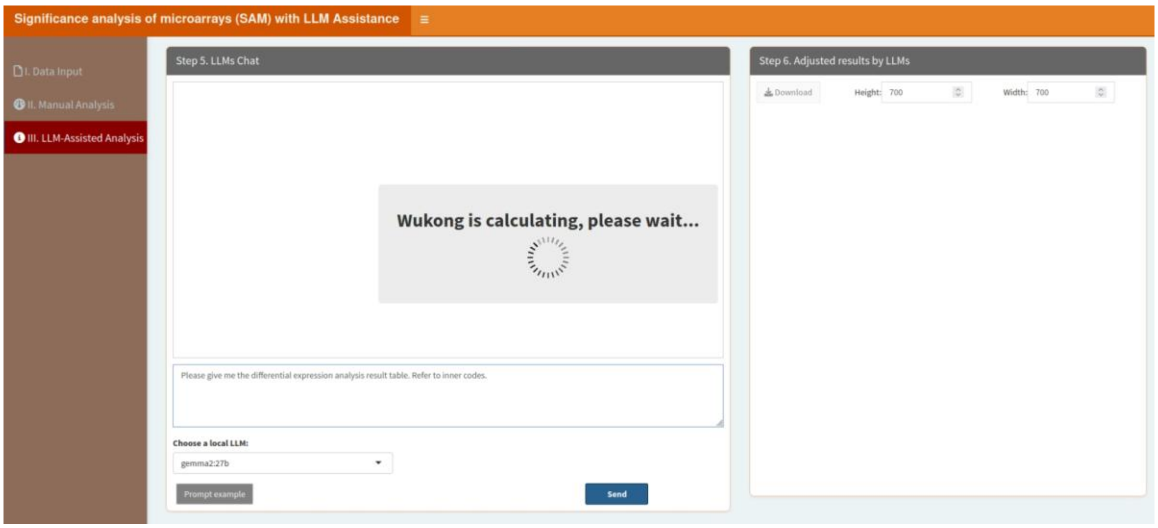
In the “4.1. Tables” part, users can click “Download” button to download the table, which contains the results:



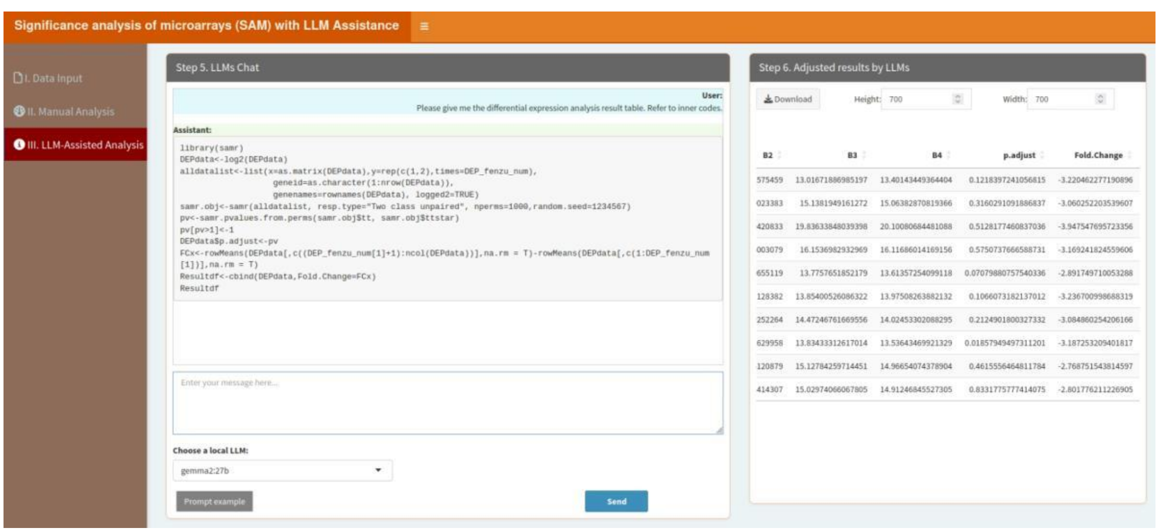
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please give me the differential expression analysis result table. Refer to inner codes." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Please give me the differential expression analysis result table. =>", it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click "Send" button. It will learn like below:

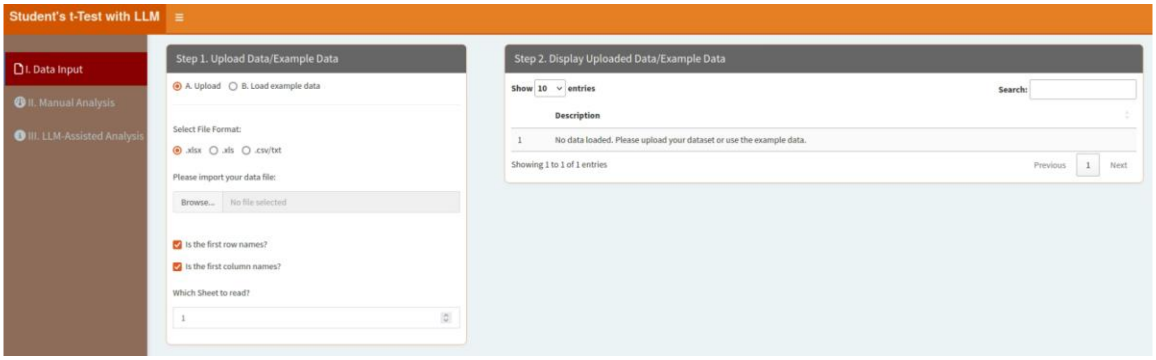


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



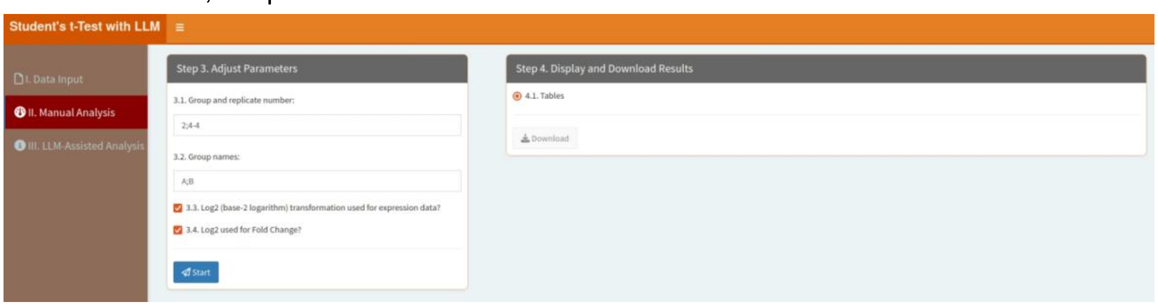
5.5. Student's t-Test

Student's t-Test is a fundamental statistical test for comparing two groups. Users can configure sample information and Log2 (base-2 logarithm) transformation through the classic framework, while local LLMs can aid in refining the analysis and understanding the results. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

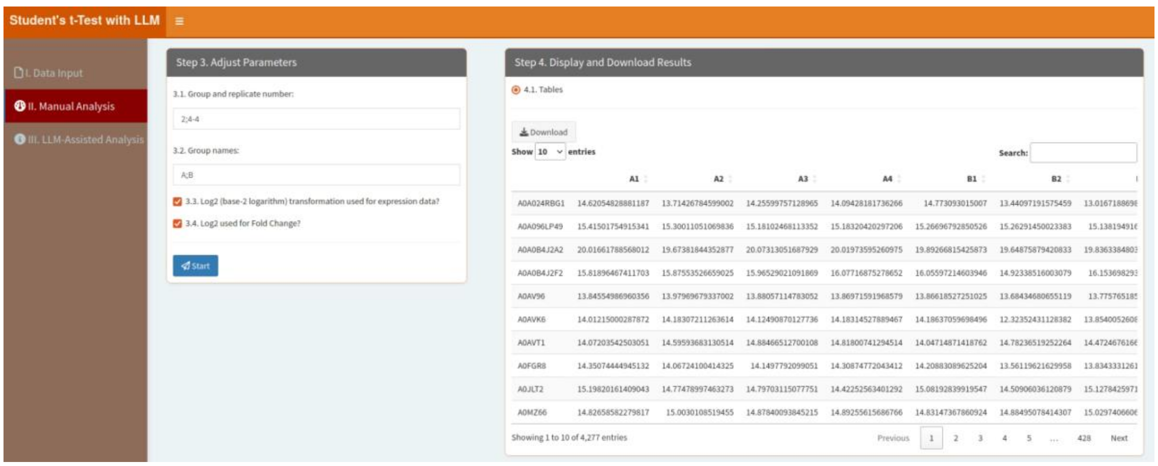
3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Log2 (base-2 logarithm) transformation used for expression data? If true, the data will be transformed to the logarithmic scale with base 2.

3.4. Log2 used for Fold Change? If true, the Fold Change is calculated with log2, for example, Fold Change = $\log2(\text{Experiment}_{mean}/\text{Control}_{mean})$, otherwise, Fold Change = $\text{Experiment}_{mean}/\text{Control}_{mean}$. Experiment_{mean} is the mean value of the Experiment group, Control_{mean} is the mean value of the Control group.

After setting parameters, click "Start" button, the results will be shown as below:

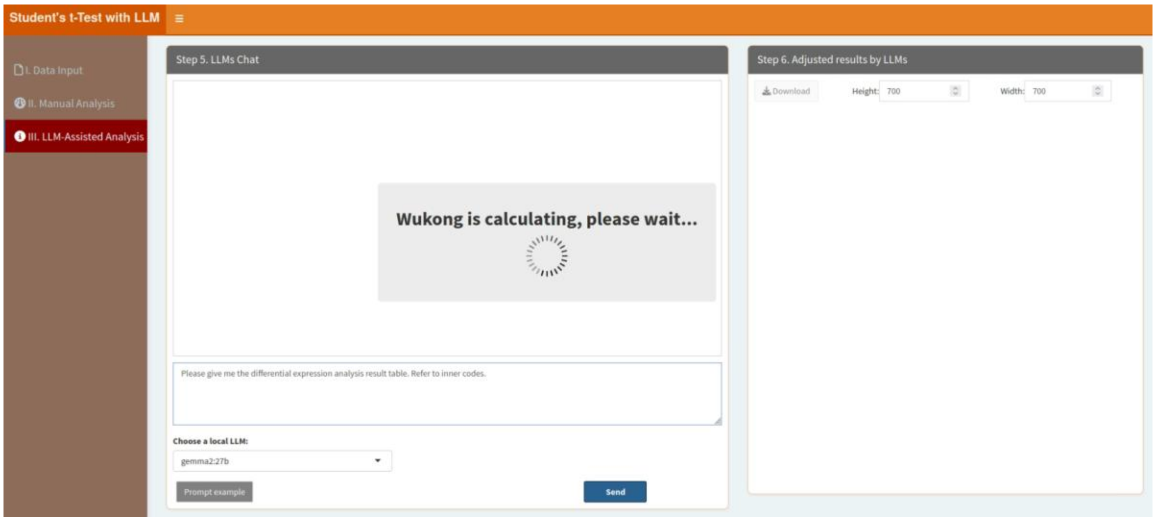
In the "4.1. Tables" part, users can click "Download" button to download the table, which contains the results:



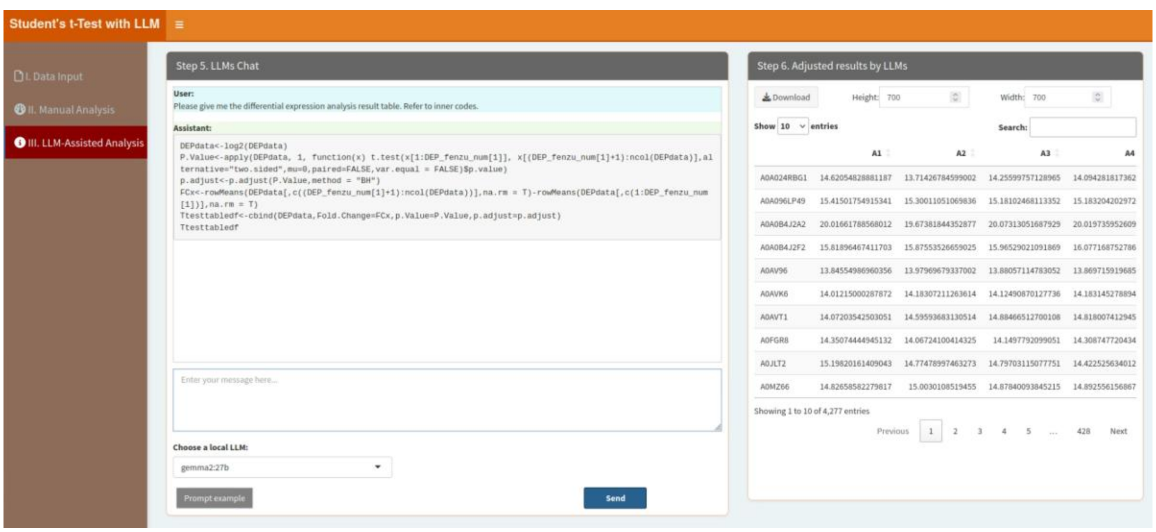
"III. LLM-Assisted Analysis" means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt "Please give me the differential expression analysis result table. Refer to inner codes." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please give me the differential expression analysis result table. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

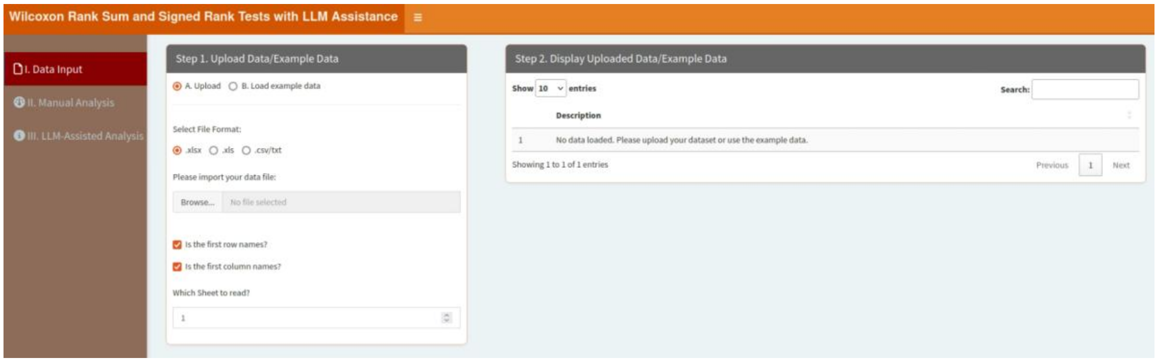


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



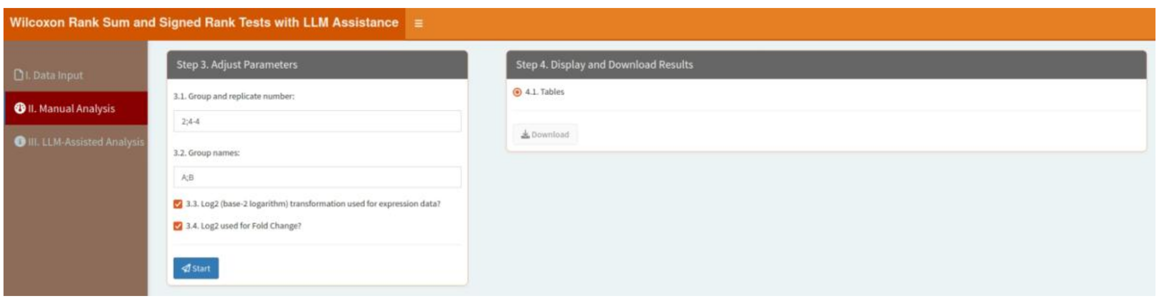
5.6. Wilcoxon Rank Sum and Signed Rank Tests

These non-parametric tests are ideal for comparing groups without assuming normality. Users can customize test settings through the classic framework and rely on local LLMs to improve result interpretation and analysis. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

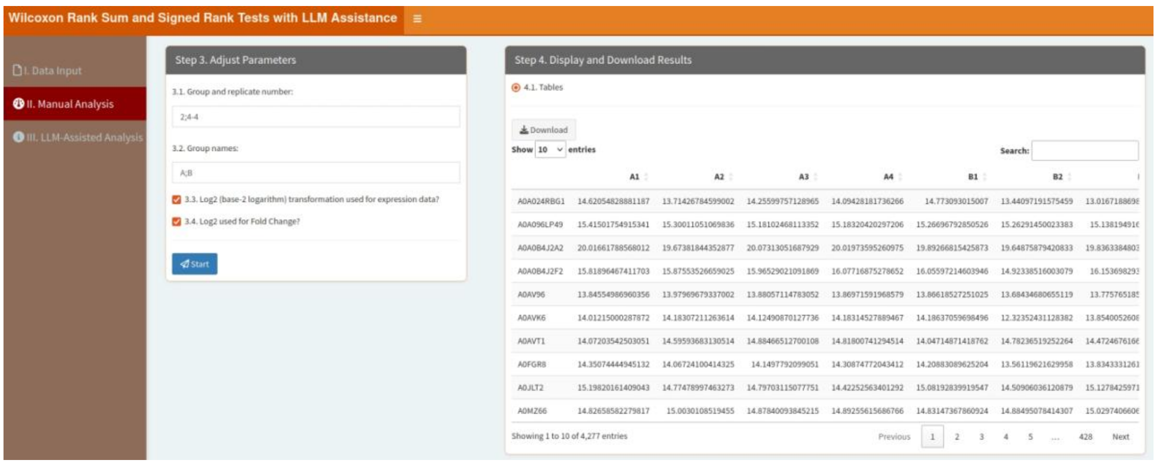
3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Log2 (base-2 logarithm) transformation used for expression data? If true, the data will be transformed to the logarithmic scale with base 2.

3.4. Log2 used for Fold Change? If true, the Fold Change is calculated with log2, for example, Fold Change = $\log2(\text{Experiment}_{mean}/\text{Control}_{mean})$, otherwise, Fold Change = $\text{Experiment}_{mean}/\text{Control}_{mean}$. Experiment_{mean} is the mean value of the Experiment group, Control_{mean} is the mean value of the Control group.

After setting parameters, click “Start” button, the results will be shown as below:

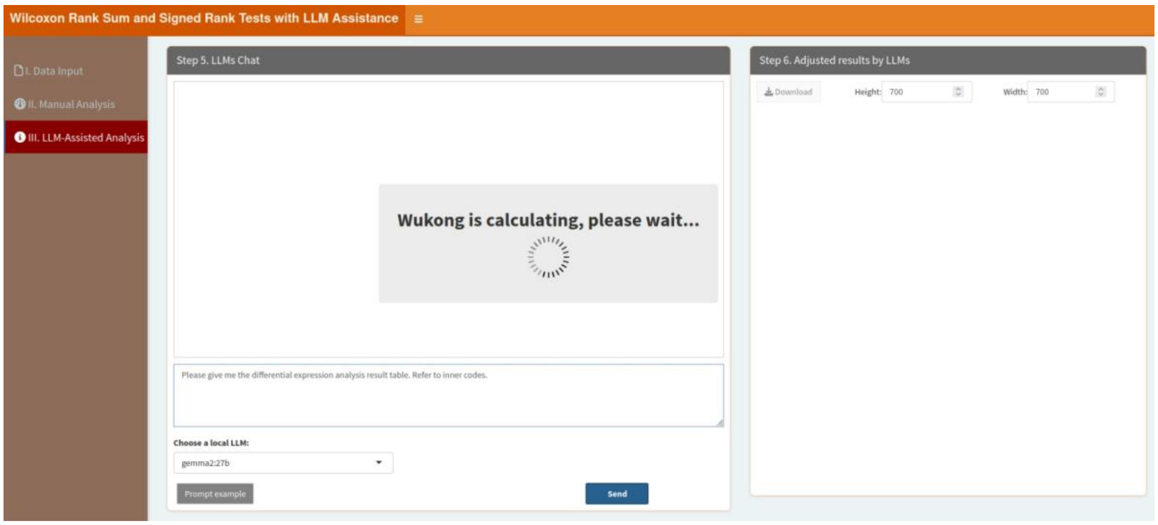
In the “4.1. Tables” part, users can click “Download” button to download the table, which contains the results:



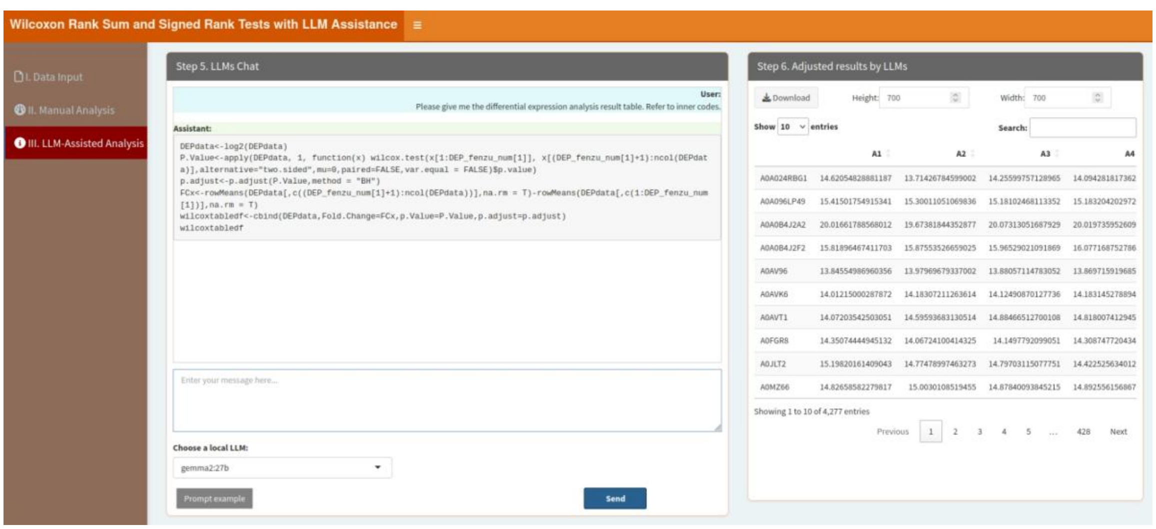
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please give me the differential expression analysis result table. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please give me the differential expression analysis result table. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

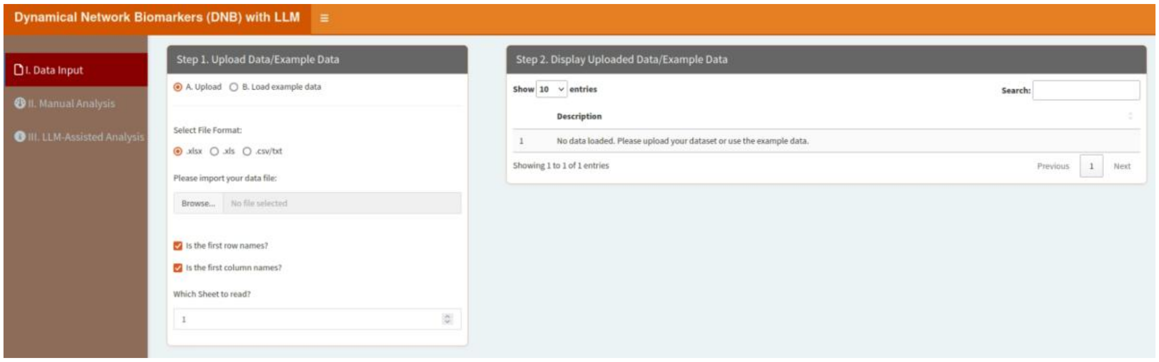


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



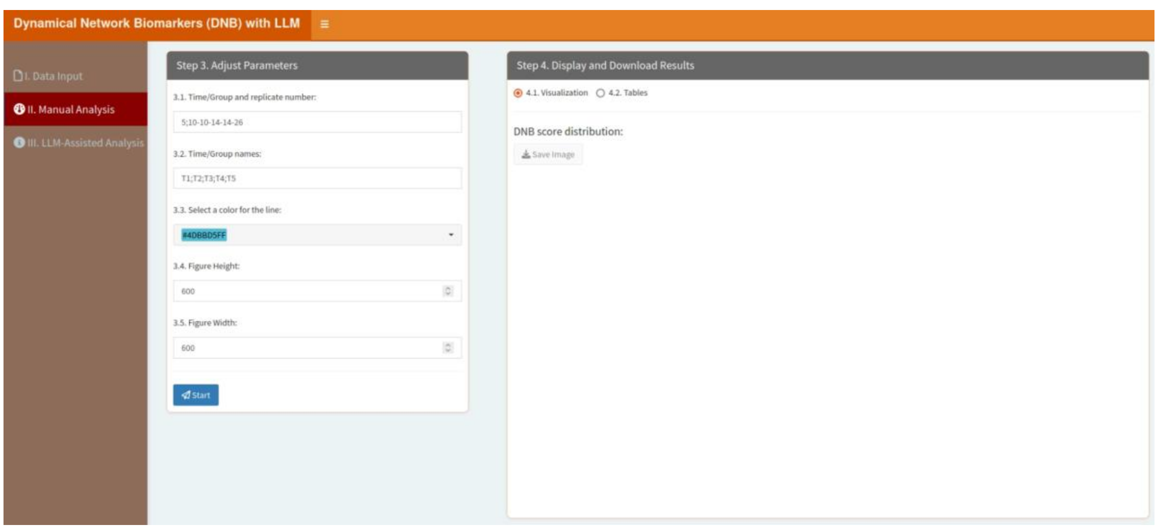
5.7. Dynamic Network Biomarkers (DNBs)

Dynamic Network Biomarkers help identify potential biomarkers in biological systems. Users can fine-tune network parameters using the classic framework, while local LLMs provide enhanced insights into this model. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Time/Group and replicate number: Type in the timepoint/group number and replicate number here. Please note, the timepoint/group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two timepoints/groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups/timepoints with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 5 timepoints and (10, 10, 14, 14, 26) replicates in each group, so here is “5;10-10-14-14-26”.

3.2. Time/Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 5 timepoints in the example data, you can type in "T1;T2;T3;T4;T5".

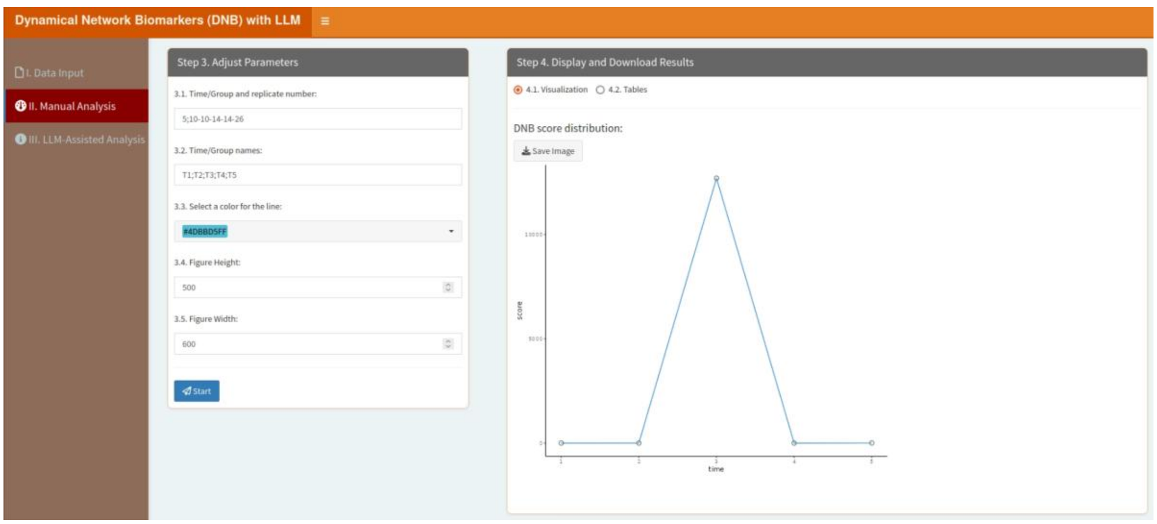
3.3. Select a color for the line: Select a color for the line in the figure.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

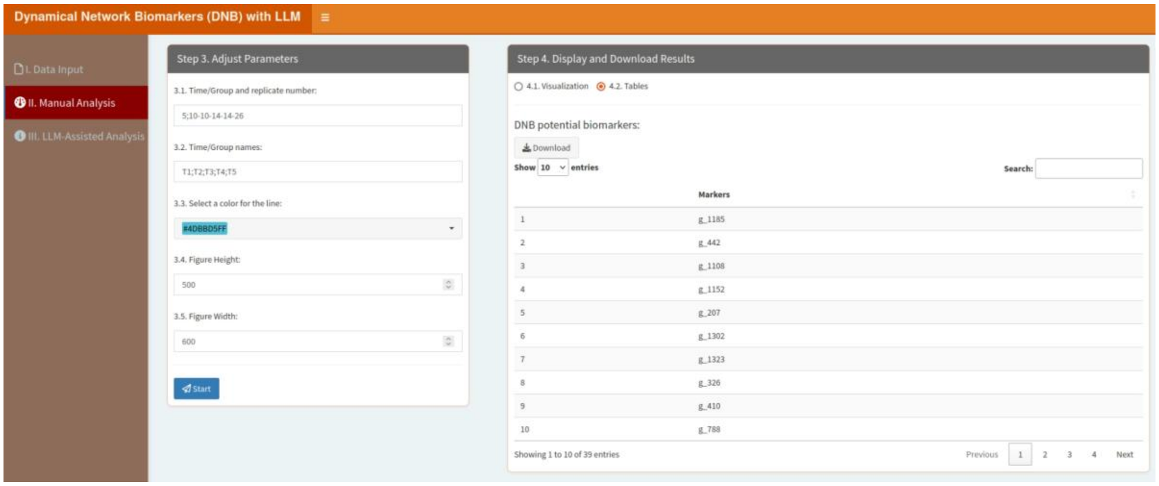
3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



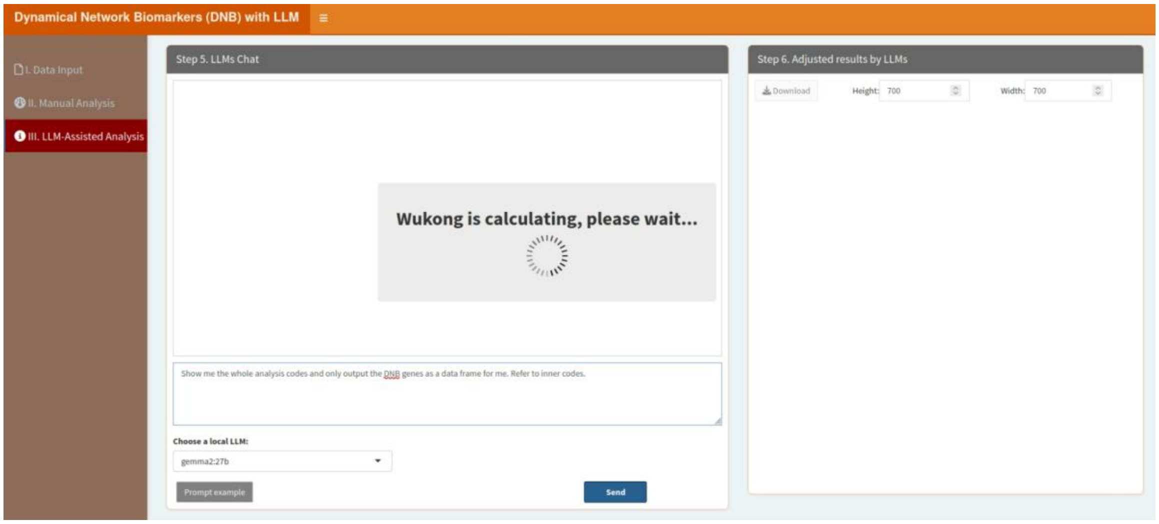
In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the DNB potential biomarkers:



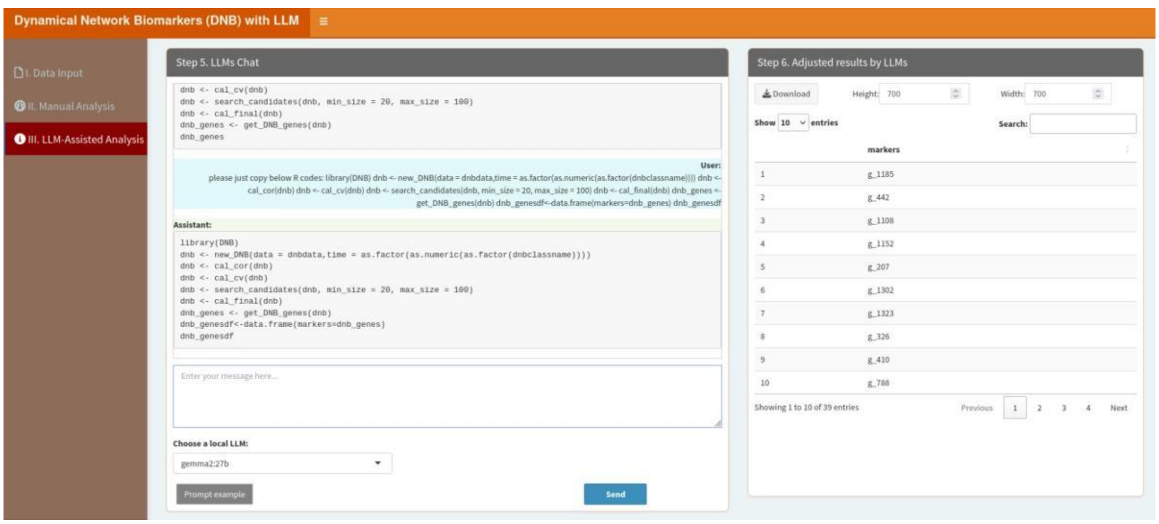
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Show me the whole analysis codes and only output the results. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Show me the whole analysis codes and only output the results. =>”, it will similarly prompt LLMs to learn the internal R codes and produce theresult table.

After type the prompt, click “Send” button. It will learn like below:

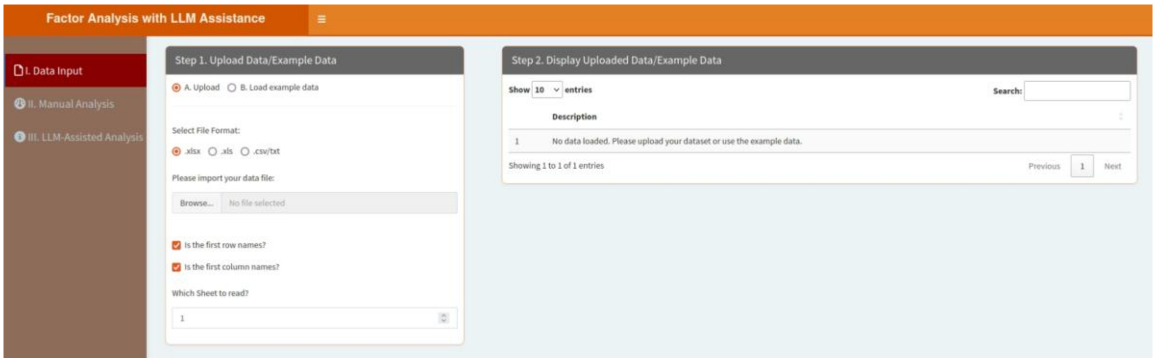


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



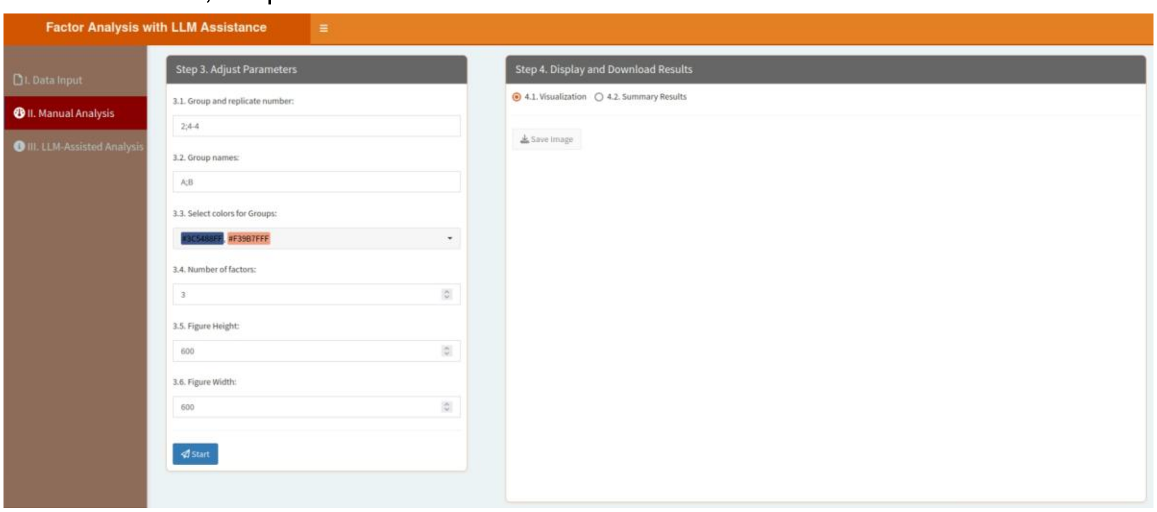
5.8. Factor Analysis

Factor analysis reduces data dimensionality by identifying latent variables. Users can adjust the number of factors and sample information via the classic framework, and local LLMs help refine the analysis and interpret the factors. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

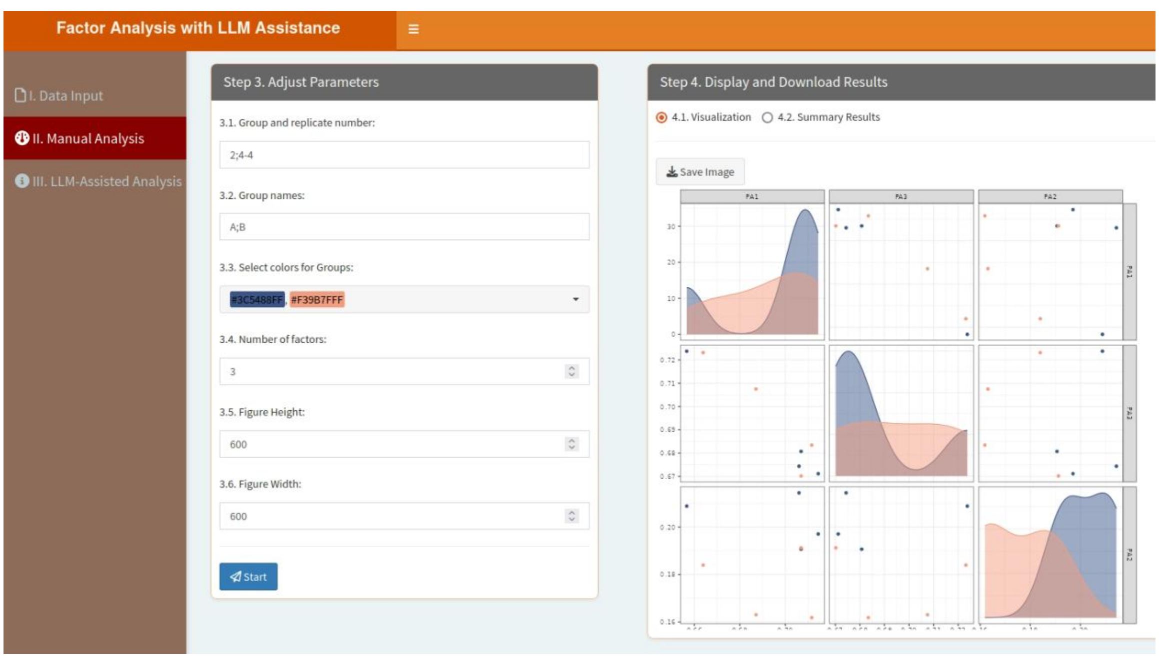
The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



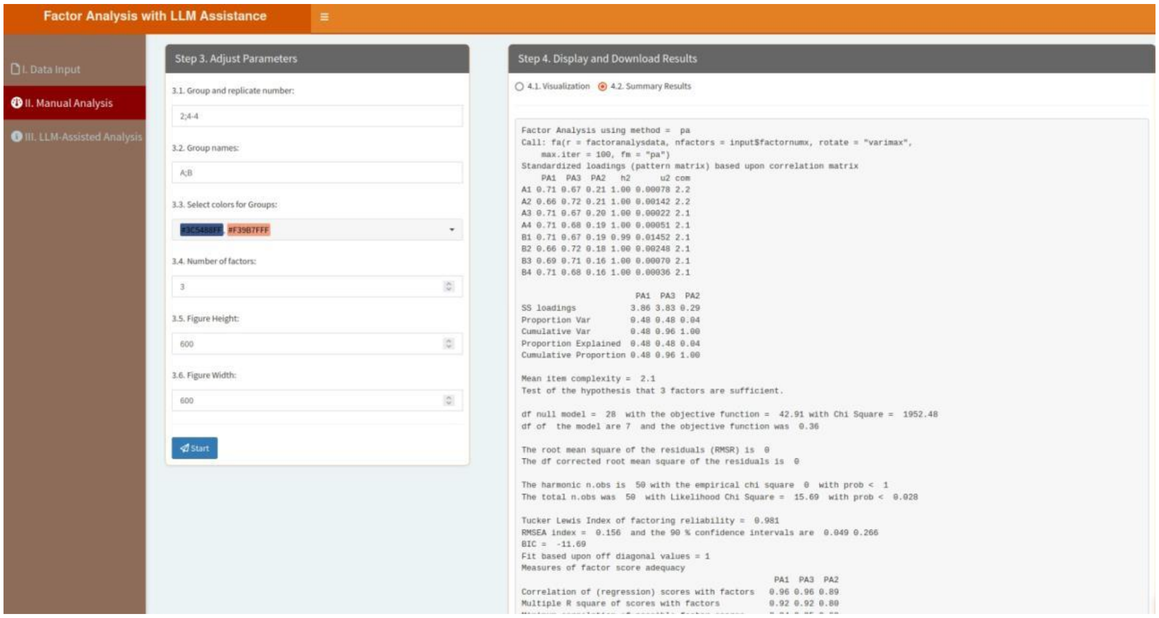
3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.
3.3. Select colors for groups: This parameter lets you choose specific colors for each group. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of groups, ensuring that the visual representation is both informative and aesthetically pleasing.
3.4. Number of factors: Number of factors to extract.
3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



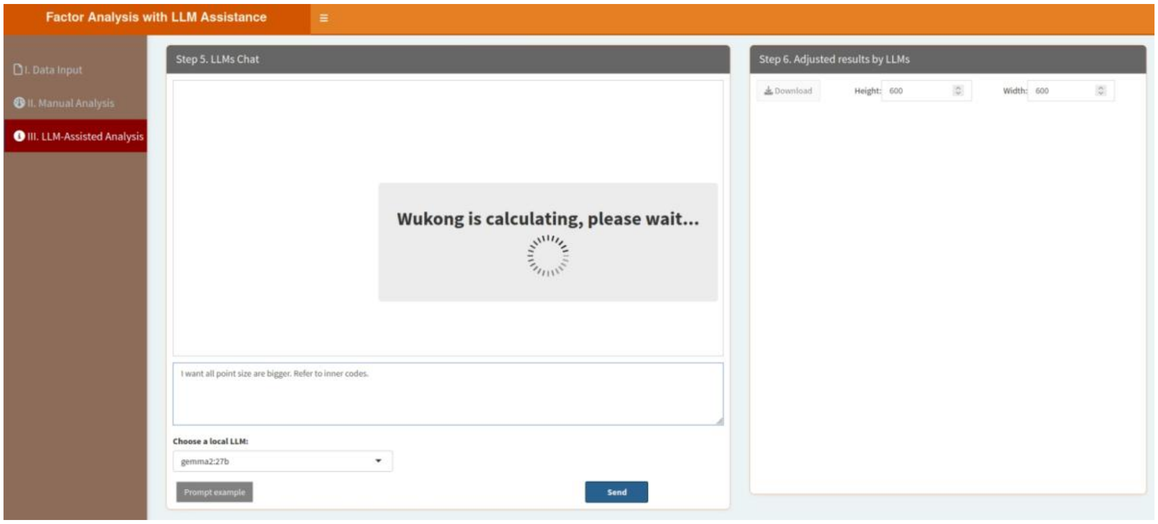
In the “4.2. Summary Results” part, users can click “Download” button to download the table, which contains the results summarized from the factor analysis model:



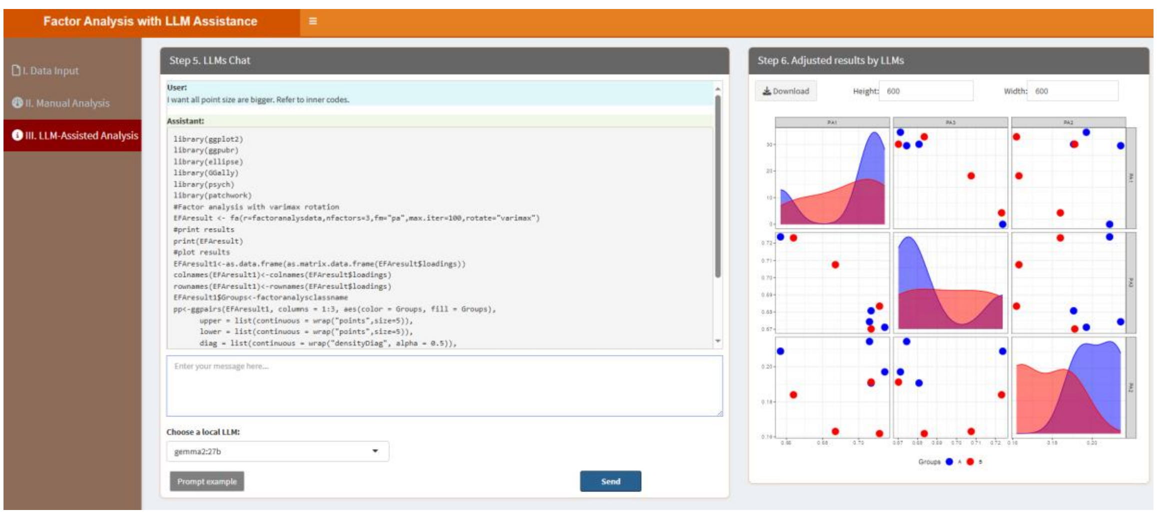
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I want all point size are bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I want all point size are bigger. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

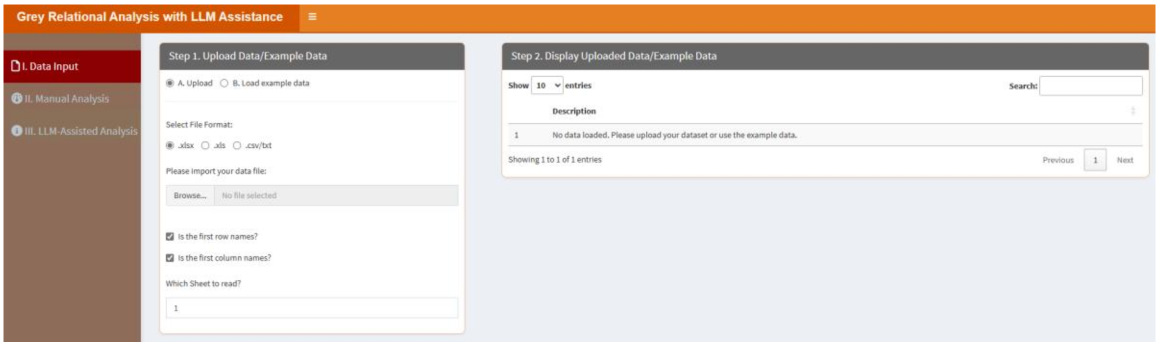


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



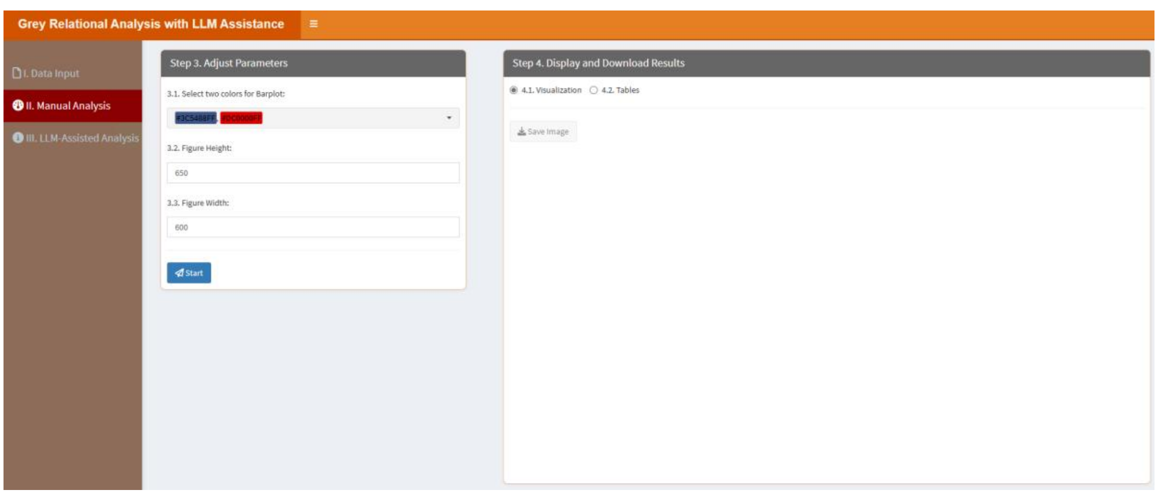
5.9. Grey Relational Analysis (GRA)

GRA evaluates relationships between variables in complex systems. Users can modify basic parameters to obtain the results directly through the classic framework, while local LLMs enhance result interpretation and analysis refinement. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



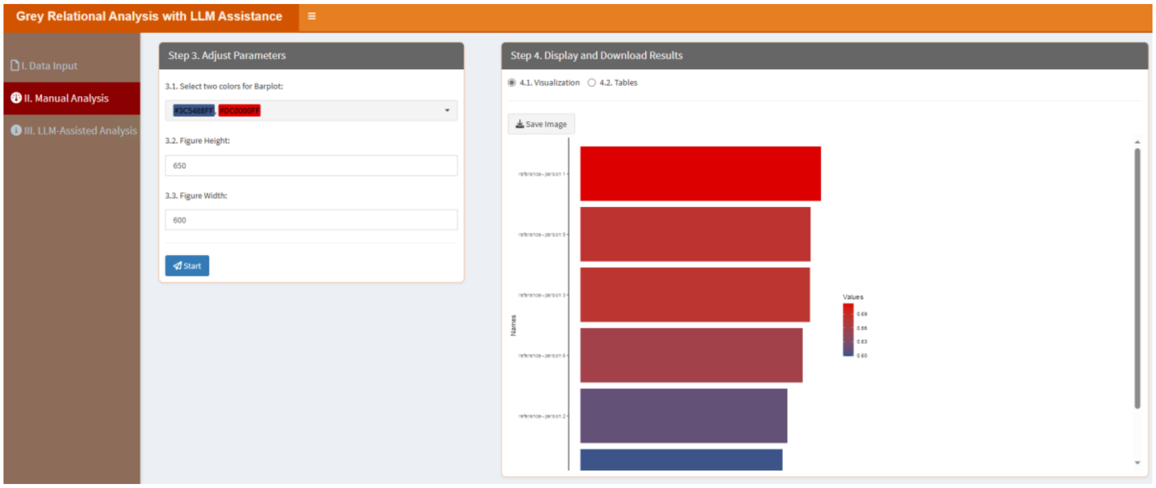
3.1. Select two colors for Barplot: This parameter lets you choose two colors for barplot, as the “4.1. Visualization” show the results using a barplot by default. The first color is for the lowest value and the second color is for the highest value.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

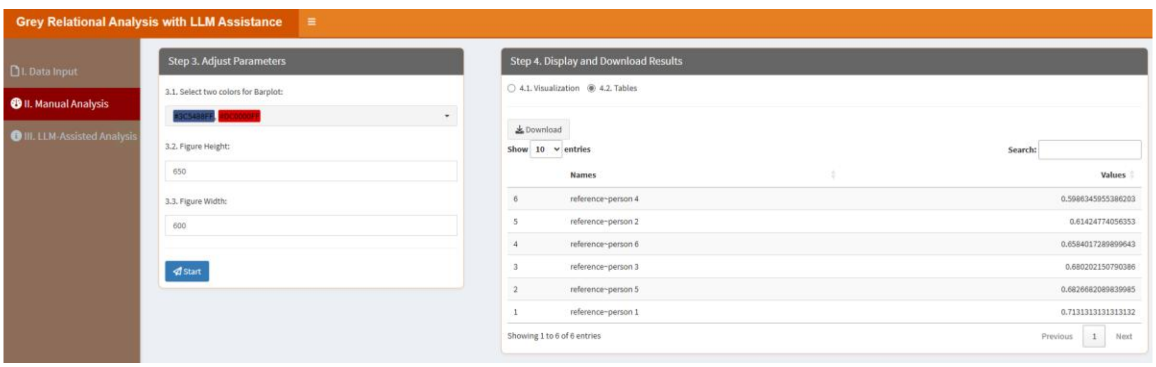
3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure

in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



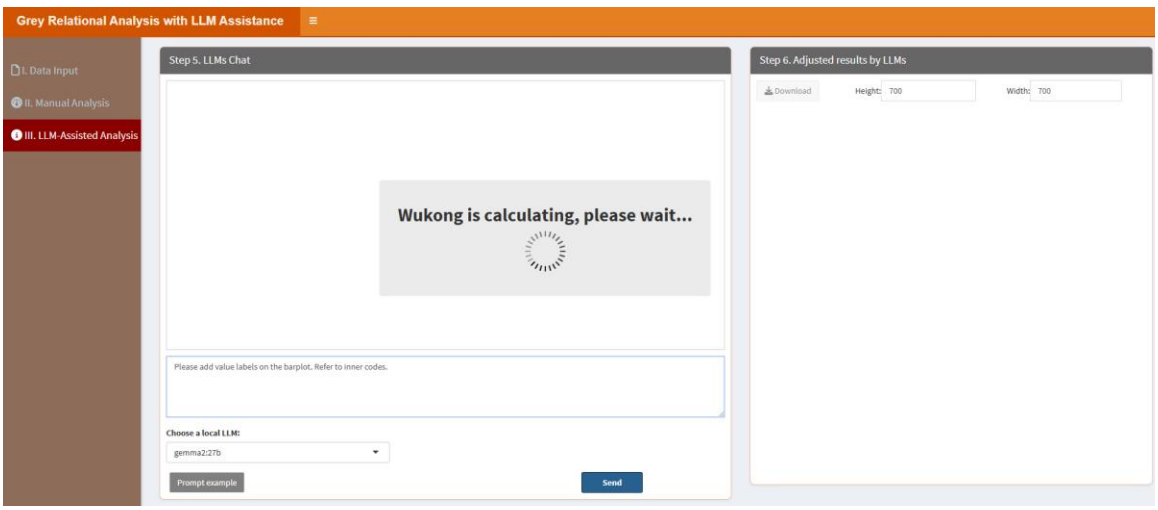
In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the results from the grey relational analysis:



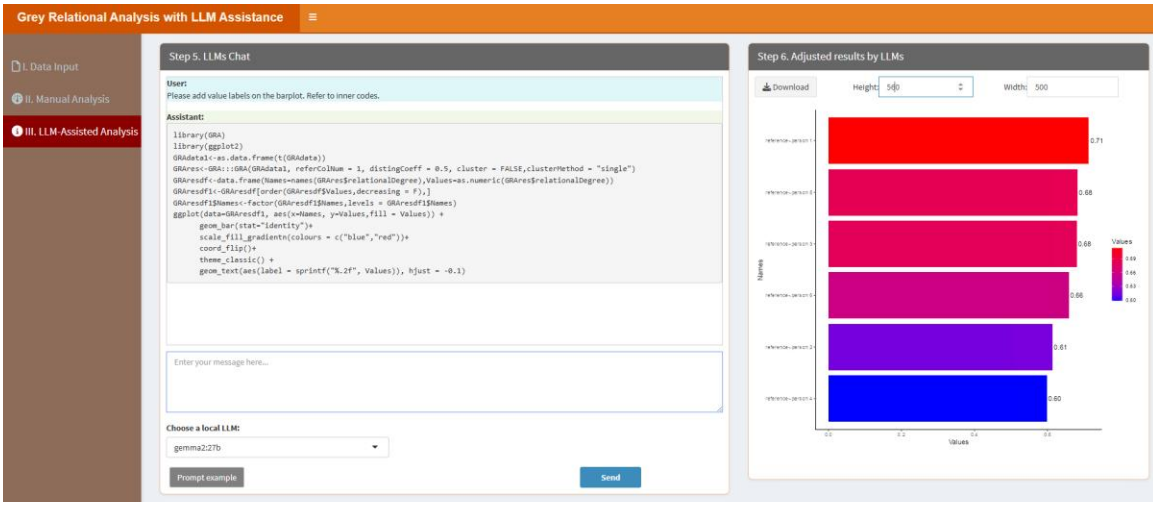
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please add value labels on the barplot. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please add value labels on the barplot. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

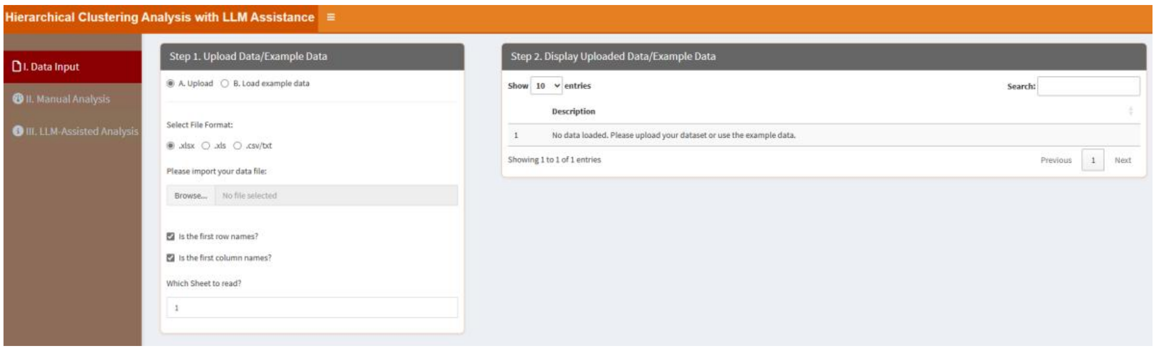


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.


Grey Relational Analysis with LLM Assistance
I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis
Step 5. LLMs Chat
User:
Please add value labels on the barplot. Refer to inner codes.
Assistant:
library(GRA)
library(ggplot2)
GRAdata1<-as.data.frame(t(GRAdata))
GRAres<-GRA::GRA(GRAdata1, referColNum = 1, distingCoeff = 0.5, cluster = FALSE,clusterMethod = "single")
GRAresdf<-data.frame(Names=names(GRAres$relationalDegree),Values=as.numeric(GRAres$relationalDegree))
GRAresdf1<-GRAresdf[order(GRAresdf$Values,decreasing = F),]
GRAresdf1$Names<-factor(GRAresdf1$Names,levels = GRAresdf1$Names)
ggplot(data=GRAresdf1, aes(x=Names, y=Values,fill = Values)) +
geom_bar(stat="identity")+
scale_fill_gradientn(colours = c("blue","red"))+
coord_flip()+
theme_classic() +
geom_text(aes(label = sprintf("%.2f", Values)), hjust = -0.1)
Enter your message here...
Choose a local LLM:
gemma2:27b
Prompt example
Send
Step 6. Adjusted results by LLMs
Download
Height:
Width: 500
0.71
0.68
0.68
0.66
0.61
0.60
Values
Names

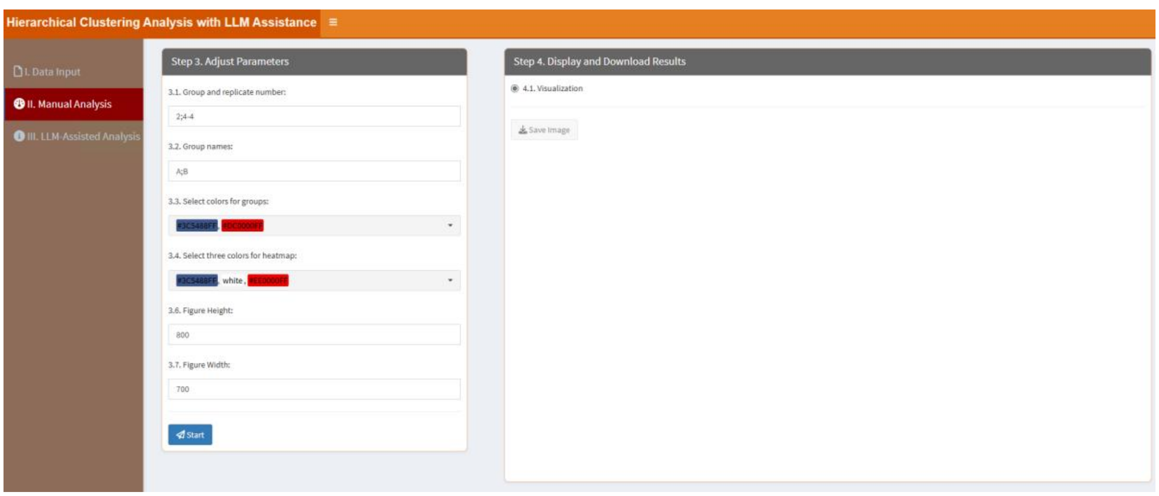
5.10. Hierarchical Cluster Analysis (HCA)

HCA organizes data into a hierarchy of clusters. Users can select linkage methods and distance metrics via the classic framework, and local LLMs assist in refining cluster interpretation. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control"

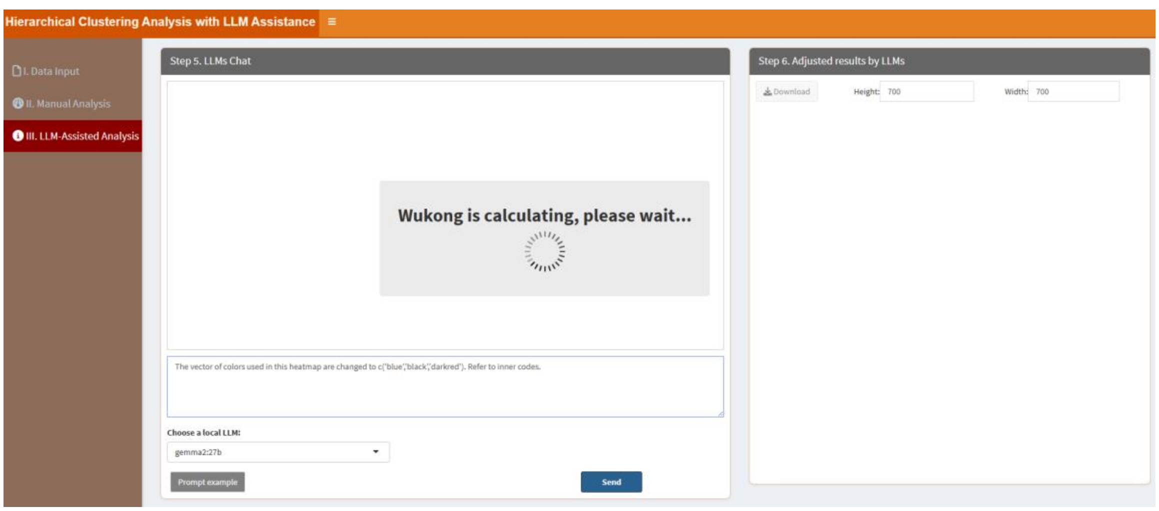
and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.
3.3. Select colors for groups: This parameter lets you choose specific colors for each group. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of groups, ensuring that the visual representation is both informative and aesthetically pleasing.
3.4. Select three colors for heatmap: Colors for the heatmap plot. By default, there are three color names (#3C5488FF, white, #EE0000FF) here for the lowest (#3C5488FF), middle (white), highest (#EE0000FF) values repectively.
3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

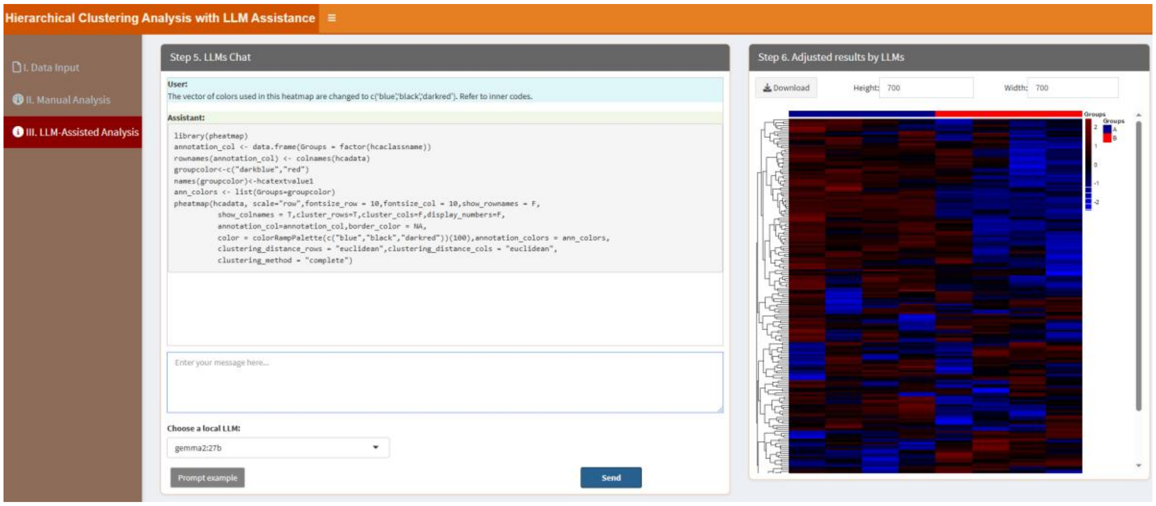
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “The vector of colors used in this heatmap are changed to c('blue','black','darkred'). Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “The vector of colors used in this heatmap are changed to c('blue','black','darkred'). =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



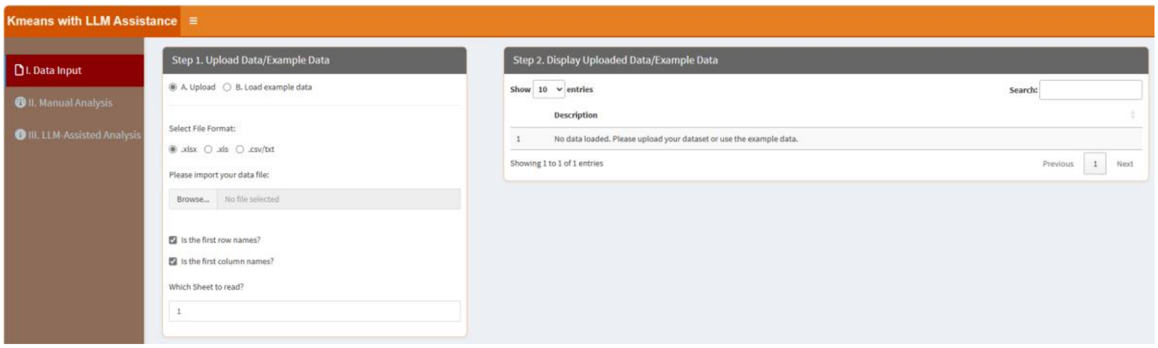
The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



Hierarchical Clustering Analysis with LLM Assistance
I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis
Step 5. LLMs Chat
User:
The vector of colors used in this heatmap are changed to c('blue','black','darkred'). Refer to inner codes.
Assistant:
library(pheatmap)
annotation_col <- data.frame(Groups = factor(hcaclassname))
rownames(annotation_col) <- colnames(hcadata)
groupcolor<-c("darkblue","red")
names(groupcolor)<-hcatextvalue1
ann_colors <- list(Groups=groupcolor)
pheatmap(hcadata, scale="row",fontsize_row = 10,fontsize_col = 10,show_rownames = F,
show_colnames = T,cluster_rows=T,cluster_cols=F,display_numbers=F,
annotation_col=annotation_col,border_color = NA,
color = colorRampPalette(c("blue","black","darkred"))(100),annotation_colors = ann_colors,
clustering_distance_rows = "euclidean",clustering_distance_cols = "euclidean",
clustering_method = "complete")
Enter your message here...
Choose a local LLM:
gemma2:27b
Prompt example
Send
Step 6. Adjusted results by LLMs
Download
Height: 700
Width: 700
Groups

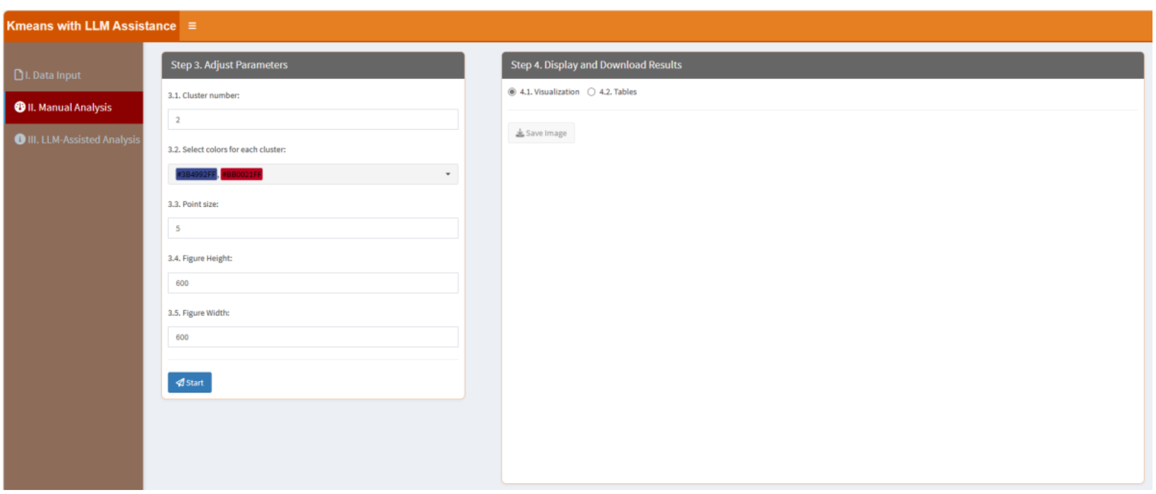
5.11. K-means Clustering

K-means clustering partitions data into distinct groups. Users can adjust the number of clusters and initialization methods using the classic framework, while local LLMs help refine cluster analysis and provide insights. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Cluster number: Maximum cluster number to evaluate.

3.2. Select colors for each cluster: This parameter lets you choose specific colors for each cluster. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of clusters, ensuring that the visual representation is both informative and aesthetically pleasing.

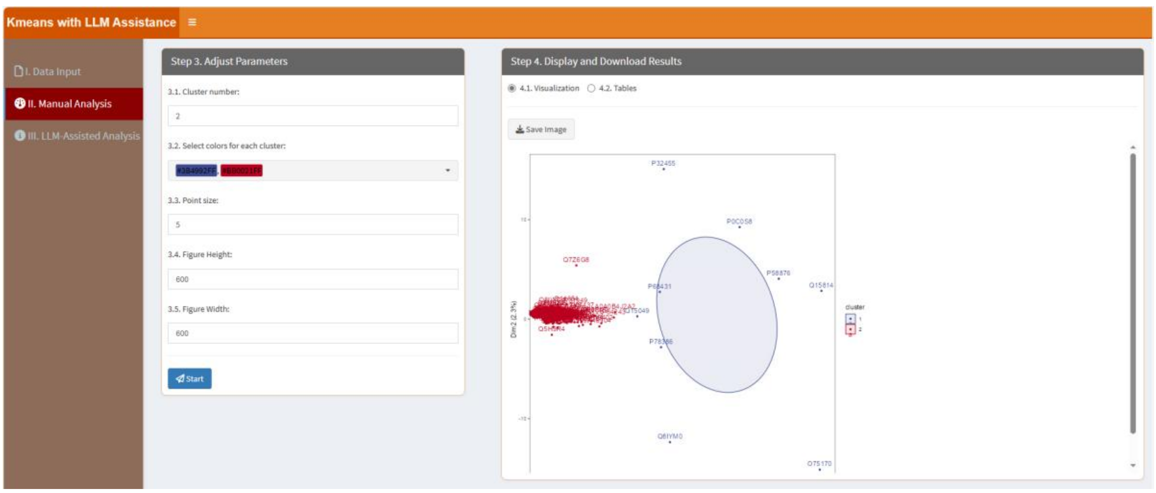
3.3. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

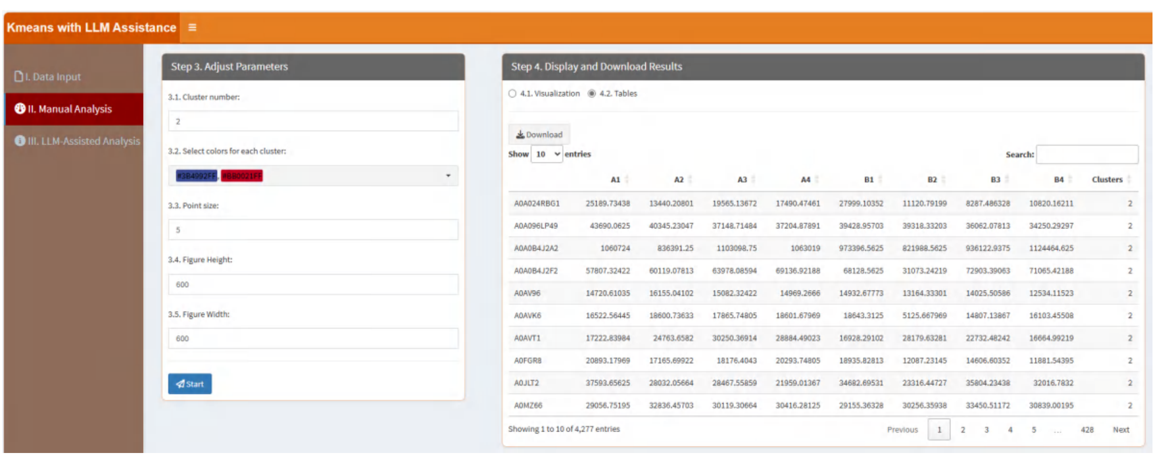
3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click "Start" button, the results will be shown as below:

In the "4.1. Visualization" part, users can click "Save image" button to download the figure:



In the "4.2. Tables" part, users can click "Download" button to download the table, which contains the results:

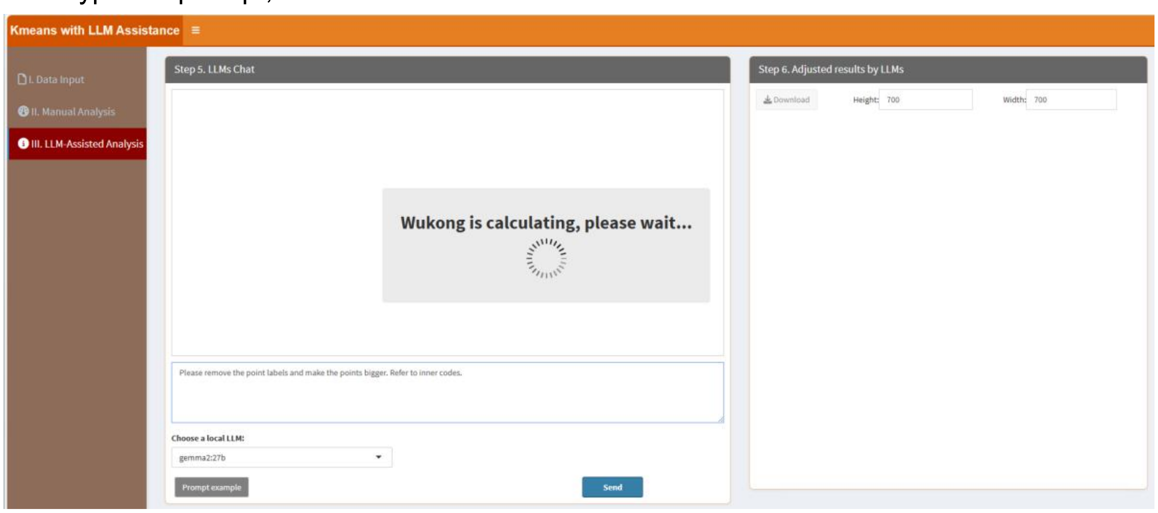


"III. LLM-Assisted Analysis" means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..."

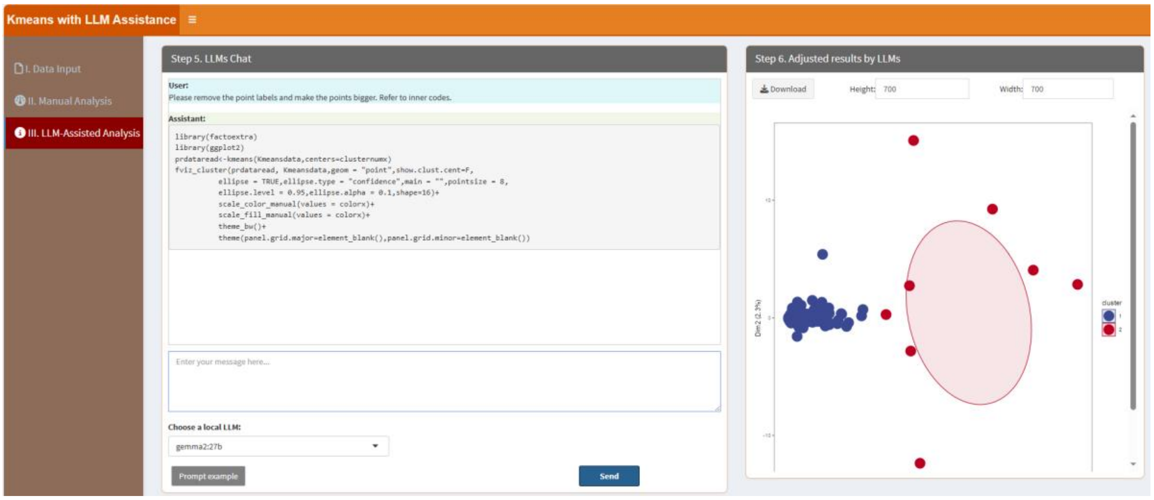
enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt "Please remove the point labels and make the points bigger. Refer to inner codes." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Please remove the point labels and make the points bigger. =>", it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click "Send" button. It will learn like below:

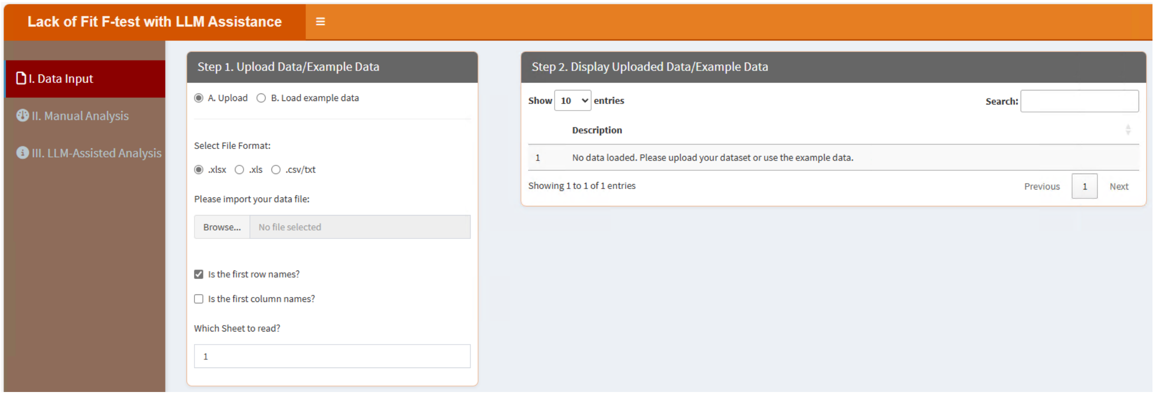


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.


Kmeans with LLM Assistance
I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis
Step 5. LLMs Chat
User:
Please remove the point labels and make the points bigger. Refer to inner codes.
Assistant:
library(factoextra)
library(ggplot2)
prdataread<-kmeans(Kmeansdata,centers=clusternumx)
fviz_cluster(prdataread, Kmeansdata,geom = "point",show.clust.cent=F,
ellipse = TRUE,ellipse.type = "confidence",main = "",pointsize = 8,
ellipse.level = 0.95,ellipse.alpha = 0.1,shape=16)+
scale_color_manual(values = colorx)+
scale_fill_manual(values = colorx)+
theme_bw()+
theme(panel.grid.major=element_blank(),panel.grid.minor=element_blank())
Enter your message here...
Choose a local LLM:
gemma2:27b
Prompt example
Send
Step 6. Adjusted results by LLMs
Download
Height: 700
Width: 700
cluster
1
2

5.12. Lack of Fit F-test

This test evaluates the adequacy of regression models. Users can configure model parameters through the classic framework, while local LLMs assist in interpreting the results and refining the analysis. After clicking the “Start” button, it will be started in a new window like below:

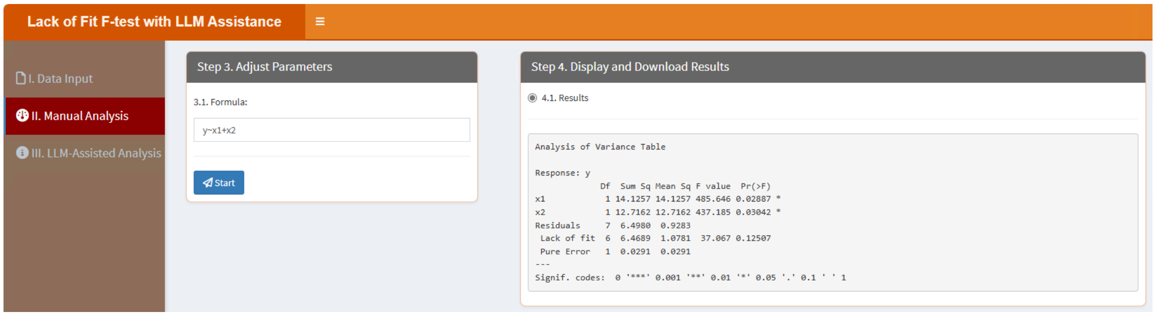


The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

3.1.Formula: A symbolic description of the model to be fitted. For example, in the example data, the dependent variable is y, the independent variables are x1 and x2, so the formula should be y~x1+x2.

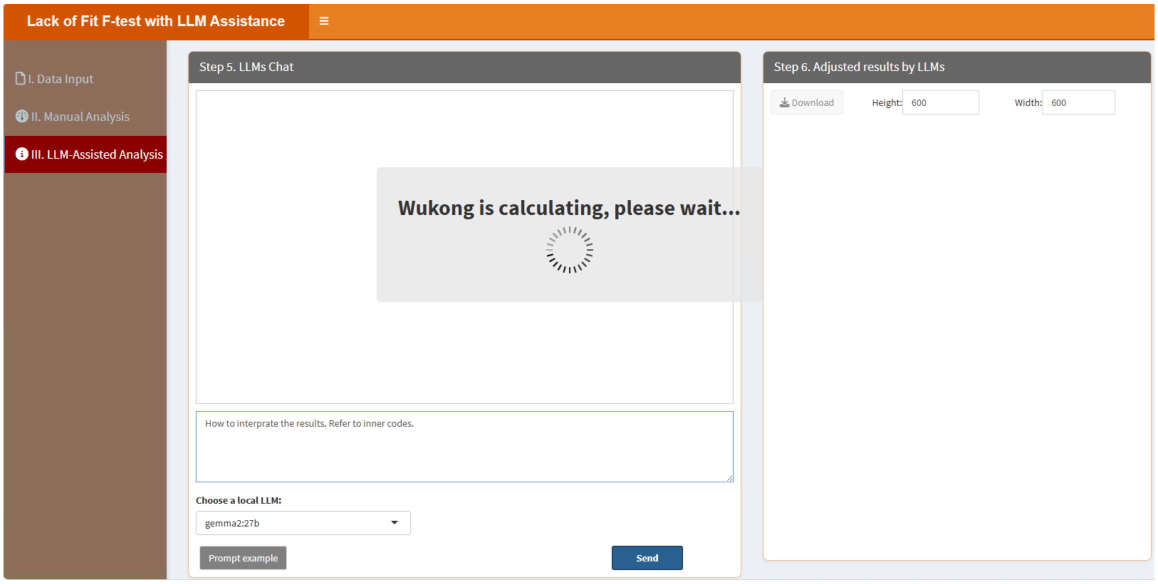
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



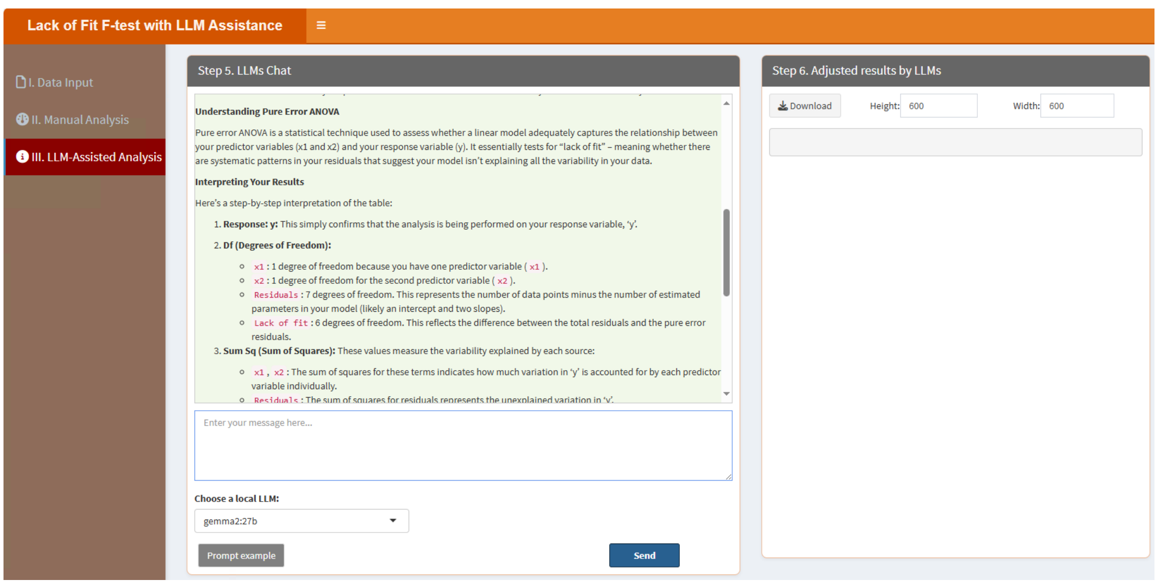
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “How to interprate the results. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “How to interprate the results. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

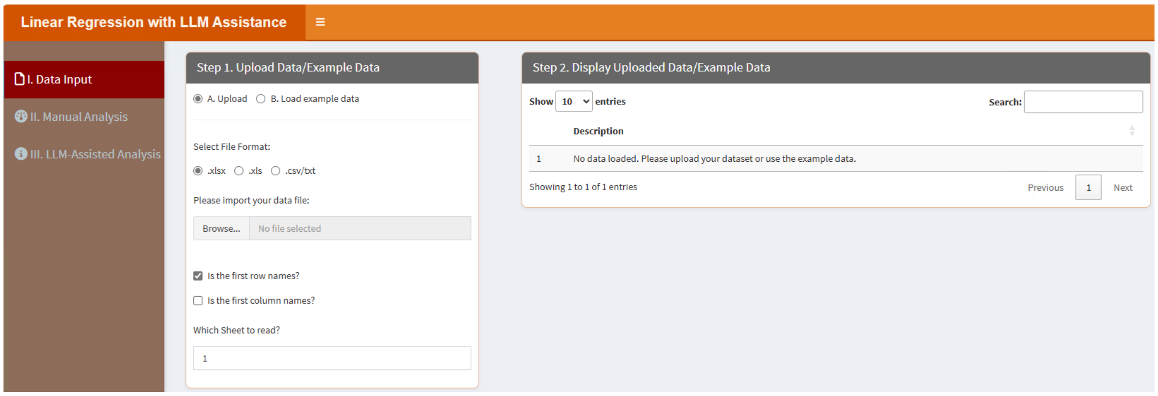


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



5.13. Linear Regression

Linear regression models relationships between variables. Users can set predictors and response variables via the classic framework, and local LLMs provide enhanced insights into model interpretation. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

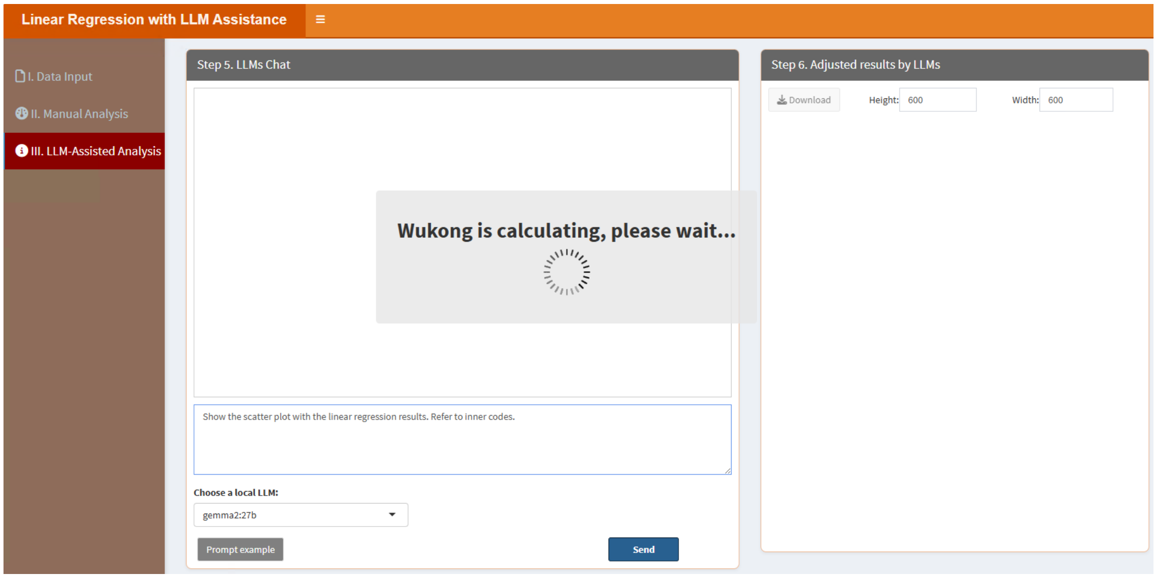
3.1.Formula: A symbolic description of the model to be fitted. For example, in the example data, the dependent variable is y, the independent variable is x, so the formula should be y~x.

After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Results” part, users can check the linear regression results:

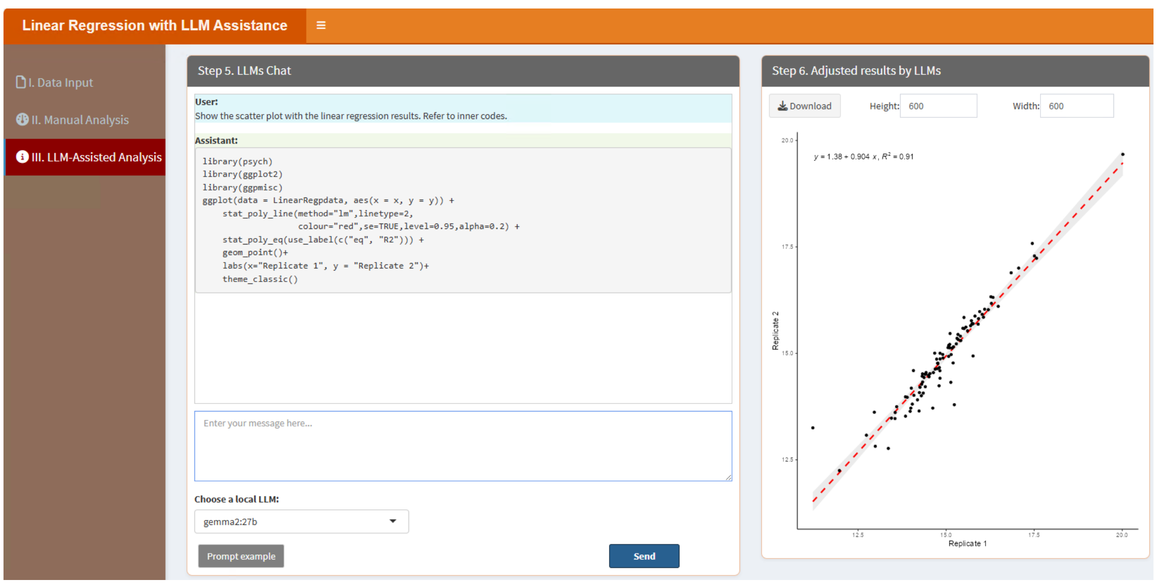
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Show the scatter plot with the linear regression results. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Show the scatter plot with the linear regression results. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

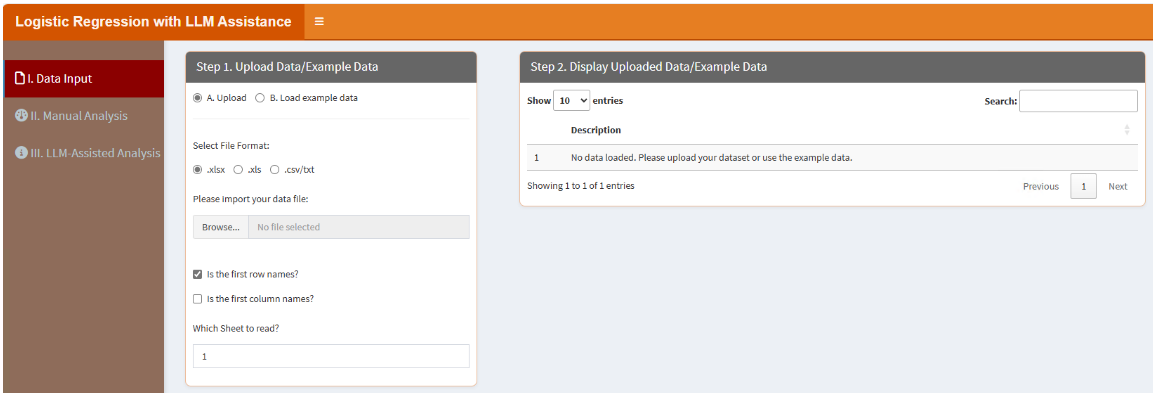


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



5.14. Logistic Regression

Logistic regression predicts binary outcomes. Users can configure model parameters through the classic framework, while local LLMs refine the analysis and improve interpretability. After clicking the “Start” button, it will be started in a new window like below:



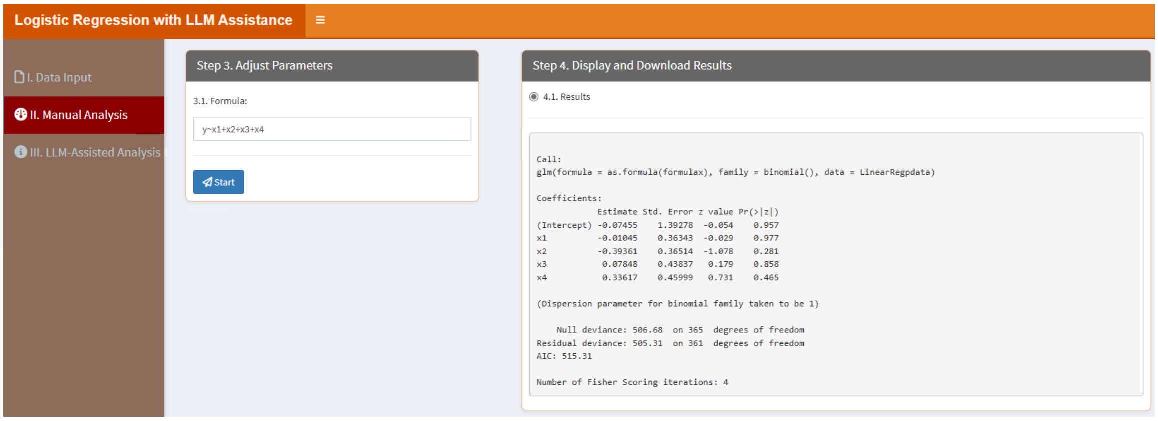
The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

3.1.Formula: A symbolic description of the model to be fitted. For example, in the example data, the dependent variable is y, the independent variables are x1, x2, x3 and x4, so the formula should be y~x1+x2+x3+x4.

After setting parameters, click “Start” button, the results will be shown as below:

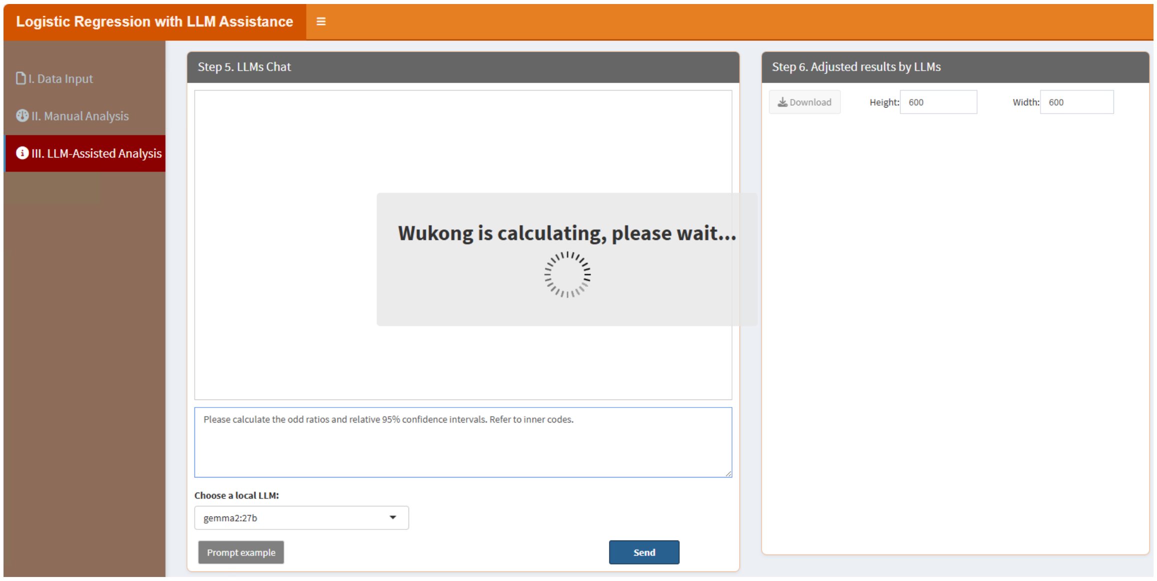
In the “4.1. Results” part, users can check the logistic regression results:



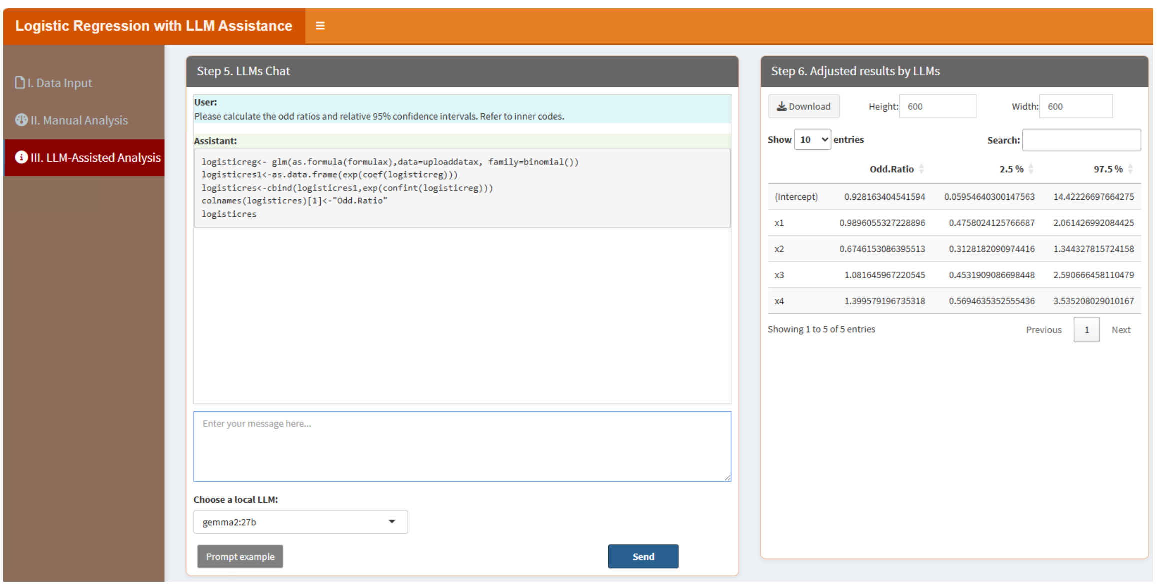
"III. LLM-Assisted Analysis" means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt "Please calculate the odd ratios and relative 95% confidence intervals. Refer to inner codes." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Please calculate the odd ratios and relative 95% confidence intervals. =>", it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click "Send" button. It will learn like below:

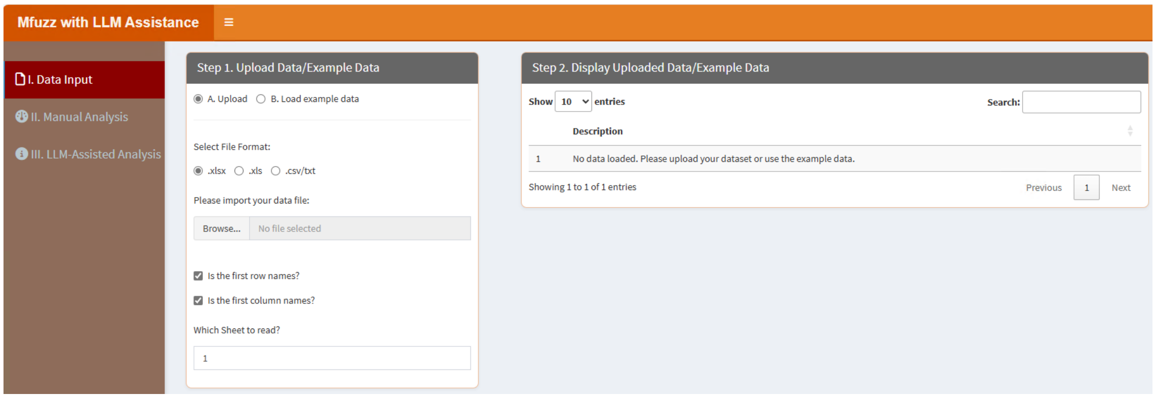


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



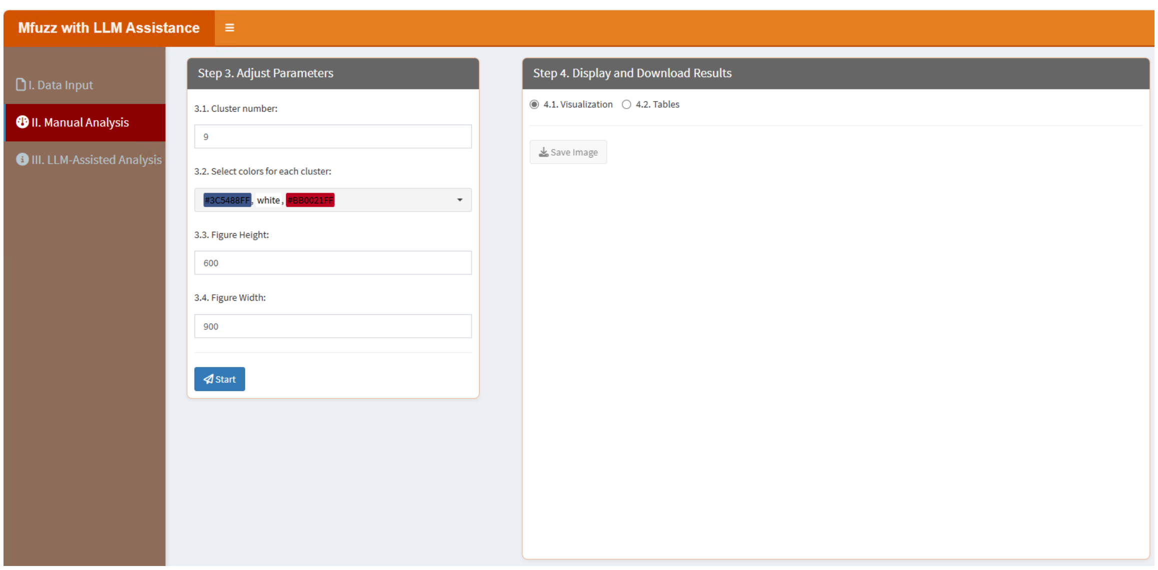
5.15. Mfuzz

Mfuzz is used for clustering time-series data. Users can adjust fuzzification parameters and cluster numbers through the classic framework, while local LLMs help refine the analysis and interpret clusters. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

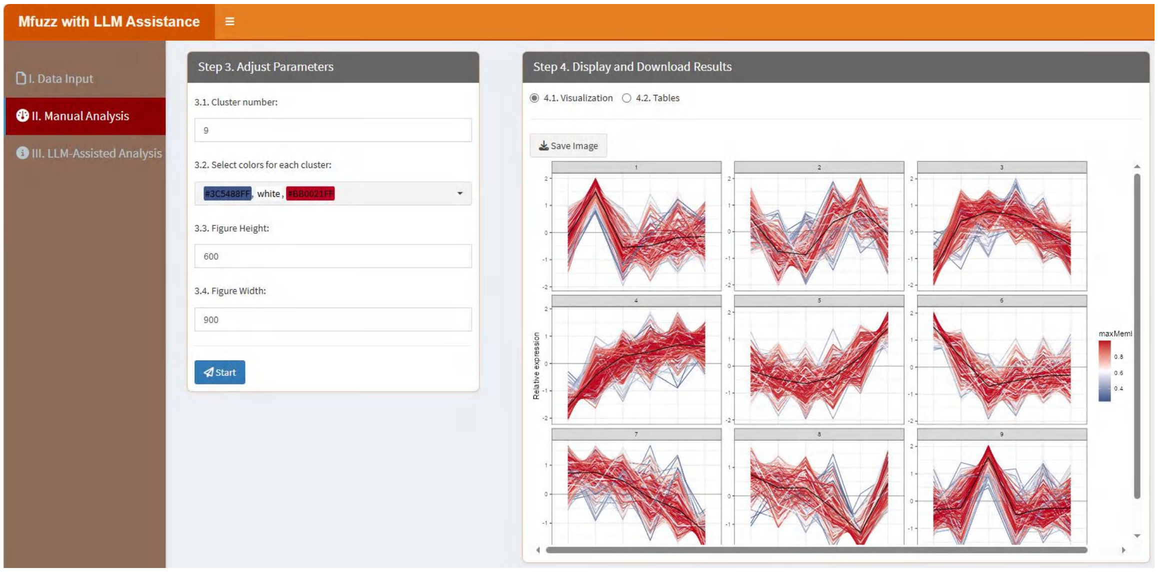


3.1. Cluster number: maximum cluster number to evaluate.

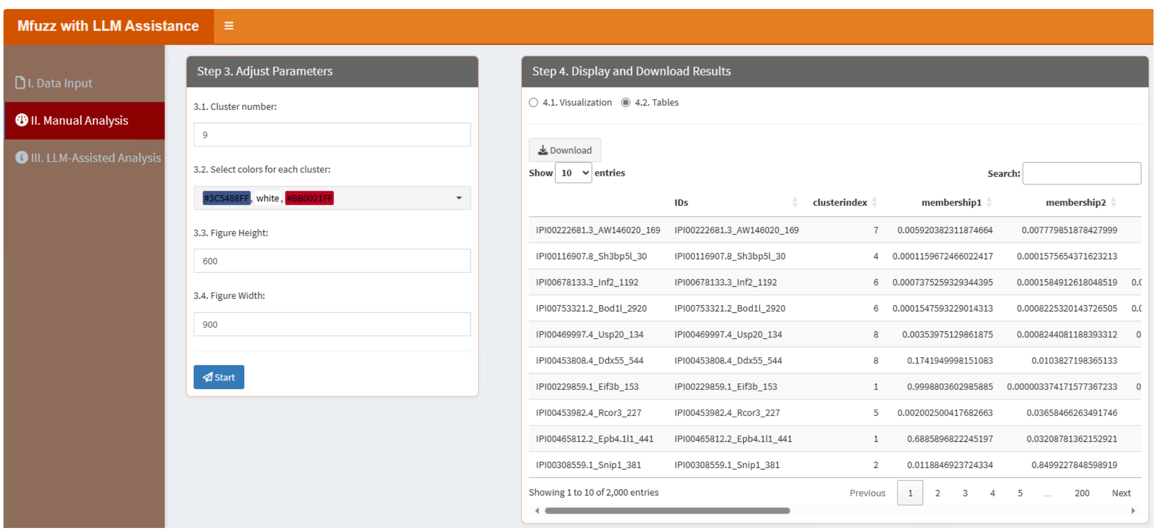
3.2. Select colors for each cluster: This parameter lets you choose specific colors for each cluster. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of clusters, ensuring that the visual representation is both informative and aesthetically pleasing.

3.3. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.4. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



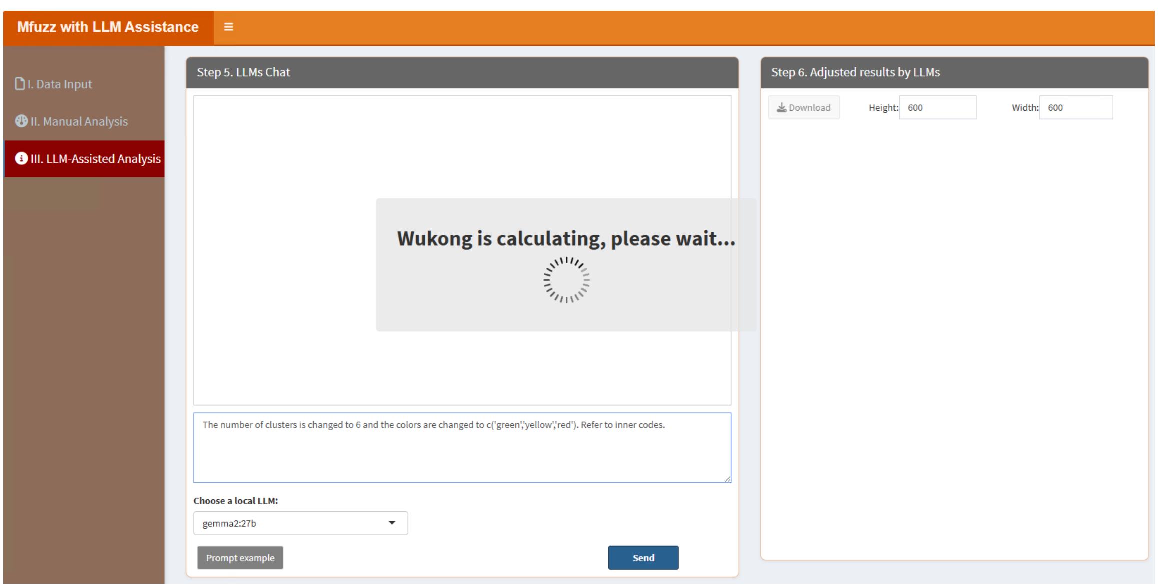
In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the results from the grey relational analysis:



“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

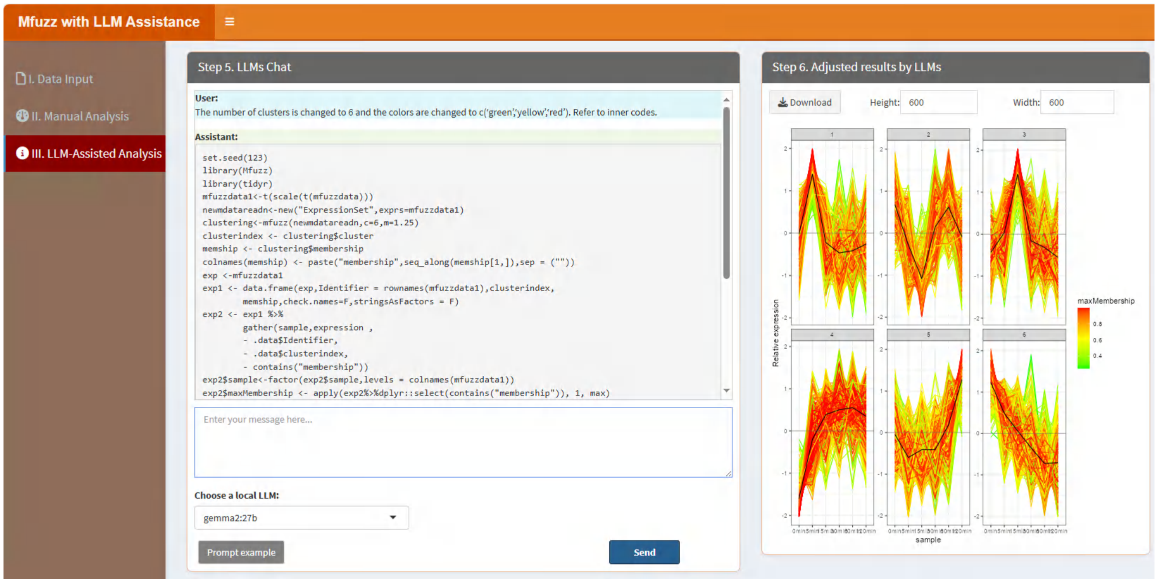
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “The number of clusters is changed to 6 and the colors are changed to c('green','yellow','red'). Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “The number of clusters is changed to 6 and the colors are changed to c('green','yellow','red'). =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



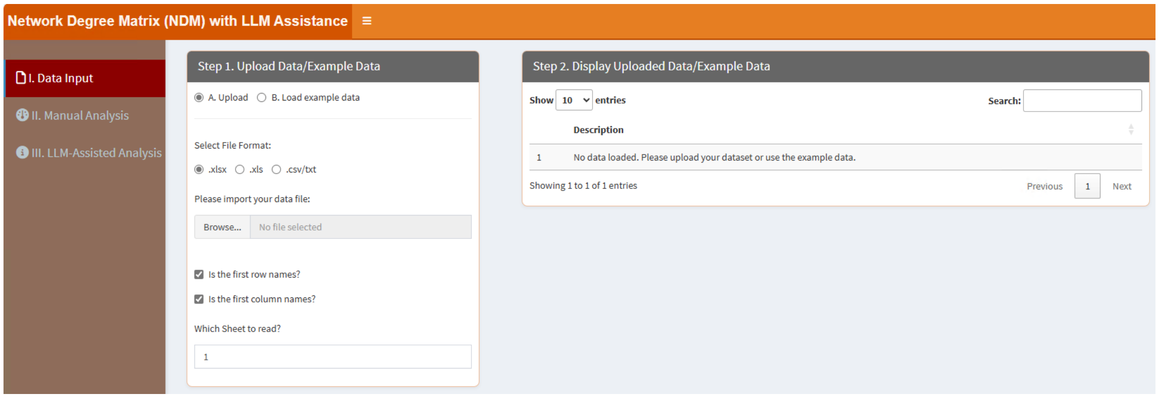
The results are shown as below, then users can download the results by clicking the

“Download” button in the “Step 6. Adjust results by LLMs” section.



5.16. Network Degree Matrix (NDM)

NDM analyzes connectivity in biological networks. Users can customize network parameters using the classic framework, and local LLMs provide deeper insights into the results. After clicking the "Start" button, it will be started in a new window like below:

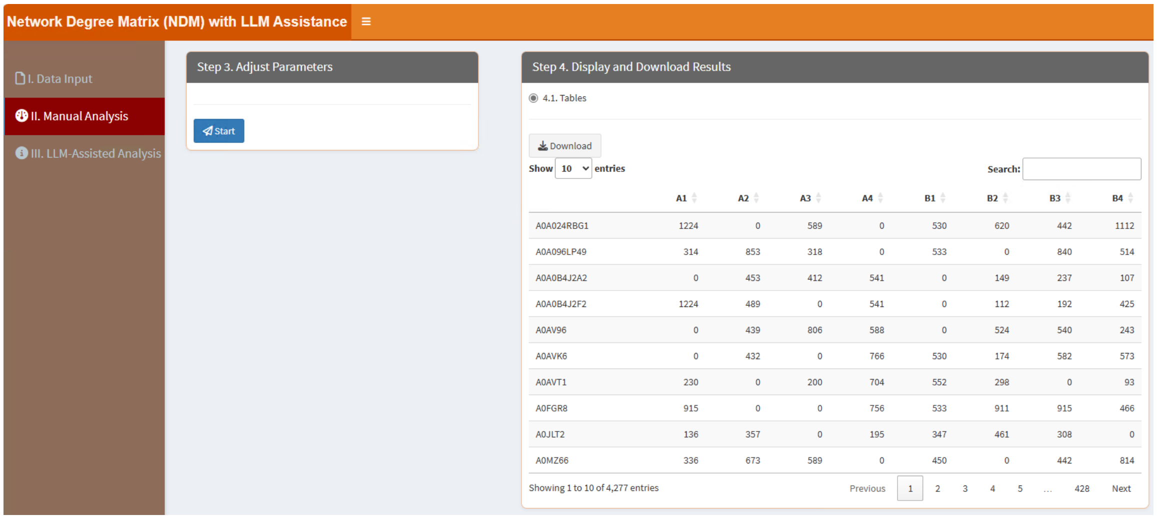


The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.

After setting parameters, click "Start" button, the results will be shown as below:

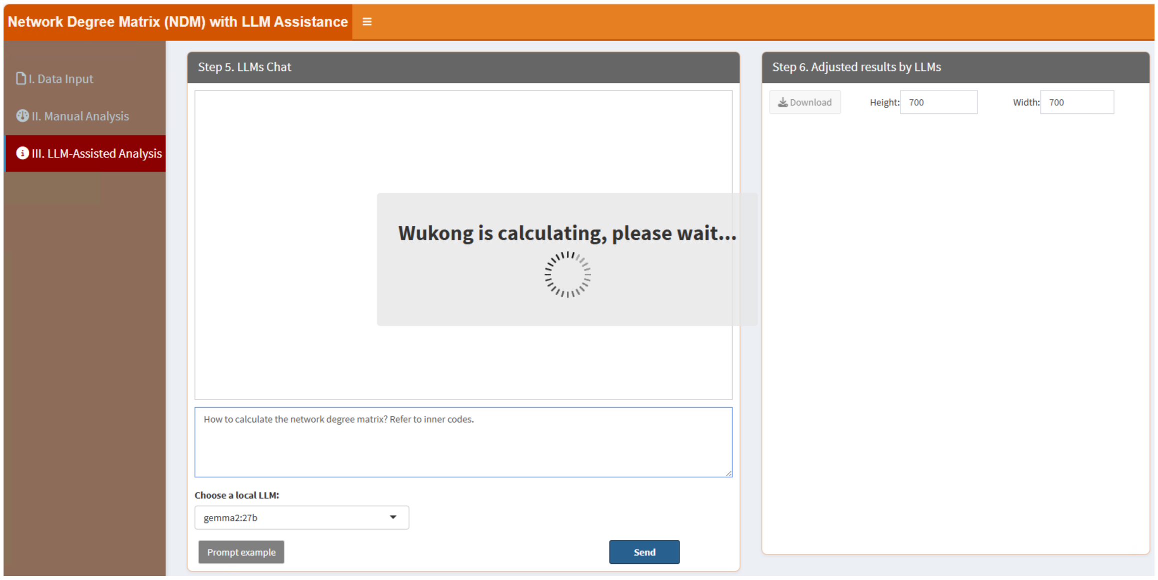
In the "4.1. Tables" part, users can click "Download" button to download the table, which contains the results from this module:



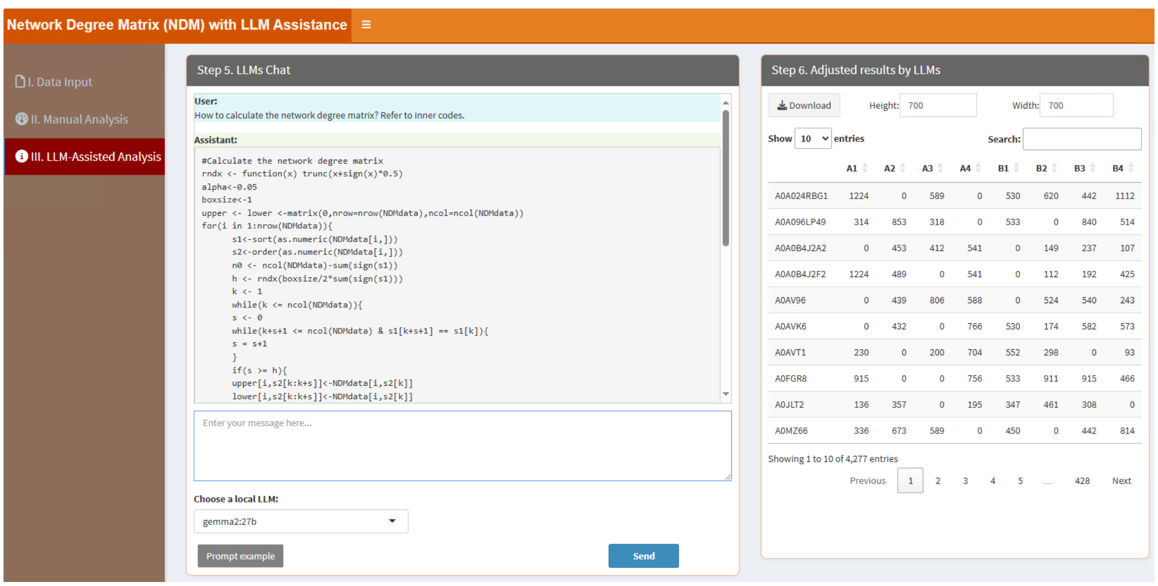
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “How to calculate the network degree matrix?Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “How to calculate the network degree matrix? =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

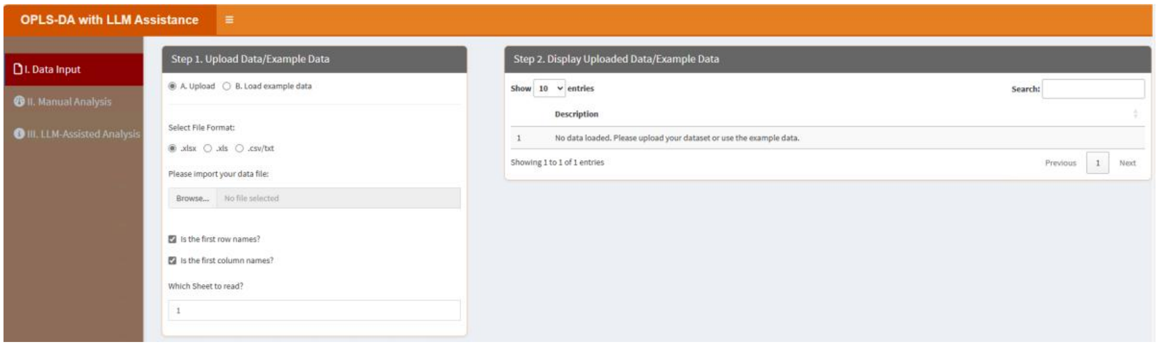


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



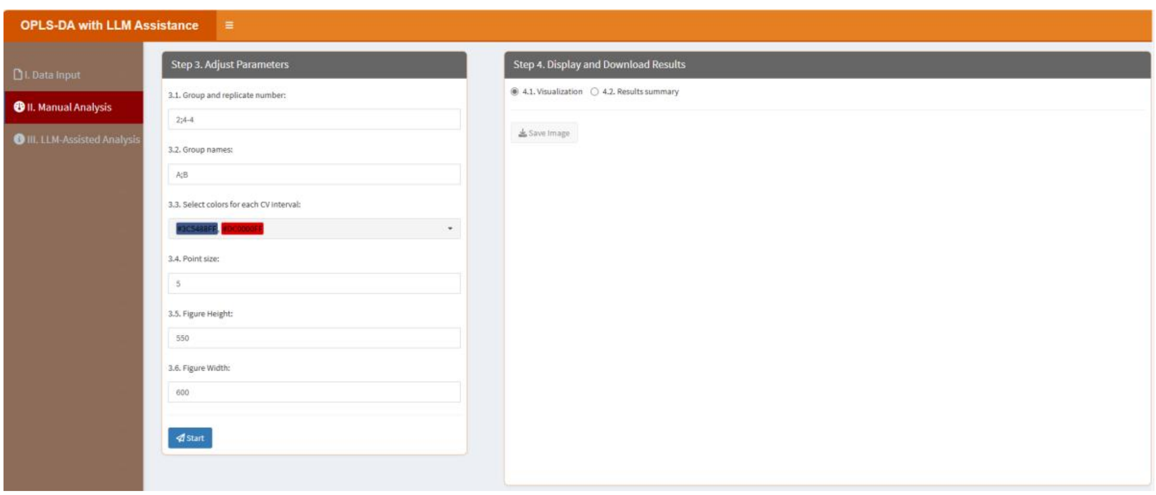
5.17. OPLS-DA (Orthogonal Partial Least Squares Discriminant Analysis)

OPLS-DA is used for classification and biomarker discovery. Users can configure model parameters via the classic framework, while local LLMs enhance the interpretation of the results. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control"

and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

*3.3. Select colors for each CV interva*l: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

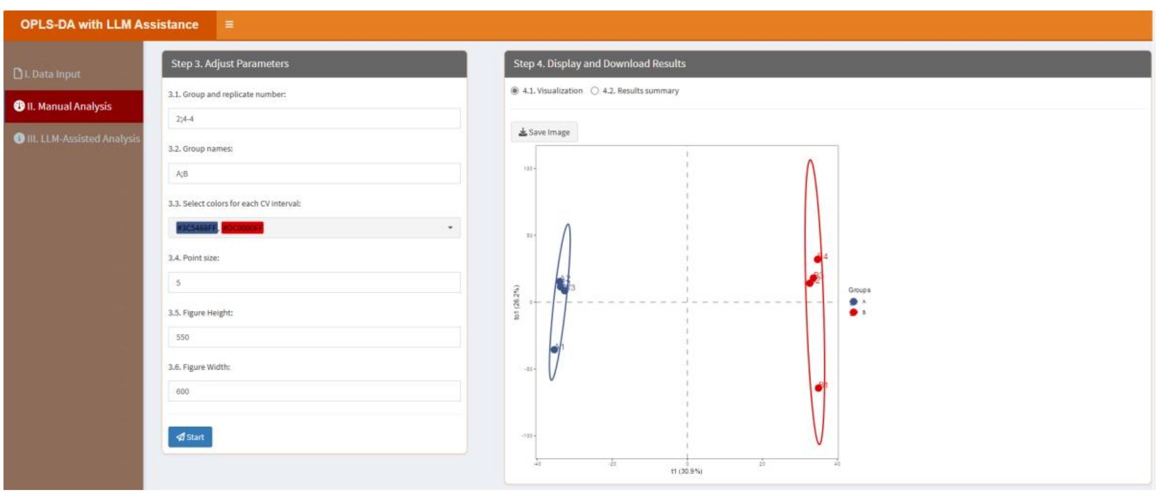
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

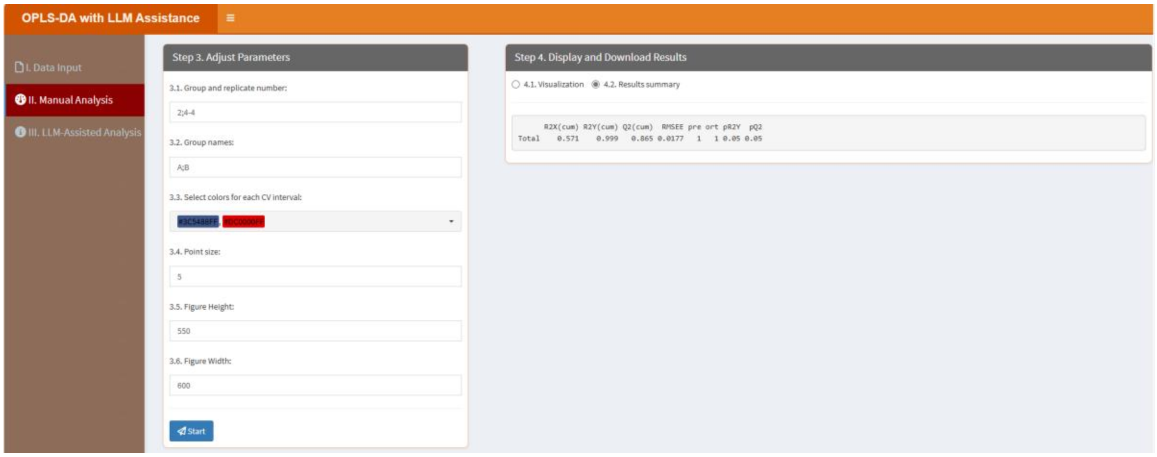
3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



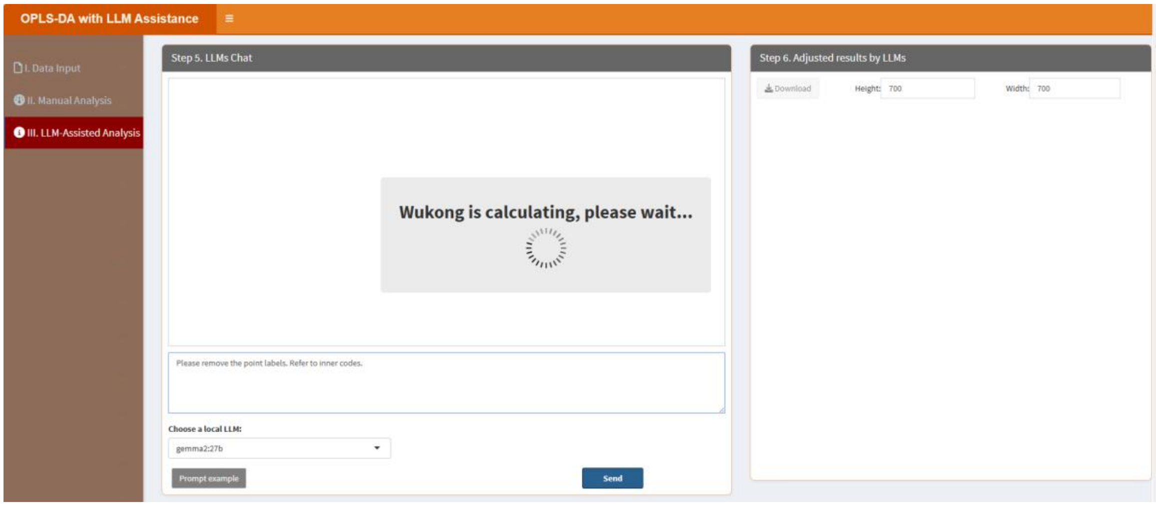
In the “4.2. Results summary” part, users can check the model summary results:



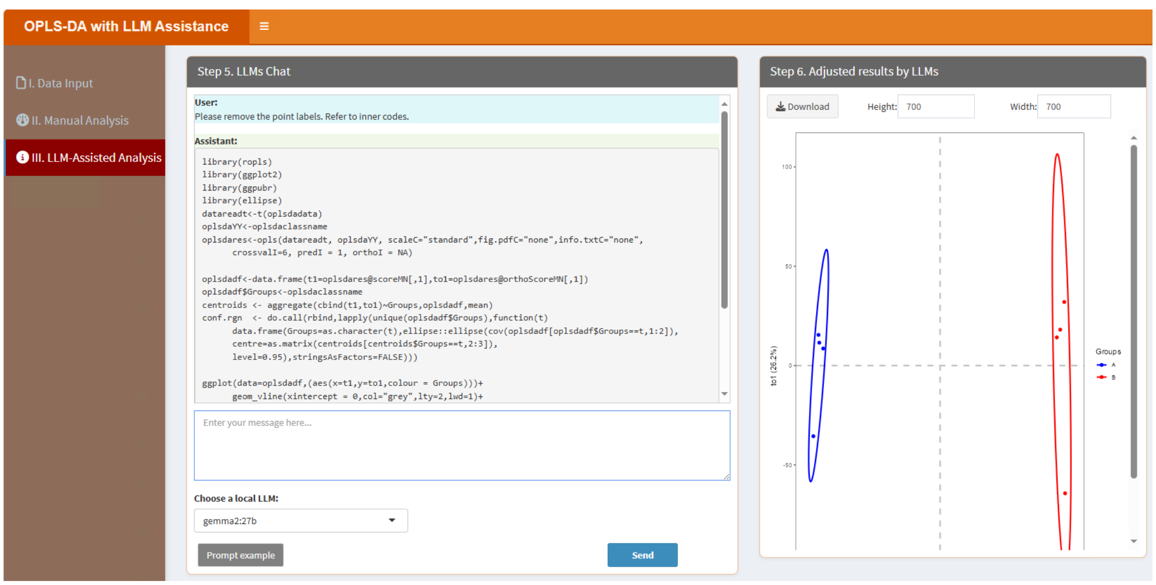
"III. LLM-Assisted Analysis" means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt "Please remove the point labels. Refer to inner codes." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Please remove the point labels. =>", it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click "Send" button. It will learn like below:

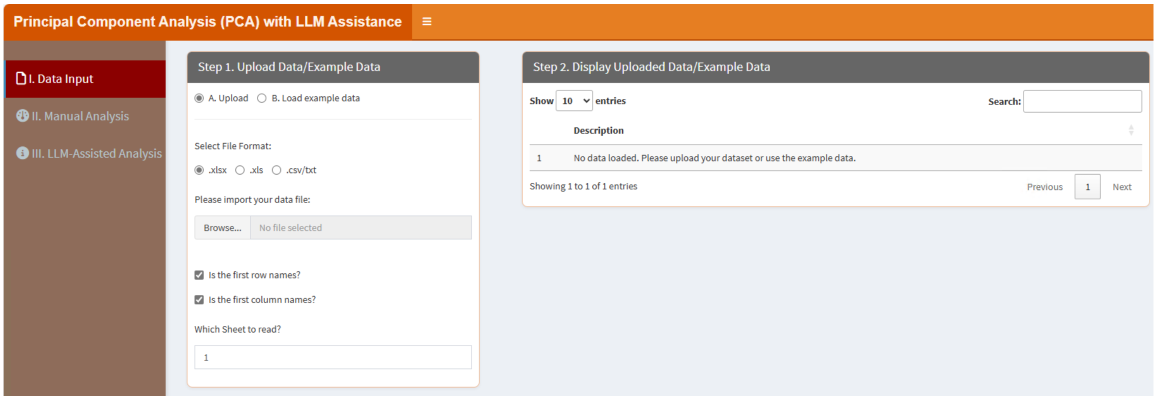


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



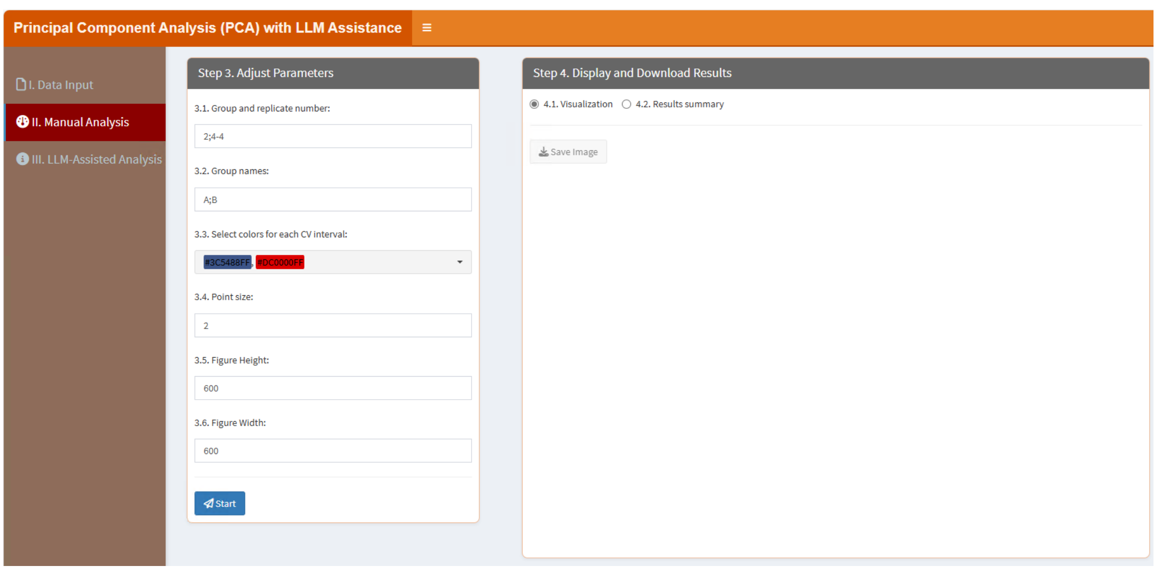
5.18. Principal Component Analysis (PCA)

PCA is an unsupervised dimensionality reduction technique. Users can adjust the number of components using the classic framework, and local LLMs assist in refining the analysis and interpreting the results. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Select colors for each CV interval: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

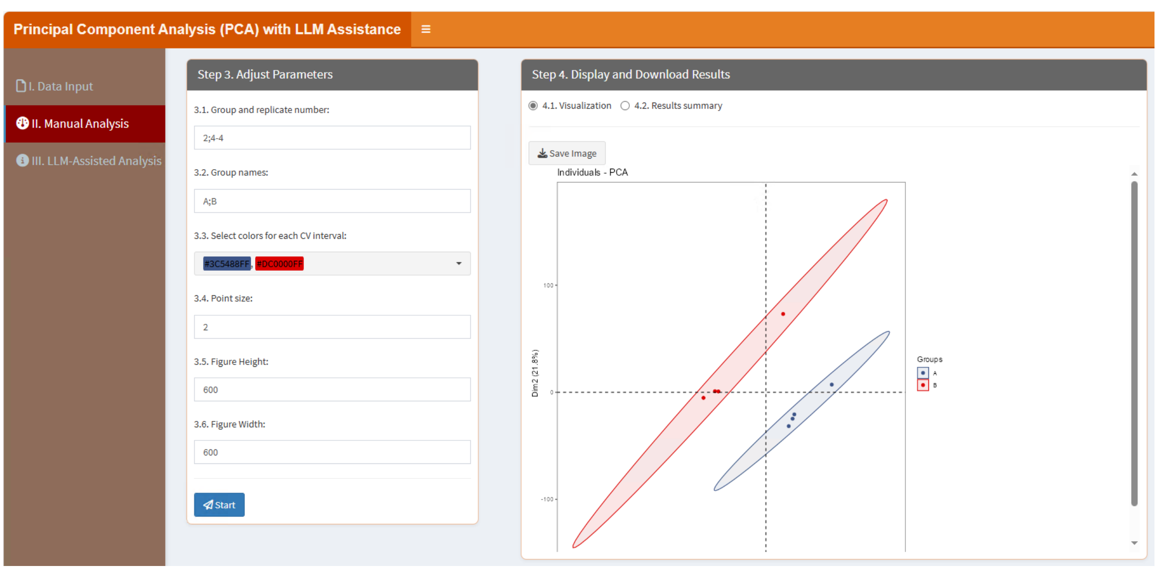
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

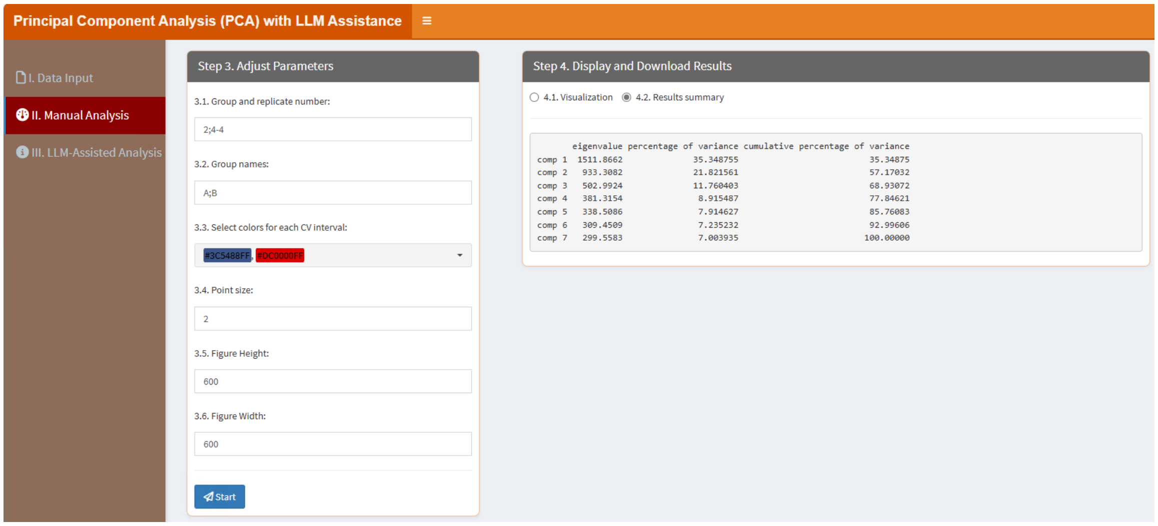
3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



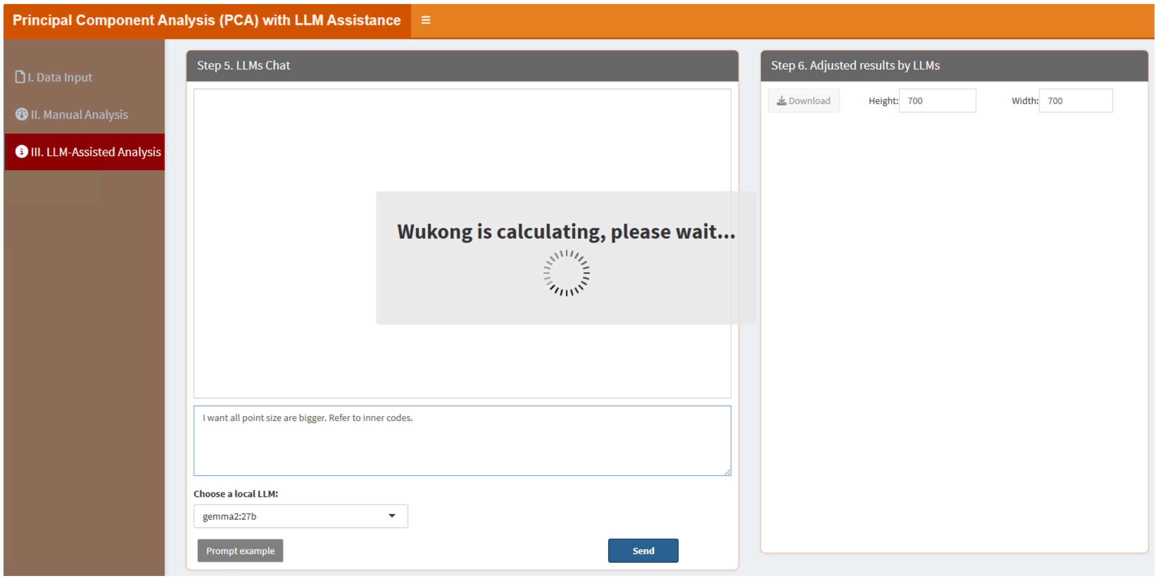
In the “4.2. Results summary” part, users can check the model summary results:



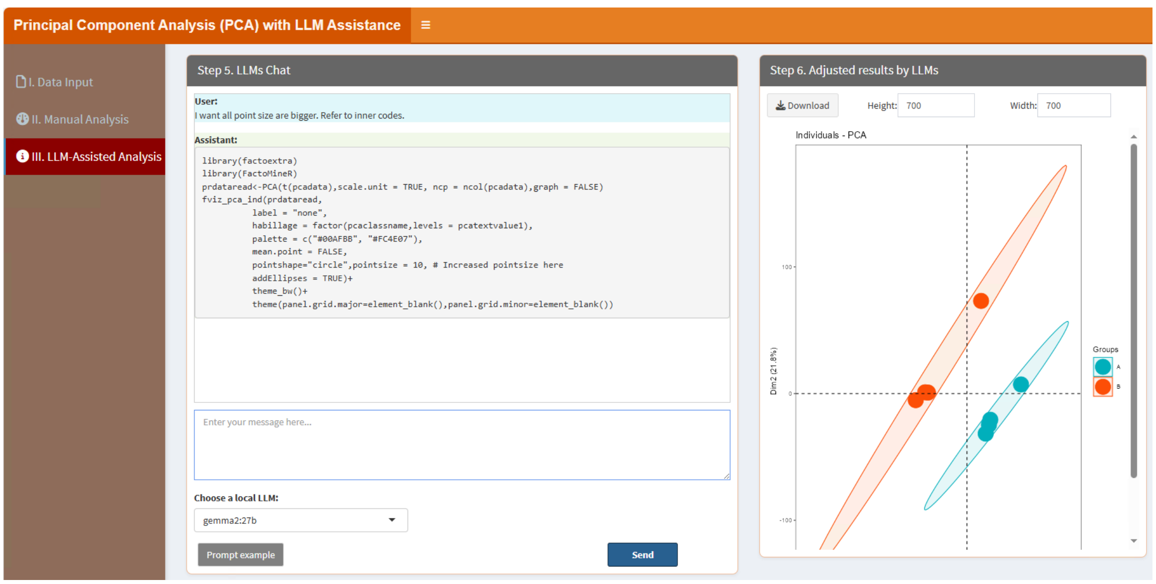
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I want all point size are bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I want all point size are bigger. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

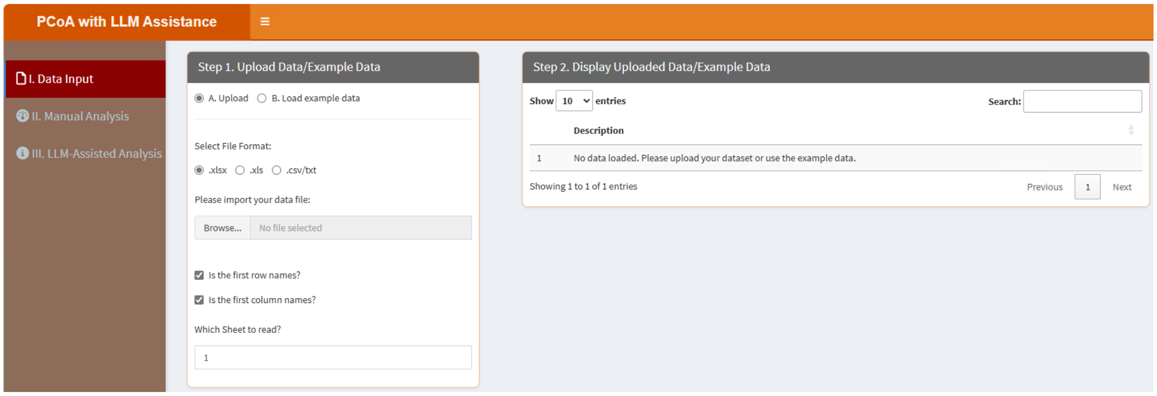


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



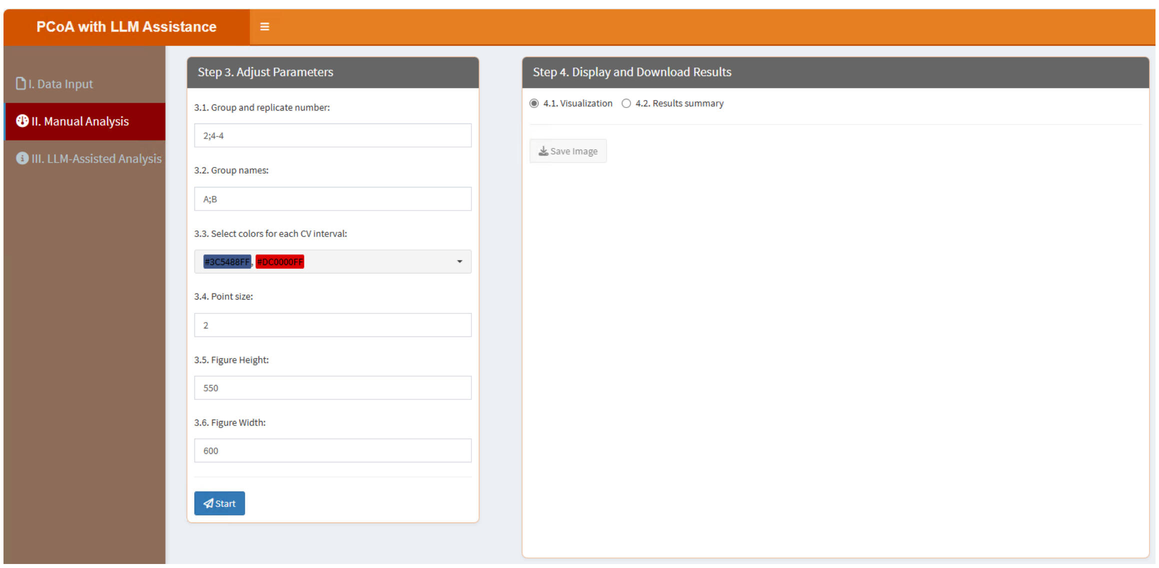
5.19. Principal Coordinates Analysis (PCoA)

PCoA visualizes distances between samples. Users can set distance metrics via the classic framework, while local LLMs help refine the analysis and provide insights. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Select colors for each CV interval: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

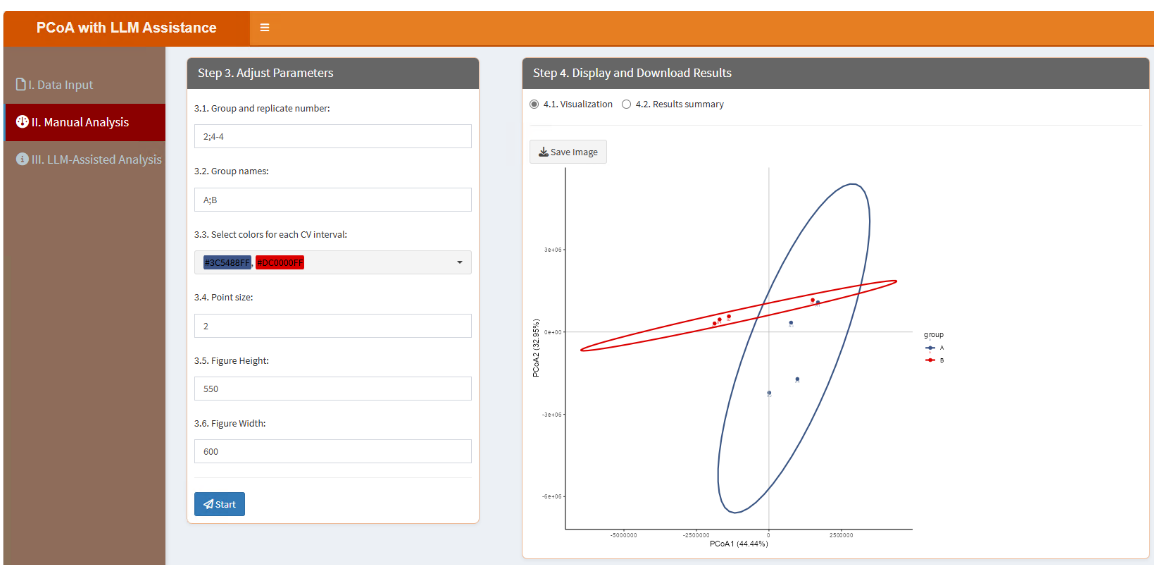
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

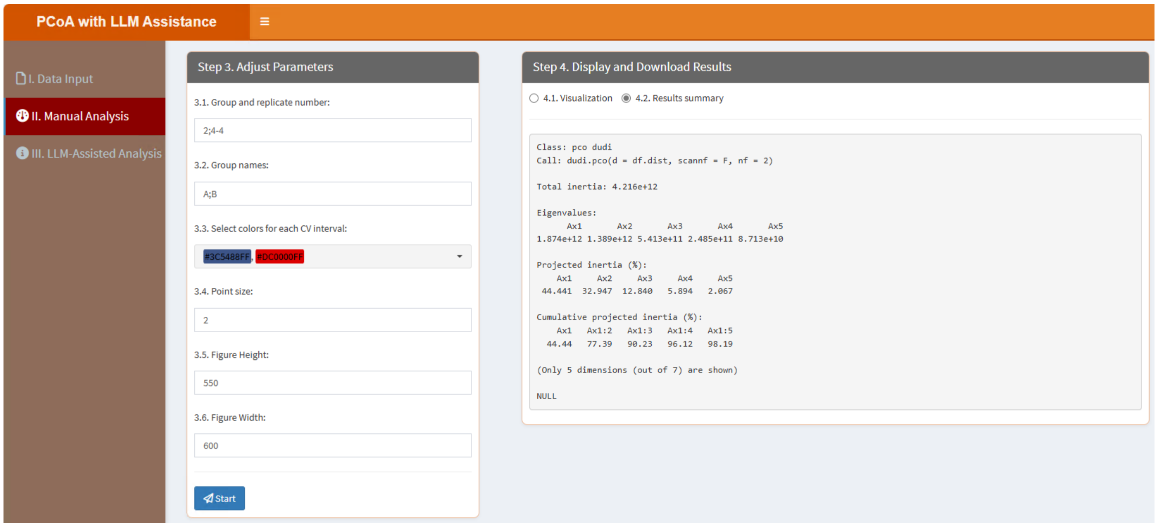
3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



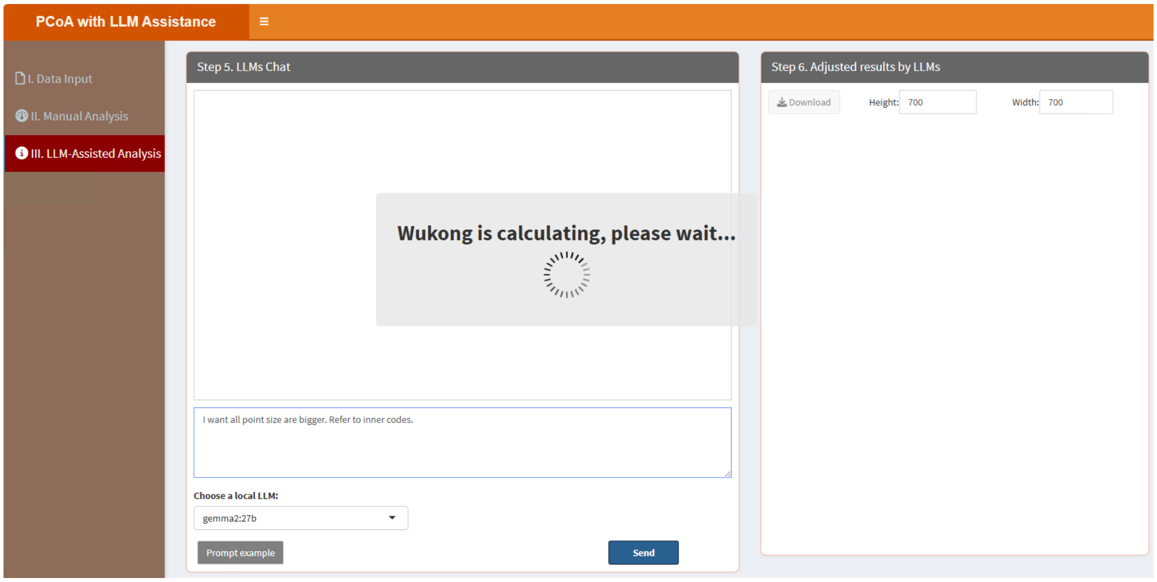
In the “4.2. Results summary” part, users can check the model summary results:



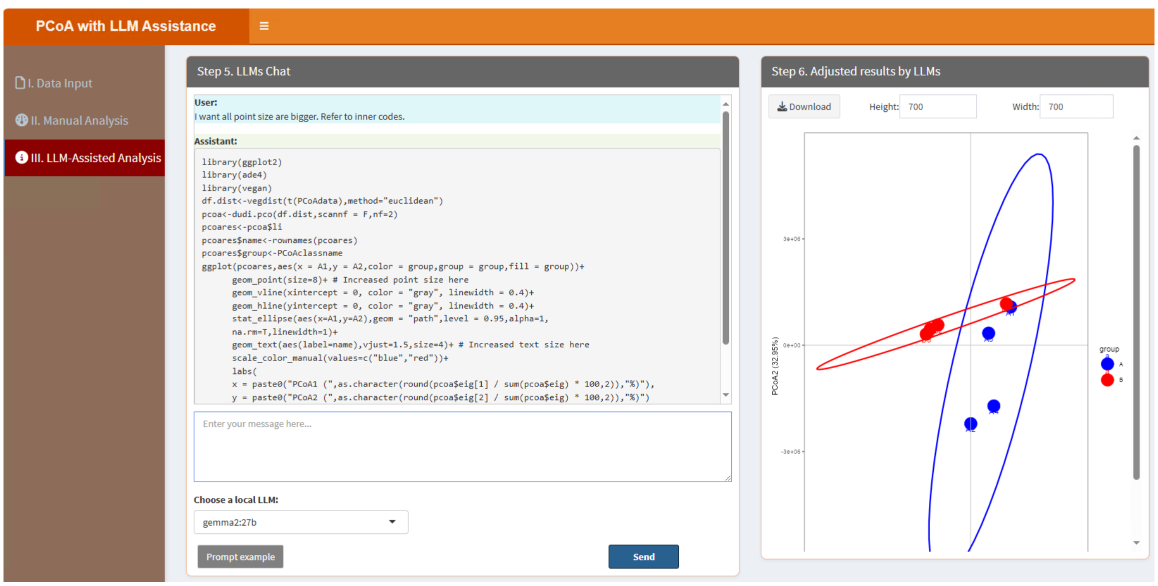
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I want all point size are bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I want all point size are bigger. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

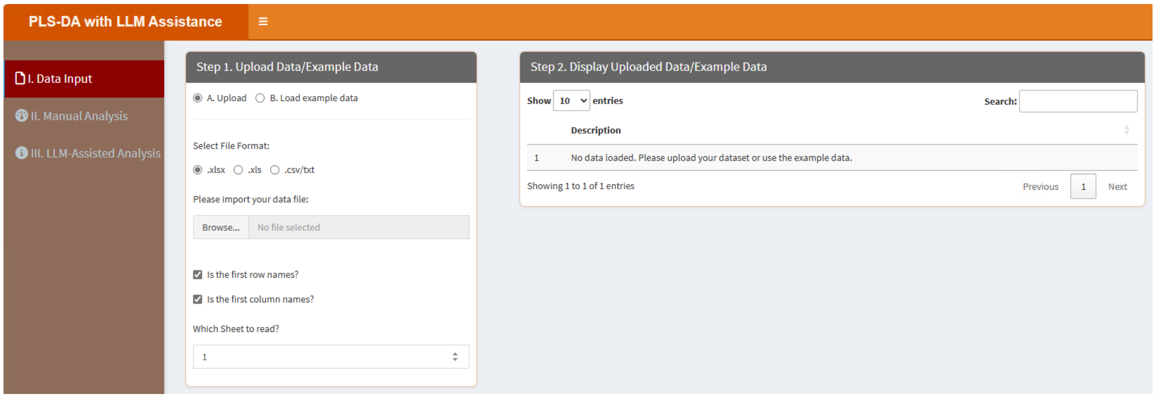


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



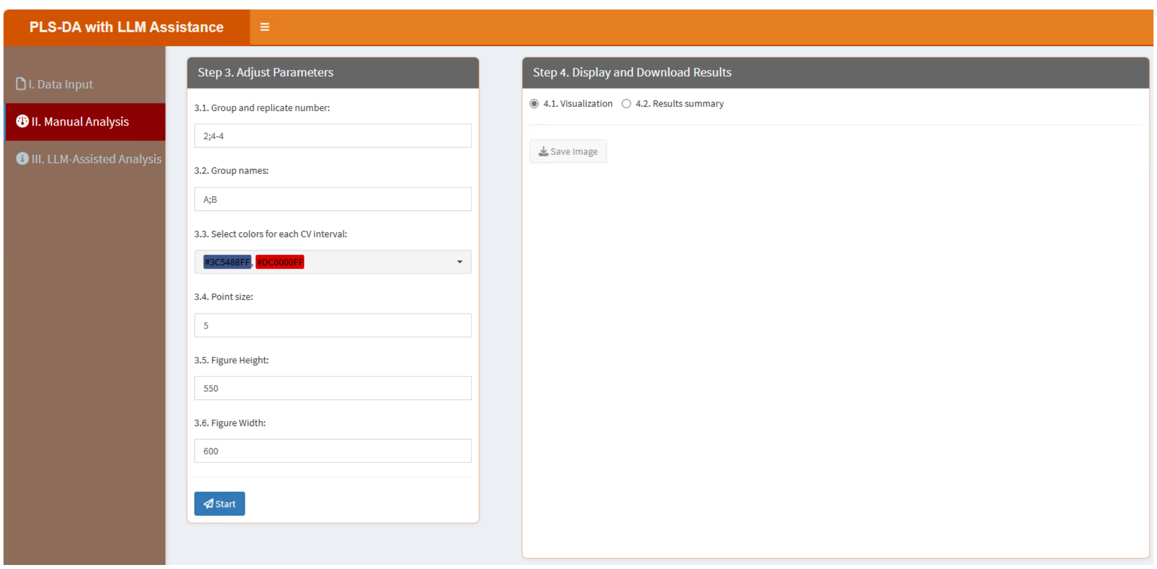
5.20. PLS-DA (Partial Least Squares Discriminant Analysis)

PLS-DA is used for classification in high-dimensional data. Users can configure model parameters through the classic framework, while local LLMs improve result interpretation. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Select colors for each CV interval: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

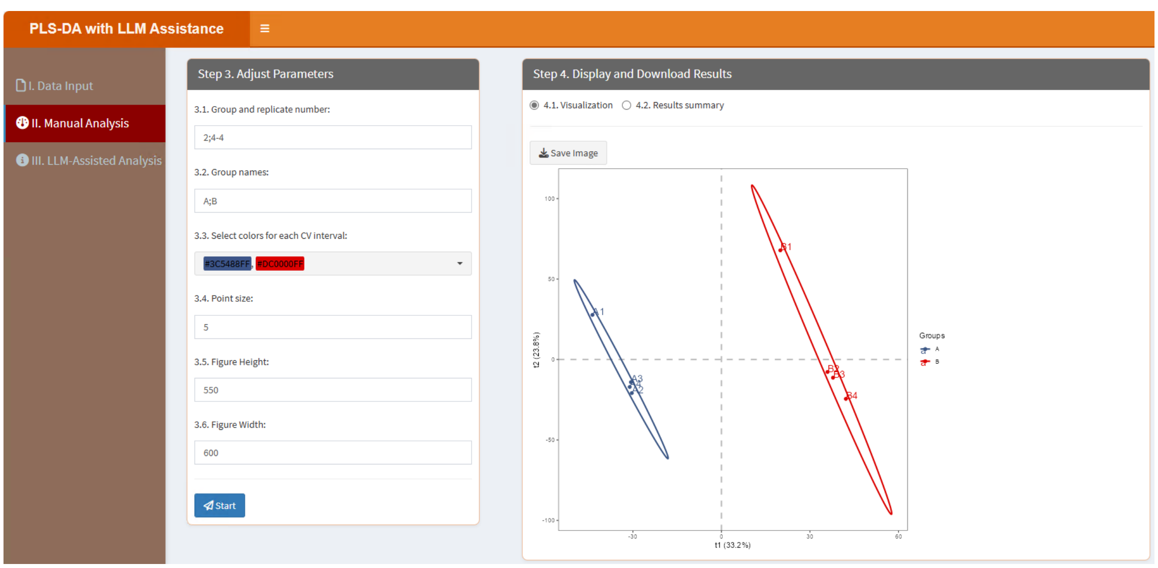
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

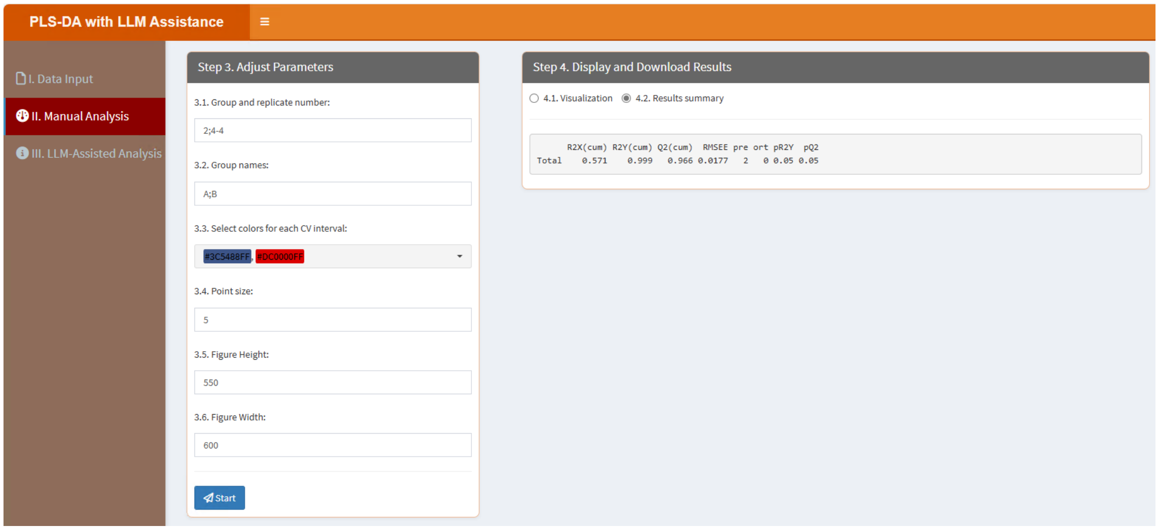
3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



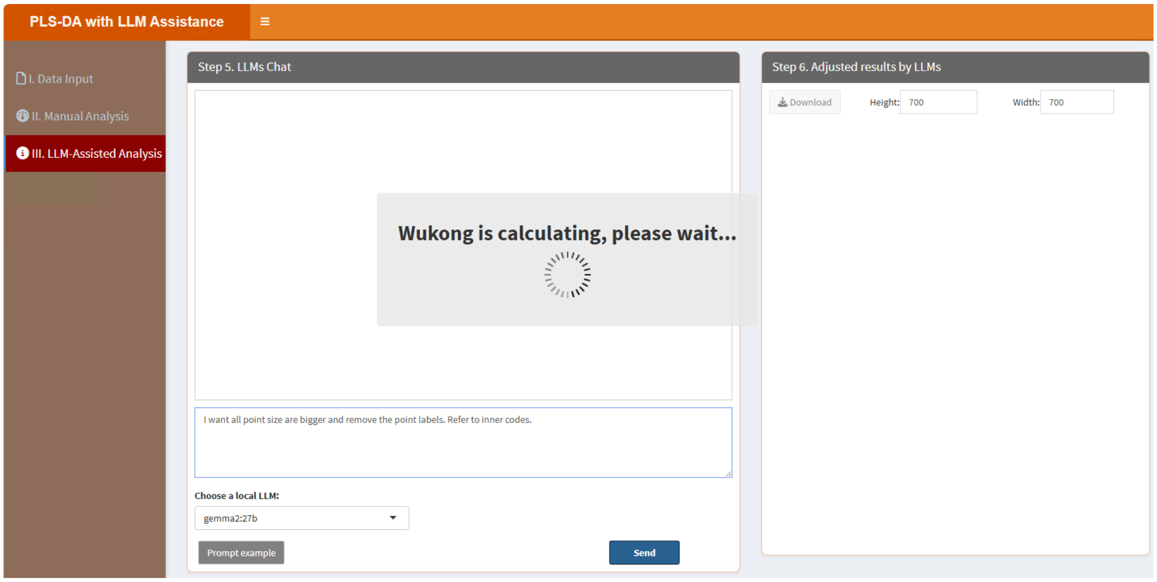
In the “4.2. Results summary” part, users can check the model summary results:



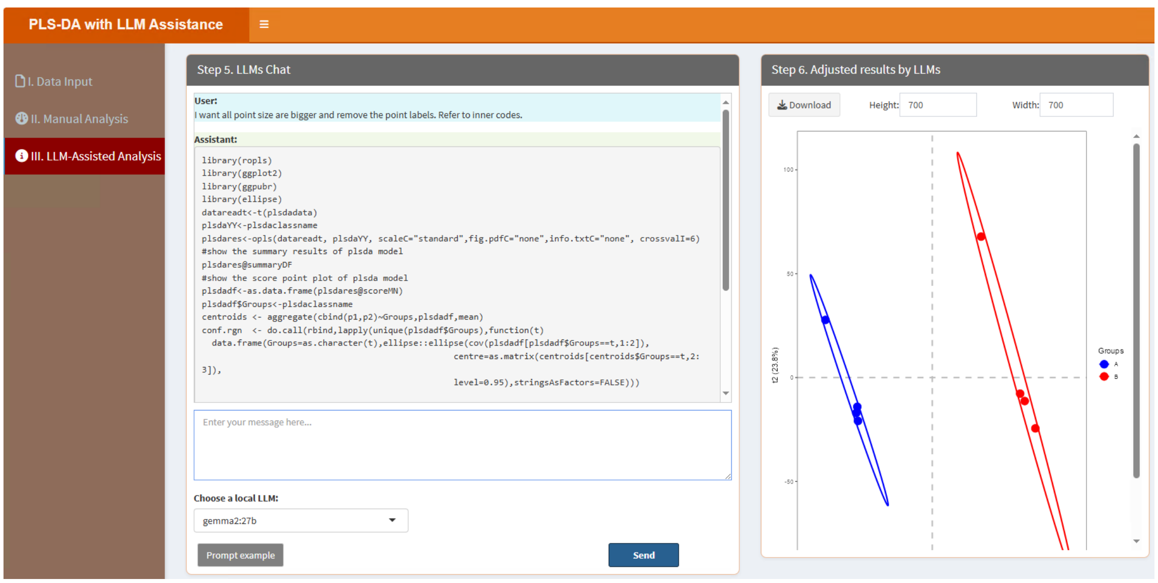
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I want all point size are bigger and remove the point labels. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I want all point size are bigger and remove the point labels. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

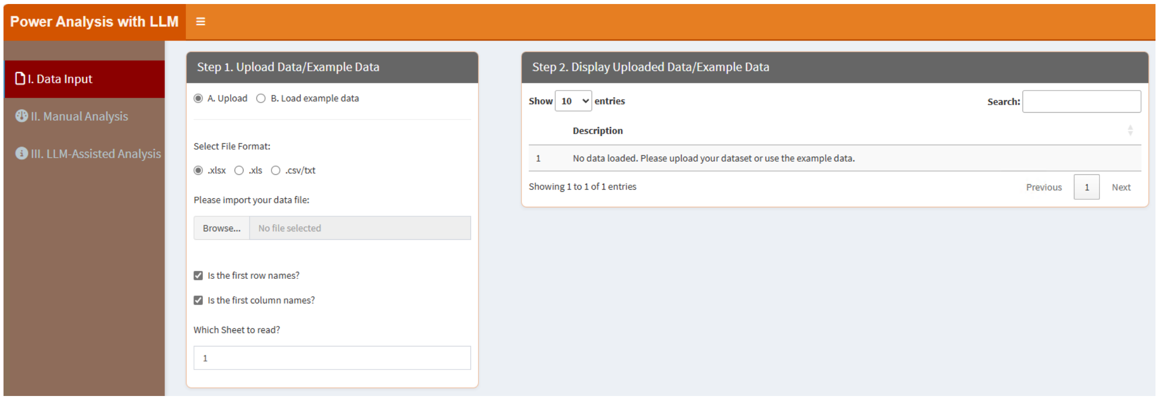


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



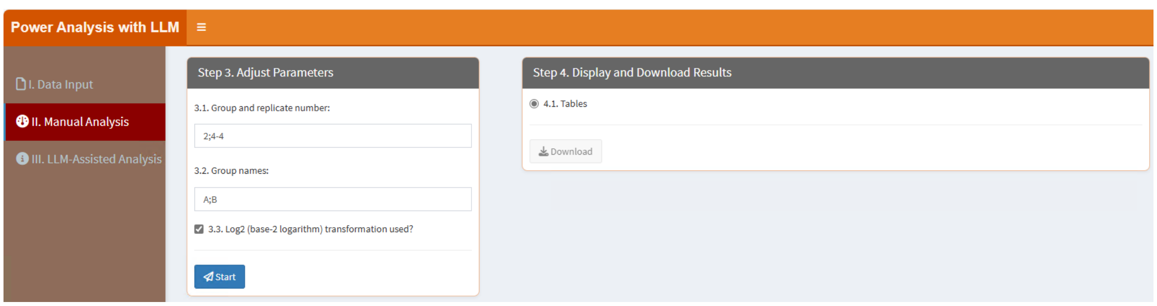
5.21. Power Analysis

Power analysis estimates sample sizes for statistical tests. Users can adjust effect sizes and significance levels via the classic framework, while local LLMs refine the analysis and provide recommendations. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

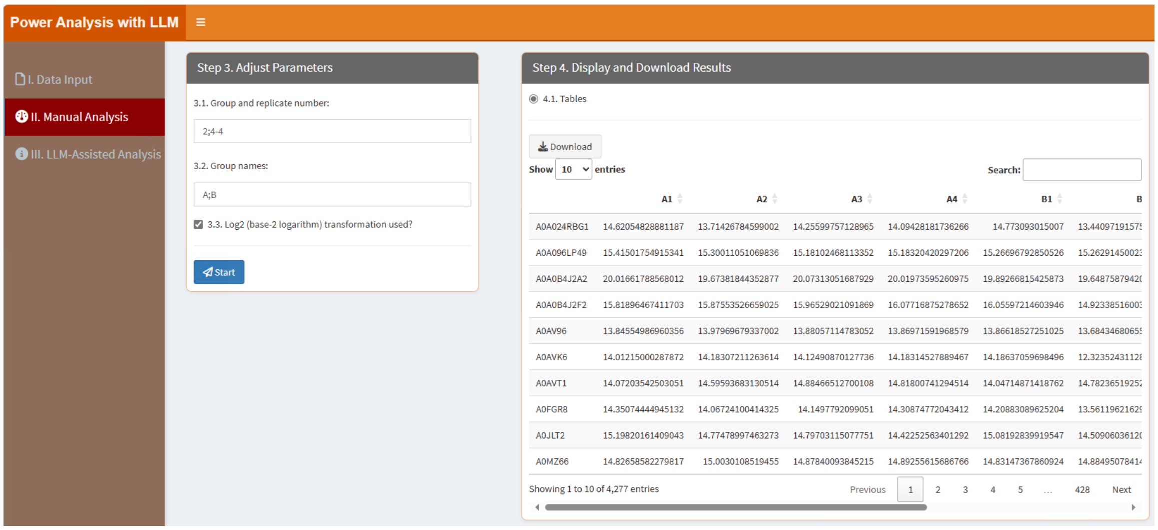


3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

3.3. Log2 (base-2 logarithm) transformation used? if true, the data will be transformed to the logarithmic scale with base 2.

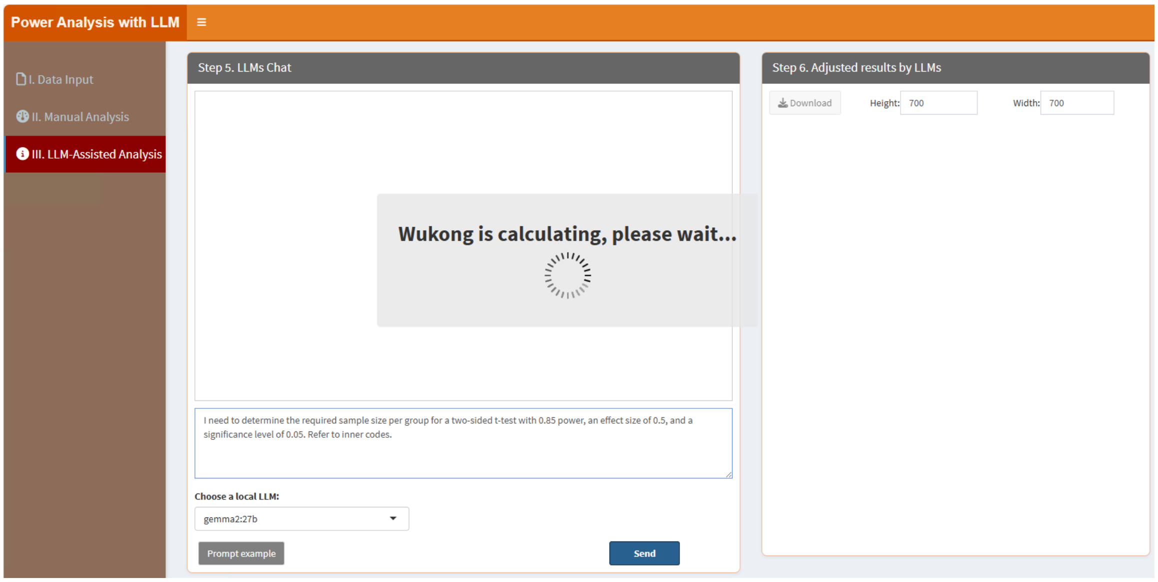
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Tables” part, users can click “Download” button to download the table, which contains the results:



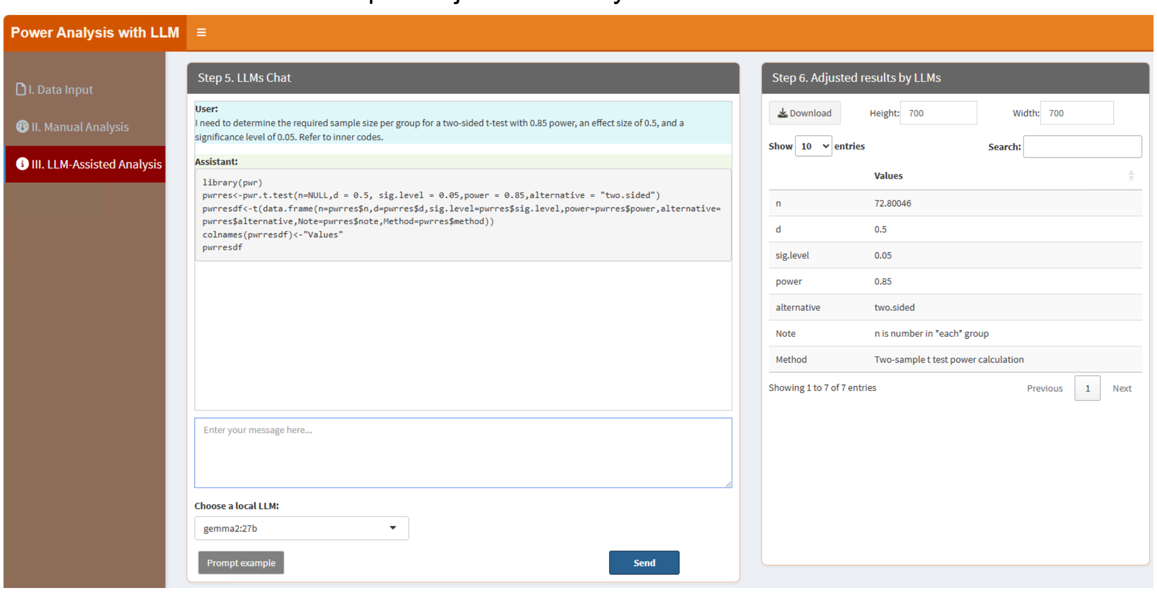
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I need to determine the required sample size per group for a two-sided t-test with 0.85 power, an effect size of 0.5, and a significance level of 0.05. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I need to determine the required sample size per group for a two-sided t-test with 0.85 power, an effect size of 0.5, and a significance level of 0.05. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

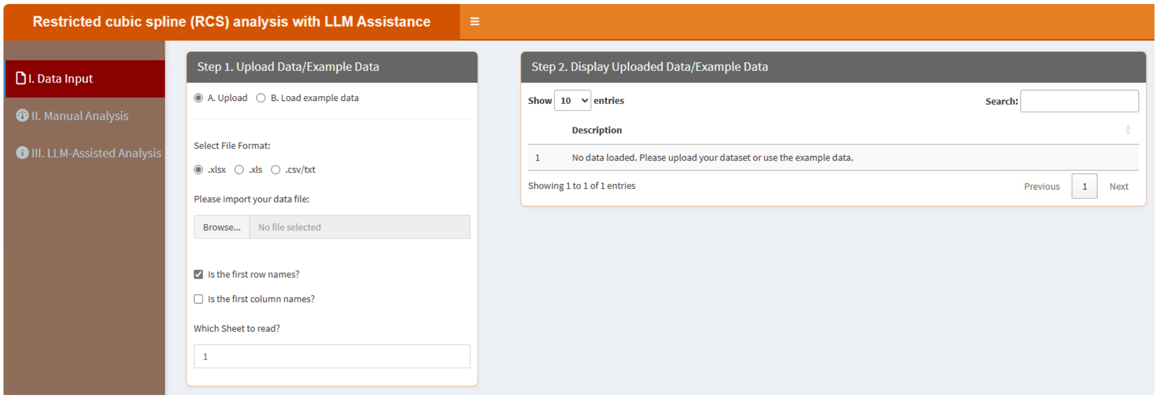


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



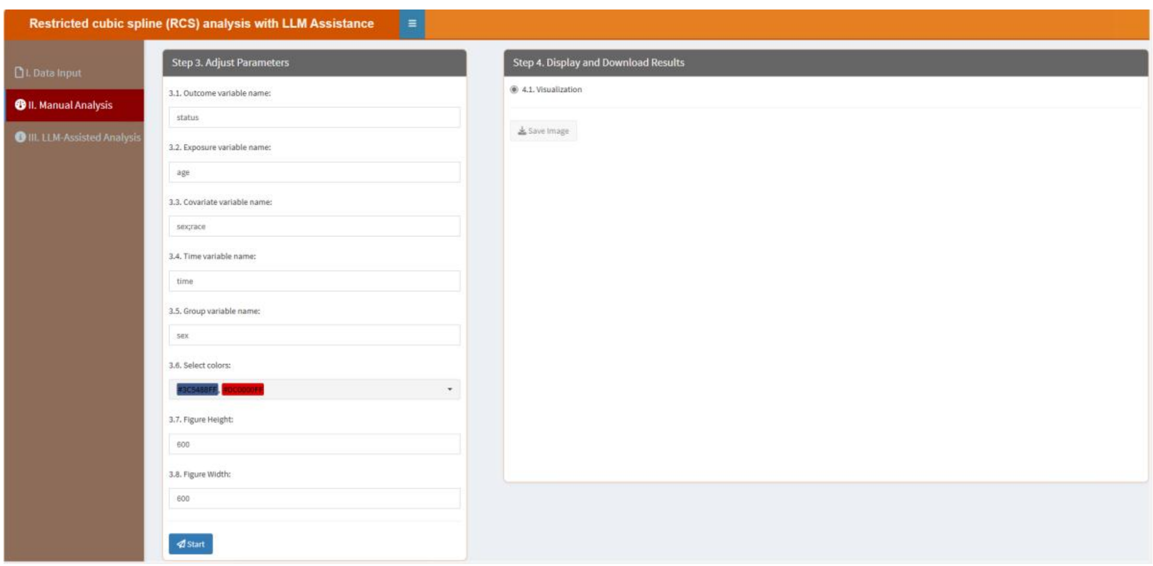
5.22. Restricted Cubic Spline (RCS) Analysis

RCS analysis models non-linear relationships. Users can configure spline parameters through the classic framework, and local LLMs assist in refining the analysis. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Outcome variable name: The name of outcome variable in the data.

3.2. Exposure variable name: The name of exposure variable in the data.

3.3. Covariate variable name: The names of covariate variables in the data.

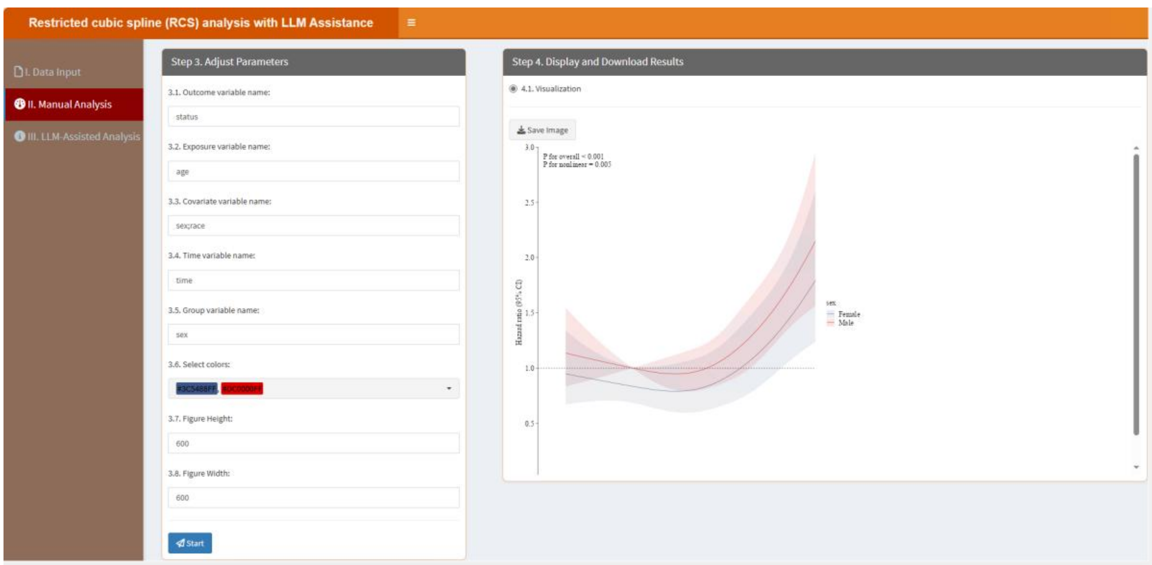
3.4. Time variable name: The name of time variable in the data, for Cox regressions.

3.5. Group variable name: The name of group variable in the data.

3.6. Select colors: This parameter lets you choose specific colors for each group. A color picker

is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of groups, ensuring that the visual representation is both informative and aesthetically pleasing.
3.7. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.8. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click "Start" button, the results will be shown as below:
In the "4.1. Visualization" part, users can click "Save image" button to download the figure:

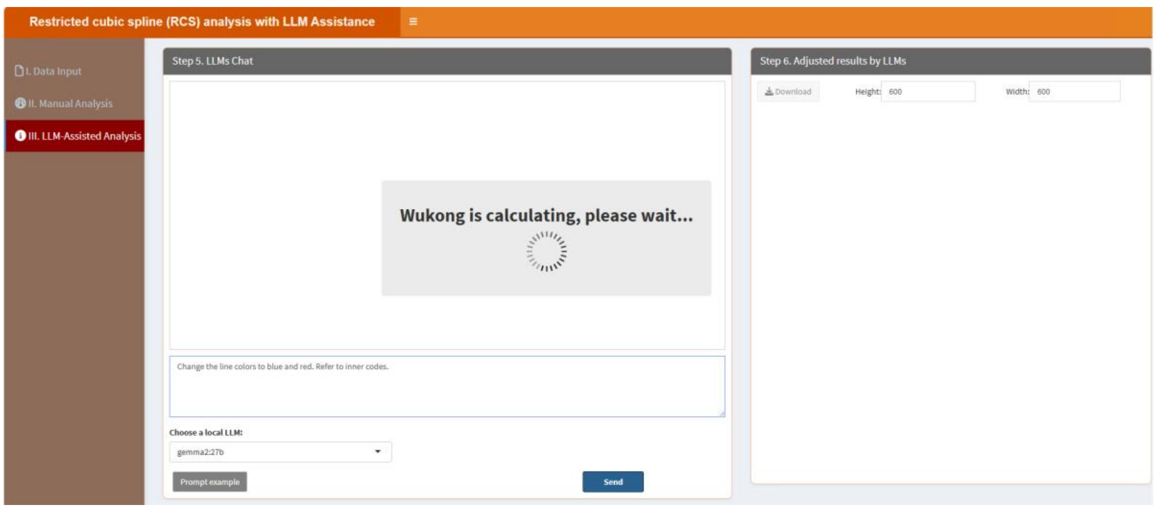


"III. LLM-Assisted Analysis" means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

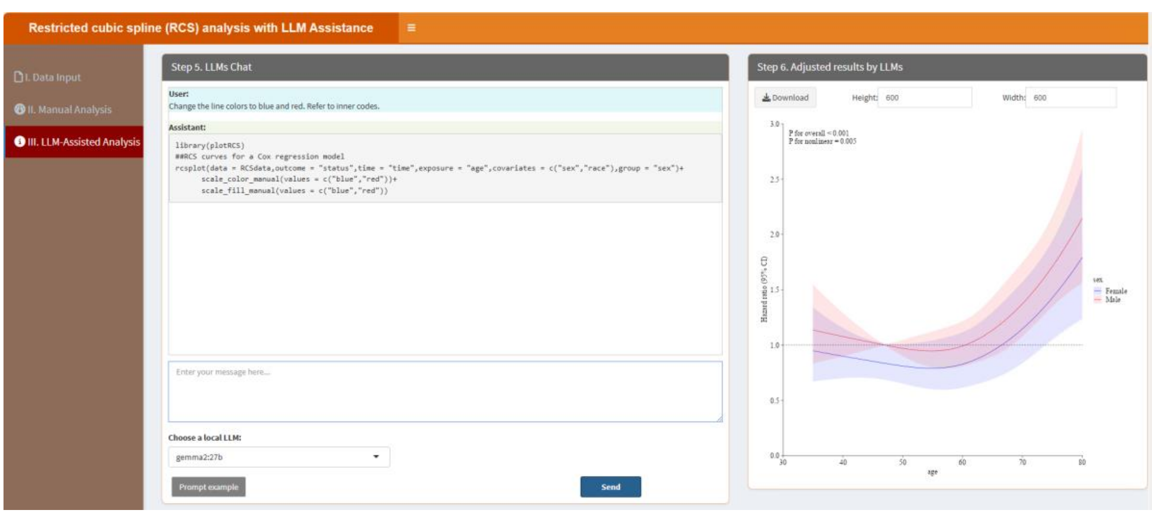
When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of

LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Change the line colors to blue and red. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Change the line colors to blue and red. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

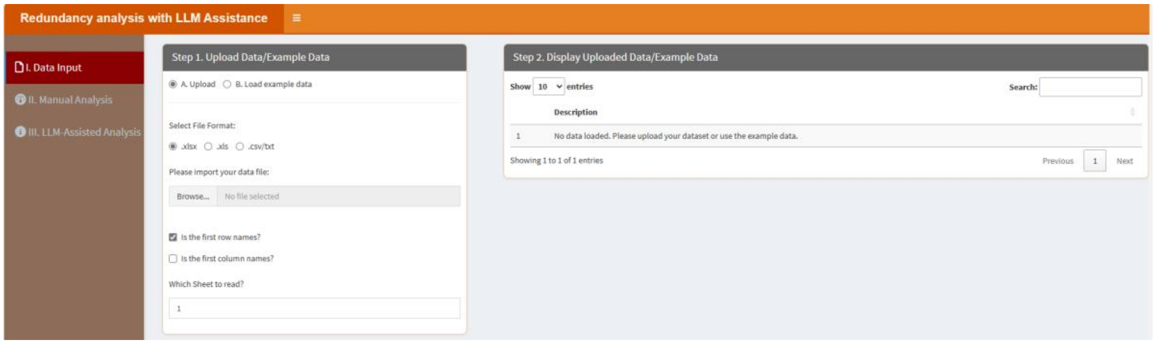


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



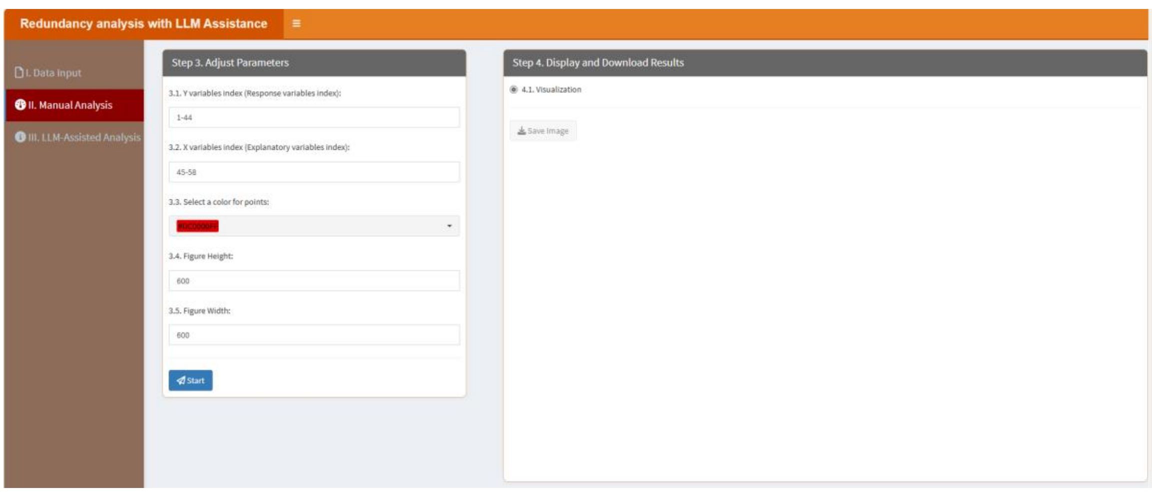
5.23. Redundancy Analysis (RDA)

RDA evaluates relationships between response and explanatory variables. Users can adjust parameters through the classic framework, while local LLMs enhance result interpretation. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Y variables index (Response variables index): The column index for response variables. For example, in the example data, 1-44 columns are response variables, thus here should be 1-44.

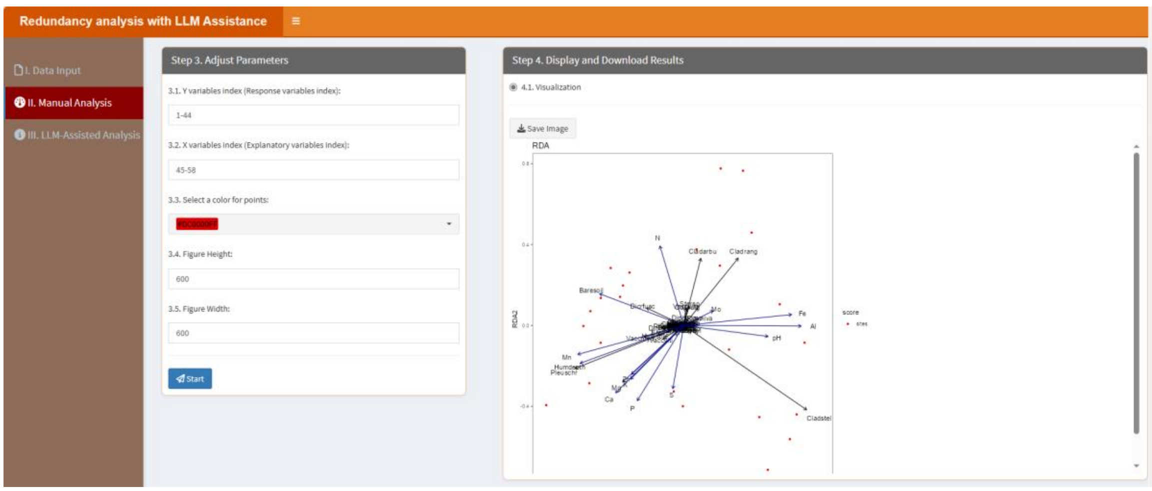
3.2. X variables index (Explanatory variables index): The column index for explanatory variables. For example, in the example data, 45-58 columns are explanatory variables, thus here should be 45-58.

3.3. Select a color for points: This parameter lets you choose specific colors for each point. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select one color for the point, ensuring that the visual representation is both informative and aesthetically pleasing.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

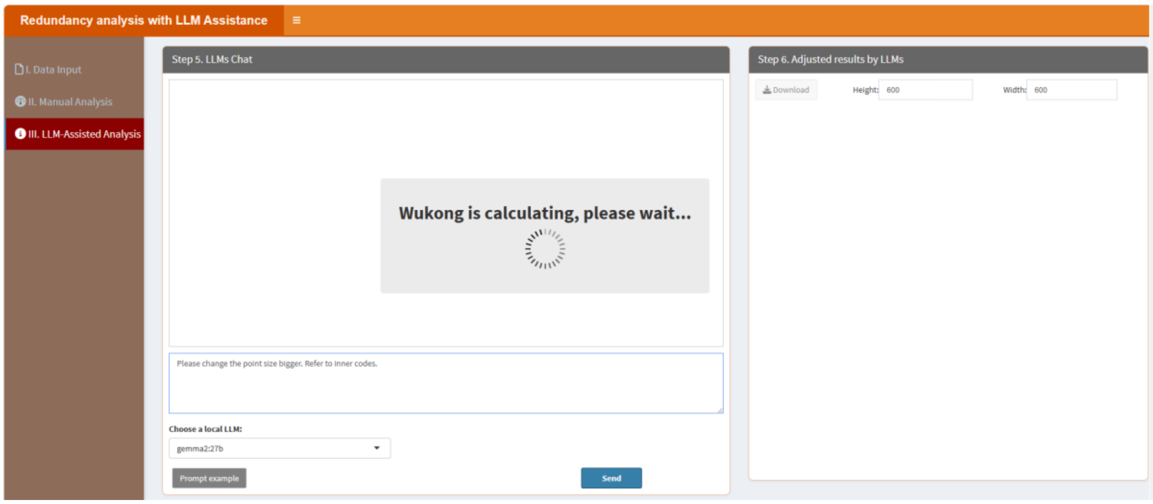
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



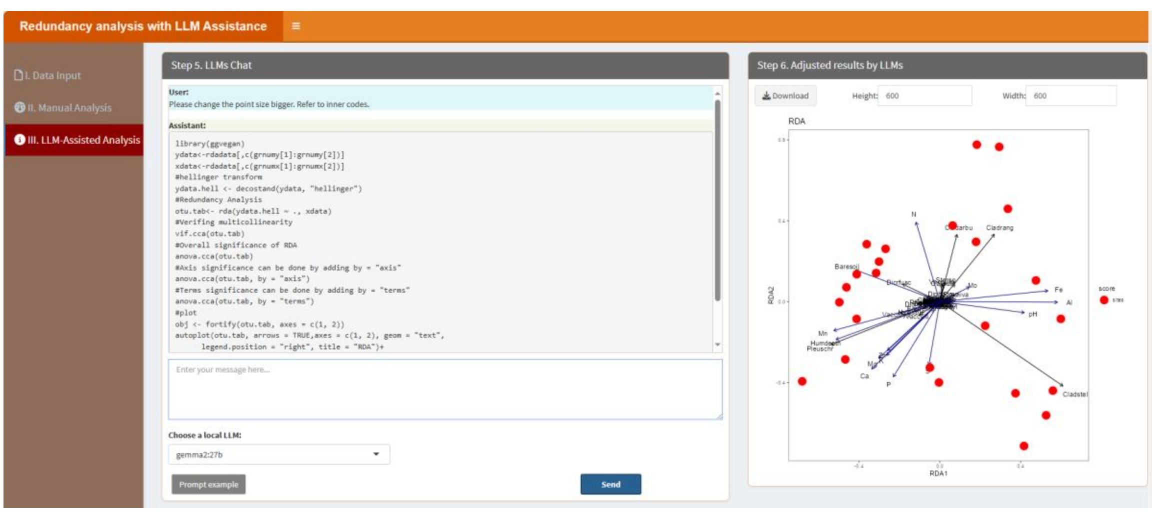
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the point size bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the point size bigger. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

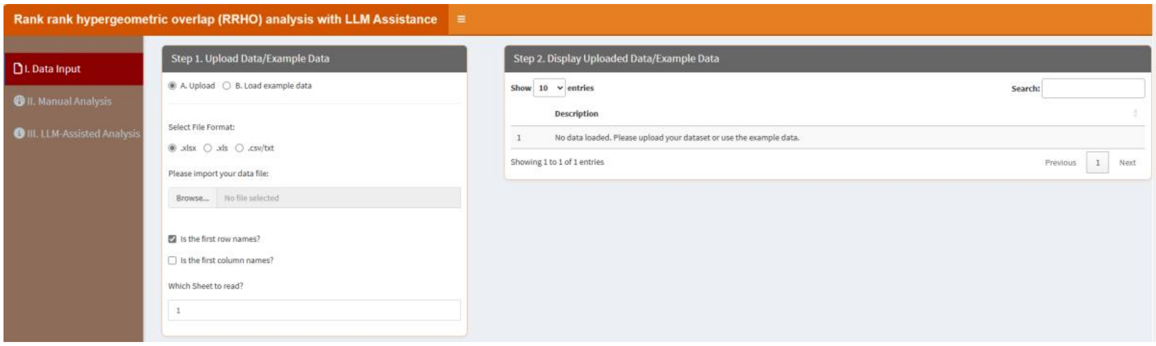


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



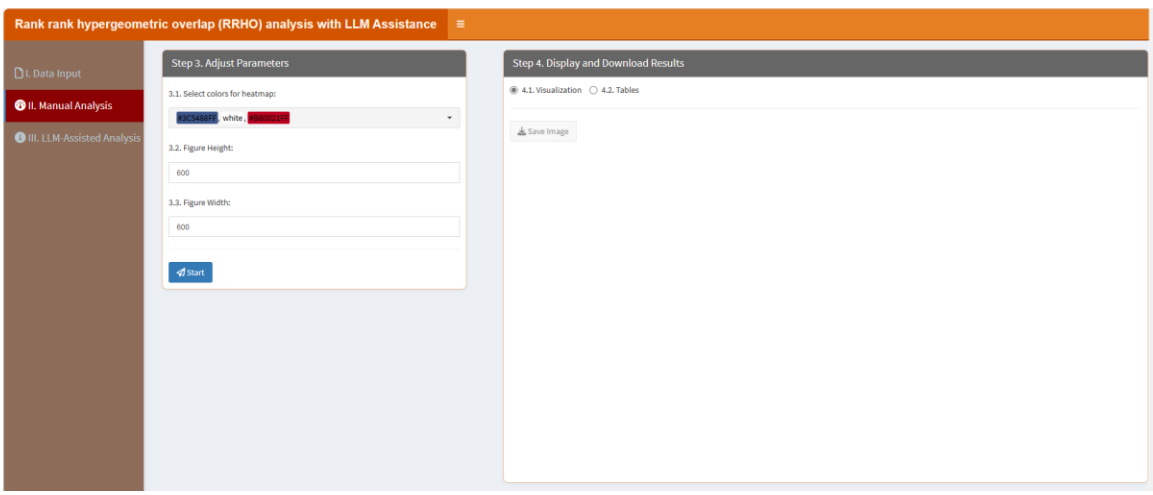
5.24. Rank Rank Hypergeometric Overlap (RRHO) Analysis

RRHO analysis identifies overlap between ranked datasets. Users can adjust ranking parameters via the classic framework, and local LLMs refine the analysis and provide insights. After clicking the "Start" button, it will be started in a new window like below:



The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.

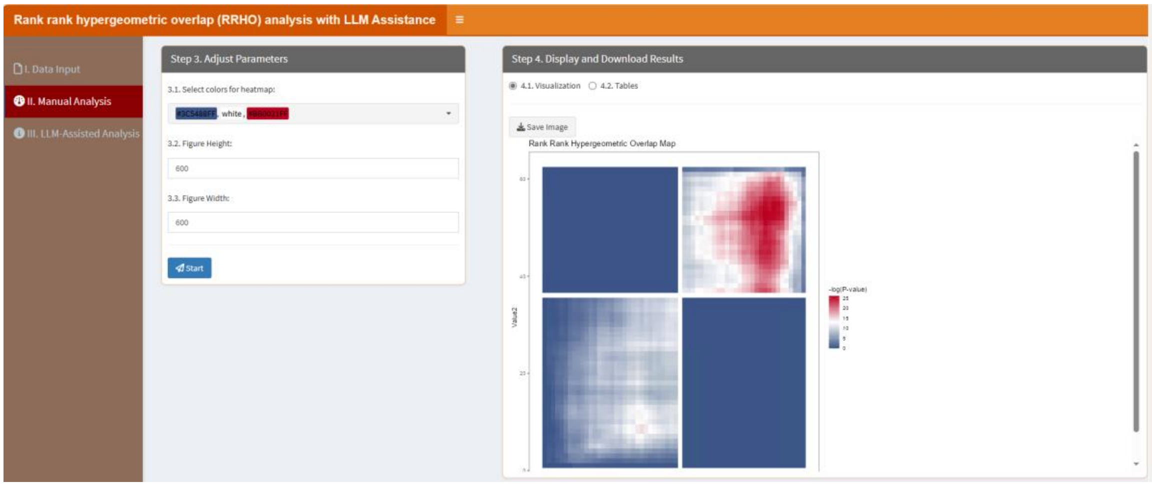


3.1. Select colors for heatmap: Colors for the heatmap plot. By default, there are three color names (#3C5488FF, white, #BB0021FF) here for the lowest (#3C5488FF), middle (white), highest (#BB0021FF) values repectively.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

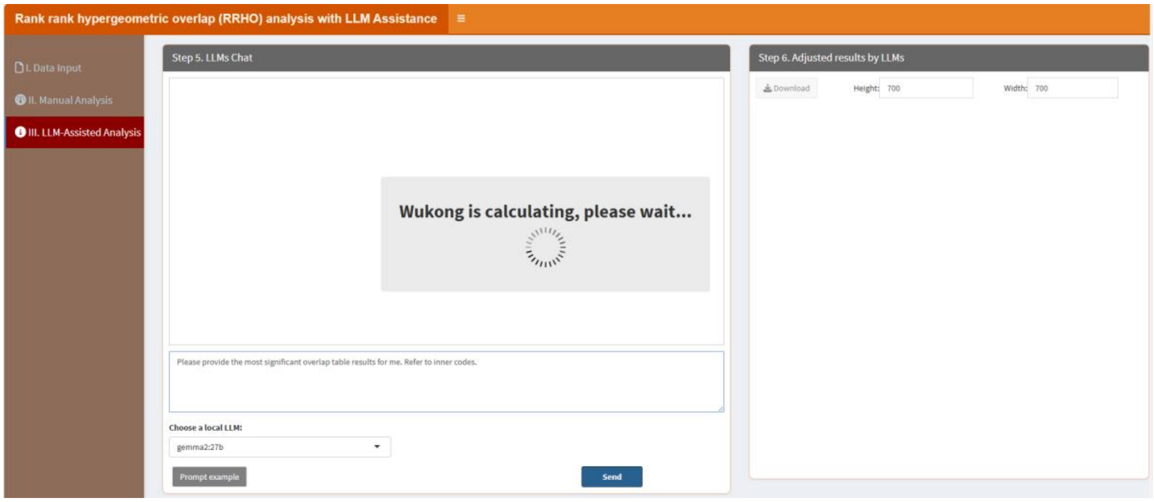
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



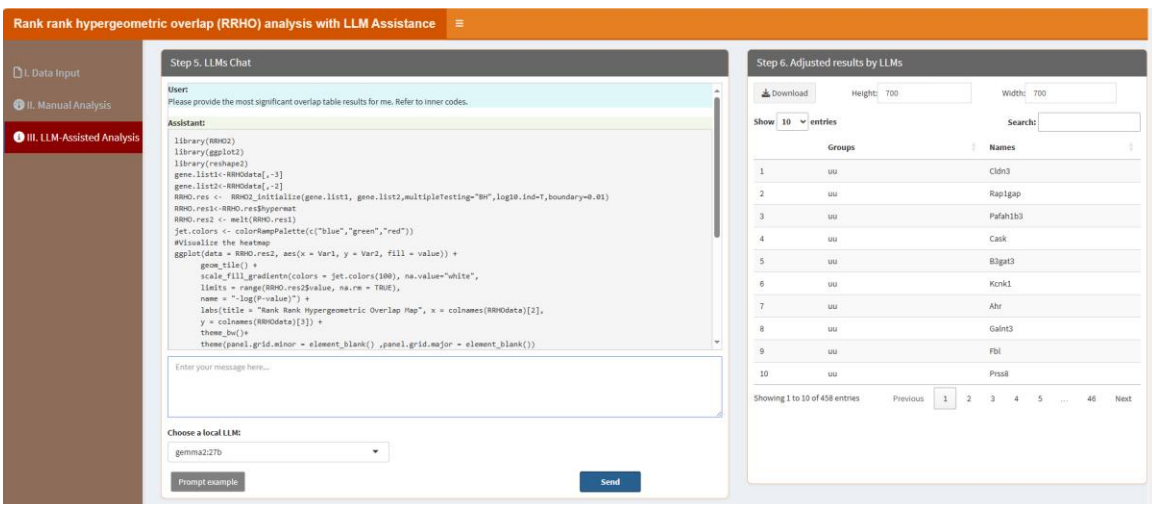
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please provide the most significant overlap table results for me. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please provide the most significant overlap table results for me. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

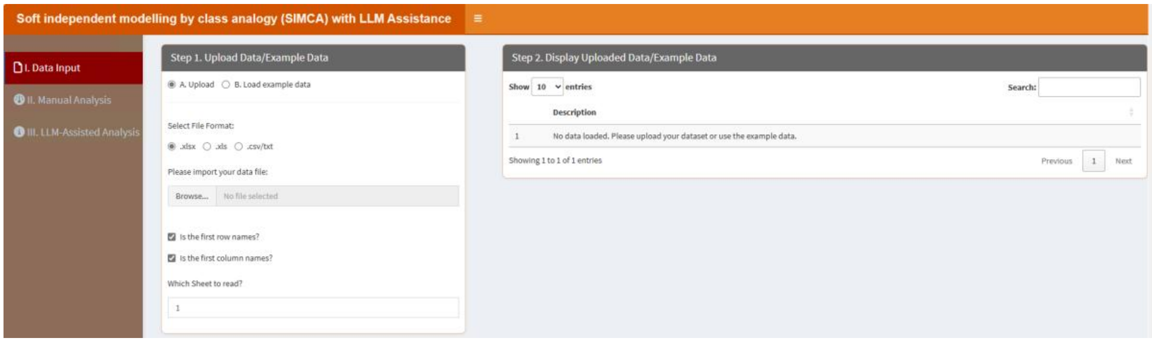


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



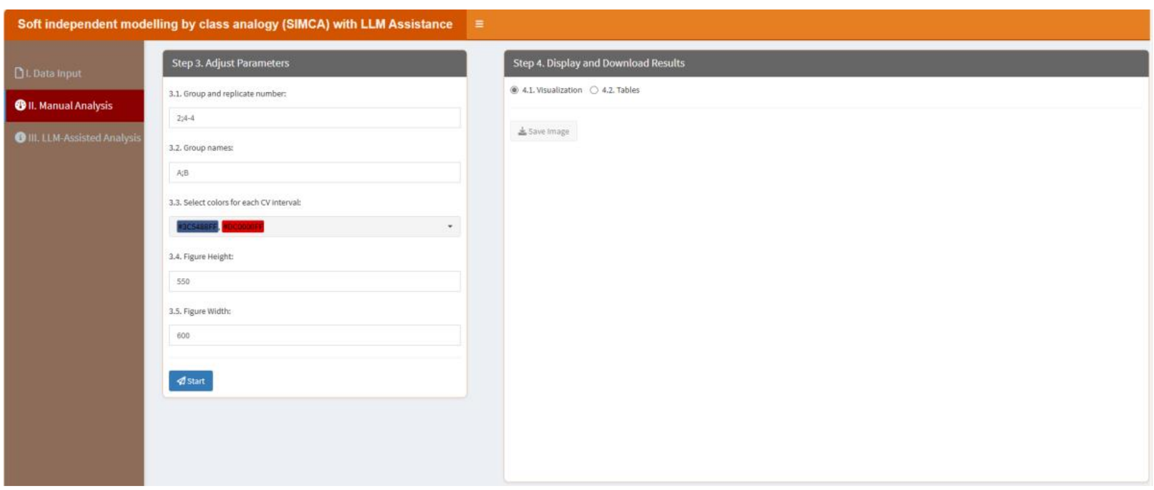
5.25. SIMCA (Soft Independent Modelling by Class Analogy)

SIMCA is used for classification in multivariate data. Users can configure model parameters through the classic framework, while local LLMs improve result interpretation. After clicking the "Start" button, it will be started in a new window like below:



The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.



3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control"

and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

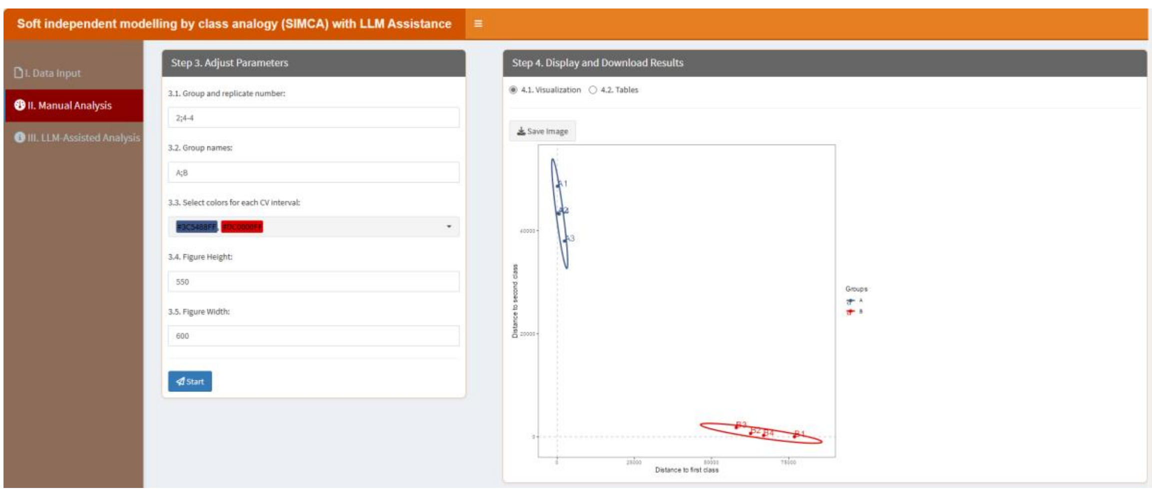
*3.3. Select colors for each CV interva*l: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

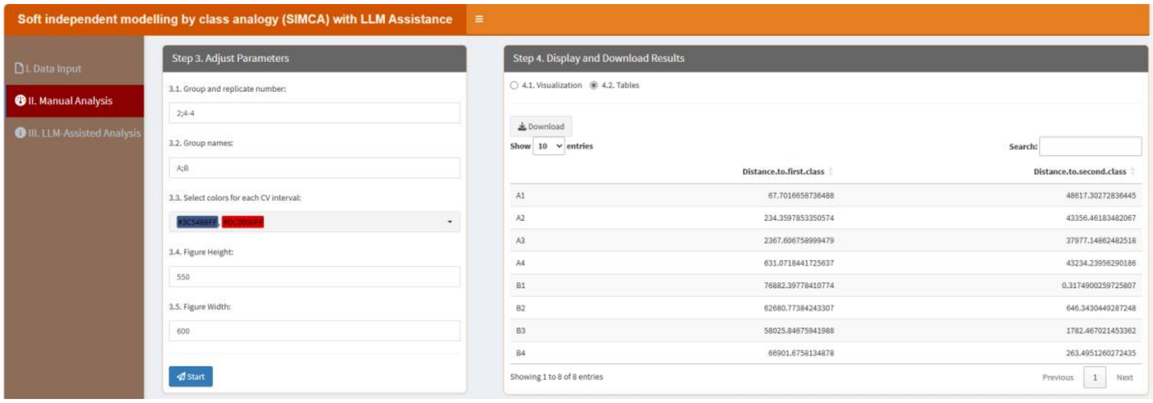
3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



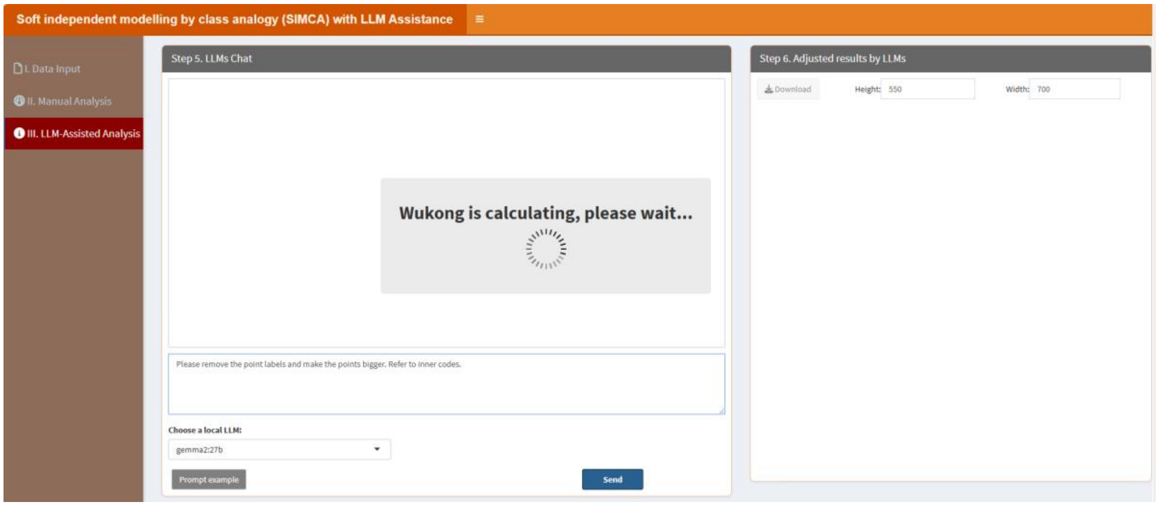
In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the CV of each group:



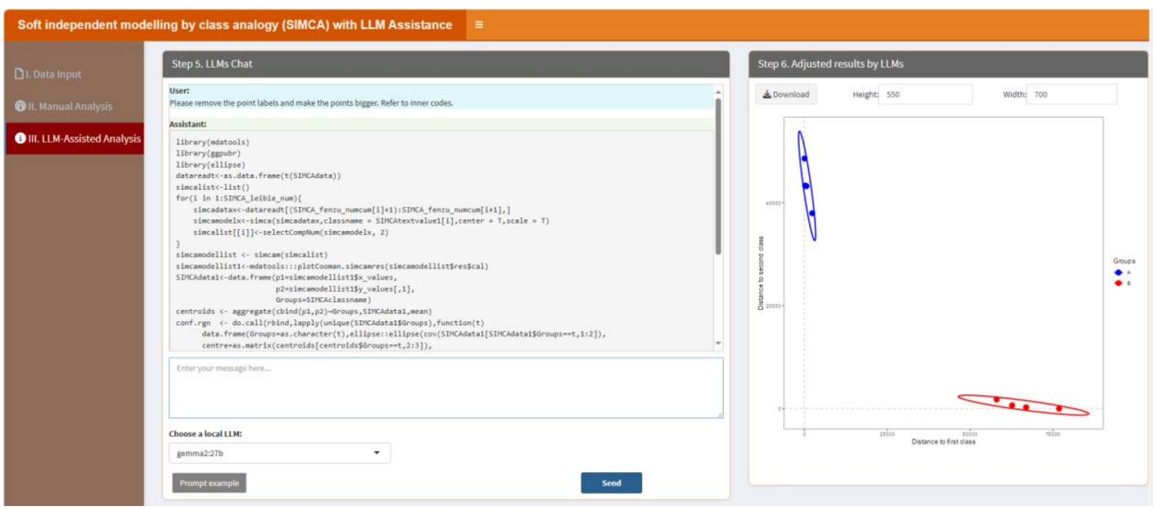
"III. LLM-Assisted Analysis" means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt "Please remove the point labels and make the points bigger. Refer to inner codes." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Please remove the point labels and make the points bigger. =>", it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click "Send" button. It will learn like below:

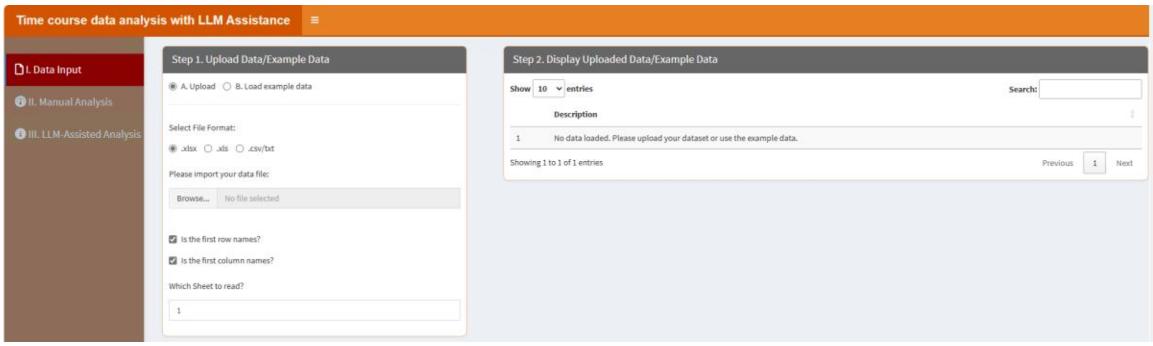


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



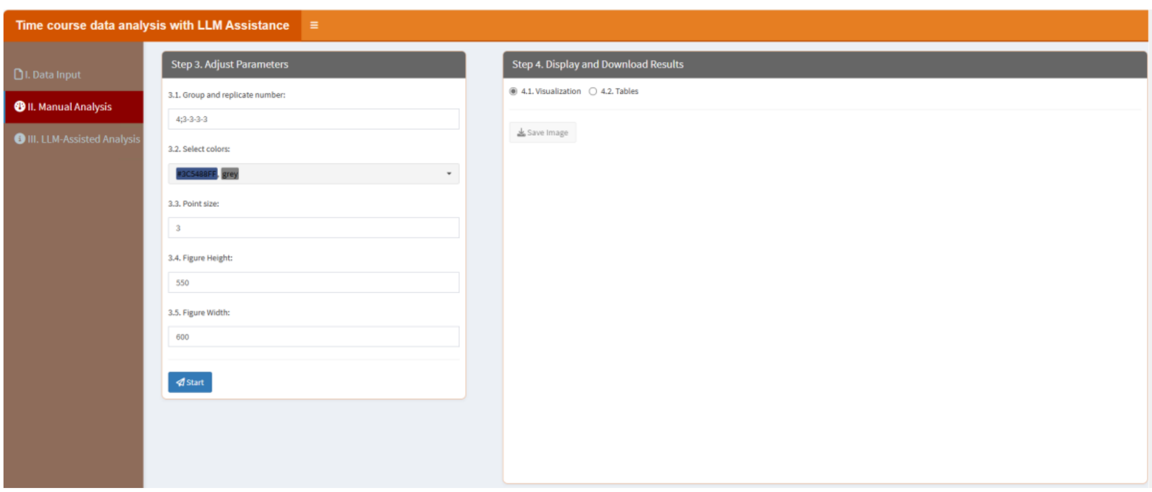
5.26. Time Course Data Analysis

This module analyzes time-series data. Users can adjust parameters like time intervals through the classic framework, while local LLMs assist in refining the analysis and interpreting trends. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

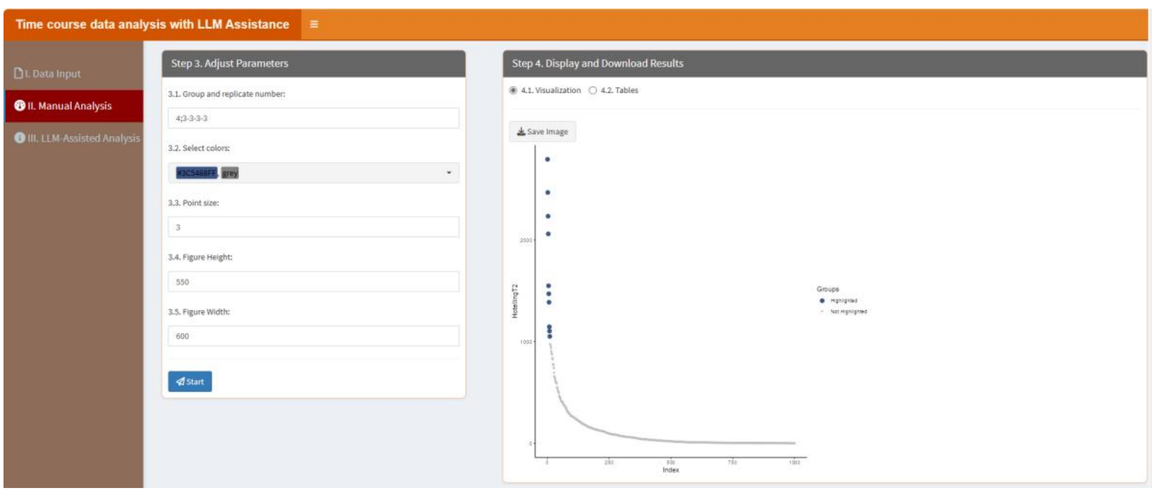


3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

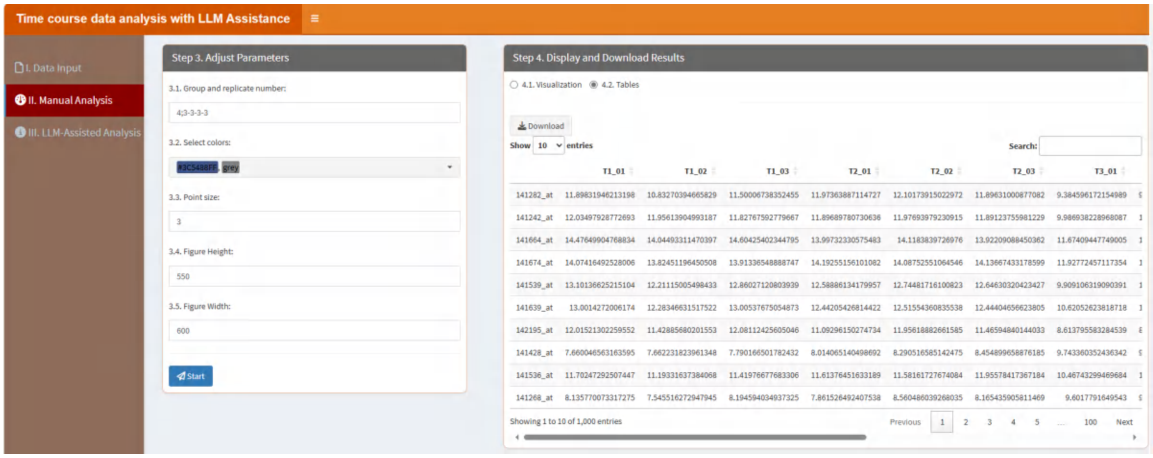
3.2. Select colors: This parameter lets you choose specific colors for each group (Highlighted and Not highlighted). A color picker is provided, offering a variety of predefined palettes such

as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select two colors for each group (Highlighted and Not highlighted), ensuring that the visual representation is both informative and aesthetically pleasing.
3.3. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.
3.4. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



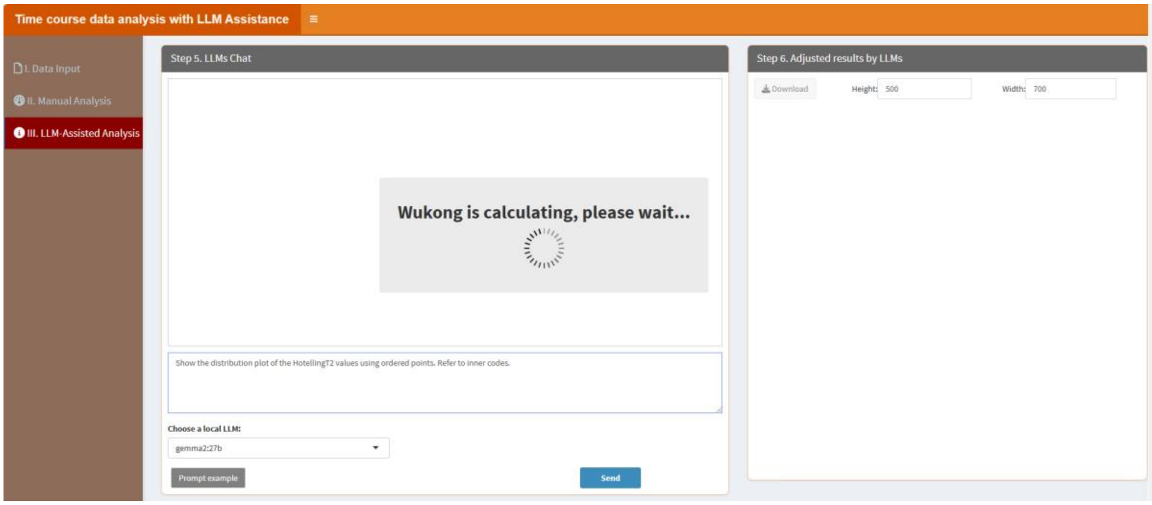
In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the CV of each group:



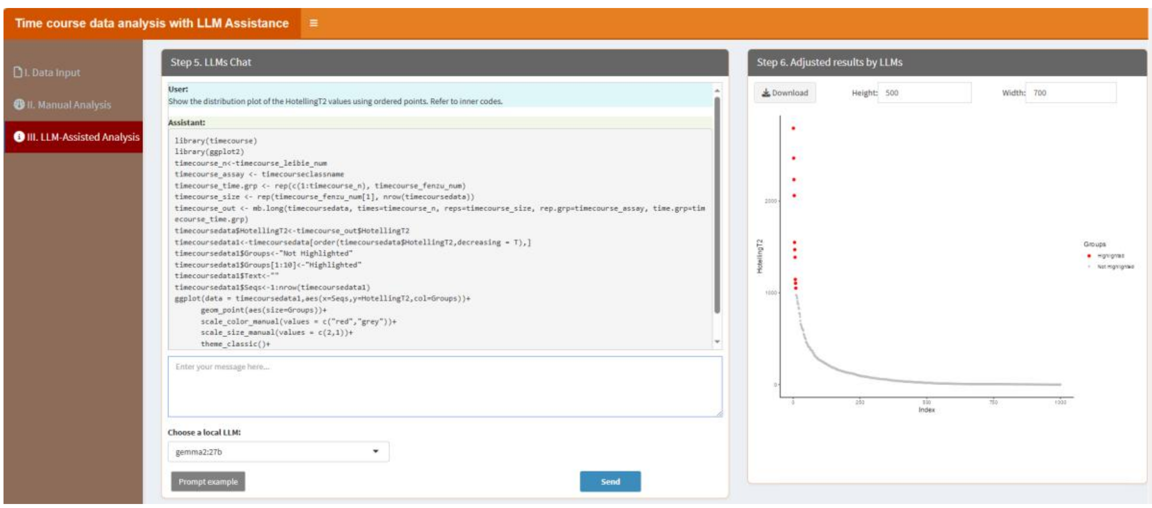
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Show the distribution plot of the HotellingT2 values using ordered points. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Show the distribution plot of the HotellingT2 values using ordered points. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

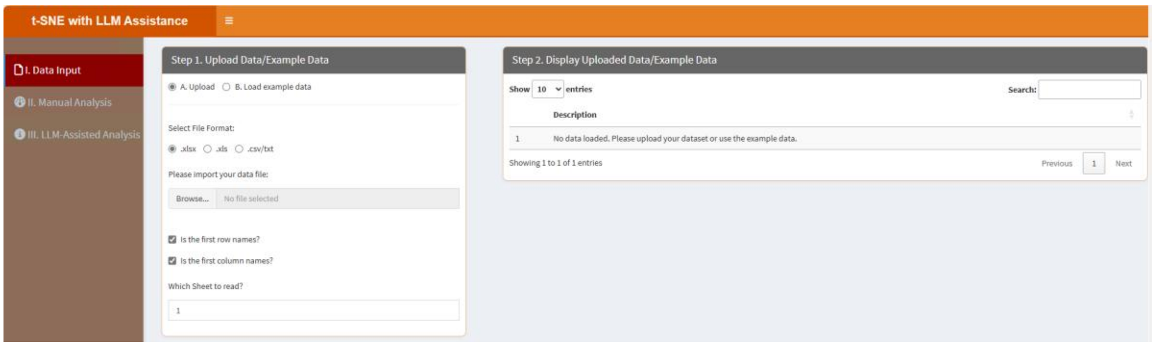


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



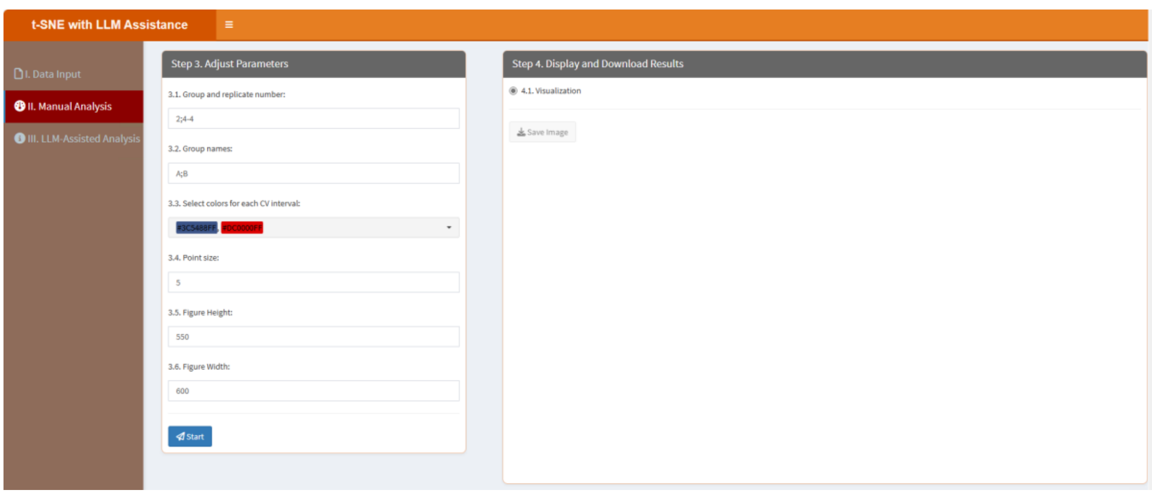
5.27. t-SNE (t-Distributed Stochastic Neighbor Embedding)

t-SNE visualizes high-dimensional data in 2D or 3D. Users can configure perplexity and learning rates via the classic framework, while local LLMs enhance interpretability. After clicking the "Start" button, it will be started in a new window like below:



The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.



3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control"

and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

*3.3. Select colors for each CV interva*l: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

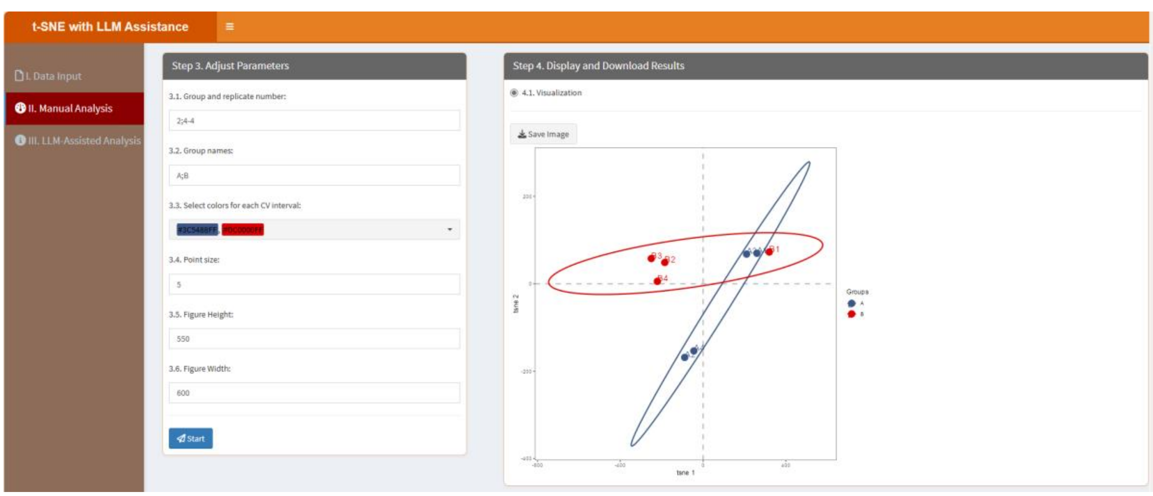
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

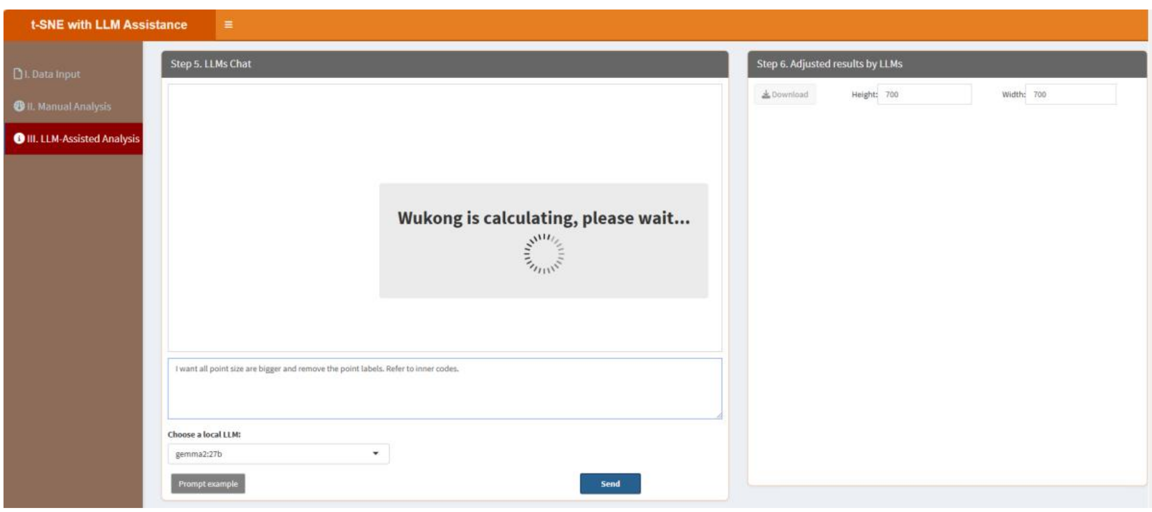
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



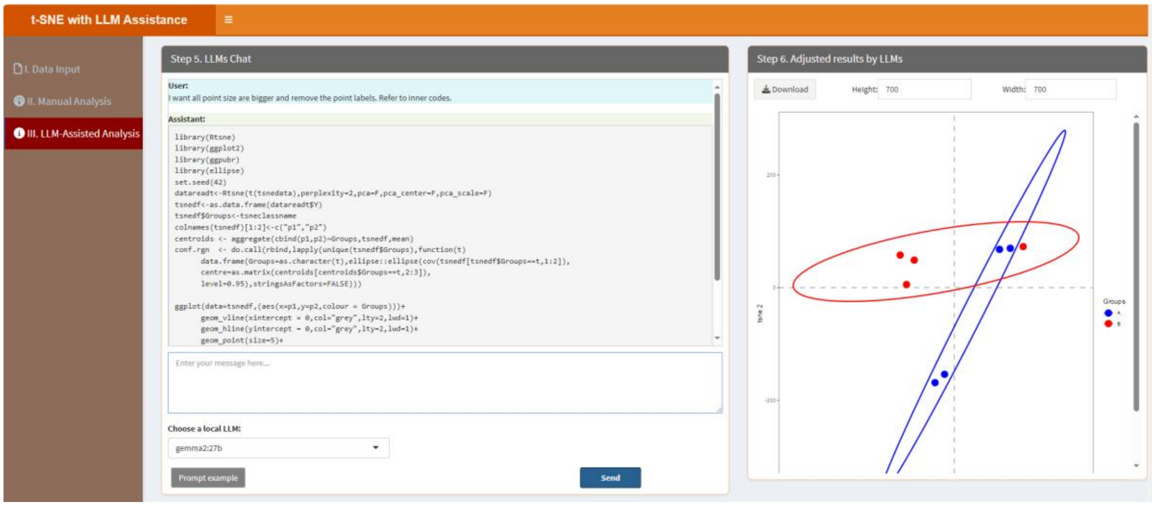
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I want all point size are bigger and remove the point labels. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I want all point size are bigger and remove the point labels. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:

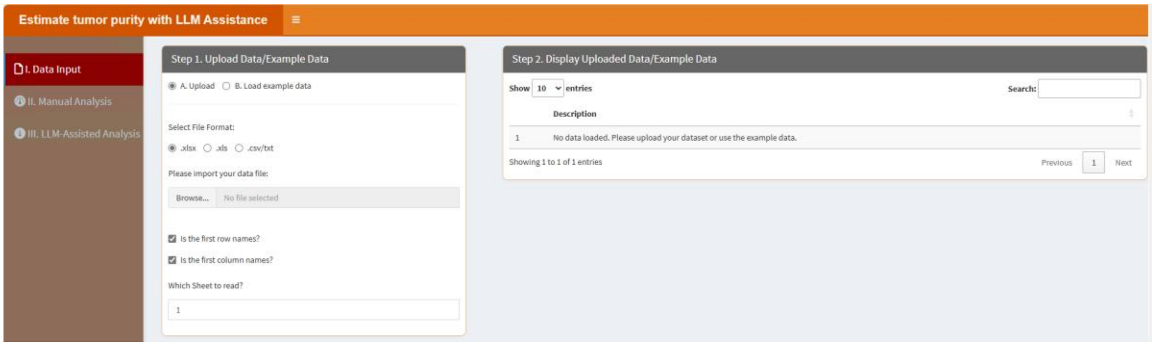


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.


t-SNE with LLM Assistance
I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis
Step 5. LLMs Chat
User:
I want all point size are bigger and remove the point labels. Refer to inner codes.
Assistant:
library(Rtsne)
library(ggplot2)
library(ggpubr)
library(ellipse)
set.seed(42)
datareadt<-Rtsne(t(tsnedata),perplexity=2,pca=F,pca_center=F,pca_scale=F)
tsnedf<-as.data.frame(datareadt$Y)
tsnedf$Groups<-tsneclassname
colnames(tsnedf)[1:2]<-c("p1","p2")
centroids <- aggregate(cbind(p1,p2)~Groups,tsnedf,mean)
conf.rgn <- do.call(rbind,lapply(unique(tsnedf$Groups),function(t)
data.frame(Groups=as.character(t),ellipse::ellipse(cov(tsnedf[tsnedf$Groups==t,1:2]),
centre=as.matrix(centroids[centroids$Groups==t,2:3]),
level=0.95),stringsAsFactors=FALSE)))
ggplot(data=tsnedf,(aes(x=p1,y=p2,colour = Groups)))+
geom_vline(xintercept = 0,col="grey",lty=2,lwd=1)+
geom_hline(yintercept = 0,col="grey",lty=2,lwd=1)+
geom_point(size=5)+
Enter your message here...
Choose a local LLM:
gemma2:27b
Prompt example
Send
Step 6. Adjusted results by LLMs
Download
Height: 700
Width: 700
Groups

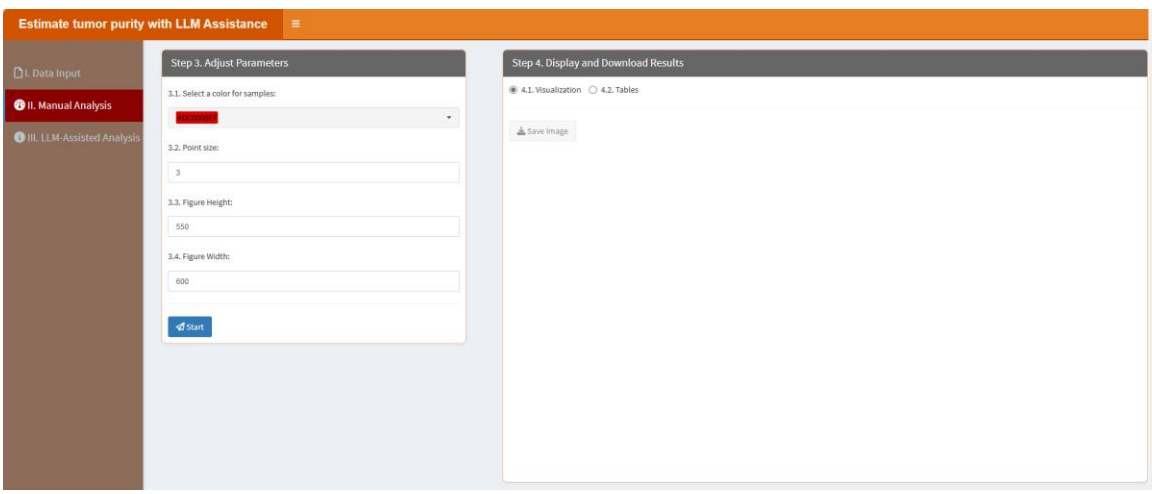
5.28. Tumor Purity Estimation

Tumor purity estimation quantifies tumor content in samples. Users can adjust estimation parameters through the classic framework, and local LLMs provide refined insights into the results. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

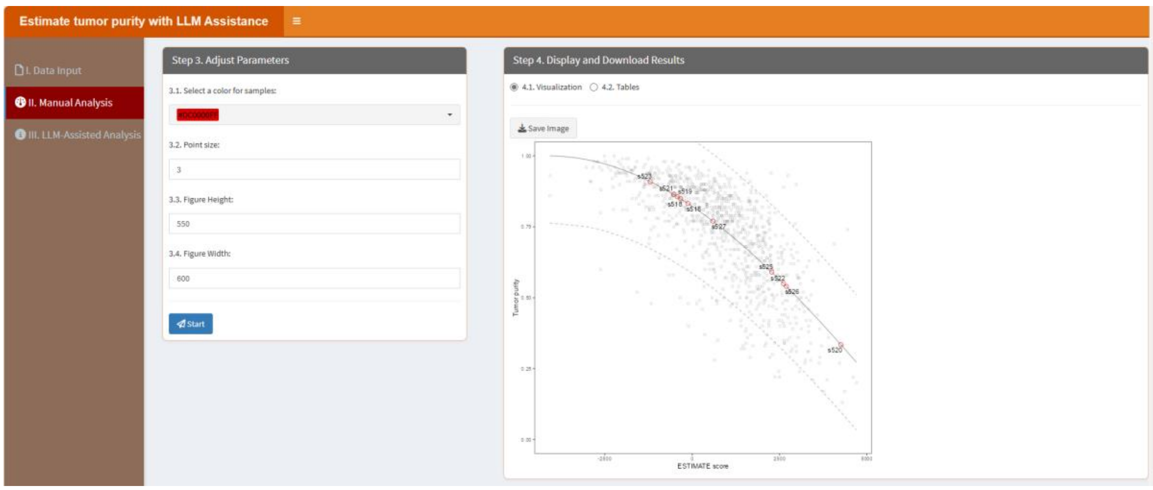


3.1. Select a color for samples: This parameter lets you choose a specific color for each sample point. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select one color for the sample points, ensuring that the visual representation is both informative and aesthetically pleasing.

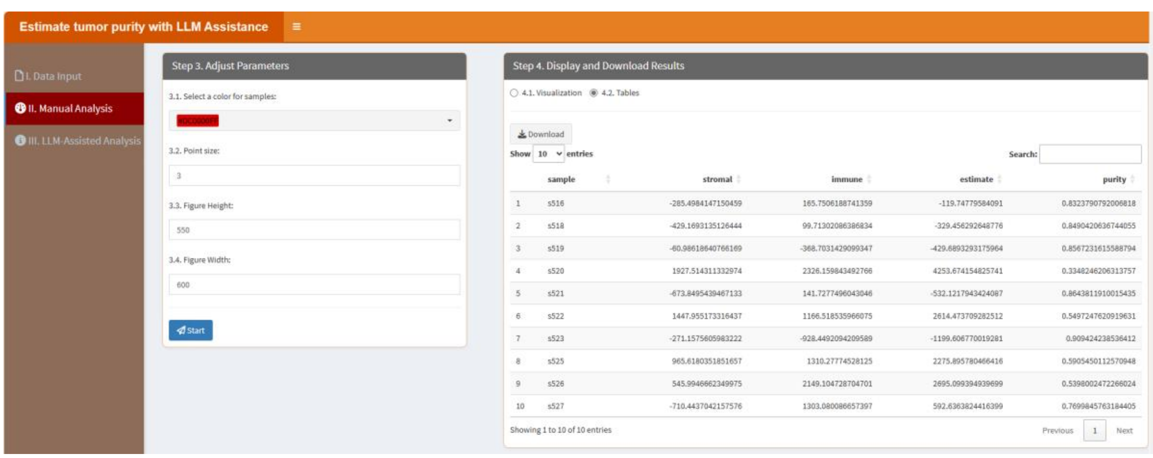
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



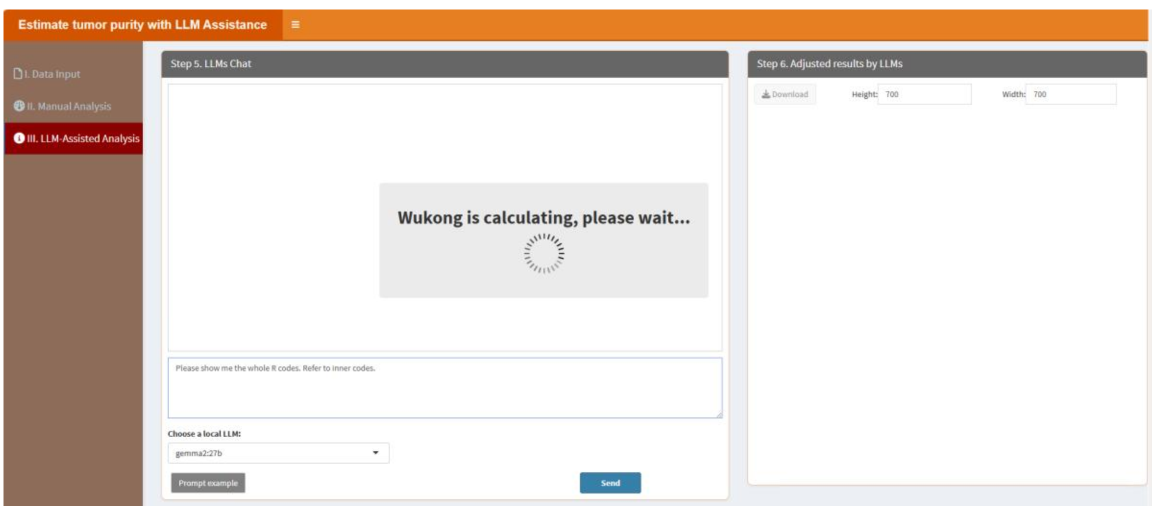
In the “4.2. Tables” part, users can click “Download” button to download the table, which contains the CV of each group:



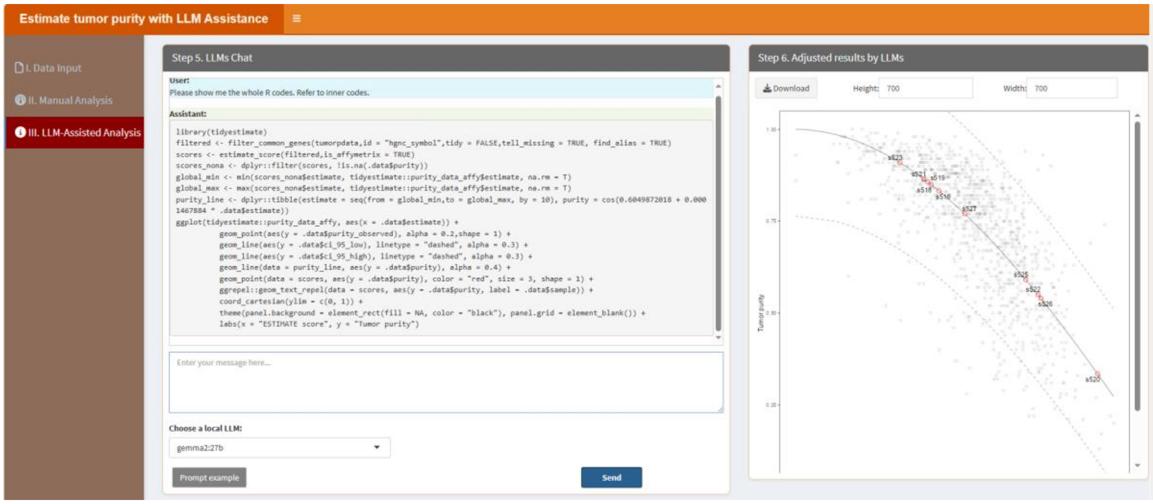
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please show me the whole R codes. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please show me the whole R codes. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



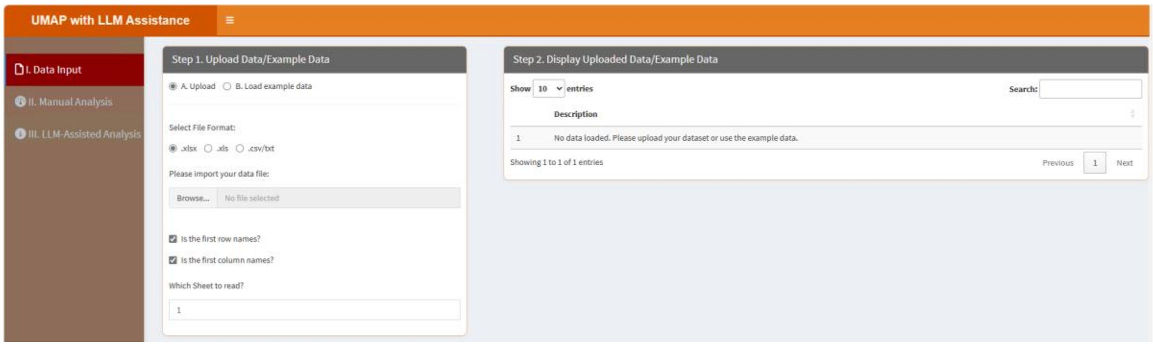
The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



Estimate tumor purity with LLM Assistance
I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis
Step 5. LLMs Chat
User:
Please show me the whole R codes. Refer to inner codes.
Assistant:
library(tidyestimate)
filtered <- filter_common_genes(tumorpdata,id = "hgnc_symbol",tidy = FALSE,tell_missing = TRUE, find_alias = TRUE)
scores <- estimate_score(filtered,is_affymetrix = TRUE)
scores_nona <- dplyr::filter(scores, !is.na(.data$purity))
global_min <- min(scores_nona$estimate, tidyestimate::purity_data_affy$estimate, na.rm = T)
global_max <- max(scores_nona$estimate, tidyestimate::purity_data_affy$estimate, na.rm = T)
purity_line <- dplyr::tibble(estimate = seq(from = global_min,to = global_max, by = 10), purity = cos(0.6049872018 + 0.0001467884 * .data$estimate))
ggplot(tidyestimate::purity_data_affy, aes(x = .data$estimate)) +
geom_point(aes(y = .data$purity_observed), alpha = 0.2,shape = 1) +
geom_line(aes(y = .data$ci_95_low), linetype = "dashed", alpha = 0.3) +
geom_line(aes(y = .data$ci_95_high), linetype = "dashed", alpha = 0.3) +
geom_line(data = purity_line, aes(y = .data$purity), alpha = 0.4) +
geom_point(data = scores, aes(y = .data$purity), color = "red", size = 3, shape = 1) +
ggrepel::geom_text_repel(data = scores, aes(y = .data$purity, label = .data$sample)) +
coord_cartesian(ylim = c(0, 1)) +
theme(panel.background = element_rect(fill = NA, color = "black"), panel.grid = element_blank()) +
labs(x = "ESTIMATE score", y = "Tumor purity")
Enter your message here...
Choose a local LLM:
gemma2:27b
Prompt example
Send
Step 6. Adjusted results by LLMs
Download
Height: 700
Width: 700
Tumor purity

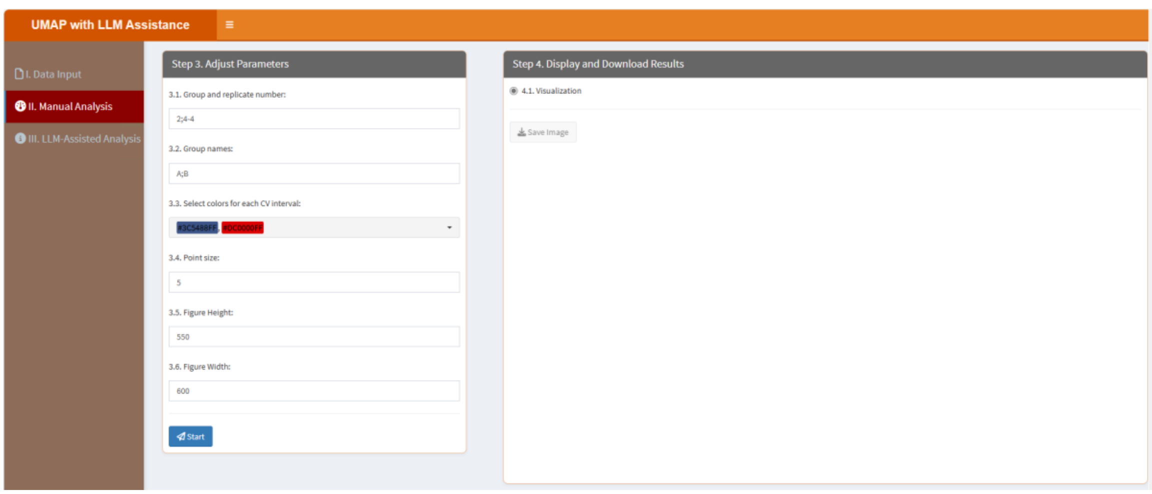
5.29. UMAP (Uniform Manifold Approximation and Projection)

UMAP is a dimensionality reduction technique for visualizing complex data. Users can configure parameters like neighbors and minimum distances via the classic framework, while local LLMs assist in refining the analysis. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Group and replicate number: This parameter allows you to define the number of groups and the number of replicates for each group in your dataset. The input format requires the group number and replicate numbers to be separated by a semicolon (;), while the replicates for each group are separated by dashes (-). For instance, if you have two groups with three replicates each, you would input "2;3-3". Similarly, for three groups with five replicates each, the input would be "3;5-5-5". This structured format ensures clarity when specifying experimental setups.

3.2. Group names: This parameter is used to assign names to the groups in your dataset. The names are separated by semicolons (;). For example, if you have two groups named "Control" and "Experiment," you would input "Control;Experiment". This helps in clearly labeling and distinguishing the groups during data analysis and visualization.

*3.3. Select colors for each CV interva*l: This parameter lets you choose specific colors to represent different CV intervals in your visualization. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of CV intervals, ensuring that the visual representation is both informative and aesthetically pleasing.

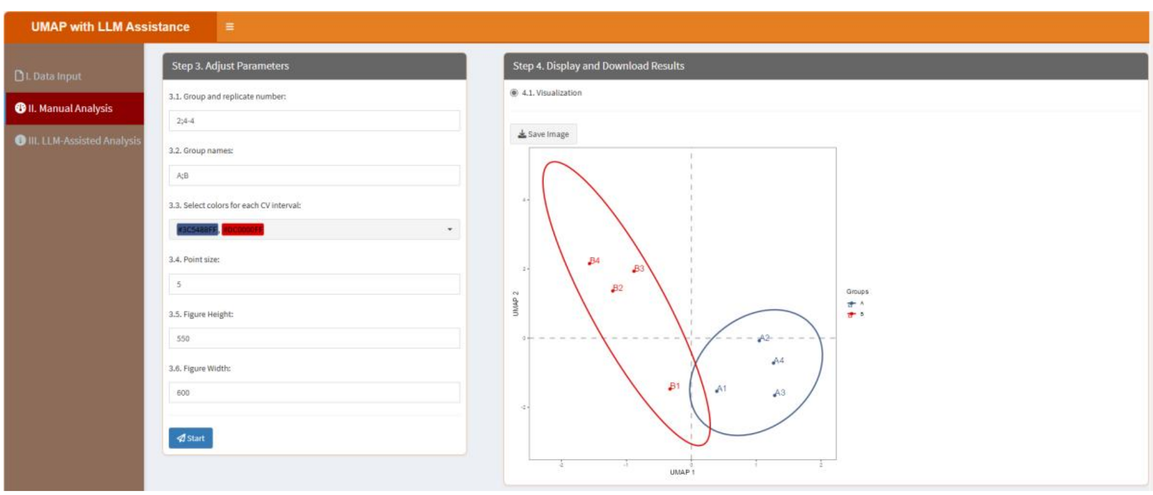
3.4. Point size: This parameter allows you to adjust the size of the points in the visualization. By specifying a numeric value, you can control how large or small the points appear, which can be useful for emphasizing certain data points or ensuring clarity when dealing with dense datasets.

3.5. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

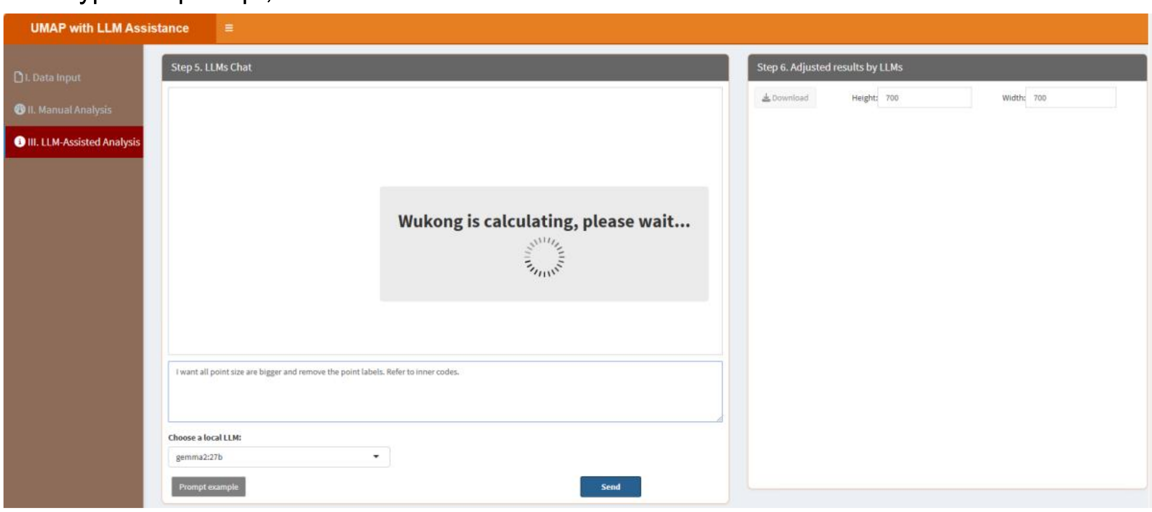
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



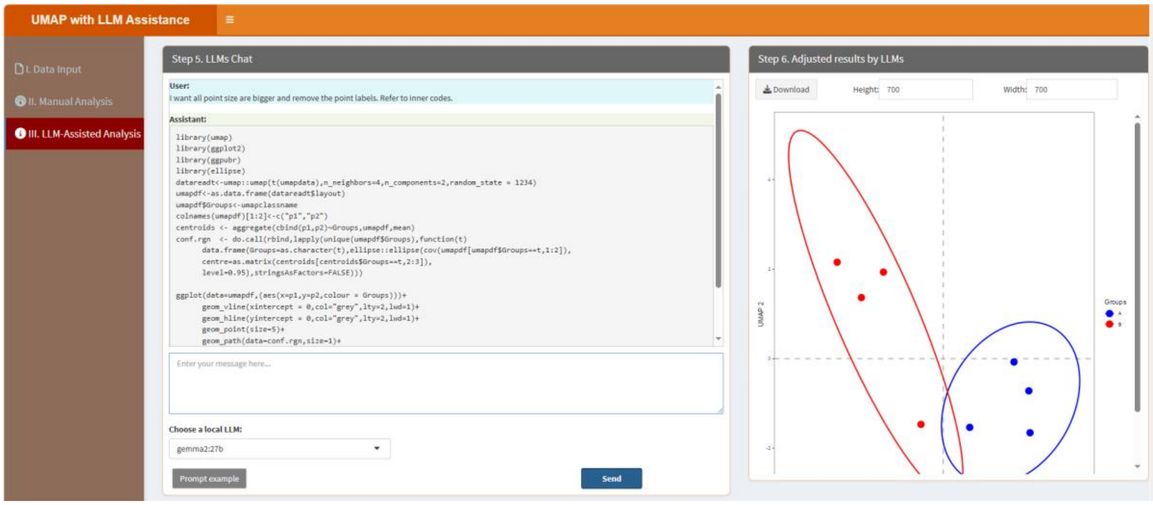
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I want all point size are bigger and remove the point labels. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I want all point size are bigger and remove the point labels. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



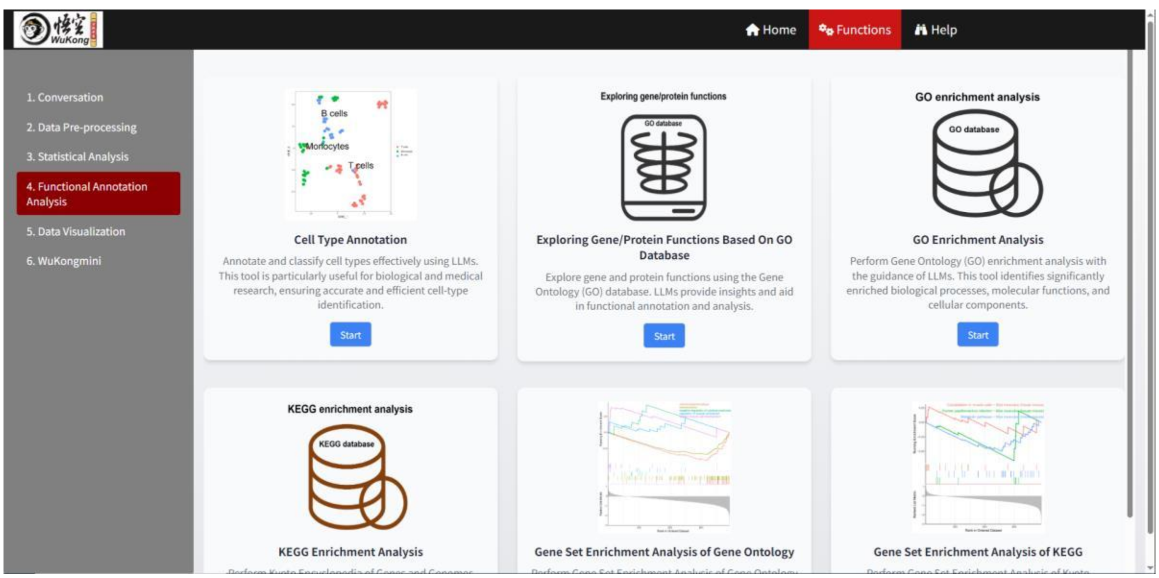
The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



UMAP with LLM Assistance
I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis
Step 5. LLMs Chat
User:
I want all point size are bigger and remove the point labels. Refer to inner codes.
Assistant:
library(umap)
library(ggplot2)
library(ggpubr)
library(ellipse)
datareadt<-umap::umap(t(umapdata),n_neighbors=4,n_components=2,random_state = 1234)
umapdf<-as.data.frame(datareadt$layout)
umapdf$Groups<-umapclassname
colnames(umapdf)[1:2]<-c("p1","p2")
centroids <- aggregate(cbind(p1,p2)~Groups,umapdf,mean)
conf.rgn <- do.call(rbind,lapply(unique(umapdf$Groups),function(t)
data.frame(Groups=as.character(t),ellipse::ellipse(cov(umapdf[umapdf$Groups==t,1:2]),
centre=as.matrix(centroids[centroids$Groups==t,2:3]),
level=0.95),stringsAsFactors=FALSE)))
ggplot(data=umapdf,(aes(x=p1,y=p2,colour = Groups)))+
geom_vline(xintercept = 0,col="grey",lty=2,lwd=1)+
geom_hline(yintercept = 0,col="grey",lty=2,lwd=1)+
geom_point(size=5)+
geom_path(data=conf.rgn,size=1)+
Enter your message here...
Choose a local LLM:
gemma2:27b
Prompt example
Send
Step 6. Adjusted results by LLMs
Download
Height: 700
Width: 700
UMAP 2
Groups
A
B

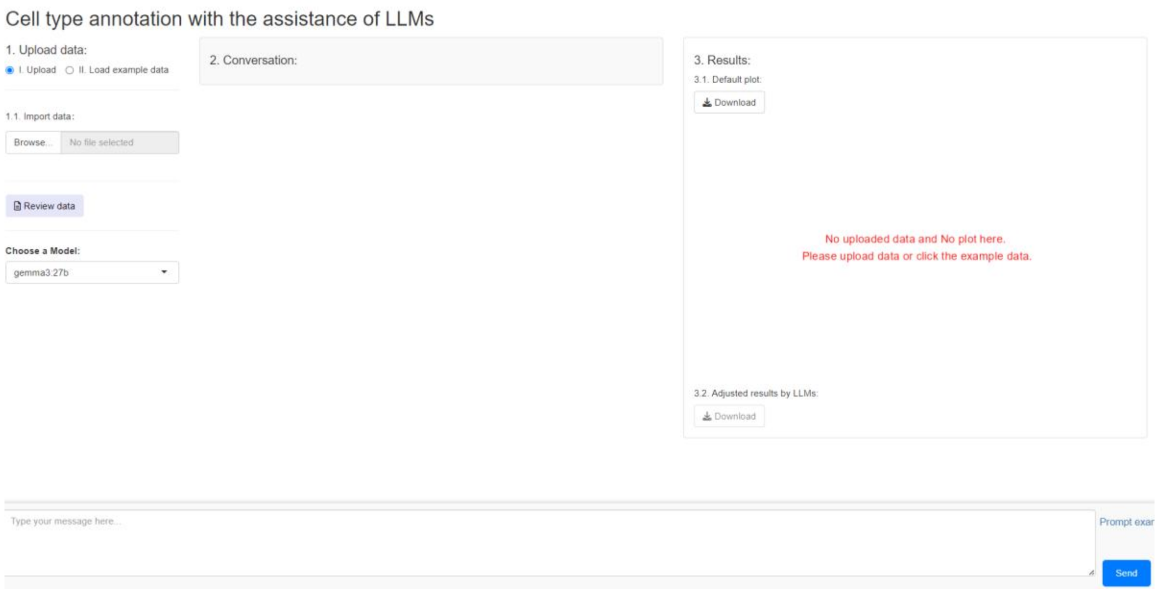
6. Functional Annotation Analysis

Functional Annotation Analysis in proteomics is a critical approach for understanding the biological significance of genes and proteins by exploring their functions, associated pathways, and enriched biological processes. With the integration of LLMs, researchers can not only obtain classic enrichment results but also deeply explore and interpret these findings with enhanced precision and efficiency. LLMs assist in cell type annotation, enabling accurate classification and identification of cell types, which is particularly valuable in biological and medical research. They also facilitate the exploration of gene and protein functions through the Gene Ontology (GO) database, providing insights into molecular functions, biological processes, and cellular components. Additionally, LLMs enhance GO and KEGG enrichment analyses, guiding researchers in identifying significantly enriched pathways, annotating enriched gene sets, and deriving meaningful biological insights. By contextualizing findings and suggesting potential implications, LLMs empower researchers to better understand complex biological relationships and design informed experimental validations or research directions.

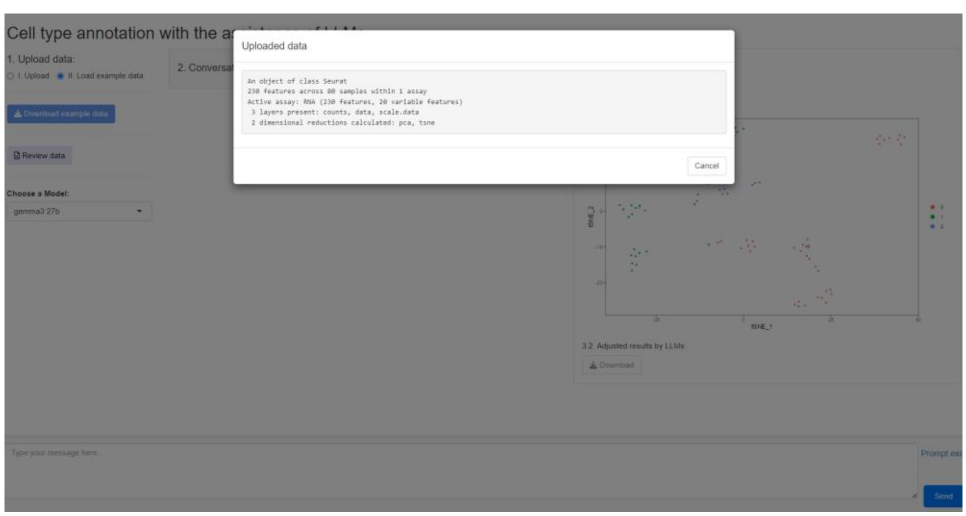


6.1. Cell type annotation

Annotate and classify cell types effectively using the local LLMs. This tool is particularly useful for biological and medical research, ensuring accurate and efficient cell-type identification. Users can adjust parameters to refine their analysis and leverage LLMs to enhance the accuracy of their results. After clicking the “Start” button, it will be started in a new window like below:

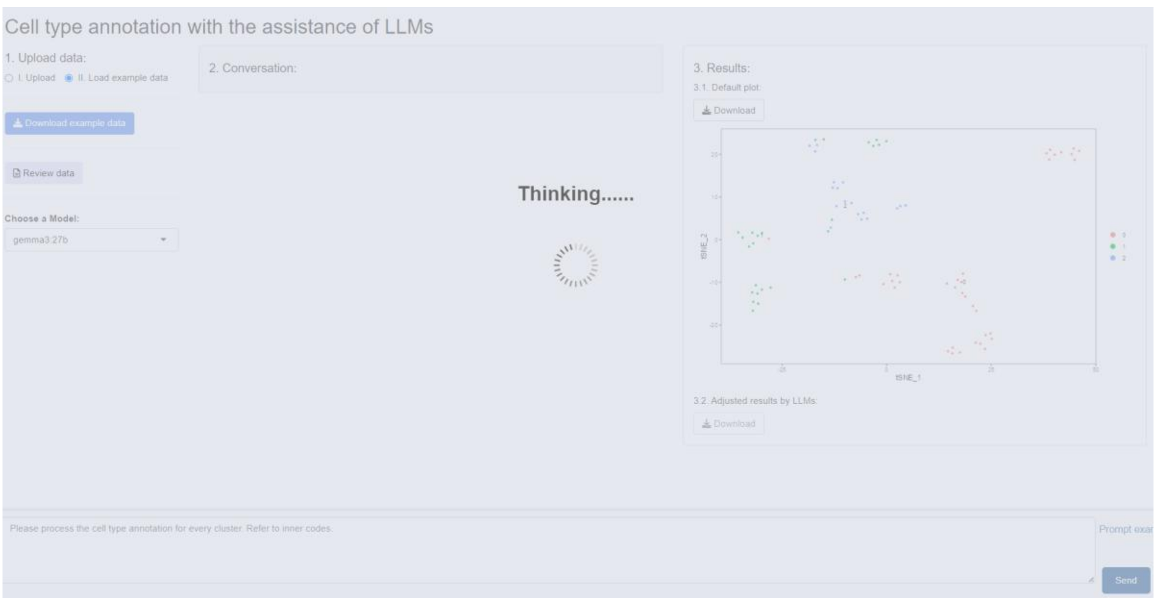


6.1.1. Upload data. Herein, users can upload their data (.RDS format) from “I. Upload”. Please note the data here should be an object of class Seurat, which means users should process their single cell data in advance using Seurat package and save the results without annotation into a .RDS file. In addition, users can select “II. Load example data” and click the button “Review data” to check the example data. Next, users can select a local LLM (e.g. gemma3:27b). As below:

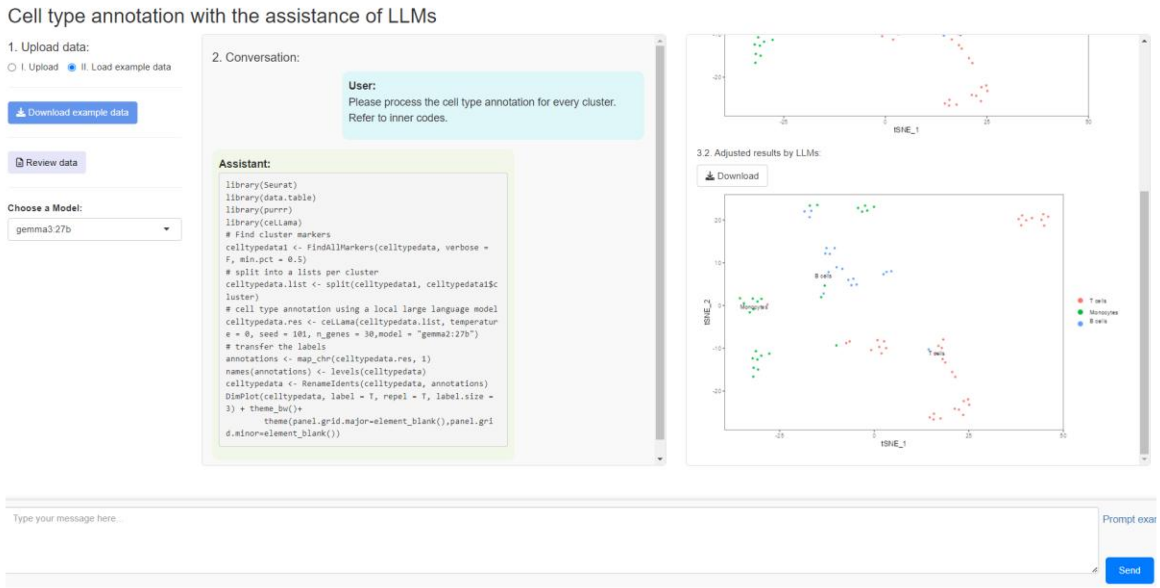


6.1.2. Conversation. Here displays the conversation history, while "Type your message here..." enables users to type messages or questions in the provided text box. When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results. For instance, the example prompt “Please process the cell type annotation for every cluster. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please process the cell type annotation for every cluster. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the results.

After type the prompt, click “Send” button. It will learn like below:



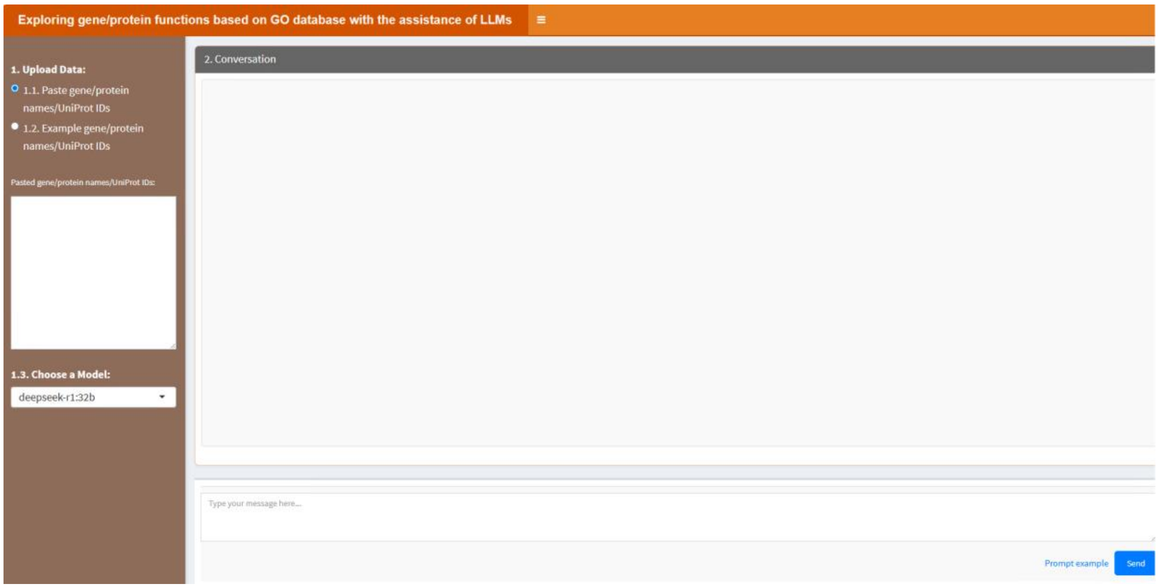
6.1.3. Results. Here shows two results: 3.1. Default plot shows the cell distribution without cell annotation. 3.2. Adjusted results by LLMs means that this panel shows the results (figures or tables) based on the conversation in the "2. Conversation". The results are shown as below.



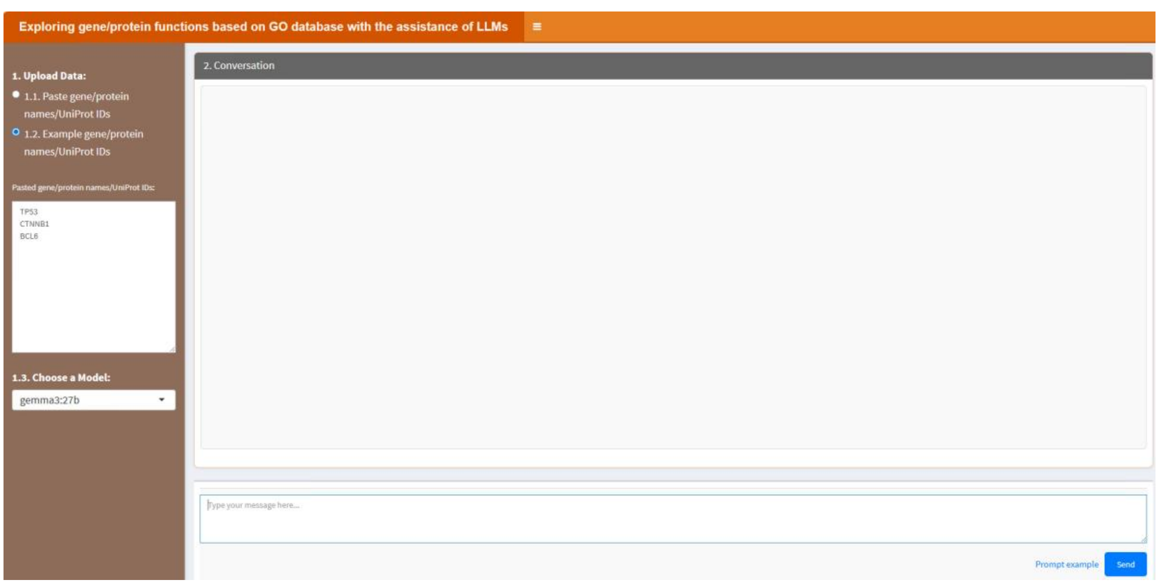
Cell type annotation with the assistance of LLMs
1. Upload data:
I. Upload
II. Load example data
Download example data
Review data
Choose a Model:
gemma3:27b
2. Conversation:
User:
Please process the cell type annotation for every cluster.
Refer to inner codes.
Assistant:
library(Seurat)
library(data.table)
library(purrr)
library(ceLLama)
# Find cluster markers
celltypedata1 <- FindAllMarkers(celltypedata, verbose = F, min.pct = 0.5)
# split into a lists per cluster
celltypedata.list <- split(celltypedata1, celltypedata1$cluster)
# cell type annotation using a local large language model
celltypedata.res <- ceLLama(celltypedata.list, temperature = 0, seed = 101, n_genes = 30,model = "gemma2:27b")
# transfer the labels
annotations <- map_chr(celltypedata.res, 1)
names(annotations) <- levels(celltypedata)
celltypedata <- RenameIdents(celltypedata, annotations)
DimPlot(celltypedata, label = T, repel = T, label.size = 3) + theme_bw()+
theme(panel.grid.major=element_blank(),panel.grid.minor=element_blank())
tSNE_1
3.2. Adjusted results by LLMs:
Download
tSNE_2
B cells
Monocytes
T cells
Type your message here...
Send

6.2. Exploring gene/protein functions based on GO database

Explore gene and protein functions using the GO database with the help of local LLMs. This tool allows users to adjust parameters for detailed functional annotation and analysis while leveraging LLMs to provide deeper insights into gene and protein roles. After clicking the “Start” button, it will be started in a new window like below:



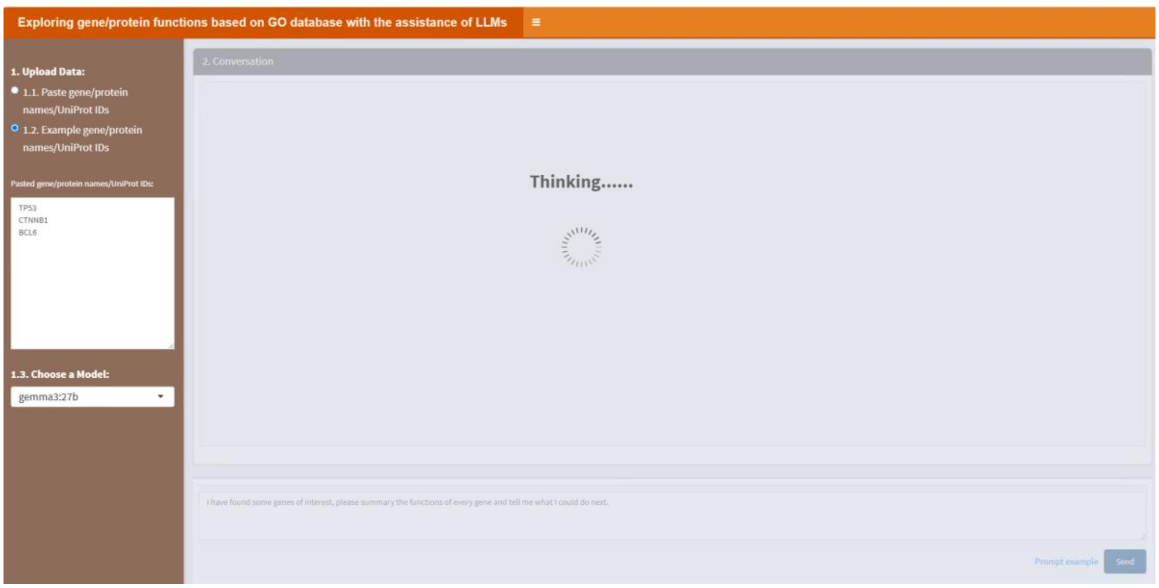
In the “1. Upload Data”, users can paste gene/protein names, or UniProt IDs of interest (e.g. those proteins obtained from differential expression analysis) in the below box when selecting “1.1. Paste gene/protein names/UniProt IDs”. Or, users can select “1.2. Example gene/protein names/UniProt IDs” to check the example, as below:



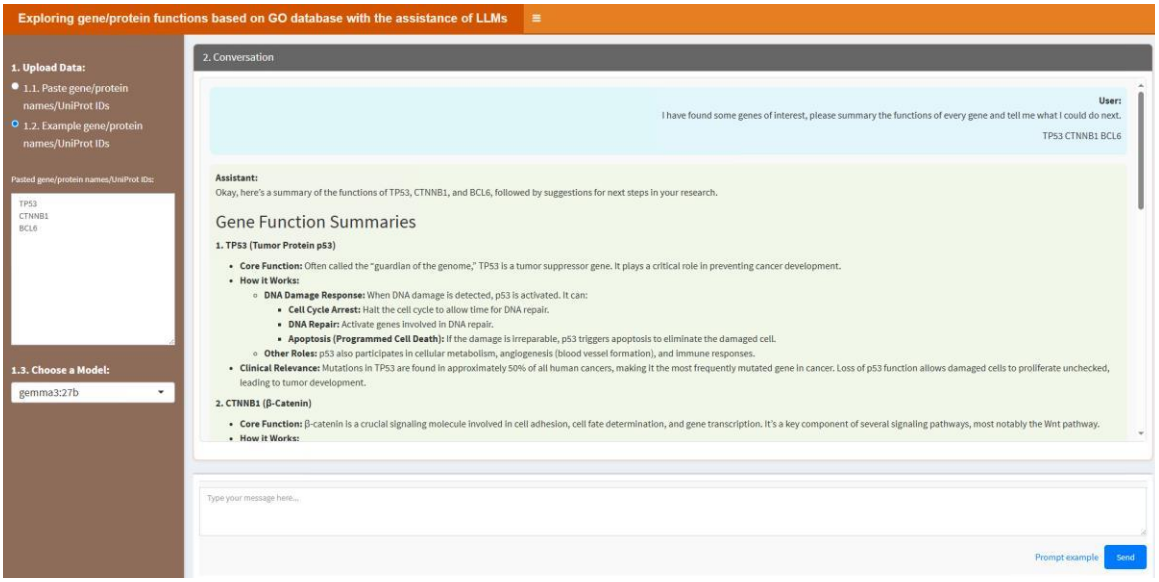
Then, users can select a pre-installed LLM from the "1.3. Choose a Model" option for their interactions (please see below “How to install LLMs locally?” part).

In the “2. Conversation”, it displays the conversation history, while "Type your message here..." enables users to type messages or questions in the provided text box. When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. For instance, the example prompt “I have found some genes of interest, please summary the functions of every gene and tell me what I could do next.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I have found some genes of interest, please summary the functions of every gene and tell me what I could do next. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

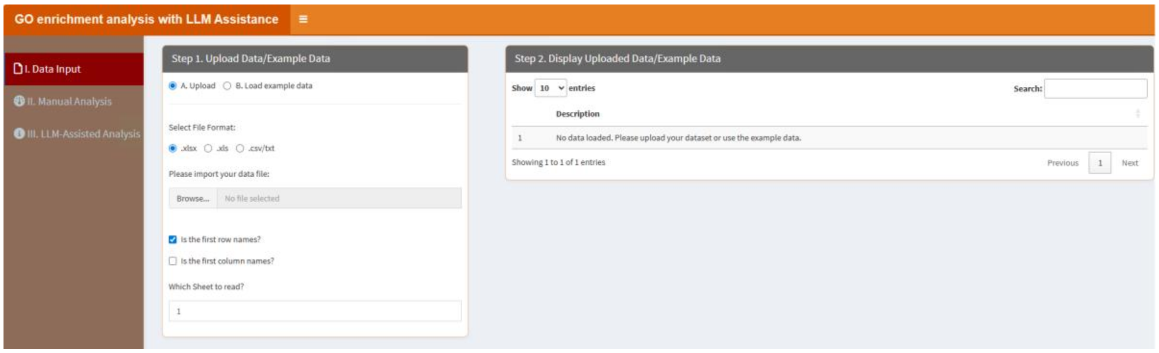


The results are shown as below.


Exploring gene/protein functions based on GO database with the assistance of LLMs
1. Upload Data:
1.1. Paste gene/protein names/UniProt IDs
1.2. Example gene/protein names/UniProt IDs
Pasted gene/protein names/UniProt IDs:
TP53
CTNNB1
BCL6
1.3. Choose a Model:
gemma3:27b
2. Conversation
User:
I have found some genes of interest, please summary the functions of every gene and tell me what I could do next.
TP53 CTNNB1 BCL6
Assistant:
Okay, here's a summary of the functions of TP53, CTNNB1, and BCL6, followed by suggestions for next steps in your research.
Gene Function Summaries
1. TP53 (Tumor Protein p53)
Core Function: Often called the "guardian of the genome," TP53 is a tumor suppressor gene. It plays a critical role in preventing cancer development.
How it Works:
DNA Damage Response: When DNA damage is detected, p53 is activated. It can:
Cell Cycle Arrest: Halt the cell cycle to allow time for DNA repair.
DNA Repair: Activate genes involved in DNA repair.
Apoptosis (Programmed Cell Death): If the damage is irreparable, p53 triggers apoptosis to eliminate the damaged cell.
Other Roles: p53 also participates in cellular metabolism, angiogenesis (blood vessel formation), and immune responses.
Clinical Relevance: Mutations in TP53 are found in approximately 50% of all human cancers, making it the most frequently mutated gene in cancer. Loss of p53 function allows damaged cells to proliferate unchecked, leading to tumor development.
2. CTNNB1 (β-Catenin)
Core Function: β-catenin is a crucial signaling molecule involved in cell adhesion, cell fate determination, and gene transcription. It's a key component of several signaling pathways, most notably the Wnt pathway.
How it Works:
Type your message here...
Prompt example
Send

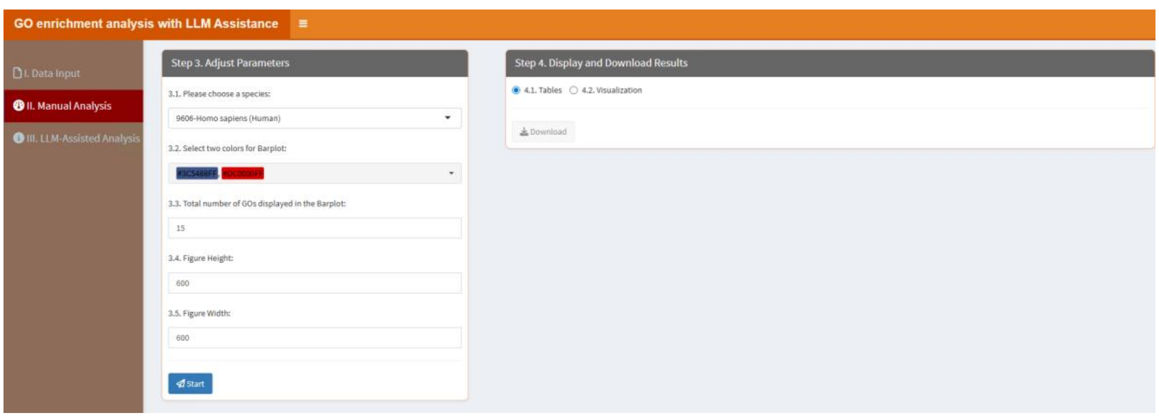
6.3. GO enrichment analysis

Perform GO (Gene Ontology) enrichment analysis by adjusting parameters through a classic framework and refining results with local LLMs. This tool identifies significantly enriched biological processes, molecular functions, and cellular components, offering enhanced interpretability and accuracy. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



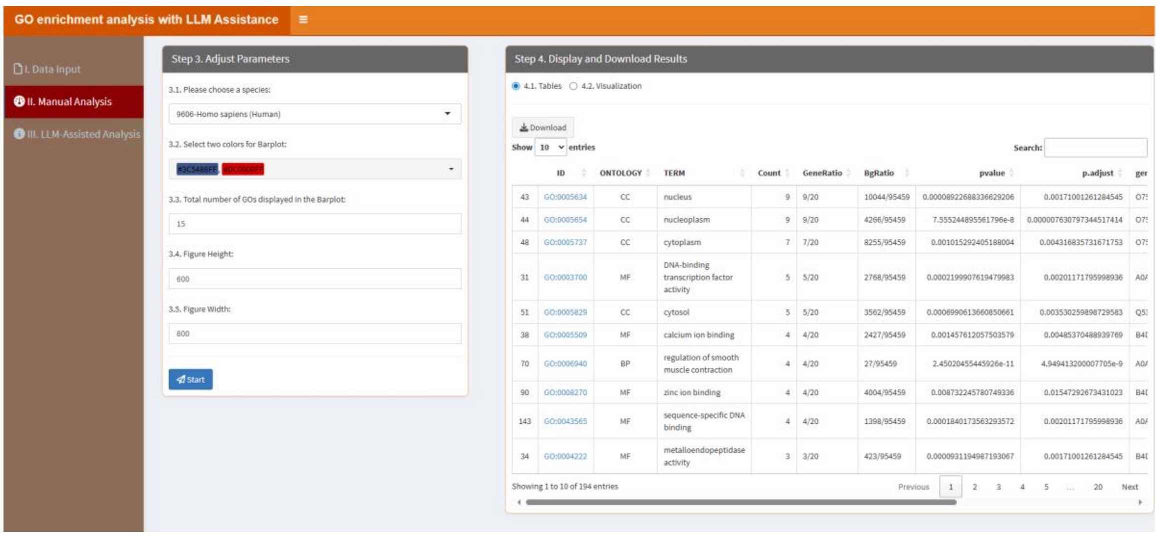
3.1. Please choose a species: Select the reference species for GO term mapping and annotation.

3.2. Select two colors for Barplot: Choose two distinct colors to visually represent GO term categories in the bar plot output. These colors will be applied in a gradient or alternating format to help distinguish between different GO terms or significance levels.

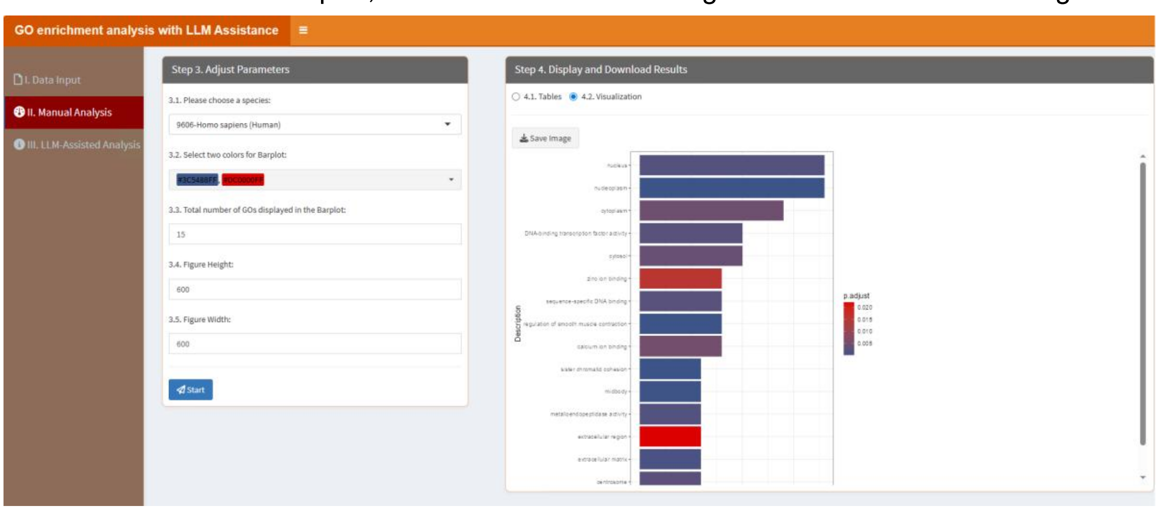
3.3. Total number of GOs displayed in the Barplot: Specify the maximum number of enriched GO terms to be visualized in the bar plot. The default value here is 15, users can adjust this based on their preferences or dataset size.

3.4. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Tables” part, users can click “Download” button to download the table:



In the “4.2. Visualization” part, users can click “Save Image” button to download the figure:

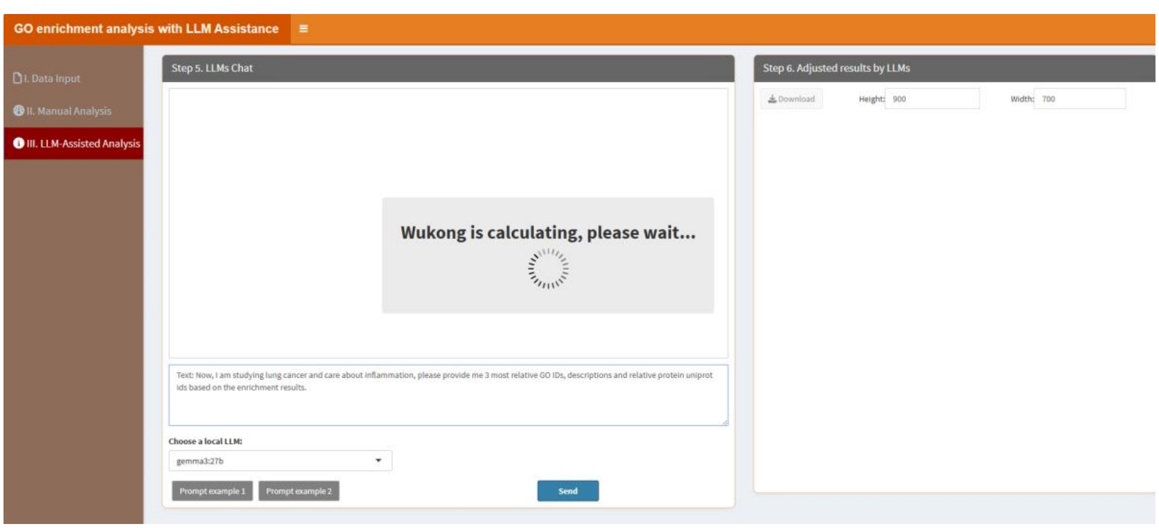


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..."

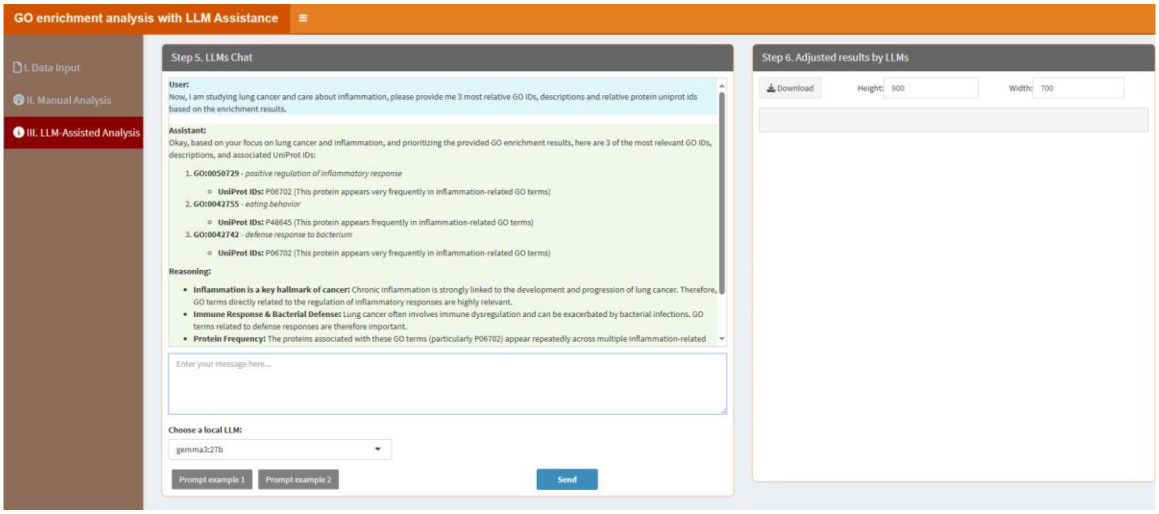
enables users to type messages or questions in the provided text box. Two types of prompts are supported: prompts beginning with "Text:" are used to request interpretation or explanation of the enrichment results (e.g. summarizing biological processes or clarifying GO term relevance), while prompts beginning with "Plot:" allow users to customize the barplot visualization (e.g. modifying colors, sorting order, or number of displayed GO terms). Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt 1 "Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results. =>", it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the first prompt, click "Send" button. It will learn like below:

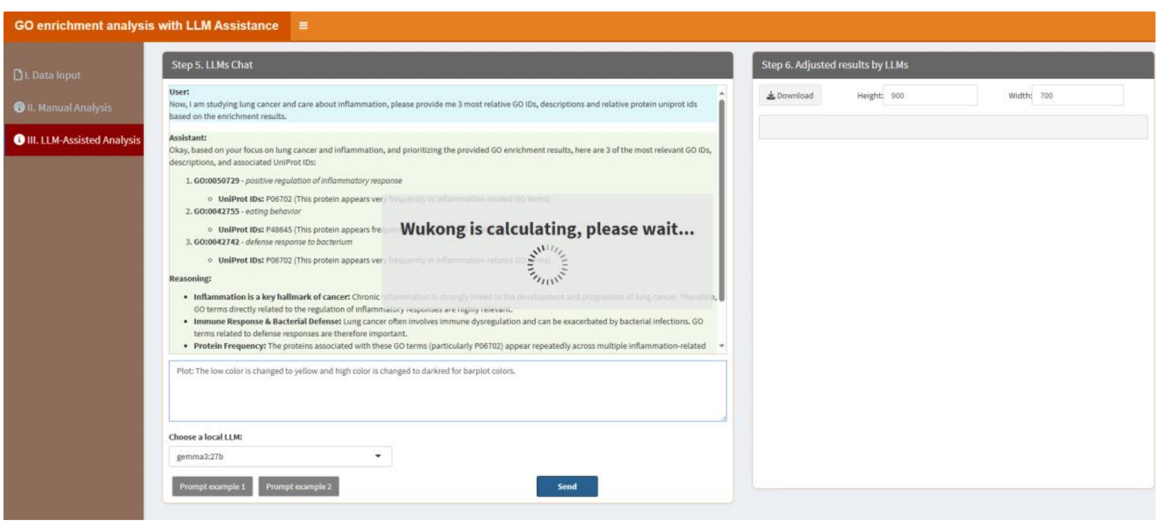


The results are shown as below:

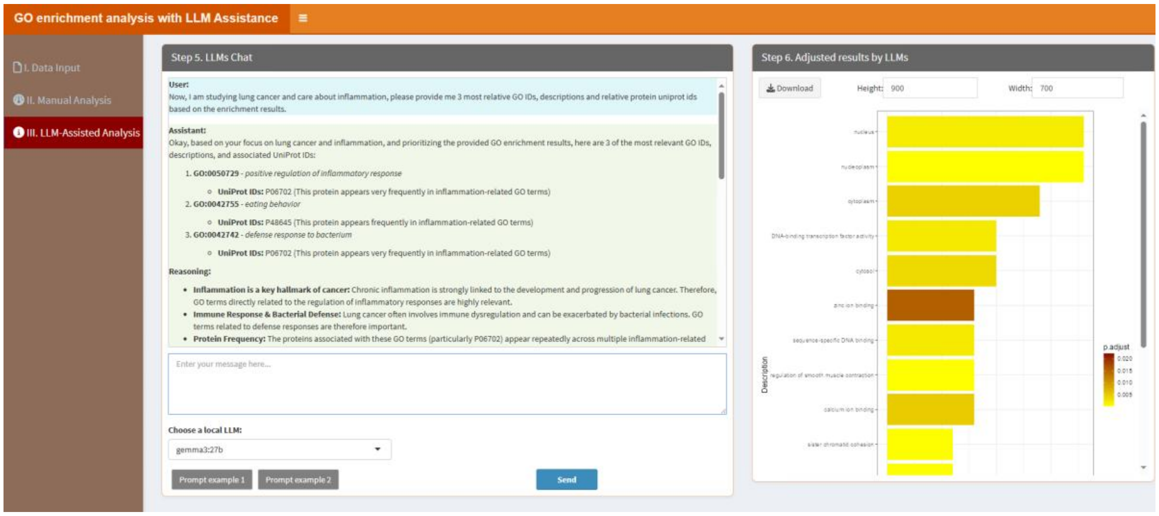


On the other hand, the example prompt 2 "Plot: The low color is changed to yellow and high color is changed to darkred for barplot colors." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Plot: The low color is changed to yellow and high color is changed to darkred for barplot colors. =>", it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the second prompt, click "Send" button. It will learn like below:

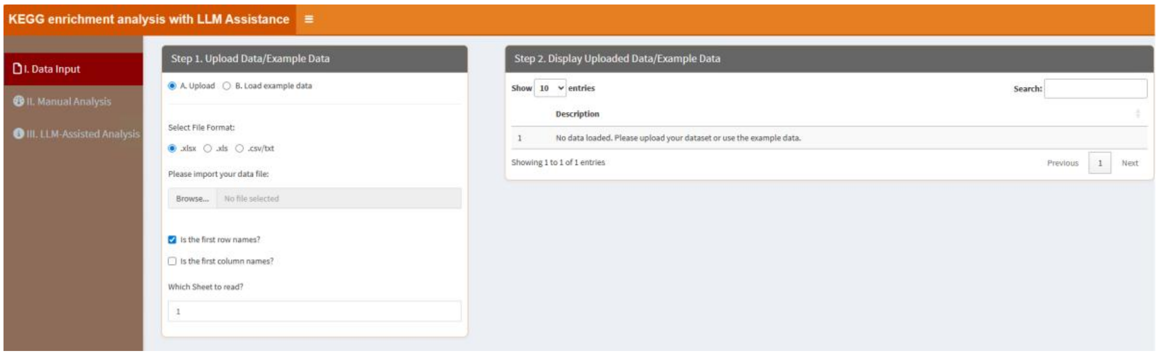


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.


GO enrichment analysis with LLM Assistance
I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis
Step 5. LLMs Chat
User:
Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results.
Assistant:
Okay, based on your focus on lung cancer and inflammation, and prioritizing the provided GO enrichment results, here are 3 of the most relevant GO IDs, descriptions, and associated UniProt IDs:
1. GO:0050729 - positive regulation of inflammatory response
UniProt IDs: P06702 (This protein appears very frequently in inflammation-related GO terms)
2. GO:0042755 - eating behavior
UniProt IDs: P48645 (This protein appears frequently in inflammation-related GO terms)
3. GO:0042742 - defense response to bacterium
UniProt IDs: P06702 (This protein appears very frequently in inflammation-related GO terms)
Reasoning:
Inflammation is a key hallmark of cancer: Chronic inflammation is strongly linked to the development and progression of lung cancer. Therefore, GO terms directly related to the regulation of inflammatory responses are highly relevant.
Immune Response & Bacterial Defense: Lung cancer often involves immune dysregulation and can be exacerbated by bacterial infections. GO terms related to defense responses are therefore important.
Protein Frequency: The proteins associated with these GO terms (particularly P06702) appear repeatedly across multiple inflammation-related
Enter your message here...
Choose a local LLM:
gemma3:27b
Prompt example 1
Prompt example 2
Send
Step 6. Adjusted results by LLMs
Download
Height: 900
Width: 700
nucleus
nucleoplasm
cytoplasm
DNA-binding transcription factor activity
cytosol
zinc ion binding
sequence-specific DNA binding
regulation of smooth muscle contraction
calcium ion binding
sister chromatid cohesion
Description
p.adjust
0.020
0.015
0.010
0.005

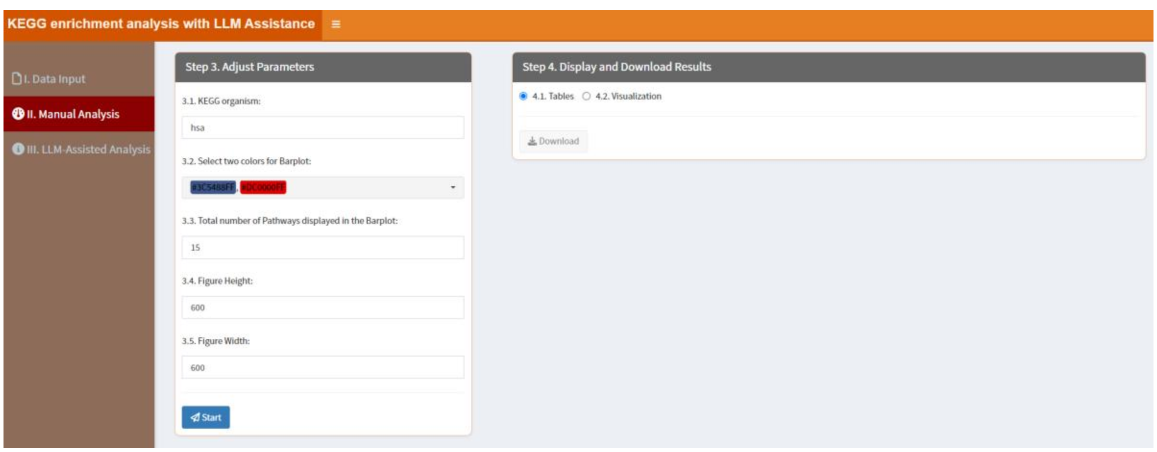
6.4. KEGG enrichment analysis

Perform KEGG (Kyoto Encyclopedia of Genes and Genomes) enrichment analysis by customizing parameters in a classic framework and utilizing local LLMs to refine pathway enrichment results. This tool helps identify enriched pathways and biological processes with improved precision and usability. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. KEGG organism: Type the organism code (e.g. hsa for Homo sapiens, mmu for Mus musculus) corresponding to the dataset being analyzed.

3.2. Select two colors for Barplot: Choose two distinct colors to visually represent pathways in the bar plot output. These colors will be applied in a gradient or alternating format to help distinguish between different pathways or significance levels.

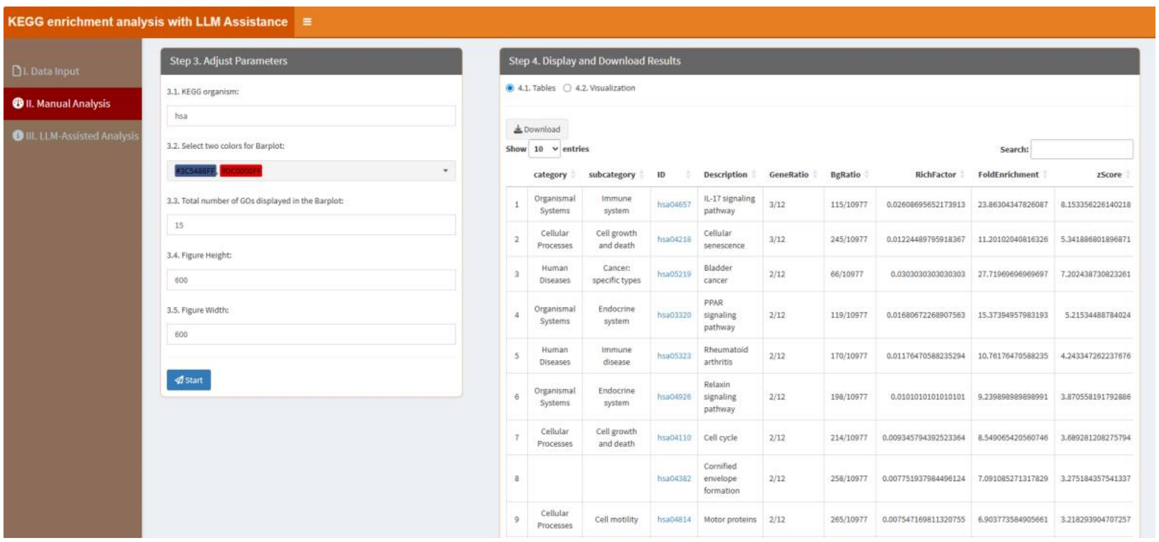
3.3. Total number of Pathways displayed in the Barplot: Specify the maximum number of enriched GO terms to be visualized in the bar plot. The default value here is 15, users can adjust this based on their preferences or dataset size.

3.4. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

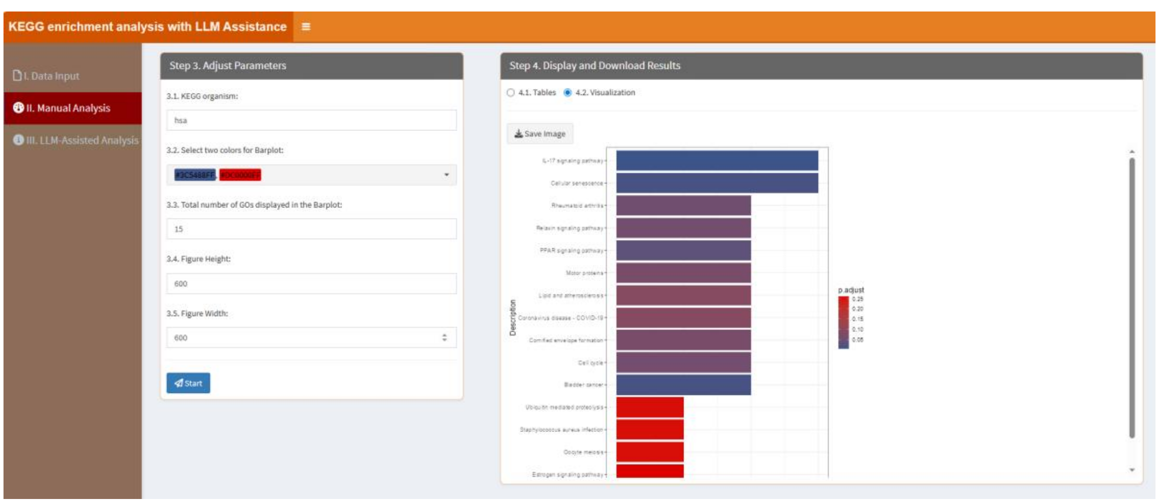
3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Tables” part, users can click “Download” button to download the table:



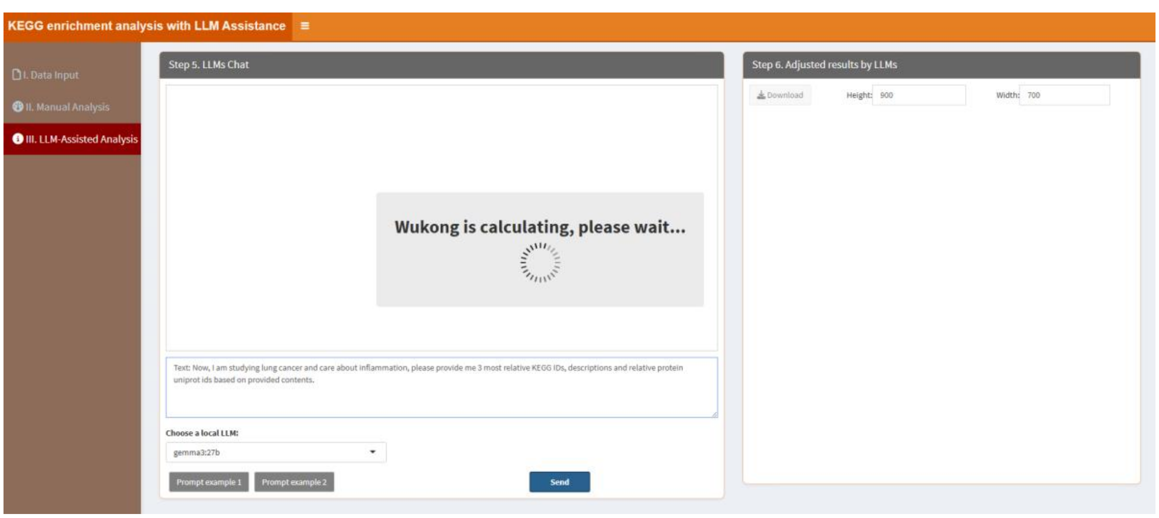
In the “4.2. Visualization” part, users can click “Save image” button to download the figure:



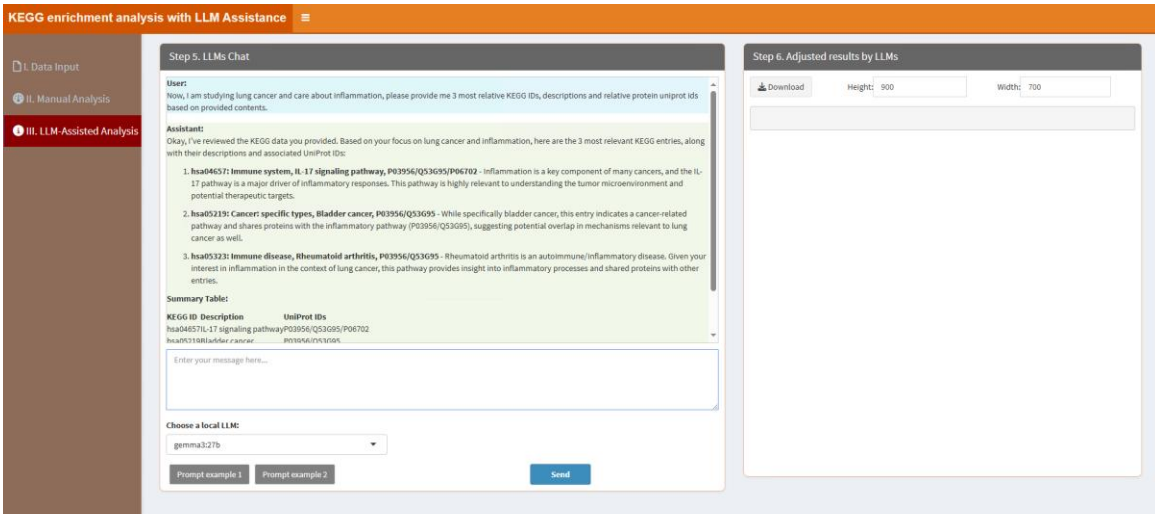
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Two types of prompts are supported: prompts beginning with “Text:” are used to request interpretation or explanation of the enrichment results (e.g. summarizing pathways relevance), while prompts beginning with “Plot:” allow users to customize the barplot visualization (e.g. modifying colors, sorting order, or number of displayed pathways). Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt 1 “Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative KEGG IDs, descriptions and relative protein uniprot ids based on provided contents.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative KEGG IDs, descriptions and relative protein uniprot ids based on provided contents. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the first prompt click “Send” button. It will learn like below:

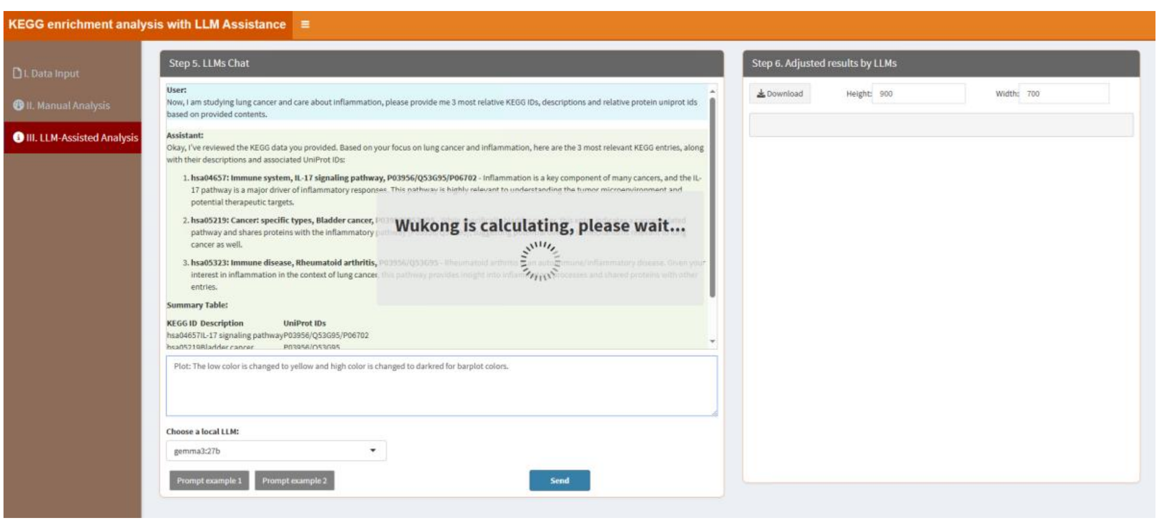


The results are shown as below:

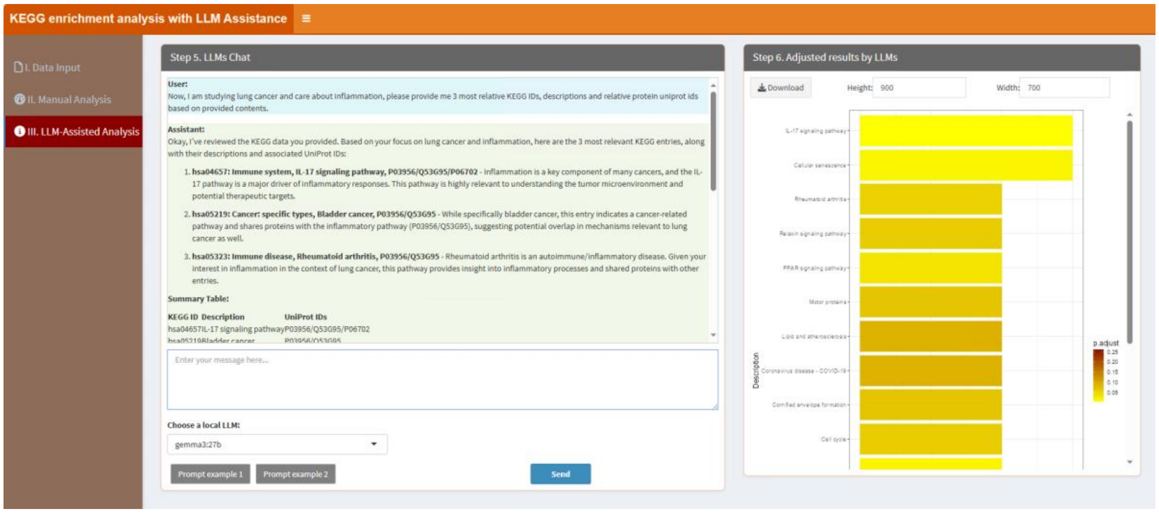


On the other hand, the example prompt 2 “Plot: The low color is changed to yellow and high color is changed to darkred for barplot colors. ”instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Plot: The low color is changed to yellow and high color is changed to darkred for barplot colors.=>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the second prompt, click “Send” button. It will learn like below:



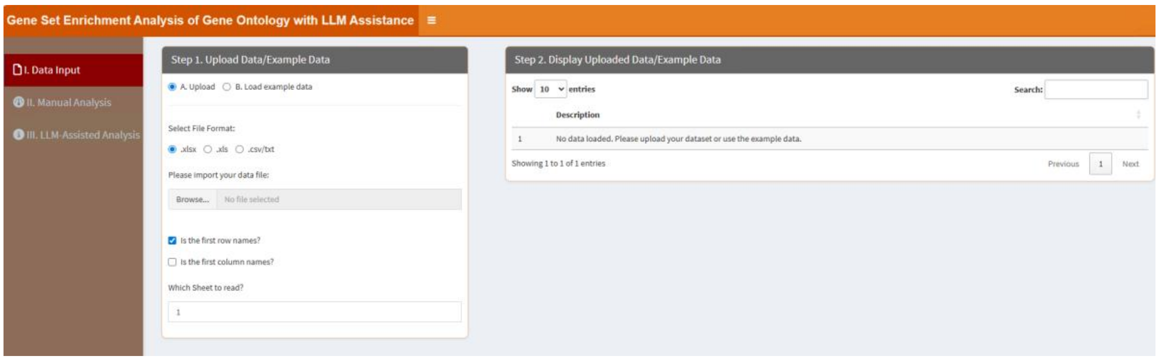
The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



KEGG enrichment analysis with LLM Assistance
I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis
Step 5. LLMs Chat
User:
Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative KEGG IDs, descriptions and relative protein uniprot ids based on provided contents.
Assistant:
Okay, I've reviewed the KEGG data you provided. Based on your focus on lung cancer and inflammation, here are the 3 most relevant KEGG entries, along with their descriptions and associated UniProt IDs:
1. hsa04657: Immune system, IL-17 signaling pathway, P03956/Q53G95/P06702 - Inflammation is a key component of many cancers, and the IL-17 pathway is a major driver of inflammatory responses. This pathway is highly relevant to understanding the tumor microenvironment and potential therapeutic targets.
2. hsa05219: Cancer: specific types, Bladder cancer, P03956/Q53G95 - While specifically bladder cancer, this entry indicates a cancer-related pathway and shares proteins with the inflammatory pathway (P03956/Q53G95), suggesting potential overlap in mechanisms relevant to lung cancer as well.
3. hsa05323: Immune disease, Rheumatoid arthritis, P03956/Q53G95 - Rheumatoid arthritis is an autoimmune/inflammatory disease. Given your interest in inflammation in the context of lung cancer, this pathway provides insight into inflammatory processes and shared proteins with other entries.
Summary Table:
KEGG ID Description UniProt IDs
hsa04657IL-17 signaling pathwayP03956/Q53G95/P06702
Enter your message here...
Choose a local LLM:
gemma3:27b
Prompt example 1
Prompt example 2
Send
Step 6. Adjusted results by LLMs
Download
Height: 900
Width: 700
IL-17 signaling pathway
Cellular senescence
Rheumatoid arthritis
Relaxin signaling pathway
PPAR signaling pathway
Motor proteins
Lipid and atherosclerosis
Coronavirus disease - COVID-19
Cornified envelope formation
Cell cycle
Description
p.adjust
0.25
0.20
0.15
0.10
0.05

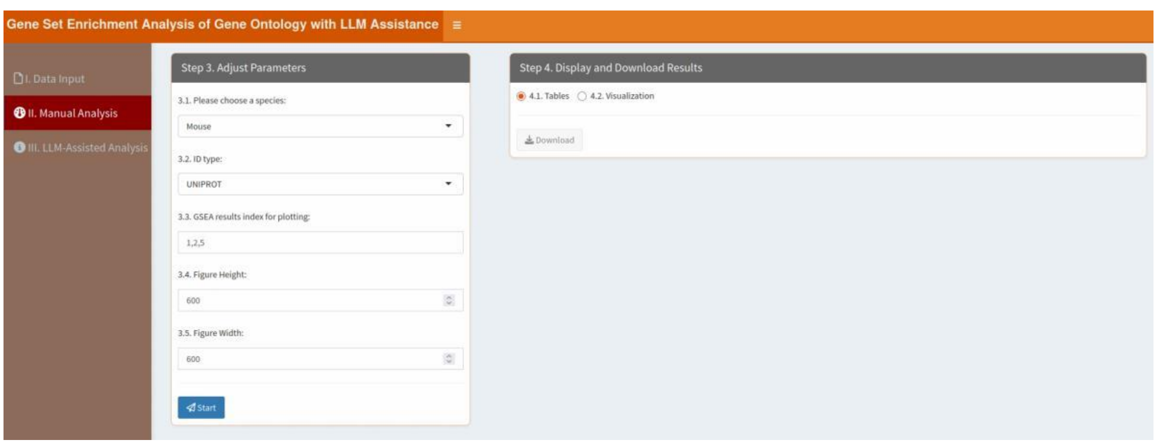
6.5. Gene set enrichment analysis of GO

Perform gene set enrichment analysis based on the GO database by adjusting parameters through a classic framework and leveraging local LLMs to refine results. LLMs assist in interpreting enriched gene sets, prioritizing GO terms, and deriving meaningful biological insights. They also provide suggestions for experimental validation or further research. After clicking the "Start" button, it will be started in a new window like below:



The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.



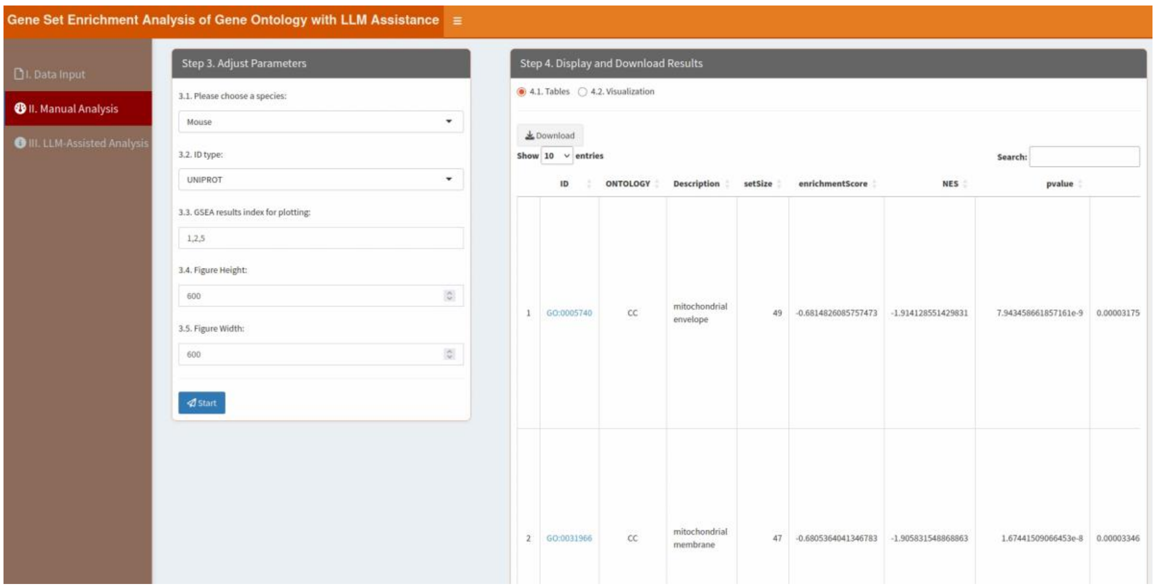
3.1. Please choose a species: Select the reference species for GO term mapping and annotation.

3.2. ID type: Select the type of gene/protein identifiers used in your dataset (e.g. UniProt ID, Entrez ID).

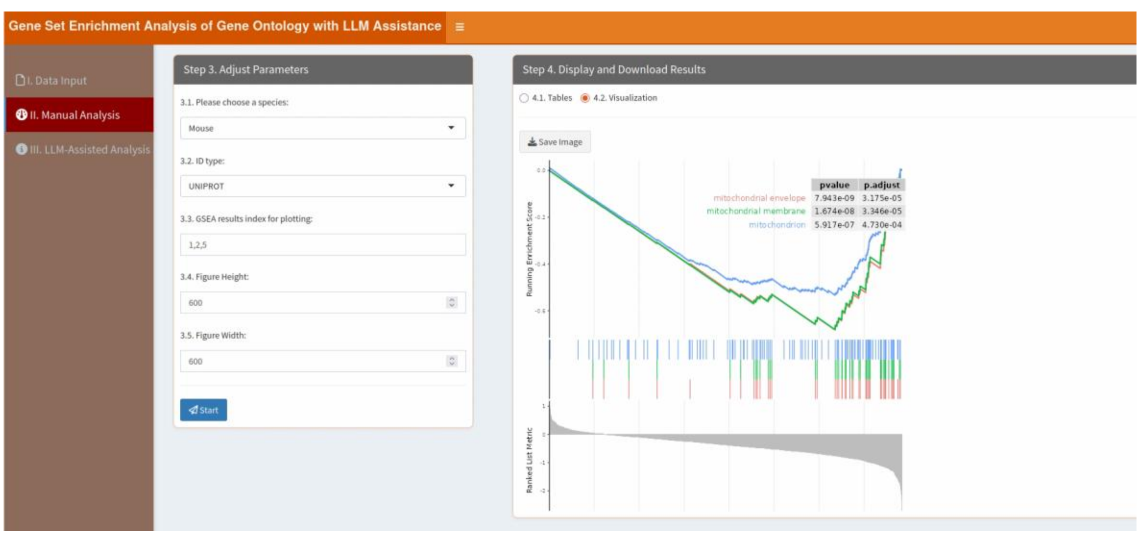
3.3. GSEA results index for plotting: Specify the index to visualize enriched GO terms. Enter '1' here to plot the first enrichment result, '1,2' to plot the first one and the second one, and so on. If you enter '1-3', it means plotting the first to the third results on one plot.

3.4. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Tables” part, users can click “Download” button to download the table:



In the “4.2. Visualization” part, users can click “Save Image” button to download the figure:

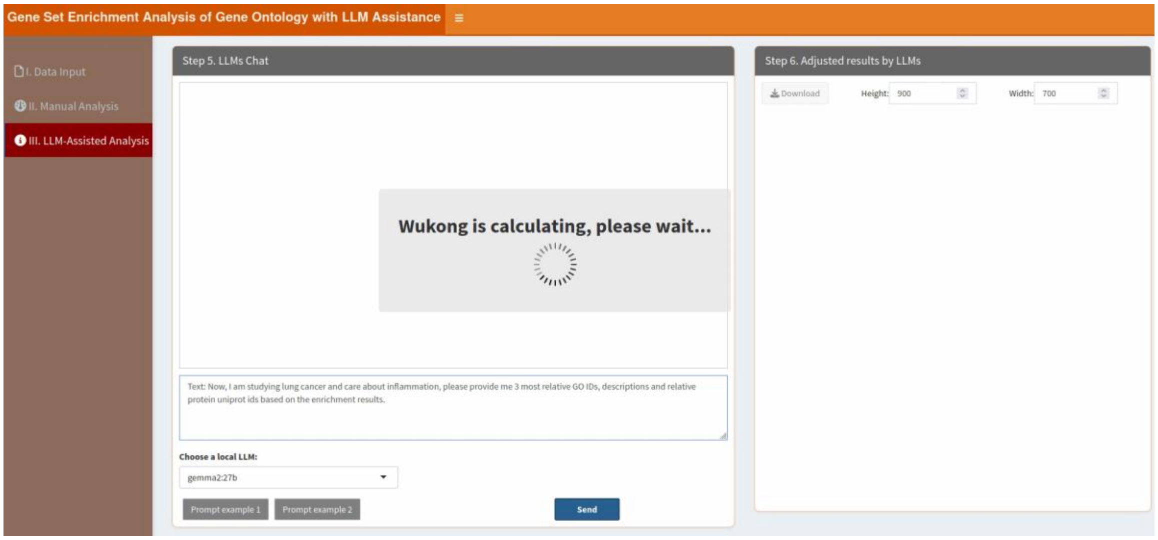


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..."

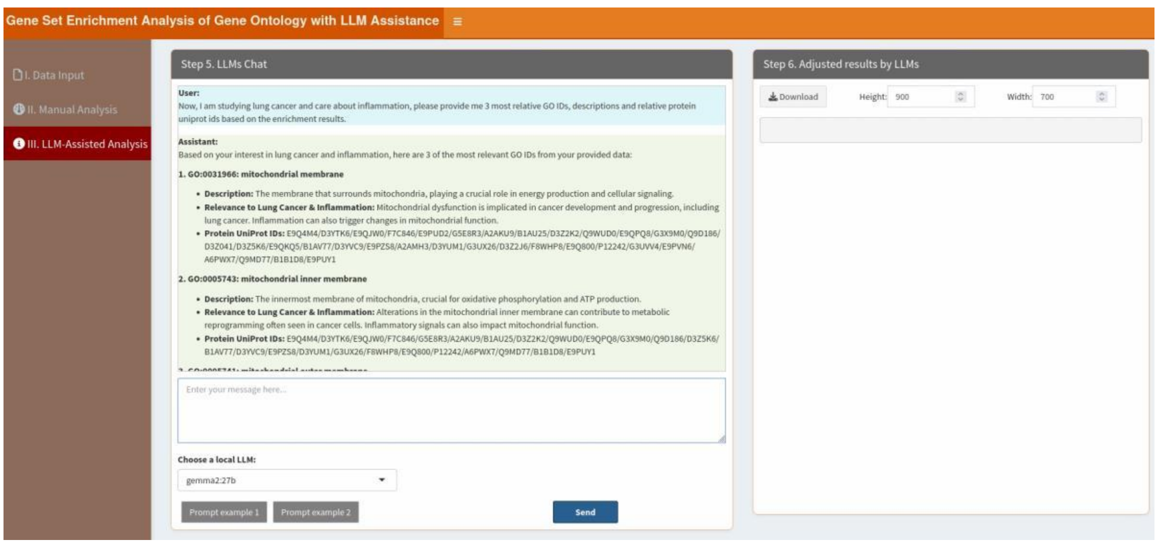
enables users to type messages or questions in the provided text box. Two types of prompts are supported: prompts beginning with “Text:” are used to request interpretation or explanation of the enrichment results (e.g. summarizing biological processes or clarifying GO term relevance), while prompts beginning with “Plot:” allow users to customize the barplot visualization (e.g. modifying colors, sorting order, or number of displayed GO terms). Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt 1 “Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the first prompt, click “Send” button. It will learn like below:

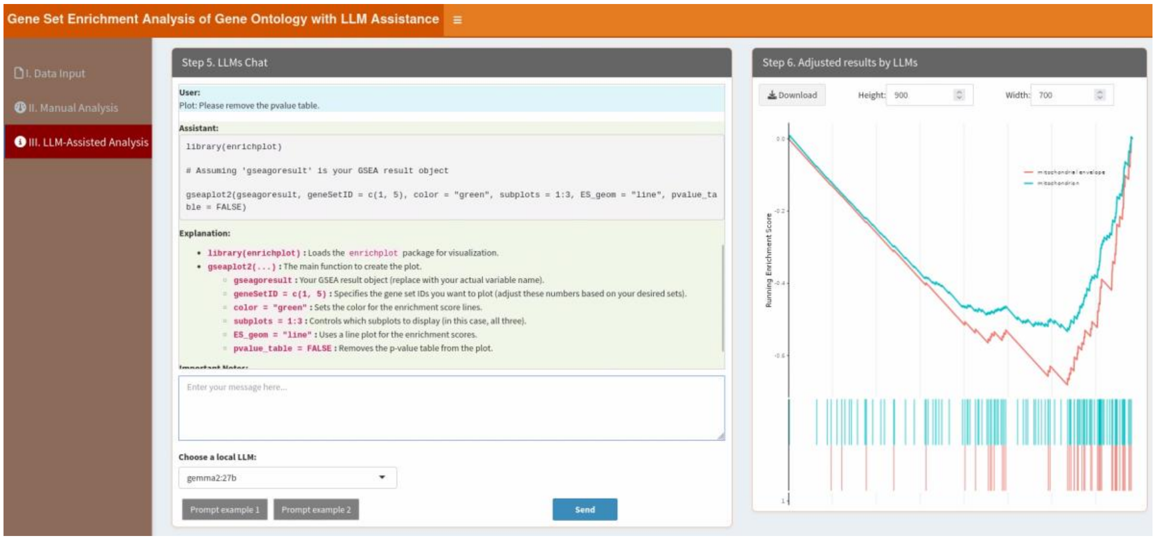


The results are shown as below:



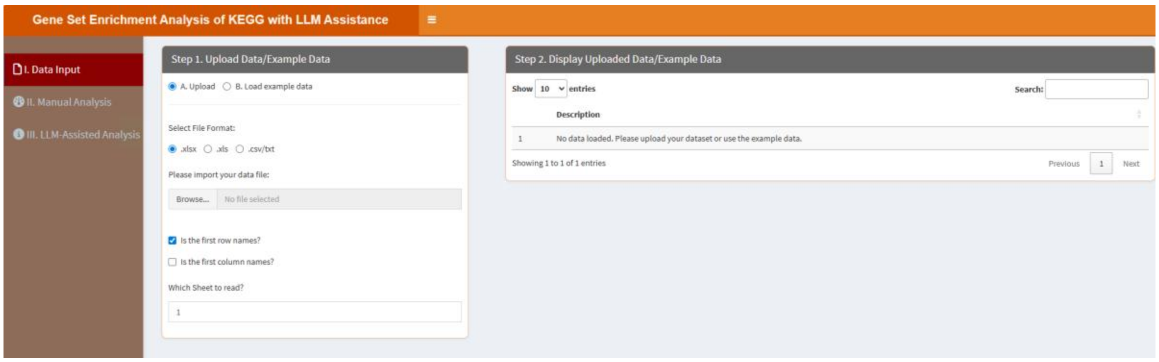
On the other hand, the example prompt 2 “Plot: Please remove the pvalue table. ”instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Plot: Please remove the pvalue table.=>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt example 2, click “Send” button, the results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.


Gene Set Enrichment Analysis of Gene Ontology with LLM Assistance
I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis
Step 5. LLMs Chat
User:
Plot: Please remove the pvalue table.
Assistant:
library(enrichplot)
# Assuming 'gseagoresult' is your GSEA result object
gseaplot2(gseagoresult, geneSetID = c(1, 5), color = "green", subplots = 1:3, ES_geom = "line", pvalue_table = FALSE)
Explanation:
library(enrichplot) : Loads the enrichplot package for visualization.
gseaplot2(...) : The main function to create the plot.
gseagoresult : Your GSEA result object (replace with your actual variable name).
geneSetID = c(1, 5) : Specifies the gene set IDs you want to plot (adjust these numbers based on your desired sets).
color = "green" : Sets the color for the enrichment score lines.
subplots = 1:3 : Controls which subplots to display (in this case, all three).
ES_geom = "line" : Uses a line plot for the enrichment scores.
pvalue_table = FALSE : Removes the p-value table from the plot.
Enter your message here...
Choose a local LLM:
gemma2:27b
Prompt example 1
Prompt example 2
Send
Step 6. Adjusted results by LLMs
Download
Height: 900
Width: 700
Running Enrichment Score

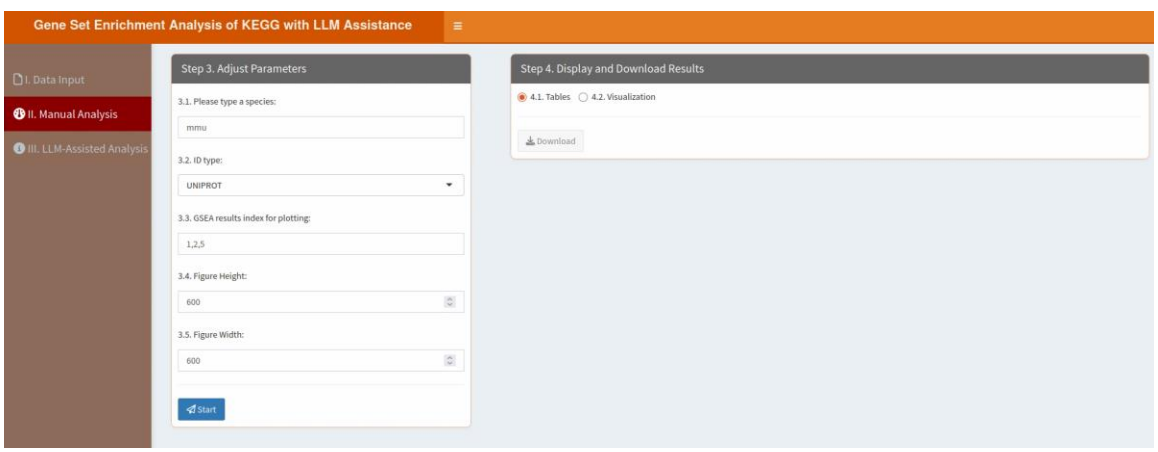
6.6. Gene set enrichment analysis of KEGG

Perform gene set enrichment analysis based on the KEGG database using a classic framework for parameter adjustments and local LLMs for enhanced interpretation. LLMs help identify key biological pathways, contextualize findings, and suggest potential implications or regulatory mechanisms for further exploration. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



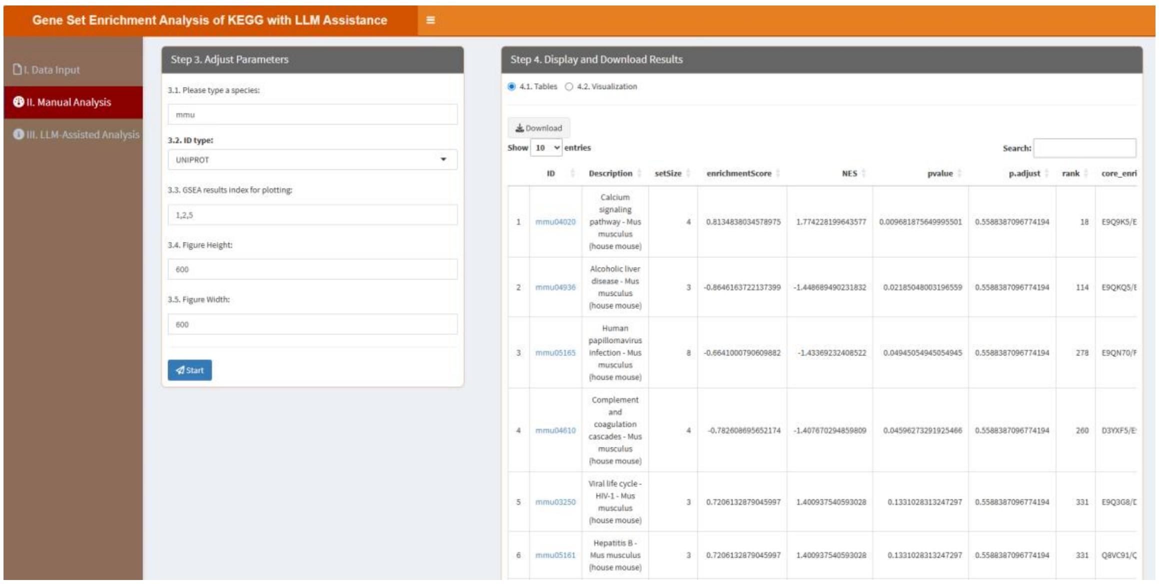
3.1. Please type a species: Type the organism code (e.g., hsa for Homo sapiens, mmu for Mus musculus) corresponding to the dataset being analyzed.

3.2. ID type: Select the type of gene/protein identifiers used in your dataset (e.g., UniProt ID, Entrez ID).

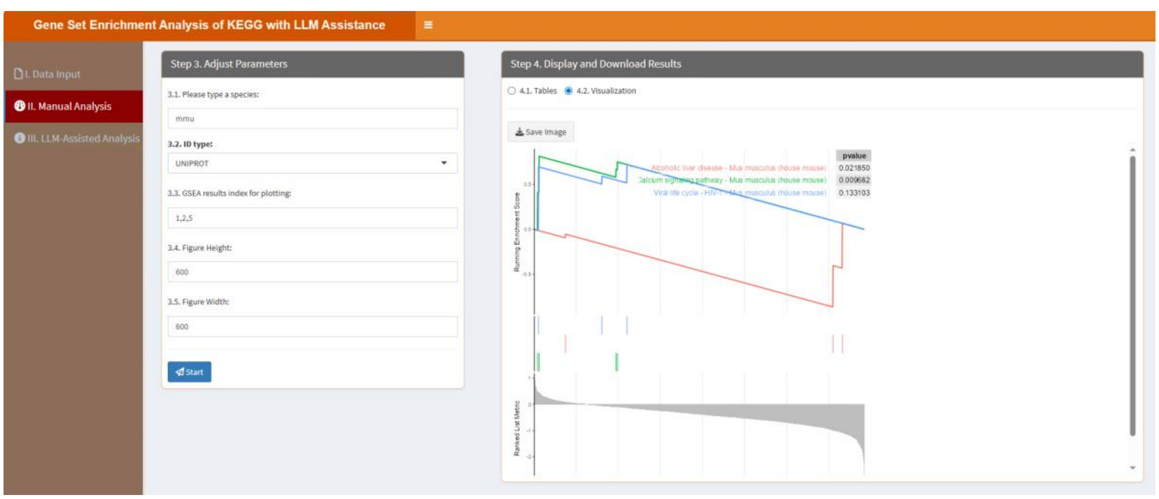
3.3. GSEA results index for plotting: Specify the index to visualize enriched GO terms. Enter '1' here to plot the first enrichment result, '1,2' to plot the first one and the second one, and so on. If you enter '1-3', it means plotting the first to the third results on one plot.

3.4. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Tables” part, users can click “Download” button to download the table:



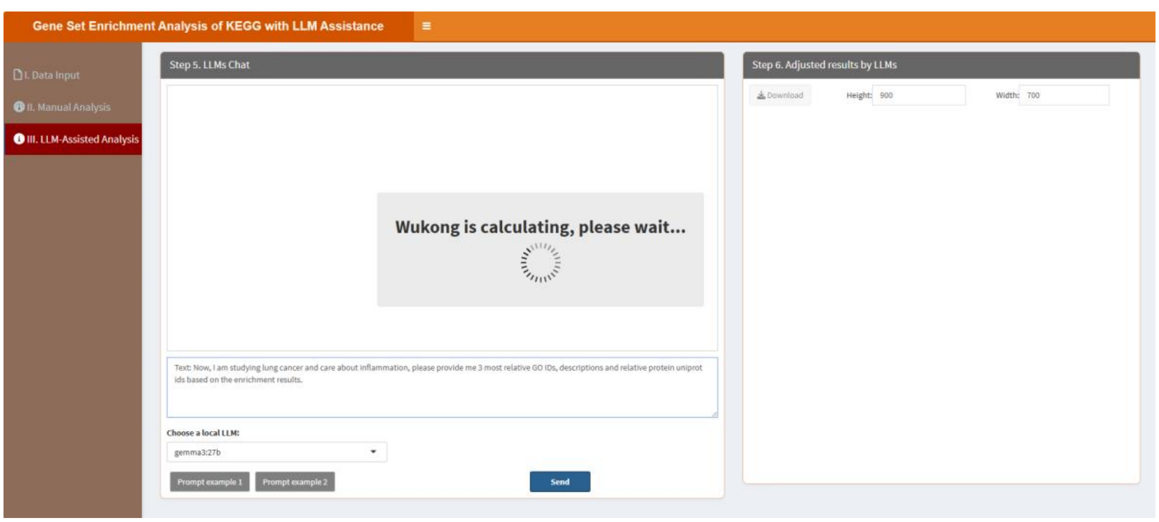
In the “4.2. Visualization” part, users can click “Save Image” button to download the figure:



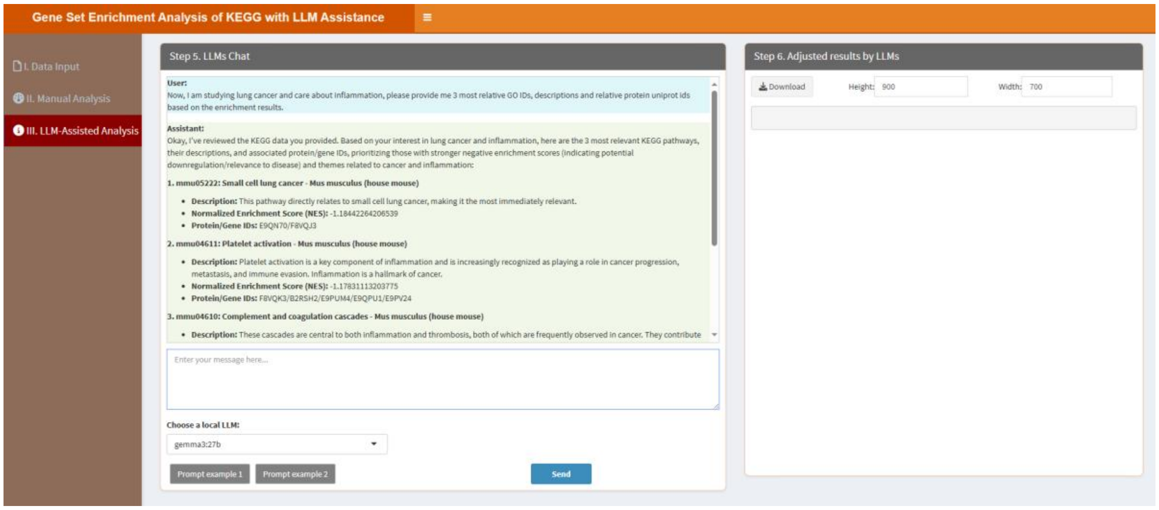
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Two types of prompts are supported: prompts beginning with “Text:” are used to request interpretation or explanation of the enrichment results (e.g. summarizing pathways relevance), while prompts beginning with “Plot:” allow users to customize the barplot visualization (e.g. modifying colors, sorting order, or number of displayed pathways). Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt 1 “Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the first prompt, click “Send” button. It will learn like below:

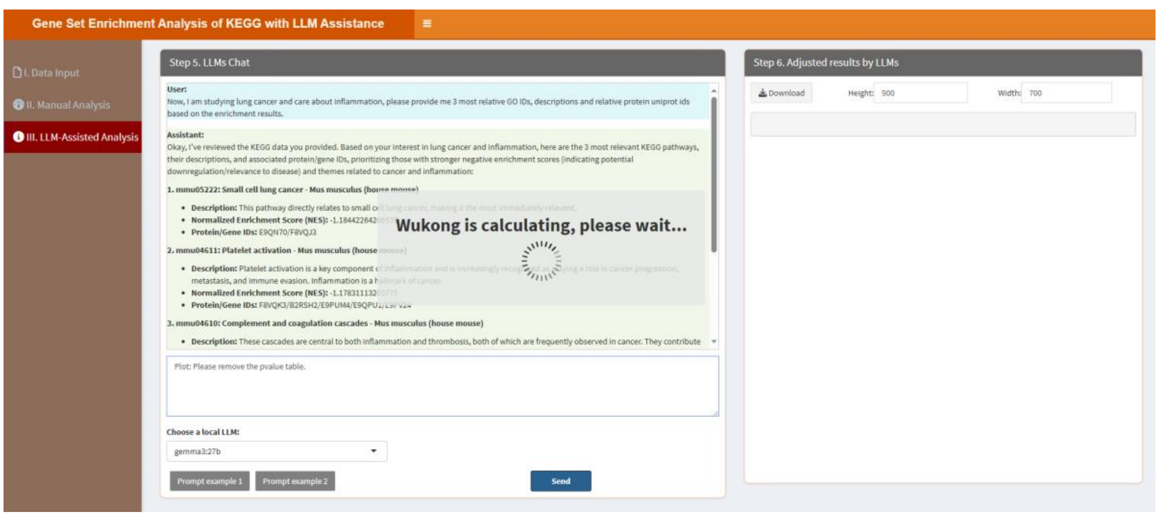


The results are shown as below:

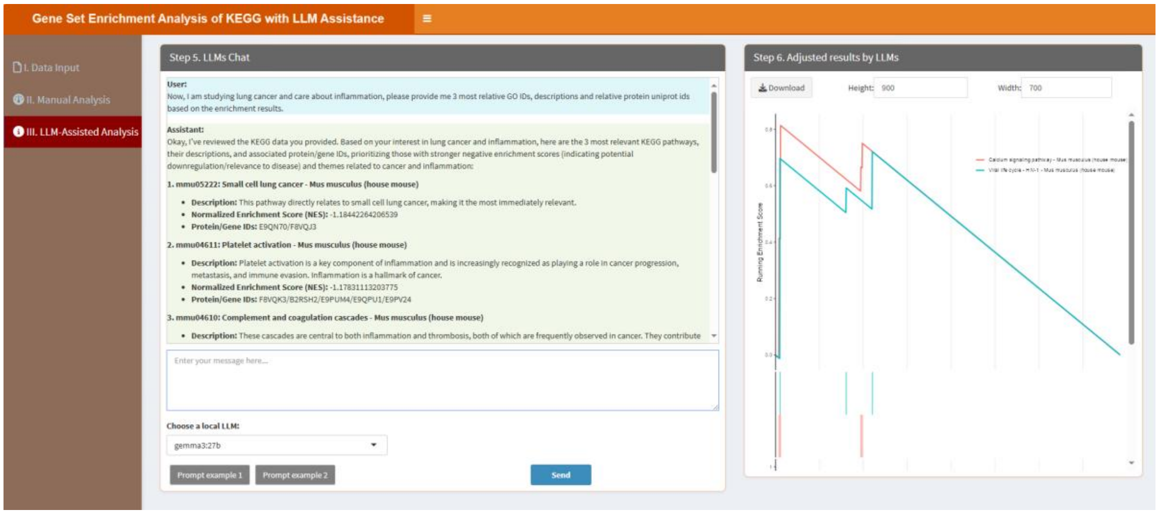


On the other hand, the example prompt 2 “Plot: Please remove the pvalue table. ”instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Plot: Please remove the pvalue table.=>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the second prompt, click “Send” button. It will learn like below:



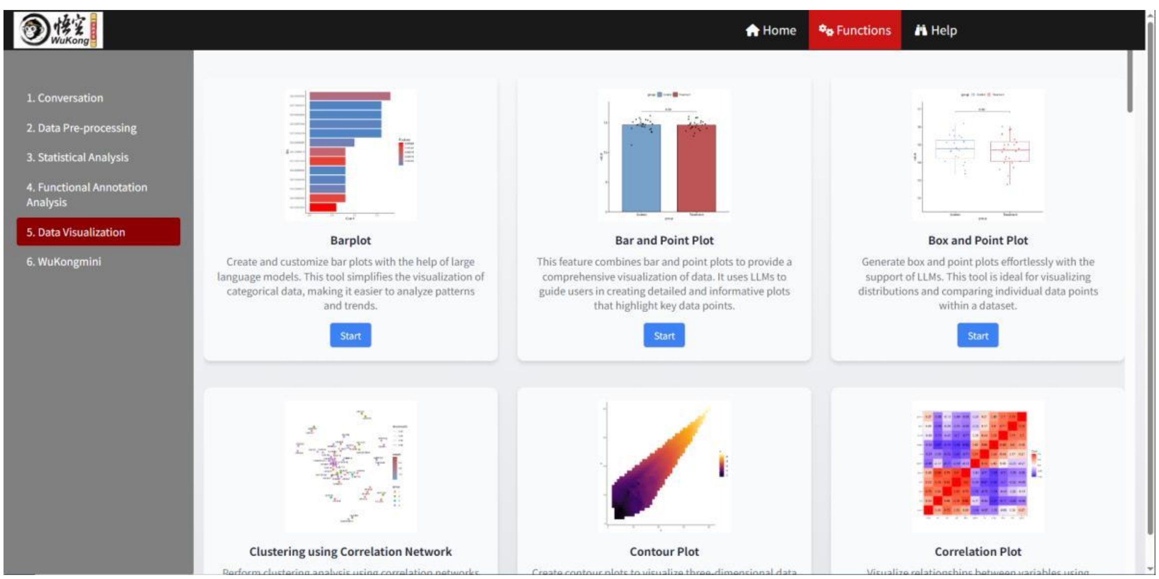
The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



Gene Set Enrichment Analysis of KEGG with LLM Assistance
I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis
Step 5. LLMs Chat
User:
Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results.
Assistant:
Okay, I've reviewed the KEGG data you provided. Based on your interest in lung cancer and inflammation, here are the 3 most relevant KEGG pathways, their descriptions, and associated protein/gene IDs, prioritizing those with stronger negative enrichment scores (indicating potential downregulation/relevance to disease) and themes related to cancer and inflammation:
1. mmu05222: Small cell lung cancer - Mus musculus (house mouse)
Description: This pathway directly relates to small cell lung cancer, making it the most immediately relevant.
Normalized Enrichment Score (NES): -1.18442264206539
Protein/Gene IDs: E9QN70/F8VQJ3
2. mmu04611: Platelet activation - Mus musculus (house mouse)
Description: Platelet activation is a key component of inflammation and is increasingly recognized as playing a role in cancer progression, metastasis, and immune evasion. Inflammation is a hallmark of cancer.
Normalized Enrichment Score (NES): -1.17831113203775
Protein/Gene IDs: F8VQK3/B2RSH2/E9PUM4/E9QPU1/E9PV24
3. mmu04610: Complement and coagulation cascades - Mus musculus (house mouse)
Description: These cascades are central to both inflammation and thrombosis, both of which are frequently observed in cancer. They contribute
Enter your message here...
Choose a local LLM:
gemma3:27b
Prompt example 1
Prompt example 2
Send
Step 6. Adjusted results by LLMs
Download
Height: 900
Width: 700
Running Enrichment Score

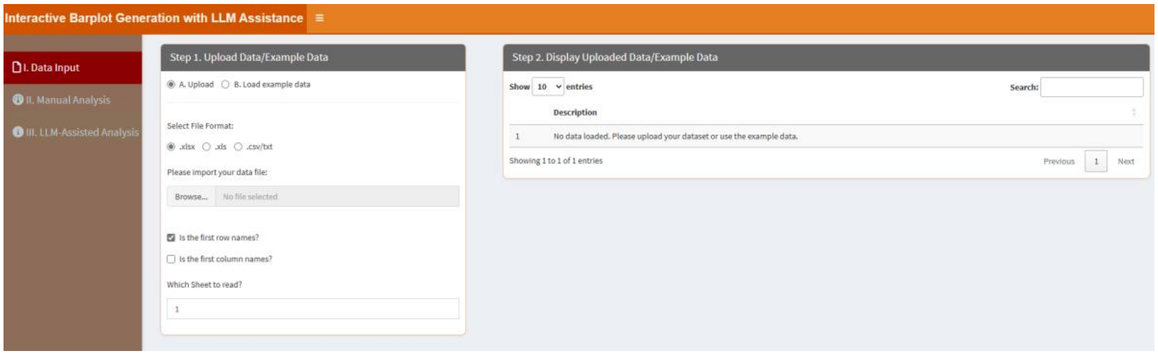
7. Data Visualization

Data visualization in proteomics is essential for interpreting complex datasets, and a wide range of visualization modules is available to meet diverse analytical needs. Users can customize their plots by adjusting parameters through a classic framework while leveraging local large language models (LLMs) to refine visualizations based on inner R codes. These modules include bar plots, bar and point plots, and box and point plots for categorical and distributional data. Clustering can be performed using correlation networks, while contour plots and correlation plots reveal relationships in multidimensional data. Advanced modules like correlation network plots, cross error bar plots, and dendrograms (including ggtree-based dendrograms) enhance hierarchical and statistical analysis. Users can create diverging bar plots, funnel plots, protein/DNA sequence logo plots, colored scatterplots, and combined histogram and density plots for specialized insights. Additionally, lollipop charts, marginal histogram/boxplots, Nightingale rose diagrams, pair point line plots, pie plots, radar charts, rain cloud plots, rank point plots, ridge plots, and ROC plots provide tailored visualizations for different data types. Sankey charts, scatter plots with ellipses, survival plots, ternary plots, UpSet plots, Venn plots, violin plots, volcano plots, and word clouds further expand the toolkit for exploring and presenting data. By combining user-friendly parameter adjustments with LLM-driven guidance, these visualization modules empower researchers to uncover patterns, trends, and relationships within proteomics data efficiently and effectively.



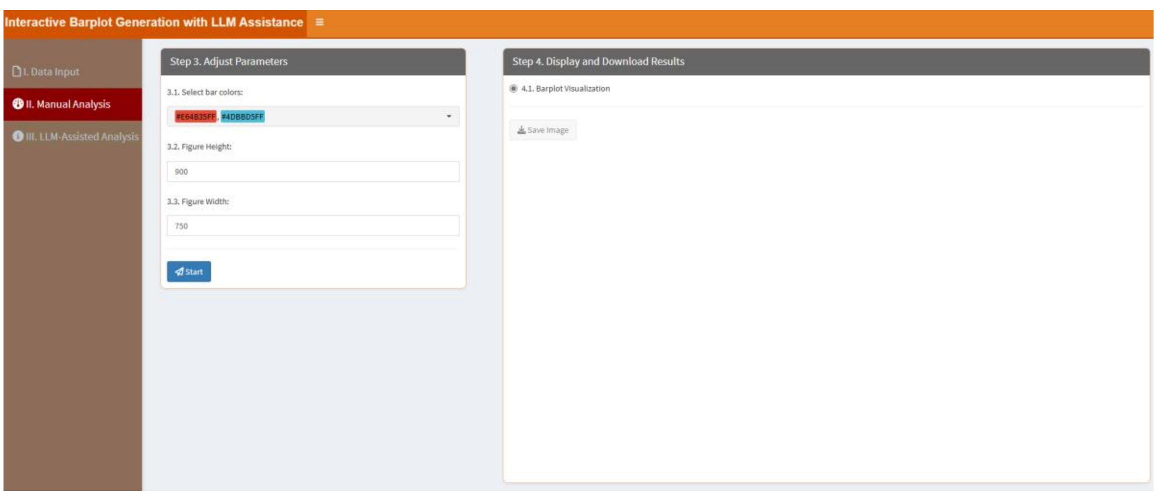
7.1. Barplot

Users can create and customize bar plots to visualize categorical data by adjusting parameters through the classic framework. Additionally, local LLMs can be leveraged to refine the plot for clearer insights and better representation of patterns and trends. After clicking the "Start" button, it will be started in a new window like below:



The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.

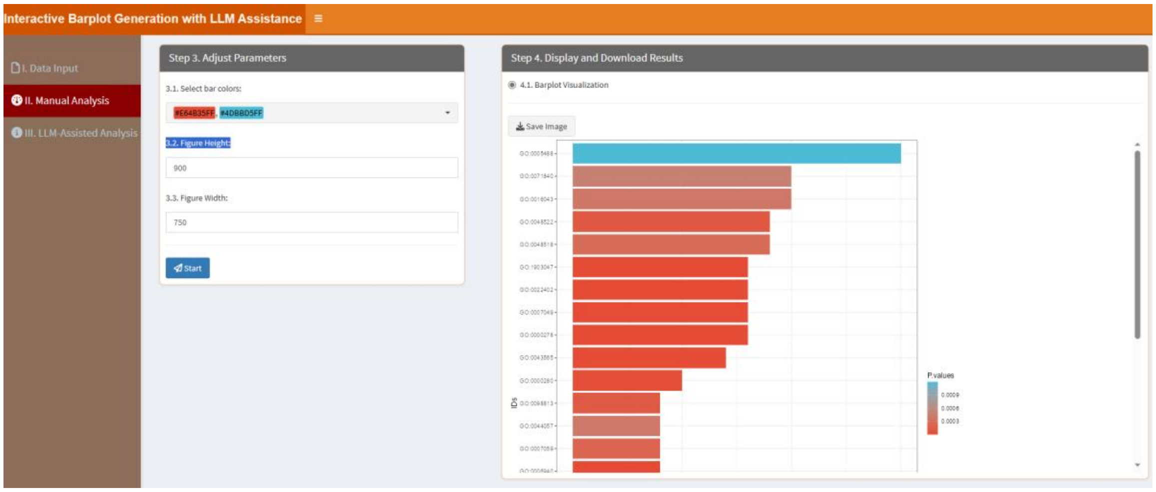


3.1. Select bar colors: Choose two distinct colors for the bar plot, one is for the lowest value and the other is for the highest value in the uploaded data.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

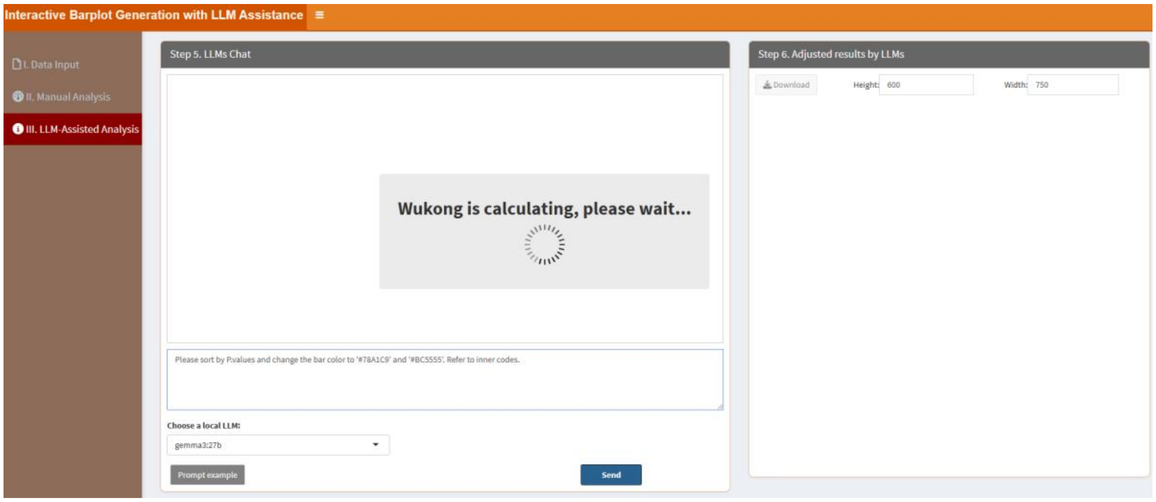
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



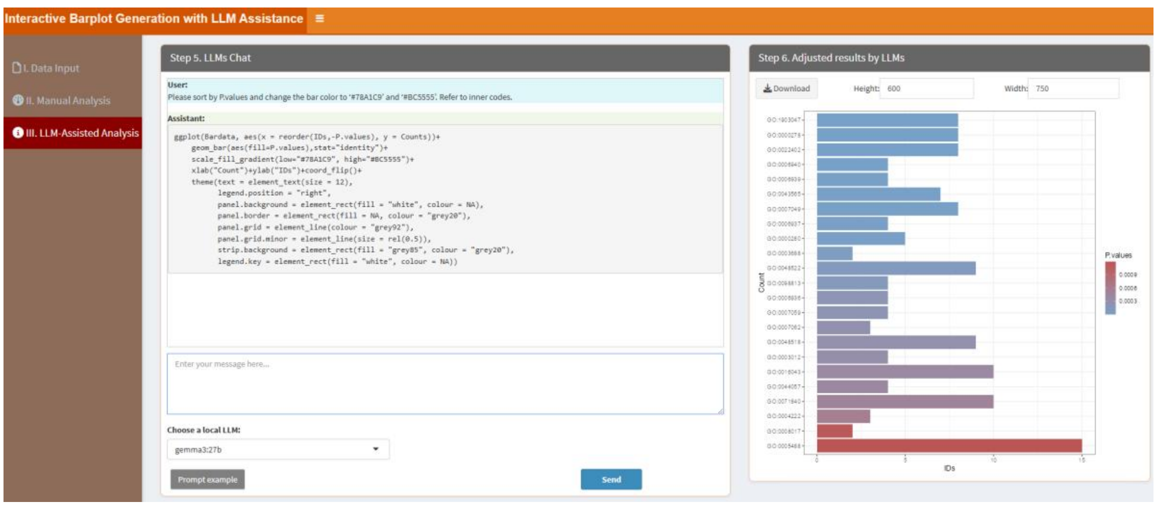
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please sort by P.values and change the bar color to '#78A1C9' and '#BC5555'. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please sort by P.values and change the bar color to '#78A1C9' and '#BC5555'. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

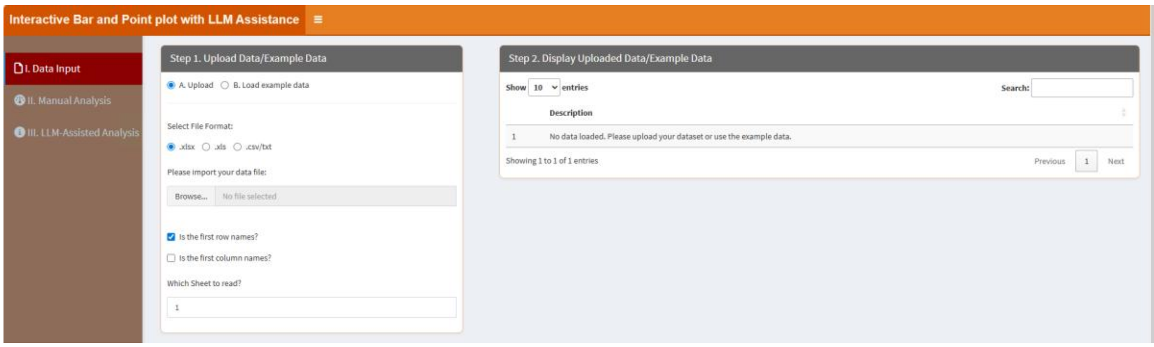


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



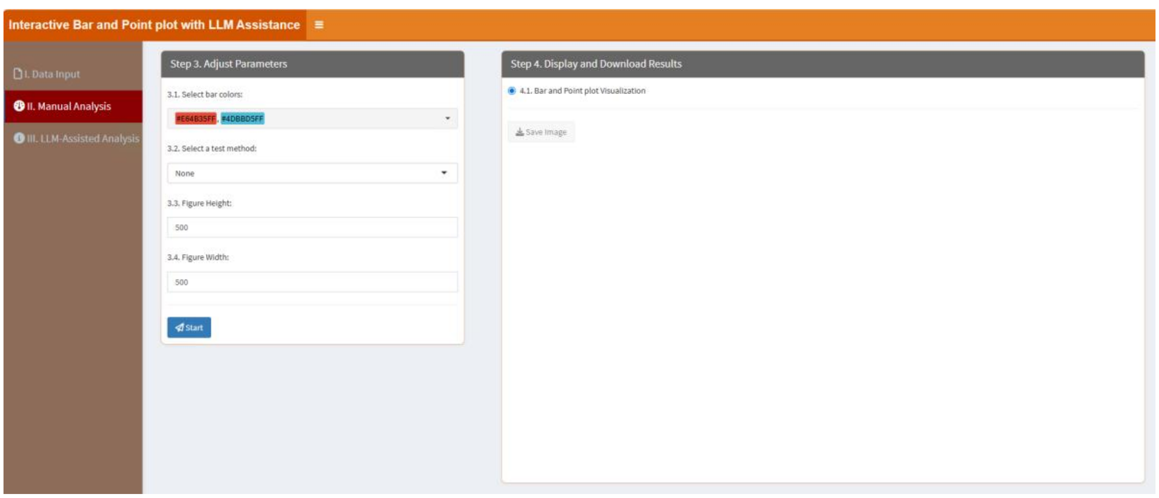
7.2. Bar and Point plot

This module allows users to combine bar and point plots for comprehensive data visualization. By adjusting parameters through the classic framework and utilizing local LLMs, users can refine the plot to highlight key data points with precision. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

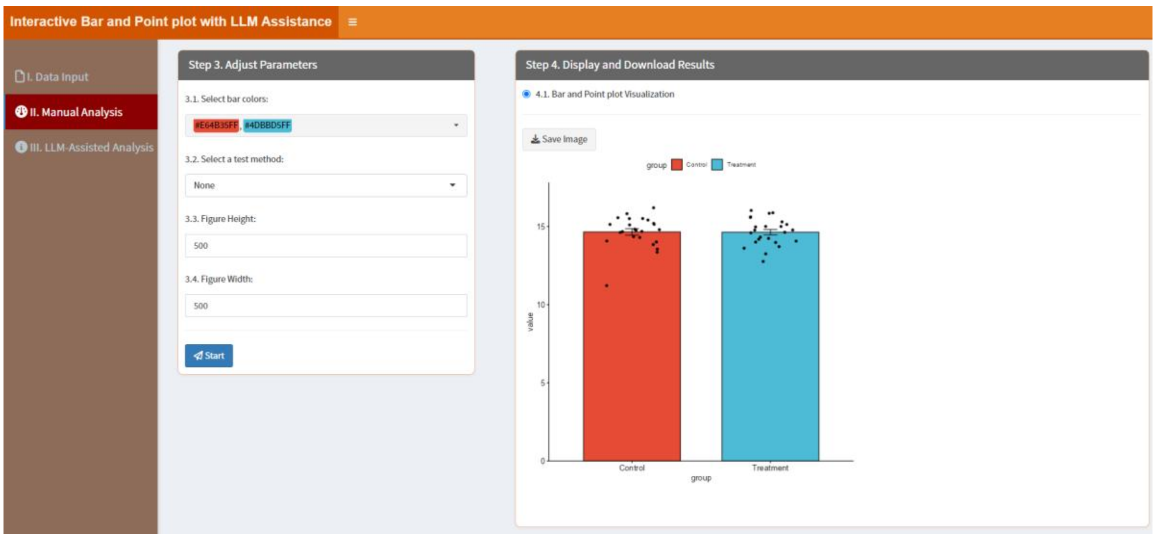


3.1. Select bar colors: Choose two distinct colors for the bar plot, one is for the lowest value and the other is for the highest value in the uploaded data.

3.2. Select a test method: Specifies the statistical test to be used for comparing groups in bar and point plots. Selecting "None" means no statistical test will be performed, and the plot will not display any p-values. Choosing "t_test" applies the Student's t-test, which compares the means of two groups. The "wilcox_test" option performs the Wilcoxon rank-sum test (Mann-Whitney U test). The "anova_test" conducts a one-way ANOVA, suitable for comparing the means of three or more groups.

3.2. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

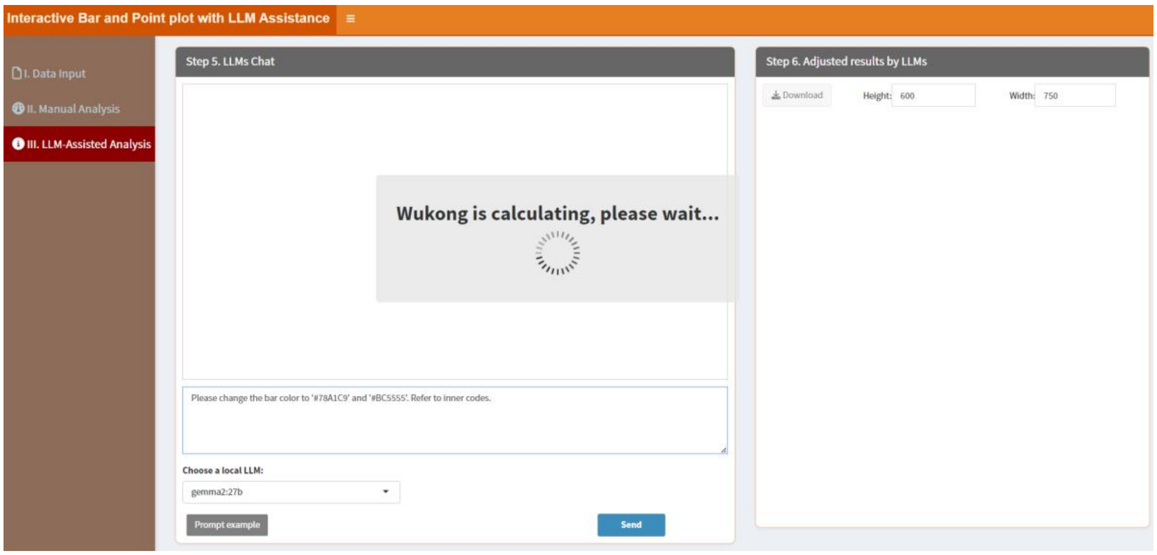
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



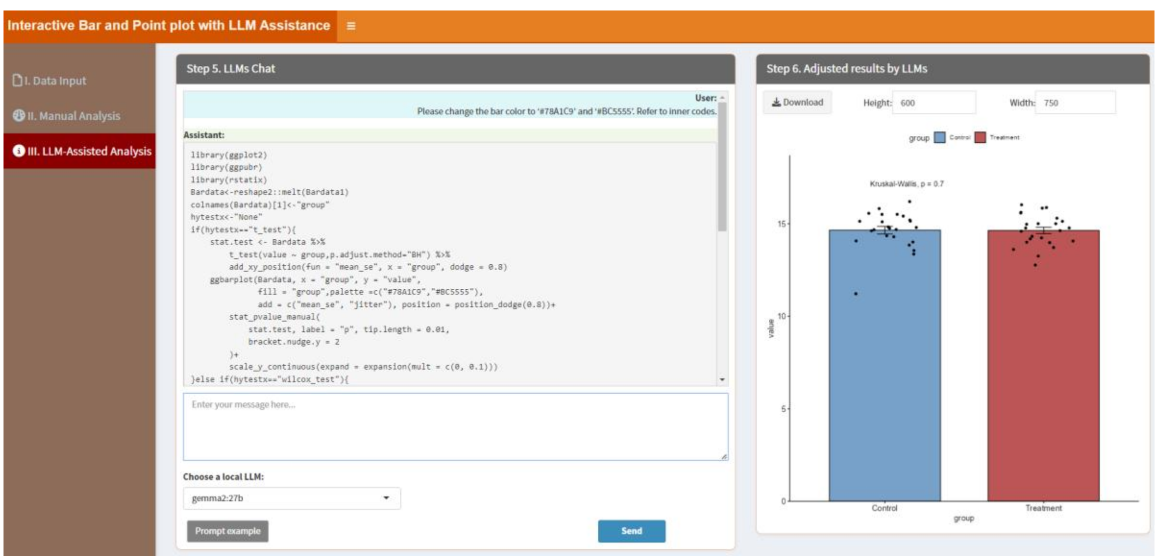
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the bar color to '#78A1C9' and '#BC5555'. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the bar color to '#78A1C9' and '#BC5555'. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

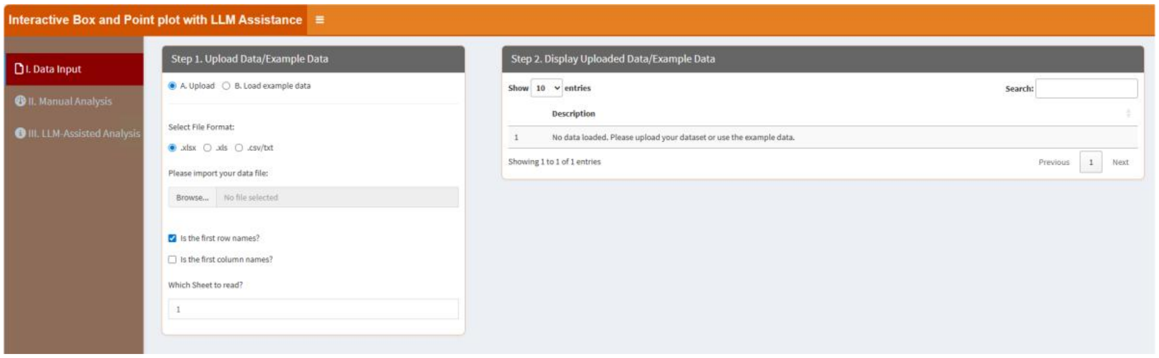


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section



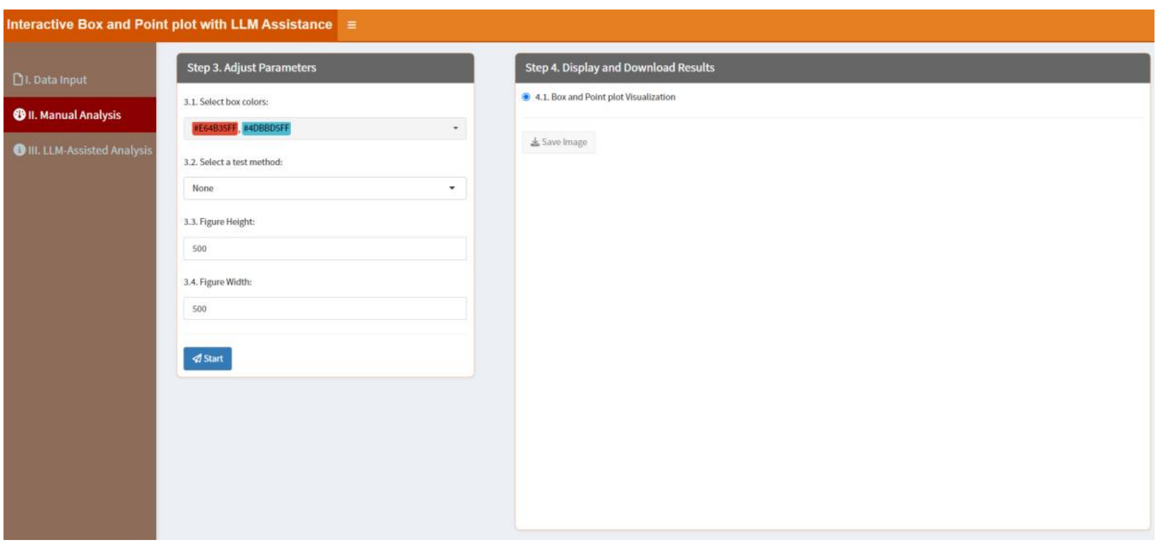
7.3. Box and Point plot

Box and point plots can be generated to visualize data distributions and compare individual data points. Users can adjust parameters via the classic framework and use local LLMs to enhance the plot's clarity and detail. After clicking the "Start" button, it will be started in a new window like below:



The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.



3.1. Select box colors: Choose two distinct colors for the box plot, one is for the lowest value and the other is for the highest value in the uploaded data.

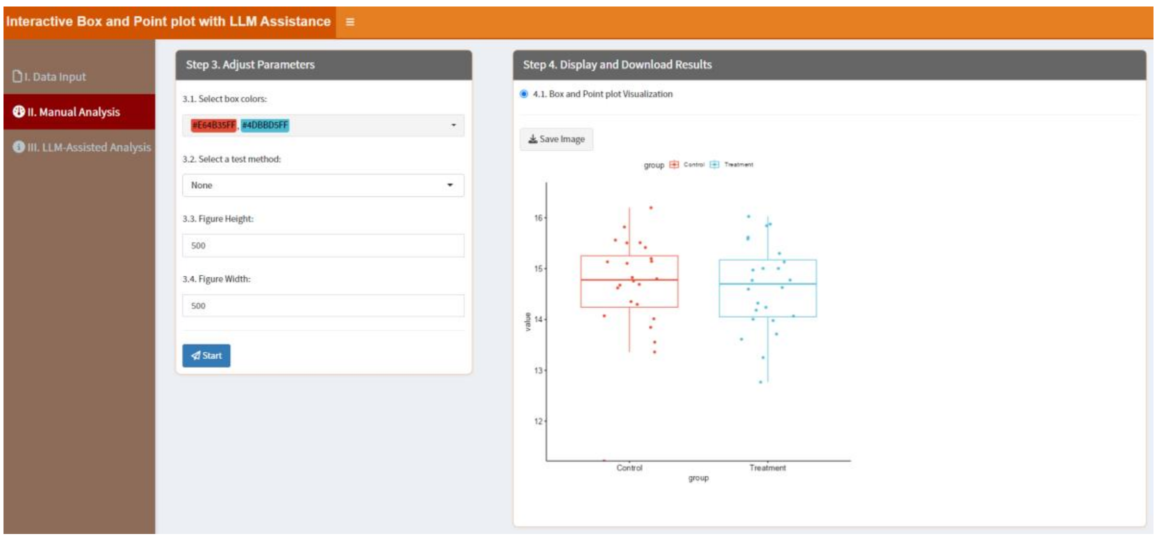
3.2. Select a test method: Specifies the statistical test to be used for comparing groups in box and point plots. Selecting "None" means no statistical test will be performed, and the plot will not display any p-values. Choosing "t_test" applies the Student's t-test, which compares the means of two groups. The "wilcox_test" option performs the Wilcoxon rank-sum test (Mann-Whitney U test). The "anova_test" conducts a one-way ANOVA, suitable for comparing the means of three or more groups.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

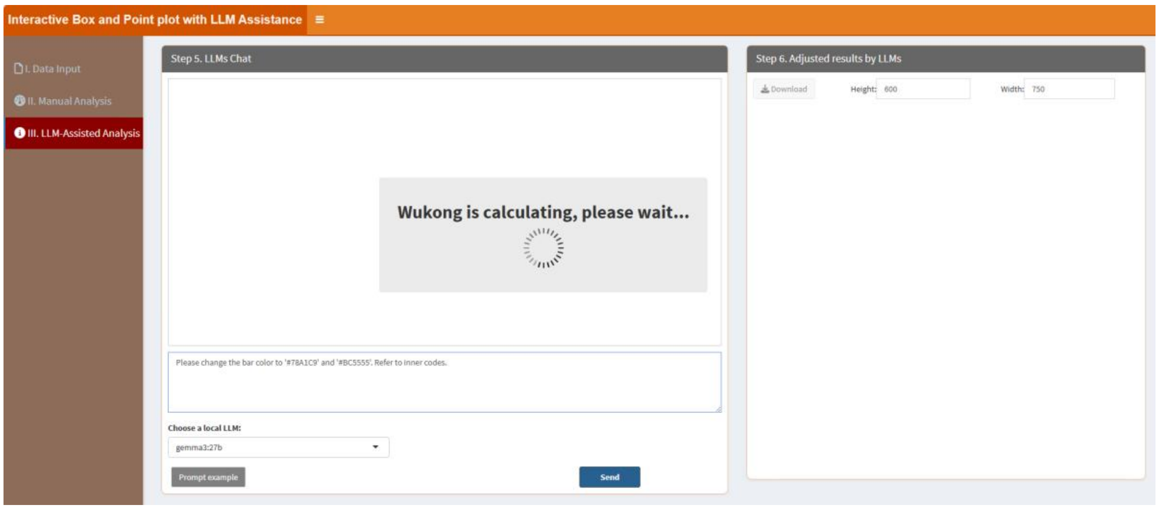
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



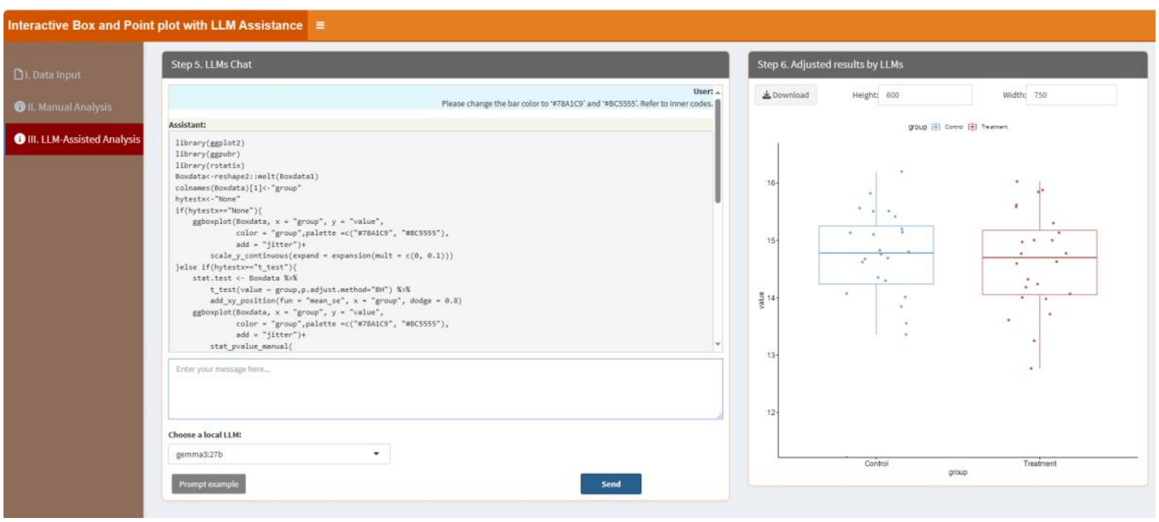
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the bar color to '#78A1C9' and '#BC5555'. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Please change the bar color to '#78A1C9' and '#BC5555'. =>", it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click "Send" button. It will learn like below:

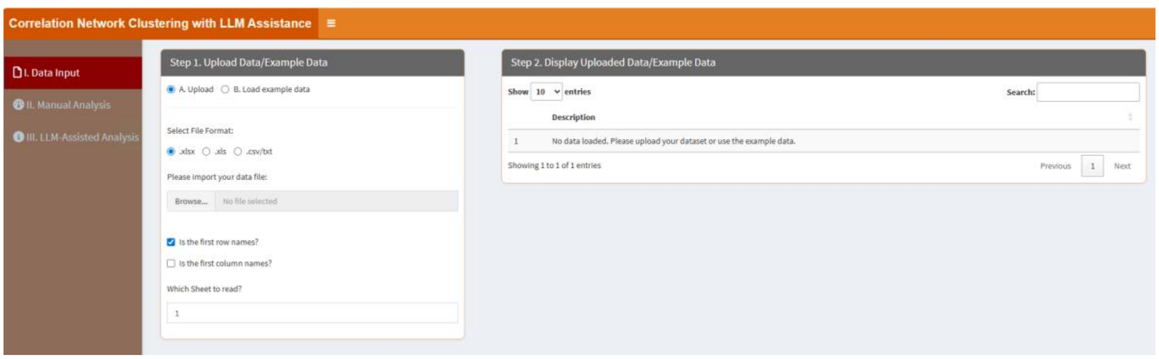


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



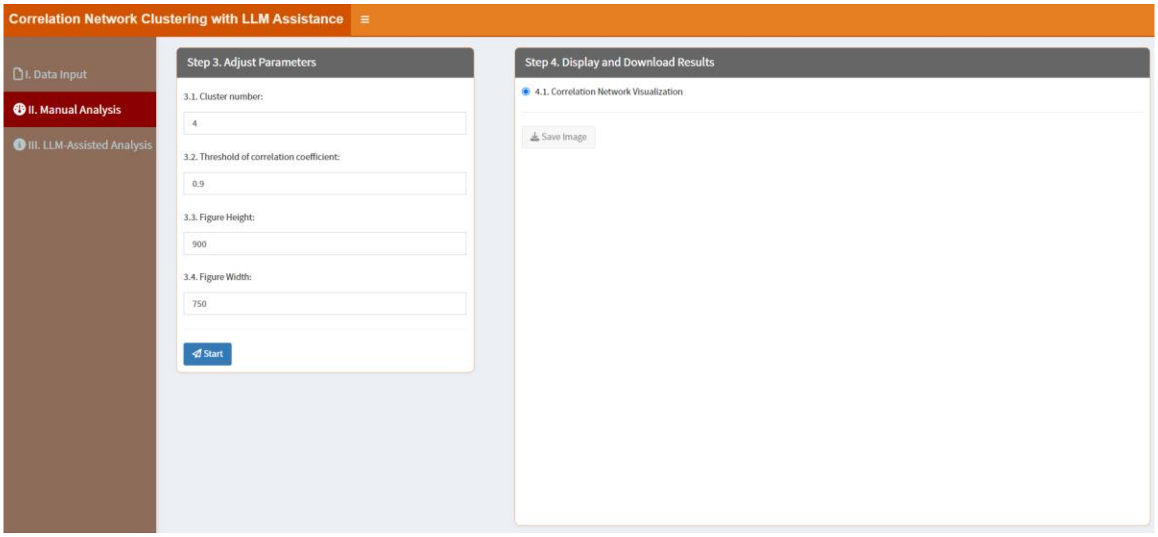
7.4. Clustering using Correlation Network

This module enables clustering analysis based on correlation networks combined with K-means clustering. First, pairwise correlation coefficients are calculated between all objects in the input dataset. Correlation values below a user-defined threshold are filtered out to retain only strong associations. A correlation network is then constructed, where nodes represent individual objects and edges reflect significant correlations. K-means clustering is subsequently applied to partition the data into a specified number of groups. Cluster assignments are mapped onto the network, allowing nodes to be color-coded by cluster membership. Finally, the network is visualized with edge widths and colors representing correlation strength, and node labels and groupings highlighting cluster structures within the network. Users can adjust settings through the classic framework and rely on local LLMs to refine the visualization, uncovering relationships and groupings within complex datasets. After clicking the "Start" button, it will be started in a new window like below:



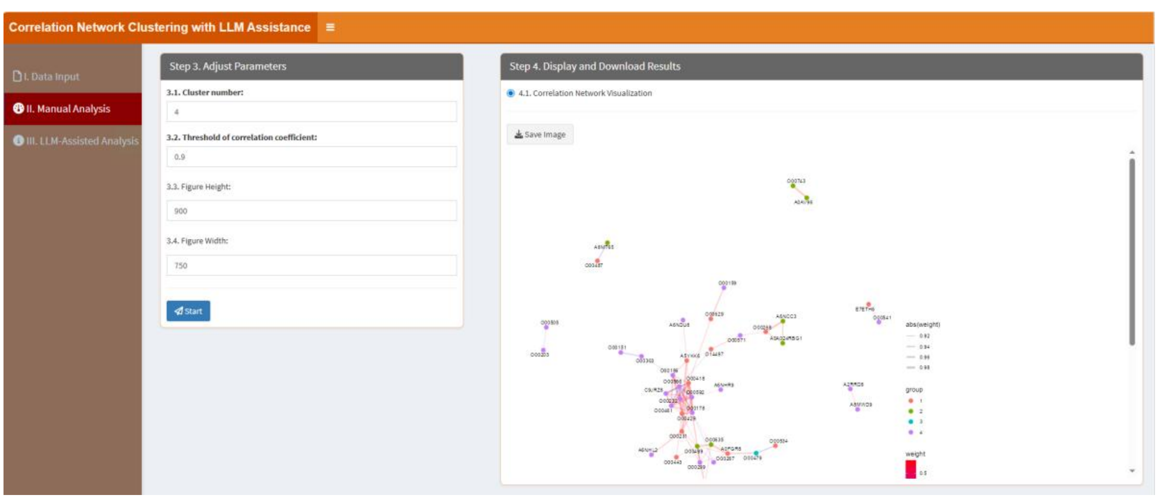
The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.



3.1. Cluster number: This parameter Specifies the number of clusters for the K-means method.
3.2. Threshold of correlation coefficient: This parameter defines the minimum absolute value of the correlation coefficient for an edge to be included in the network.
3.3. Figure Height: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.4. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

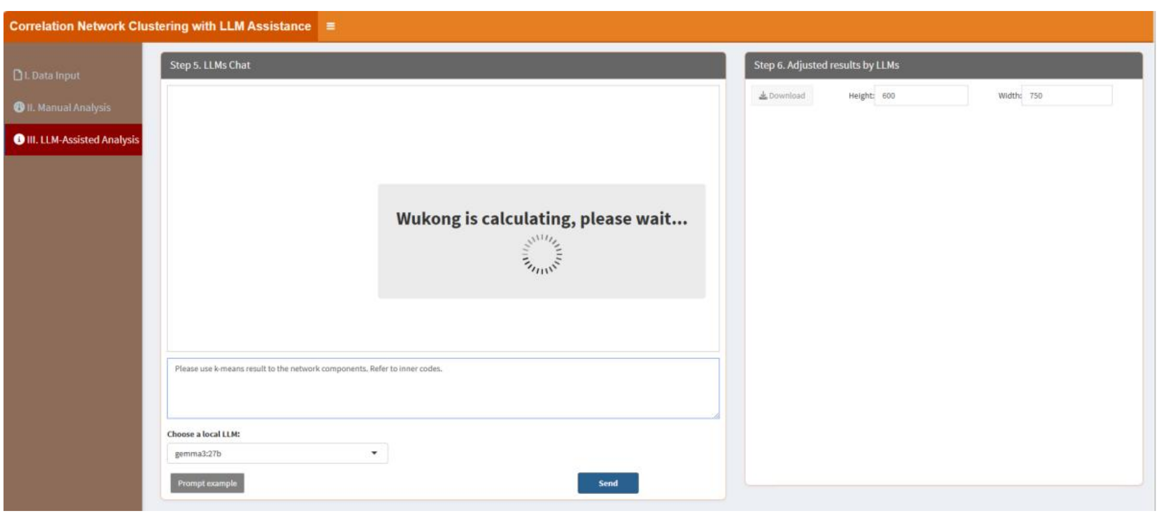


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..."

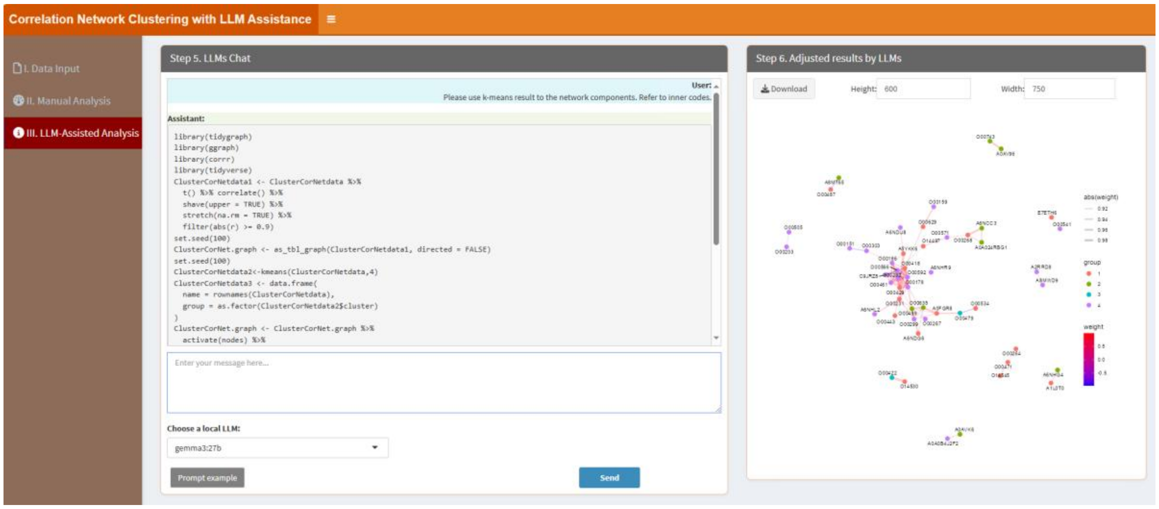
enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please use k-means result to the network components. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please use k-means result to the network components. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

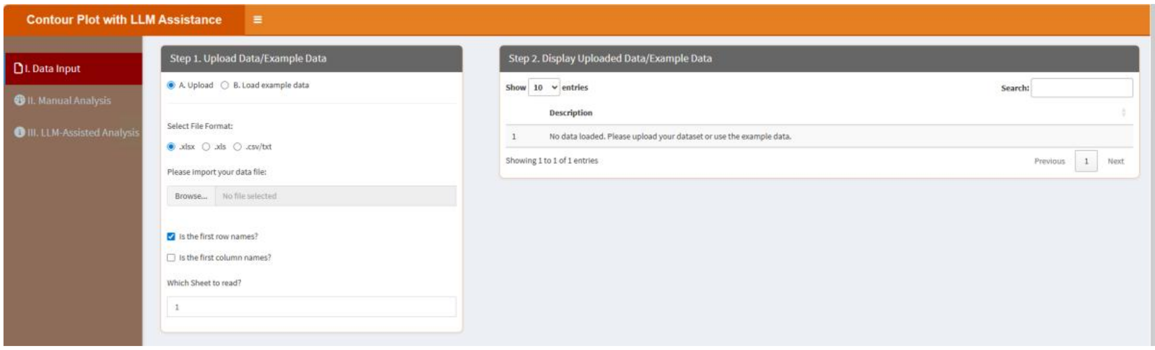


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.


Correlation Network Clustering with LLM Assistance
I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis
Step 5. LLMs Chat
User:
Please use k-means result to the network components. Refer to inner codes.
Assistant:
library(tidygraph)
library(ggraph)
library(corrr)
library(tidyverse)
ClusterCorNetdata1 <- ClusterCorNetdata %>%
t() %>% correlate() %>%
shave(upper = TRUE) %>%
stretch(na.rm = TRUE) %>%
filter(abs(r) >= 0.9)
set.seed(100)
ClusterCorNet.graph <- as_tbl_graph(ClusterCorNetdata1, directed = FALSE)
set.seed(100)
ClusterCorNetdata2<-kmeans(ClusterCorNetdata,4)
ClusterCorNetdata3 <- data.frame(
name = rownames(ClusterCorNetdata),
group = as.factor(ClusterCorNetdata2$cluster)
)
ClusterCorNet.graph <- ClusterCorNet.graph %>%
activate(nodes) %>%
Enter your message here...
Choose a local LLM:
gemma3:27b
Prompt example
Send
Step 6. Adjusted results by LLMs
Download
Height:
600
Width:
750
abs(weight)
group
weight

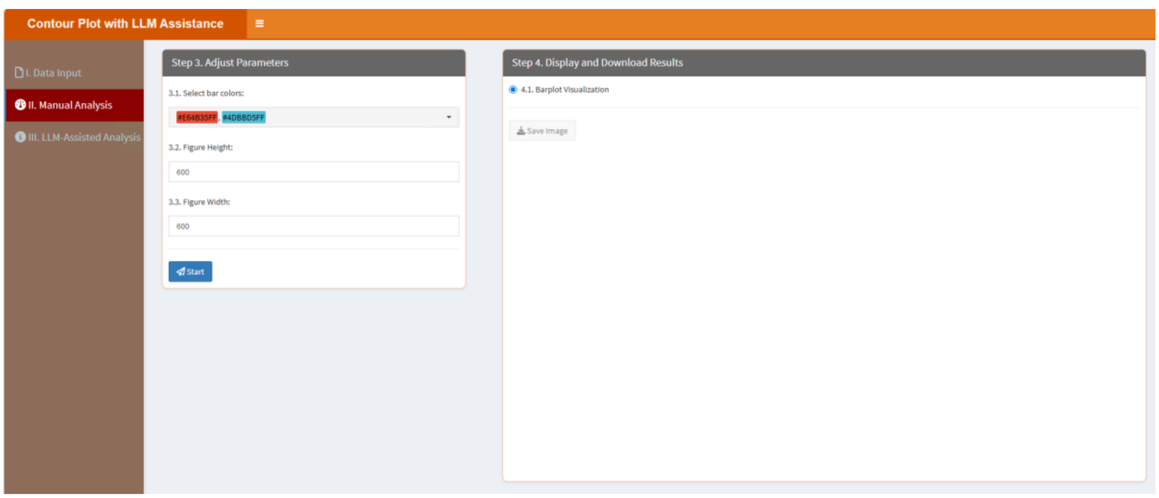
7.5. Contour Plot

Contour plots can be used to visualize three-dimensional data in two dimensions. Users can customize the plot through the classic framework and leverage local LLMs to ensure clear and informative results. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



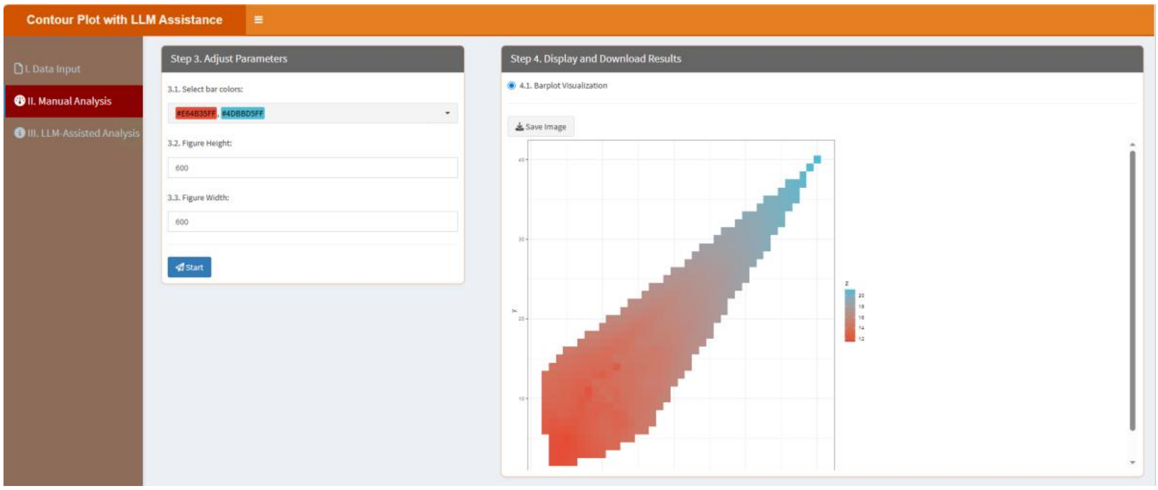
3.1. Select bar colors: Choose two distinct colors for the bar plot, one is for the lowest value and the other is for the highest value in the uploaded data.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

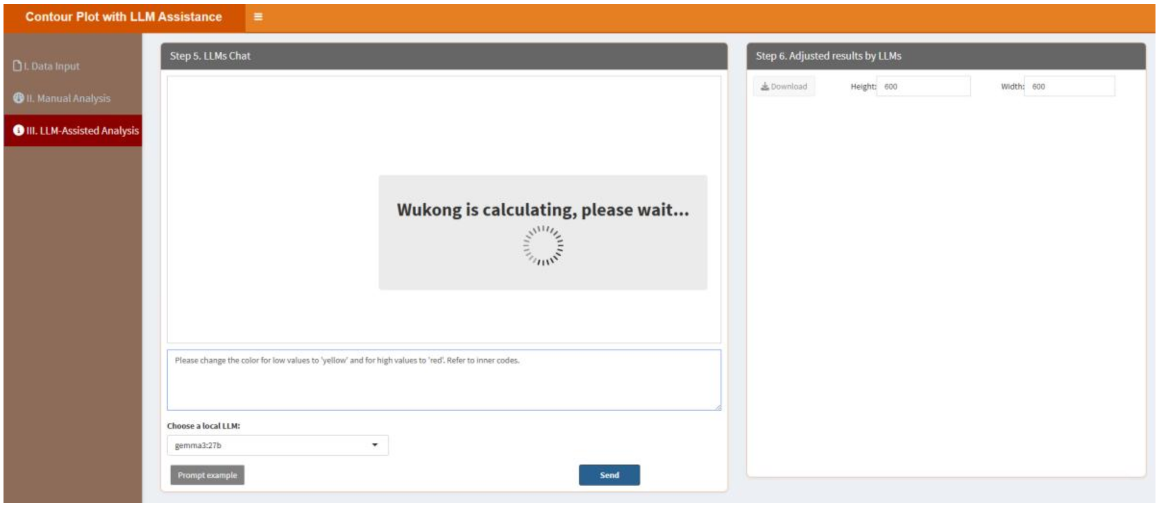
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



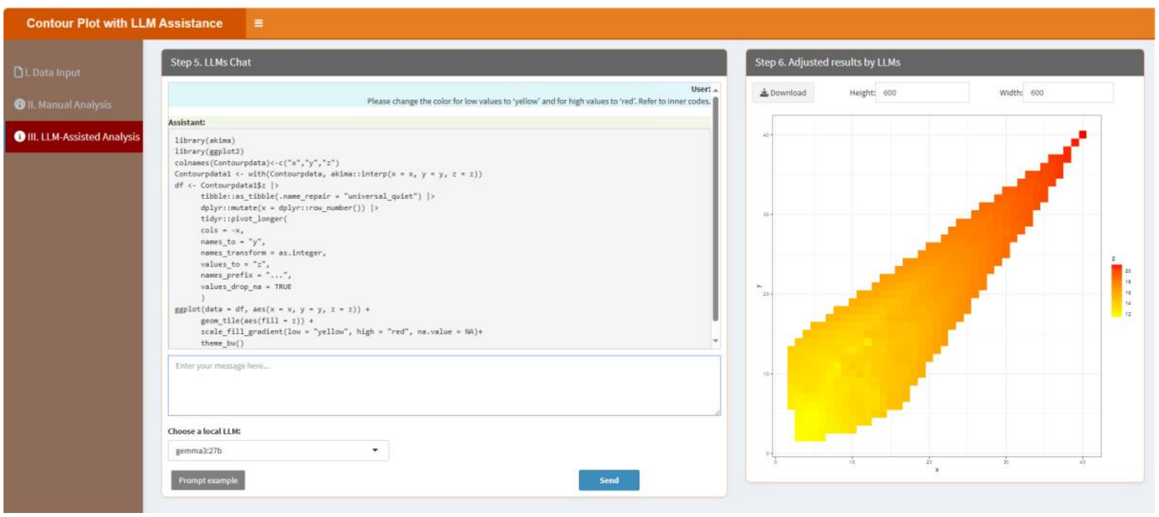
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the color for low values to 'yellow' and for high values to 'red'.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the color for low values to 'yellow' and for high values to 'red'. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

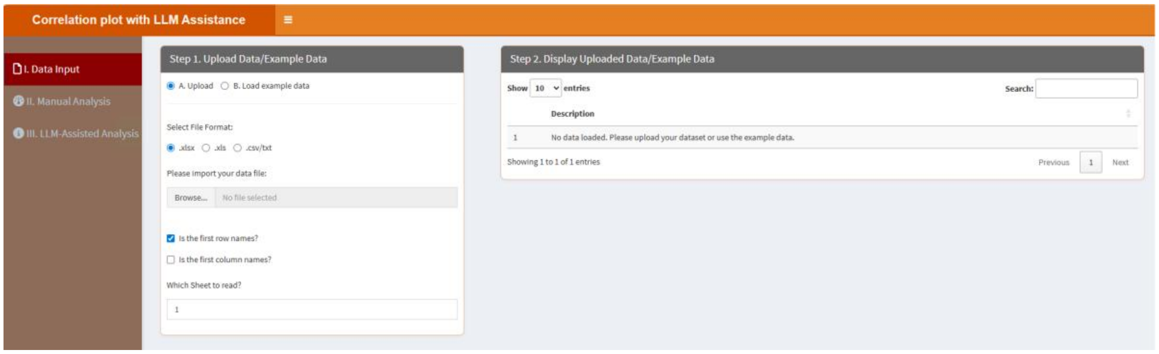


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



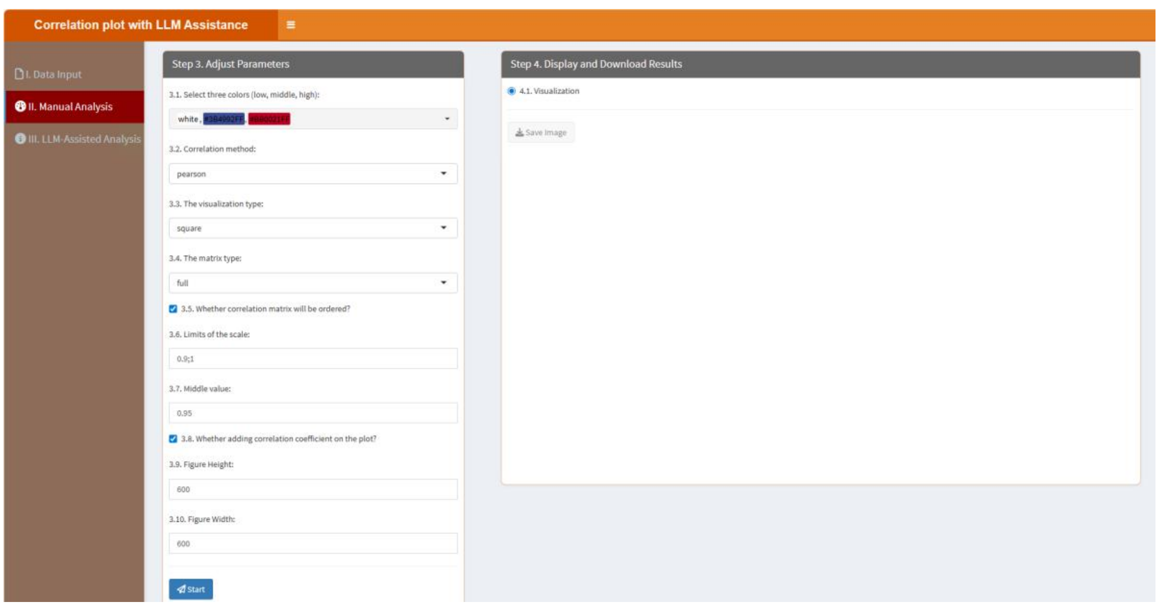
7.6. Correlation plot

This module helps users visualize relationships between variables using correlation plots. By adjusting parameters in the classic framework and refining with local LLMs, users can ensure accurate and insightful representations. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

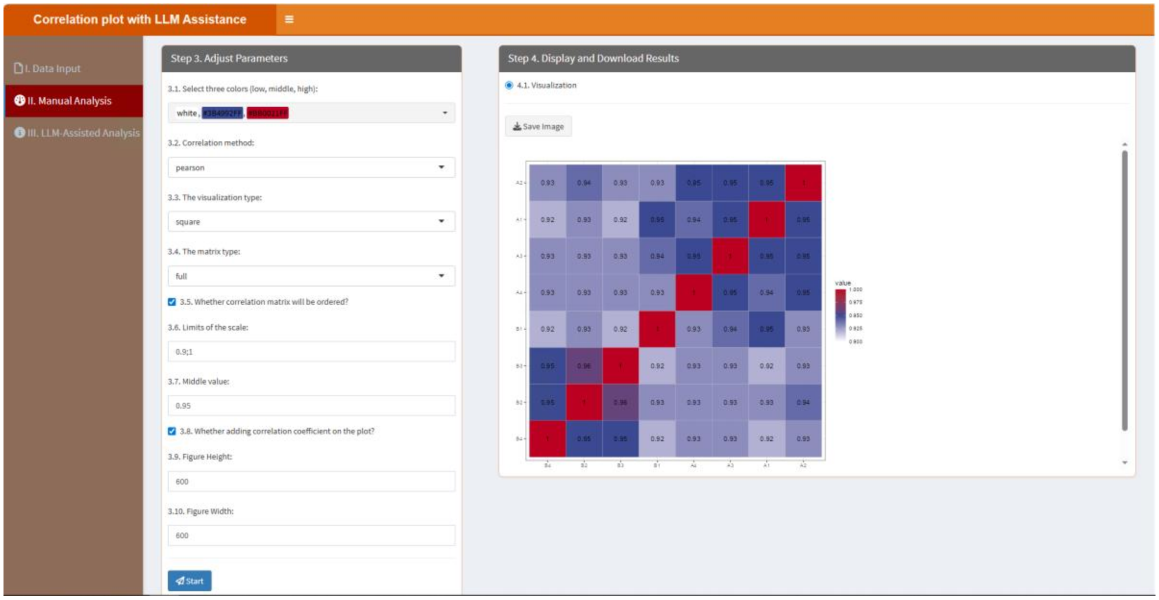


3.1. Select three colors (low, middle, high): Colors for the correlation plot. By default, there are three color names (#3C5488FF, white, #EE0000FF) here for the lowest (#3C5488FF), middle (white), highest (#EE0000FF) values repectively.

3.2. Correlation method: This setting determines the statistical method used to calculate the correlation coefficients between variables. Available options typically include Pearson,

Spearman and Kendall.
3.3. The visualization type: This parameter controls how the correlations are visually represented in the plot. Users can choose from multiple styles such as circle, square.
3.4. The matrix type: This setting specifies which part of the correlation matrix is shown in the plot. Users can choose full, upper, or lower matrix views.
3.5. Whether correlation matrix will be ordered? This parameter allows users to reorder the matrix using hierarchical clustering to group variables with similar correlation patterns together. Enabling this option helps reveal clusters and hidden relationships more clearly by arranging correlated variables adjacent to each other. If disabled, the matrix retains the original input variable order, which may be preferable in some analytical contexts where order is predefined.
3.6. Limits of the scale: This option defines the minimum and maximum values of the color scale used in the correlation plot, typically ranging from -1 to 1. Customizing these limits allows users to zoom in on specific ranges of interest (e.g., focusing only on high correlations by setting limits to 0.9 and 1.0).
3.7. Middle value: This parameter sets the central reference point of the color gradient, typically defaulting to 0, which represents no correlation. Adjusting the middle value can be useful when the user wants to shift the visual emphasis in asymmetric correlation distributions.
3.8. Whether adding correlation coefficient on the plot: This toggle determines whether the numerical correlation coefficients are overlaid on the visualization. When enabled, each matrix cell will display its exact correlation value, enhancing precision and analytical utility.
3.9. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.10. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

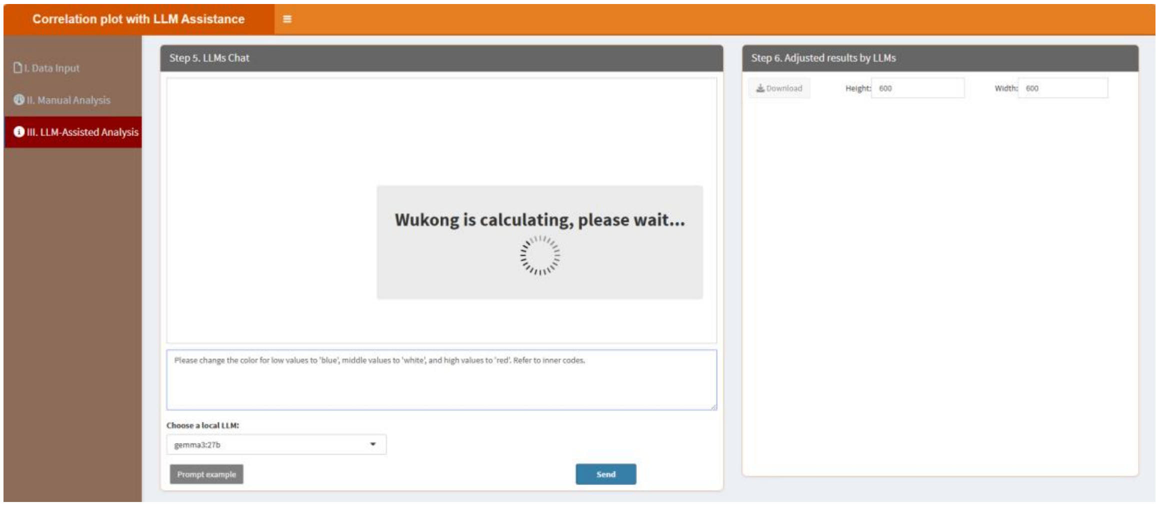
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



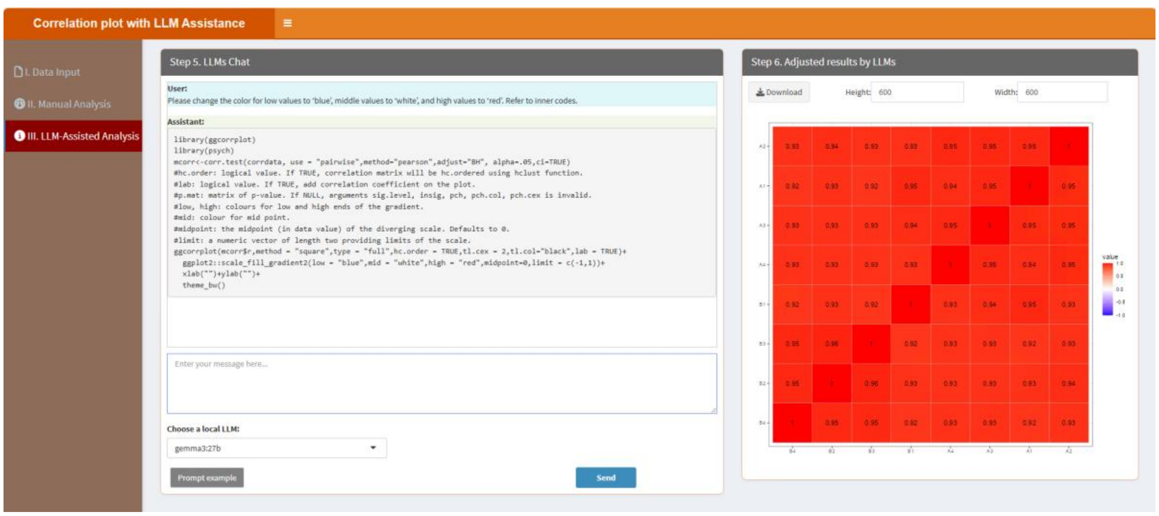
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the color for low values to 'blue', middle values to 'white', and high values to 'red'. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the color for low values to 'blue', middle values to 'white', and high values to 'red'. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

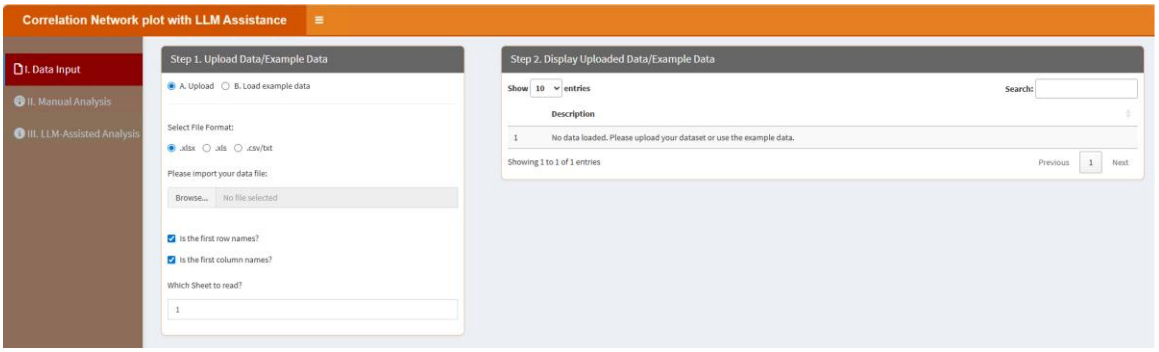


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



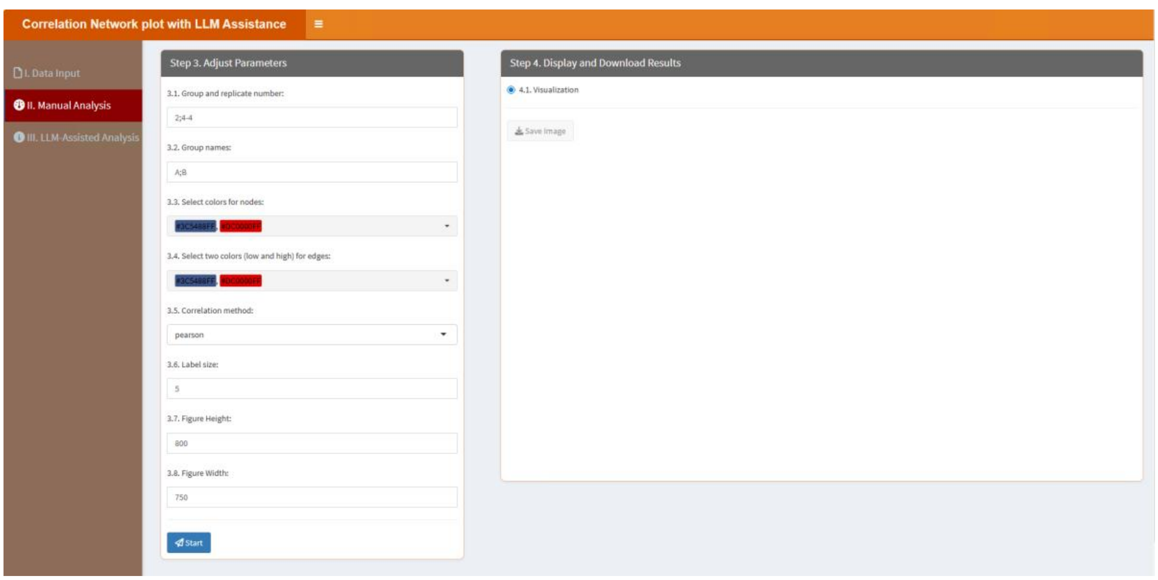
7.7. Correlation Network plot

Users can construct correlation network plots to visualize complex relationships between variables. The classic framework allows parameter adjustments, while local LLMs refine the visualization for better insights. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

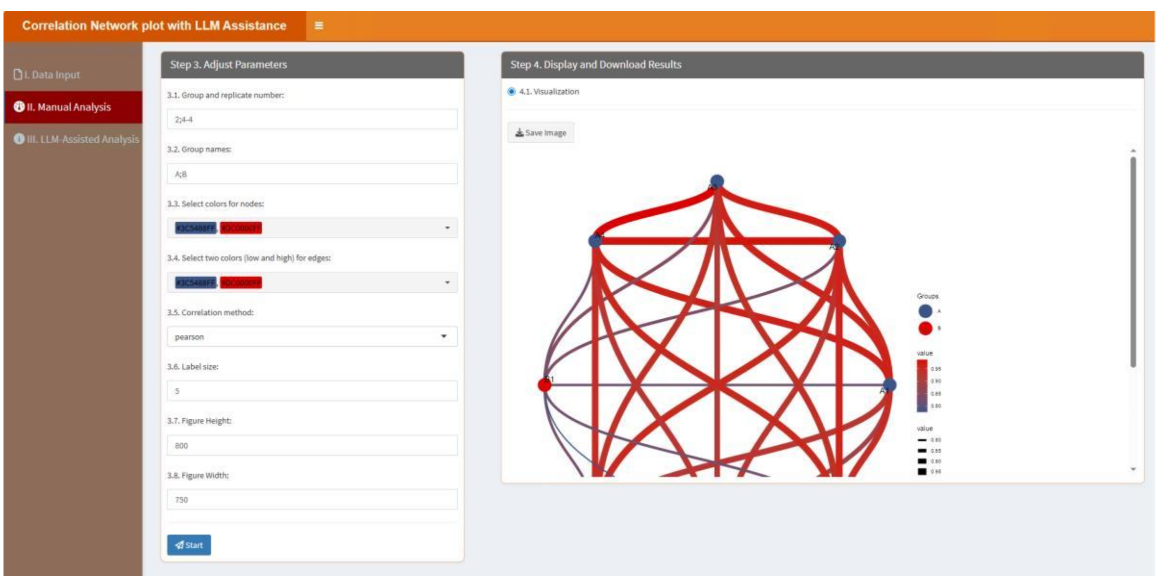
The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Group and replicate number: Type in the group number and replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 2 groups and 4 replicates in each group, so here is “2;4-4”.

3.2. Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 2 groups in the example data, you can type in "A;B".
3.3. Select colors for nodes: This parameter allows users to assign specific colors to the nodes (representing variables) of each group in the network plot. The color number should be equal to the group number.
3.4. Select two colors (low and high) for edges: This option determines the color gradient for the edges (lines connecting nodes) based on correlation strength. The user selects two colors: one for low correlation values and one for high values.
3.5. Correlation method: This setting determines the statistical method used to calculate the correlation coefficients between variables. Available options typically include Pearson, Spearman and Kendall.
3.6. Label size: This parameter controls the font size of the node labels in the network plot.
3.7. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.8. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

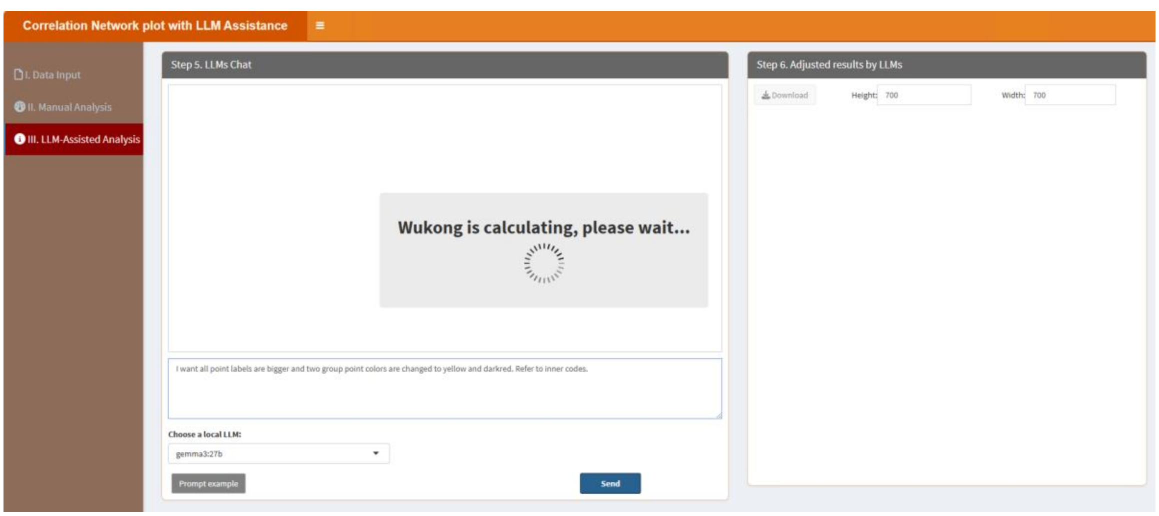
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



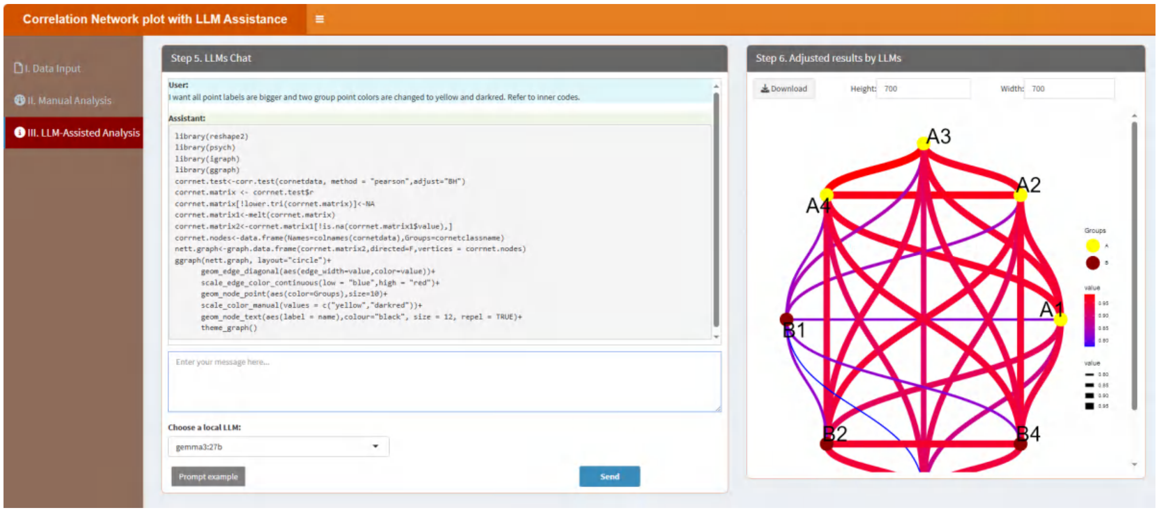
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I want all point labels are bigger and two group point colors are changed to yellow and darkred'. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I want all point labels are bigger and two group point colors are changed to yellow and darkred. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



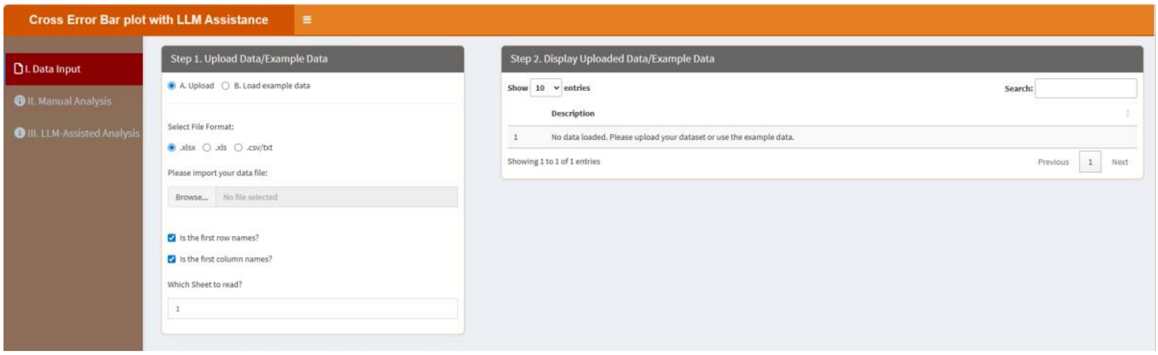
The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



Correlation Network plot with LLM Assistance
I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis
Step 5. LLMs Chat
User:
I want all point labels are bigger and two group point colors are changed to yellow and darkred. Refer to inner codes.
Assistant:
library(reshape2)
library(psych)
library(igraph)
library(ggraph)
cornnet.test<-corr.test(cornetdata, method = "pearson",adjust="BH")
cornnet.matrix <- cornnet.test$r
cornnet.matrix[!lower.tri(cornnet.matrix)]<-NA
cornnet.matrix1<-melt(cornnet.matrix)
cornnet.matrix2<-cornnet.matrix1[!is.na(cornnet.matrix1$value),]
cornnet.nodes<-data.frame(Names=colnames(cornetdata),Groups=cornetclassname)
nett.graph<-graph.data.frame(cornnet.matrix2,directed=F,vertices = cornnet.nodes)
ggraph(nett.graph, layout="circle")+
geom_edge_diagonal(aes(edge_width=value,color=value))+
scale_edge_color_continuous(low = "blue",high = "red")+
geom_node_point(aes(color=Groups),size=10)+
scale_color_manual(values = c("yellow","darkred"))+
geom_node_text(aes(label = name),colour="black", size = 12, repel = TRUE)+
theme_graph()
Enter your message here...
Choose a local LLM:
gemma3:27b
Prompt example
Send
Step 6. Adjusted results by LLMs
Download
Height: 700
Width: 700
A3
A2
A4
A1
B1
B2
B4
Groups
value

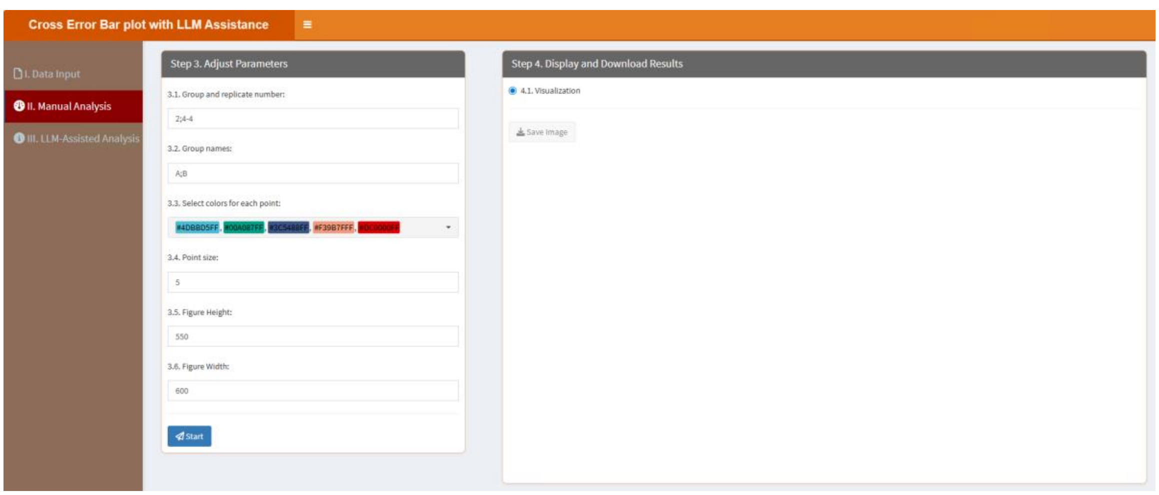
7.8. Cross Error Bar Plot

This module enables users to design cross error bar plots to represent data variability. Parameters can be adjusted through the classic framework, and local LLMs help refine the plot for precision and clarity. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Group and replicate number: Type in the group number and replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 2 groups and 4 replicates in each group, so here is “2;4-4”.

3.2. Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 2 groups in the example data, you can type in "A;B".

3.3. Select colors for each point: One color for one point.

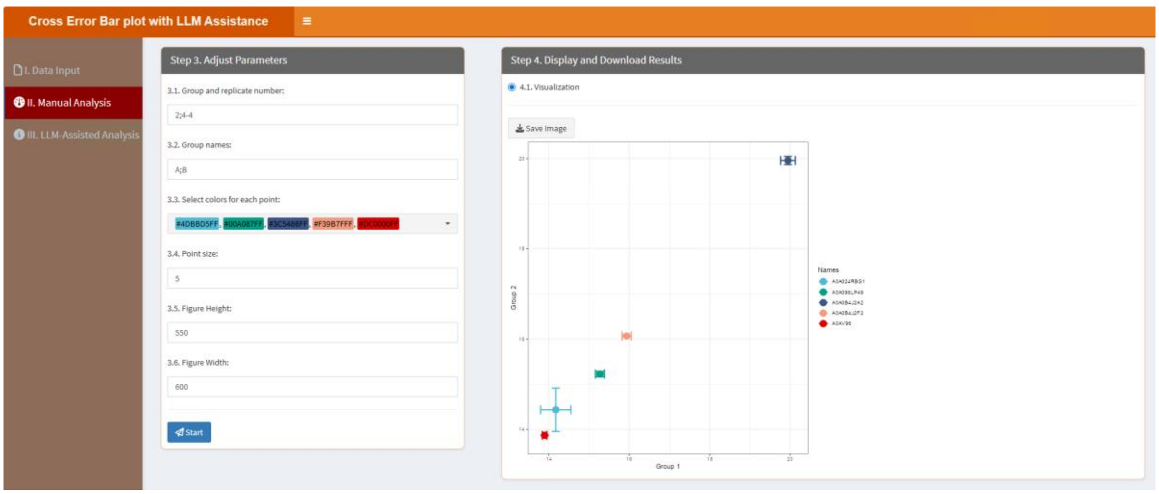
3.4. Point size: This setting controls the size of the data points shown in the plot.

3.5. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.6. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

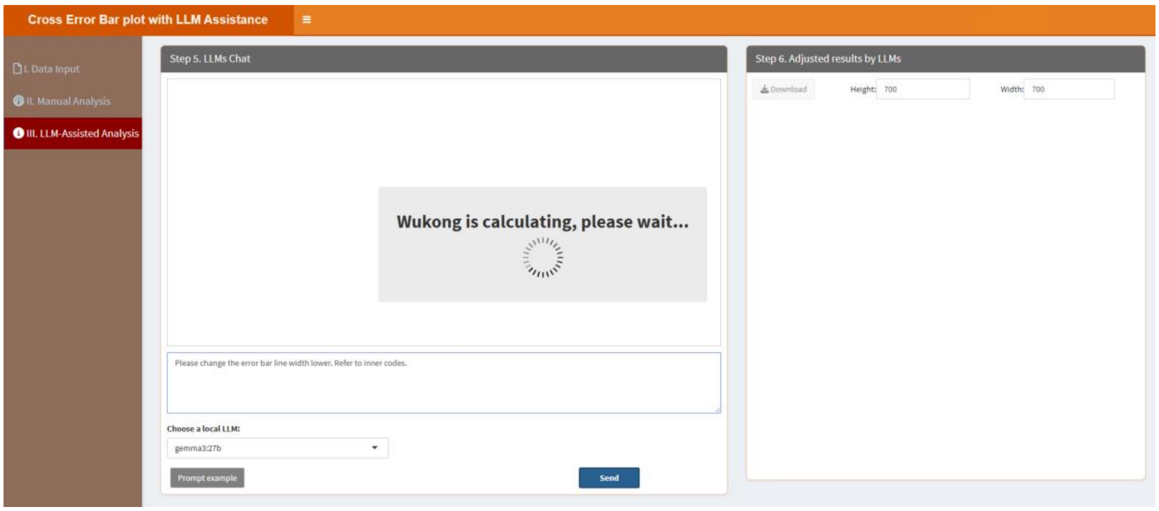
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



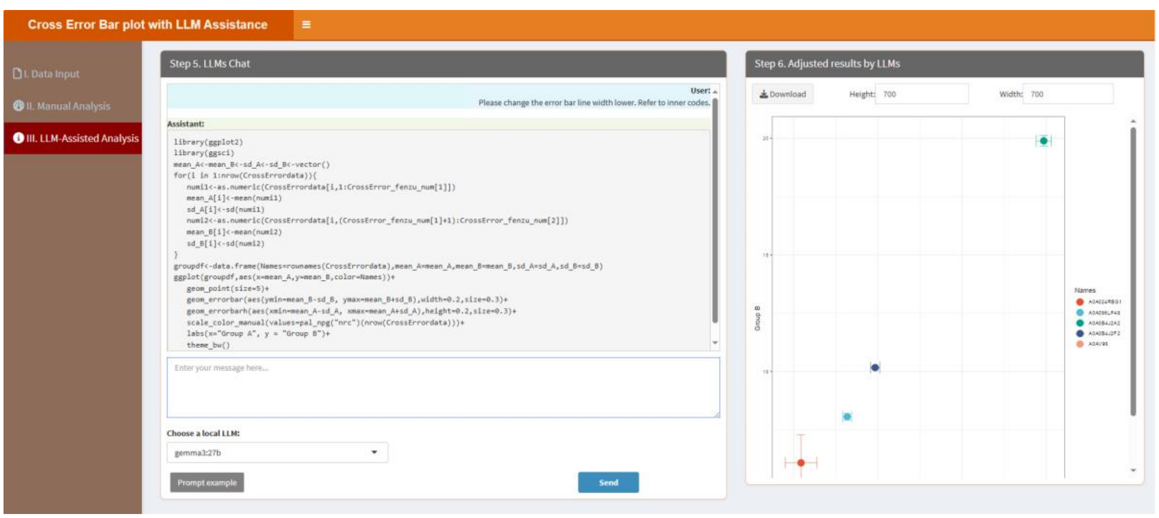
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the error bar line width lower. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the error bar line width lower. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

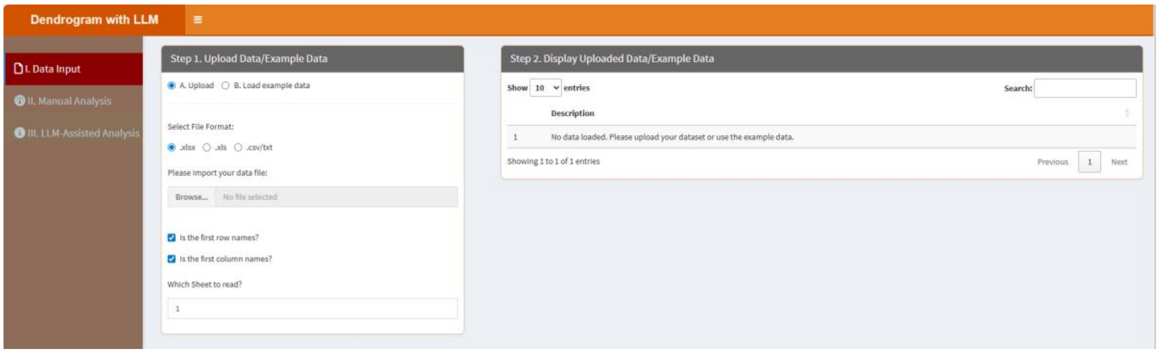


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



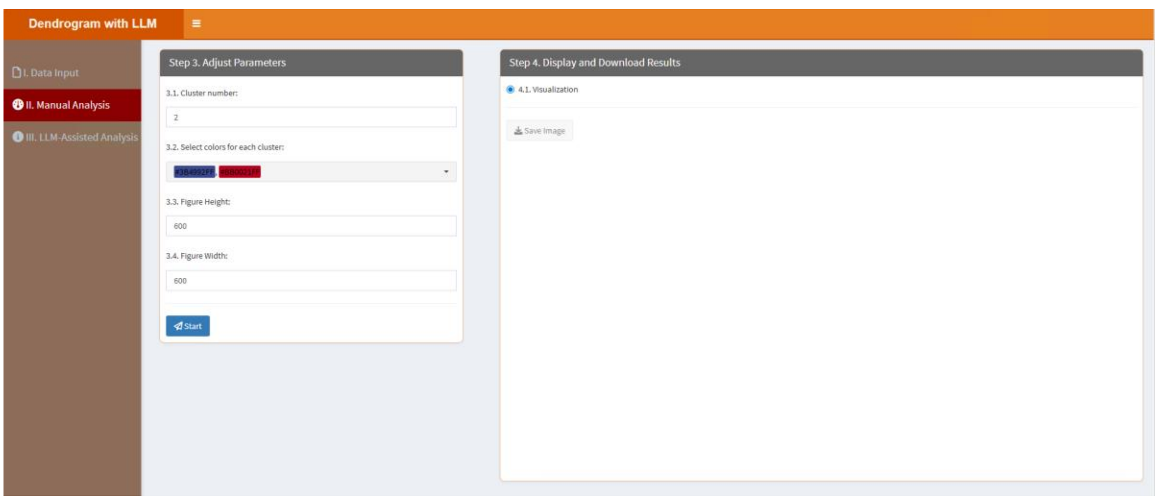
7.9. Dendrogram

Dendrograms are used for hierarchical clustering analysis. Users can customize the plot through the classic framework and enhance it with local LLMs to create detailed and interpretable visualizations. After clicking the "Start" button, it will be started in a new window like below:



The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.



3.1. Cluster number: This parameter allows users to specify the number of clusters into which the data should be grouped.

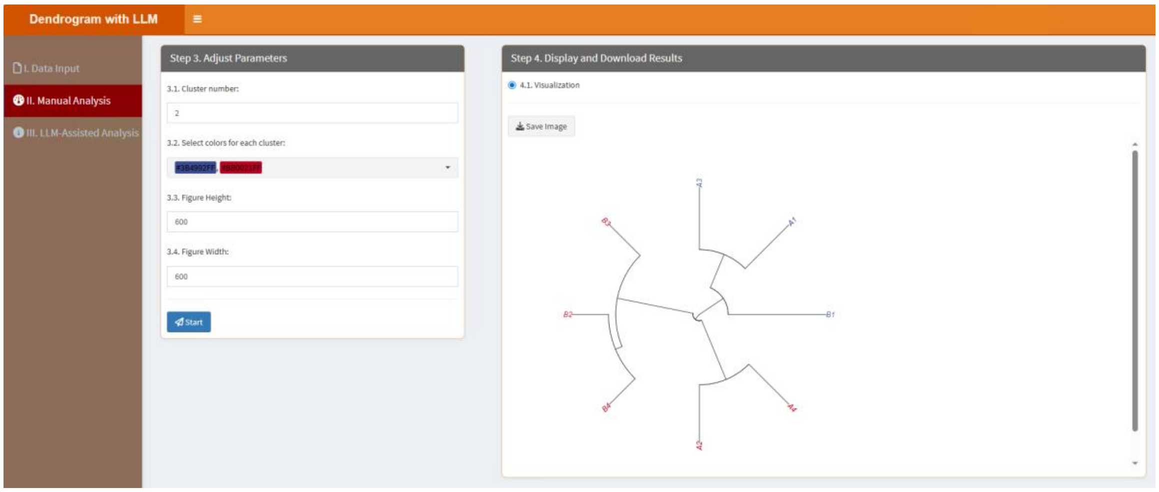
3.2. Select colors for each cluster: This setting lets users assign specific colors to each identified cluster for better visual distinction.

3.3. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.4. Figure Width: Similar to the figure height, this parameter specifies the width of the figure

in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

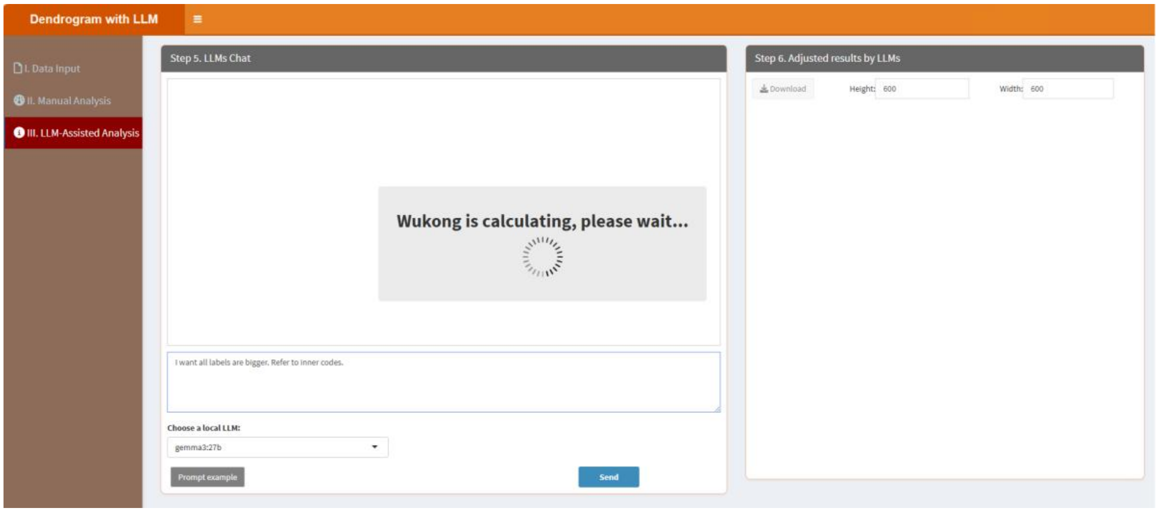
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



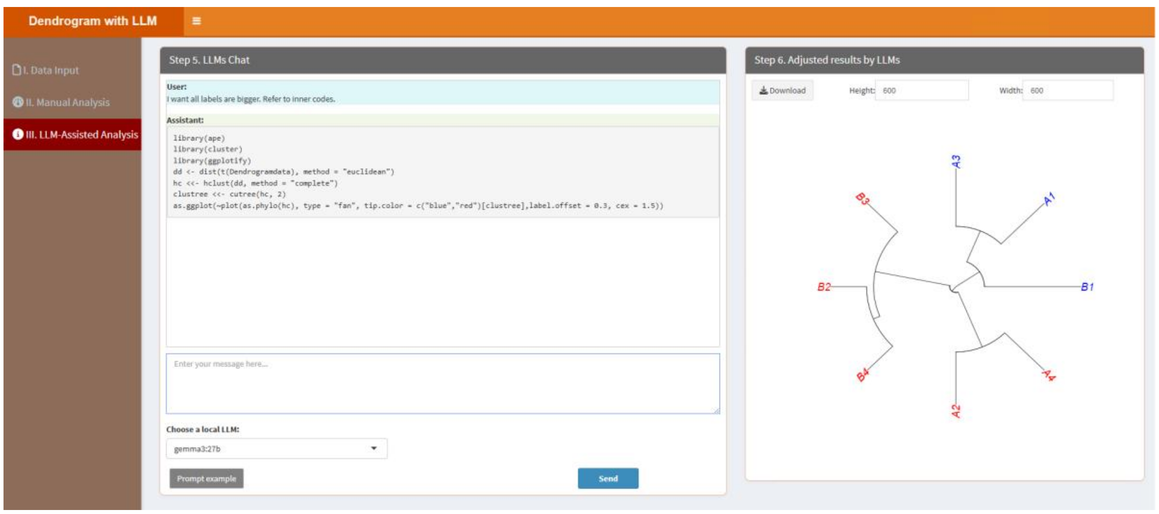
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “I want all labels are bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “I want all labels are bigger. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

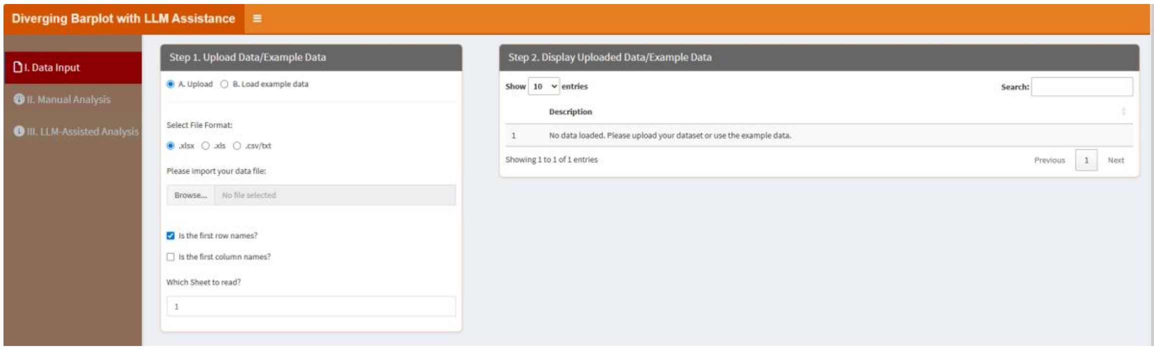


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



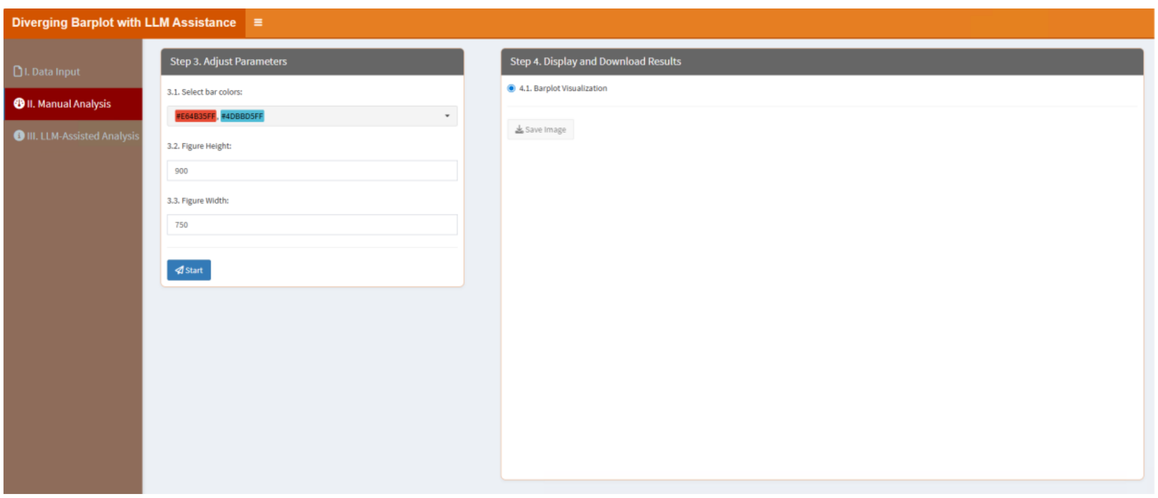
7.10. Diverging Barplot

Users can create diverging bar plots to represent data deviations. By adjusting parameters in the classic framework and refining with local LLMs, users can visualize positive and negative trends effectively. After clicking the "Start" button, it will be started in a new window like below:



The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.



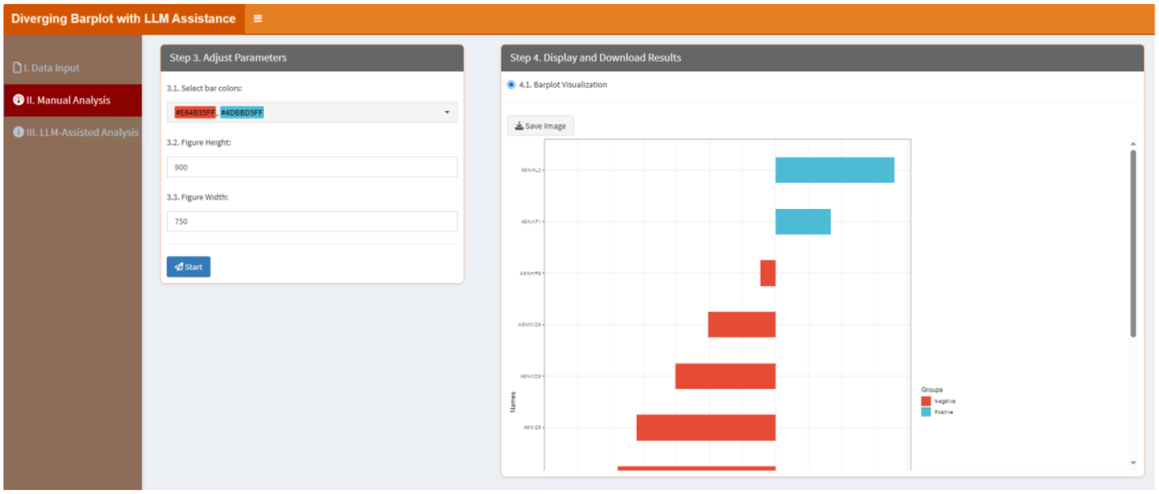
3.1. Select bar colors: Choose two distinct colors for the bar plot, one is for the negative value and the other is for the positive value in the uploaded data.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

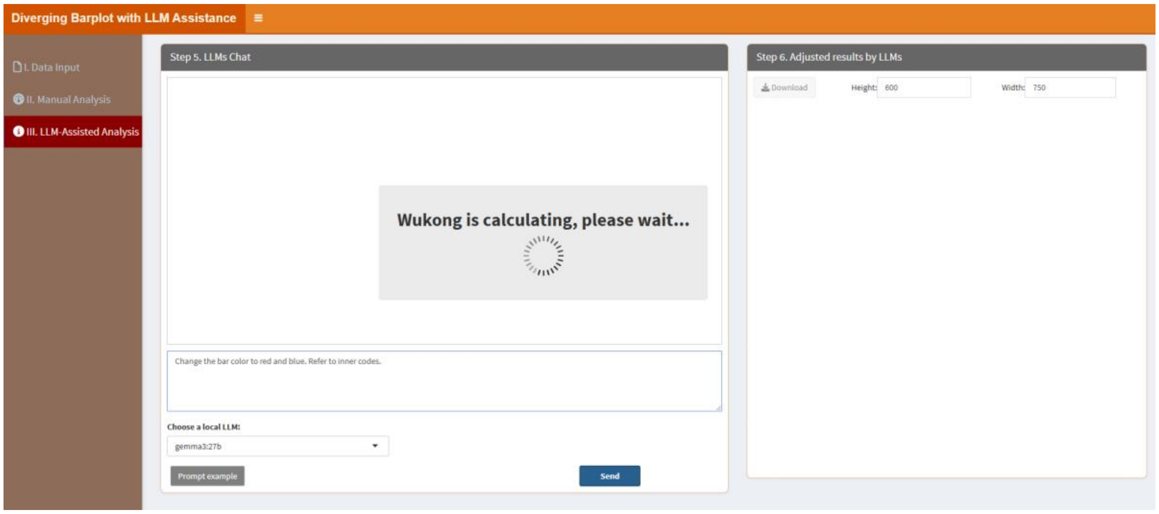
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



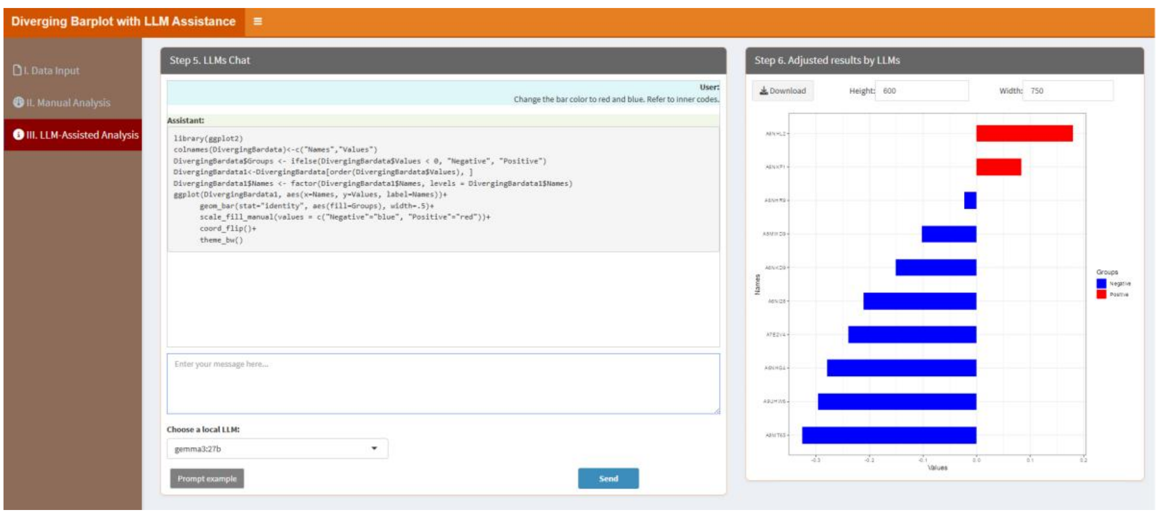
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Change the bar color to red and blue. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Change the bar color to red and blue. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

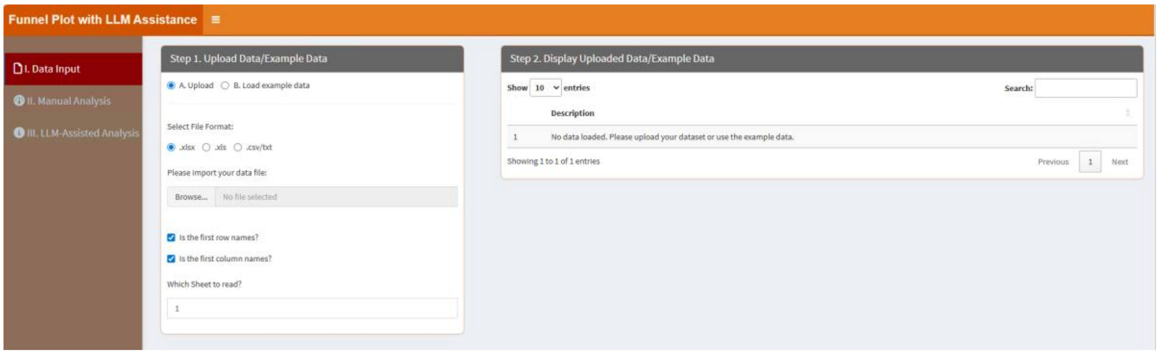


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



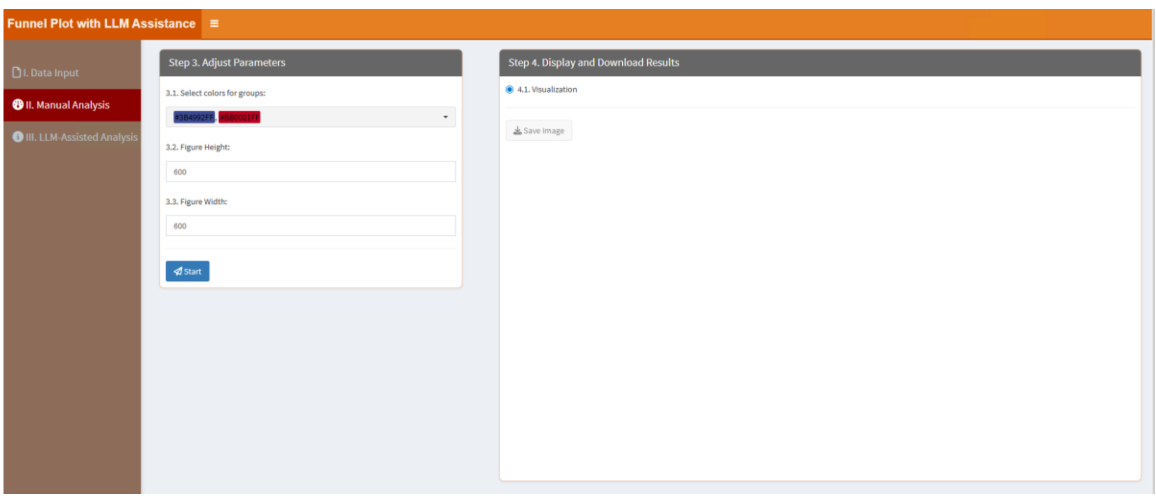
7.11. Funnel Plot

Funnel plots can be used to assess data variability and bias. Users can customize the plot using the classic framework and rely on local LLMs to generate clear and informative visualizations. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



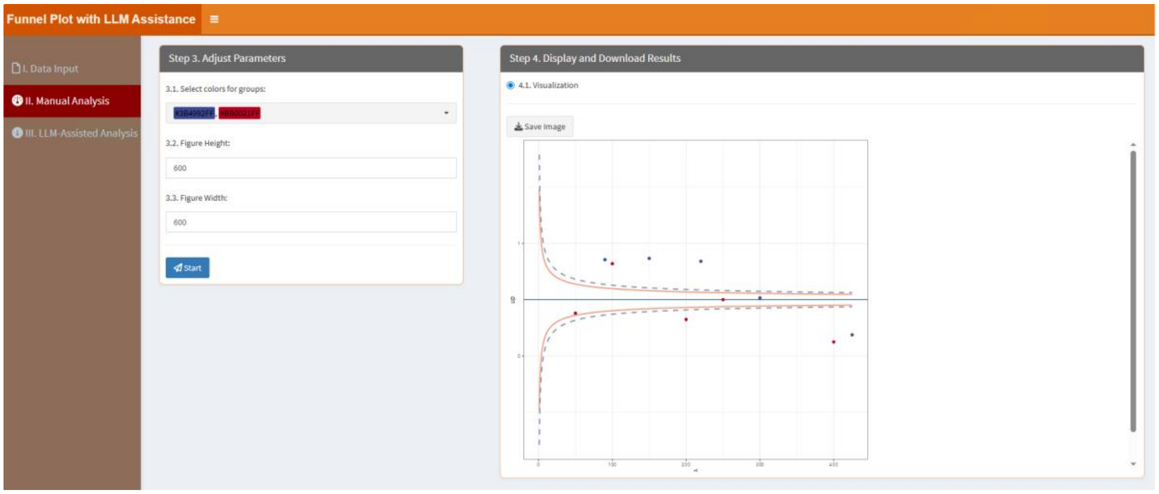
3.1. Select colors for groups: This parameter lets you choose specific colors for each group. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of groups, ensuring that the visual representation is both informative and aesthetically pleasing.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure

in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

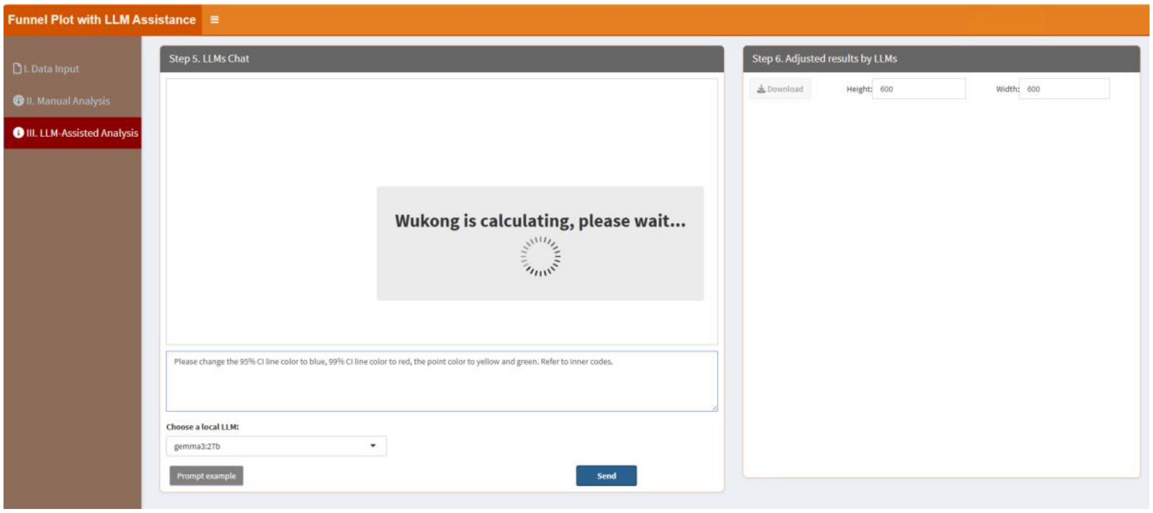
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



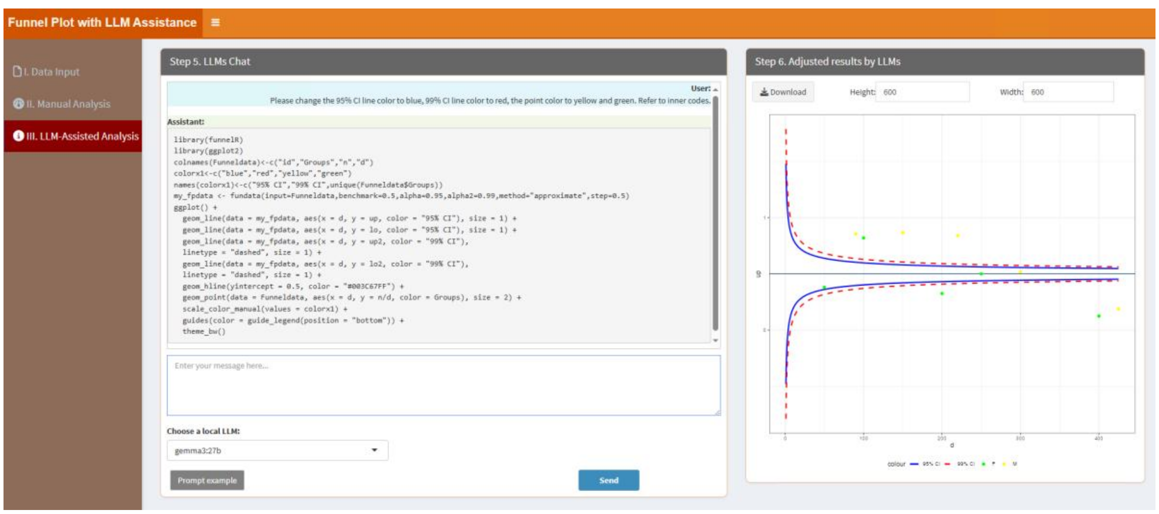
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the 95% CI line color to blue, 99% CI line color to red, the point color to yellow and green. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the 95% CI line color to blue, 99% CI line color to red, the point color to yellow and green. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

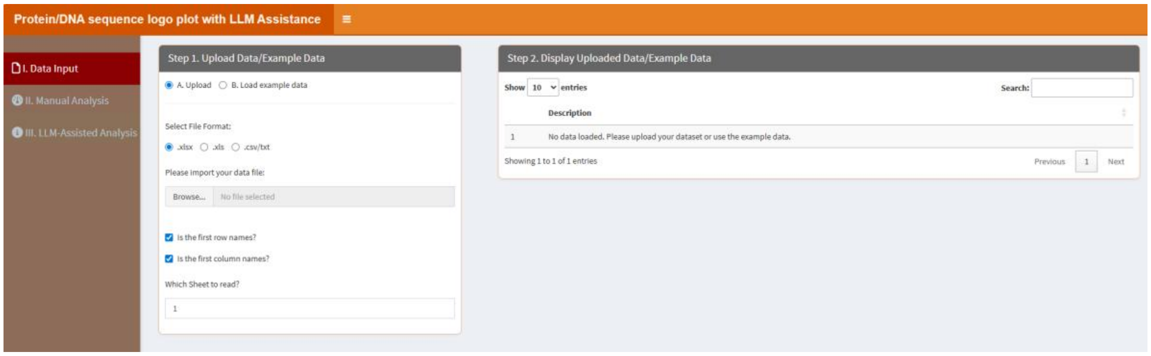


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



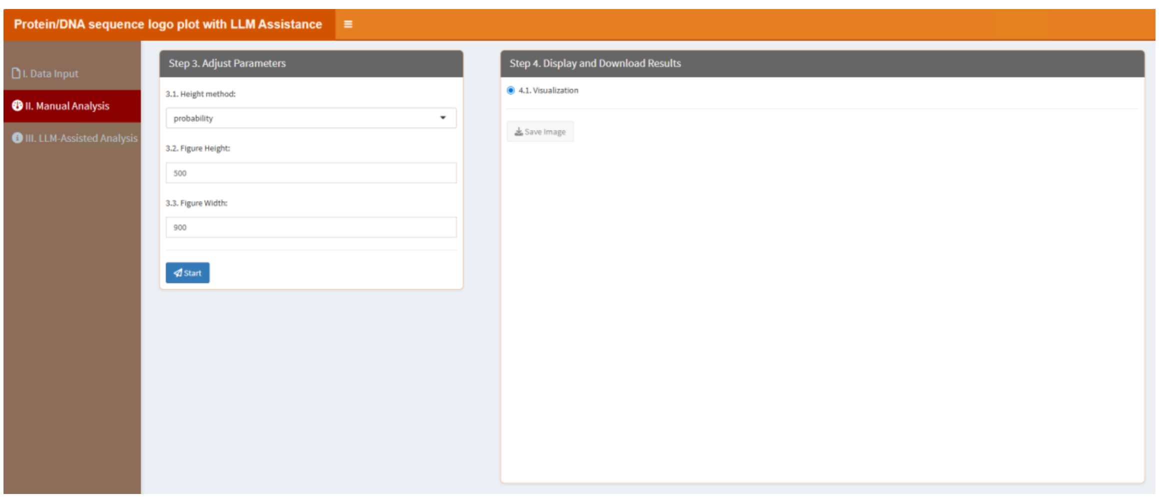
7.12. Protein/DNA sequence logo plot

This module allows users to generate sequence logo plots for protein or DNA sequences. Parameters can be adjusted in the classic framework, and local LLMs simplify the process to highlight conserved regions effectively. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

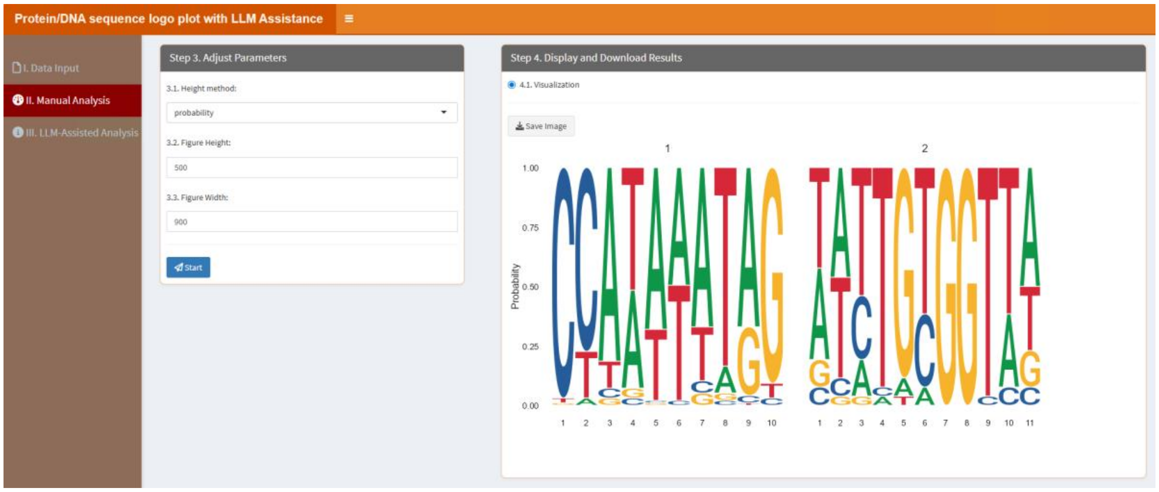


3.1. Height method: Height method, can be one of "bits" or "probability".

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

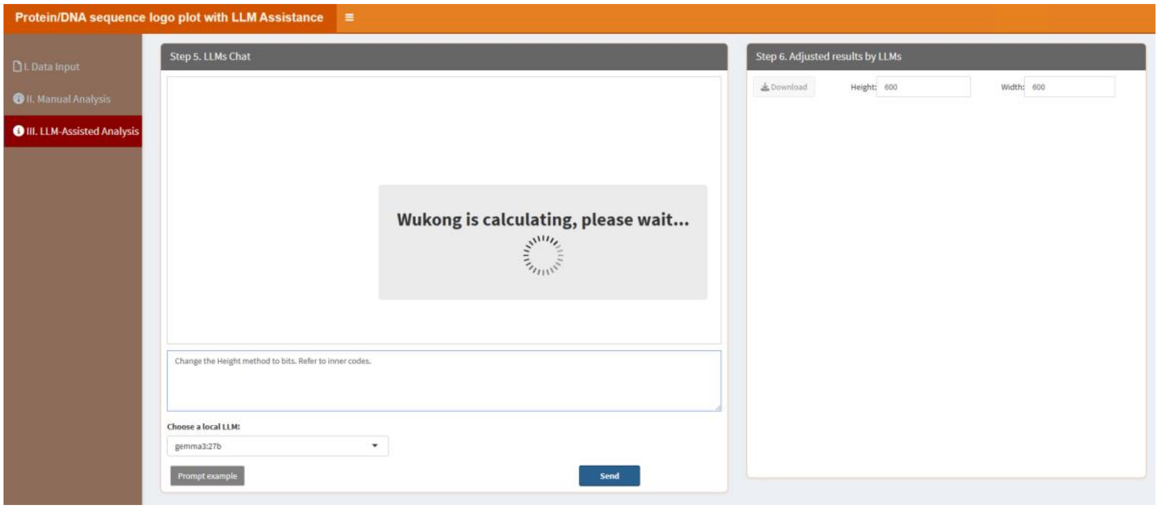
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



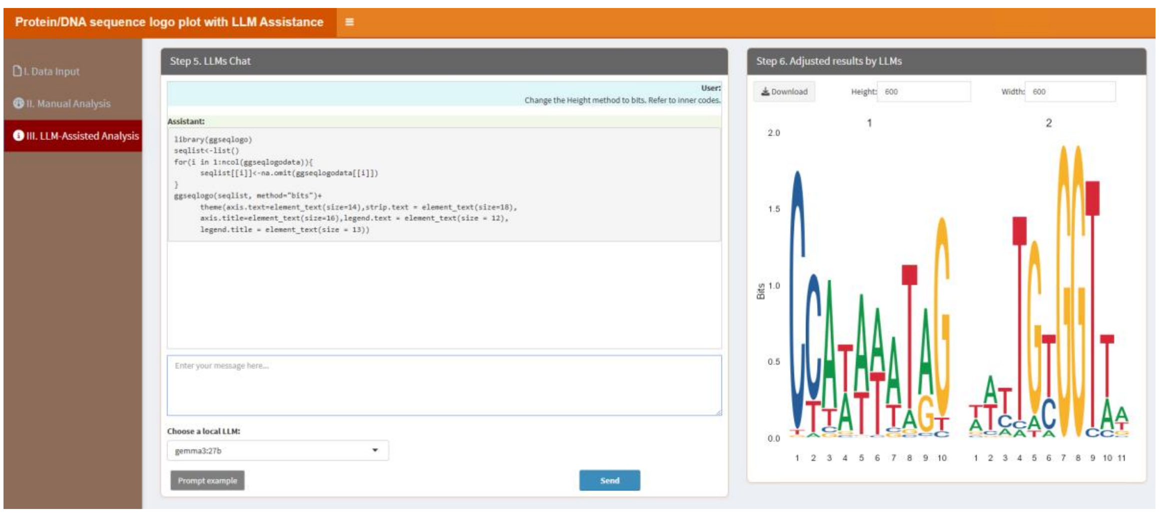
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Change the Height method to bits. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Change the Height method to bits. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

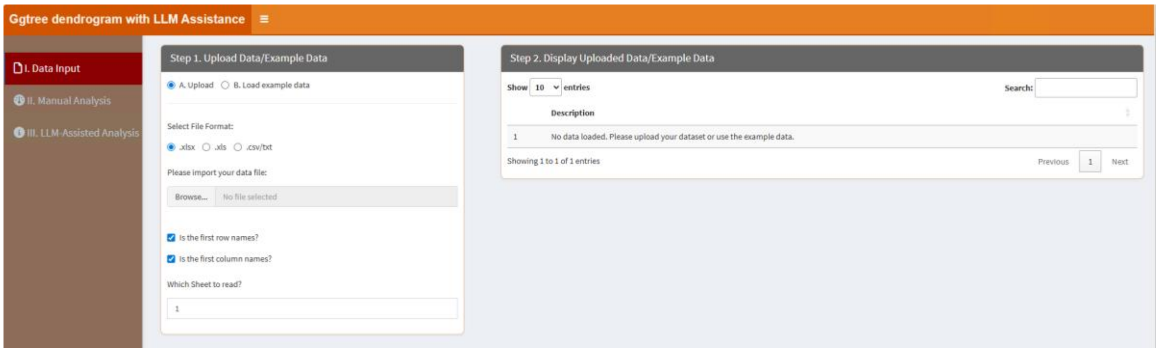


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



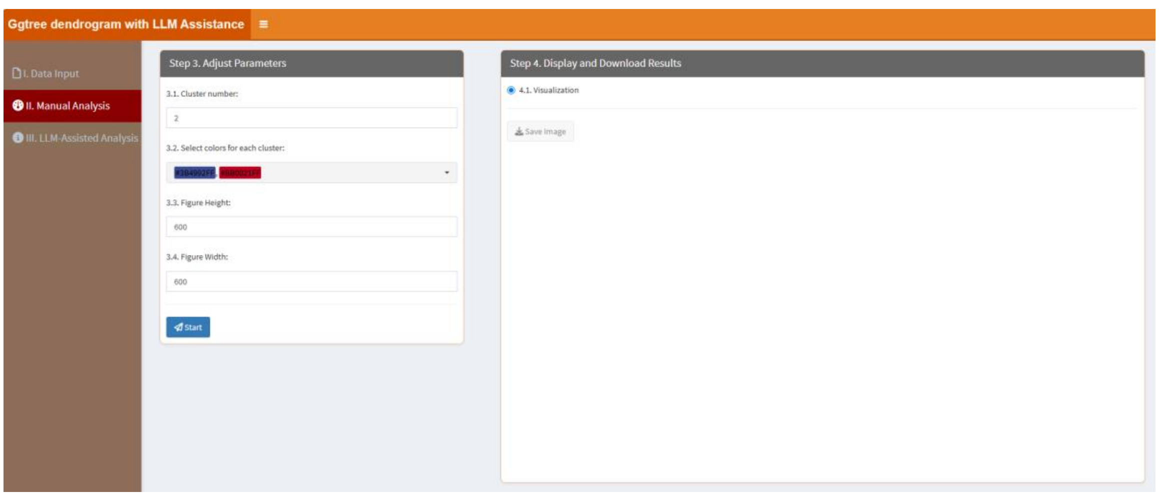
7.13. Dendrogram using ggtree package

Users can create advanced dendrograms using the ggtree package for hierarchical clustering visualization. The classic framework supports customization, while local LLMs assist in refining the plot for publication-ready results. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Cluster number: Maximum cluster number to evaluate.

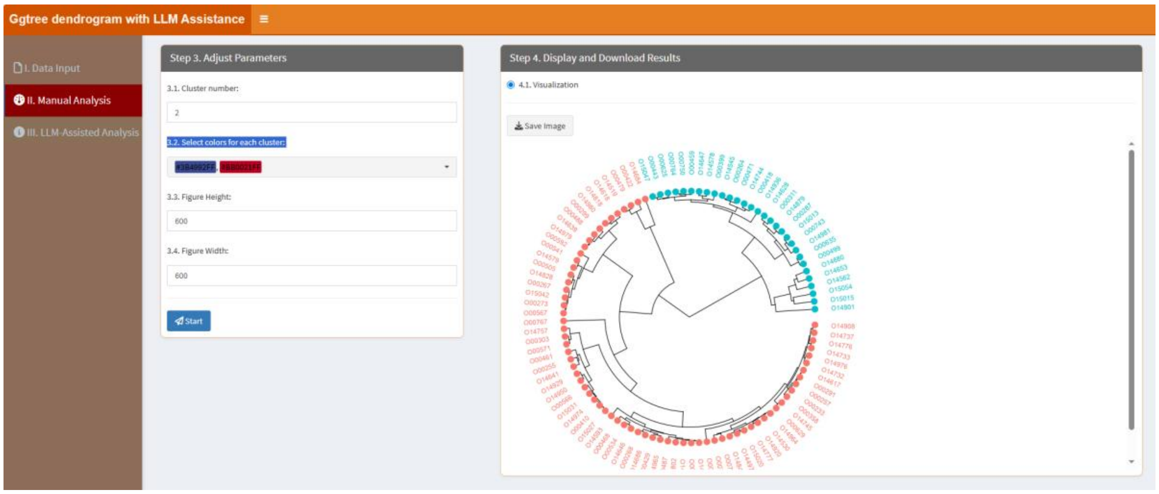
3.2. Select colors for each cluster: This parameter lets you choose specific colors for each cluster. A color picker is provided, offering a variety of predefined palettes such as "npg," "aaas," "nejm," and others, along with options for custom colors. You can select multiple colors to match the number of clusters, ensuring that the visual representation is both informative and aesthetically pleasing.

3.3. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.4. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

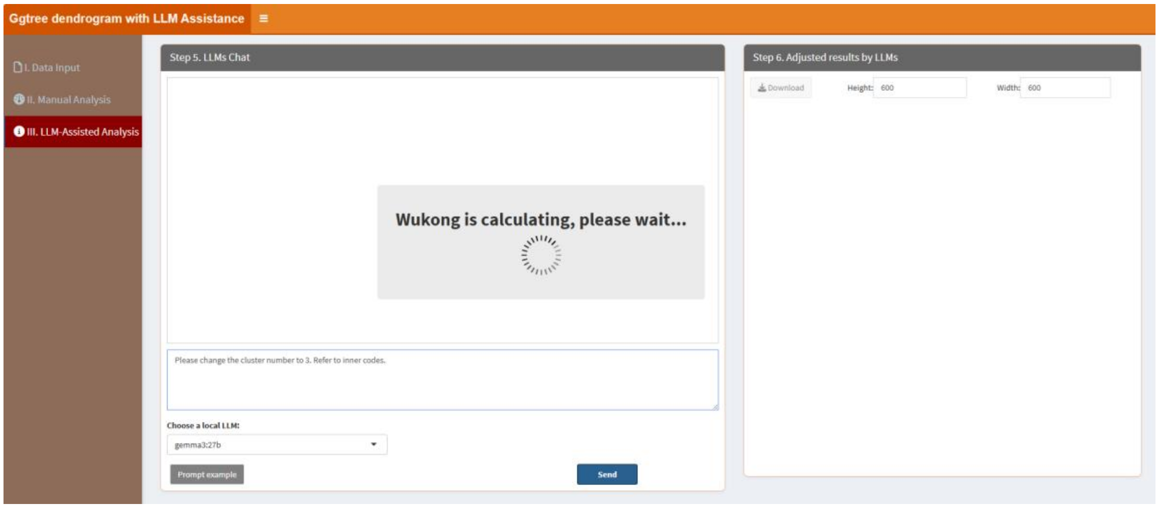


“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

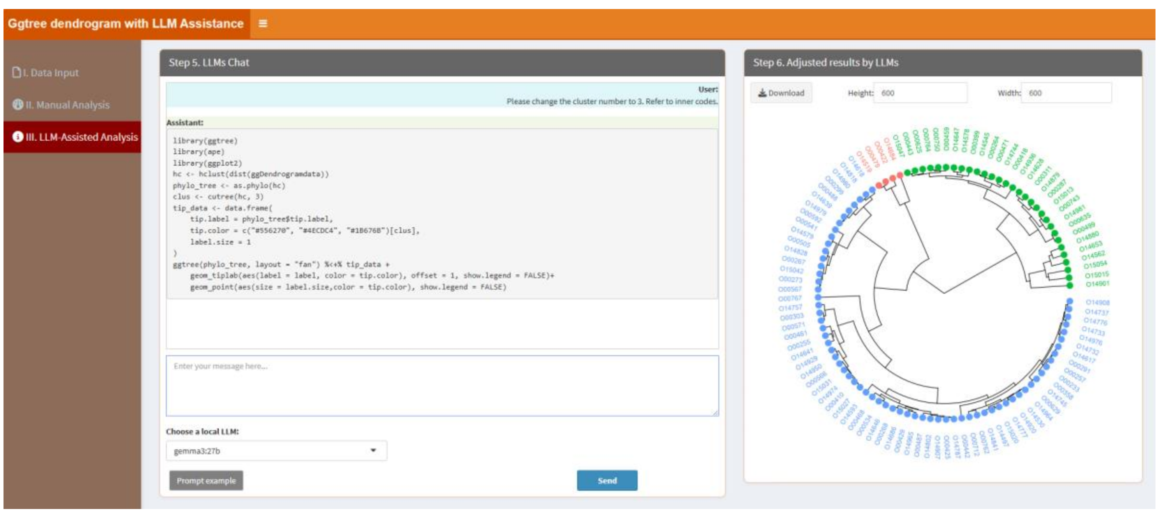
When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the cluster number to 3. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the cluster number to 3. =>”, it will similarly prompt

LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

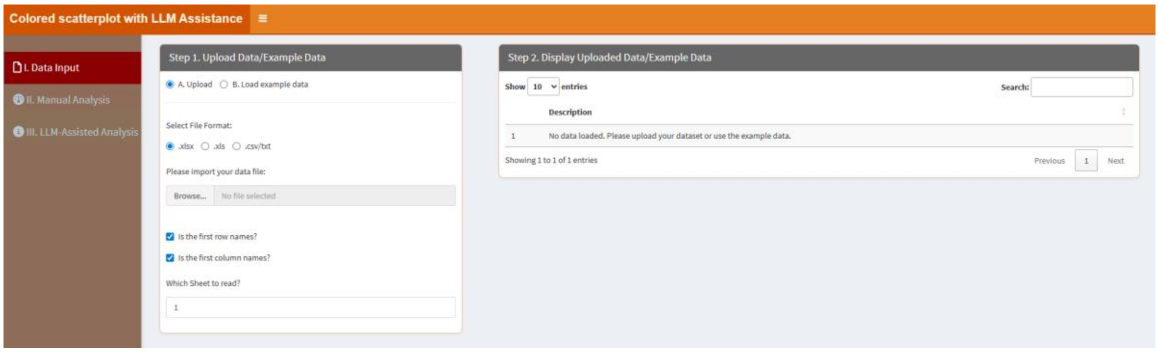


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



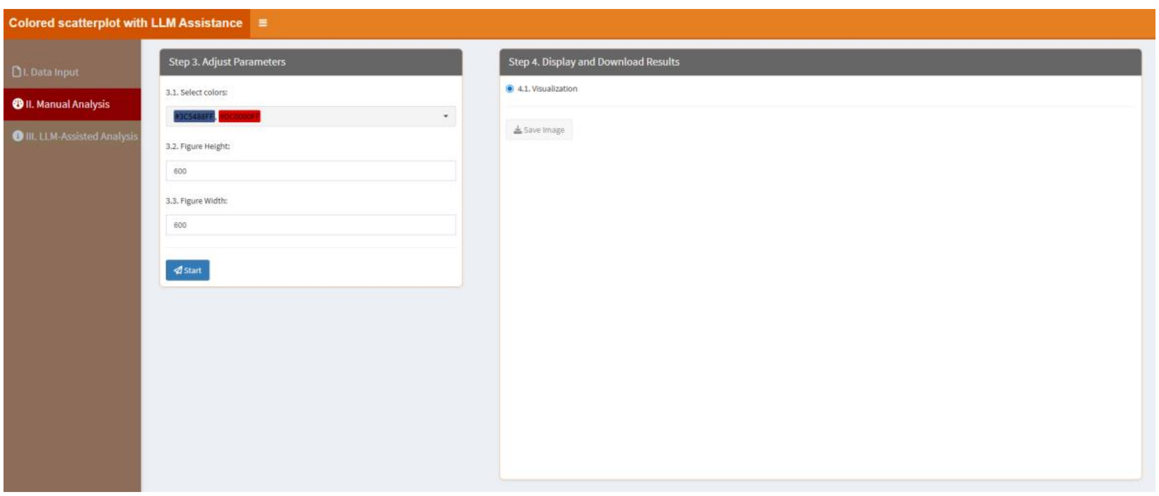
7.14. Colored scatterplot

Colored scatterplots combine scatterplots with color gradients to represent data density. Users can adjust parameters in the classic framework and use local LLMs to enhance the visualization's aesthetics and informativeness. After clicking the "Start" button, it will be started in a new window like below:



The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.



3.1. Select colors: Choose two distinct colors for the bar plot, one is for the lowest value and the other is for the highest value in the uploaded data.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

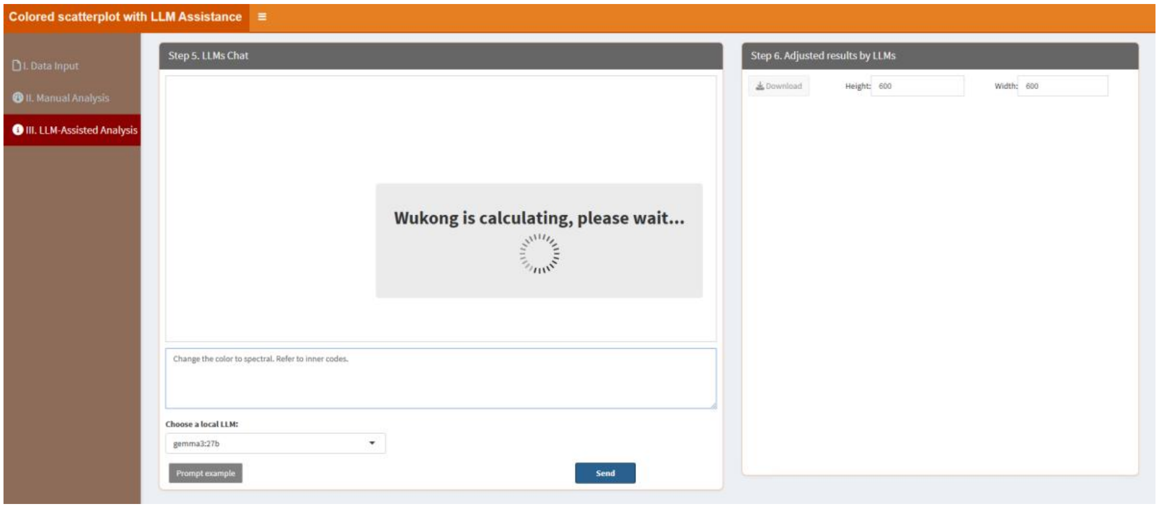
3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:

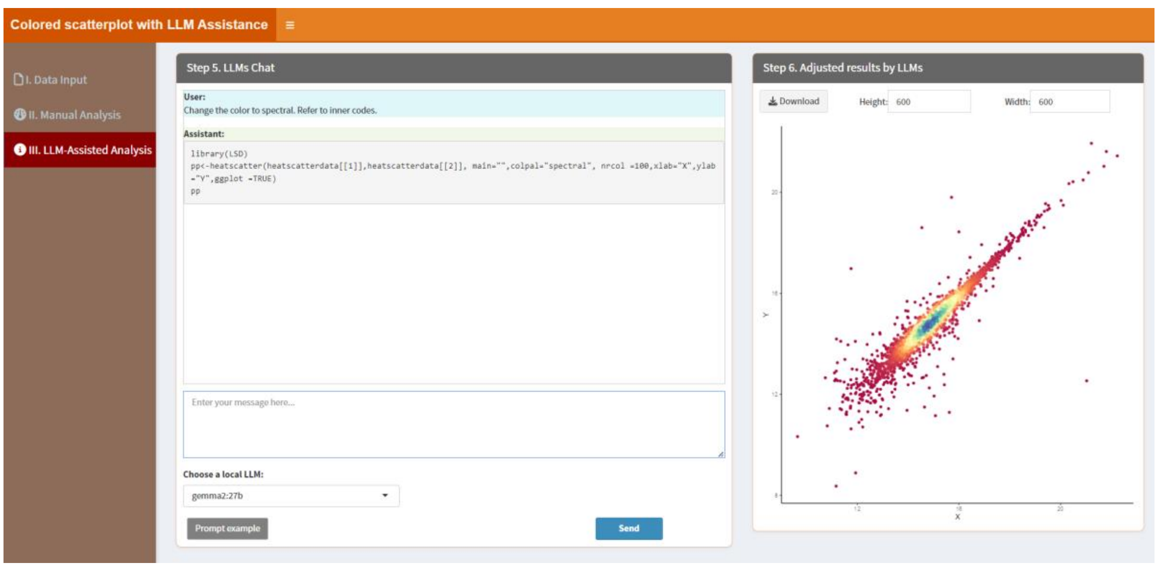
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Change the color to spectral. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Change the color to spectral. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

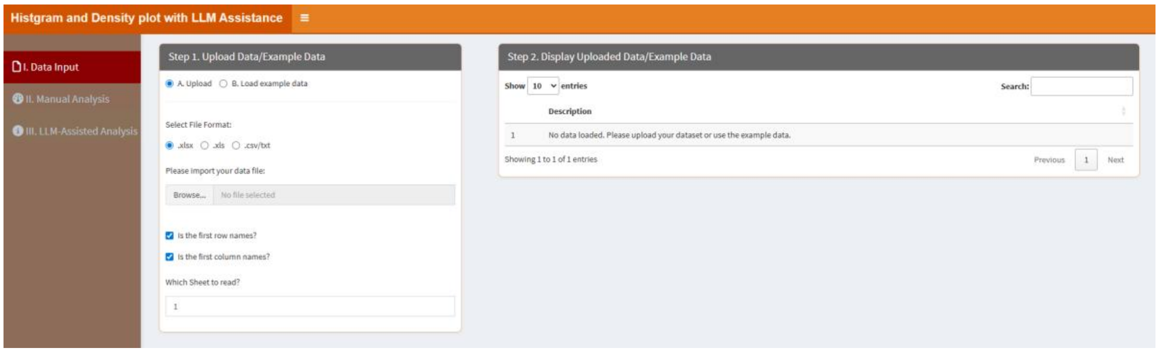


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



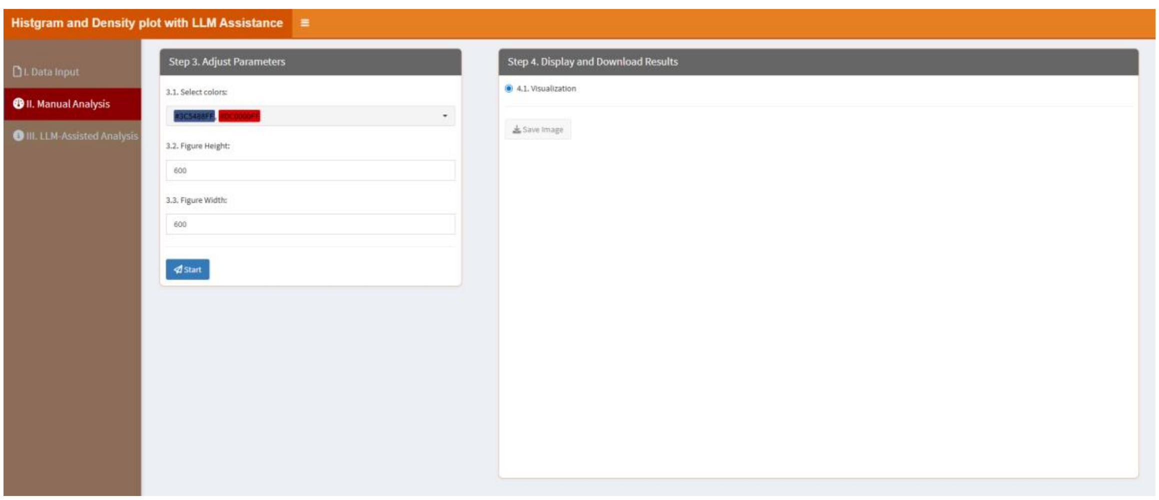
7.15. Histogram and Density plot

This module allows users to create combined histogram and density plots to visualize data distributions. By leveraging the classic framework and local LLMs, users can generate detailed and interpretable plots. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

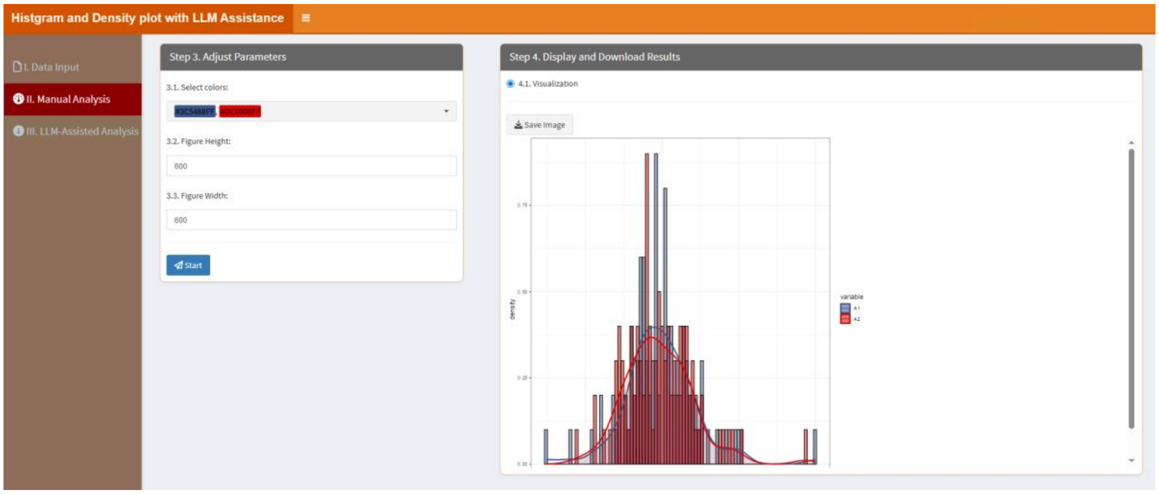


3.1. Select colors: This parameter lets you choose specific colors for each column. The color number should be equal to the column number.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

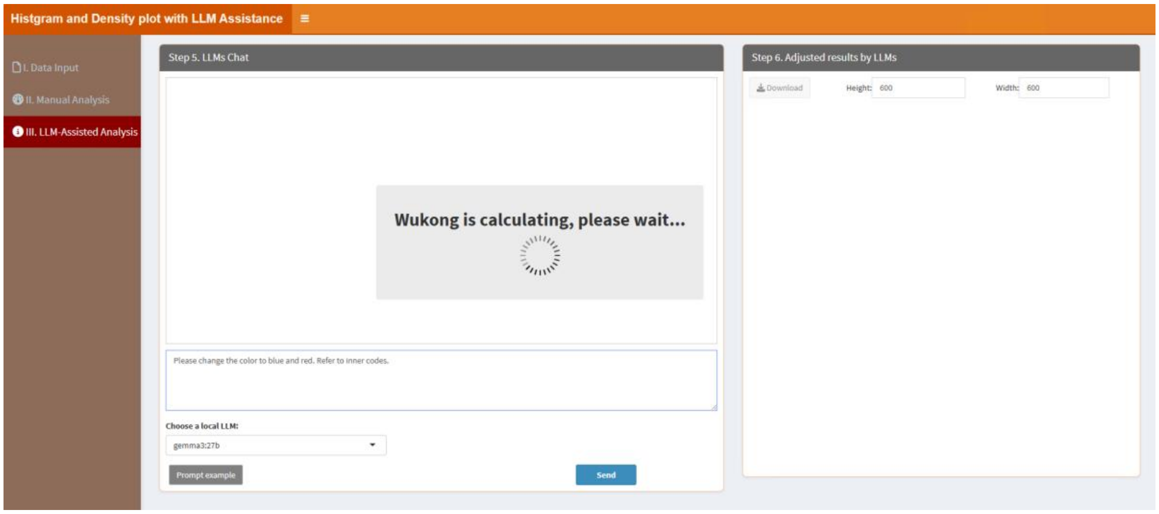
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



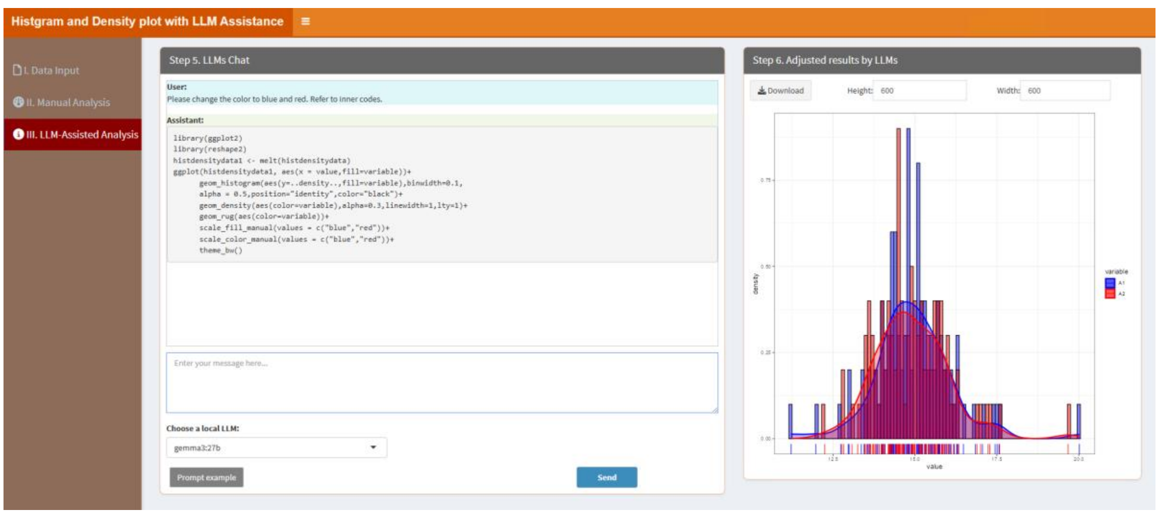
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the color to blue and red. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the color to blue and red. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

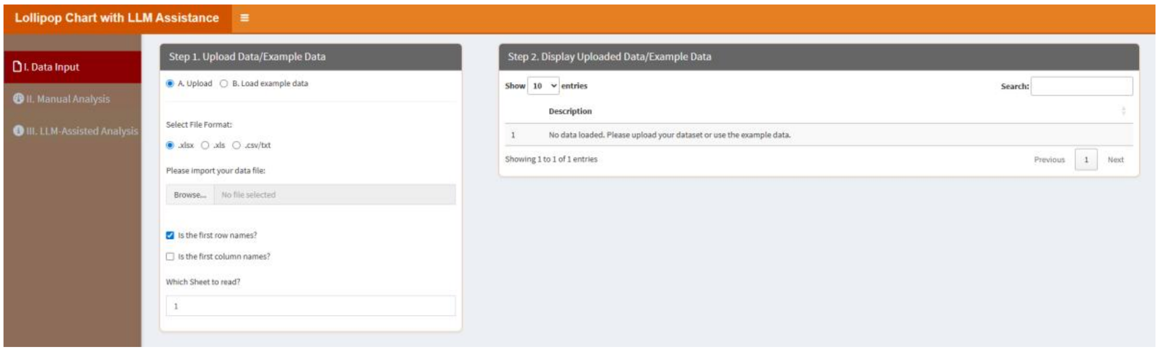


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



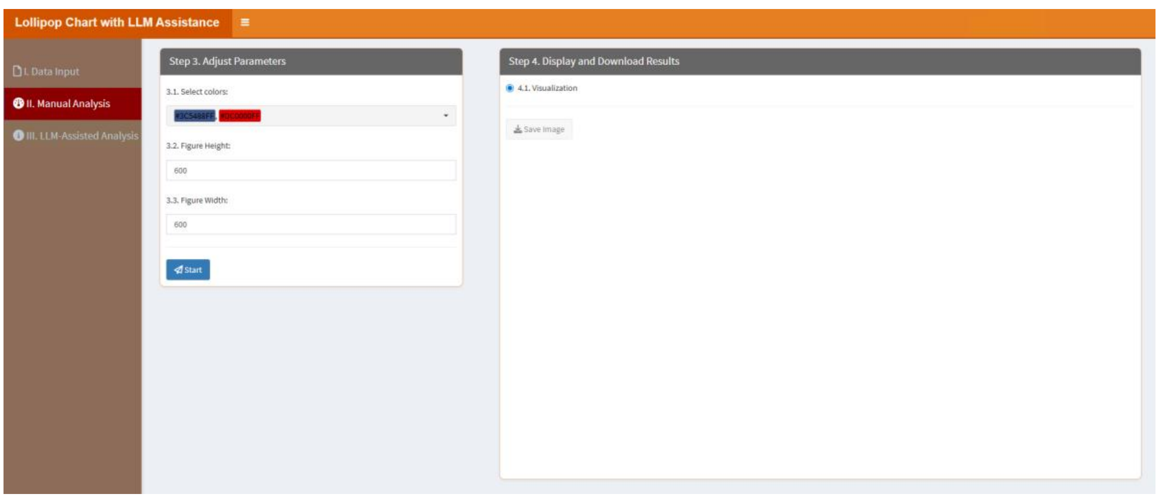
7.16. Lollipop Chart

Lollipop charts are ideal for comparing data points. Users can adjust settings through the classic framework and refine the visualization with local LLMs for clear and effective representation. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

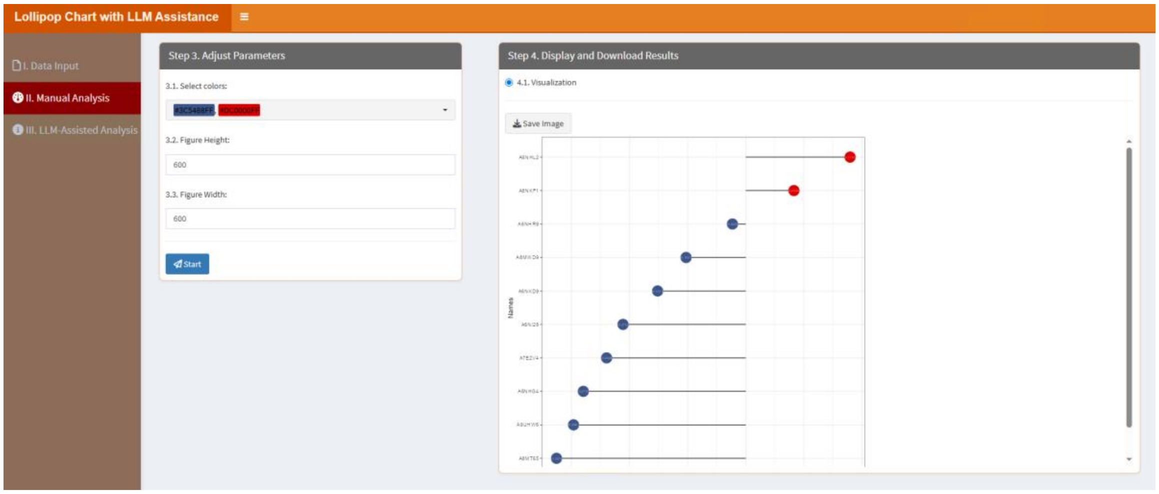


3.1. Select colors: Choose two distinct colors for the bar plot, one is for the negative value and the other is for the positive value in the uploaded data.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

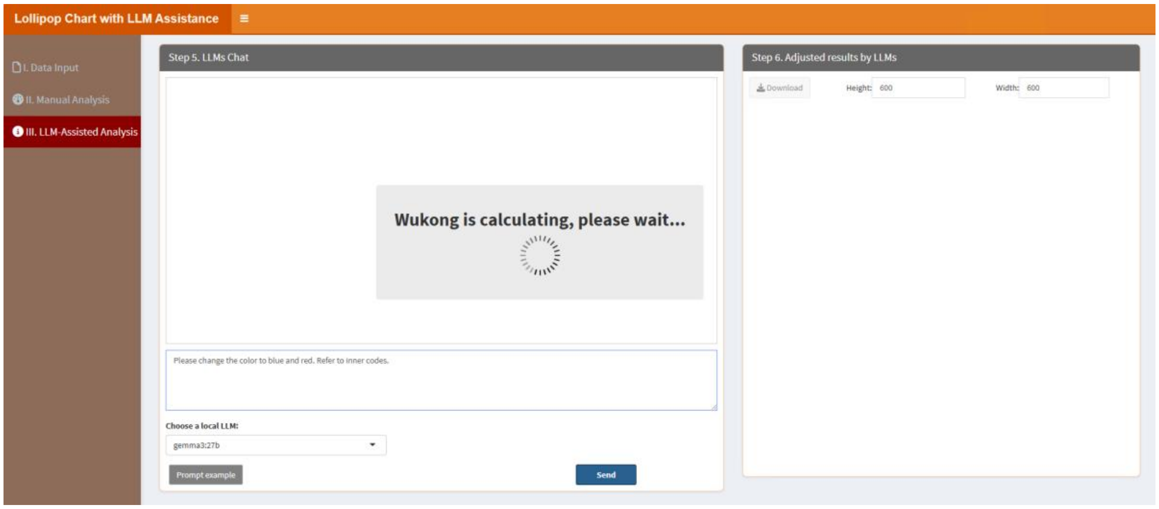
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



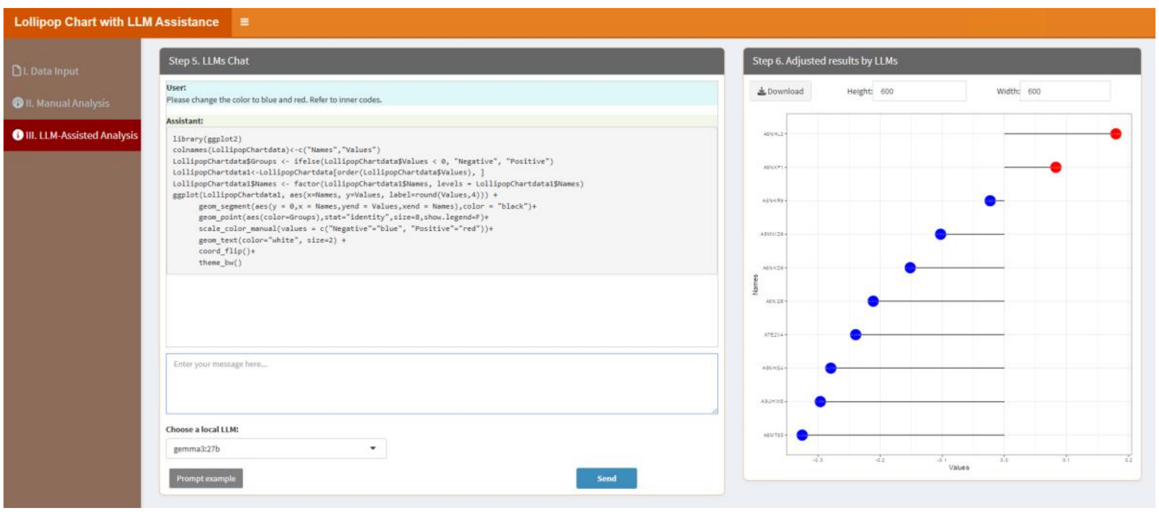
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the color to blue and red. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the color to blue and red. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

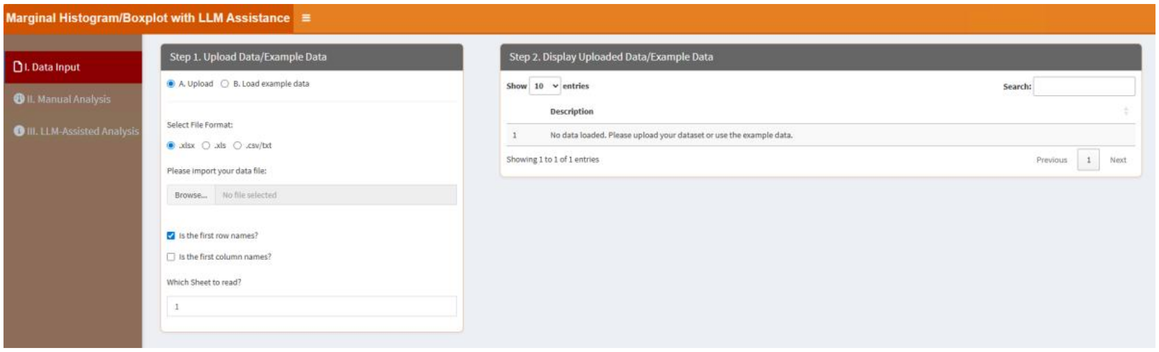


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



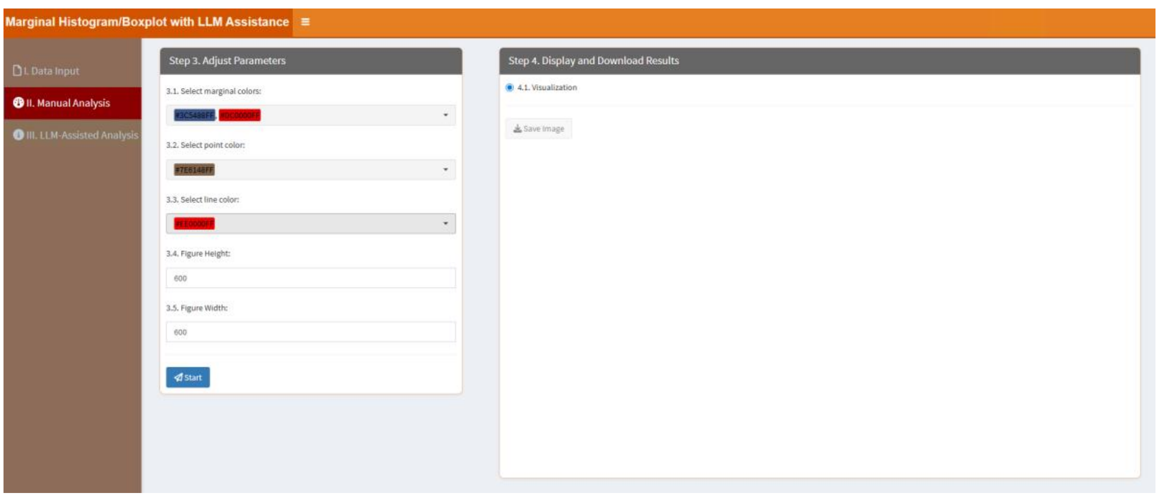
7.17. Marginal Histogram/Boxplot

Marginal histograms or boxplots can be generated alongside scatterplots to visualize data distributions. Users can customize the plot through the classic framework and enhance it with local LLMs. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Select marginal colors: Choose two distinct colors for the marginal plot.

3.2. Select point color: Choose a color for the points.

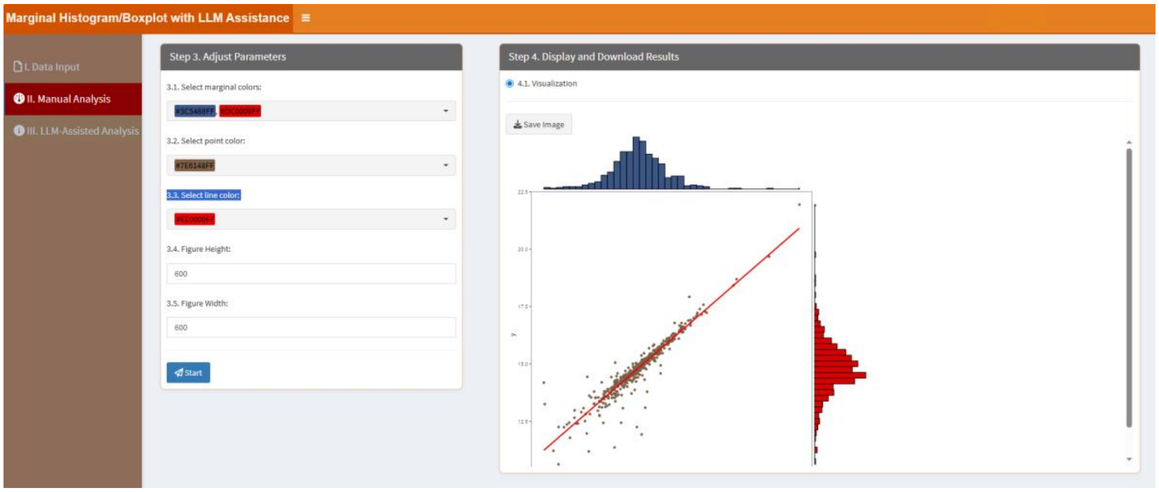
3.3. Select line color: Choose a color for the line.

3.4. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

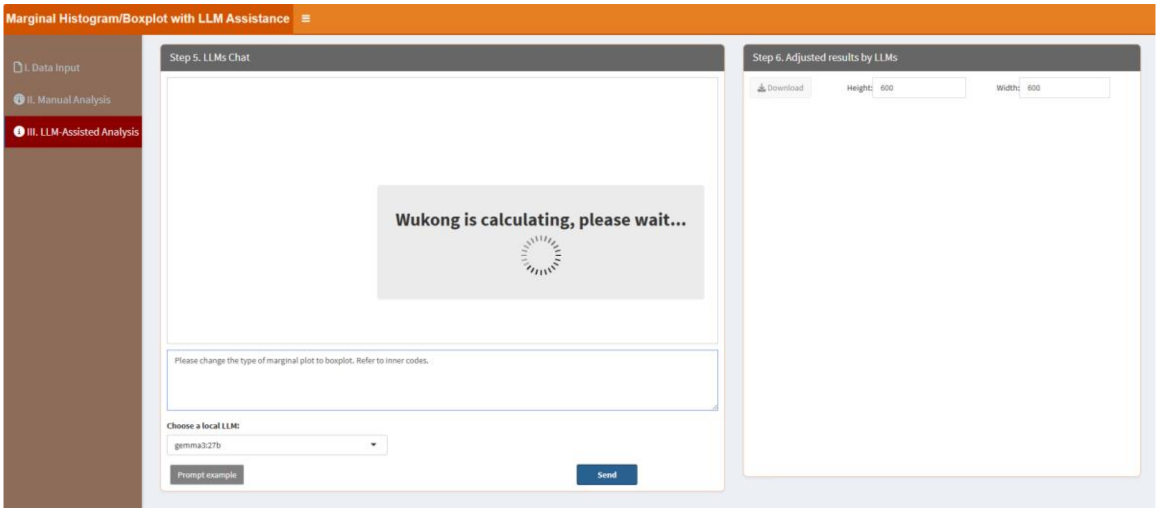
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



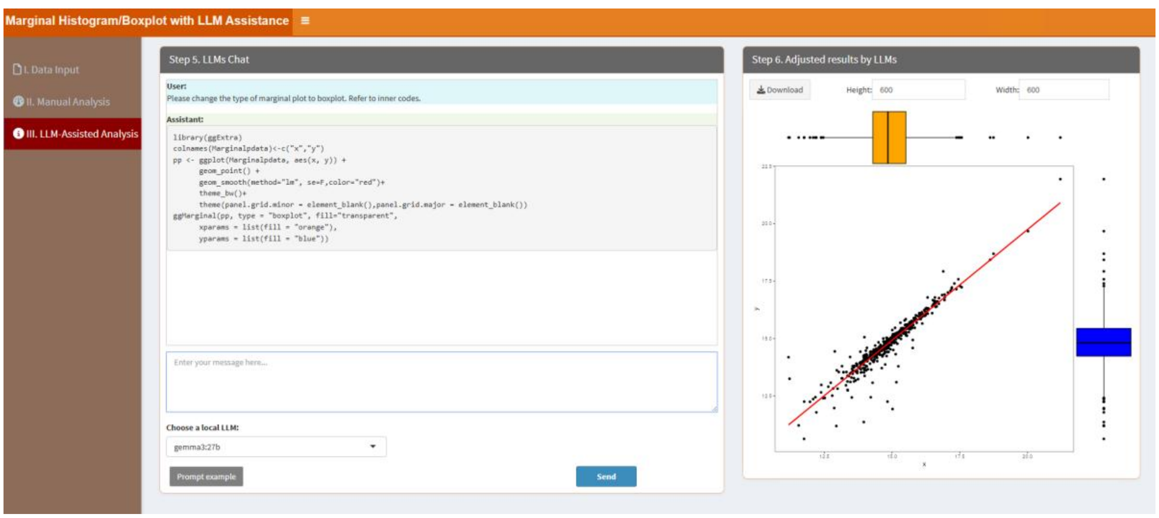
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the type of marginal plot to boxplot. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the type of marginal plot to boxplot. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

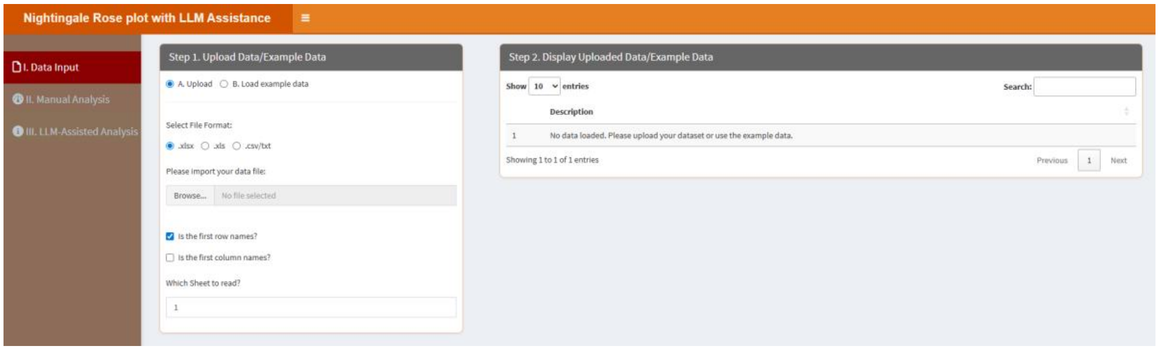


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



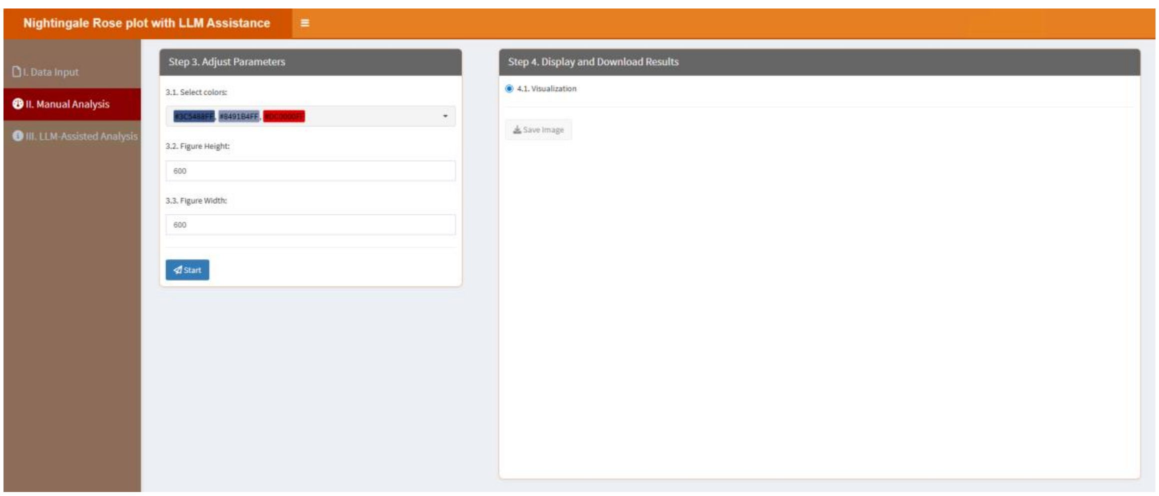
7.18. Nightingale Rose Diagram

Nightingale rose diagrams visualize circular data distributions. Users can adjust parameters in the classic framework and use local LLMs to ensure accurate and visually appealing results. After clicking the "Start" button, it will be started in a new window like below:



The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.

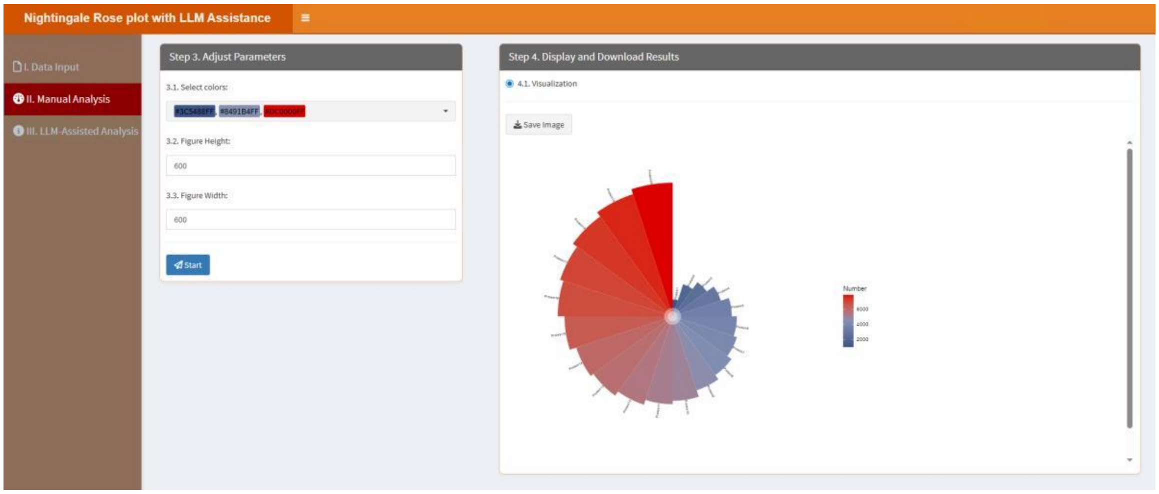


3.1. Select colors: Choose three distinct colors for the circle bar plot, one is for the lowest value, one is for the middle value and the last is for the highest value in the uploaded data.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

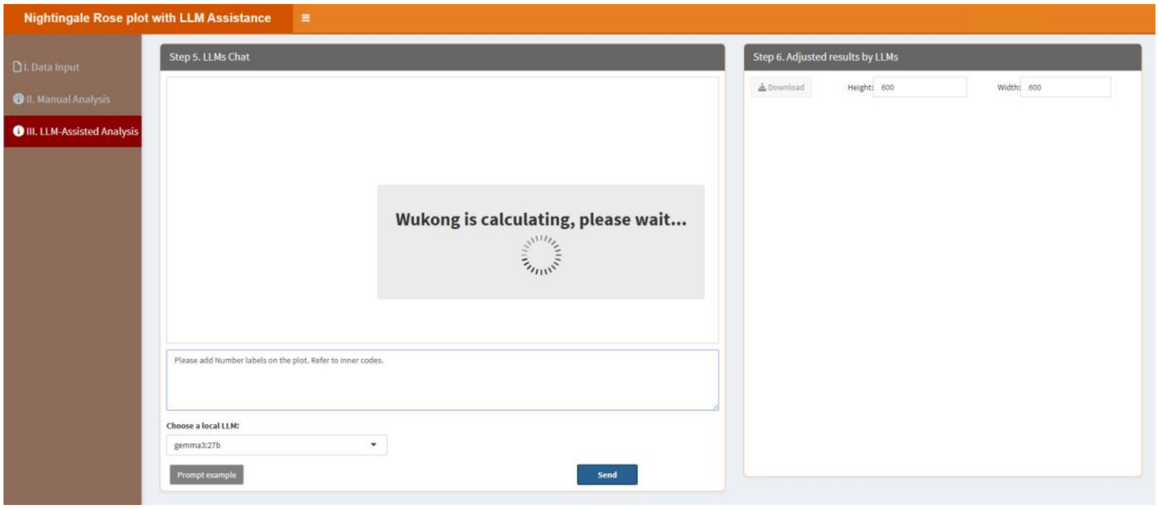
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



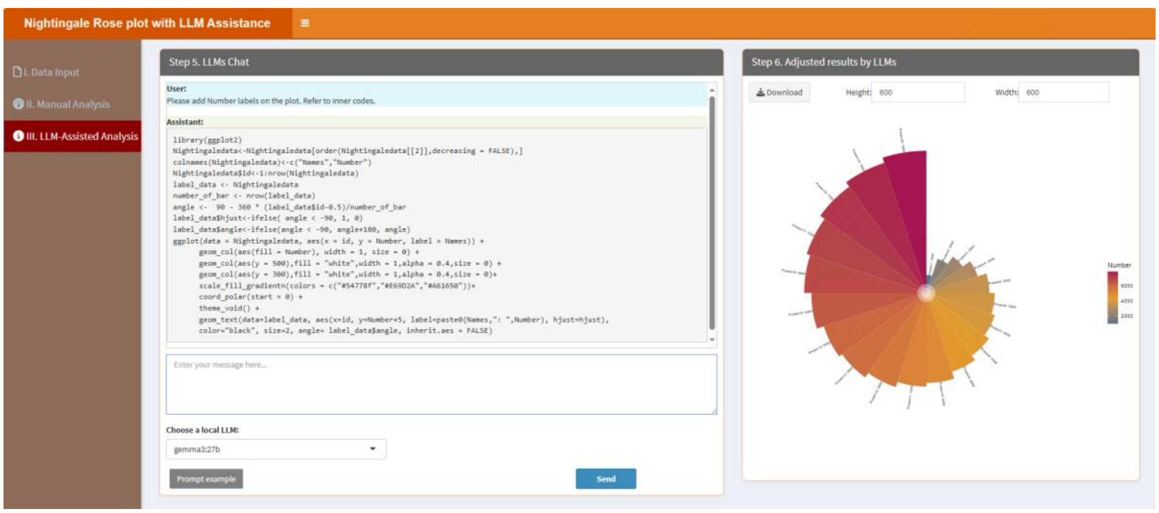
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please add Number labels on the plot. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please add Number labels on the plot. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

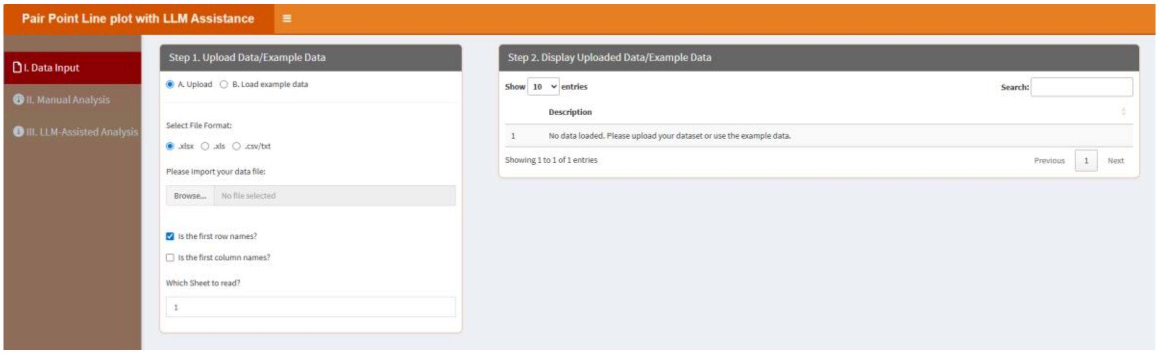


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



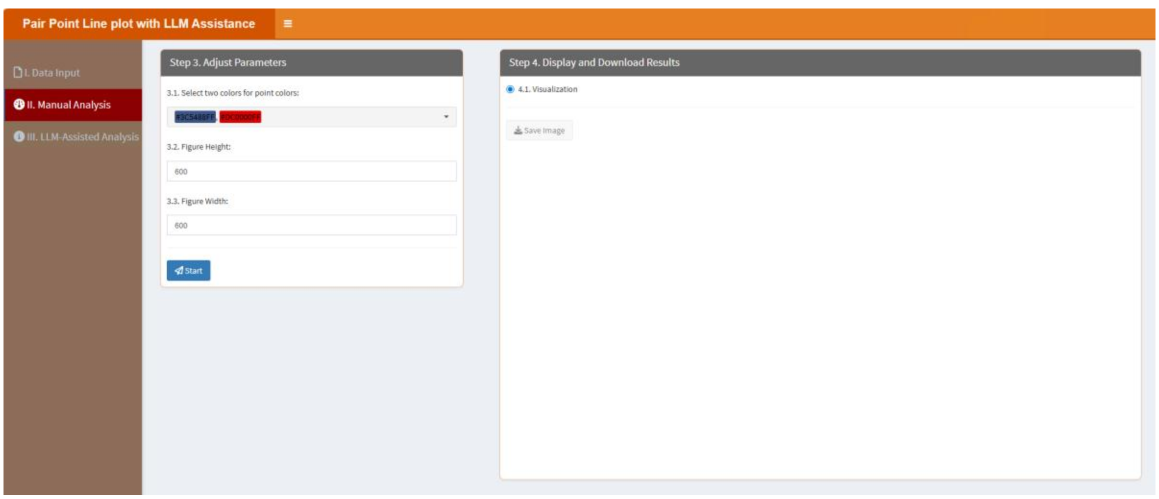
7.19. Pair Point Line plot

Pair point line plots highlight relationships between paired data points. Users can customize the plot through the classic framework and refine it with local LLMs for better clarity. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



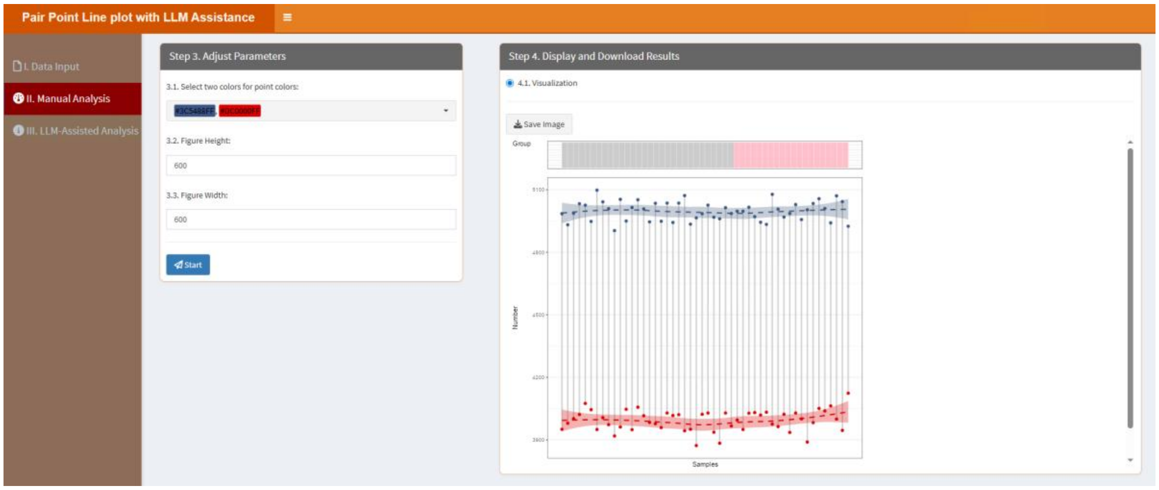
3.1. Select two colors for point colors: This parameter allows users to assign two distinct colors to visually differentiate the points representing two groups in the pair point line plot.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

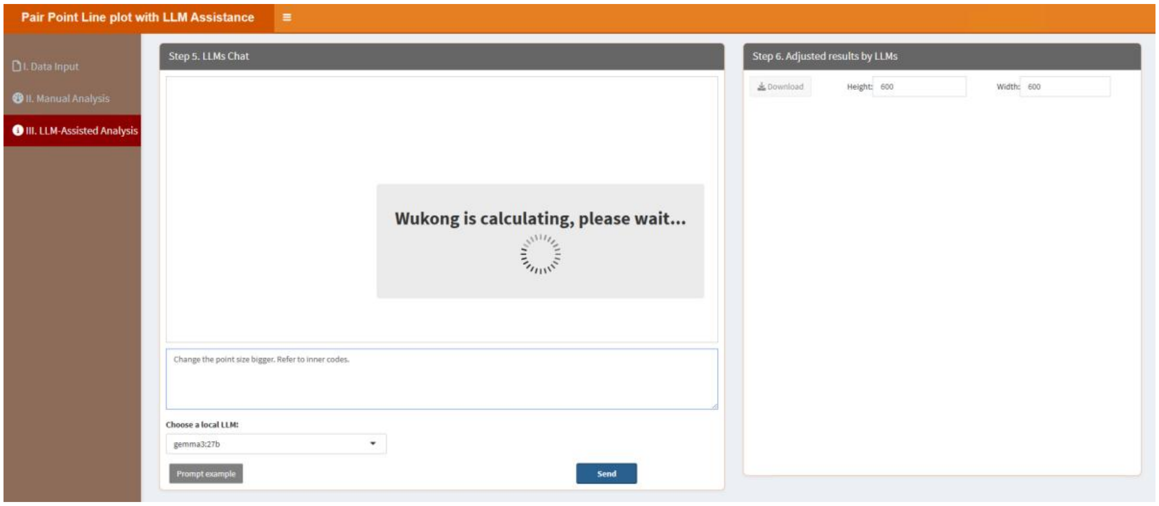
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



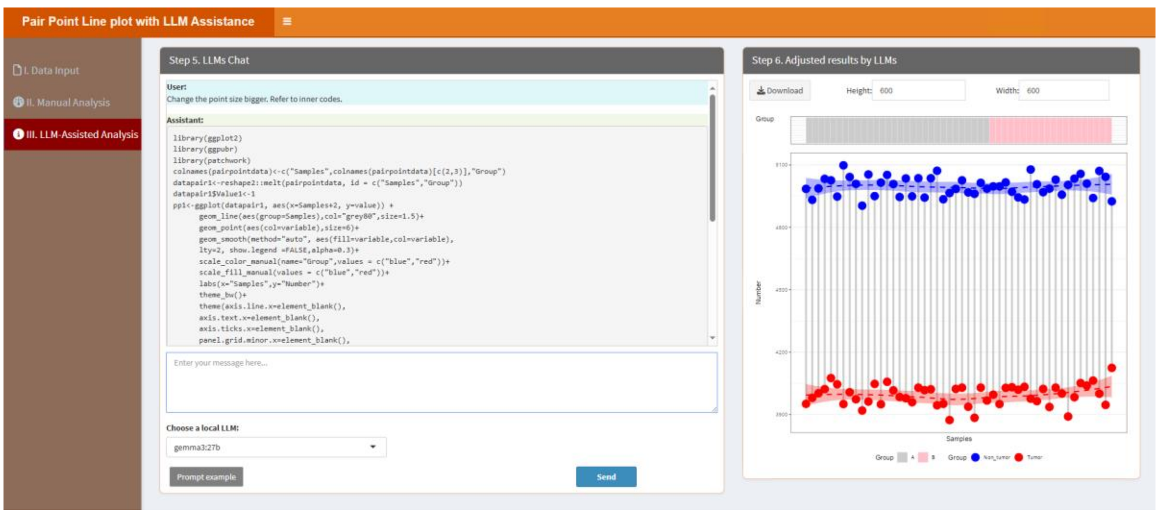
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Change the point size bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Change the point size bigger. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

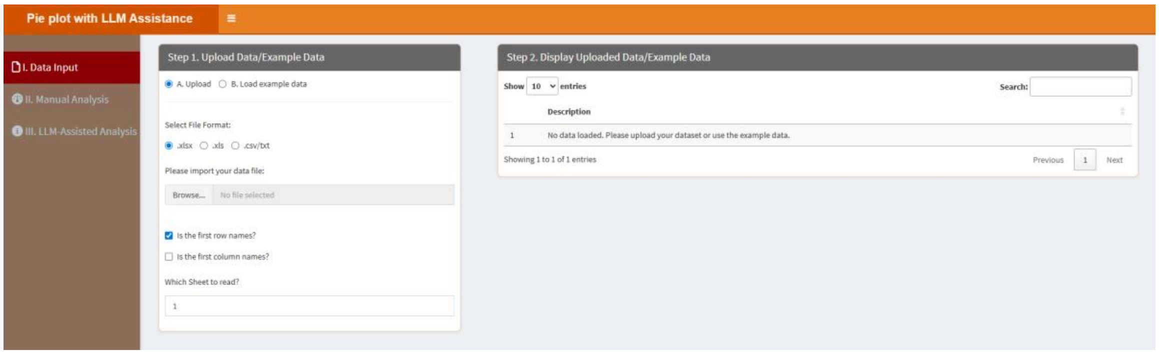


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



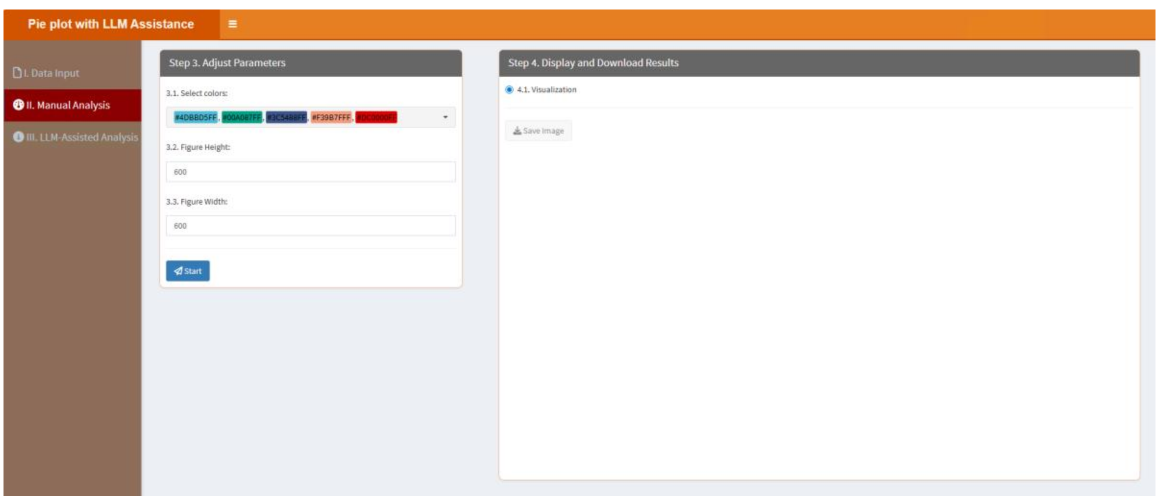
7.20. Pie plot

Pie charts can be used to visualize categorical data distributions. Users can adjust parameters in the classic framework and rely on local LLMs for customization and interpretation. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



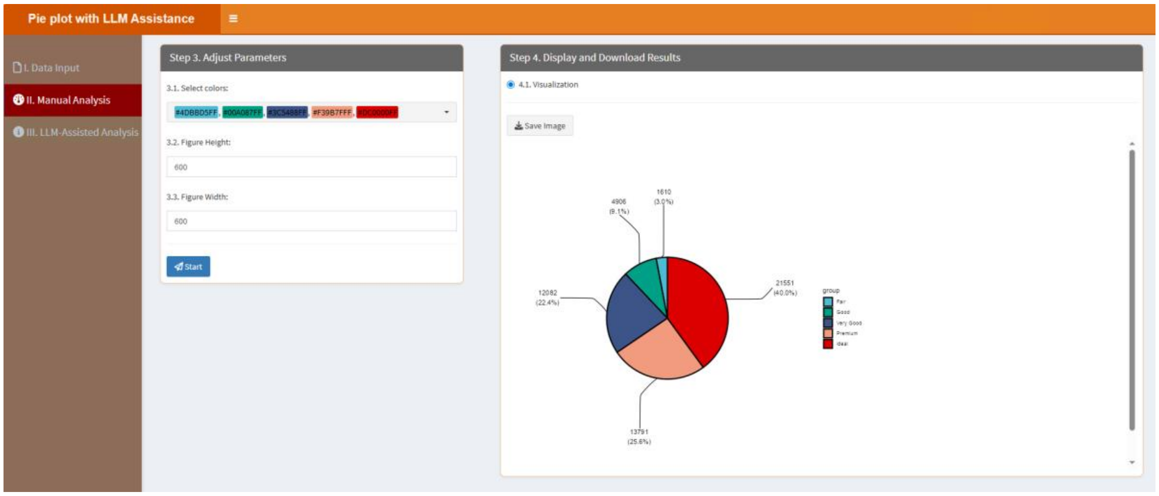
3.1. Select bar colors: This parameter allows users to choose specific colors for each category segment in the pie chart.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

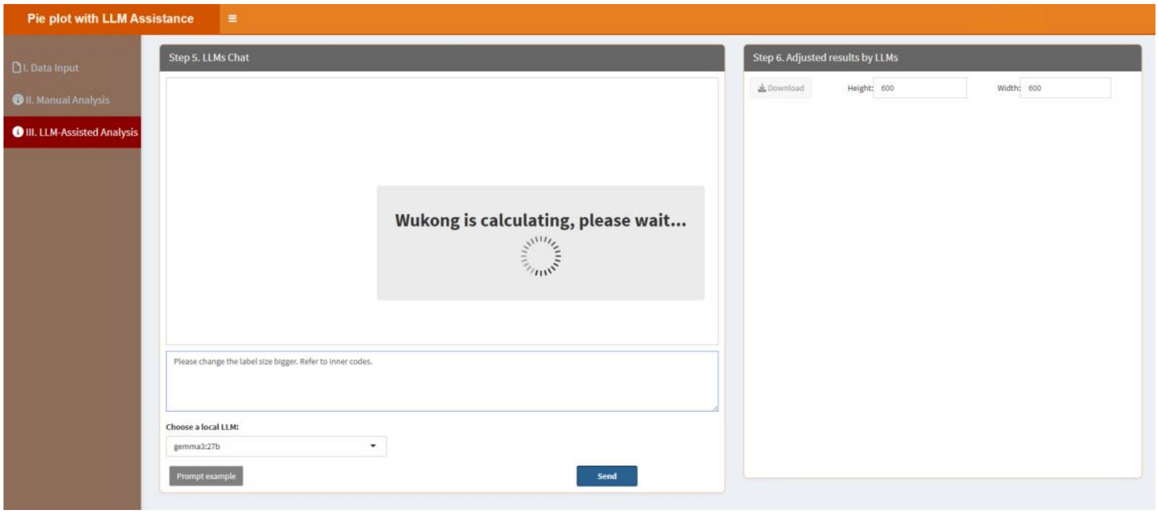
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



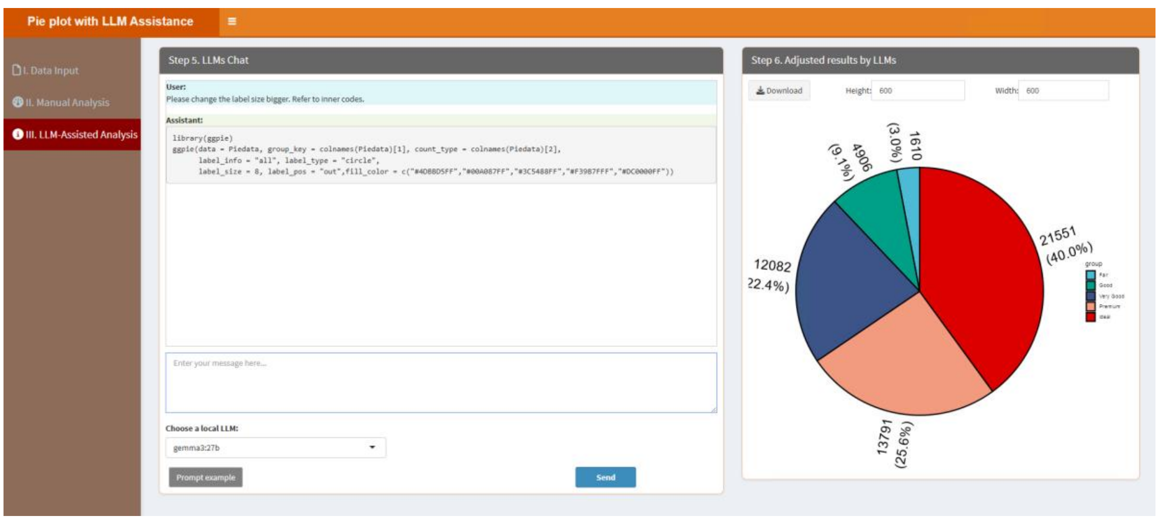
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the label size bigger. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the label size bigger. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

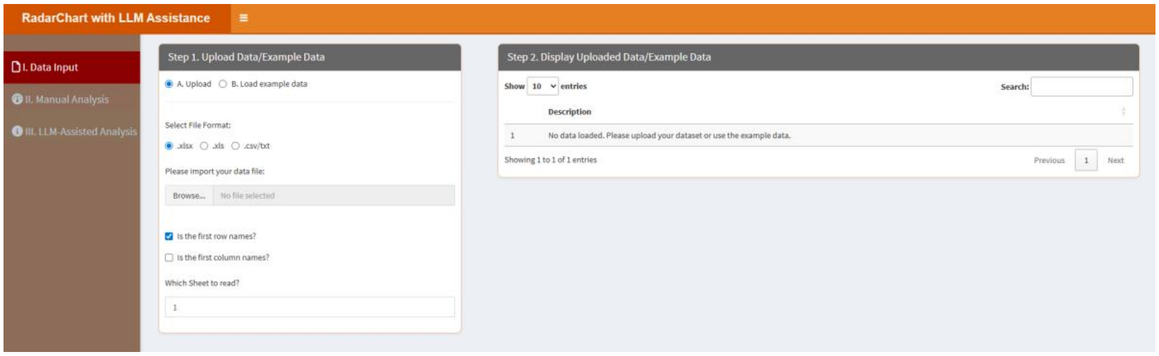


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



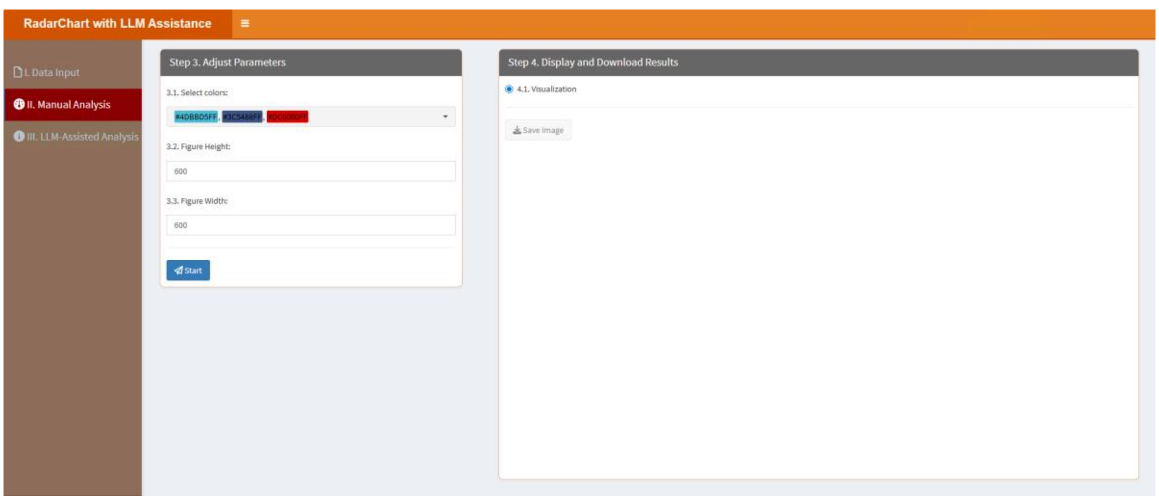
7.21. Radar Chart

Radar charts are useful for comparing multivariate data. Users can customize the plot using the classic framework and refine it with local LLMs to create informative and visually appealing results. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



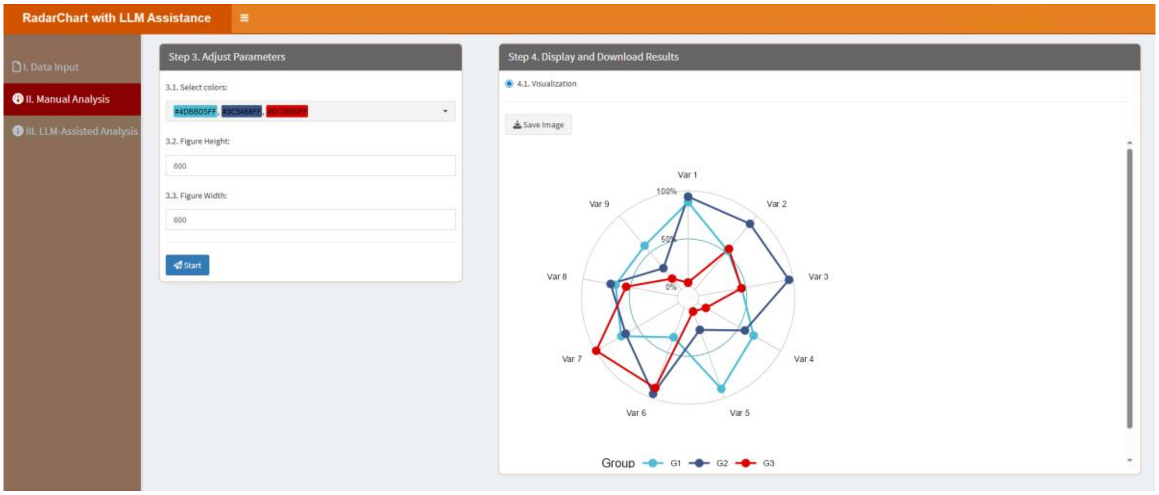
3.1. Select colors: This parameter lets you choose specific colors for each group. The color number should be equal to the group number.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

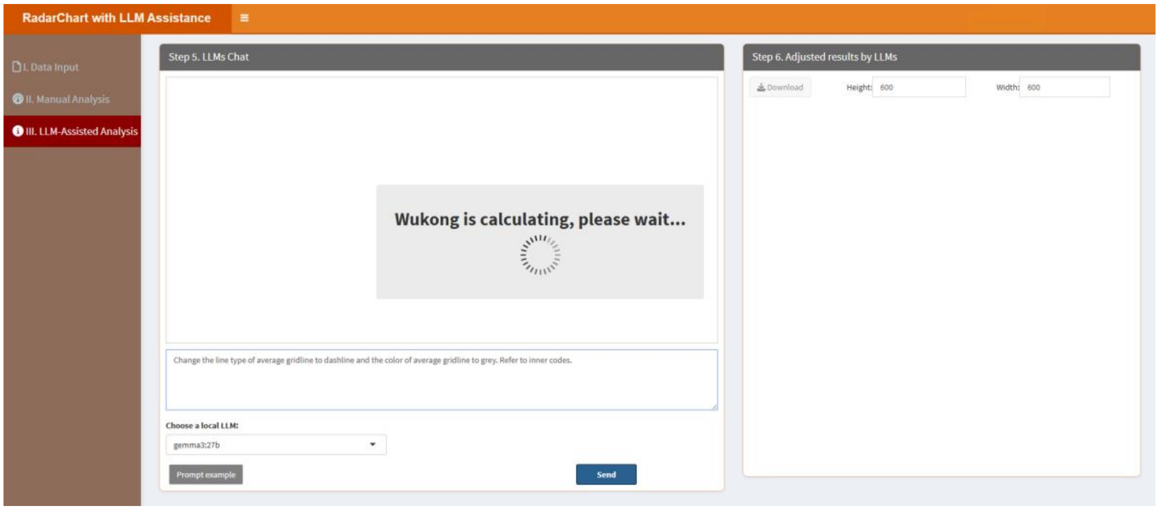
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



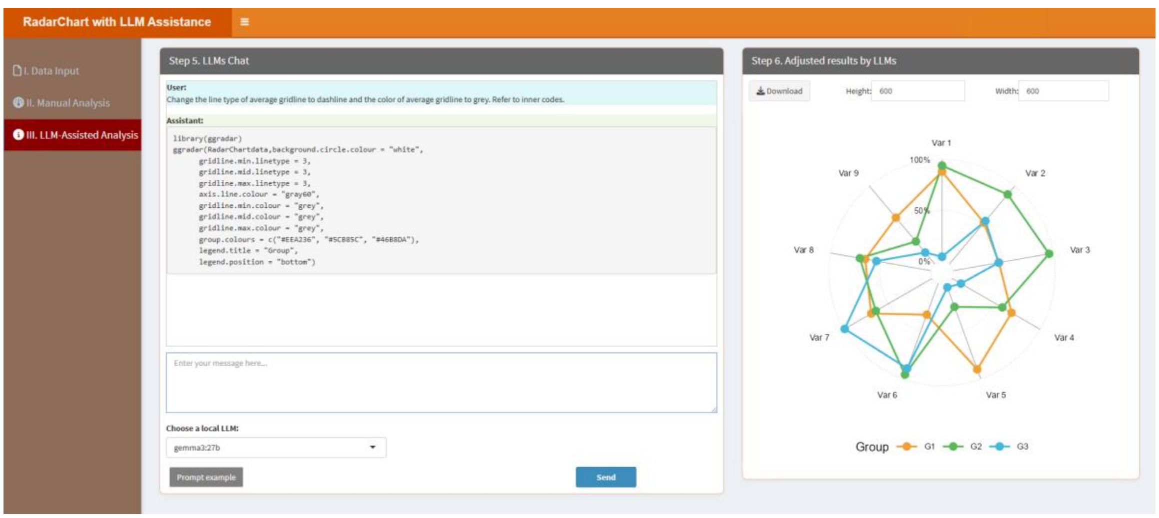
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Change the line type of average gridline to dashline and the color of average gridline to grey. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Change the line type of average gridline to dashline and the color of average gridline to grey. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

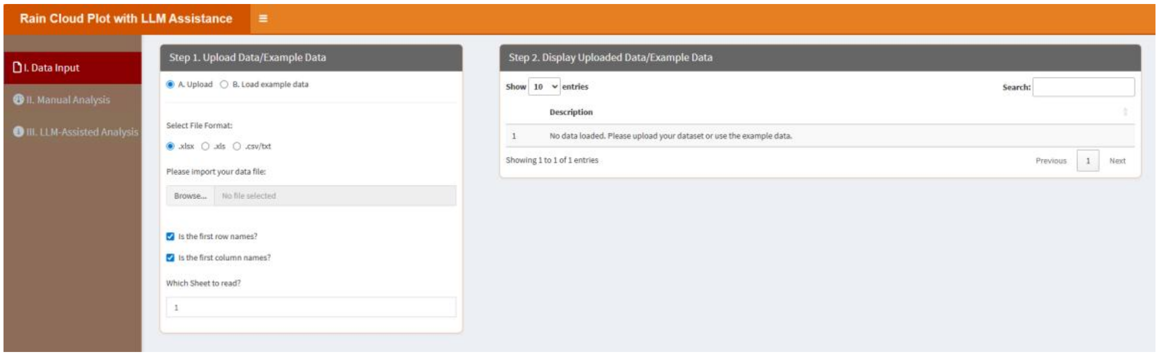


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



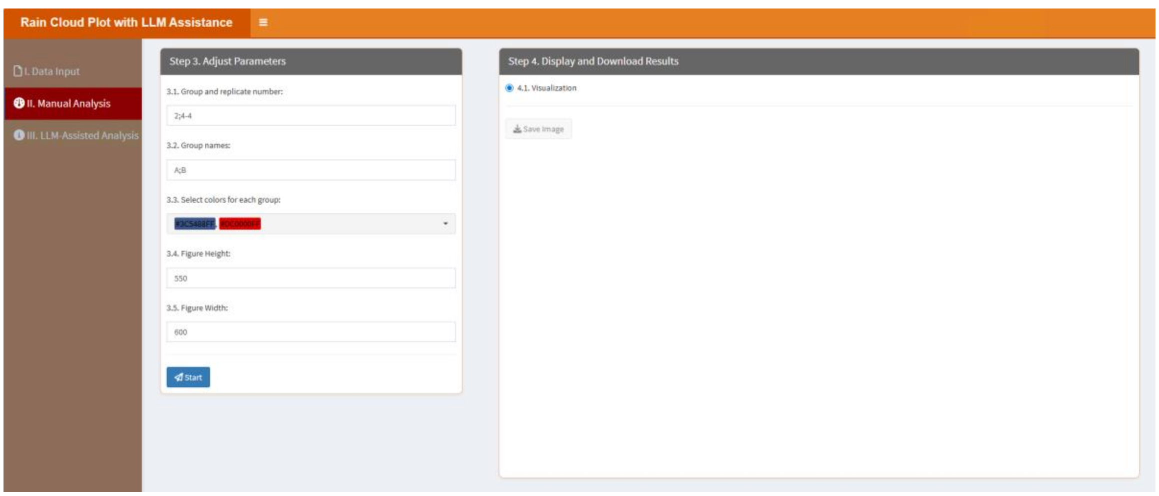
7.22. Rain Cloud Plot

Rain cloud plots combine density and scatter plots to visualize data distributions. Users can adjust parameters in the classic framework and use local LLMs to enhance the visualization. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



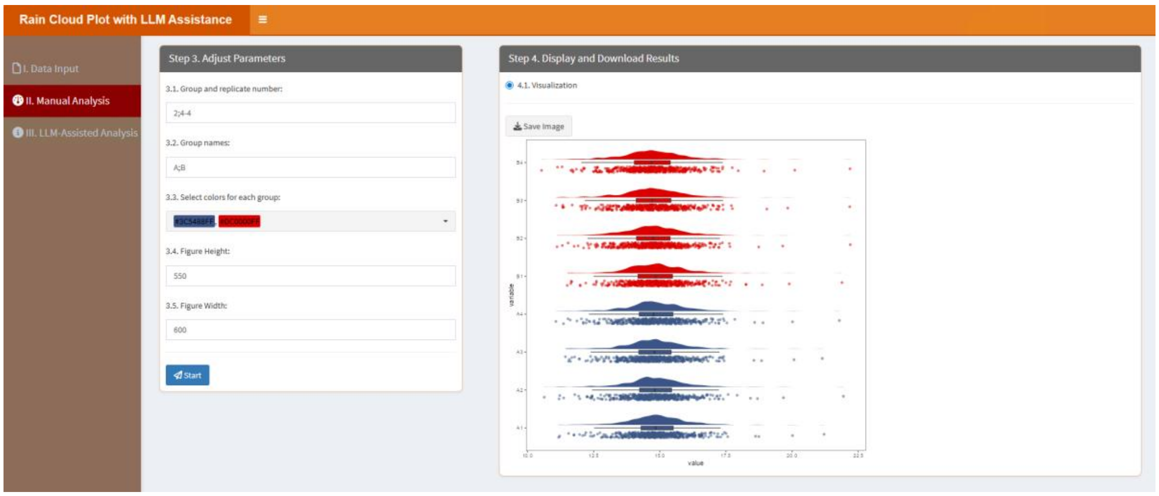
3.1. Group and replicate number: Type in the group number and replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 2 groups and 4 replicates in each group, so here is “2;4-4”.

3.2. Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 2 groups in the example data, you can type in "A;B".

3.3. Select colors for each group: This parameter lets you choose specific colors for each

group. The color number should be equal to the group number.
3.4. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

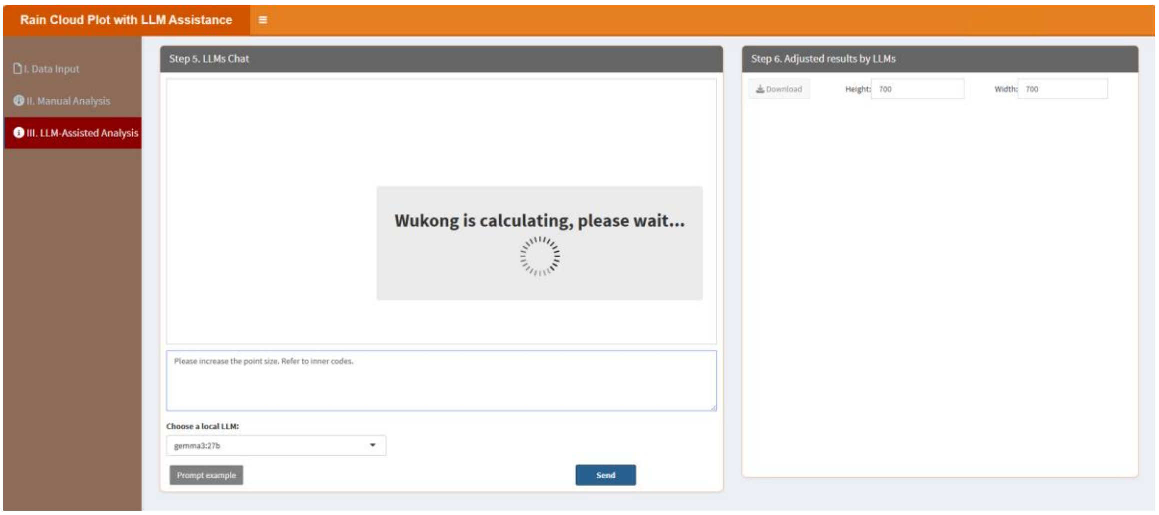
After setting parameters, click "Start" button, the results will be shown as below:
In the "4.1. Visualization" part, users can click "Save image" button to download the figure:



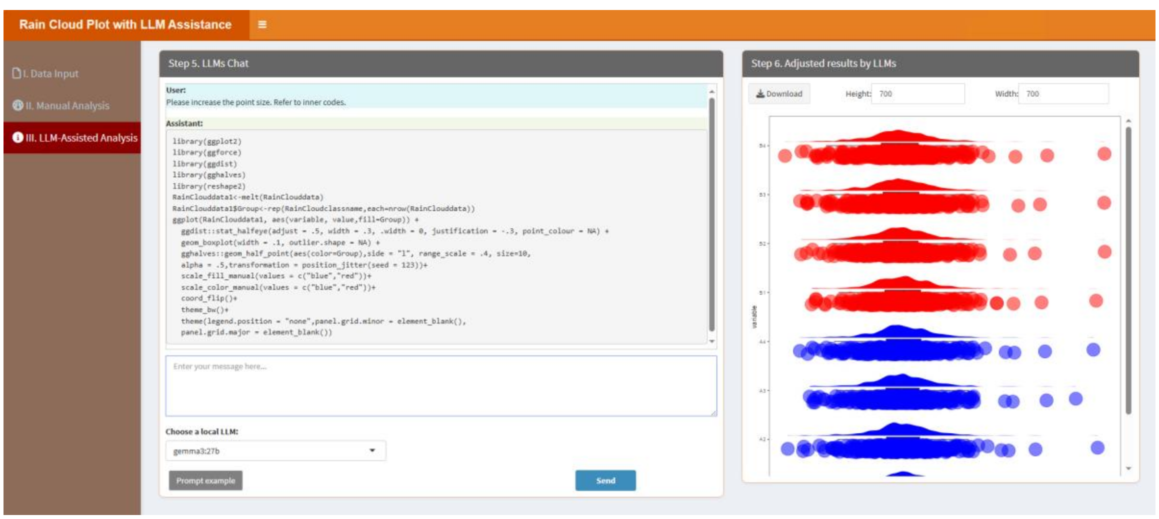
"III. LLM-Assisted Analysis" means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt "Please increase the point size. Refer to inner codes." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please increase the point size. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

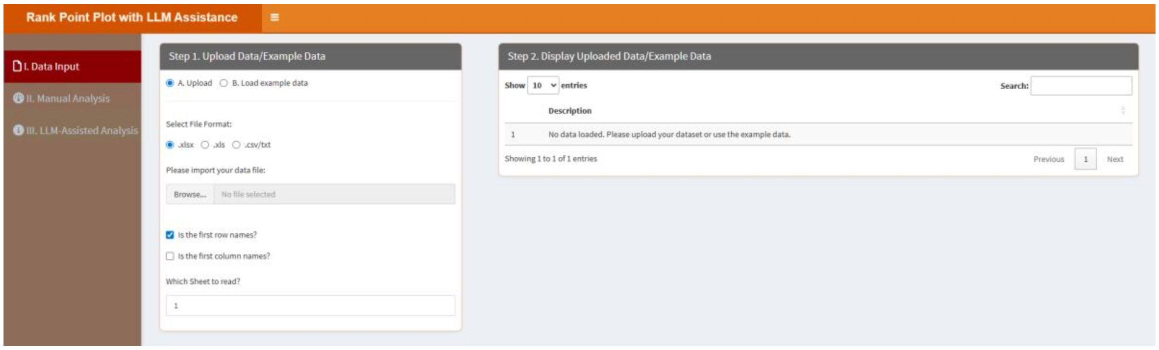


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



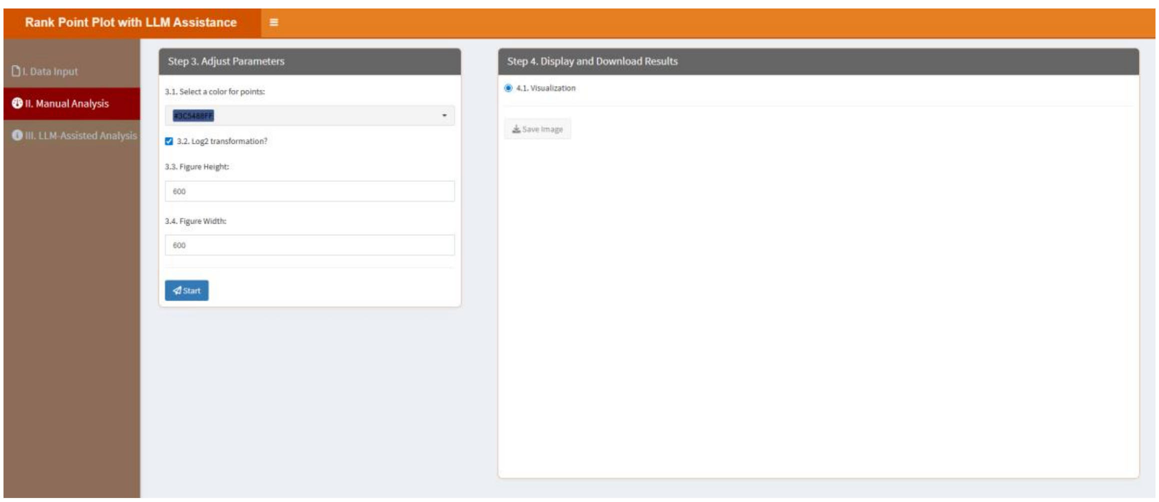
7.23. Rank Point Plot

Rank point plots highlight data rankings. Users can customize the plot through the classic framework and refine it with local LLMs for clear and effective visualization. After clicking the "Start" button, it will be started in a new window like below:



The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.



3.1. Select a color for points: Choose a color for the points.

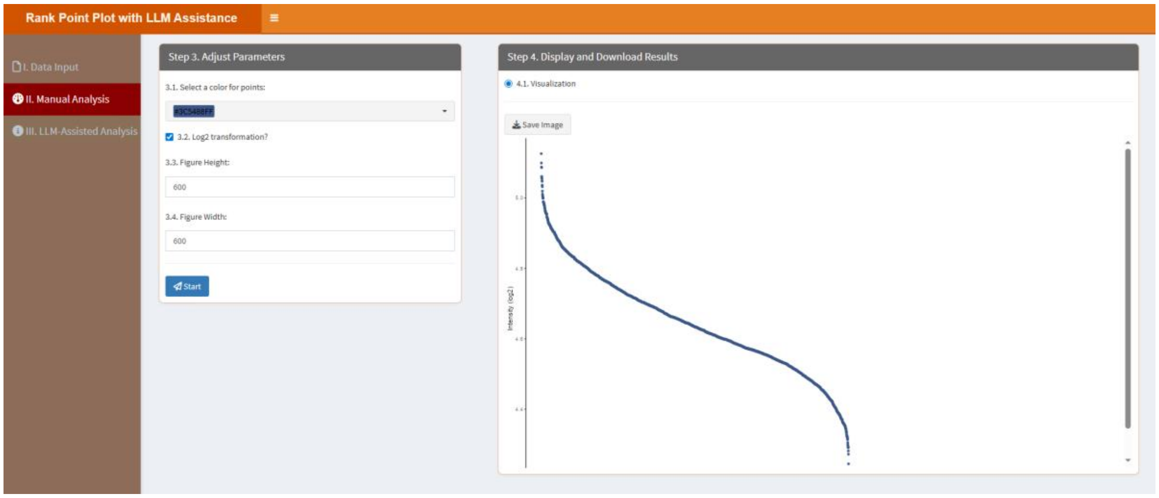
3.2. Log2 transformation? If ture, all values will be transformed using the base-2 logarithm.

3.3. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.4. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

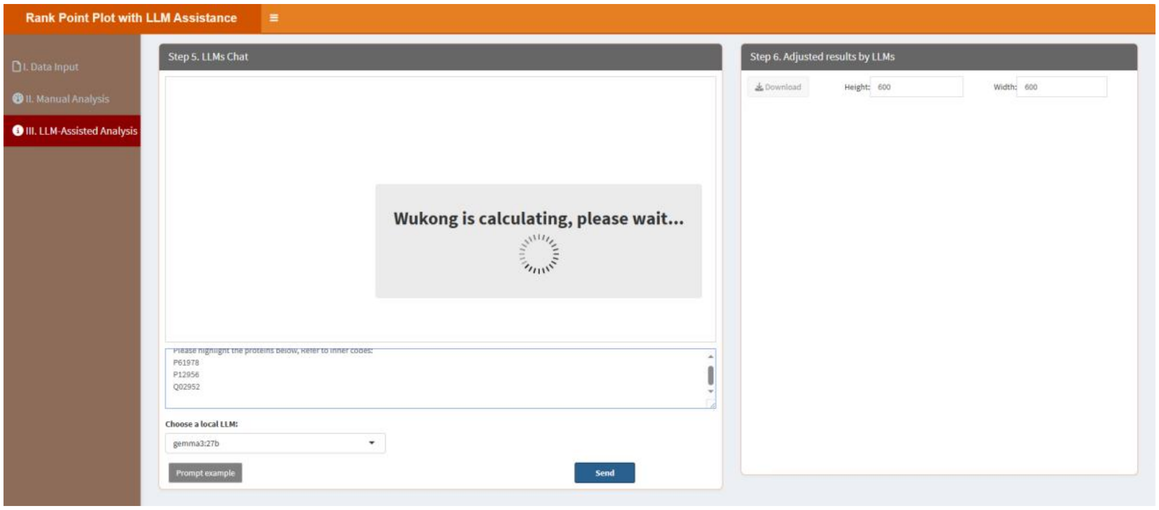
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



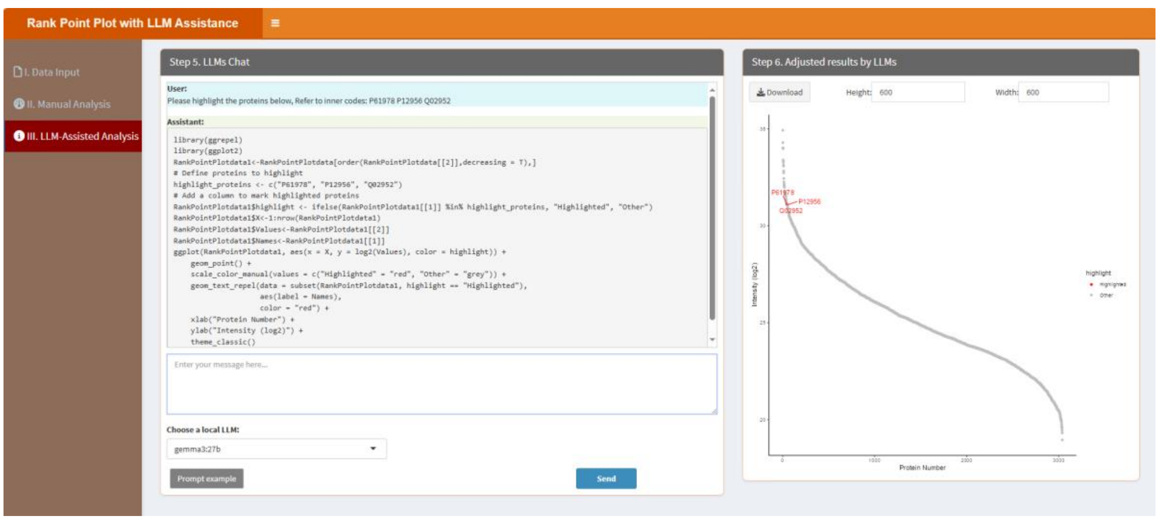
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please highlight the proteins below, Refer to inner codes: P61978, P12956, Q02952.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please highlight the proteins below, Refer to inner codes: P61978, P12956, Q02952.=>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

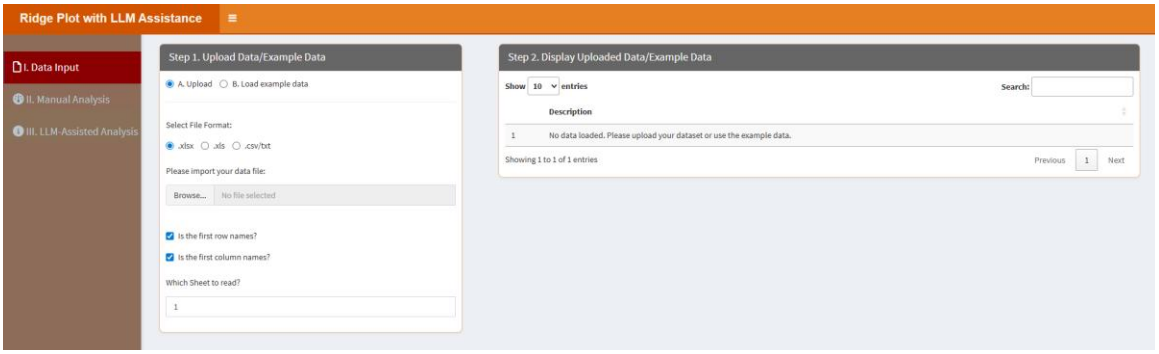


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



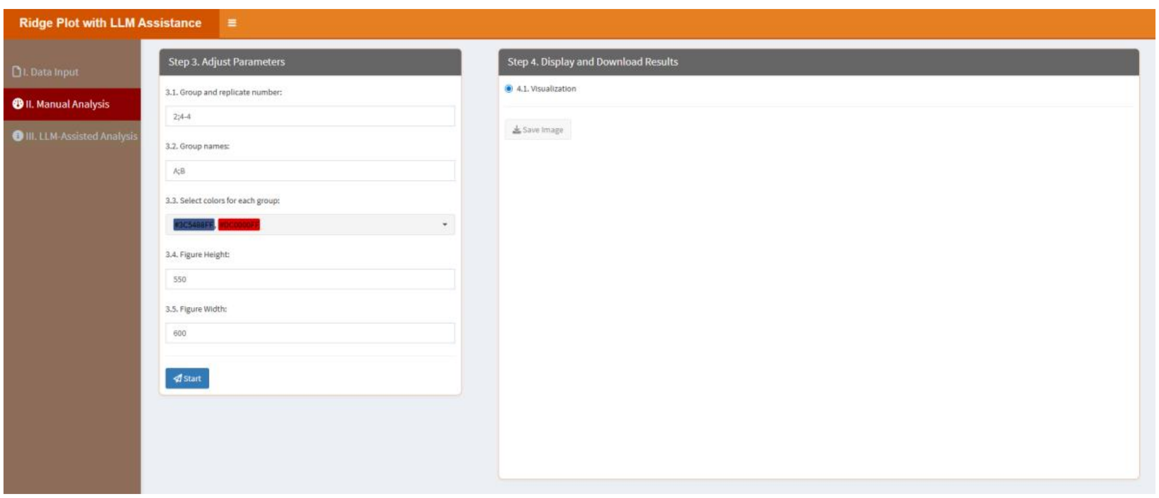
7.24. Ridge Plot

Ridge plots visualize distributions across multiple categories. Users can adjust settings using the classic framework and refine the plot with local LLMs for detailed and aesthetically pleasing results. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Group and replicate number: Type in the group number and replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 2 groups and 4 replicates in each group, so here is “2;4-4”.

3.2. Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 2 groups in the example data, you can type in "A;B".

3.3. Select colors for each group: This parameter lets you choose specific colors for each

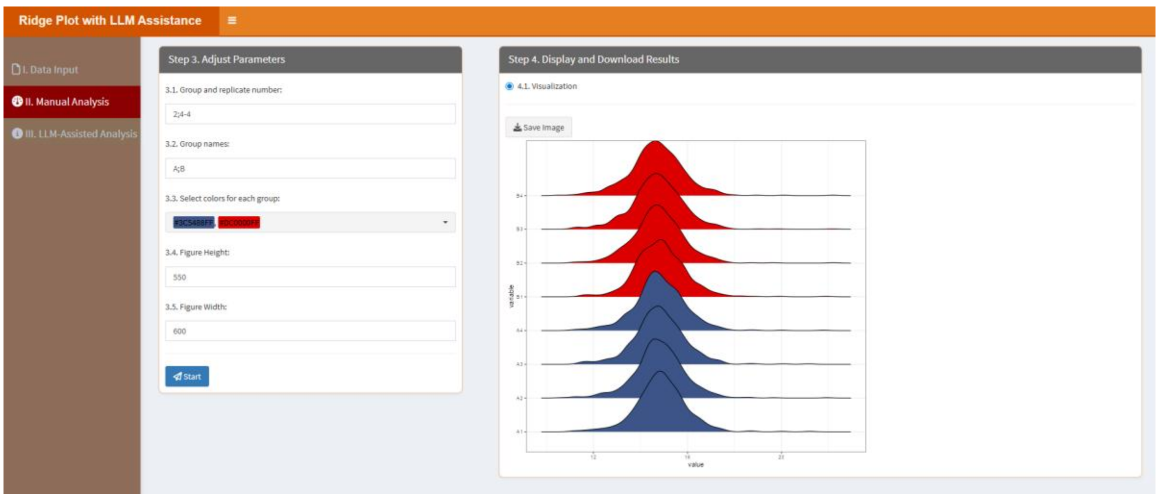
group. The color number should be equal to the group number.

3.4. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

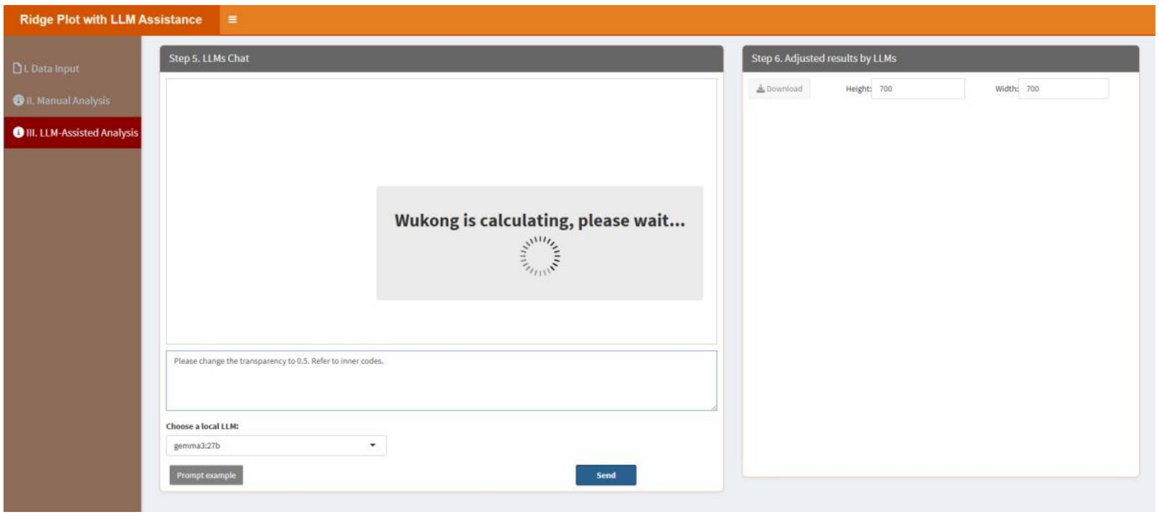
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



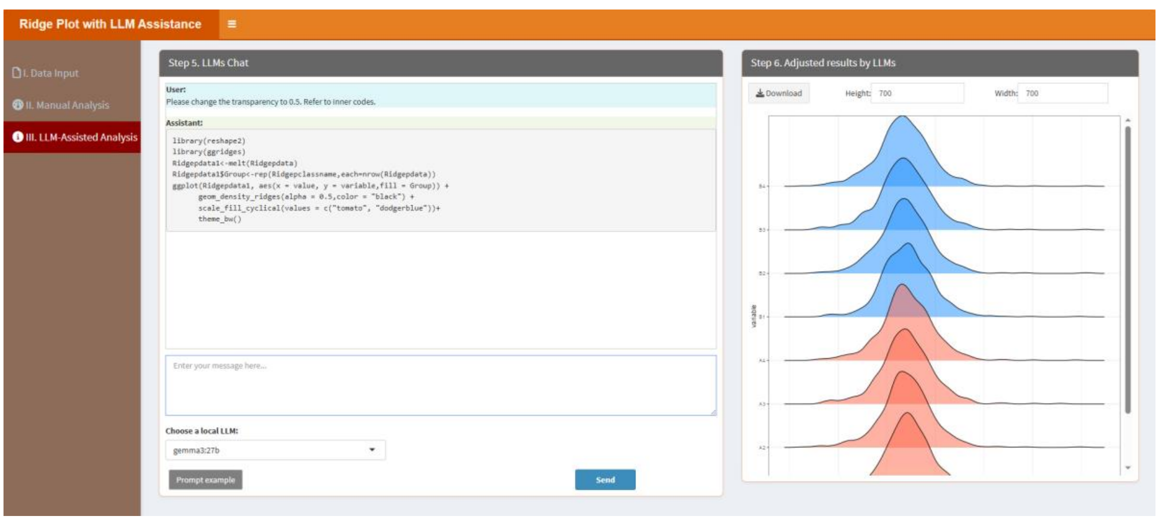
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the transparency to 0.5. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please change the transparency to 0.5. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

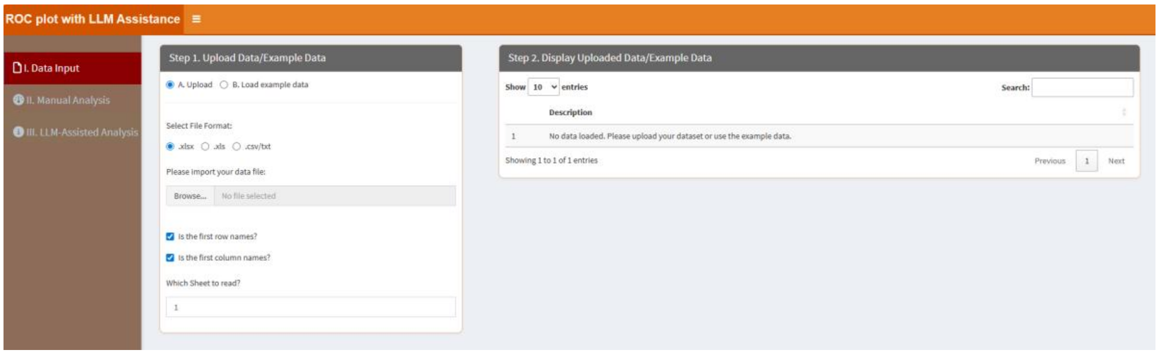


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



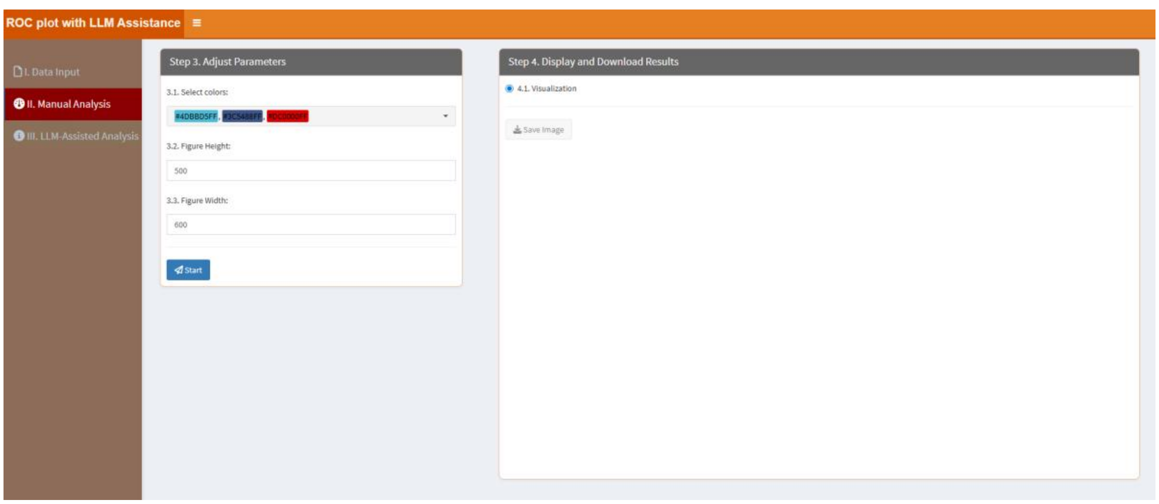
7.25. ROC plot

ROC plots evaluate classification model performance. Users can customize the plot through the classic framework and leverage local LLMs for accurate interpretation and visualization. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



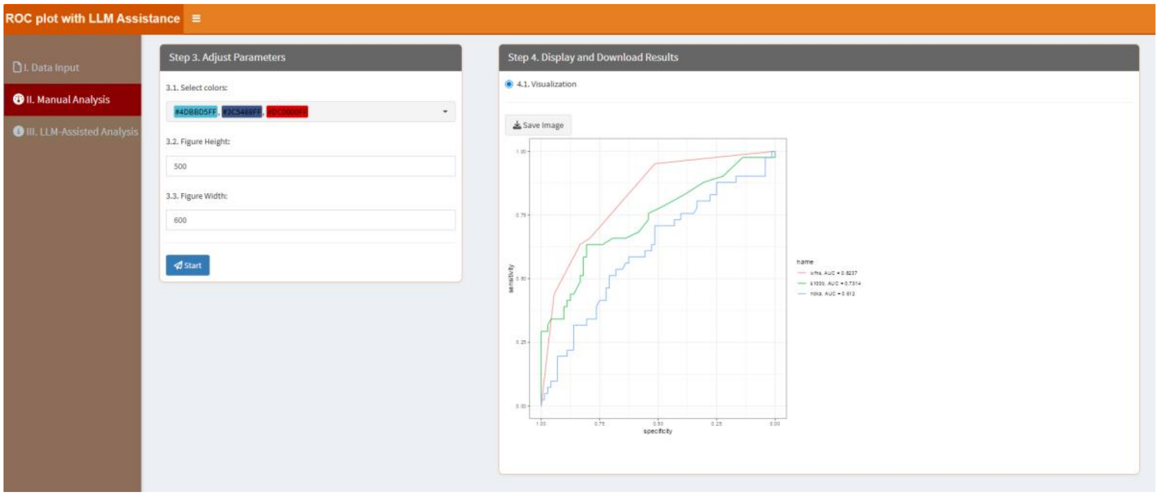
3.1. Select colors: This parameter allows users to assign specific colors to each group or variable whose ROC curve will be displayed. The number of selected colors should match the number of groups or classification targets being compared.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization,

ensuring that it accommodates the data and fits appropriately in the given space.

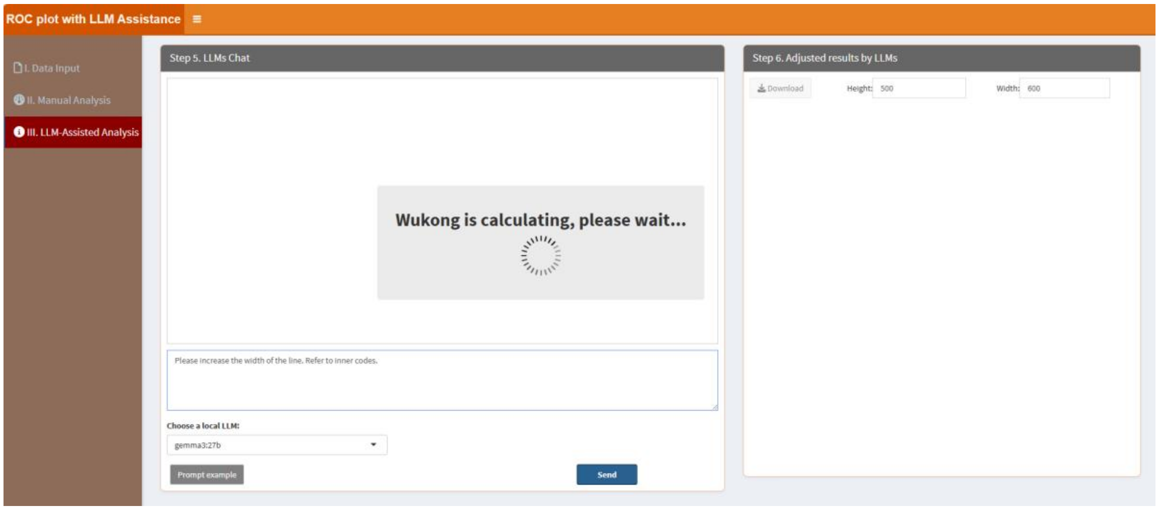
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



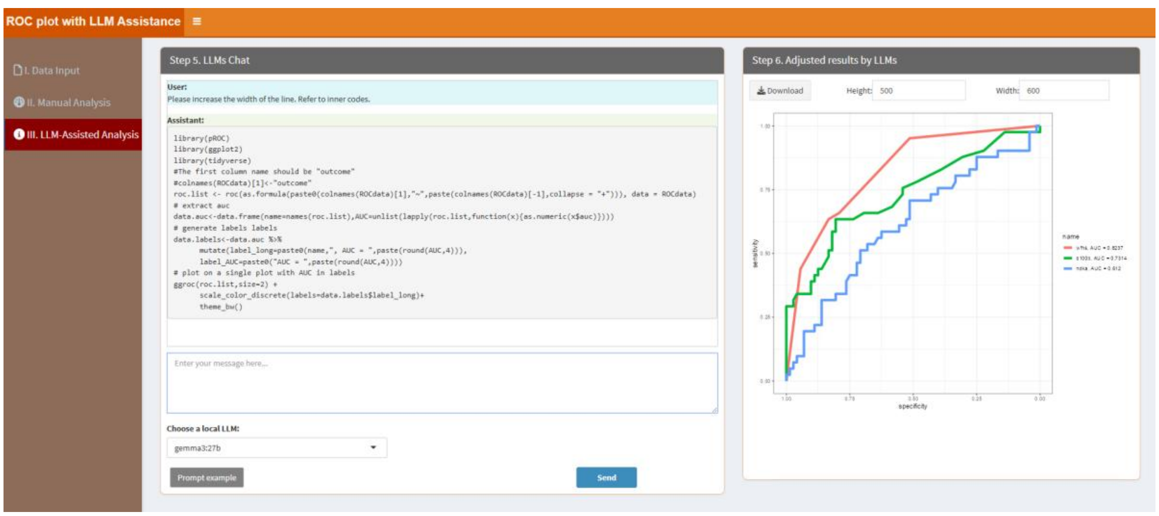
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please increase the width of the line. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please increase the width of the line. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

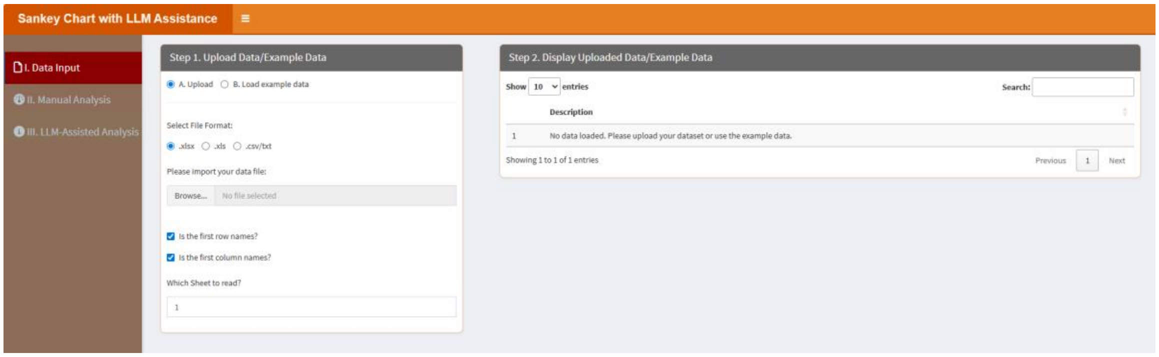


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



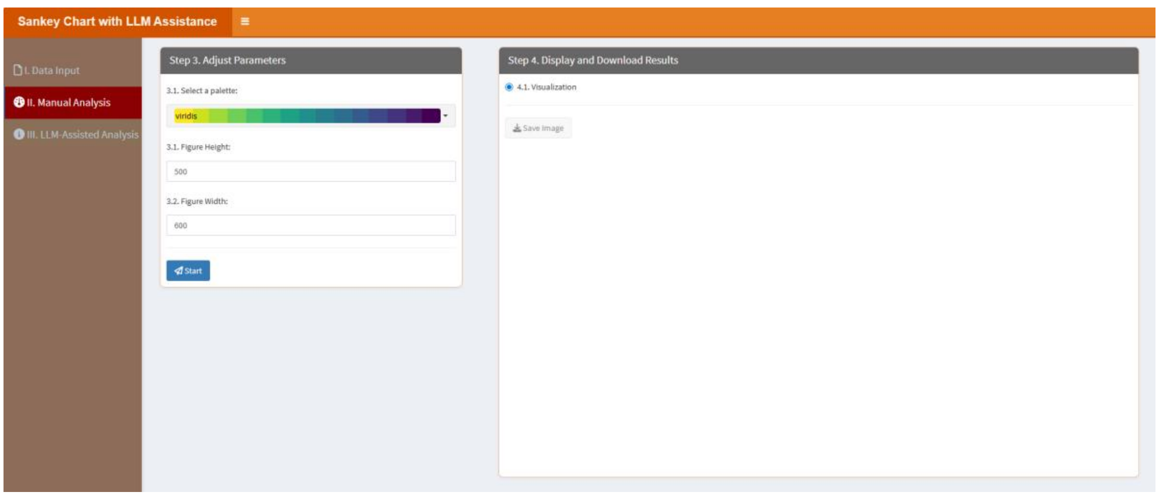
7.26. Sankey Chart

Sankey charts visualize data flows and relationships. Users can adjust parameters in the classic framework and refine the plot with local LLMs for engaging and accurate visualizations. After clicking the "Start" button, it will be started in a new window like below:



The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.



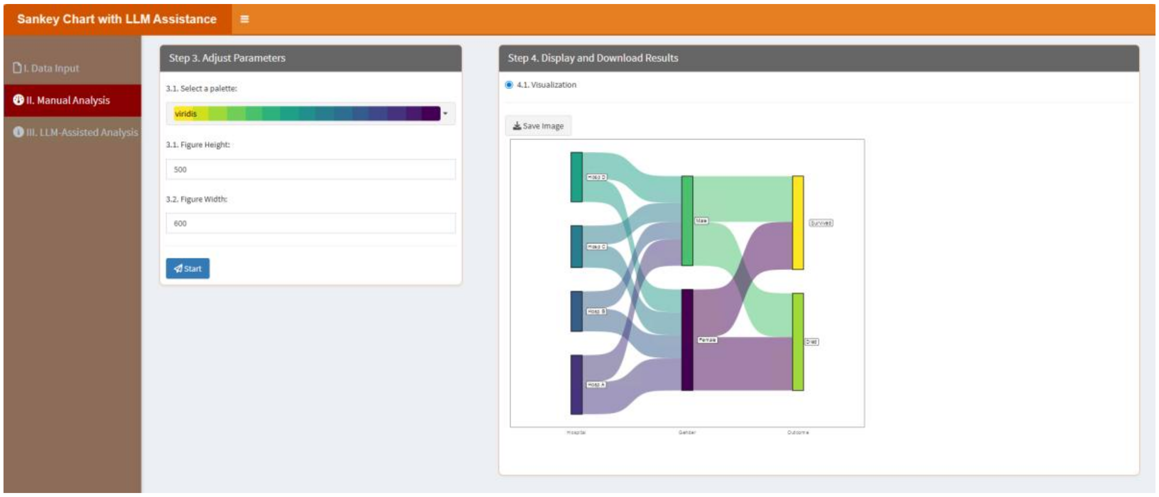
3.1. Select a palette: Choose a palette for the sankey chart.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click "Start" button, the results will be shown as below:

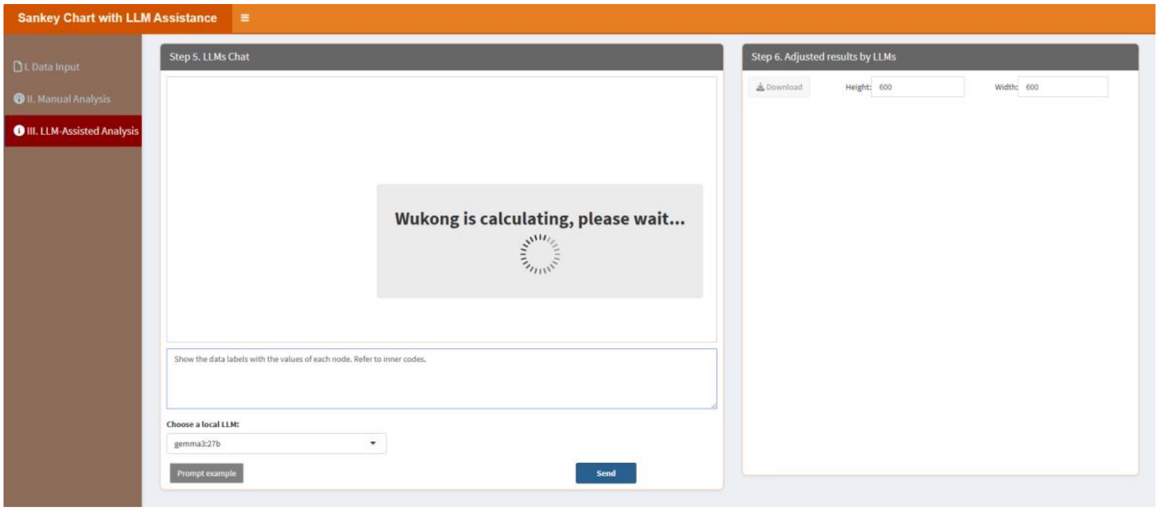
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



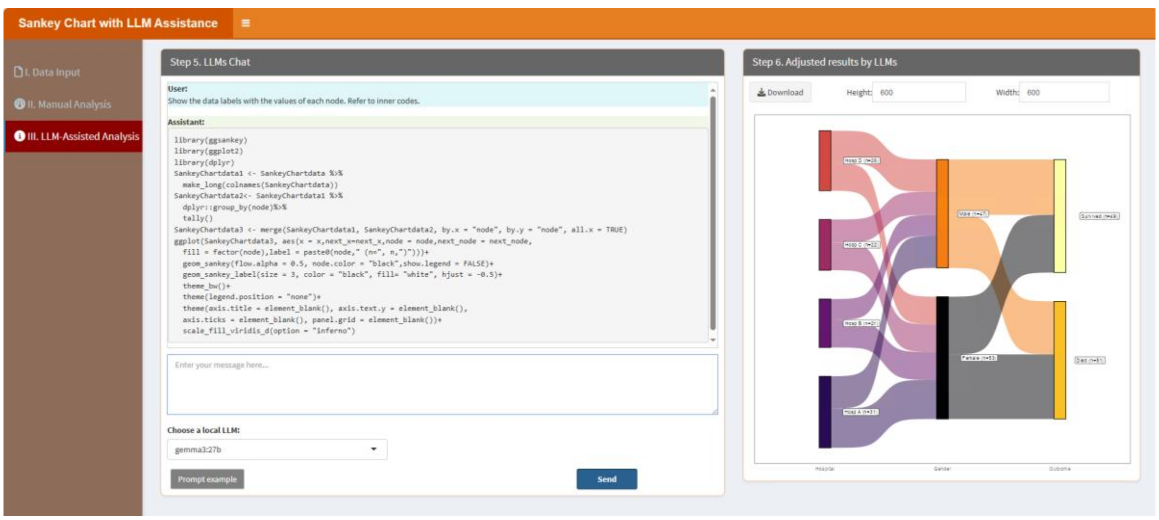
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Show the data labels with the values of each node. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Show the data labels with the values of each node. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

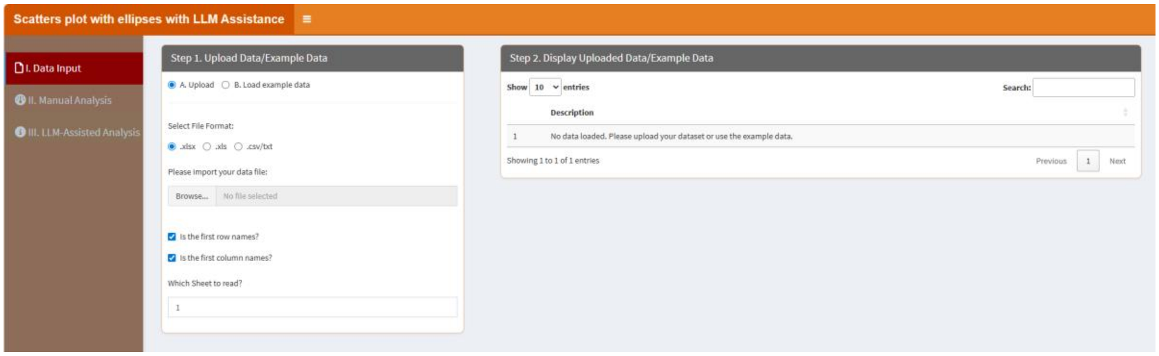


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



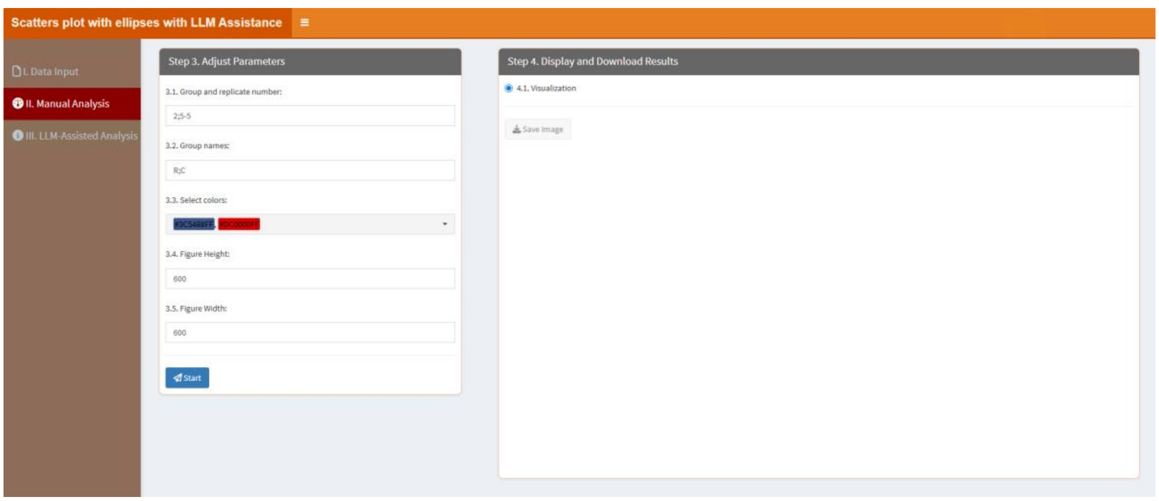
7.27. Scatter plot with ellipses

Scatter plots with ellipses highlight data groupings. Users can customize the plot through the classic framework and enhance it with local LLMs for better insights. After clicking the "Start" button, it will be started in a new window like below:



The "I. Data Input" here is similar to that in the "4.1. Coefficient of variation (CV)" above. Users can upload their data or check example data from here.

The "II. Manual Analysis" here is also similar to that in the "4.1. Coefficient of variation (CV)" above. However, the parameters here are different.



3.1. Group and replicate number: Type in the group number and replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 2 groups and 4 replicates in each group, so here is "2;4-4".

3.2. Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 2 groups in the example data, you can type in "A;B".

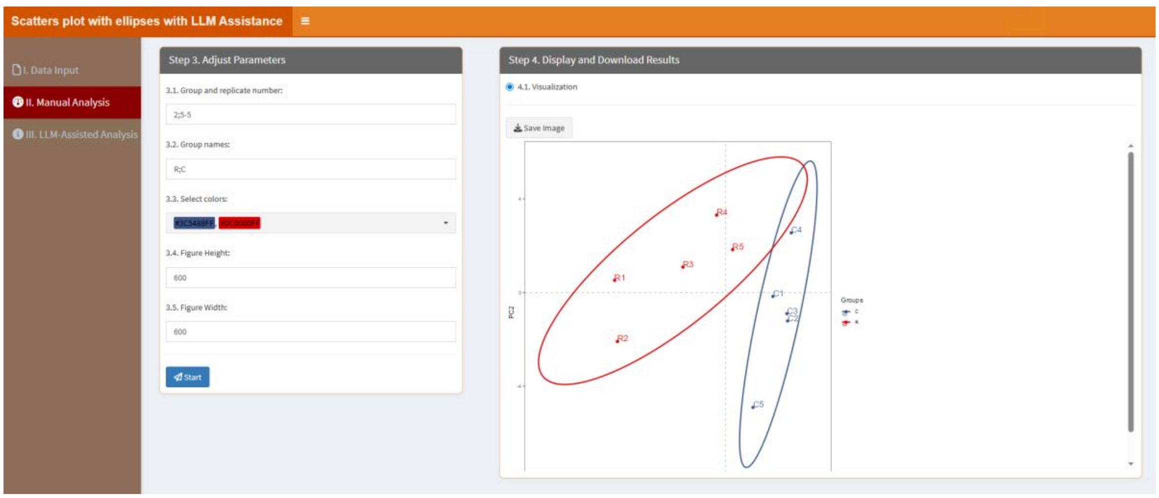
3.3. Select colors: This parameter lets you choose specific colors for each group. The color number should be equal to the group number.

3.4. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

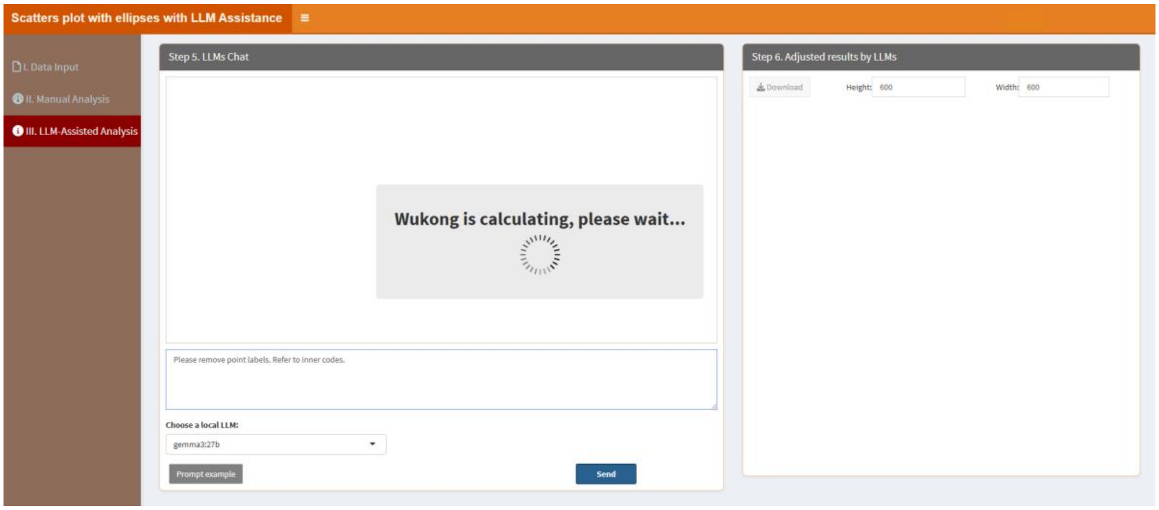
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



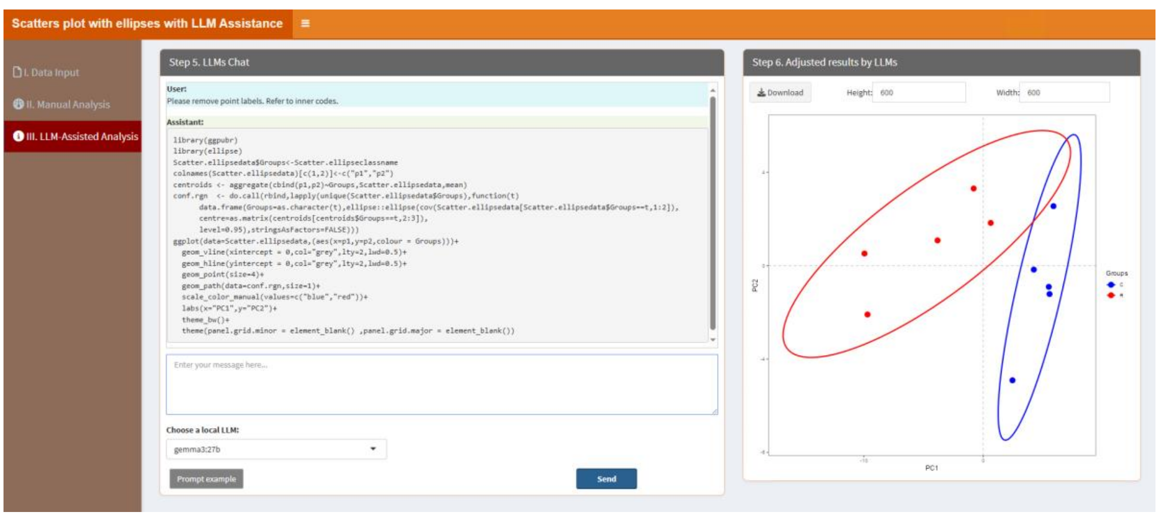
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please remove point labels. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please remove point labels. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

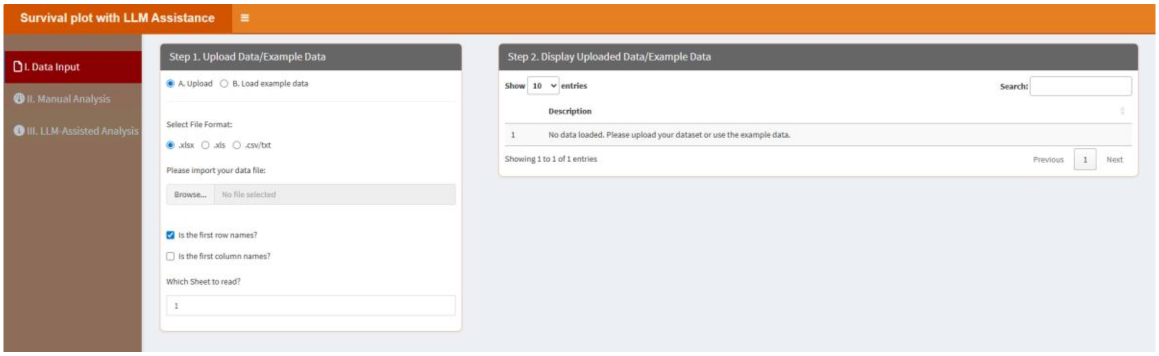


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



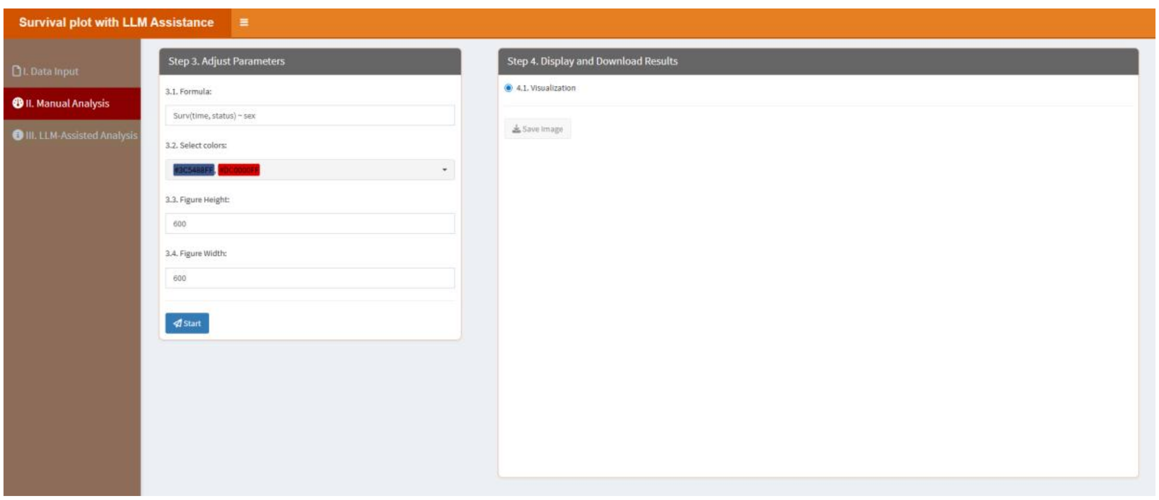
7.28. Survival Plot

Survival plots analyze time-to-event data. Users can adjust parameters in the classic framework and refine the plot with local LLMs for clear and accurate visualizations. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Formula: This parameter allows users to define the model formula for survival analysis, typically in the format *Surv(time, status) ~ group*.

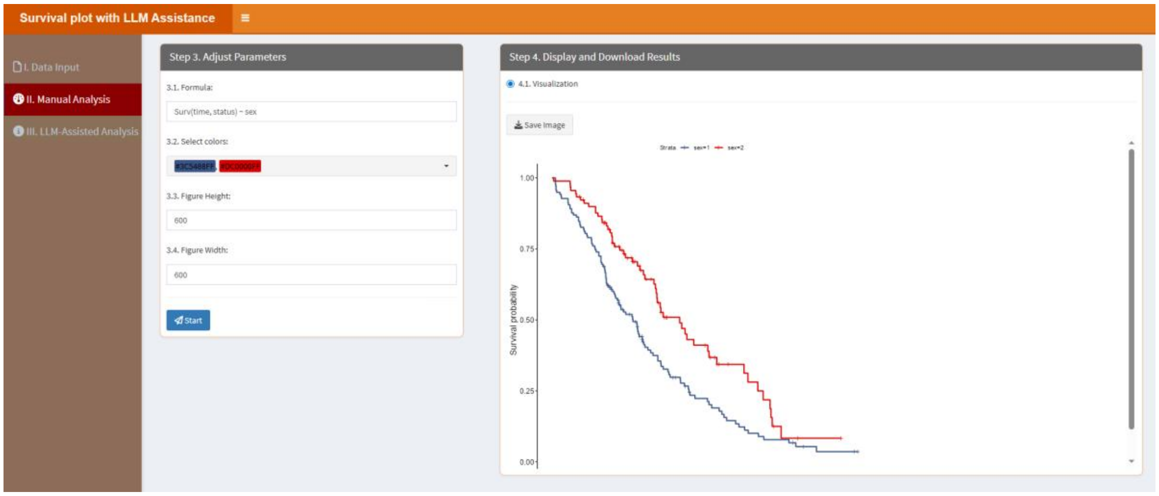
3.2. Select colors: This parameter lets you choose specific colors for each group. The color number should be equal to the group number.

3.3. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.4. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization,

ensuring that it accommodates the data and fits appropriately in the given space.

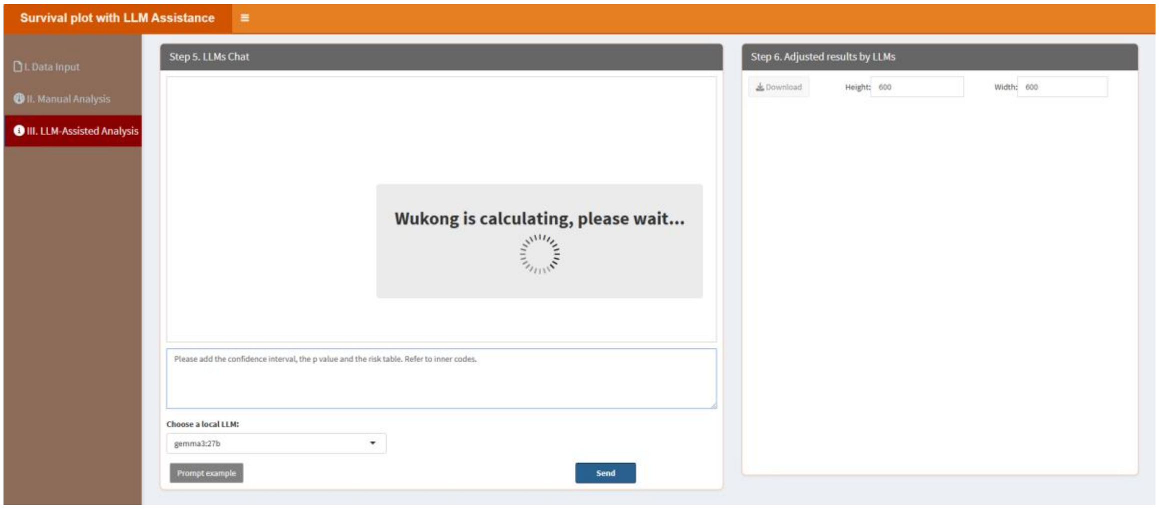
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



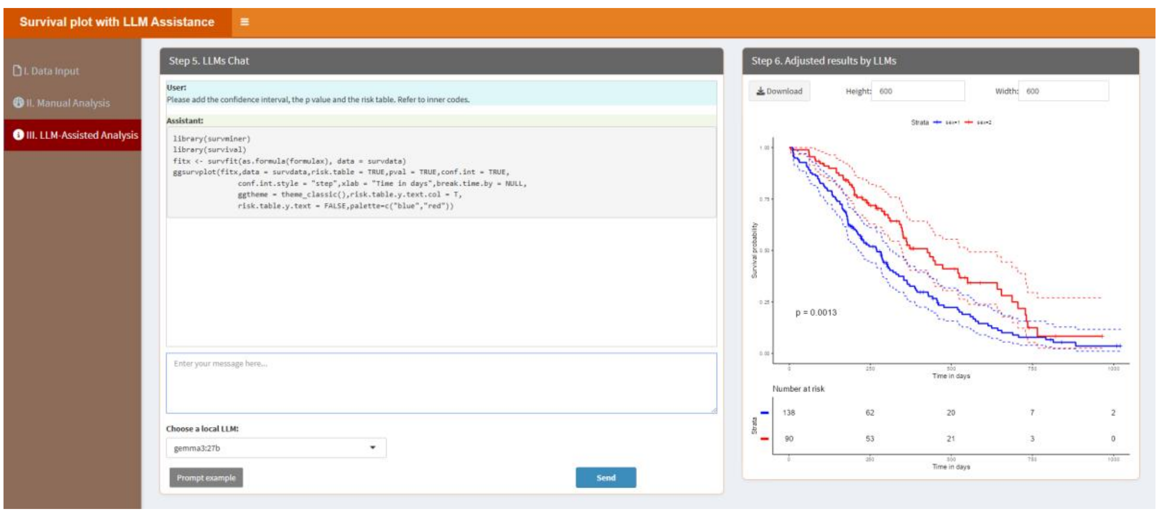
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please add the confidence interval, the p value and the risk table. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please add the confidence interval, the p value and the risk table. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

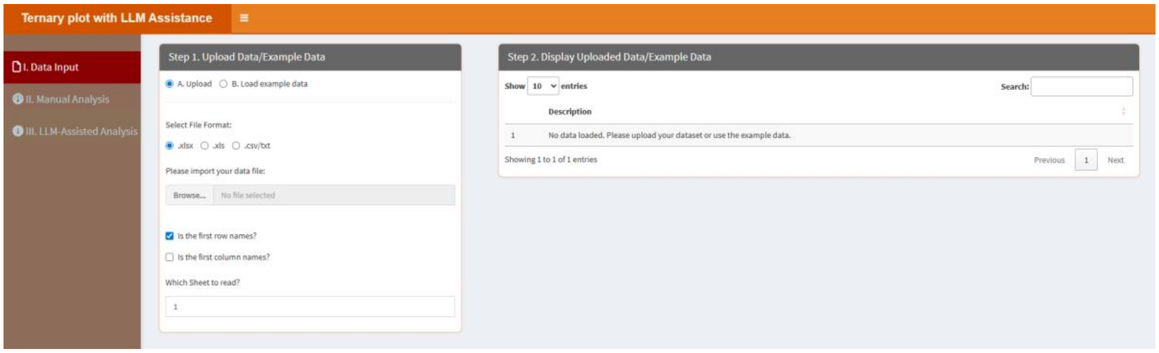


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



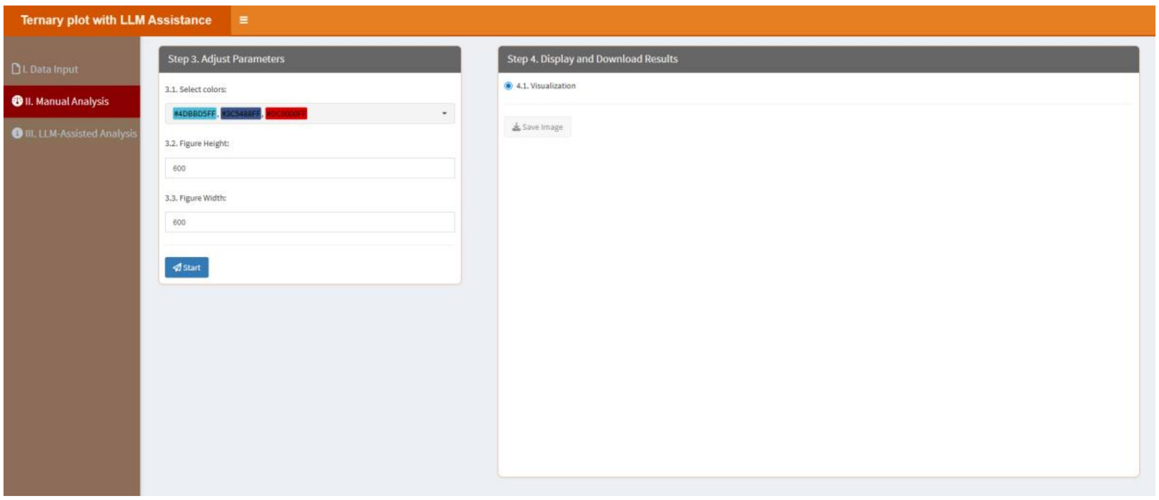
7.29. Ternary Plot

Ternary plots visualize three-component data. Users can customize the plot through the classic framework and refine it using local LLMs for accurate and visually appealing results. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.

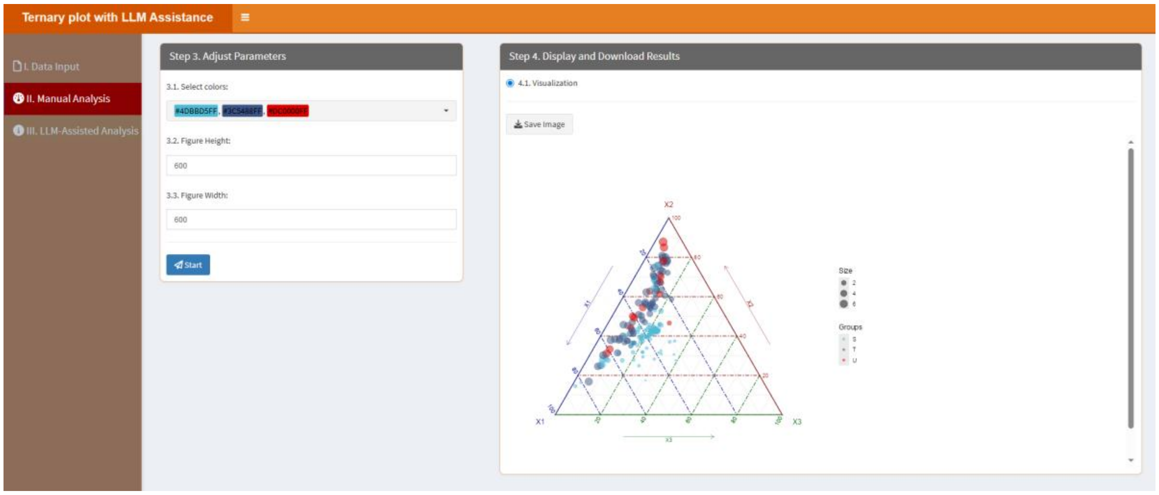


3.1. Select colors: This parameter lets you choose specific colors for each group. The color number should be equal to the group number.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

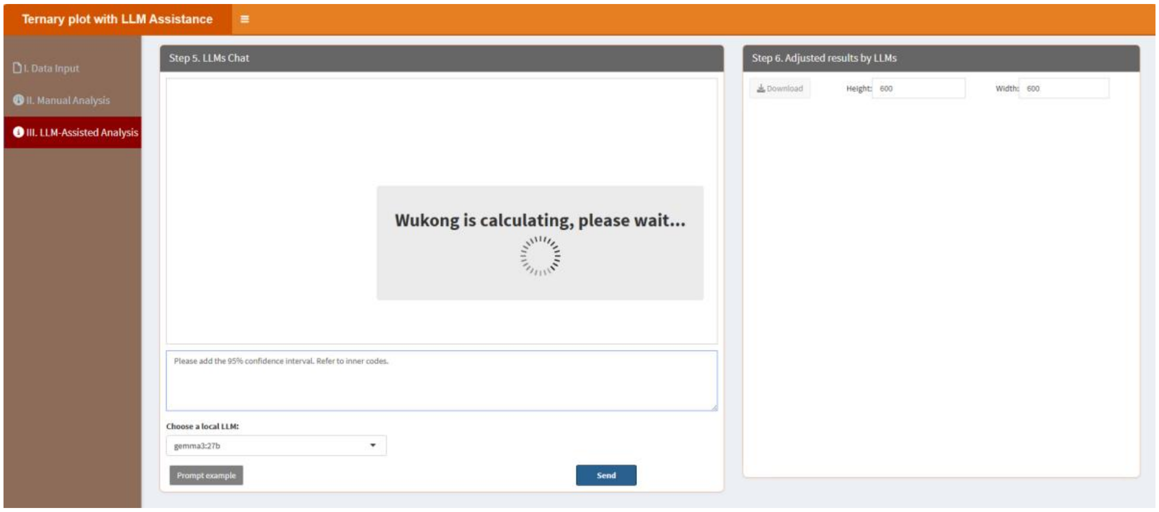
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



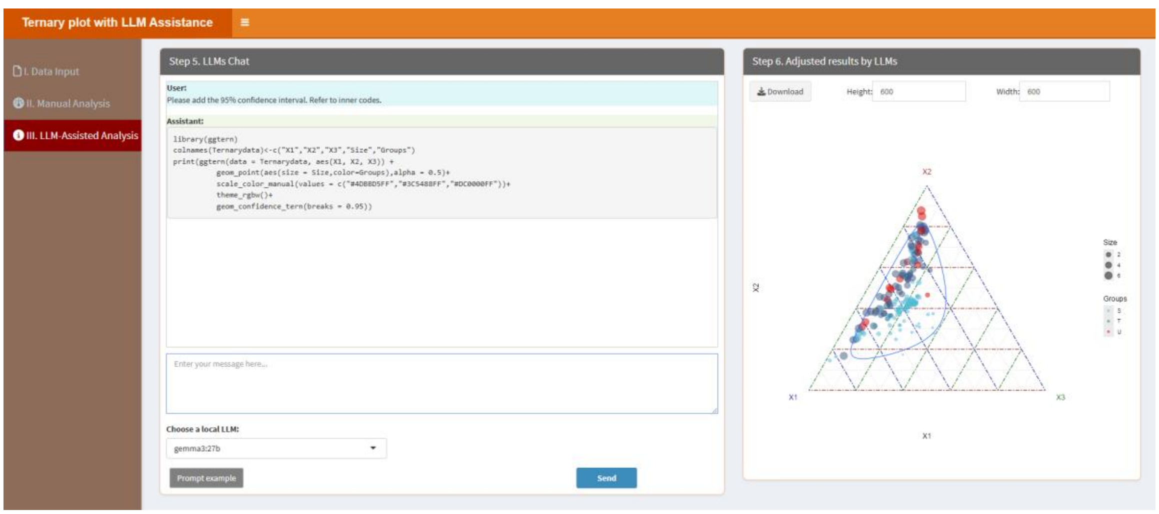
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please add the 95% confidence interval. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please add the 95% confidence interval. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

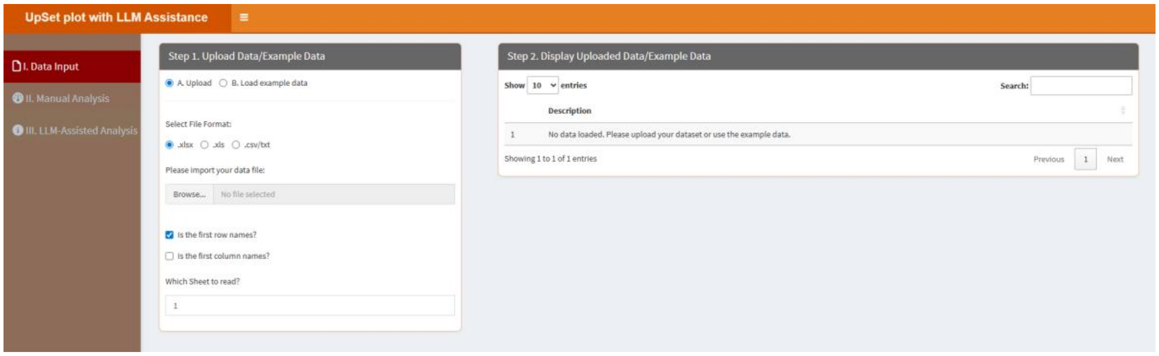


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



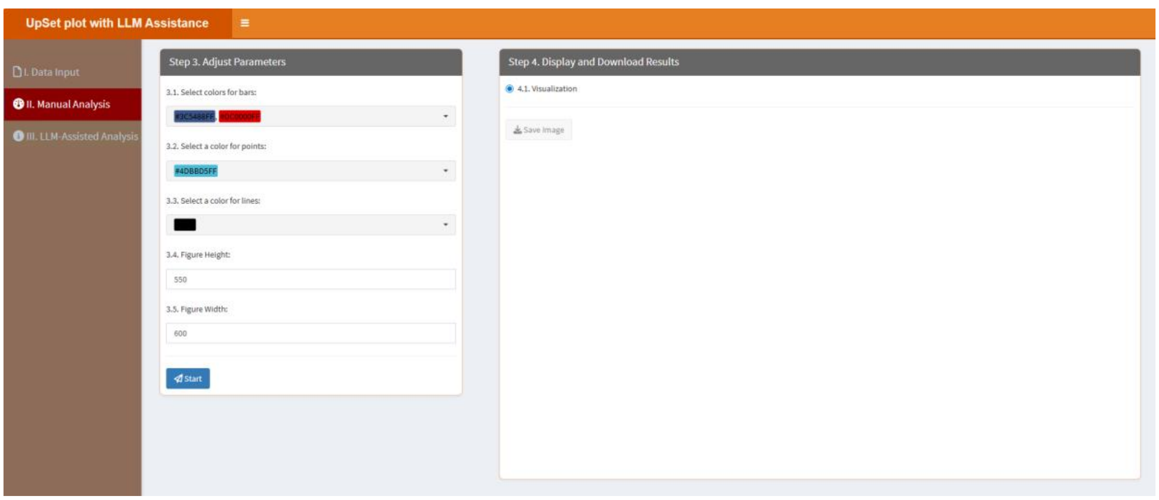
7.30. UpSet plot

UpSet plots visualize intersections between sets. Users can adjust parameters in the classic framework and refine the visualization with local LLMs for engaging and accurate results. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Select colors for bars: This parameter lets users assign specific colors to the horizontal bars that represent the size of each individual set.

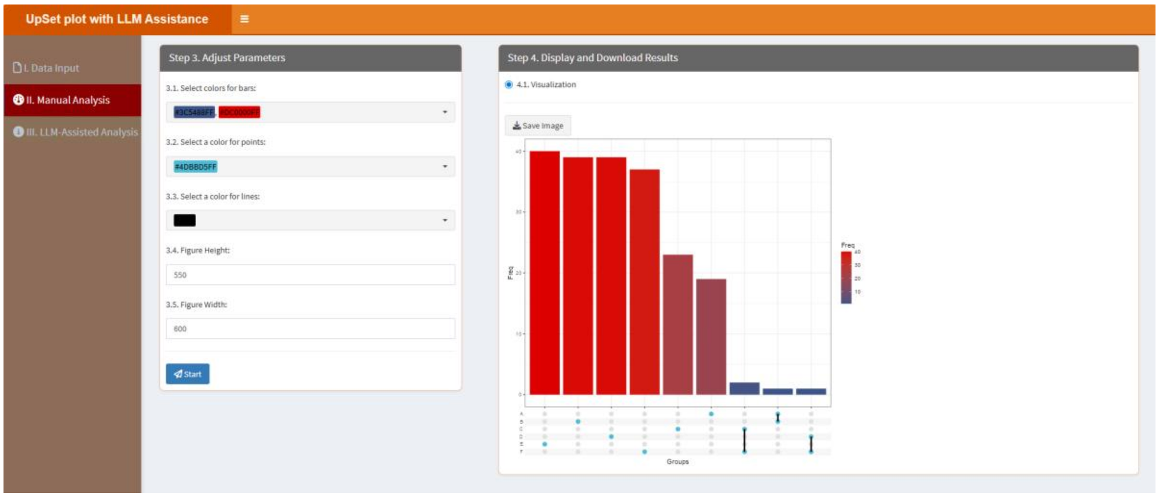
3.2. Select a color for points: This setting determines the color of the dots in the intersection matrix, where each row represents a set and each column shows a specific intersection.

3.3. Select a color for lines: This parameter allows users to define the color of the vertical lines that connect the intersection points within each column.

3.4. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

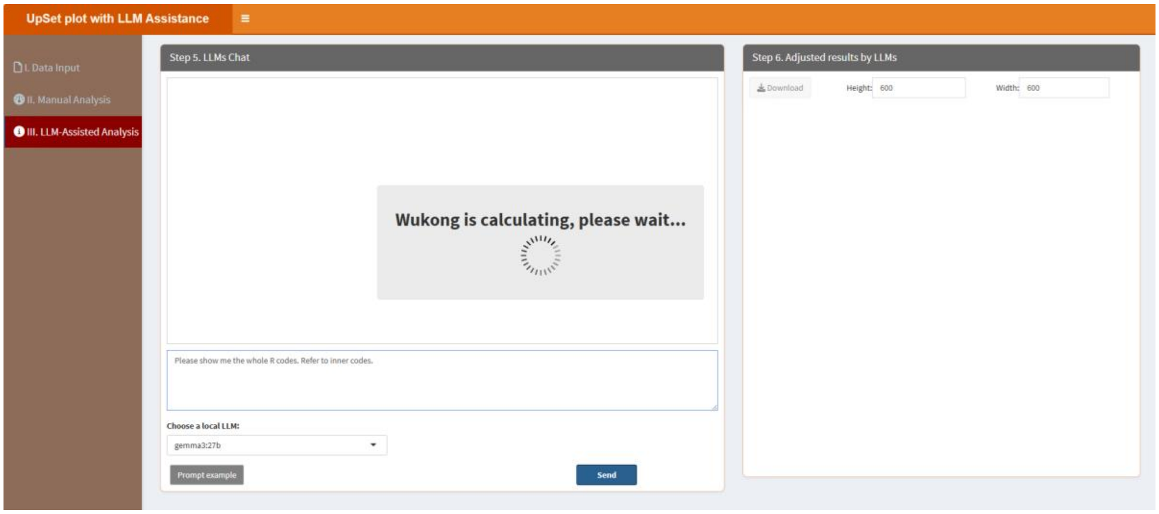
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



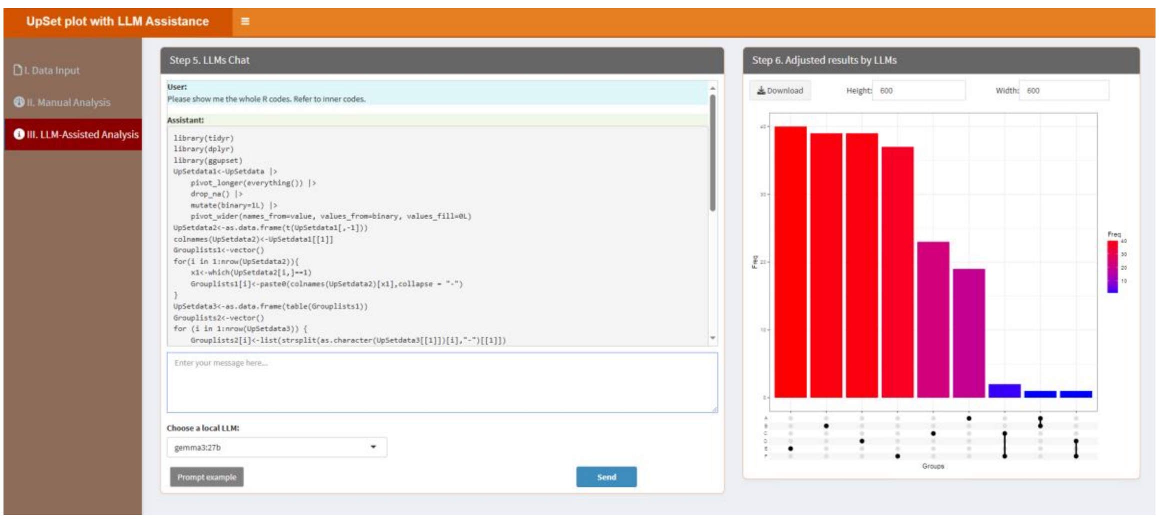
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please show me the whole R codes. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please show me the whole R codes. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

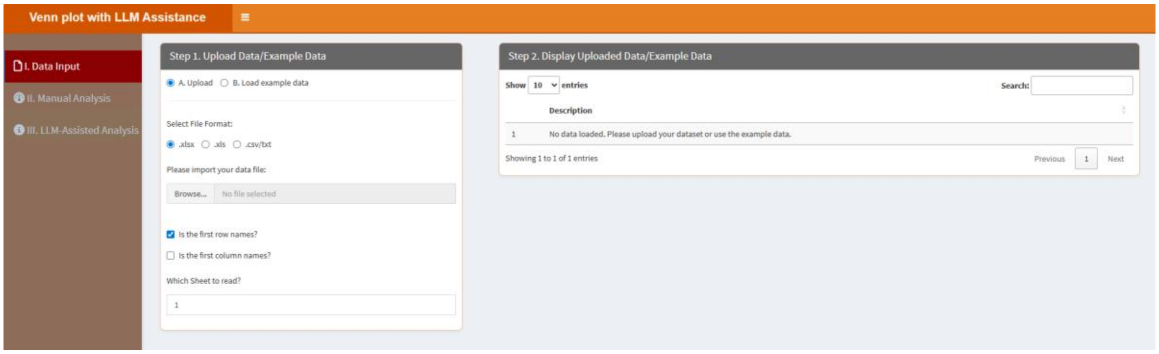


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



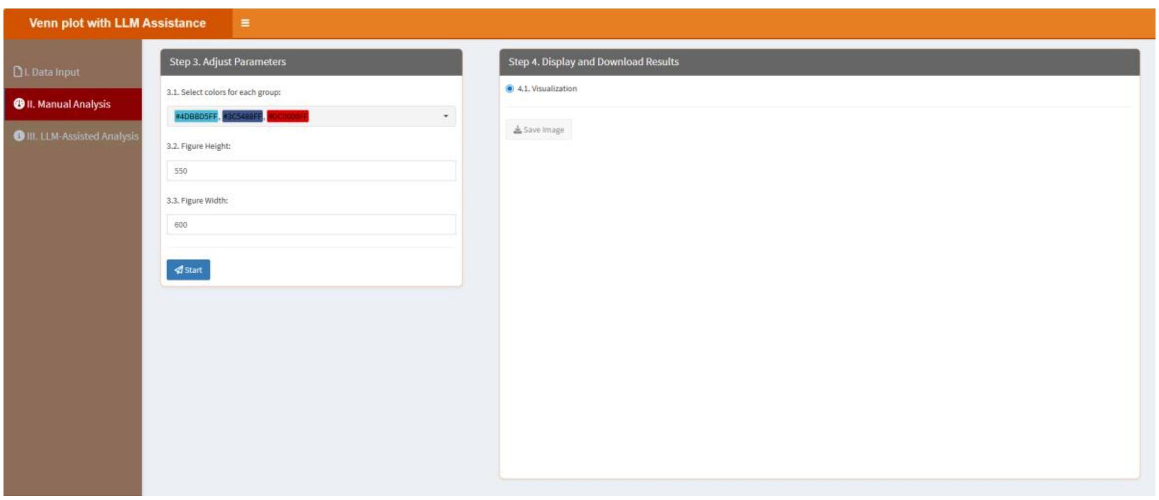
7.31. Venn plot

Venn plots display overlaps between sets. Users can customize the plot through the classic framework and enhance it with local LLMs for clear and informative diagrams. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



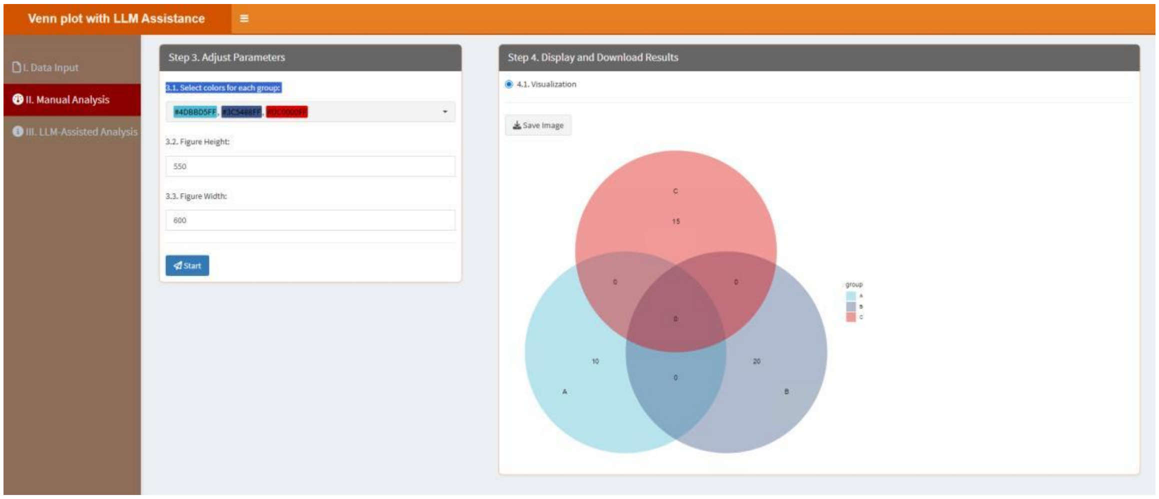
3.1. Select colors for each group: This parameter lets you choose specific colors for each group. The color number should be equal to the group number.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click “Start” button, the results will be shown as below:

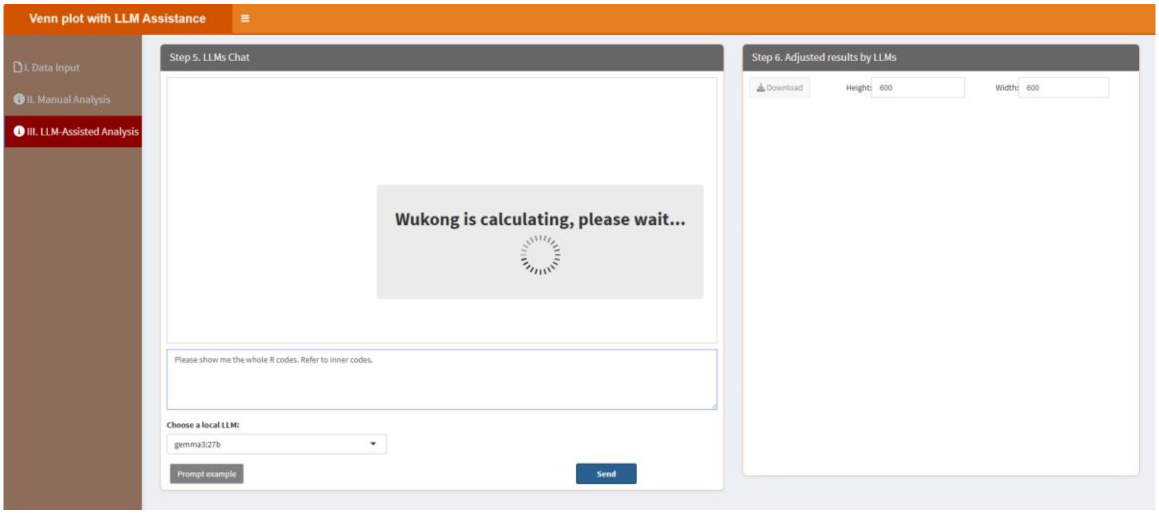
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



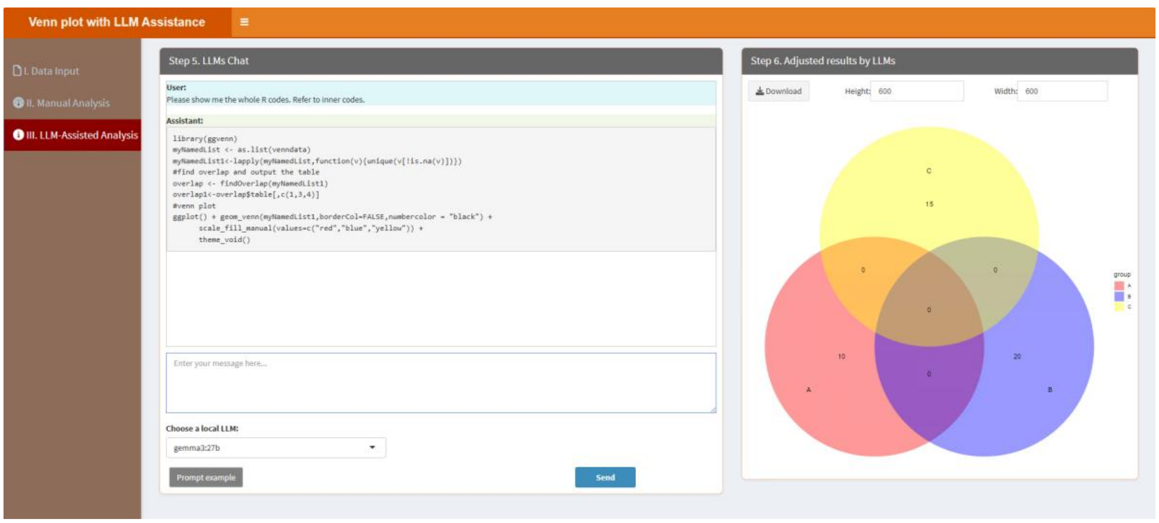
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please show me the whole R codes. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please show me the whole R codes. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

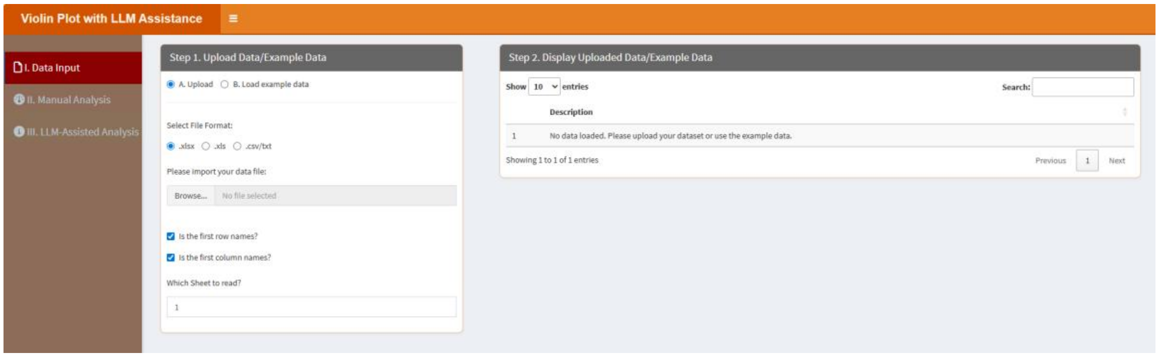


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



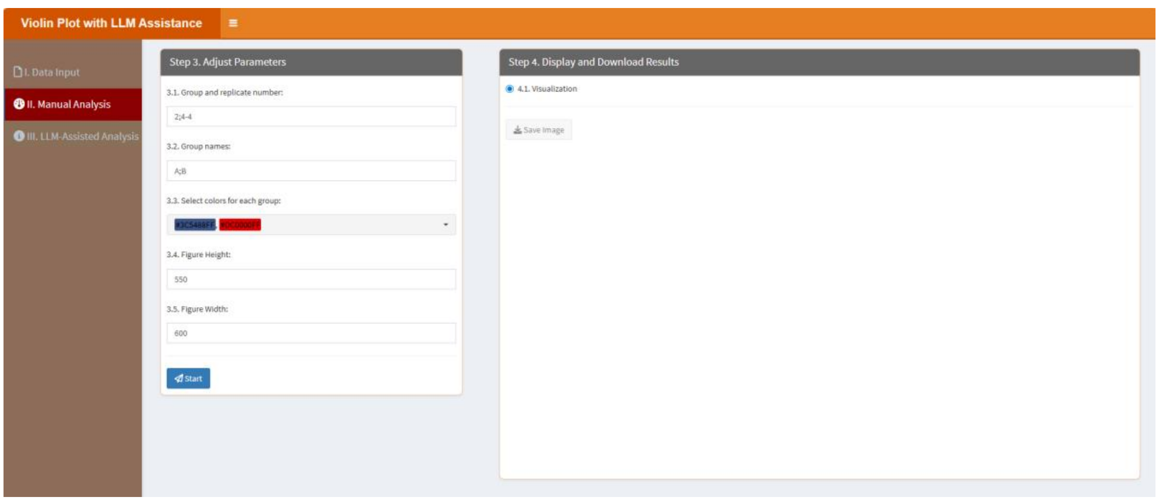
7.32. Violin Plot

Violin plots visualize data distributions. Users can adjust parameters in the classic framework and refine the plot with local LLMs for detailed and interpretable results. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



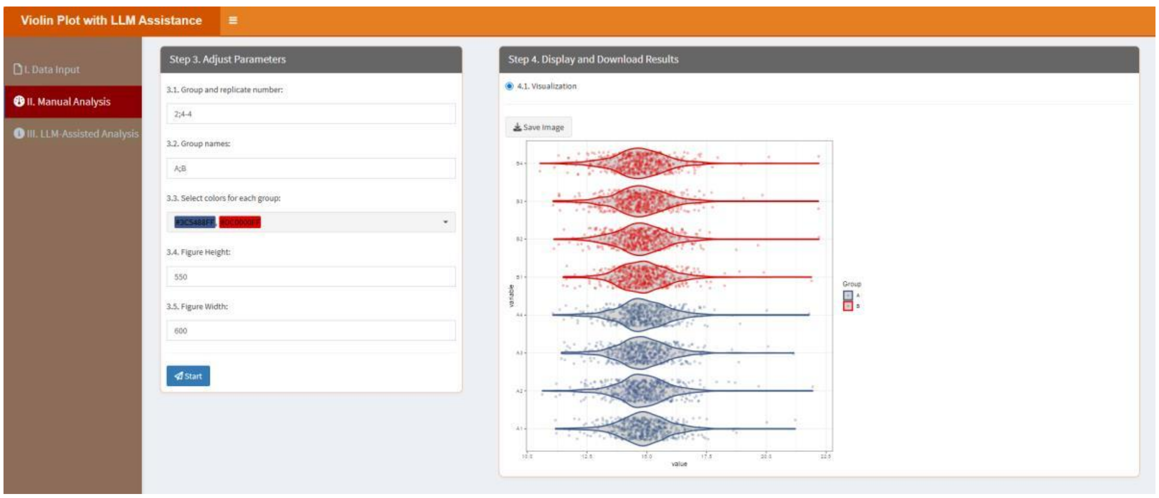
3.1. Group and replicate number: Type in the group number and replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5". The example data has 2 groups and 4 replicates in each group, so here is “2;4-4”.

3.2. Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 2 groups in the example data, you can type in "A;B".

3.3. Select colors for each group: This parameter lets you choose specific colors for each

group. The color number should be equal to the group number.
3.4. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.
3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

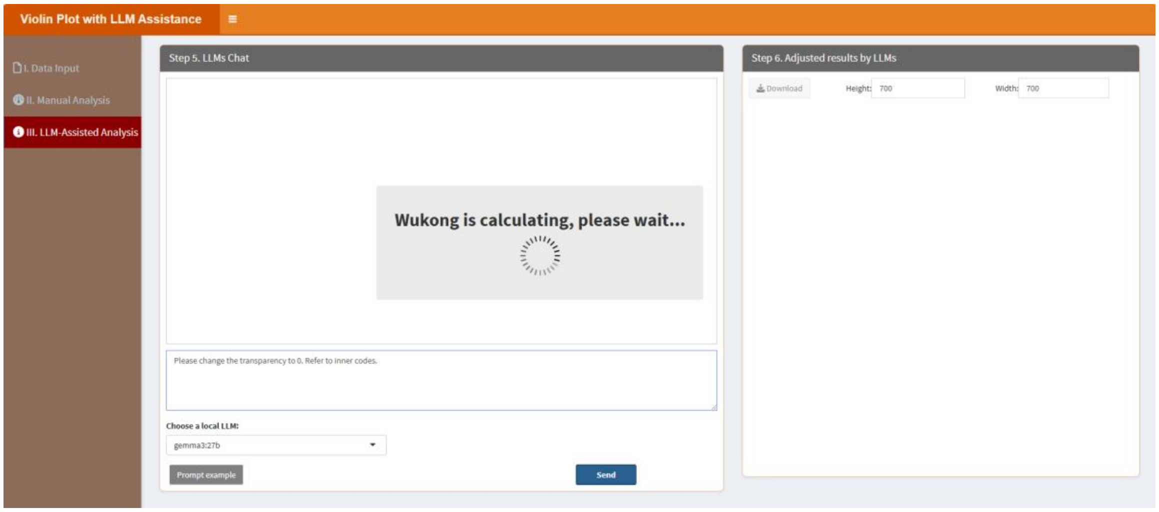
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



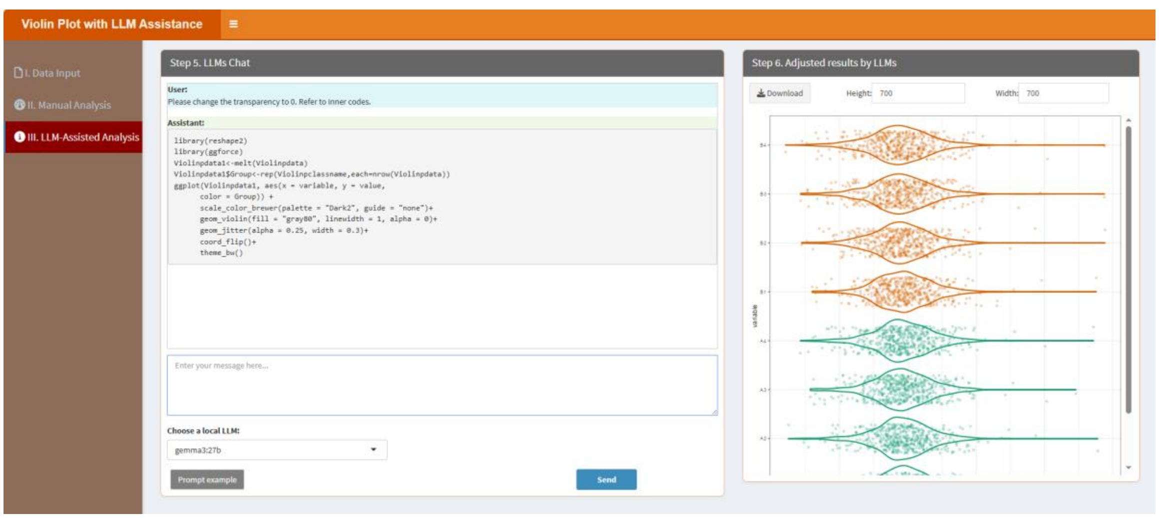
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please change the transparency to 0. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Please change the transparency to 0. =>", it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click "Send" button. It will learn like below:

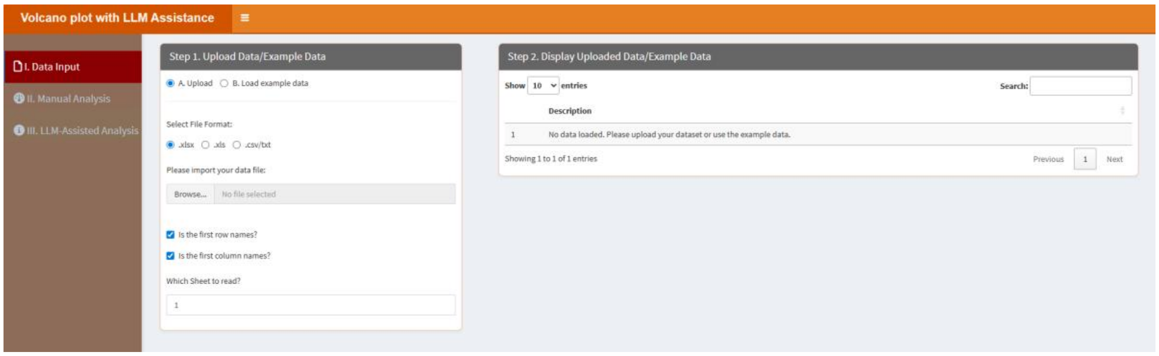


The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



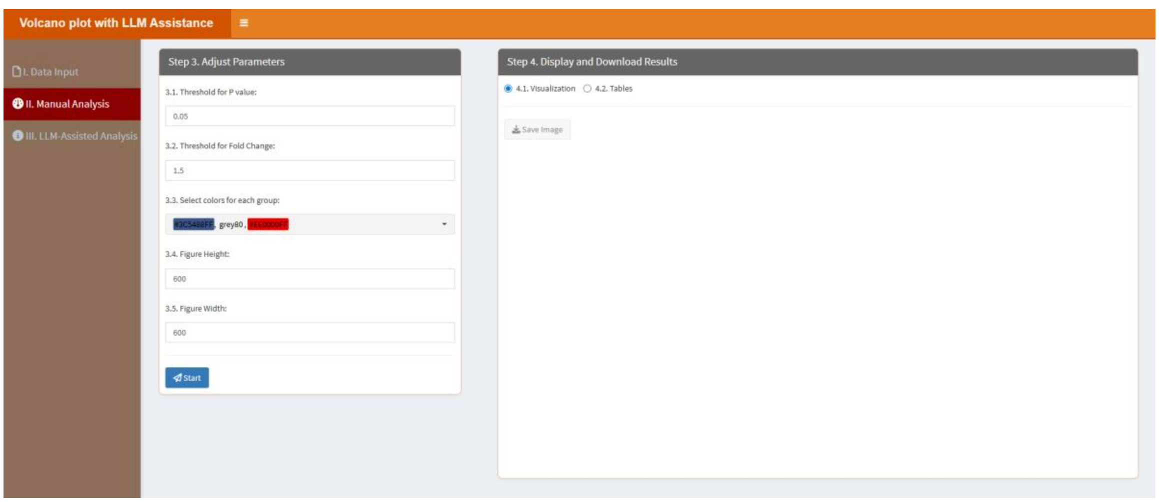
7.33. Volcano plot

Volcano plots identify significant changes in data. Users can customize the plot through the classic framework and refine it with local LLMs for clear and visually appealing results. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



3.1. Threshold for P value: This parameter allows users to define the cutoff for statistical significance. Variables with a p-value below this threshold will be considered significantly different between groups. Common values include 0.05.

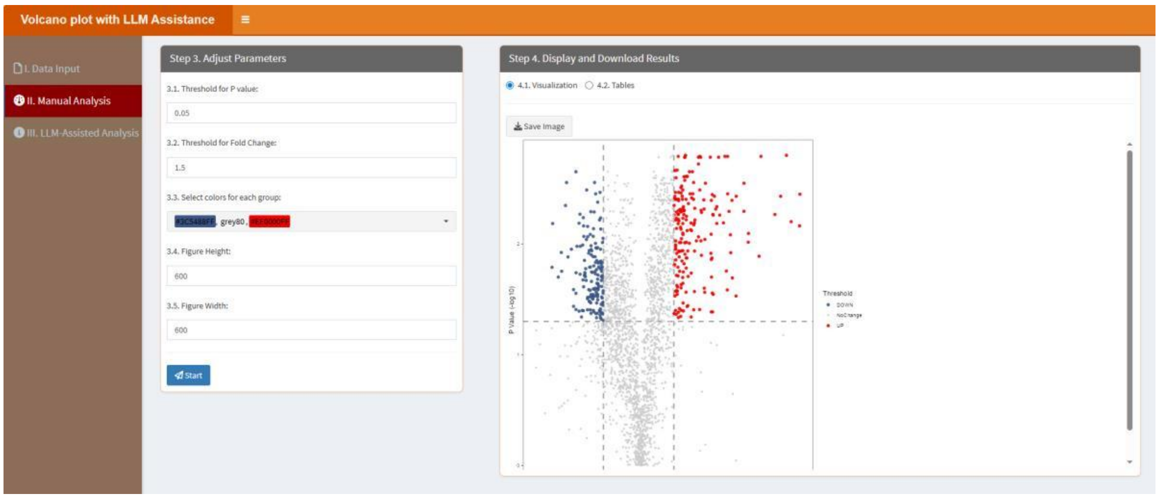
3.2. Threshold for Fold Change: This parameter sets the minimum absolute fold change required for a variable to be considered significantly upregulated or downregulated.

3.3. Select colors for each group: This setting allows users to define distinct colors for each category of data point, typically "UP", "DOWN" and "NoChange".

3.4. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your

presentation or analysis needs, ensuring it fits well within the intended display context.
3.5. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization, ensuring that it accommodates the data and fits appropriately in the given space.

After setting parameters, click "Start" button, the results will be shown as below:
In the "4.1. Visualization" part, users can click "Save image" button to download the figure:

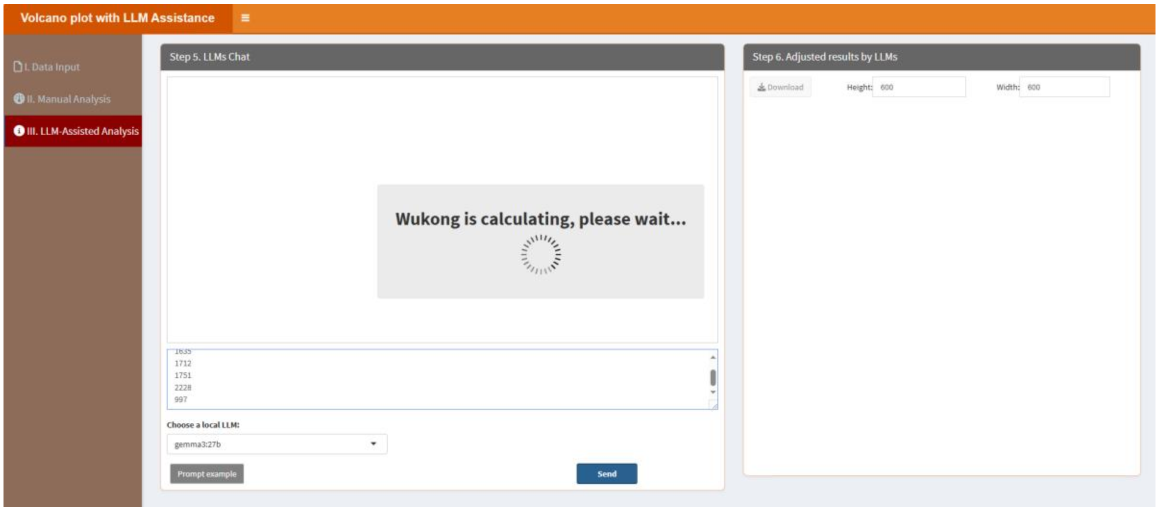


"III. LLM-Assisted Analysis" means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below "How to install LLMs locally?" part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

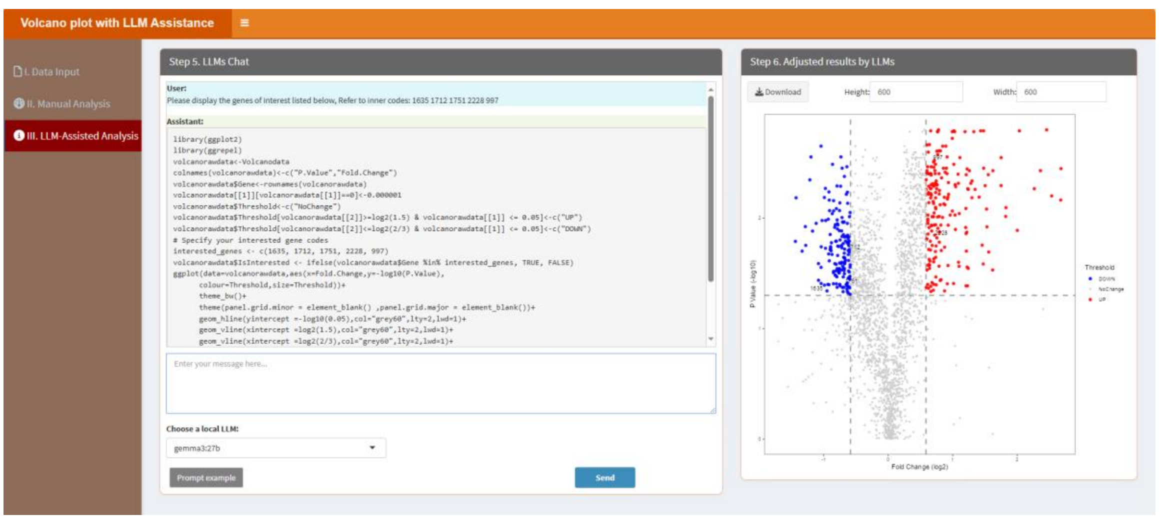
When users click the "Prompt example" button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the "II. Manual Analysis" section or to produce other desired outcomes. For instance, the example prompt "Please display the genes of interest listed below, Refer to inner codes:1635, 1712, 1751, 2228, 997." instructs LLMs to learn the internal R codes and generate the results. The key phrase "Refer to inner codes" serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label "=>" performs the same function, so if users type "Please display the genes of interest listed below, Refer to inner codes:1635, 1712, 1751, 2228, 997. =>", it will similarly prompt

LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:

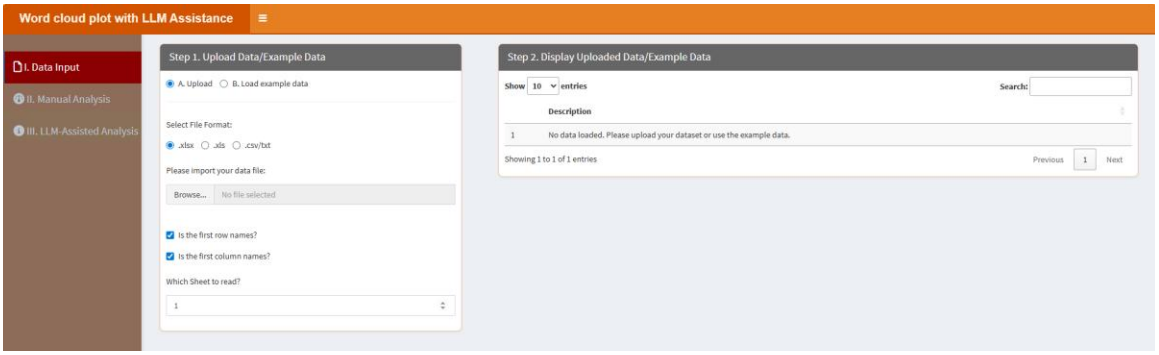


The results are shown as below, then users can download the results by clicking the “Download” button in the “Step 6. Adjust results by LLMs” section.



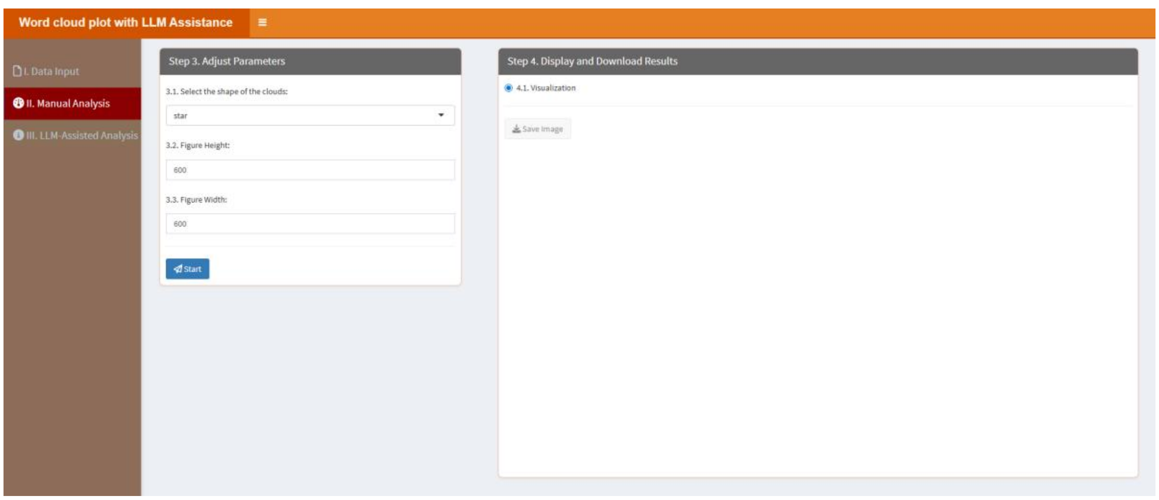
7.34. Word cloud

Word clouds visualize text data. Users can adjust parameters in the classic framework and rely on local LLMs to generate aesthetically pleasing and informative visualizations. After clicking the “Start” button, it will be started in a new window like below:



The “I. Data Input” here is similar to that in the “4.1. Coefficient of variation (CV)” above. Users can upload their data or check example data from here.

The “II. Manual Analysis” here is also similar to that in the “4.1. Coefficient of variation (CV)” above. However, the parameters here are different.



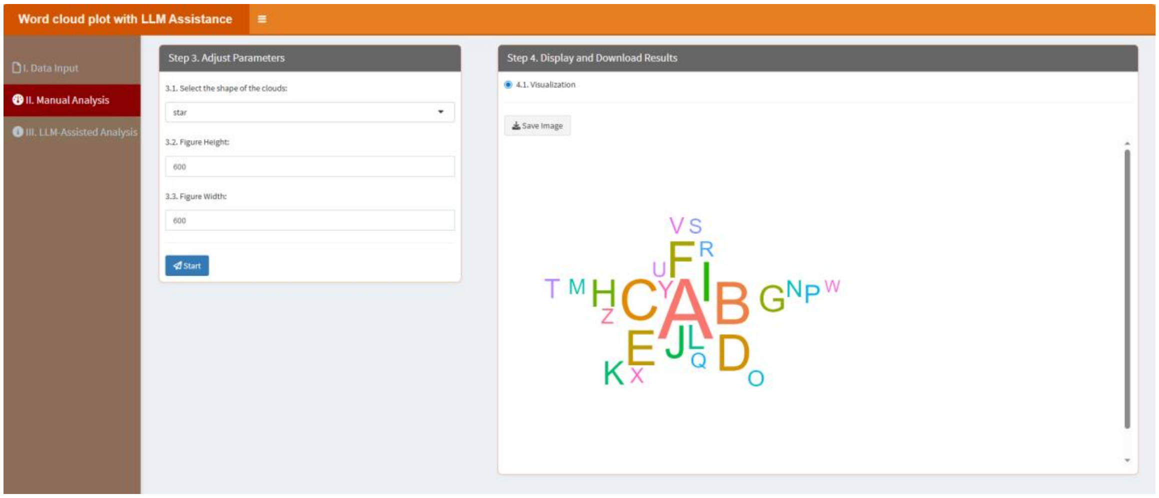
3.1. Select the shape of the clouds: This parameter allows users to choose the geometric layout of the word cloud. Options include star, circle, cardioid, diamond, and square, each offering a different aesthetic and spatial organization. For example, "circle" produces a balanced radial design, while "cardioid" creates a heart-shaped layout.

3.2. *Figure Height*: This parameter determines the height of the figure in pixels. By setting a numeric value, you can customize the vertical dimension of the output visualization to suit your presentation or analysis needs, ensuring it fits well within the intended display context.

3.3. Figure Width: Similar to the figure height, this parameter specifies the width of the figure in pixels. Adjusting this value allows you to control the horizontal dimension of the visualization,

ensuring that it accommodates the data and fits appropriately in the given space.

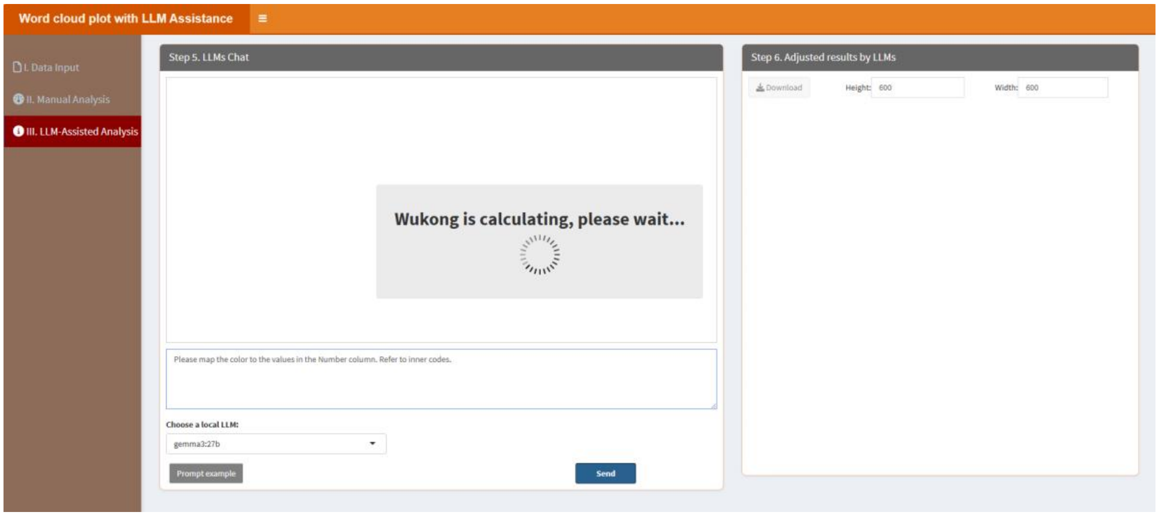
After setting parameters, click “Start” button, the results will be shown as below:
In the “4.1. Visualization” part, users can click “Save image” button to download the figure:



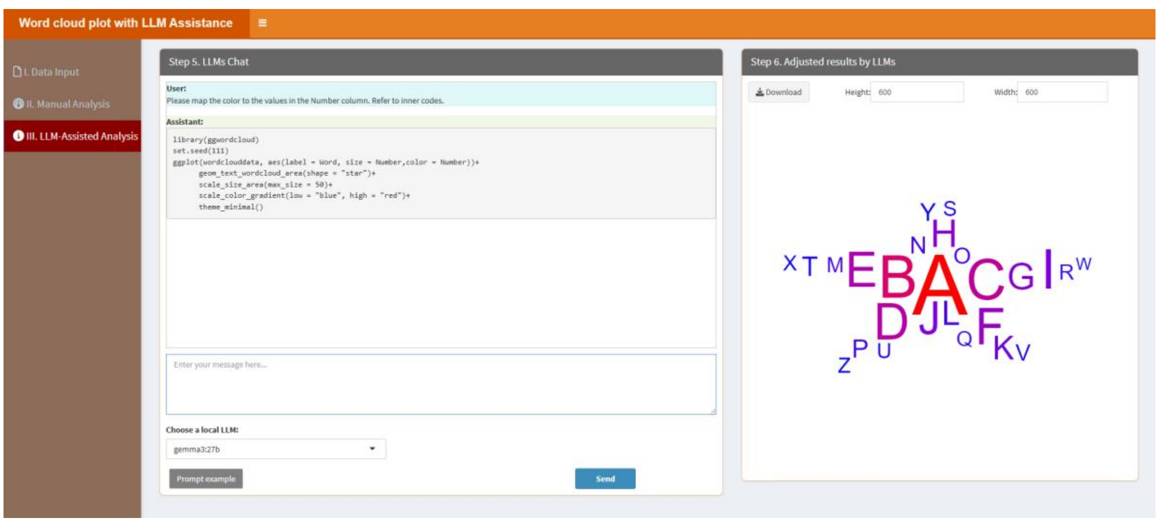
“III. LLM-Assisted Analysis” means users can analyze their data with LLM assistance. Section "Step 5. LLMs Chat" displays the conversation history, while "Enter your message here..." enables users to type messages or questions in the provided text box. Adjacent to the text box, the "Choose a local LLM" option lets users select a pre-installed LLM for their interactions (please see below “How to install LLMs locally?” part). Section "Step 6. Adjust results by LLMs" means that this panel shows the results (figures or tables) based on the conversation in the "Step 5. LLMs Chat".

When users click the “Prompt example” button, it provides a sample prompt that users can customize to create their own prompts. Additionally, we supply the R codes for the corresponding module, enabling users to study and modify the codes with the assistance of LLMs to generate results that cannot be achieved through the “II. Manual Analysis” section or to produce other desired outcomes. For instance, the example prompt “Please map the color to the values in the Number column. Refer to inner codes.” instructs LLMs to learn the internal R codes and generate the results. The key phrase “Refer to inner codes” serves as a trigger for LLMs to access and utilize the internal R codes. For convenience, the label “=>” performs the same function, so if users type “Please map the color to the values in the Number column. =>”, it will similarly prompt LLMs to learn the internal R codes and produce the result table.

After type the prompt, click “Send” button. It will learn like below:



The results are shown as below, then users can download the results by clicking the "Download" button in the "Step 6. Adjust results by LLMs" section.



8. WuKongmini

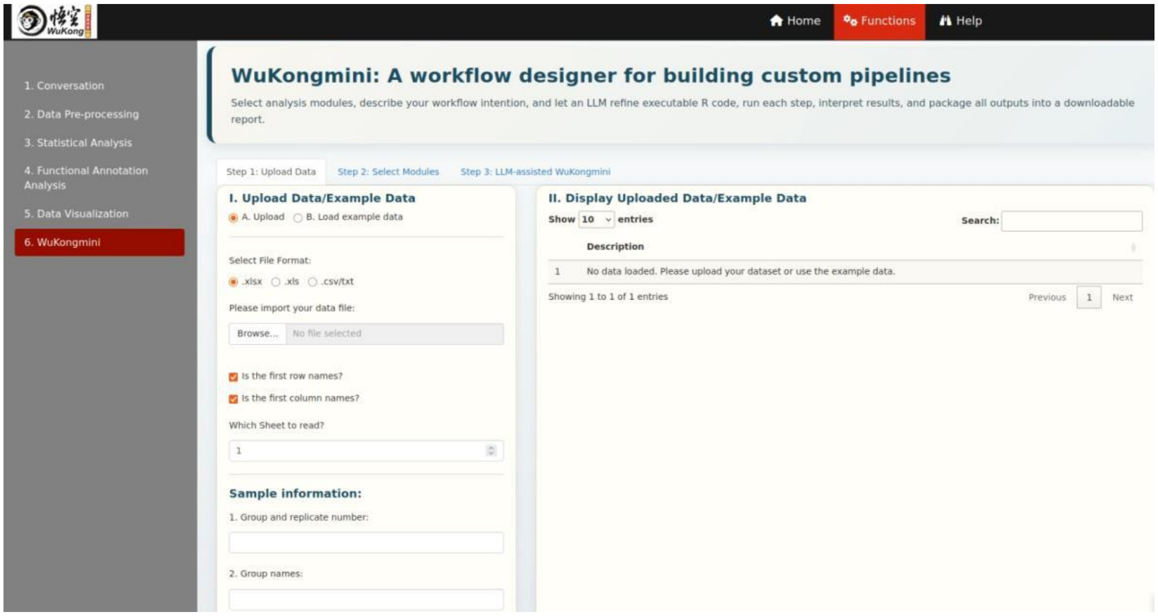
The "WuKongmini" in the WuKong platform represents a significant leap forward in the way researchers approach complex proteomics data analysis. Unlike traditional and rigid analysis pipelines, this module provides a highly interactive, modular environment where users can construct, customize and execute multi-step analytical workflows tailored to their researches. The WuKongmini is intuitively organized into a stepwise interface: users begin by uploading their own dataset or selecting from example data, and then input sample information, including biological replications, group numbers, names and species. Once data is loaded, users are presented with a categorized gallery of analytical modules, spanning data preprocessing, statistical analysis, functional annotation and data visualization. By simply clicking on the desired modules, users assemble a sequential workflow, the order of which can be easily adjusted to reflect the logical progression of their analysis. Next, users could type a brief introduction about the assembled workflow for following LLMs to learn and restrict data analysis based on the brief description. For added convenience, we have built several example pipelines for users to learn how to build their own data analysis workflow and choose directly to analyze their data.

Once a workflow is constructed, users can select from a range of installed LLMs (such as gemma3:27b, gemma2:27b) and invoke the "Run Workflow" function. Please note, here we recommend users to install large-scale LLMs. The platform then automatically chains together the selected modules, passing outputs from one step as inputs to the next, and generates refined, executable R code that adheres strictly to best practices in scientific computing. The LLMs not only ensure code clarity and reproducibility but also provide stepwise explanations of input objects and workflow logic, making the process accessible to users with varying levels of programming expertise. Those without any programming background can ignore these R codes and only check the results. Results are presented in an organized fashion, with each step's outputs, including tables, plots and statistical summaries. All results can be downloaded into a zip file.

Below are the step-by-step instructions for using this module.

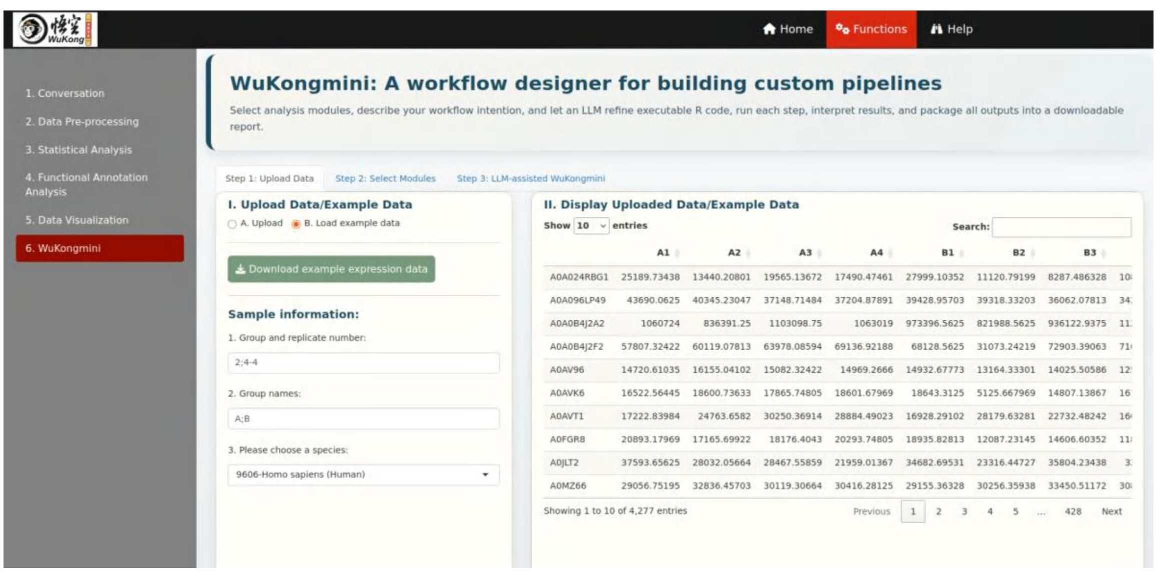
After installing the WuKong package (please refer to section 1 “How to run this tool locally?”), users can open the WuKong platform as shown below.

Next, users can click "Functions" and locate the "6. WuKongmini" module. This module includes three main steps:



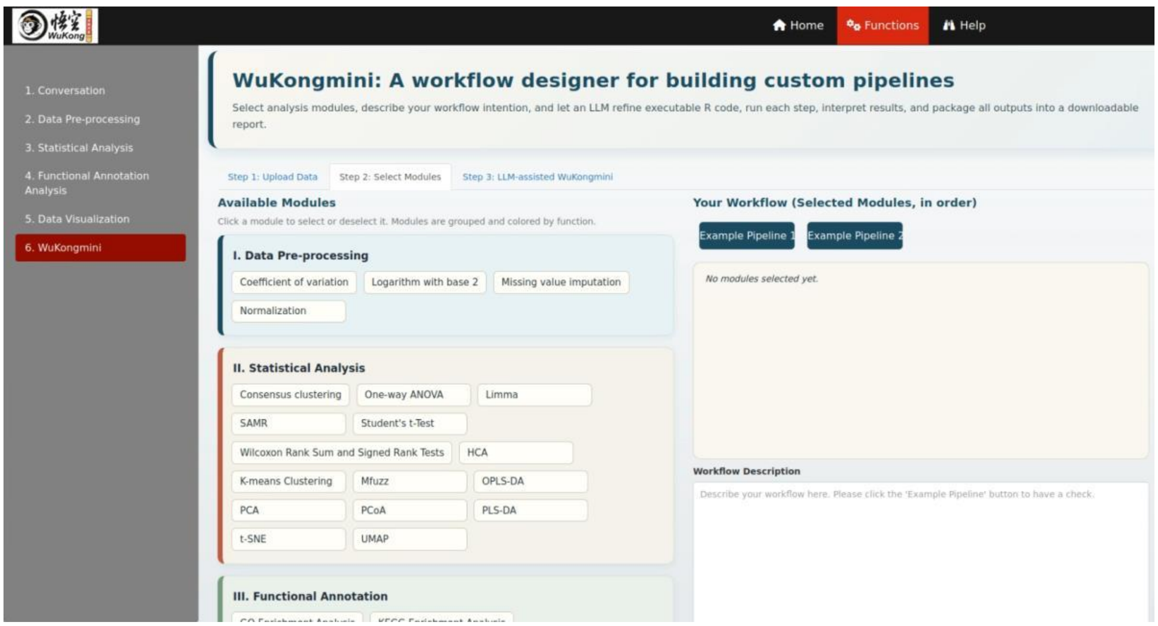
Step 1: Upload Data

In this step, users can upload their own data or load an example dataset. Please refer to section "2. Data Preparation" for details on how to format your data. If users choose "B. Load example data", the platform displays an example dataset and corresponding sample information: this includes 4,277 proteins identified by UniProt IDs and 8 samples, divided into two groups (A and B), each with 4 biological replicates.

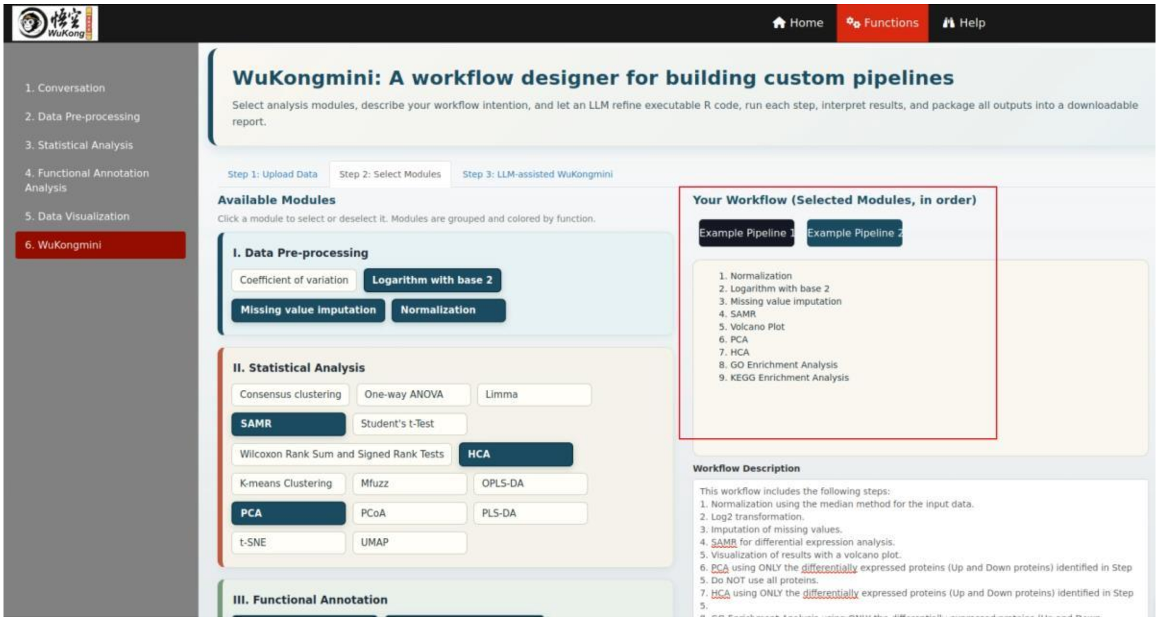


Step 2: Select Modules

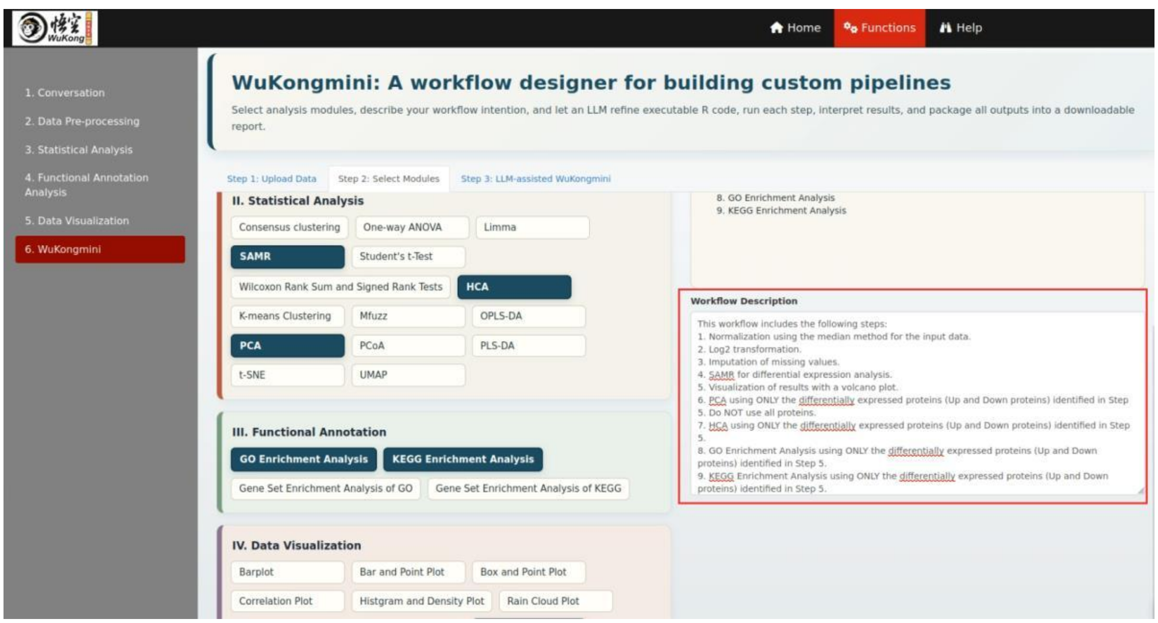
In this step, users can customize and combine various modules to process their data. These modules are categorized into four core areas as previously described: data preprocessing, statistical analysis, functional analysis and visualization. As shown below, the left panel displays all available modules, grouped and color-coded by category. Users can select any modules of interest, which will then appear in the right panel under “Your Workflow (Selected Modules, in order)”. In this section, users can also provide a brief description for each selected module in the “Workflow Description” field. These descriptions help guide the selected LLM in the next step; if left blank, the LLM will instead rely on the default R code associated with each module. Providing custom descriptions allows users to fine-tune the LLM’s behavior for more tailored analysis.



When a user clicks on a module, it is added to the “Your Workflow (Selected Modules, in order)” list on the right. Clicking the same module again will remove it from the list, allowing users to easily modify the workflow sequence.

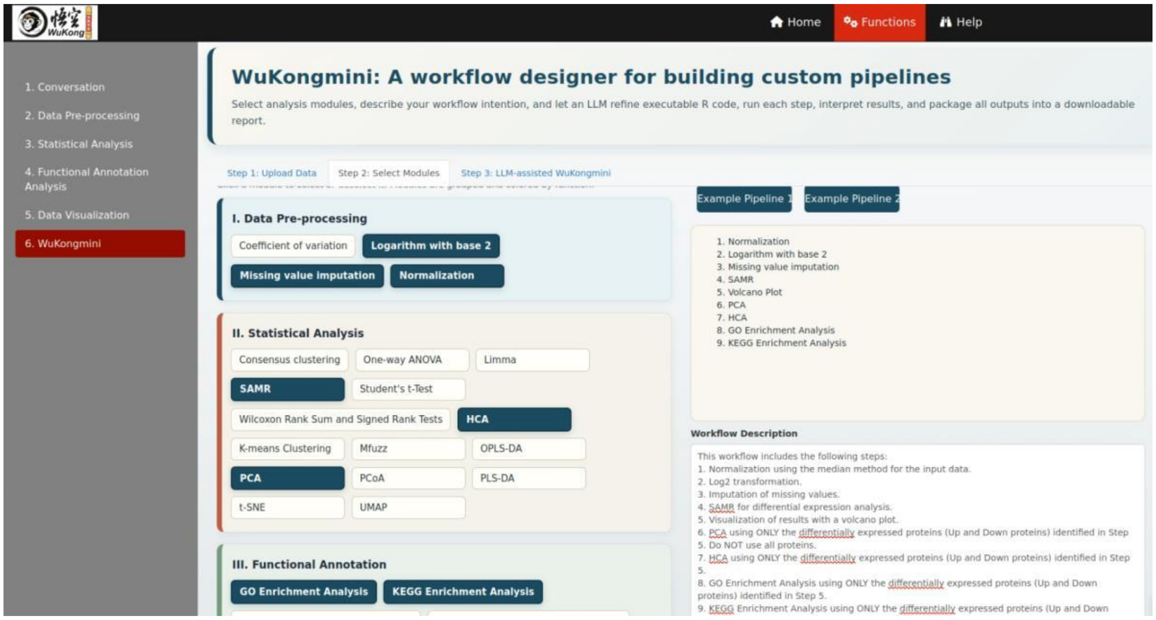


After selecting the desired modules, users are encouraged to write descriptions for each one to help the LLM better understand the context and goals of the analysis.

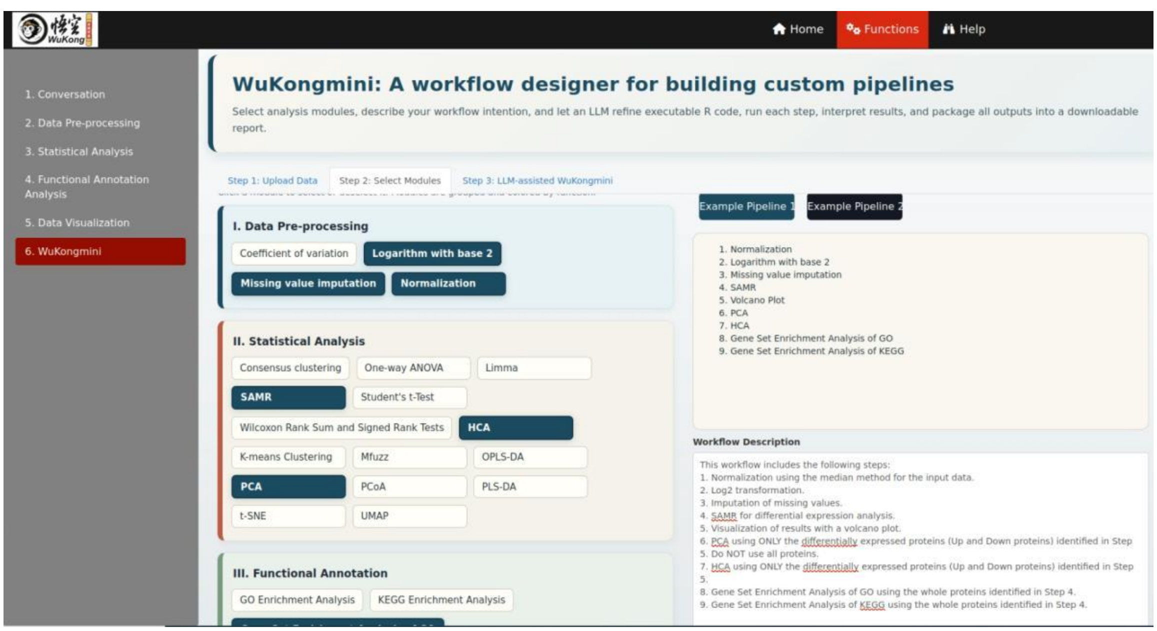


For added convenience, users can click the “Example Pipeline 1” or “Example Pipeline 2” buttons to load pre-built, classical proteomics data analysis workflows. These examples serve as templates that users can run directly or study to learn how to construct their own workflows.

Clicking the “Example Pipeline 1” button loads a predefined 9-step workflow that includes Normalization, Logarithm with base 2, Missing value imputation, SAMR, Volcano Plot, PCA, HCA, GO Enrichment Analysis, KEGG Enrichment Analysis. This pipeline comes with a sample workflow description, which helps the selected LLM in Step 3 understand and execute the analysis more effectively.

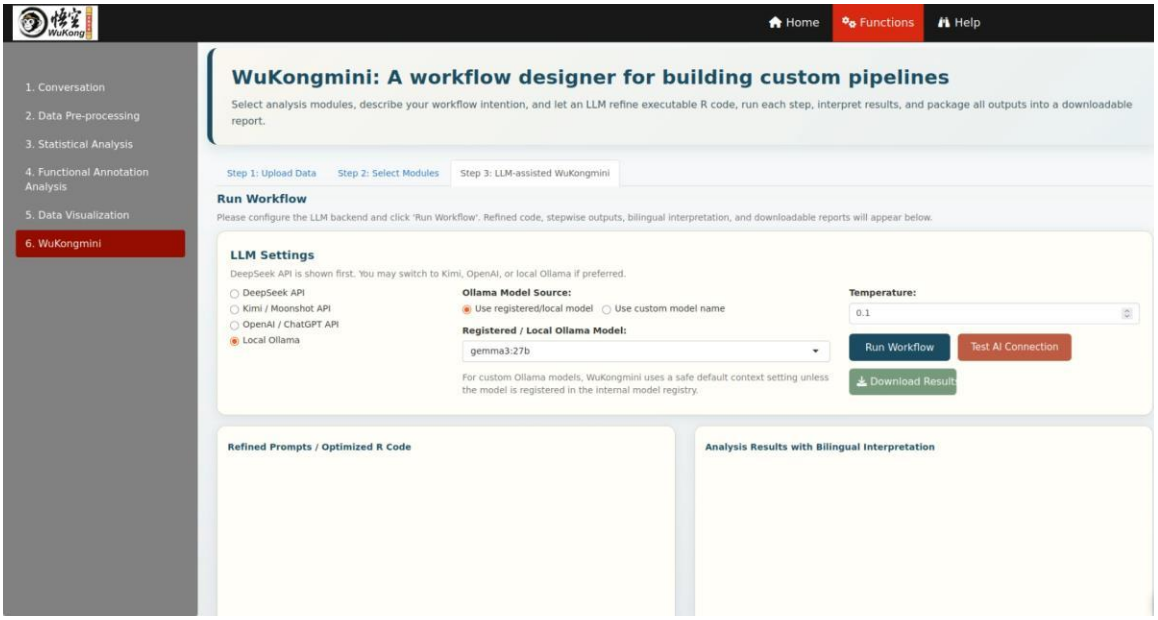


Clicking the “Example Pipeline 2” button loads a similar 9-step workflow: Normalization, Logarithm with base 2, Missing value imputation, SAMR, Volcano Plot, PCA, HCA, Gene Set Enrichment Analysis of GO, Gene Set Enrichment Analysis of KEGG. This version uses gene set enrichment analysis methods to explore protein functions. Like Pipeline 1, it includes a pre-written description that will be used by the selected LLM in Step 3 to perform the analysis.

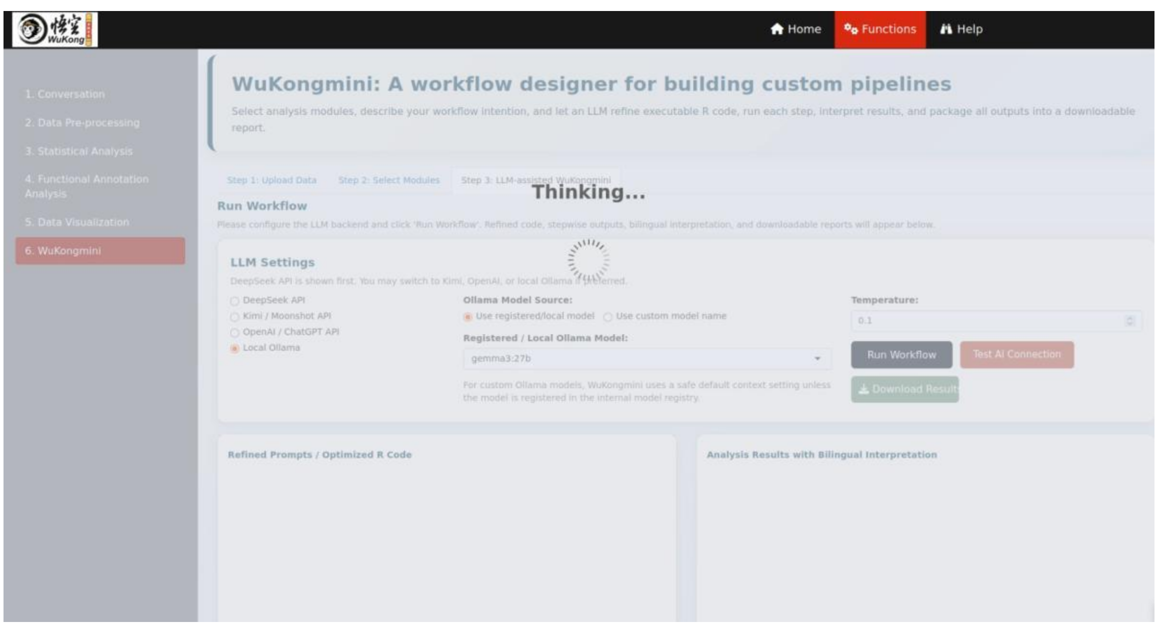


Step 3: LLM-assisted WuKongmini

In this step, the workflow constructed in Step 2 is executed with the support of a LLM, and the platform displays both the “Refined Prompts (LLM-generated R code or instructions)” and the “Analysis Results”. Users can also download all output results (e.g. tables, figures) in a ZIP file.

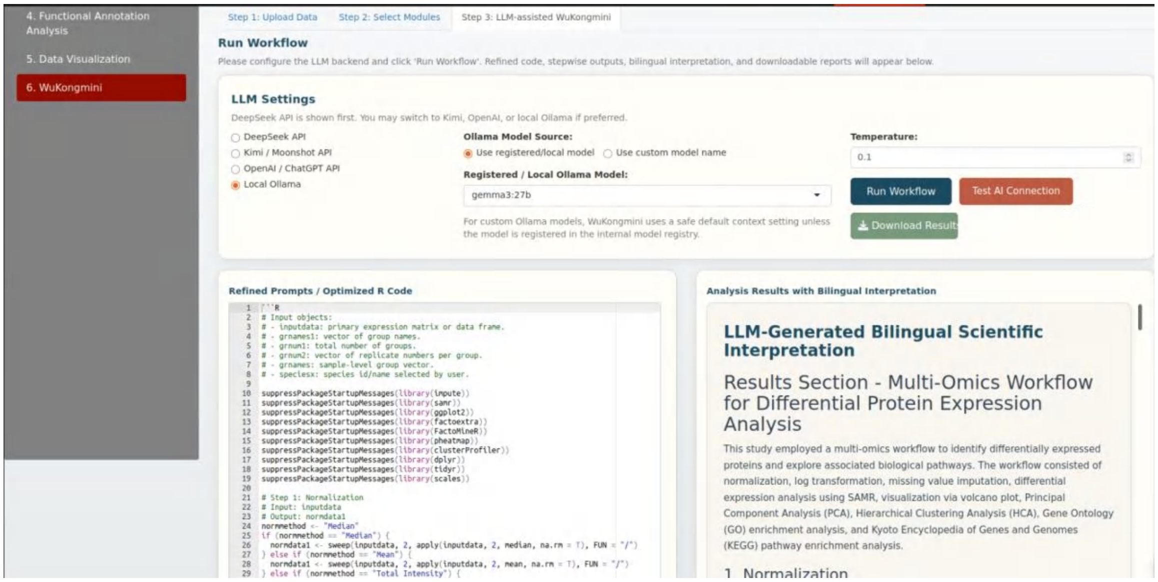


After selecting a local LLM (e.g., gemma3:27b), users simply click the “Run Workflow” button to begin execution. We strongly recommend using a large-scale local LLM for optimal understanding and handling of complex workflows.

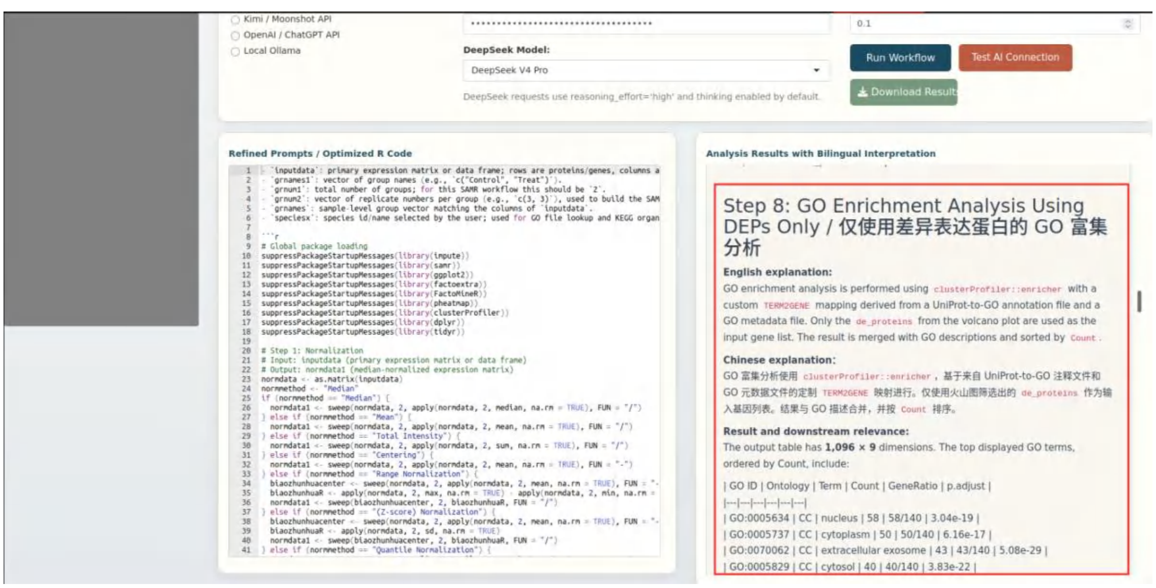


Once the workflow execution is complete, results are displayed within two main sections:

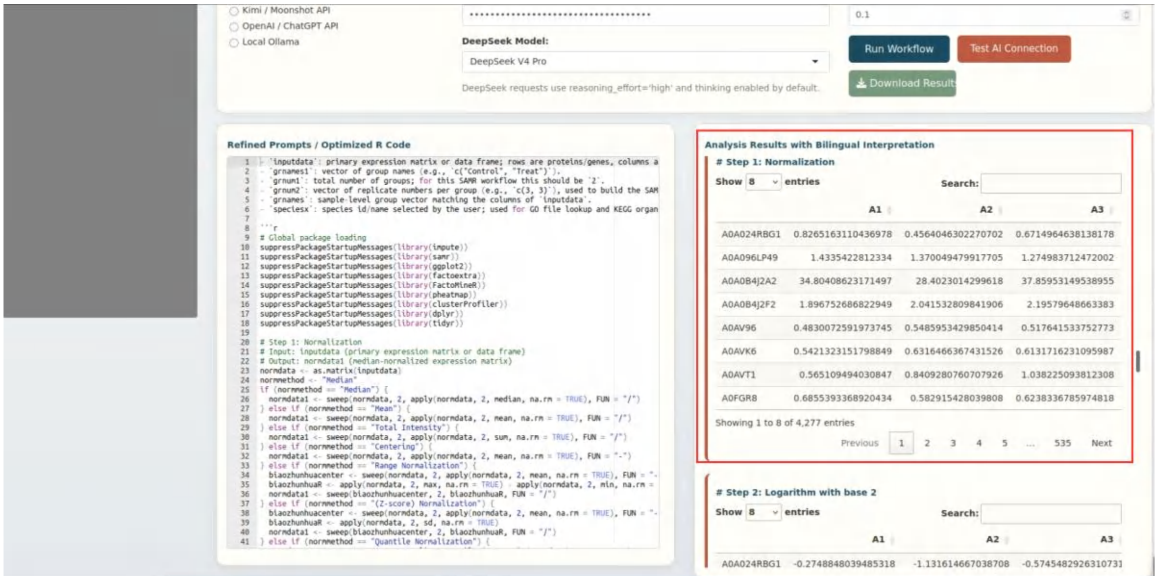
"Refined Prompts" and "Analysis Results".

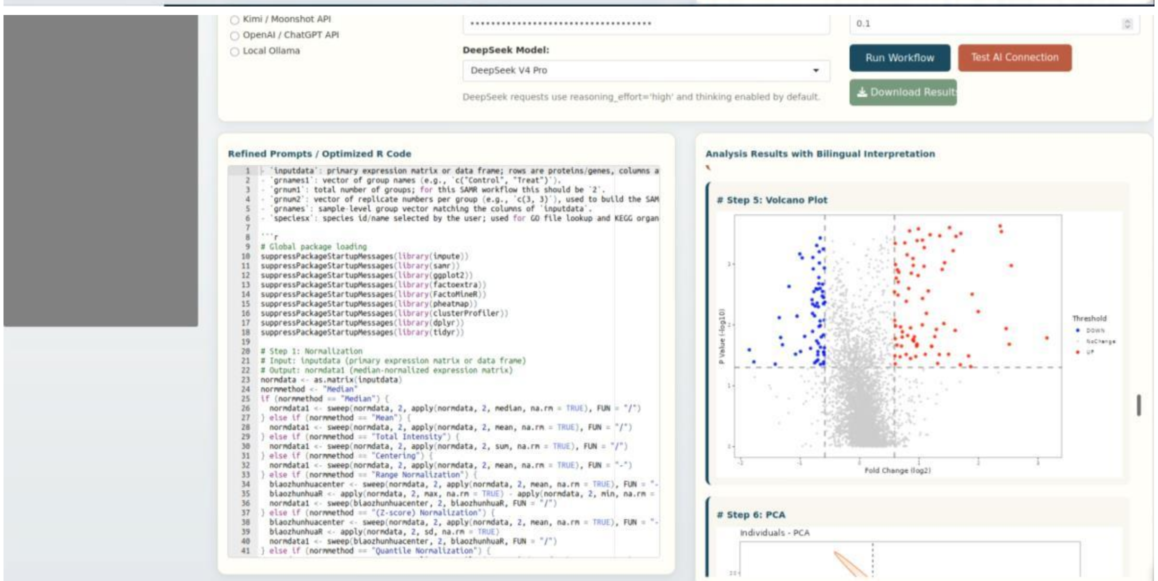


The "Refined Prompts" section contains step-by-step R code automatically generated by the LLM, aligned with the structure of the workflow built in Step 2. Users with R programming knowledge can examine these scripts to understand how the data was processed at each stage, or reuse the code for independent analysis. For users unfamiliar with coding, this section can be skipped without impacting their ability to interpret results. The "Analysis Results" section presents the full output of the workflow, beginning with a brief, LLM-generated summary of the overall analysis. Notably, this section includes a dedicated "Differentially Expressed Protein Functional Categorization", which provides insights into the biological roles and significance of differentially expressed proteins identified during the analysis.



Following this summary, the module displays the results of each individual workflow step in sequence, including figures, tables and visualizations as applicable.





Finally, users can click the “Download Results” button to export all output data in ZIP format for offline access, reporting, or further analysis.

9. Case Study

To assess the functional versatility of the WuKong platform, we utilized a publicly available proteomics dataset on lung squamous cell carcinoma from the Clinical Proteomic Tumor Analysis Consortium (CPTAC), originally published by Shankha et al. The dataset is available through ProteomeXchange with the identifier PDC000234 and includes quantitative proteomic profiles from 212 samples-comprising 110 tumor tissues and 102 matched adjacent normal tissues-with expression data for 12,759 proteins. The dataset in a .csv format can be accessed and downloaded at: https://github.com/wangshisheng/WuKong.

When opened in Excel, the dataset is displayed with protein names as rows and sample names as columns. Sample names containing ".N" indicate adjacent normal tissues, while those without ".N" represent tumor tissues. The data has been normalized and transformed with binary logarithm (i.e. log2).

	A	B	C	D	E	F	G	H	I	J	K	L
1		C3L-00081.N	C3L-00415.N	C3L-00445.N	C3L-00568.N	C3L-00603.N	C3L-00904.N	C3L-00923.N	C3L-00927.N	C3L-00993.N	C3L-01000.N	C3L-01285.N
2	ARF5	-0.04178705	0.0616689	0.1188204	-0.14546018	-0.376528442	-0.382265481	0.0538091	-0.349116468	-0.030397907	-0.19696926	-0.073082033
3	M6PR	-0.220777197	-0.13408	0.0043697	-0.204541608	-0.228215198	-0.260806008	-0.235954	0.080082231068	0.22412219226	-0.198817796	0.1494818966
4	ESRRA	-0.091096924	-0.379467	-0.191655	0.0088098161	1.1082488450	0.2625423569	-0.019013	-0.069992433	-0.347673071	0.5228739501	-0.556894679
5	FKBP4	-0.759884574	-0.577785	-0.563748	-0.728165764	-0.710225996	-0.864701272	-0.828016	-0.881461878	-0.802232243	-0.899496602	-0.558510124
6	NDUFAF7	-0.349874899	-0.15448	-0.310628	0.1543552716	0.2090691729	0.0697506118	-0.101414	-0.361786358	-0.223416964	0.1109470782	-0.163725543
7	FUCA2	-0.553674481	0.1862242	0.1159842	-0.24162283	-0.082683665	0.1562042252	0.1568698	-0.079559975	-0.186408082	-0.090084366	0.0101989914
8	DBNDD1	NA	NA	-0.280613	NA	0.8682306601	NA	NA	NA	NA	NA	NA
9	HS3ST1	NA	NA	-0.67653	NA	NA	-0.113160194	-0.406904	NA	-1.297089973	NA	-2.113925964
10	SEMA3F	-0.393563725	-0.146183	-0.858224	-0.093513459	-0.337677642	-0.063408576	-0.345781	-0.36932998	-0.026083067	0.1126936372	-0.474228132
11	CFTR	-0.053583218	NA	1.3307767	0.4041811213	-0.214732913	-0.125850491	0.3582003	0.54323236225	0.02543266954	0.3021278403	0.6201599387
12	CYP51A1	0.6739195579	0.7891086	0.7283173	0.2331819236	0.0146766964	-0.180149013	0.3393526	0.70608794965	0.37800908650	0.1486026854	0.5889865596
13	TMEM176A	0.0128708493	0.9546589	0.5412050	NA	NA	NA	NA	NA	0.33456204695	0.7310973779	0.4502381277
14	SLC7A2	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
15	HSPB6	1.4118771445	0.6357099	0.9689269	0.7812973900	1.0070038614	1.2153931119	1.5841134	0.85856087856	1.24527796560	0.7407476312	0.9526615503
16	ZNF195	NA	NA	-0.730388	-0.17861356	-1.0160149	-0.250939902	NA	-0.09584301	NA	-0.796936344	NA
17	PDK4	0.7687912299	NA	NA	NA	NA	0.4810917628	NA	1.05368894589	0.18285760730	1.3859576438	0.7018454219
18	RALA	0.2278686984	-0.223819	-0.052331	0.0096377815	0.2943083691	0.0409025077	0.1105418	0.35275462652	-0.165688395	0.6529871343	0.4257494865
19	BAIAP2L1	-0.613249719	-0.650435	-0.714799	-0.921410487	-0.710317018	-0.814561505	-0.859123	-0.683198408	-0.772054303	-0.688990346	-0.698446011
20	TMEM132A	-0.69609939	-0.29862	-0.389651	-0.480100615	-0.078622296	-0.229093712	-0.349371	-0.144116345	-0.144975472	-0.190470966	-0.187426633
21	DVL2	0.2392112220	-0.012549	-0.195581	-0.107080018	0.3153380529	0.1265513938	0.2788104	-0.120739981	0.09706727987	-0.043119898	0.2332718977
22	RPAP3	-0.312145677	-0.106894	-0.2704	-0.135310288	-0.328286379	-0.17418853	-0.360384	-0.391965697	-0.374163516	-0.406566984	-0.447510409
23	IFRD1	0.0086317974	-0.420603	0.0428456	-0.231566167	-0.105507857	-1.059716106	-0.144047	-0.310327378	-0.367867212	-0.357068619	0.1172723406
24	SKAP2	0.4613810073	0.1515485	0.4921903	0.2785908945	-0.097887607	0.1512824863	-0.040312	0.43369400927	0.39052354794	0.1598845185	0.3133841765
25	PRSS21	-0.734084935	-0.838965	-0.715417	NA	NA	0.5516044329	-0.837499	NA	-0.147712175	NA	0.0765127091
26	CX3CL1	-0.601423723	NA	-0.843388	-0.351379827	NA	0.0687328673	-0.629065	0.23122681332	-0.126179839	-0.389659171	0.4169061599
27	PRR5	-0.294196205	-0.56282	NA	-0.109954728	-0.434026138	0.0419730901	NA	-0.523889957	-0.263962429	-0.381890214	-0.13469143
28	TRAPPC6A	0.3936360317	0.1779579	0.1965846	0.2097825399	0.8705394716	0.2281736905	-0.266156	0.25107019372	0.25787227081	-0.084131333	0.1524679383
29	WDR54	0.6013513600	0.3888504	0.3657398	0.1390314186	0.5286598741	0.6554367904	0.6443141	0.07871937068	0.40467761546	0.2602207399	0.4428143647
30	SPATA20	0.1652403397	0.1651716	-0.448471	0.3127104579	0.1434197421	0.2381028172	0.0872535	-0.406106447	-0.130547405	-0.169396349	0.3183064422
31	CEACAM7	-0.285723541	-1.155254	-0.983657	NA	NA	NA	-0.637069	NA	0.44297729592	NA	-1.07952306
32	CD79B	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
33	RHBDD2	-0.013992102	-0.180187	-0.021669	0.1180826883	-0.80270776	-0.396479524	0.0763964	0.13470477537	-0.542830635	0.0378378009	-0.125432203
34	RPUSD1	NA	-0.14138	0.2061062	NA	0.6079662491	NA	-0.838217	-0.455636181	-0.478324966	0.5706292682	-1.315617855
35	TSR3	-0.611176357	0.0676637	-0.090061	-0.057761581	-1.090861126	-0.594102806	0.0461915	0.07850683203	0.04225285802	-0.717482068	-0.355245743

9.1. Preparation

Prior to commencing data analysis, users are required to ensure the following:

a. The WuKong package has been correctly installed on their computers (Refer to the section "1. How to run this tool locally?").

b. At least one Large Language Model (LLM) has been installed from Ollama (Refer to the "How to install LLMs locally?" subsection within "3. Conversation").

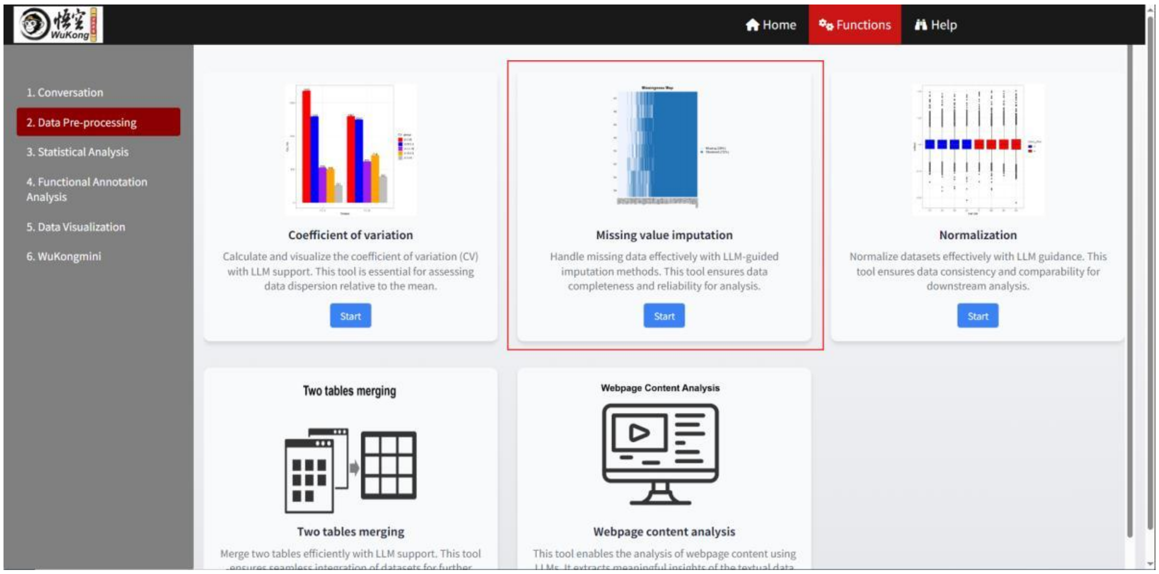
c. The data has been prepared in the appropriate format as previously described (Also refer to the "2. Data Preparation" section).

9.2. Data analysis

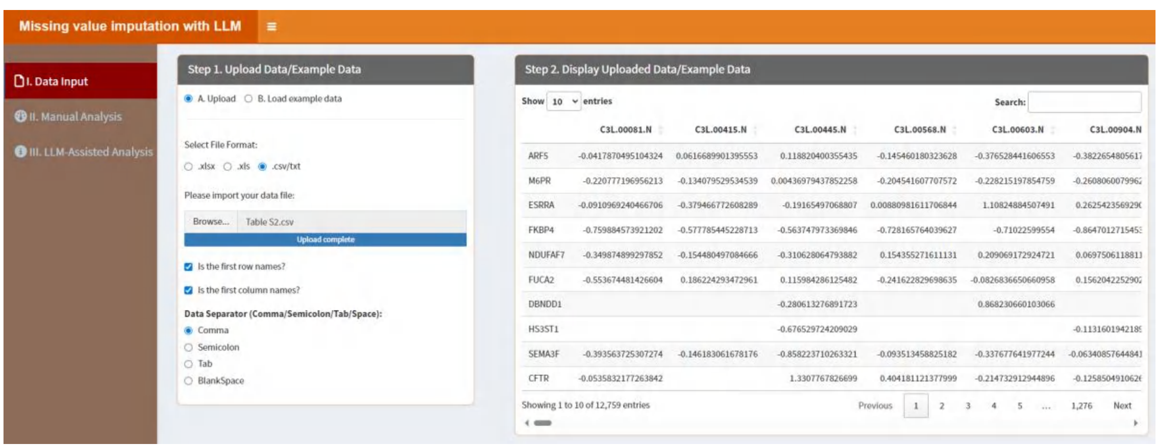
Herein, we analyzed the data above using a classical proteomics analysis pipeline, which included missing value imputation, differential protein expression analysis, volcano plot, PCA, HCA, GO and KEGG pathway enrichment analyses and post-enrichment result mining.

a. Missing value imputation

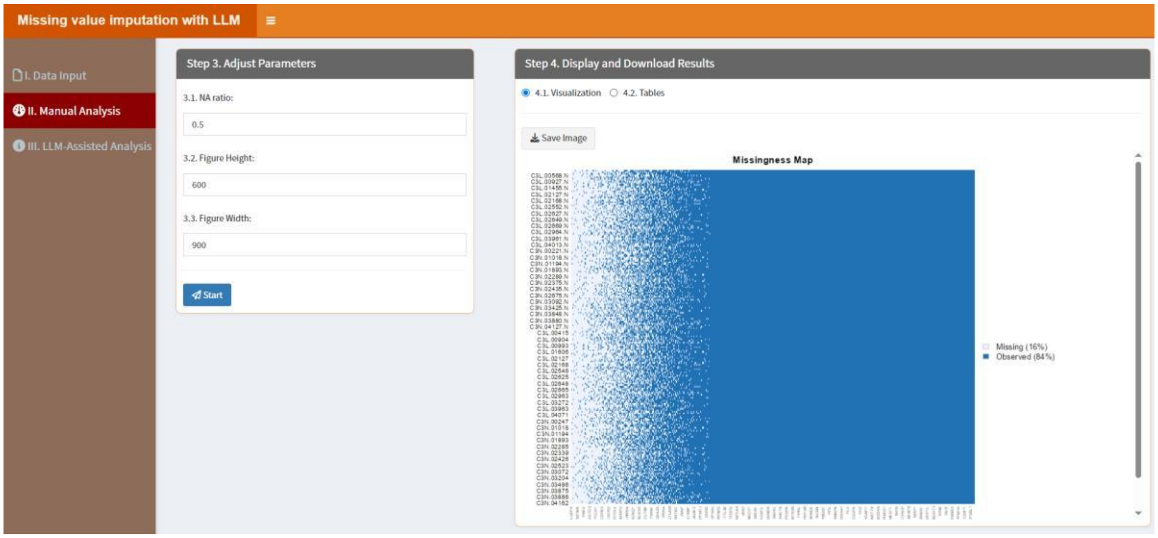
Users can find this module as below:



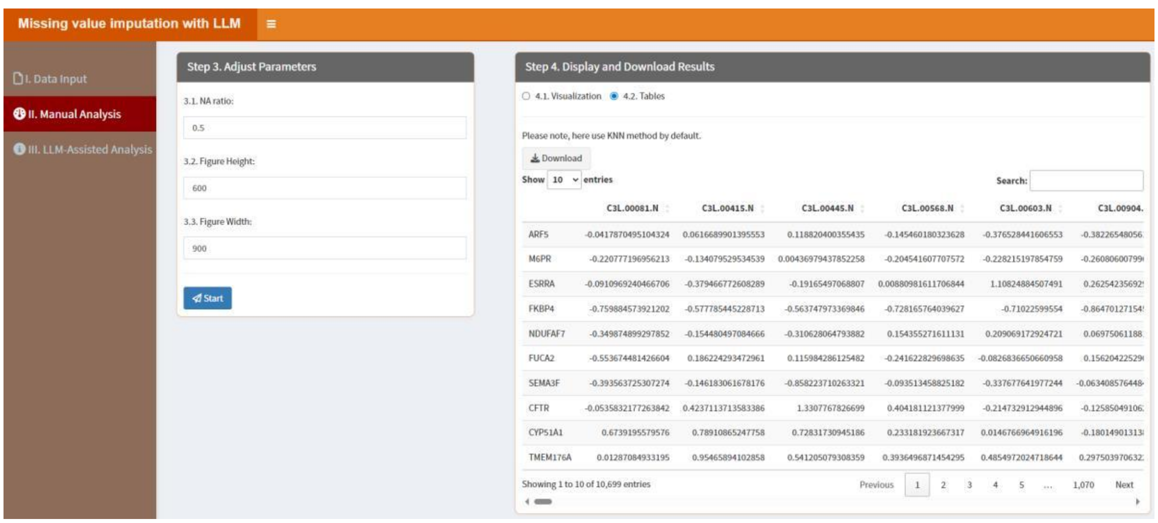
Then click "Start" button and upload the data, shown as below.



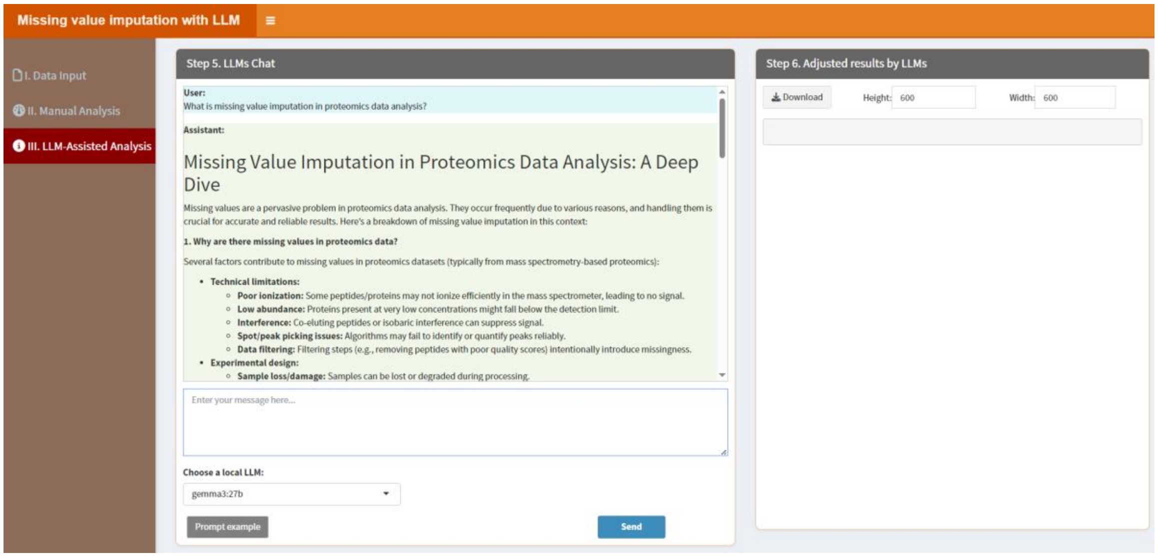
Next, click "II. Manual Analysis". The parameters here can be kept as default, and click "Start" button. The result in "4.1. Visualization" shows the missing value distribution.



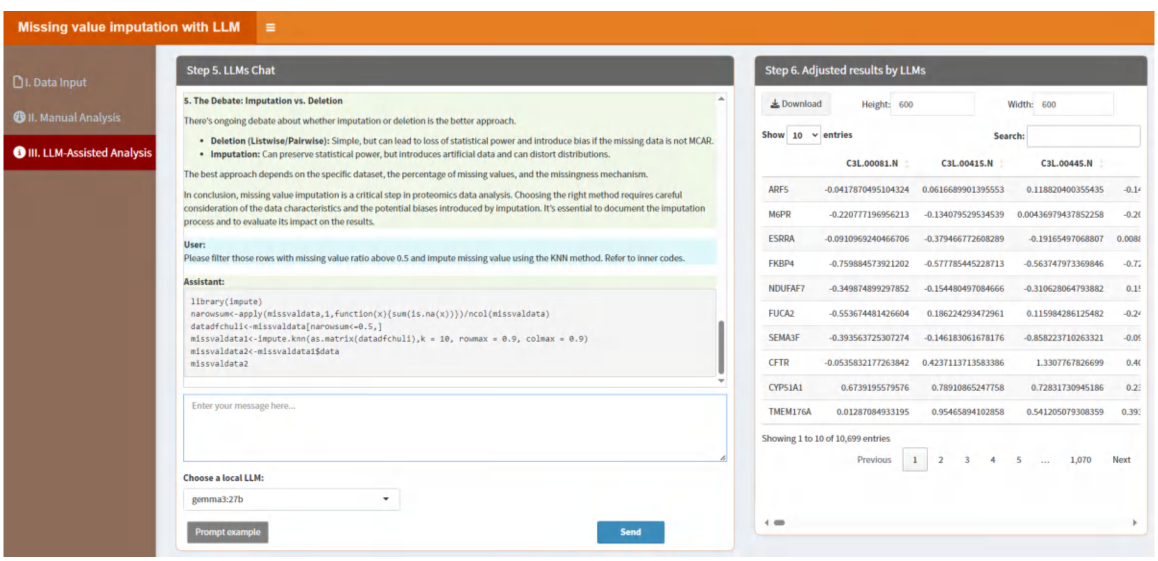
The outcomes presented in “4.2. Tables” display the imputed data generated by the K Nearest Neighbors (KNN) method as the default setting. Users are able to click the “Download” button to obtain this result for subsequent analysis.



Additionally, users can click “III. LLM-Assisted Analysis” to perform missing value imputation with the help of a LLM. For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like “What is missing value imputation in proteomics data analysis?”, as below, it will give a detailed description about missing value imputation in proteomics data analysis for users.

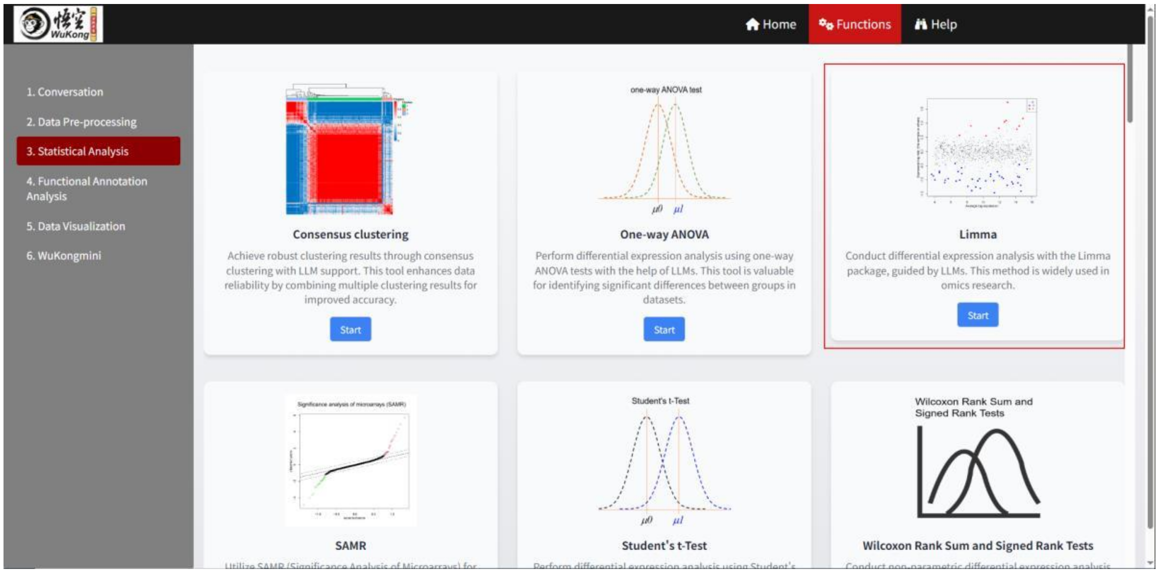


Users can input a prompt such as, “Please filter those rows with missing value ratio above 0.5 and impute missing value using the KNN method. Refer to inner codes”, where "Refer to inner codes" instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate imputed results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.

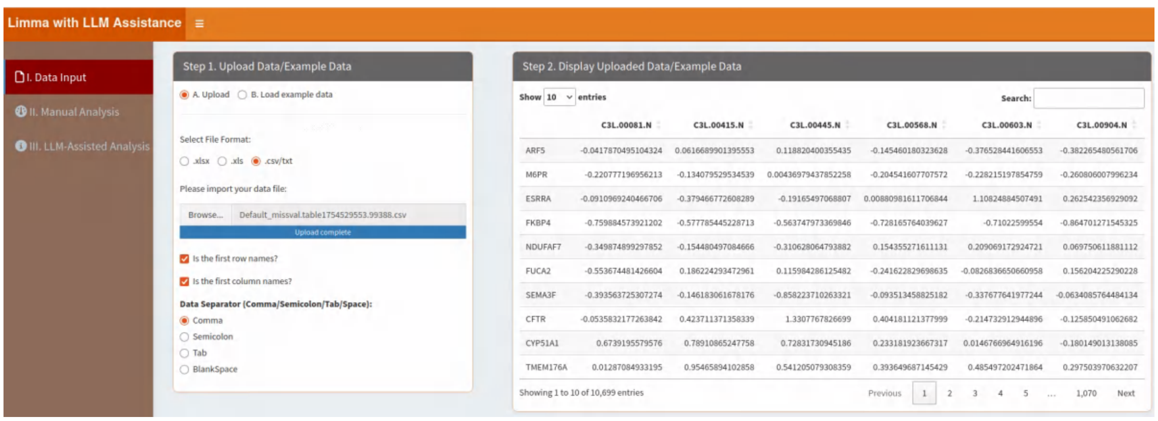


b. Differential protein expression analysis

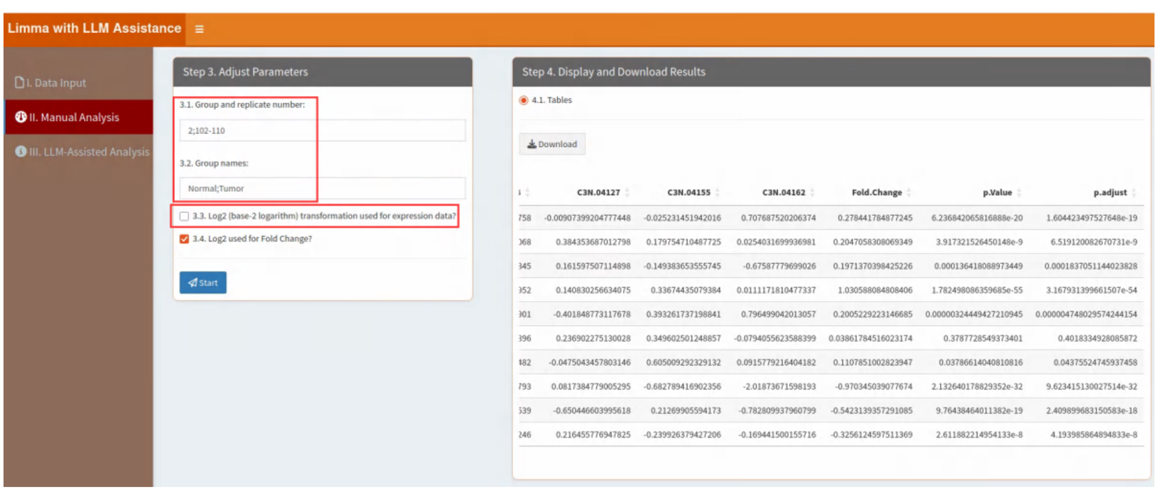
Herein, we choose Limma to process the differential protein expression analysis.



Then click "Start" button and upload the result data from the missing value imputation above, shown as below.

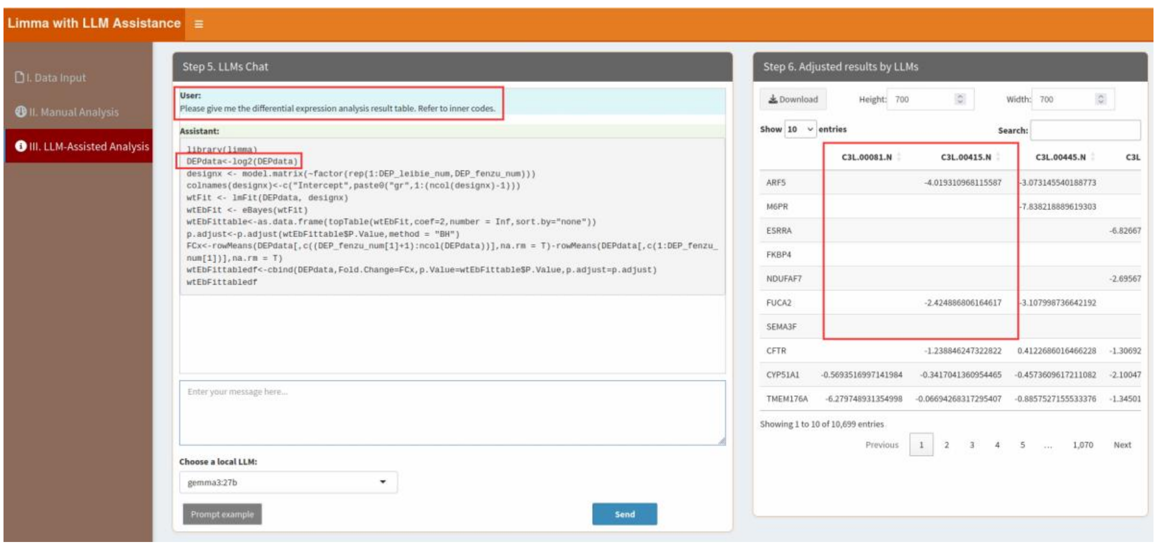


Next, click "II. Manual Analysis". We should type right parameters based on the data information, and click "Start" button. The result can be downloaded in "4.1. Tables".

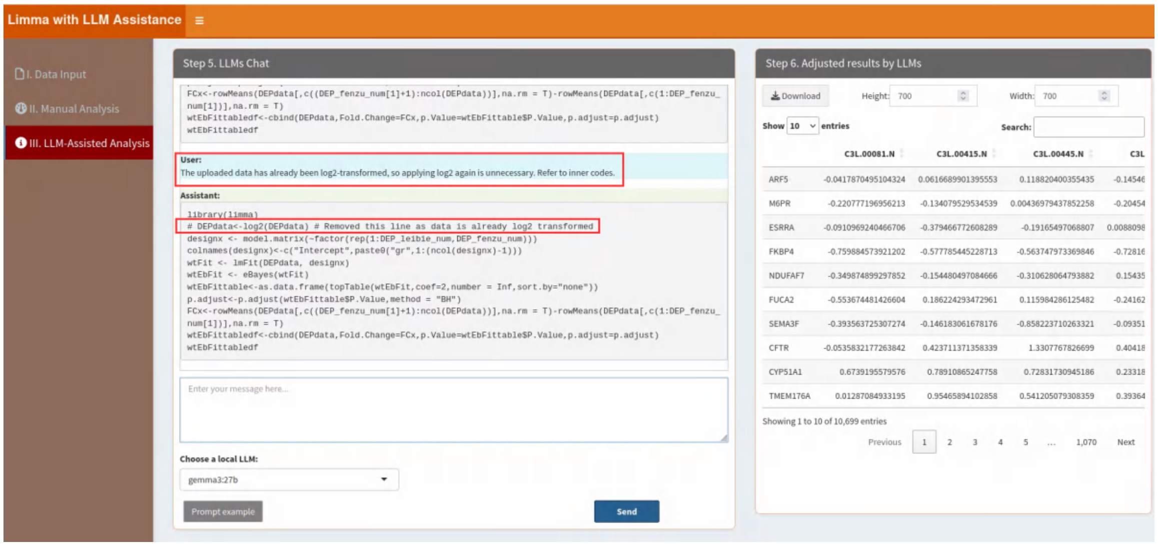


Additionally, users can click “III. LLM-Assisted Analysis” to perform this module with the help of a LLM. For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like “Please give me the differential expression analysis result table. Refer to inner codes.”, where "Refer to inner codes" instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.

For instance, as shown below, the internal code automatically applies log2 transformation to the uploaded data. If the data has already been log2-transformed beforehand, applying the transformation again would lead to incorrect results.

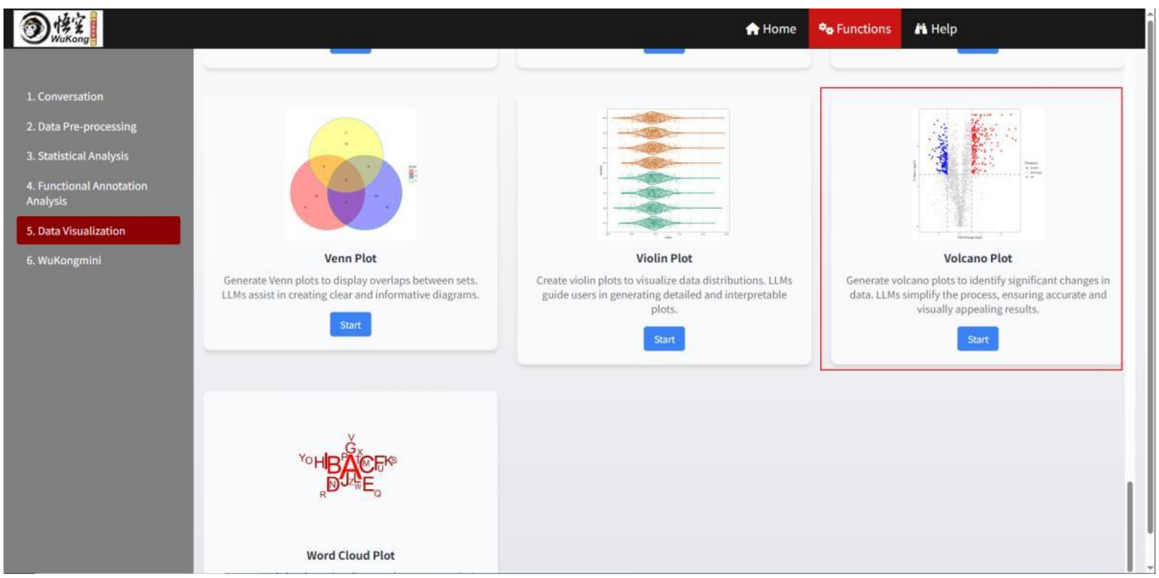


In such cases, users must revise their prompts to explicitly indicate that log2 should not be applied, ensuring the LLM interprets the data context properly.

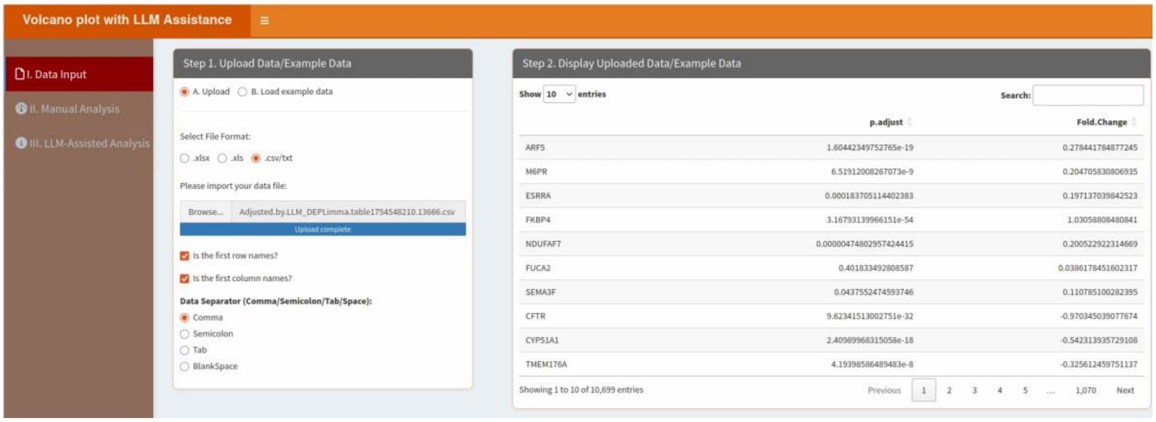


c. volcano plot

We can find this module here:

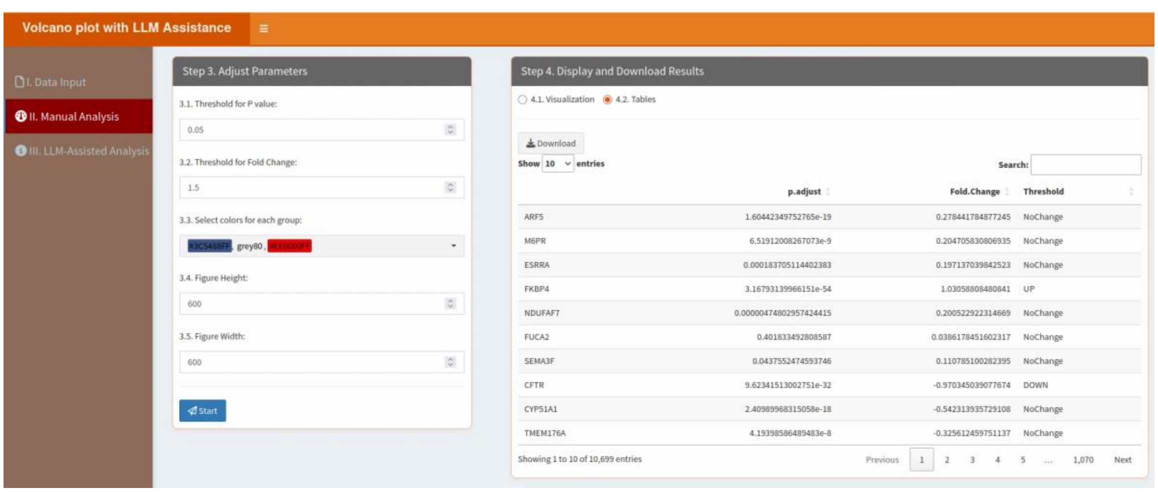


Then click "Start" button and upload the result data from above. Herein, we extracted three columns (Protein names, p.adjust and Fold.Change), shown as below.



Next, click “II. Manual Analysis”. We should type right parameters based on the data information, and click “Start” button. The result in “4.1. Visualization” shows the volcano plot.

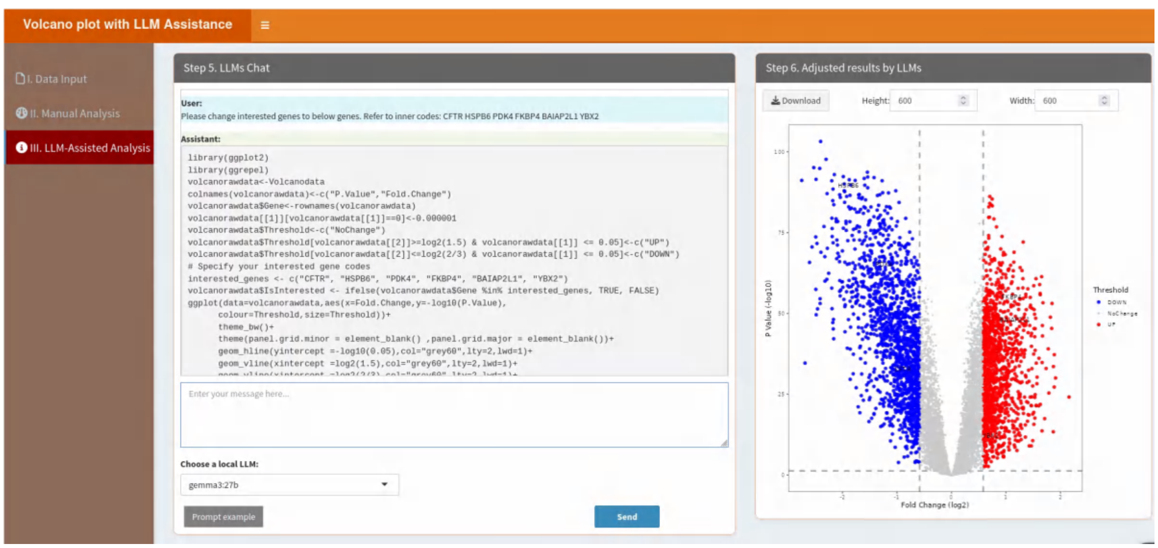
In the “4.2. Tables”, it shows the differentially expressed protein results (UP, DOWN and NoChange):



So we can click the “Download” button to save the results.

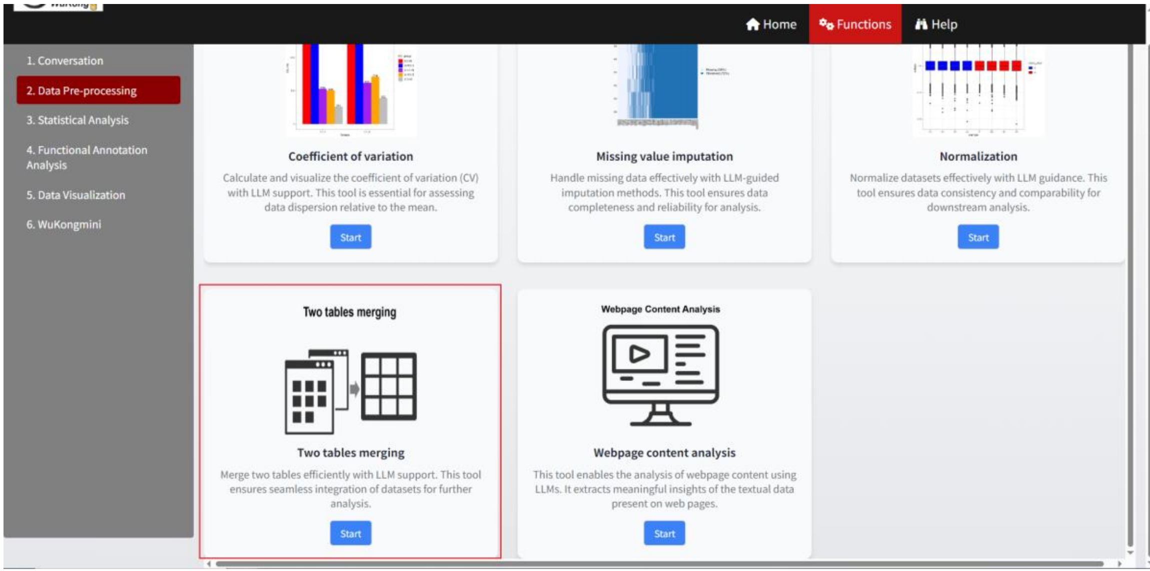
Additionally, users can click “III. LLM-Assisted Analysis” to perform this module with the help of a LLM. For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like,

“Please change interested genes to below genes. Refer to inner codes:
CFTR
HSPB6
PDK4
FKBP4
BAIAP2L1
YBX2”, where "Refer to inner codes" instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.

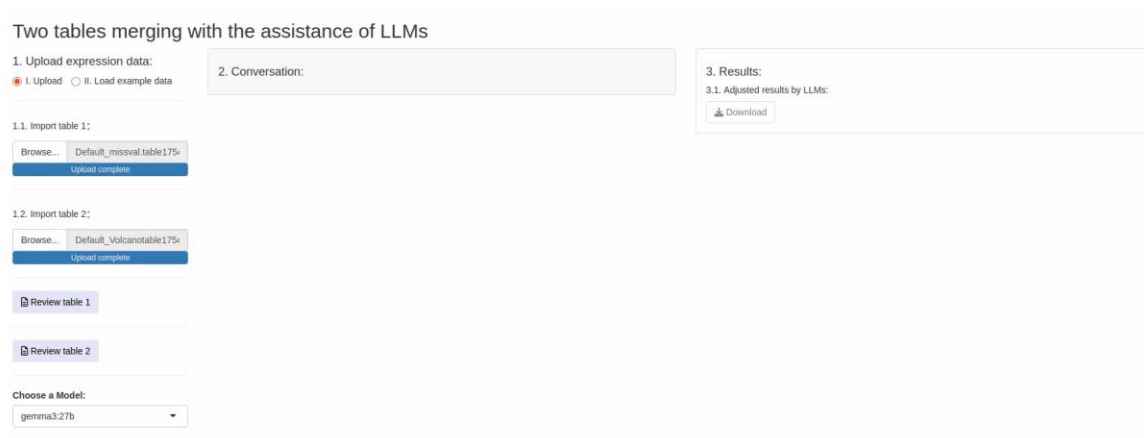


d. Two table merging

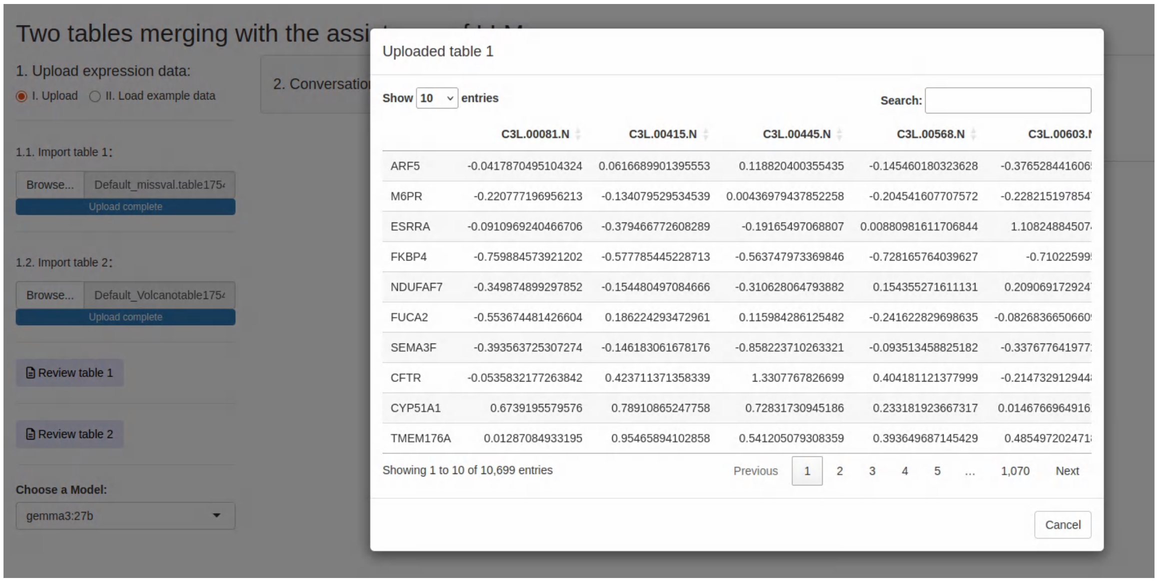
From above, we can obtain the differentially expressed proteins (UP and DOWN). So we can use this module to extract the expression matrix of these proteins. Alternatively, users may also achieve this using other analysis tools outside the platform.



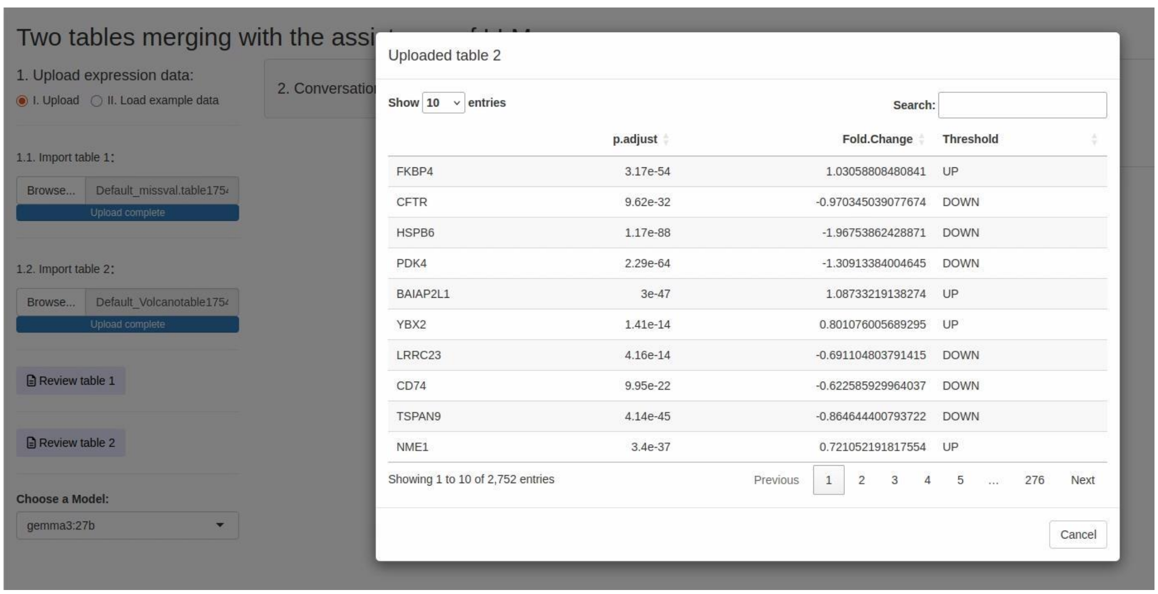
Then click “Start” button and upload two data from above: 1. Data from missing value imputation; 2. The expression matrix of differentially expressed proteins.



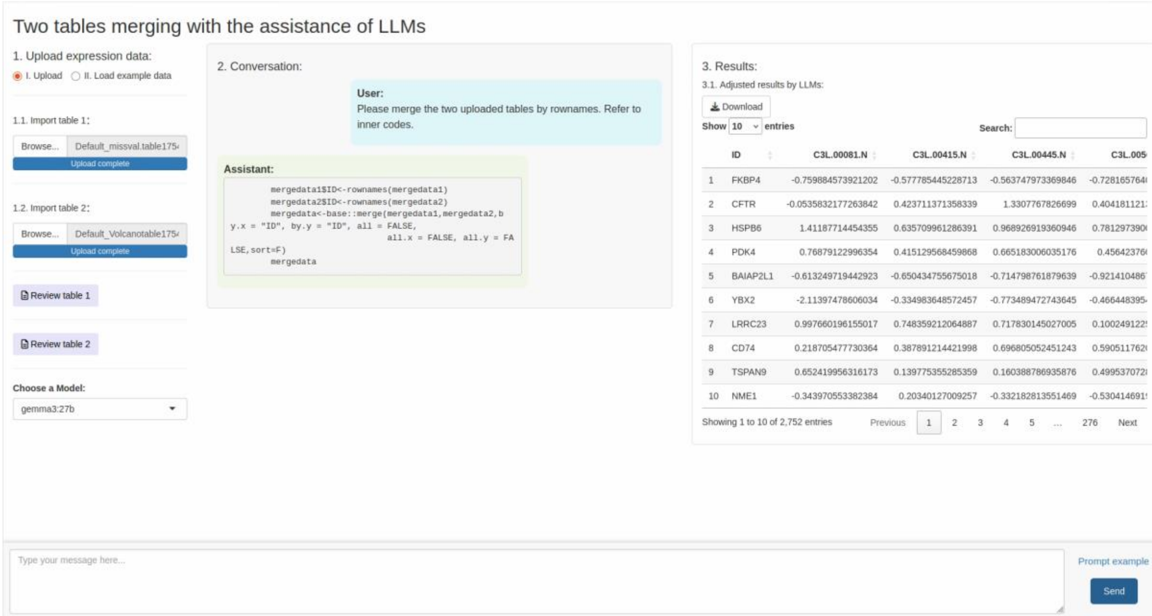
We can click the “Review table 1” button to check the first data:



Click the “Review table 2” button to check the Second data:



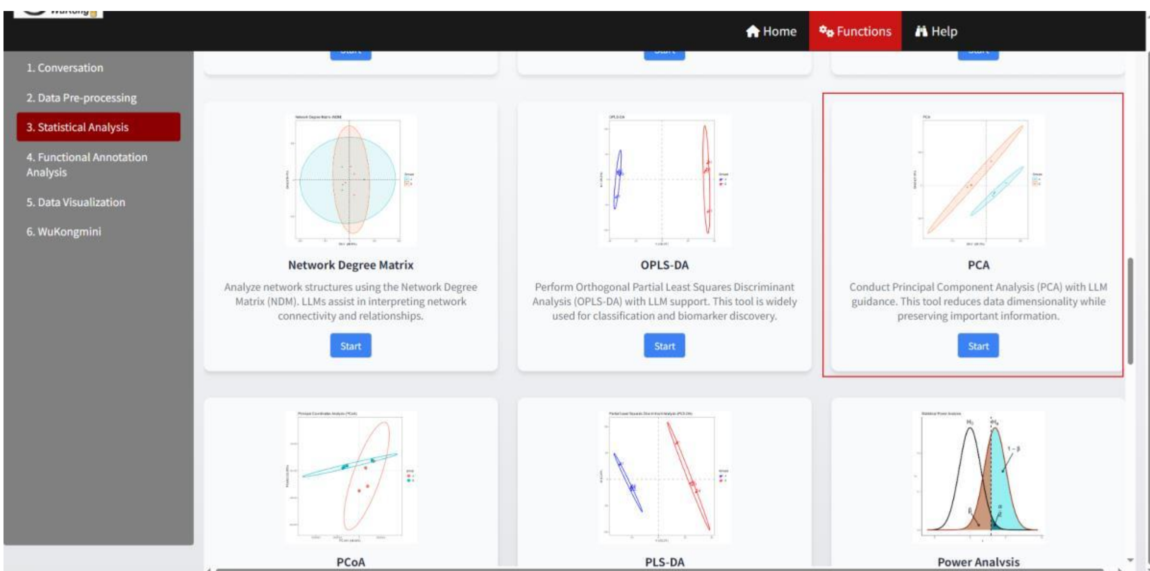
Next, users can click “Prompt example” and the “Send” button to perform this module with the help of a LLM. For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like “Please merge the two uploaded tables by rownames. Refer to inner codes.”, where "Refer to inner codes" instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.



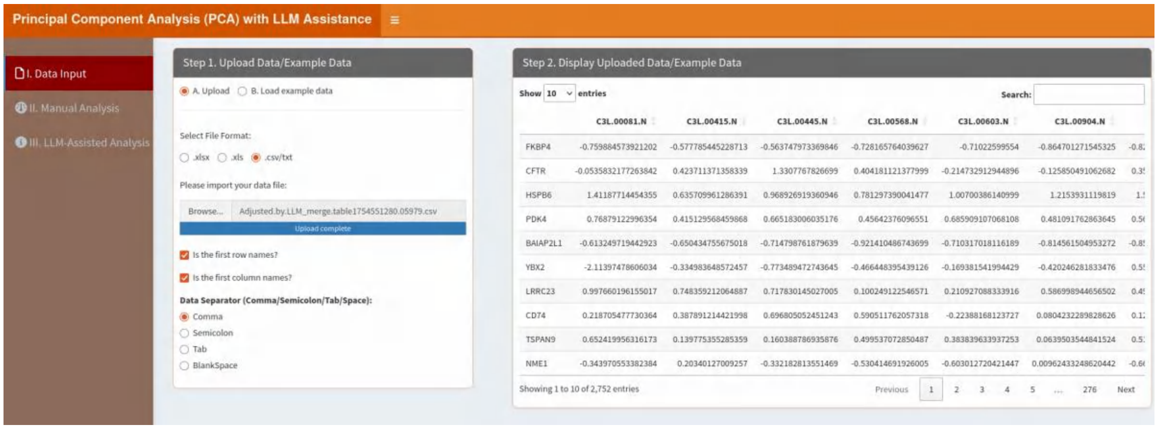
In the “3. Results” part, it shows the merged results and click the “Download” button to save the result.

e. PCA

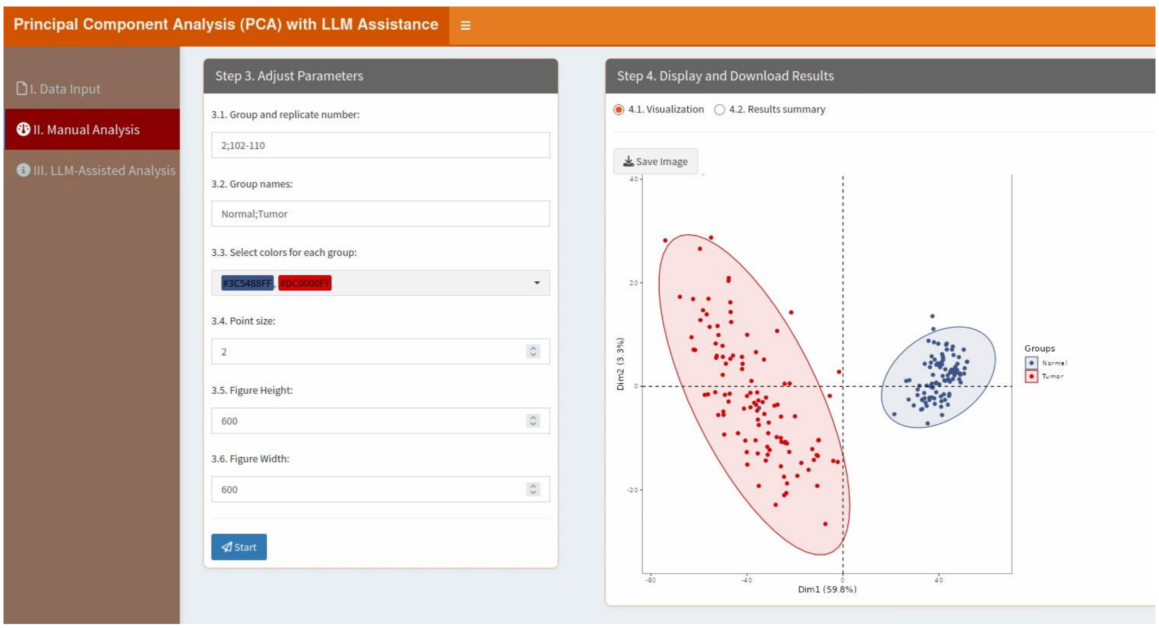
We can find this module here:



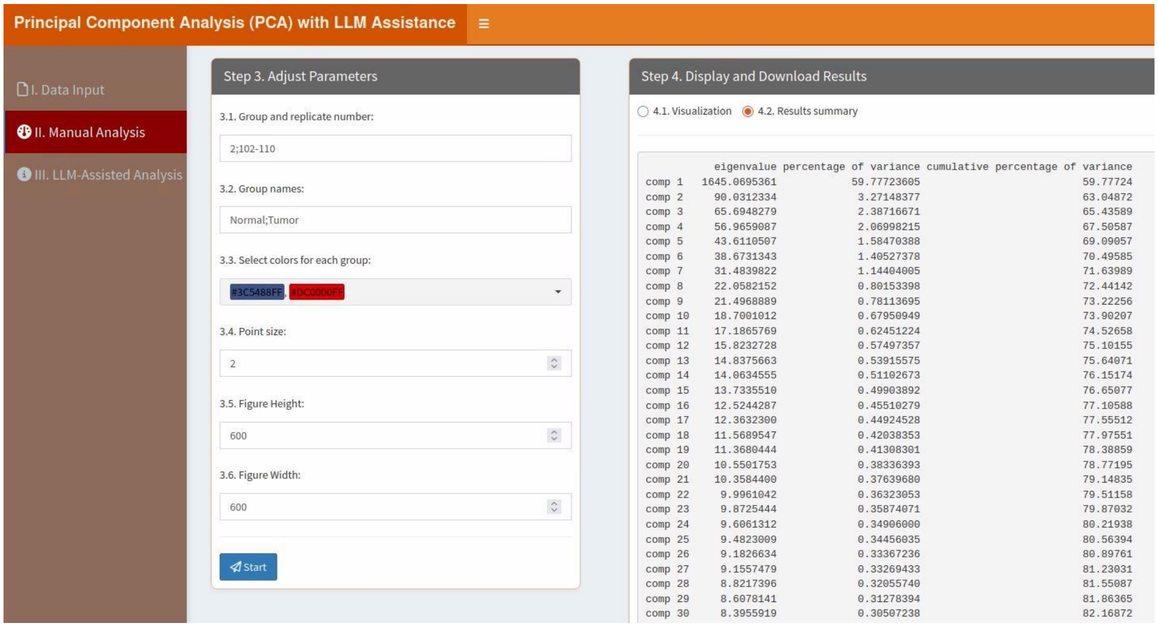
Then click “Start” button and upload the result data from above. Please note that users should remove the last three columns (p.adjust, Fold.Change, Threshold).



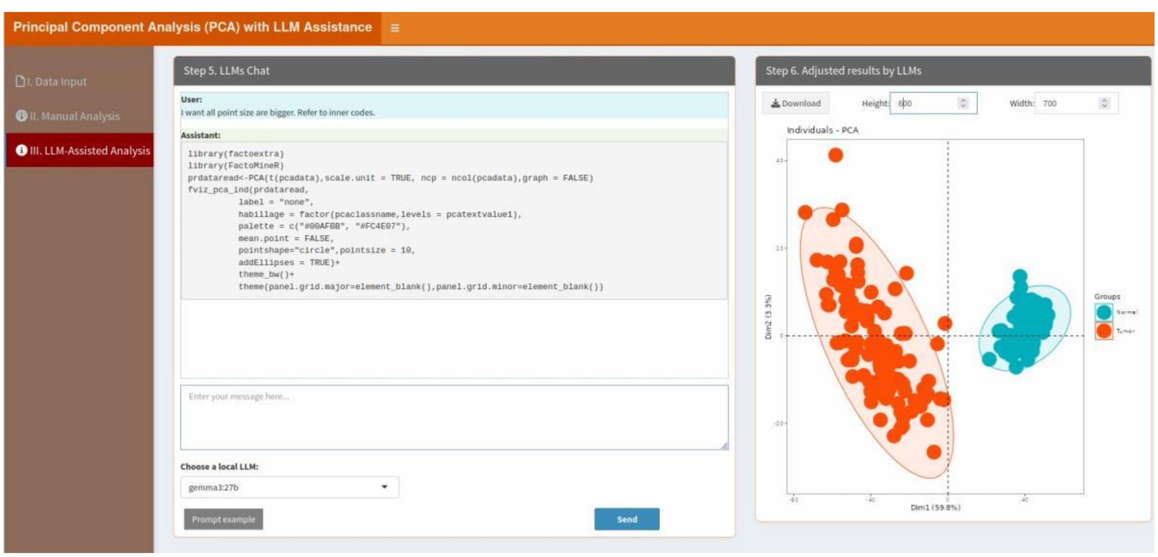
Next, click "II. Manual Analysis". We should type right parameters based on the data information, and click "Start" button. The result in "4.1. Visualization" shows the PCA plot.



In the "4.2. Results summary", it shows the summary of PCA.

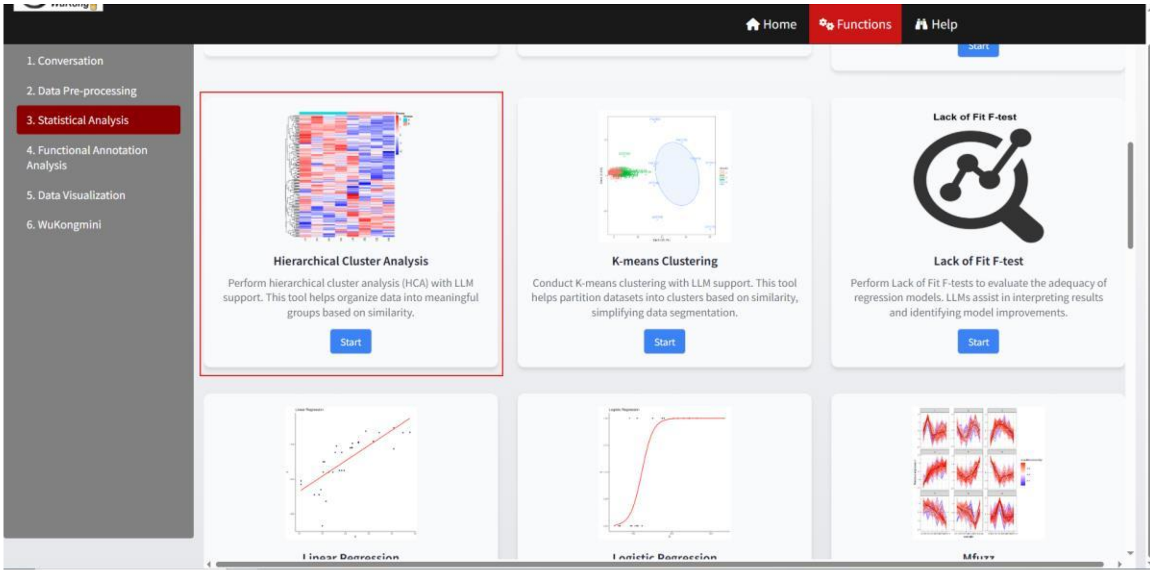


Additionally, users can click “III. LLM-Assisted Analysis” to perform this module with the help of a LLM. For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like “I want all point size are bigger. Refer to inner codes.”, where "Refer to inner codes" instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.

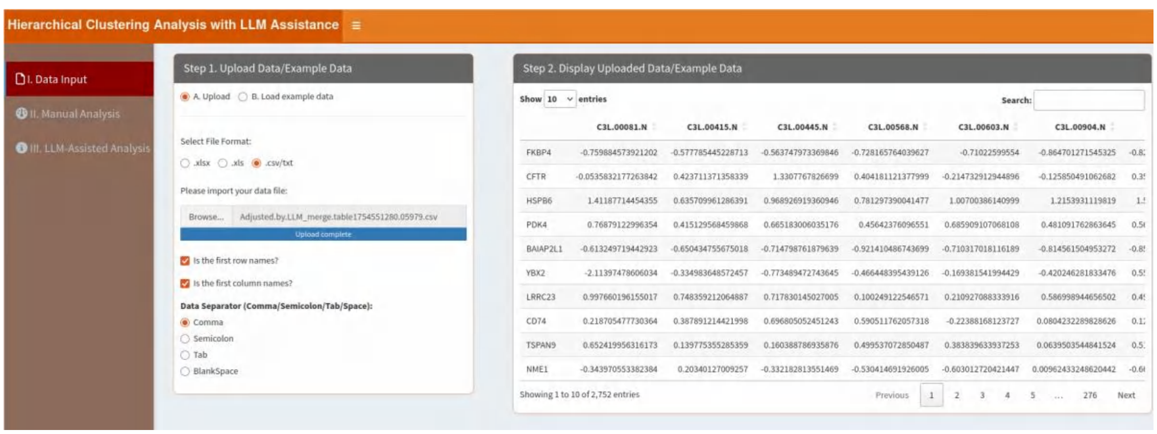


f. HCA

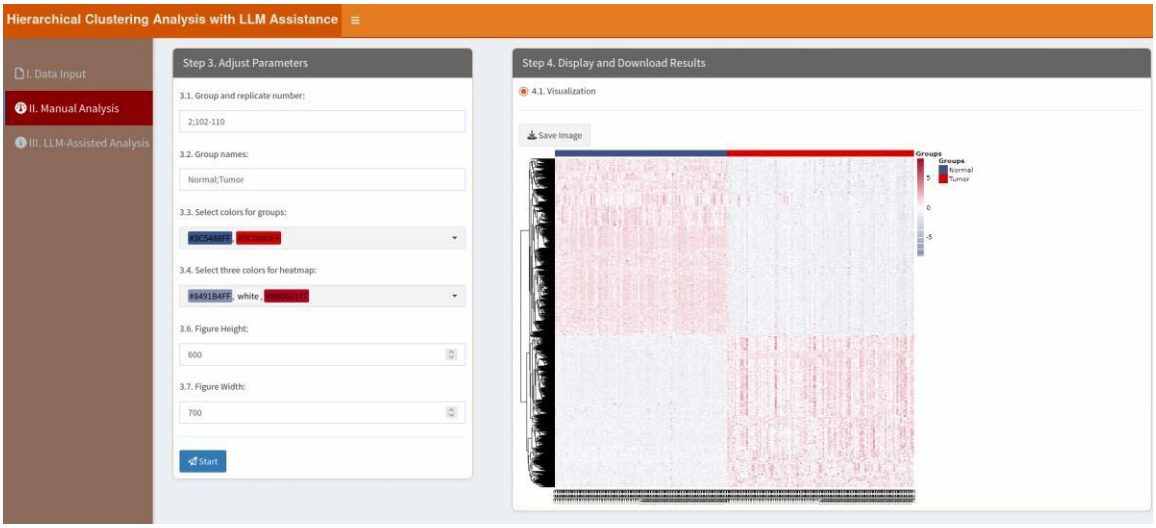
We can find this module here:



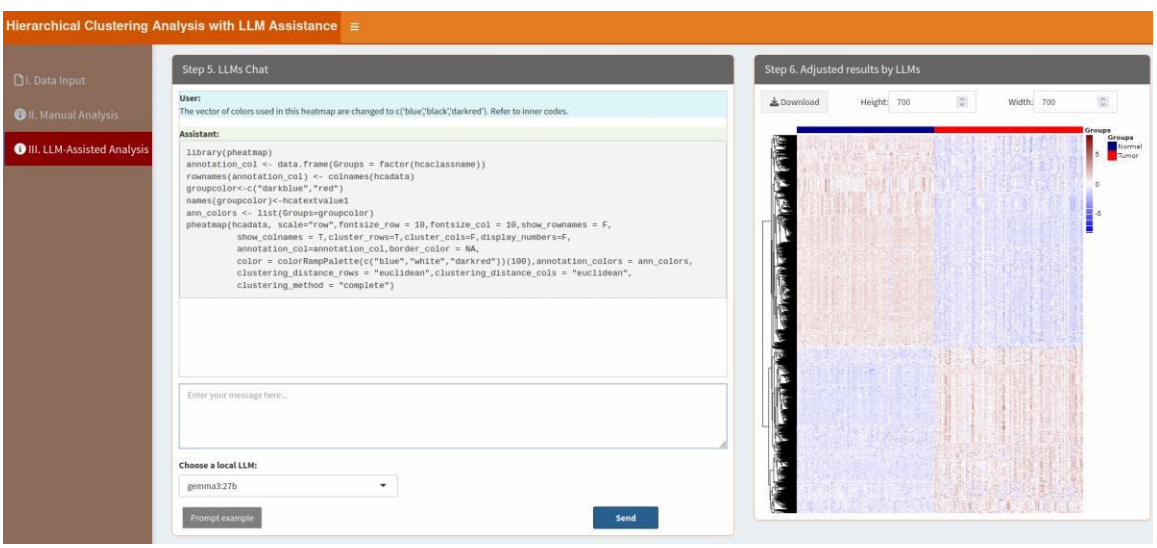
Then click “Start” button and upload the result data from the “Two table merging”. Please note that users should remove the last three columns (p.adjust, Fold.Change, Threshold).



Next, click “II. Manual Analysis”. We should type right parameters based on the data information, and click “Start” button. The result in “4.1. Visualization” shows the HCA plot.

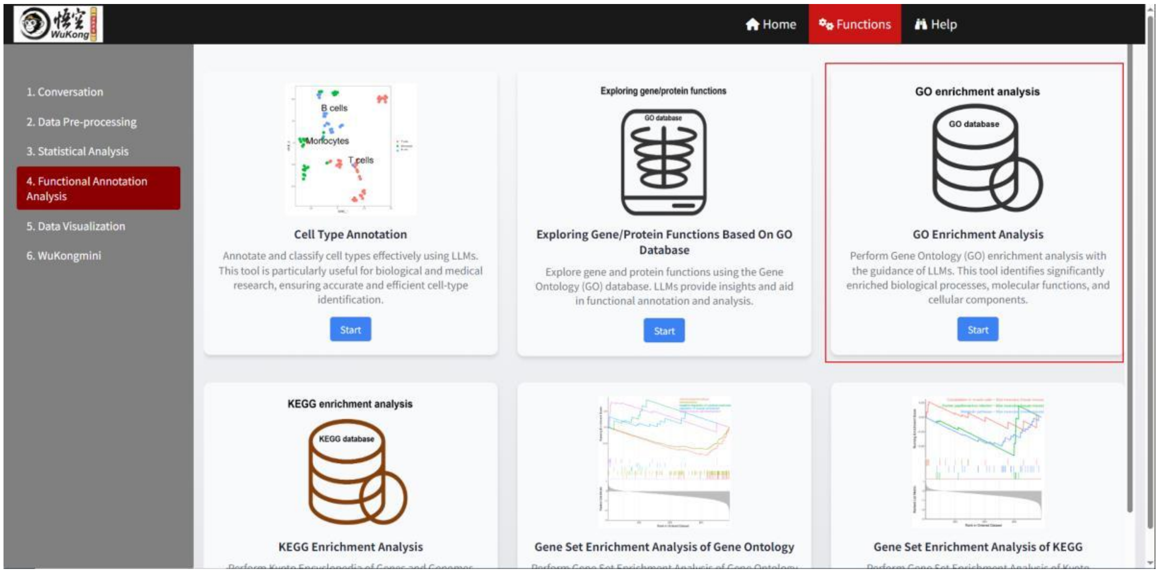


Additionally, users can click “III. LLM-Assisted Analysis” to perform this module with the help of a LLM. For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like “The vector of colors used in this heatmap are changed to c('blue','black','darkred'). Refer to inner codes.”, where "Refer to inner codes" instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.

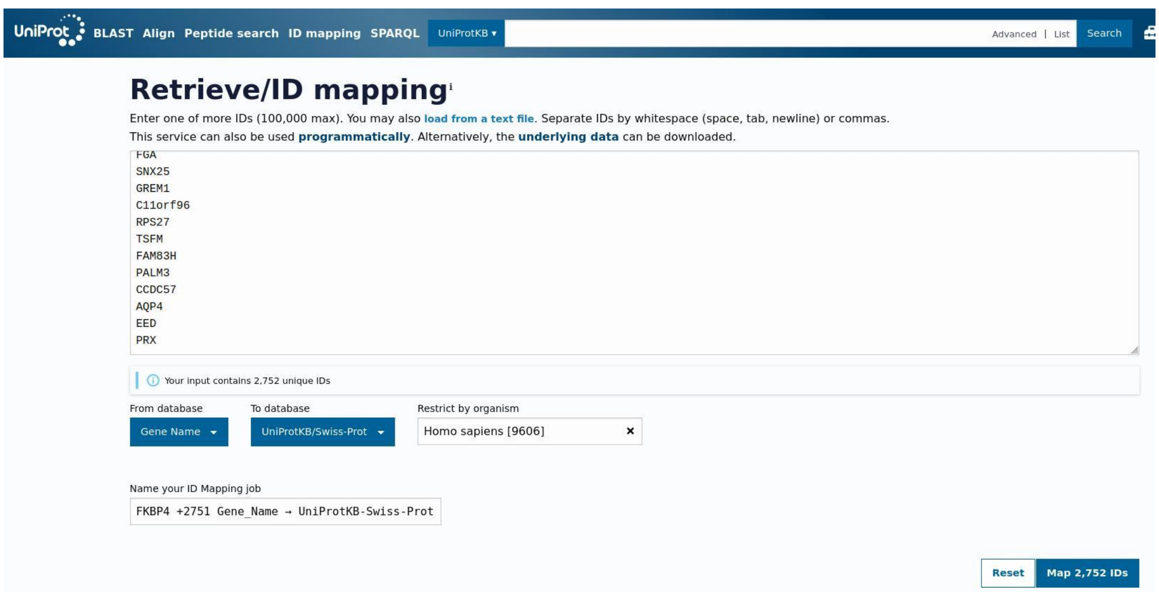


g. GO enrichment analysis

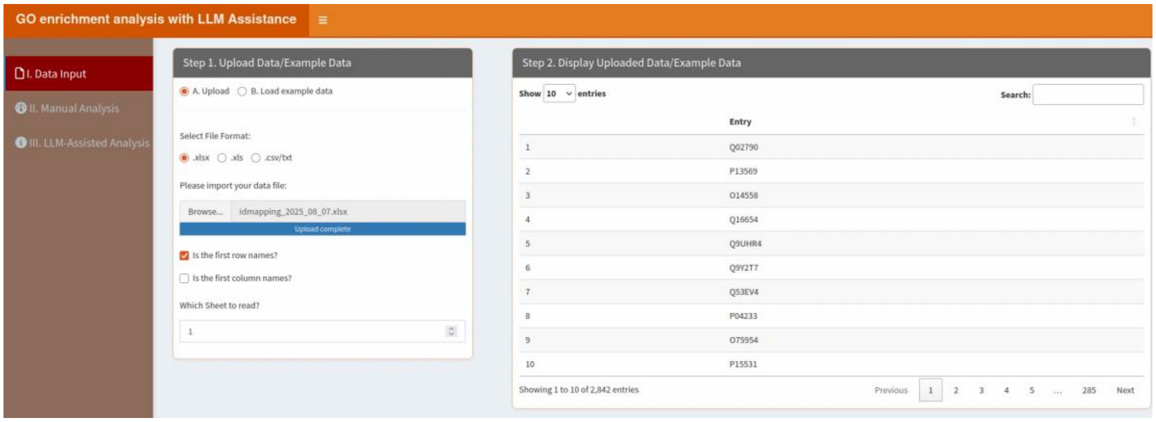
We can find this module here:



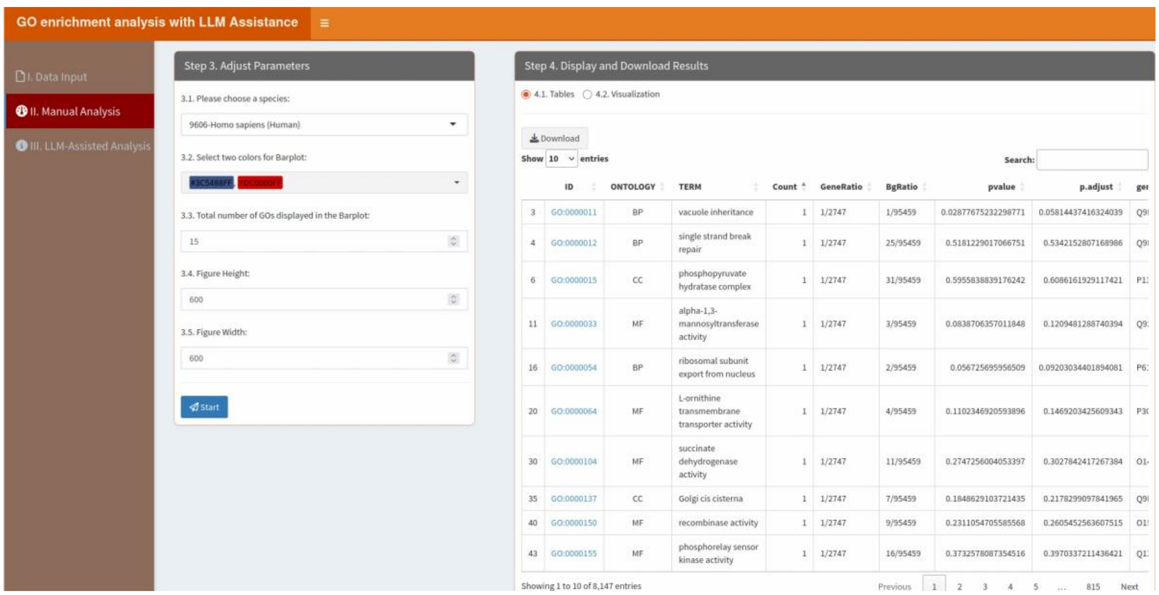
Then click “Start” button and upload the differentially expressed proteins. This module needs one column of UniProt IDs, so we need transform these protein names to UniProt IDs using the ID mapping function of the UniProt web (https://www.uniprot.org/tool-dashboard).



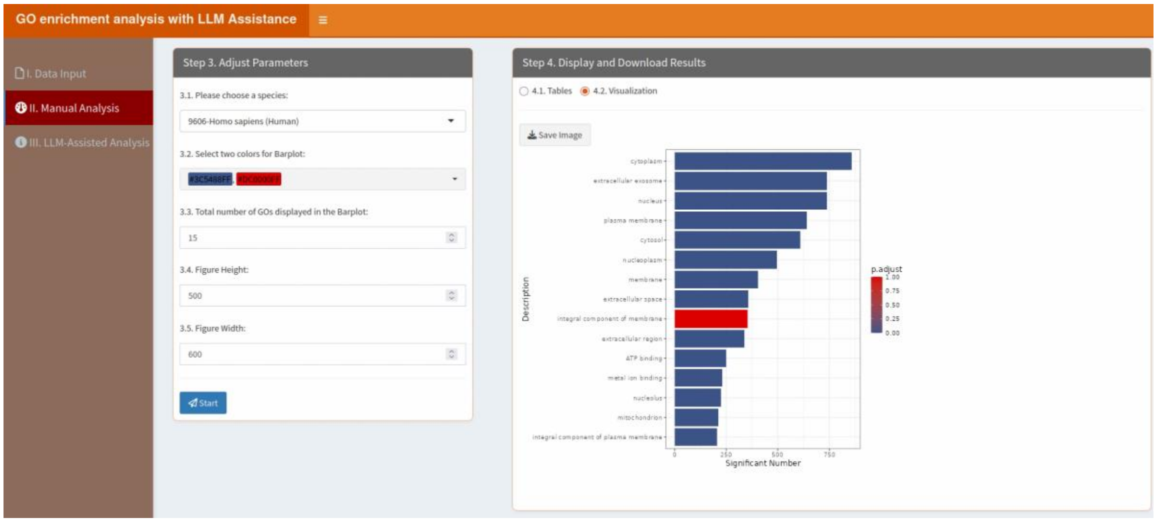
Then we can upload these UniProt IDs:



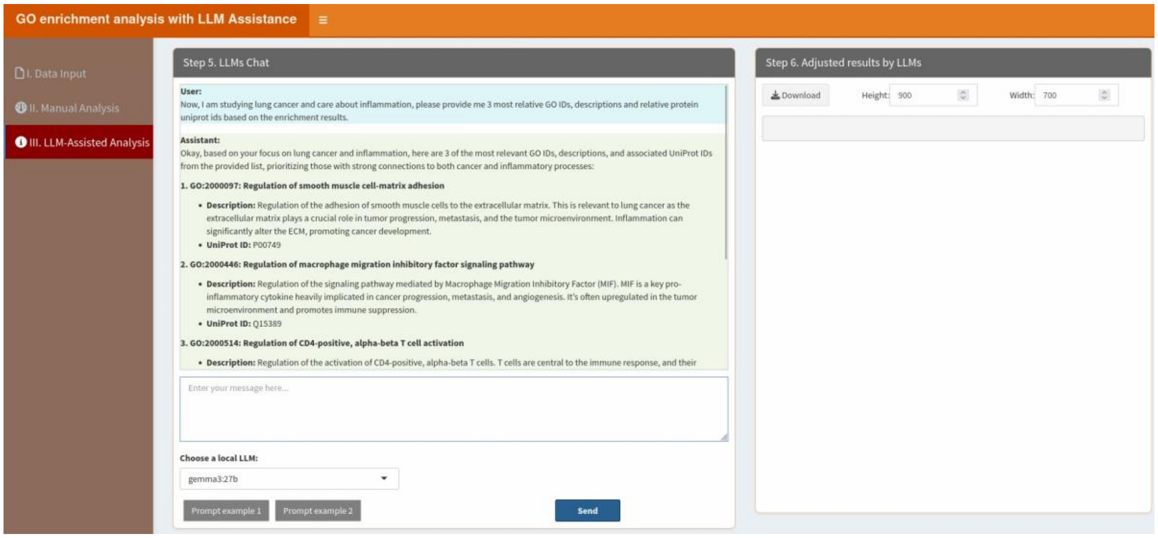
Next, click “II. Manual Analysis”. We should type right parameters based on the data information, and click “Start” button. The enrichment table is shown in the “4.1. Tables” part. Users can click the “Download” button to save the result.



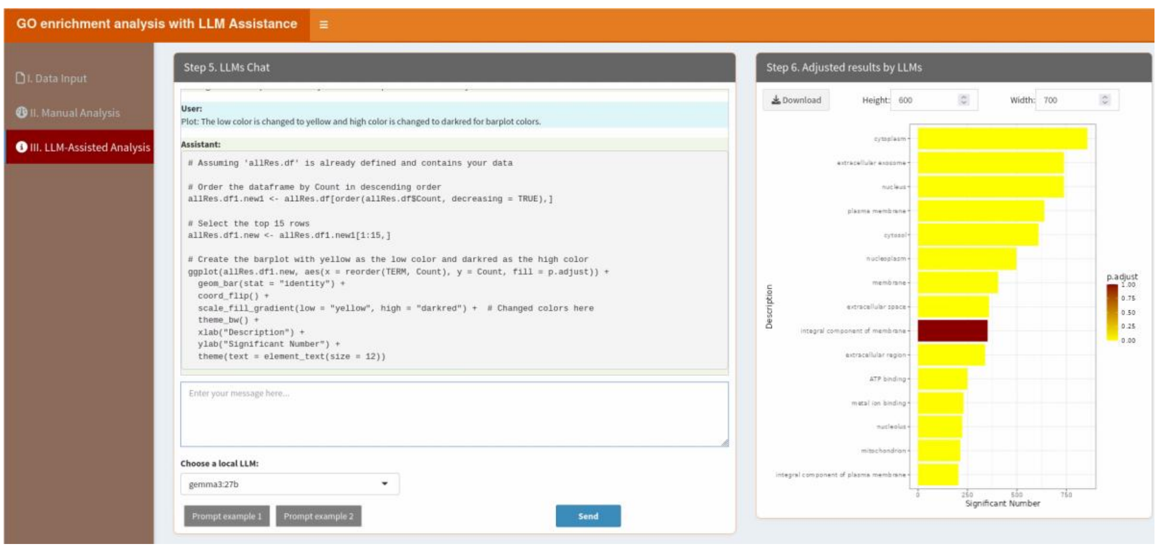
In the “4.2. Visualization” part, it shows the top 15 GOs ordered by the counts.



Additionally, users can click “III. LLM-Assisted Analysis” to perform this module with the help of a LLM. Two types of prompts are supported: prompts beginning with “Text:” are used to request interpretation or explanation of the enrichment results (e.g. summarizing pathways relevance), while prompts beginning with “Plot:” allow users to customize the barplot visualization (e.g. modifying colors, sorting order, or number of displayed pathways). For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like “Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results.”, where "Refer to inner codes" instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.

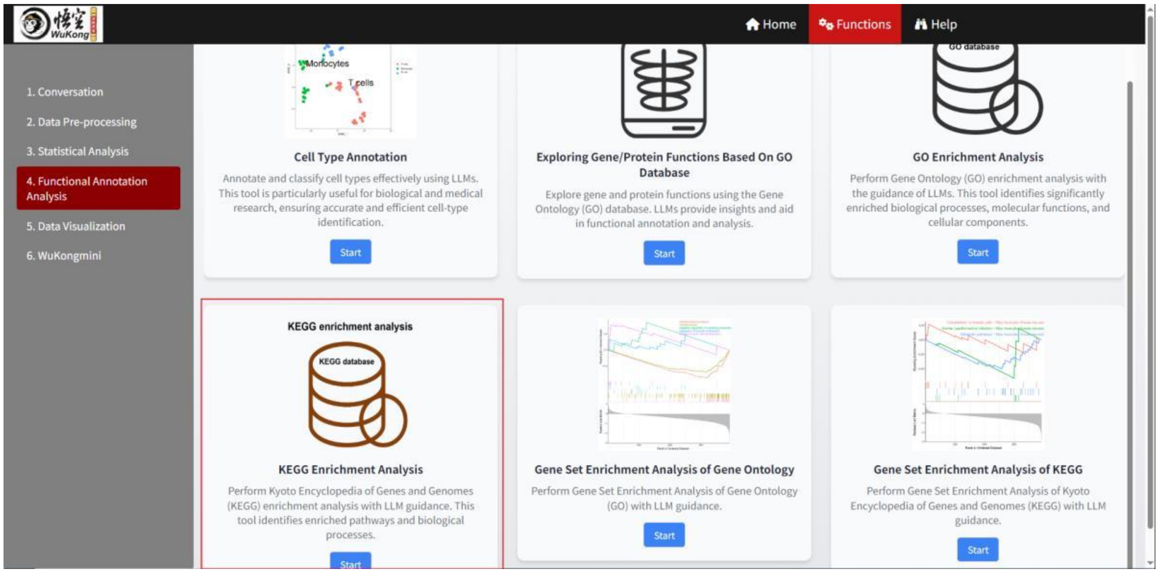


Users can also adjust the plot by type “Plot: The low color is changed to yellow and high color is changed to darkred for barplot colors.”

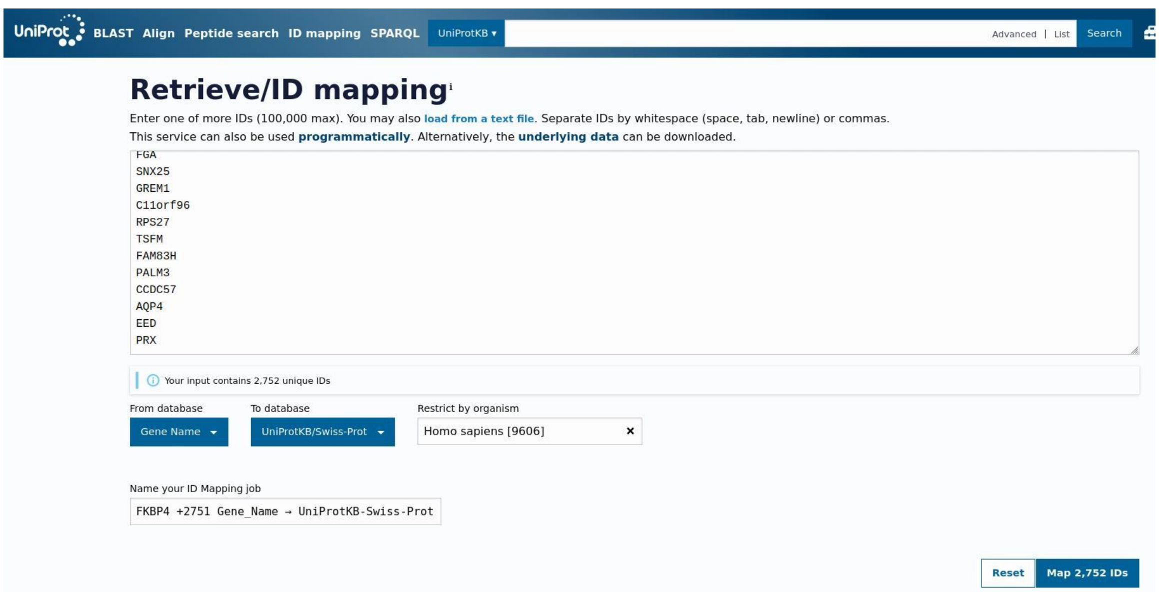


h. KEGG pathway enrichment analysis

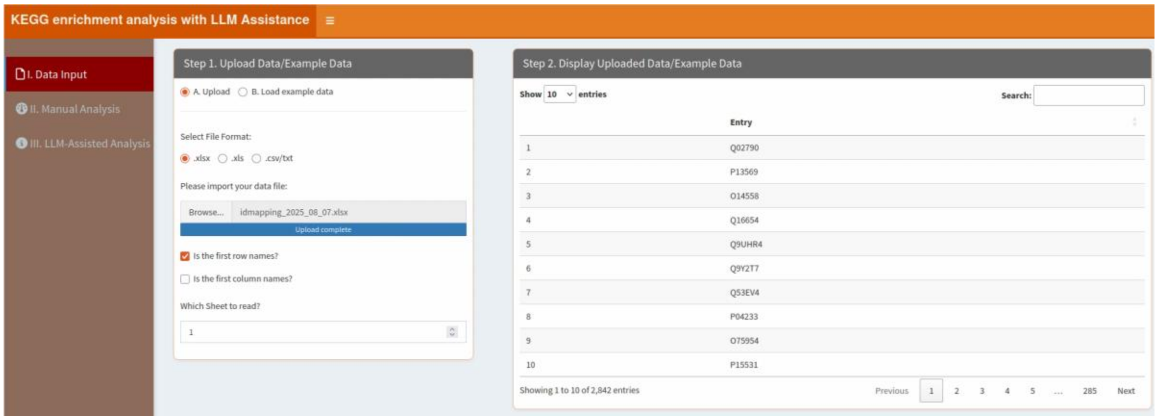
We can find this module here:



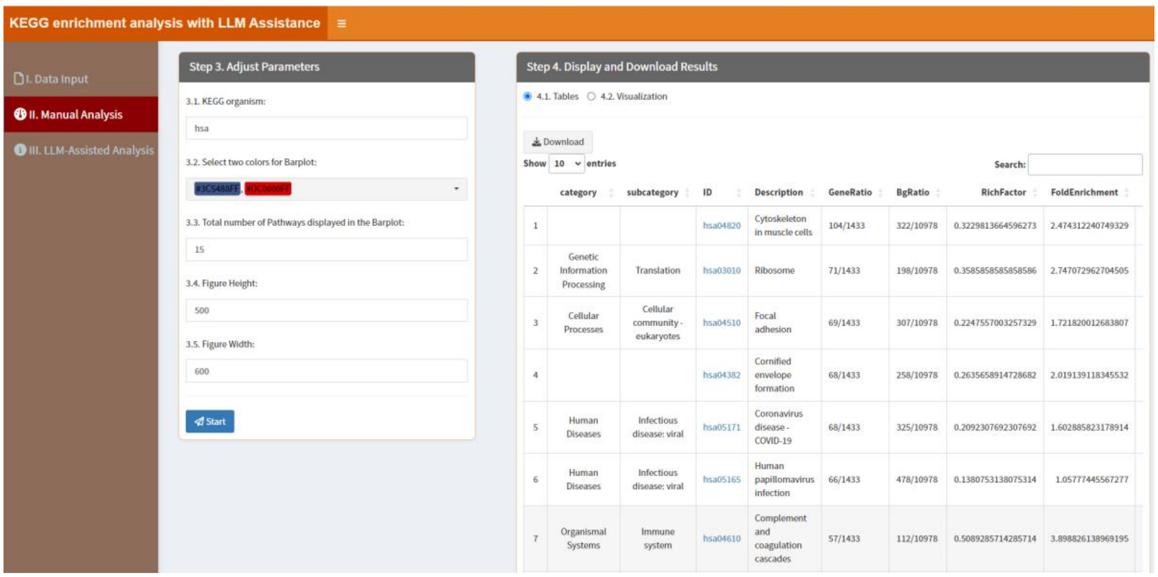
Then click “Start” button and upload the differentially expressed proteins. Same as above, this module also needs one column of UniProt IDs, so we need transform these protein names to UniProt IDs using the ID mapping function of the UniProt web (https://www.uniprot.org/tool-dashboard).



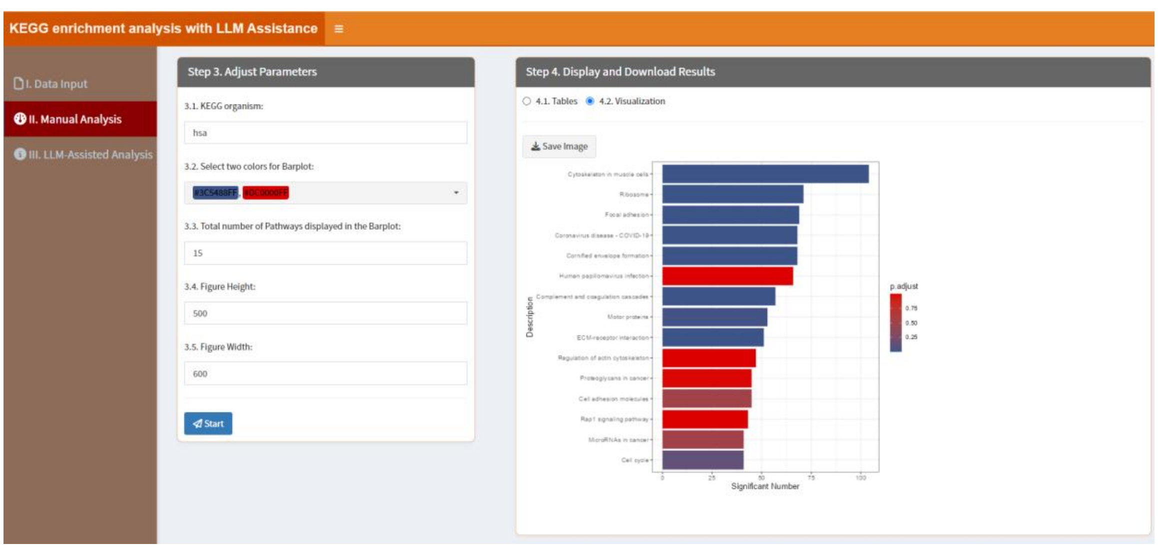
Then we can upload these UniProt IDs:



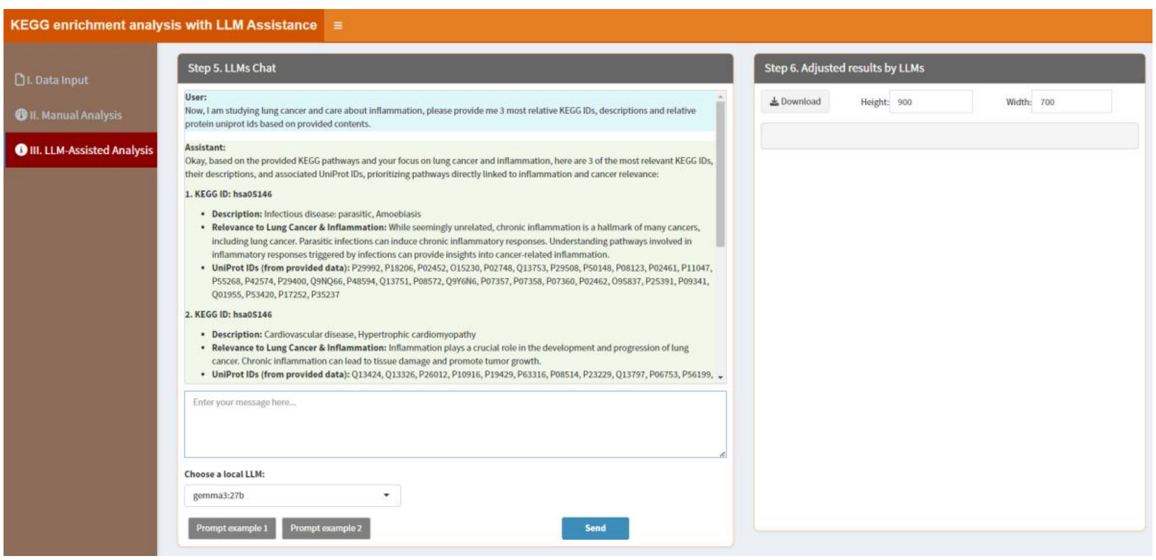
Next, click “II. Manual Analysis”. We should type right parameters based on the data information, and click “Start” button. The enrichment table is shown in the “4.1. Tables” part. Users can click the “Download” button to save the result.



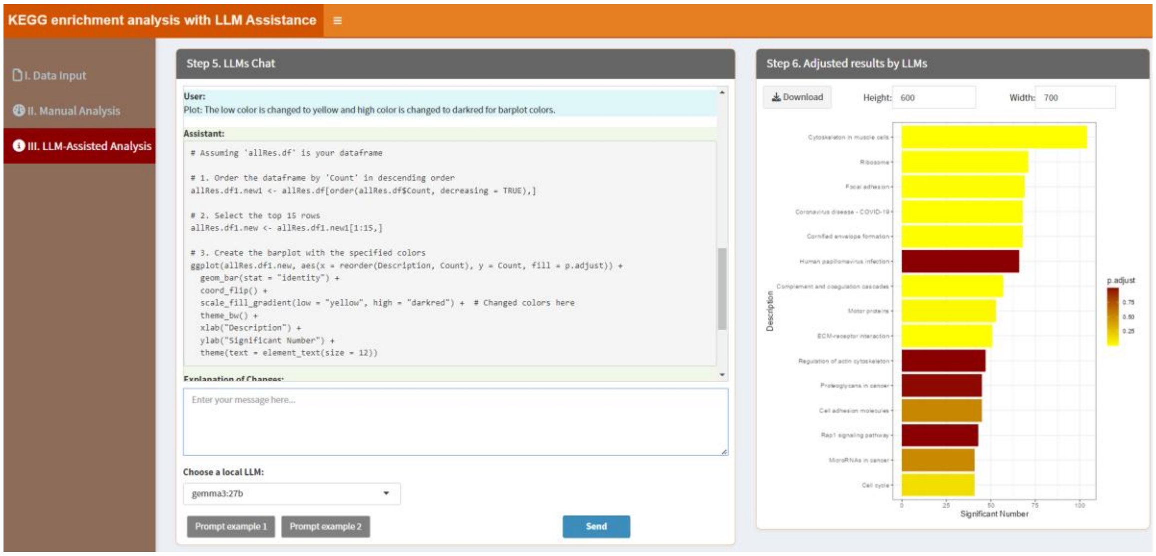
In the “4.2. Visualization” part, it shows the top 15 pathways ordered by the counts.



Additionally, users can click “III. LLM-Assisted Analysis” to perform this module with the help of a LLM. Two types of prompts are supported: prompts beginning with “Text:” are used to request interpretation or explanation of the enrichment results (e.g. summarizing pathways relevance), while prompts beginning with “Plot:” allow users to customize the barplot visualization (e.g. modifying colors, sorting order, or number of displayed pathways). For instance, after selecting an LLM (e.g. gemma3:27b, a large-scale model that requires a high-performance machine, such as one equipped with an NVIDIA 4090 GPU), users can input queries like “Text: Now, I am studying lung cancer and care about inflammation, please provide me 3 most relative GO IDs, descriptions and relative protein uniprot ids based on the enrichment results.”, where “Refer to inner codes” instructs the selected LLM to utilize built-in scripts for this analysis. In most cases, the LLM correctly interprets these built-in codes and generates appropriate R code along with accurate results, requiring no user intervention. However, if the LLM fails to apply the built-in codes effectively, users may need to revise their prompts for better clarity. Alternatively, users familiar with R can manually adjust or provide refined R code to guide the LLM toward the desired output.



Users can also adjust the plot by type “Plot: The low color is changed to yellow and high color is changed to darkred for barplot colors.”



KEGG enrichment analysis with LLM Assistance
I. Data Input
II. Manual Analysis
III. LLM-Assisted Analysis
Step 5. LLMs Chat
User:
Plot: The low color is changed to yellow and high color is changed to darkred for barplot colors.
Assistant:
# Assuming 'allRes.df' is your dataframe
# 1. Order the dataframe by 'Count' in descending order
allRes.df1.new1 <- allRes.df[order(allRes.df$Count, decreasing = TRUE),]
# 2. Select the top 15 rows
allRes.df1.new <- allRes.df1.new1[1:15,]
# 3. Create the barplot with the specified colors
ggplot(allRes.df1.new, aes(x = reorder(Description, Count), y = Count, fill = p.adjust)) +
geom_bar(stat = "identity") +
coord_flip() +
scale_fill_gradient(low = "yellow", high = "darkred") + # Changed colors here
theme_bw() +
xlab("Description") +
ylab("Significant Number") +
theme(text = element_text(size = 12))
Enter your message here...
Choose a local LLM:
gemma3:27b
Prompt example 1
Prompt example 2
Send
Step 6. Adjusted results by LLMs
Download
Height: 600
Width: 700
Cytoskeleton in muscle cells
Ribosome
Focal adhesion
Coronavirus disease - COVID-19
Cornified envelope formation
Human papillomavirus infection
Complement and coagulation cascades
Motor proteins
ECM-receptor interaction
Regulation of actin cytoskeleton
Proteoglycans in cancer
Cell adhesion molecules
Rap1 signaling pathway
MicroRNAs in cancer
Cell cycle
Description
p.adjust
0.75
0.50
0.25
0
25
50
75
100
Significant Number